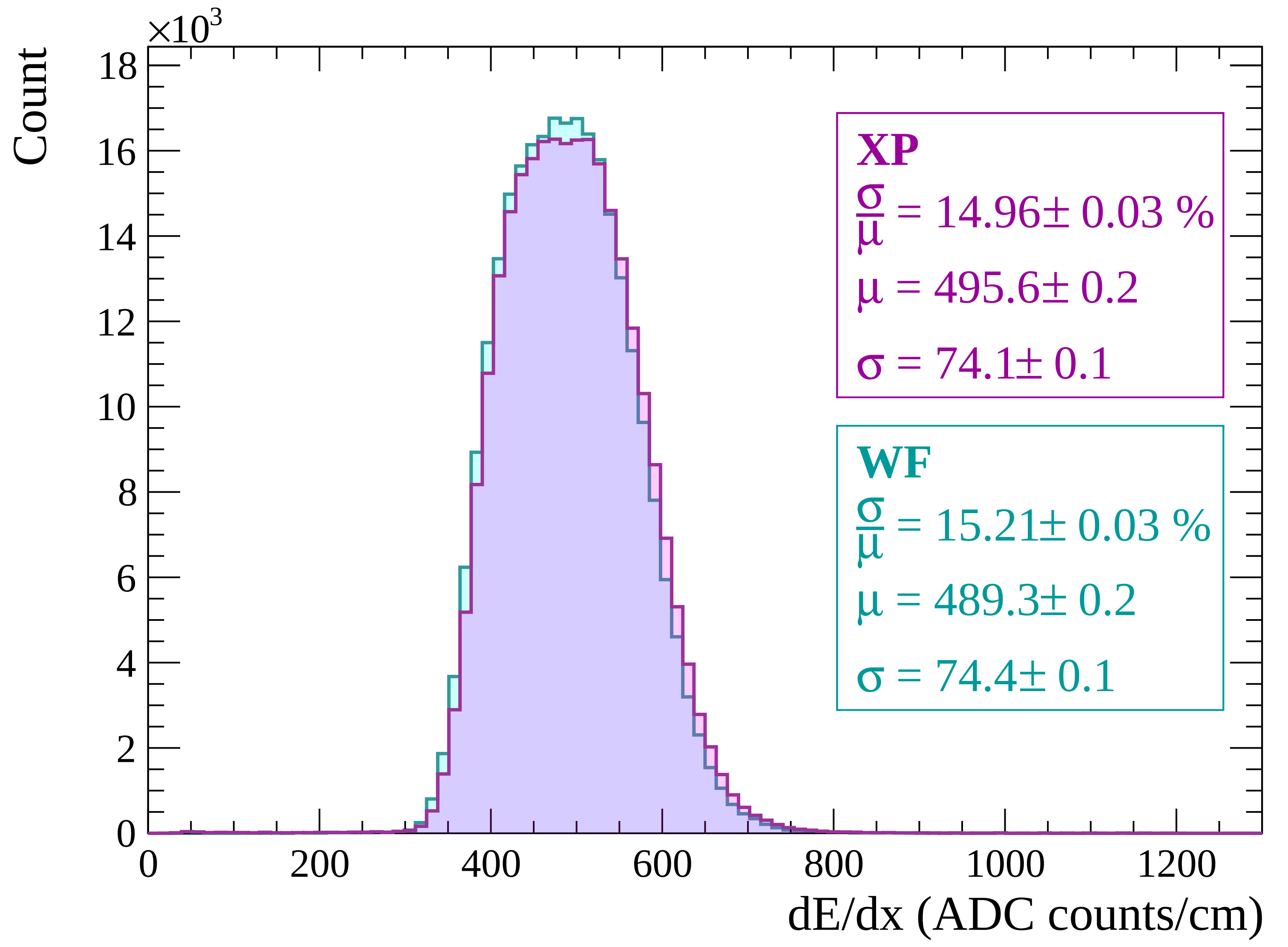
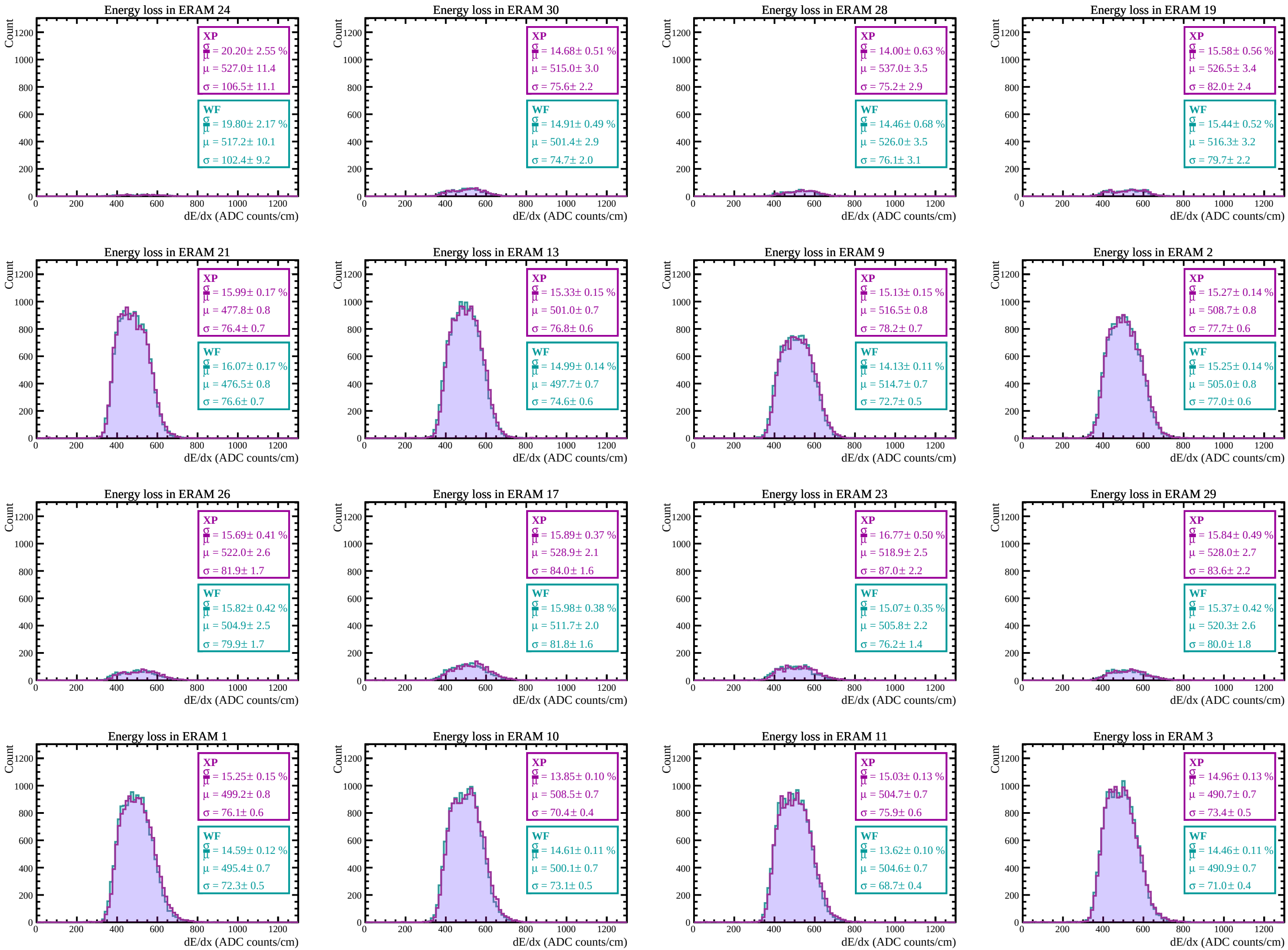
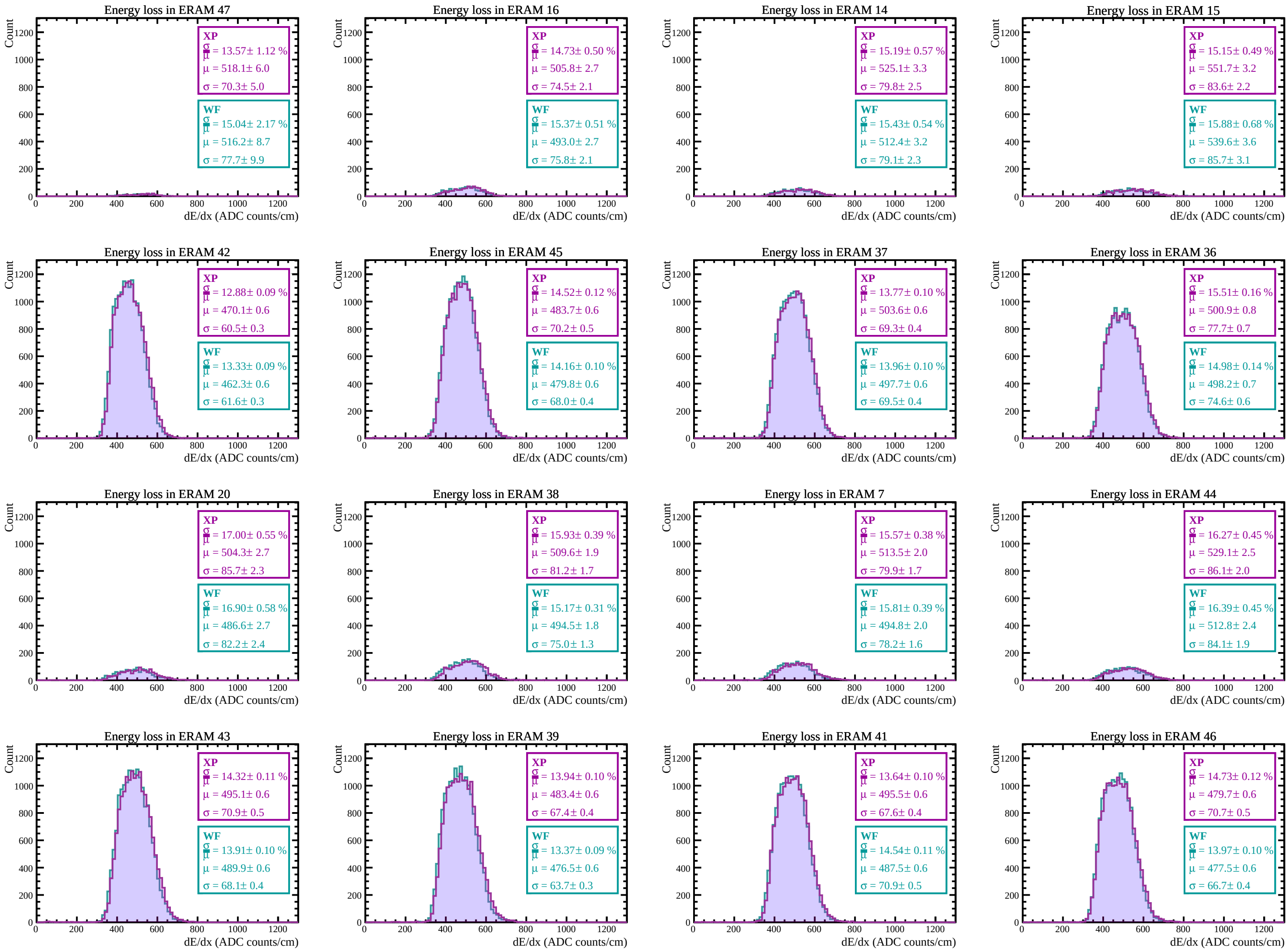


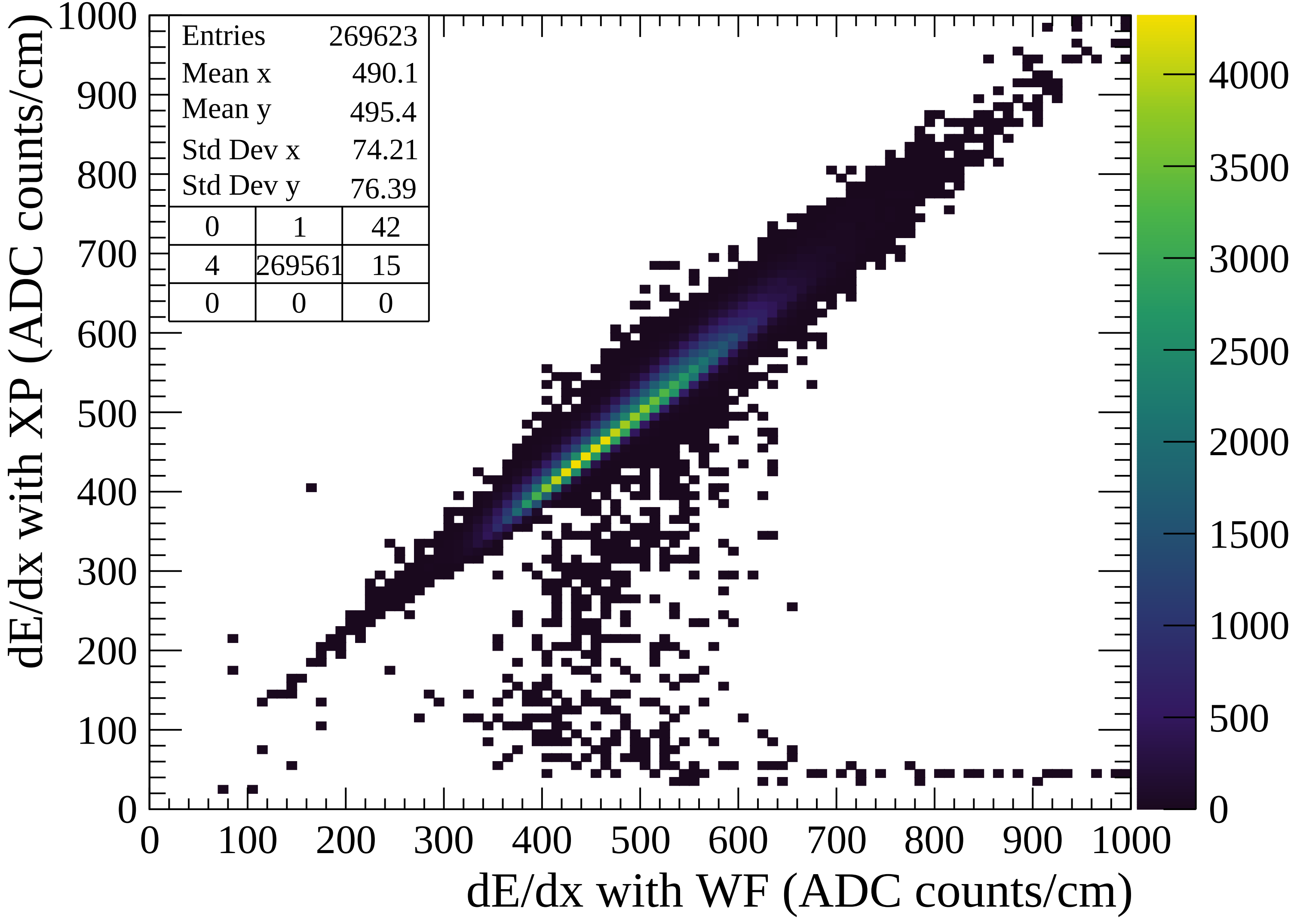
XP
$\frac{\sigma}{\mu}$ = 14.96± 0.03 %
μ = 495.6± 0.2
σ = 74.1± 0.1
WF
$\frac{\sigma}{\mu}$ = 15.21± 0.03 %
μ = 489.3± 0.2
σ = 74.4± 0.1
Count
×10³
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18



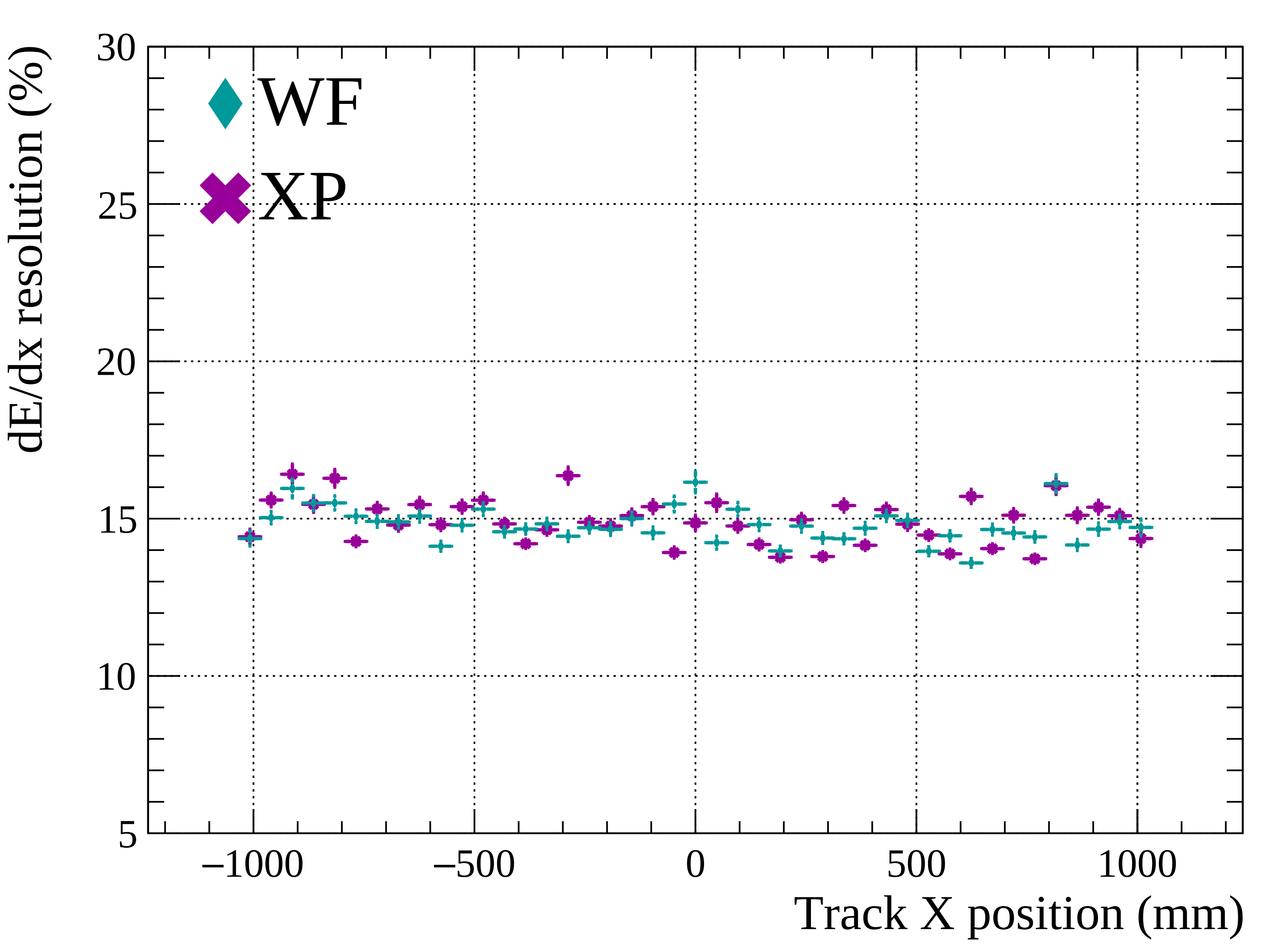
Energy loss in ERAM 24
XP
σ/μ = 20.20± 2.55 %
μ = 527.0± 11.4
σ = 106.5± 11.1
WF
σ/μ = 19.80± 2.17 %
μ = 517.2± 10.1
σ = 102.4± 9.2
Energy loss in ERAM 30
XP
σ/μ = 14.68± 0.51 %
μ = 515.0± 3.0
σ = 75.6± 2.2
WF
σ/μ = 14.91± 0.49 %
μ = 501.4± 2.9
σ = 74.7± 2.0
Energy loss in ERAM 28
XP
σ/μ = 14.00± 0.63 %
μ = 537.0± 3.5
σ = 75.2± 2.9
WF
σ/μ = 14.46± 0.68 %
μ = 526.0± 3.5
σ = 76.1± 3.1
Energy loss in ERAM 19
XP
σ/μ = 15.58± 0.56 %
μ = 526.5± 3.4
σ = 82.0± 2.4
WF
σ/μ = 15.44± 0.52 %
μ = 516.3± 3.2
σ = 79.7± 2.2
Energy loss in ERAM 21
XP
σ/μ = 15.99± 0.17 %
μ = 477.8± 0.8
σ = 76.4± 0.7
WF
σ/μ = 16.07± 0.17 %
μ = 476.5± 0.8
σ = 76.6± 0.7
Energy loss in ERAM 13
XP
σ/μ = 15.33± 0.15 %
μ = 501.0± 0.7
σ = 76.8± 0.6
WF
σ/μ = 14.99± 0.14 %
μ = 497.7± 0.7
σ = 74.6± 0.6
Energy loss in ERAM 9
XP
σ/μ = 15.13± 0.15 %
μ = 516.5± 0.8
σ = 78.2± 0.7
WF
σ/μ = 14.13± 0.11 %
μ = 514.7± 0.7
σ = 72.7± 0.5
Energy loss in ERAM 2
XP
σ/μ = 15.27± 0.14 %
μ = 508.7± 0.8
σ = 77.7± 0.6
WF
σ/μ = 15.25± 0.14 %
μ = 505.0± 0.8
σ = 77.0± 0.6
Energy loss in ERAM 26
XP
σ/μ = 15.69± 0.41 %
μ = 522.0± 2.6
σ = 81.9± 1.7
WF
σ/μ = 15.82± 0.42 %
μ = 504.9± 2.5
σ = 79.9± 1.7
Energy loss in ERAM 17
XP
σ/μ = 15.89± 0.37 %
μ = 528.9± 2.1
σ = 84.0± 1.6
WF
σ/μ = 15.98± 0.38 %
μ = 511.7± 2.0
σ = 81.8± 1.6
Energy loss in ERAM 23
XP
σ/μ = 16.77± 0.50 %
μ = 518.9± 2.5
σ = 87.0± 2.2
WF
σ/μ = 15.07± 0.35 %
μ = 505.8± 2.2
σ = 76.2± 1.4
Energy loss in ERAM 29
XP
σ/μ = 15.84± 0.49 %
μ = 528.0± 2.7
σ = 83.6± 2.2
WF
σ/μ = 15.37± 0.42 %
μ = 520.3± 2.6
σ = 80.0± 1.8
Energy loss in ERAM 1
XP
σ/μ = 15.25± 0.15 %
μ = 499.2± 0.8
σ = 76.1± 0.6
WF
σ/μ = 14.59± 0.12 %
μ = 495.4± 0.7
σ = 72.3± 0.5
Energy loss in ERAM 10
XP
σ/μ = 13.85± 0.10 %
μ = 508.5± 0.7
σ = 70.4± 0.4
WF
σ/μ = 14.61± 0.11 %
μ = 500.1± 0.7
σ = 73.1± 0.5
Energy loss in ERAM 11
XP
σ/μ = 15.03± 0.13 %
μ = 504.7± 0.7
σ = 75.9± 0.6
WF
σ/μ = 13.62± 0.10 %
μ = 504.6± 0.7
σ = 68.7± 0.4
Energy loss in ERAM 3
XP
σ/μ = 14.96± 0.13 %
μ = 490.7± 0.7
σ = 73.4± 0.5
WF
σ/μ = 14.46± 0.11 %
μ = 490.9± 0.7
σ = 71.0± 0.4
Count
dE/dx (ADC counts/cm)



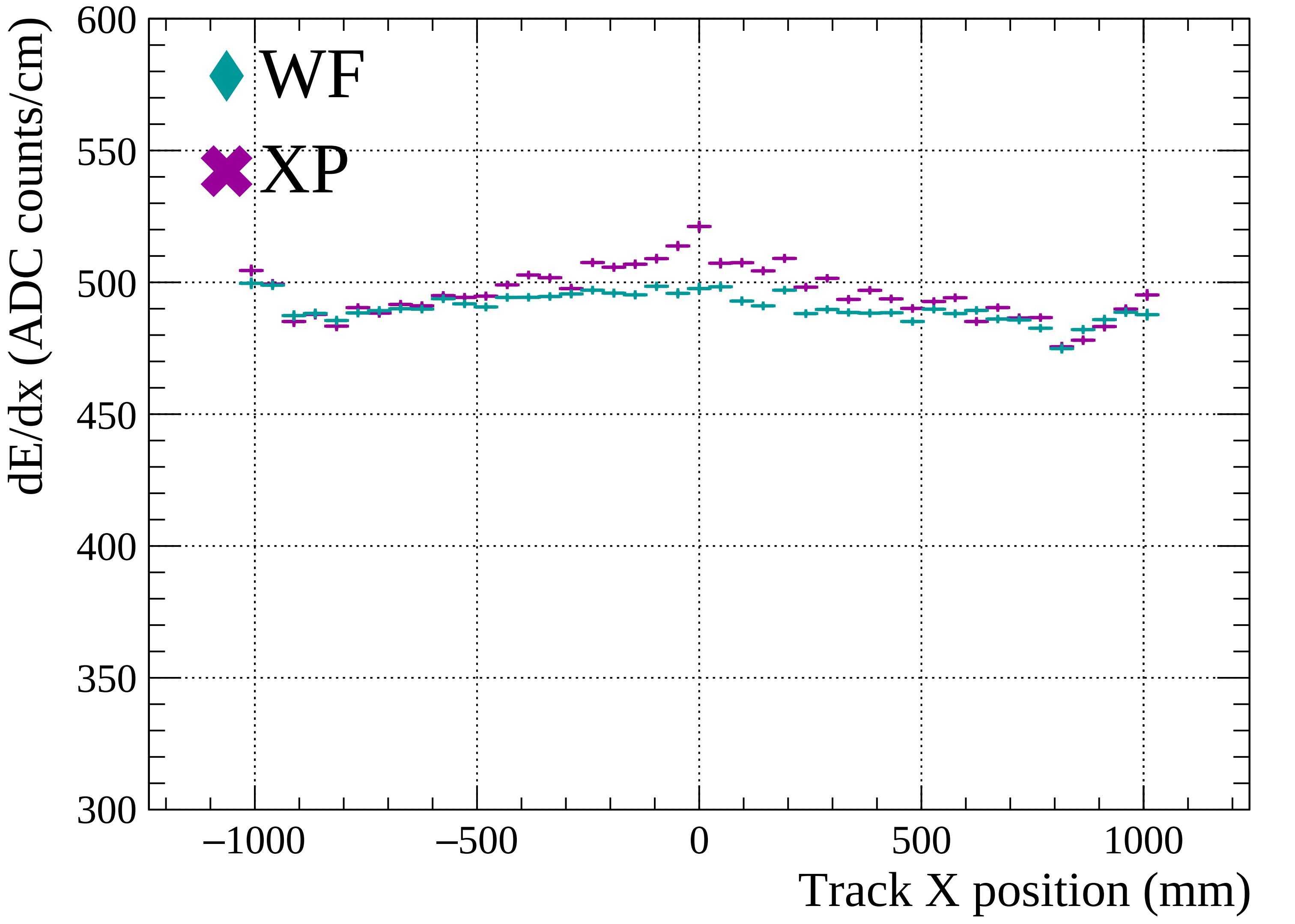
Energy loss in ERAM 47
XP σ/μ = 13.57± 1.12 % μ = 518.1± 6.0 σ = 70.3± 5.0
WF σ/μ = 15.04± 2.17 % μ = 516.2± 8.7 σ = 77.7± 9.9
Energy loss in ERAM 16
XP σ/μ = 14.73± 0.50 % μ = 505.8± 2.7 σ = 74.5± 2.1
WF σ/μ = 15.37± 0.51 % μ = 493.0± 2.7 σ = 75.8± 2.1
Energy loss in ERAM 14
XP σ/μ = 15.19± 0.57 % μ = 525.1± 3.3 σ = 79.8± 2.5
WF σ/μ = 15.43± 0.54 % μ = 512.4± 3.2 σ = 79.1± 2.3
Energy loss in ERAM 15
XP σ/μ = 15.15± 0.49 % μ = 551.7± 3.2 σ = 83.6± 2.2
WF σ/μ = 15.88± 0.68 % μ = 539.6± 3.6 σ = 85.7± 3.1
Energy loss in ERAM 42
XP σ/μ = 12.88± 0.09 % μ = 470.1± 0.6 σ = 60.5± 0.3
WF σ/μ = 13.33± 0.09 % μ = 462.3± 0.6 σ = 61.6± 0.3
Energy loss in ERAM 45
XP σ/μ = 14.52± 0.12 % μ = 483.7± 0.6 σ = 70.2± 0.5
WF σ/μ = 14.16± 0.10 % μ = 479.8± 0.6 σ = 68.0± 0.4
Energy loss in ERAM 37
XP σ/μ = 13.77± 0.10 % μ = 503.6± 0.6 σ = 69.3± 0.4
WF σ/μ = 13.96± 0.10 % μ = 497.7± 0.6 σ = 69.5± 0.4
Energy loss in ERAM 36
XP σ/μ = 15.51± 0.16 % μ = 500.9± 0.8 σ = 77.7± 0.7
WF σ/μ = 14.98± 0.14 % μ = 498.2± 0.7 σ = 74.6± 0.6
Energy loss in ERAM 20
XP σ/μ = 17.00± 0.55 % μ = 504.3± 2.7 σ = 85.7± 2.3
WF σ/μ = 16.90± 0.58 % μ = 486.6± 2.7 σ = 82.2± 2.4
Energy loss in ERAM 38
XP σ/μ = 15.93± 0.39 % μ = 509.6± 1.9 σ = 81.2± 1.7
WF σ/μ = 15.17± 0.31 % μ = 494.5± 1.8 σ = 75.0± 1.3
Energy loss in ERAM 7
XP σ/μ = 15.57± 0.38 % μ = 513.5± 2.0 σ = 79.9± 1.7
WF σ/μ = 15.81± 0.39 % μ = 494.8± 2.0 σ = 78.2± 1.6
Energy loss in ERAM 44
XP σ/μ = 16.27± 0.45 % μ = 529.1± 2.5 σ = 86.1± 2.0
WF σ/μ = 16.39± 0.45 % μ = 512.8± 2.4 σ = 84.1± 1.9
Energy loss in ERAM 43
XP σ/μ = 14.32± 0.11 % μ = 495.1± 0.6 σ = 70.9± 0.5
WF σ/μ = 13.91± 0.10 % μ = 489.9± 0.6 σ = 68.1± 0.4
Energy loss in ERAM 39
XP σ/μ = 13.94± 0.10 % μ = 483.4± 0.6 σ = 67.4± 0.4
WF σ/μ = 13.37± 0.09 % μ = 476.5± 0.6 σ = 63.7± 0.3
Energy loss in ERAM 41
XP σ/μ = 13.64± 0.10 % μ = 495.5± 0.6 σ = 67.6± 0.4
WF σ/μ = 14.54± 0.11 % μ = 487.5± 0.6 σ = 70.9± 0.5
Energy loss in ERAM 46
XP σ/μ = 14.73± 0.12 % μ = 479.7± 0.6 σ = 70.7± 0.5
WF σ/μ = 13.97± 0.10 % μ = 477.5± 0.6 σ = 66.7± 0.4
Count
dE/dx (ADC counts/cm)



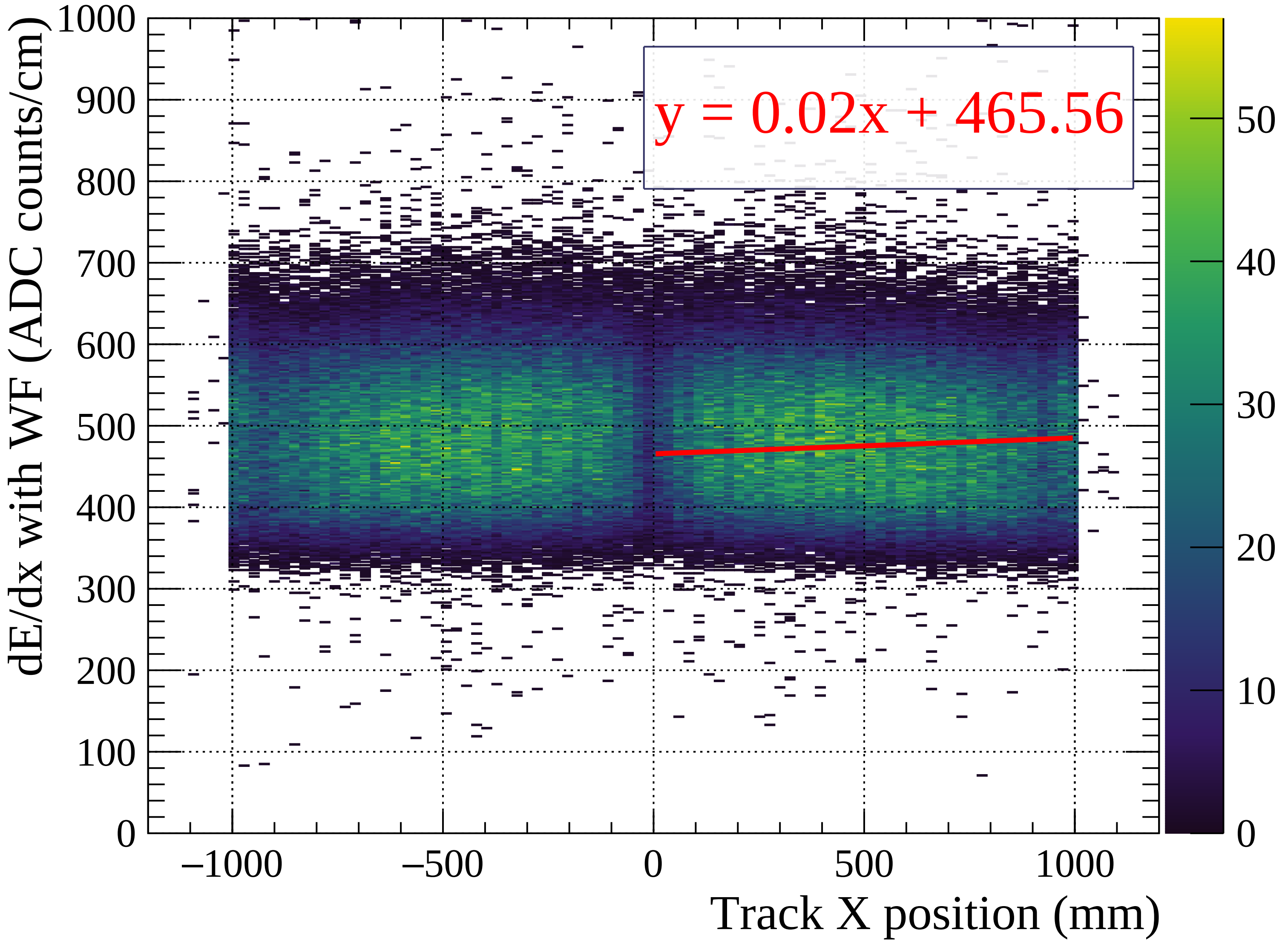
Entries 269623
Mean x 490.1
Mean y 495.4
Std Dev x 74.21
Std Dev y 76.39
0 1 42
4 269561 15
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
500
1000
1500
2000
2500
3000
3500
4000



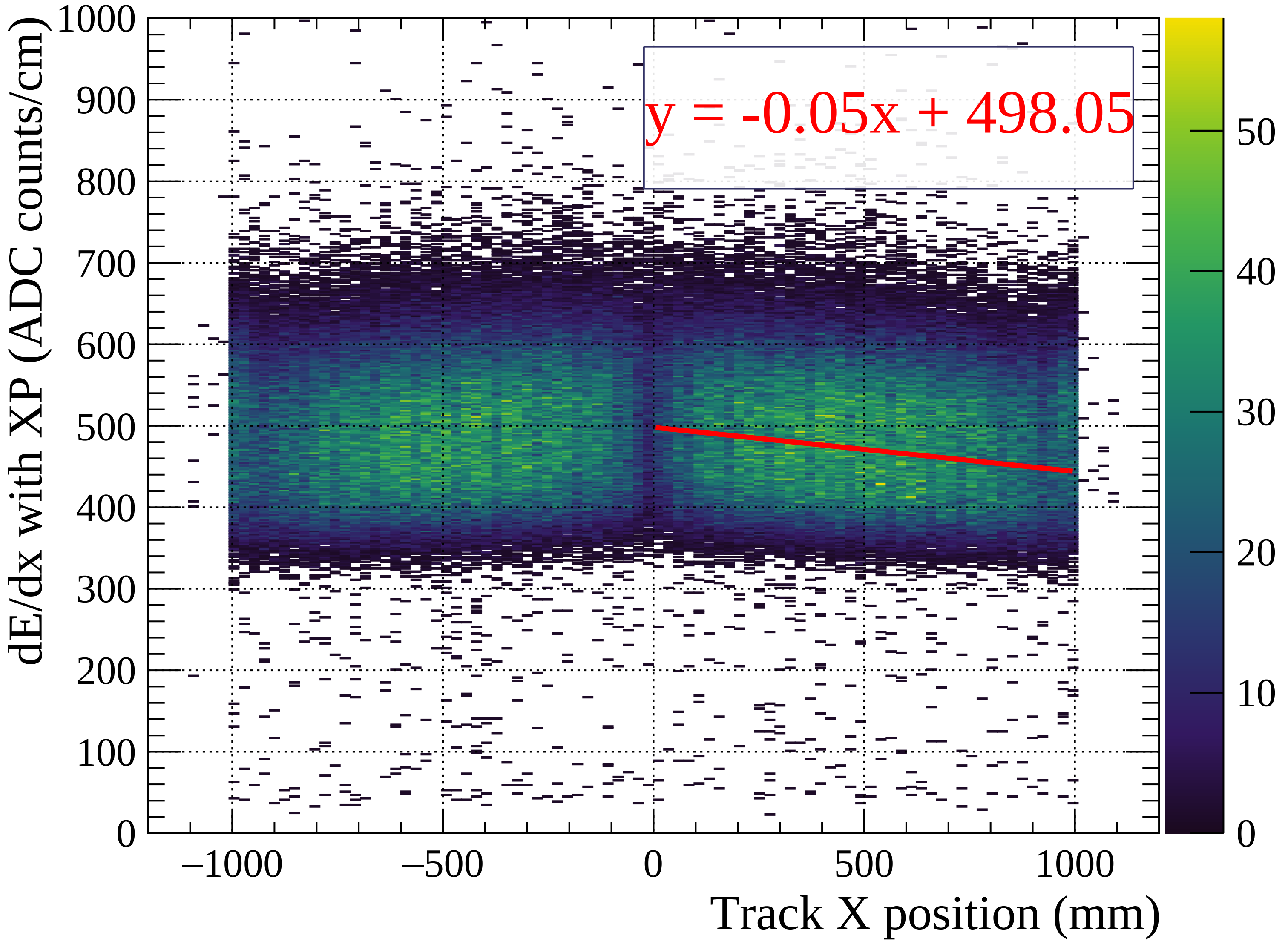
WF
XP
dE/dx resolution (%)
30
25
20
15
10
5
−1000
−500
0
500
1000
Track X position (mm)

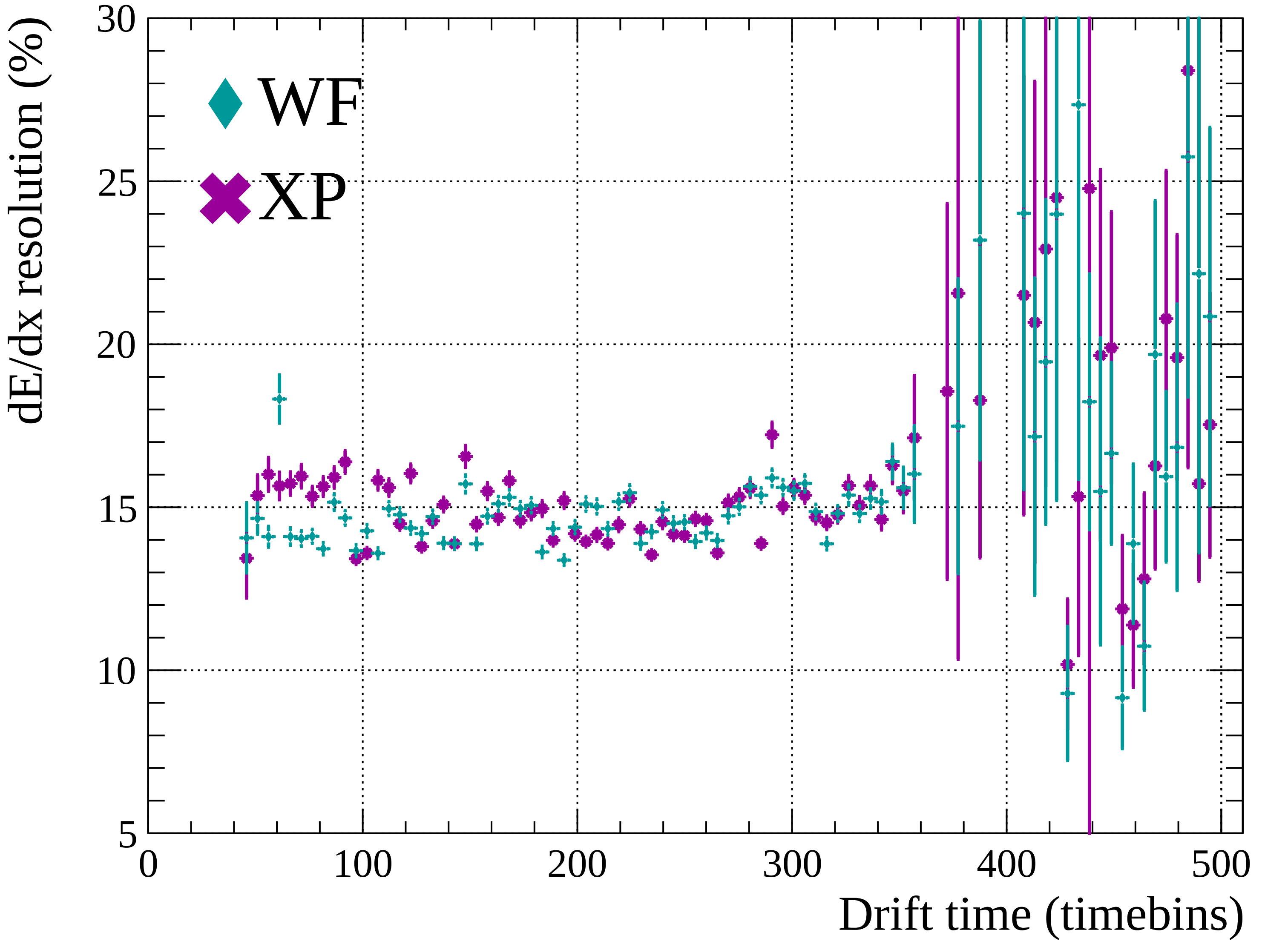


WF
XP
dE/dx (ADC counts/cm)
Track X position (mm)
600
550
500
450
400
350
300
−1000
−500
0
500
1000

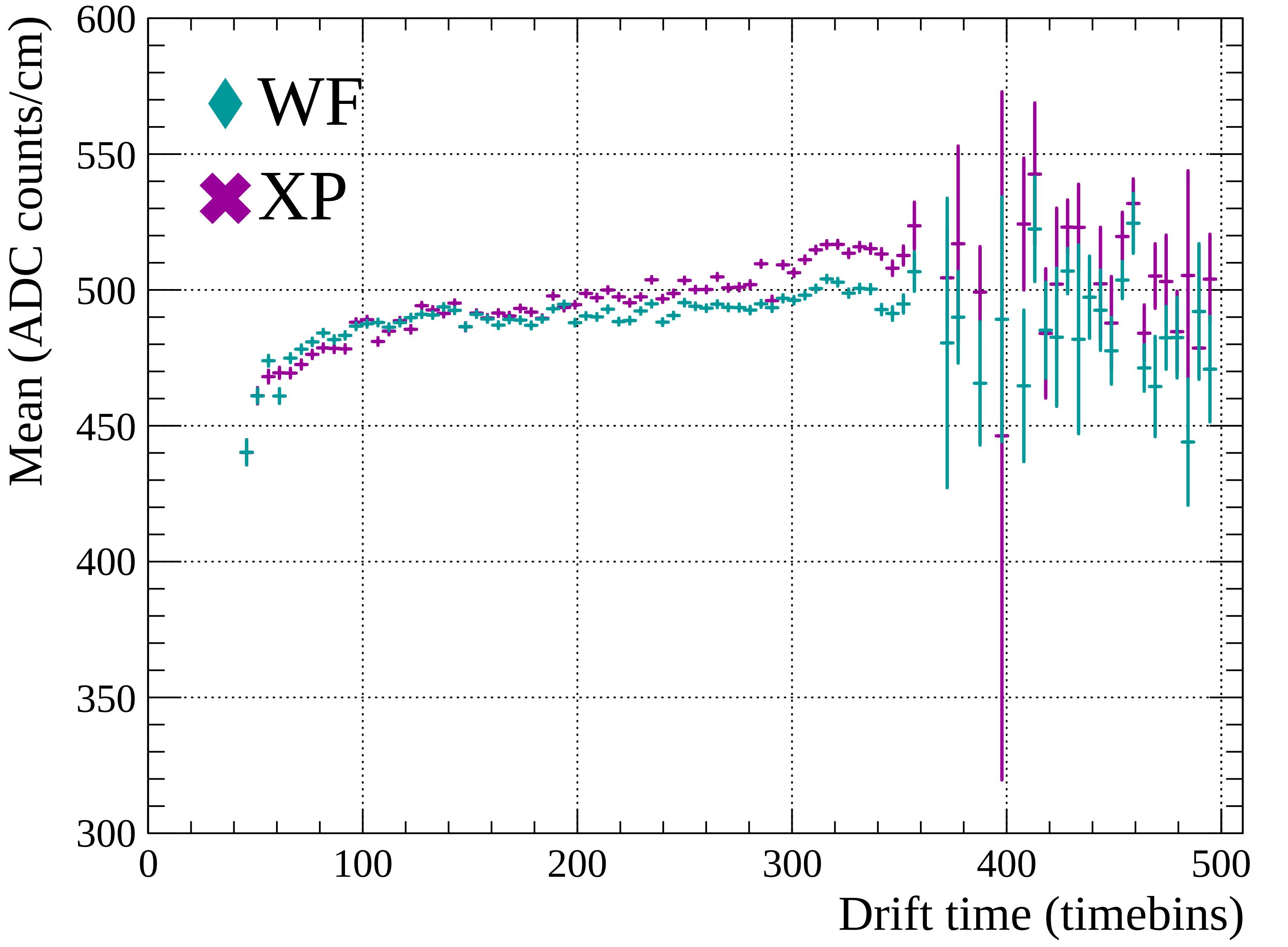


y = 0.02x + 465.56
dE/dx with WF (ADC counts/cm)
Track X position (mm)

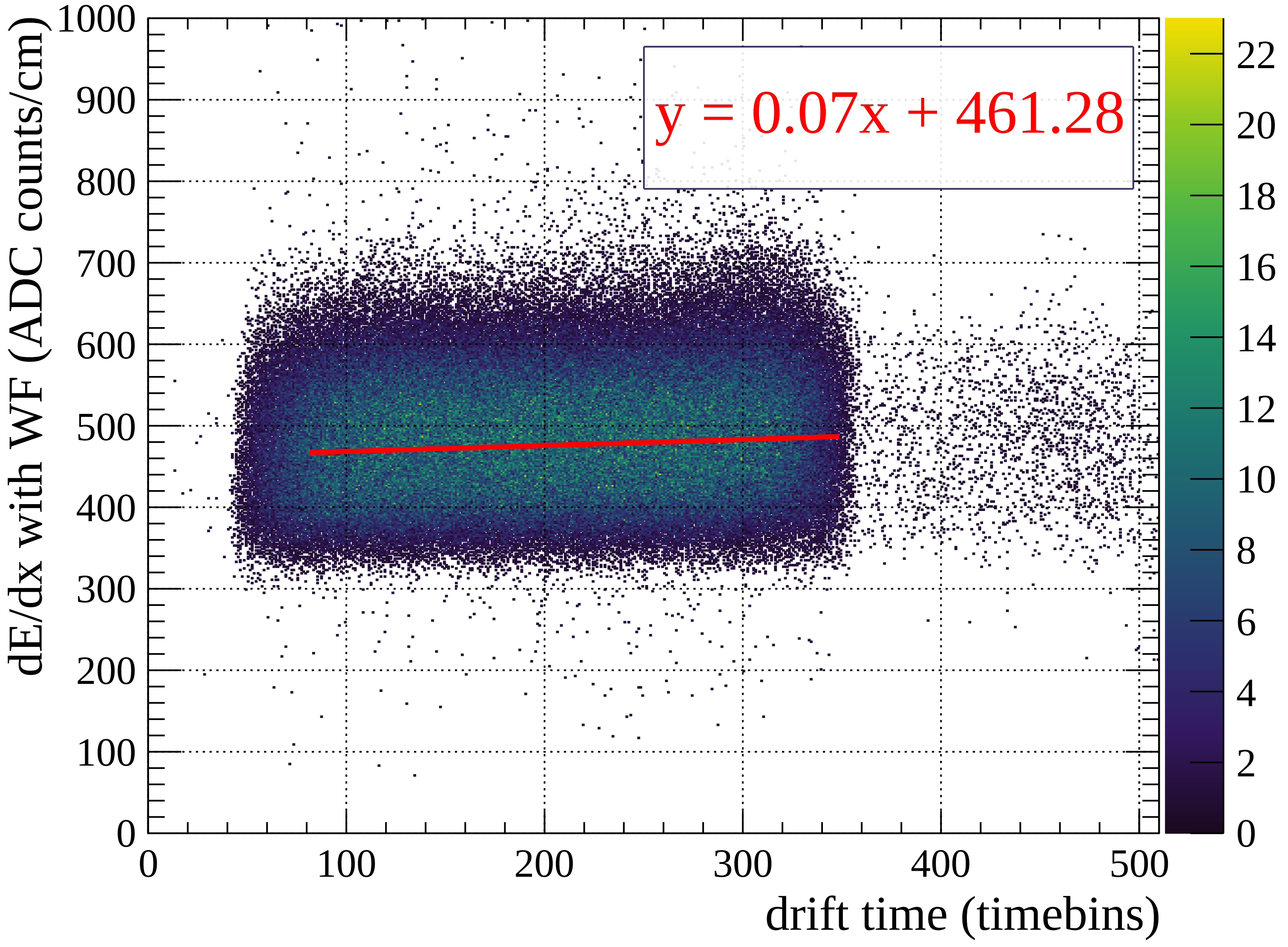

y = -0.05x + 498.05
dE/dx with XP (ADC counts/cm)
Track X position (mm)
1000
900
800
700
600
500
400
300
200
100
0
−1000
−500
0
500
1000
50
40
30
20
10
0



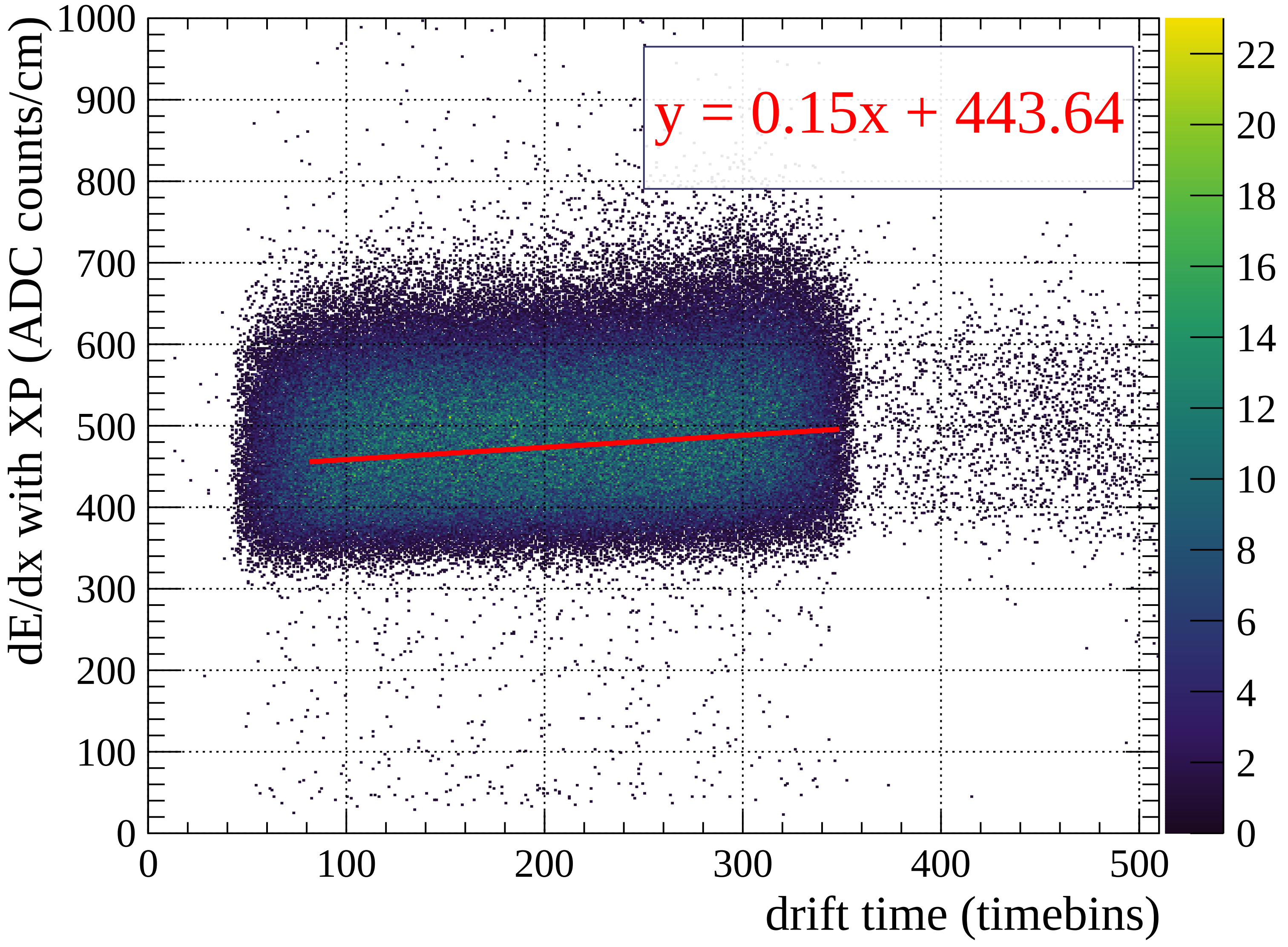
dE/dx resolution (%)
Drift time (timebins)
WF
XP
5
10
15
20
25
30
0
100
200
300
400
500



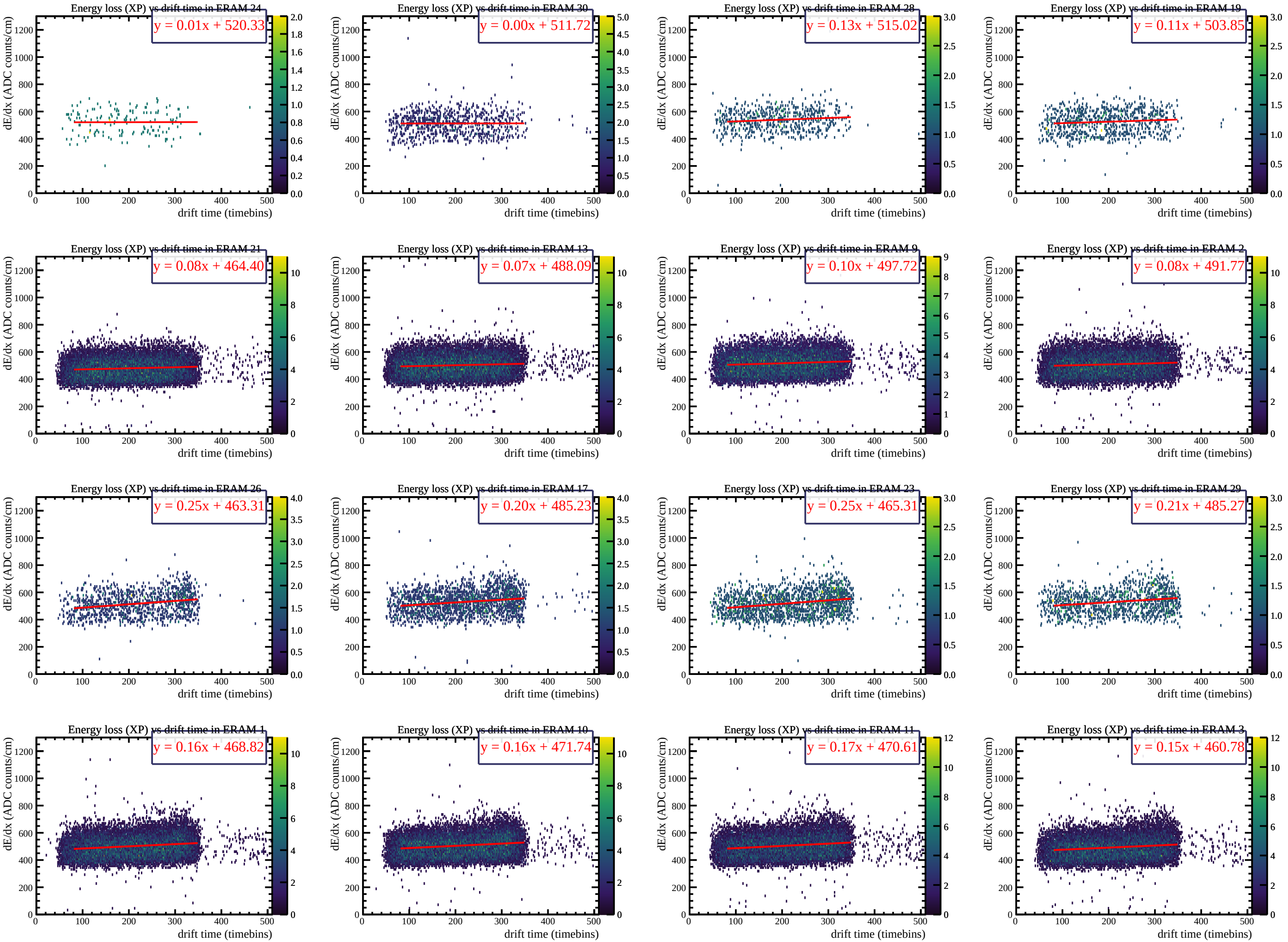
WF
XP
Mean (ADC counts/cm)
Drift time (timebins)
600
550
500
450
400
350
300
0
100
200
300
400
500

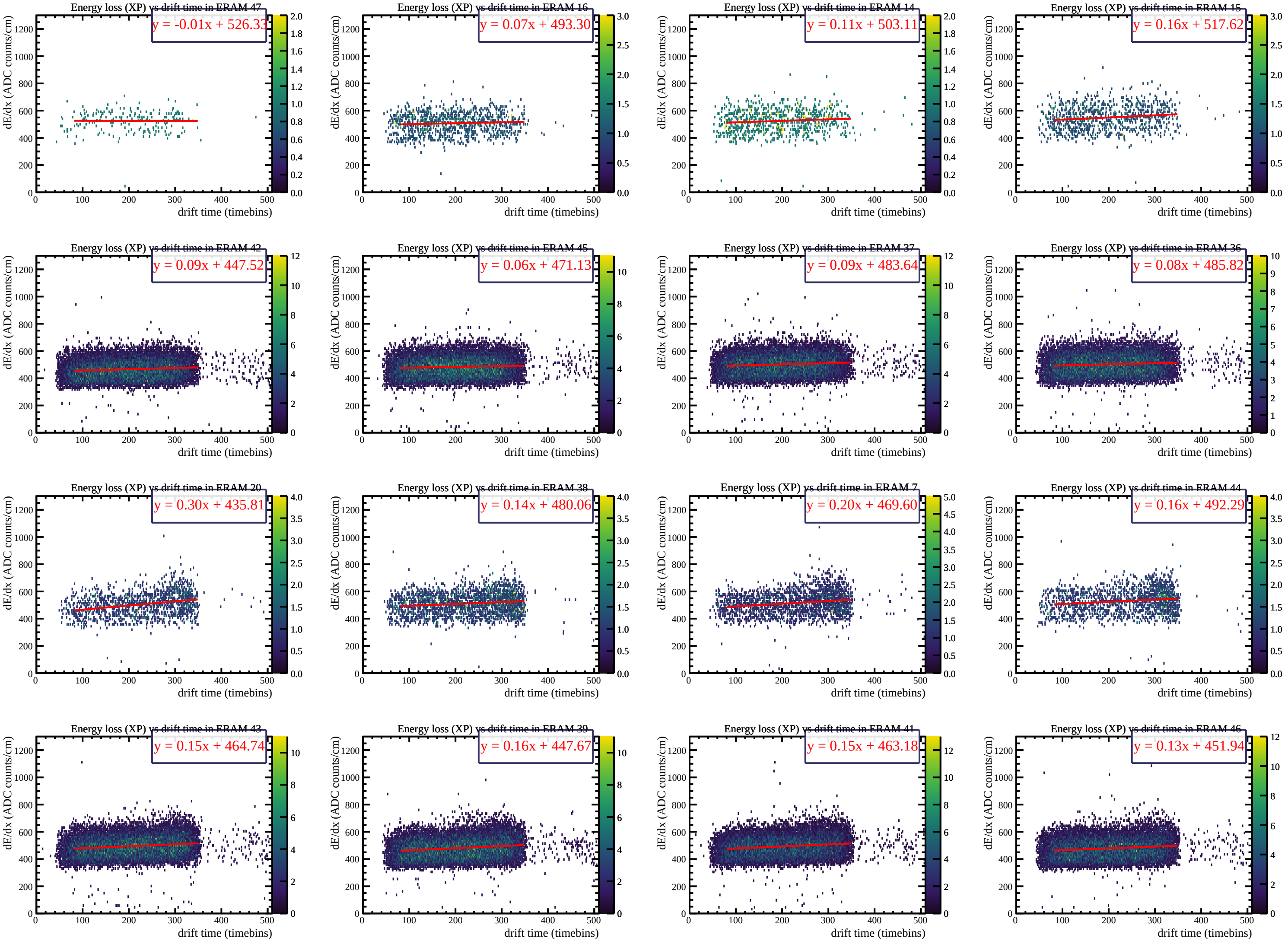


y = 0.07x + 461.28
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14
16
18
20
22

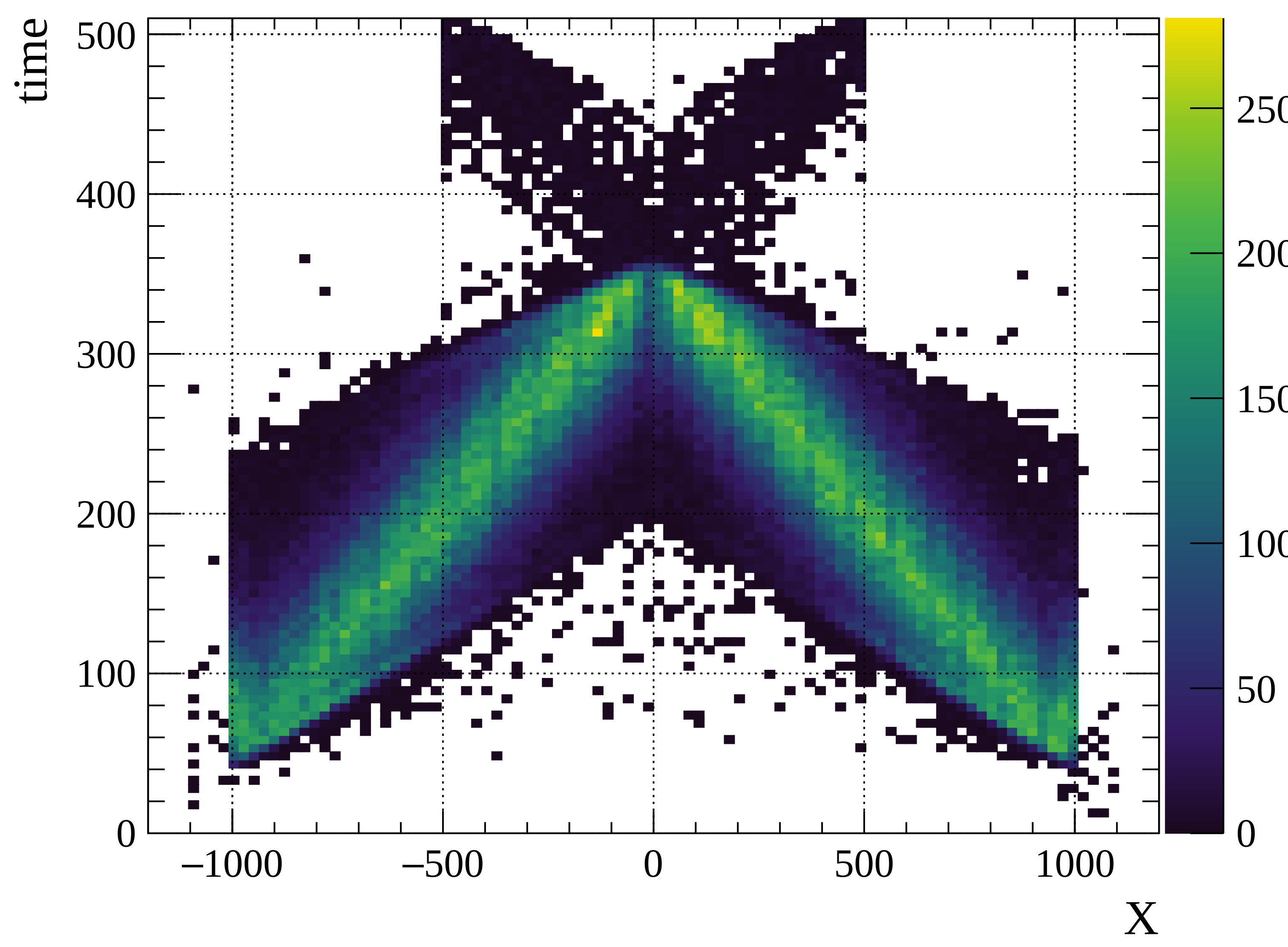


y = 0.15x + 443.64
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14
16
18
20
22

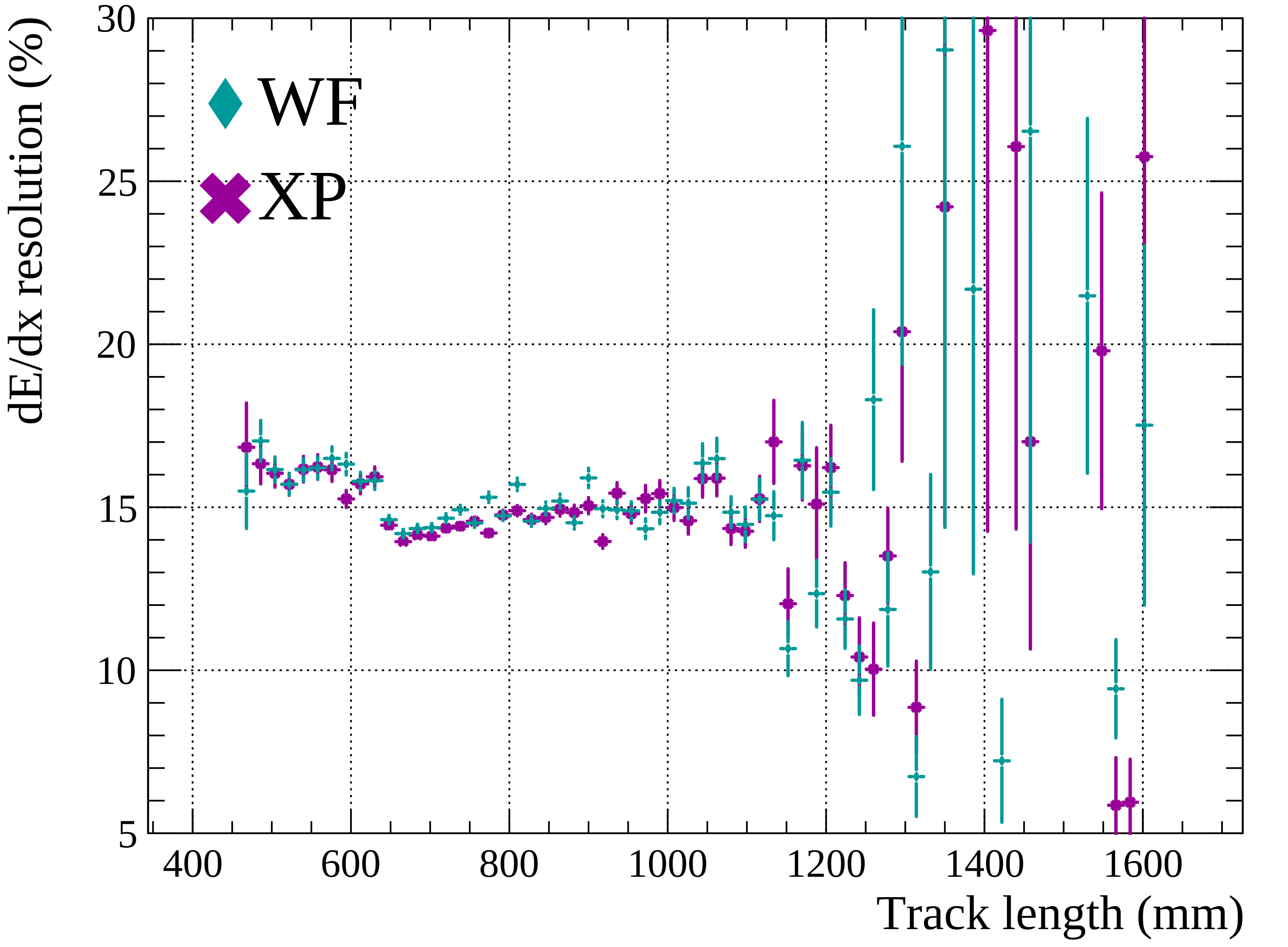

Energy loss (XP) vs drift time in ERAM 24
y = 0.01x + 520.33
Energy loss (XP) vs drift time in ERAM 30
y = 0.00x + 511.72
Energy loss (XP) vs drift time in ERAM 28
y = 0.13x + 515.02
Energy loss (XP) vs drift time in ERAM 19
y = 0.11x + 503.85
Energy loss (XP) vs drift time in ERAM 21
y = 0.08x + 464.40
Energy loss (XP) vs drift time in ERAM 13
y = 0.07x + 488.09
Energy loss (XP) vs drift time in ERAM 9
y = 0.10x + 497.72
Energy loss (XP) vs drift time in ERAM 2
y = 0.08x + 491.77
Energy loss (XP) vs drift time in ERAM 26
y = 0.25x + 463.31
Energy loss (XP) vs drift time in ERAM 17
y = 0.20x + 485.23
Energy loss (XP) vs drift time in ERAM 23
y = 0.25x + 465.31
Energy loss (XP) vs drift time in ERAM 29
y = 0.21x + 485.27
Energy loss (XP) vs drift time in ERAM 1
y = 0.16x + 468.82
Energy loss (XP) vs drift time in ERAM 10
y = 0.16x + 471.74
Energy loss (XP) vs drift time in ERAM 11
y = 0.17x + 470.61
Energy loss (XP) vs drift time in ERAM 3
y = 0.15x + 460.78
dE/dx (ADC counts/cm)
drift time (timebins)



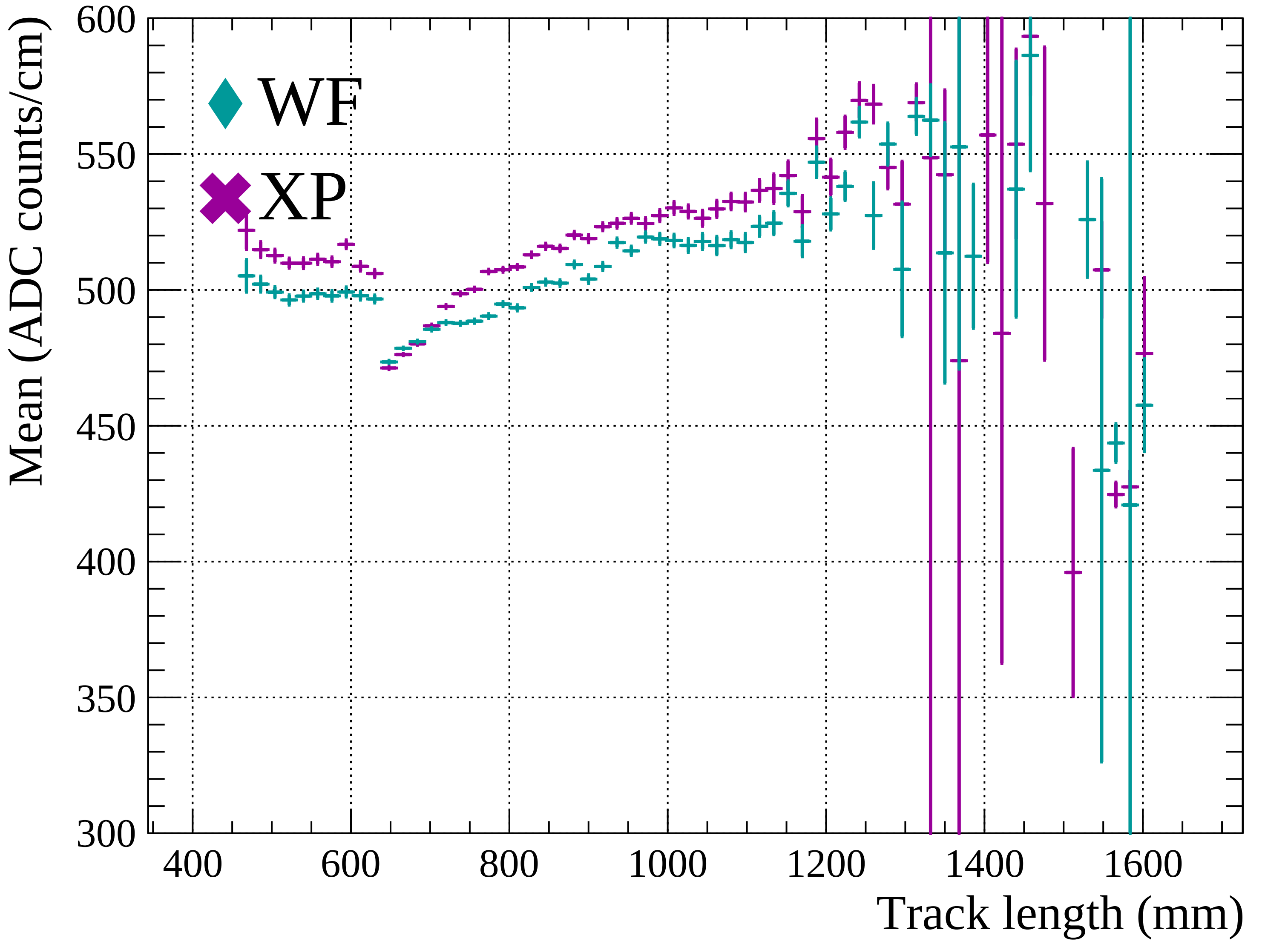
Energy loss (XP) vs drift time in ERAM 47
y = -0.01x + 526.33
Energy loss (XP) vs drift time in ERAM 16
y = 0.07x + 493.30
Energy loss (XP) vs drift time in ERAM 14
y = 0.11x + 503.11
Energy loss (XP) vs drift time in ERAM 15
y = 0.16x + 517.62
Energy loss (XP) vs drift time in ERAM 42
y = 0.09x + 447.52
Energy loss (XP) vs drift time in ERAM 45
y = 0.06x + 471.13
Energy loss (XP) vs drift time in ERAM 37
y = 0.09x + 483.64
Energy loss (XP) vs drift time in ERAM 36
y = 0.08x + 485.82
Energy loss (XP) vs drift time in ERAM 20
y = 0.30x + 435.81
Energy loss (XP) vs drift time in ERAM 38
y = 0.14x + 480.06
Energy loss (XP) vs drift time in ERAM 7
y = 0.20x + 469.60
Energy loss (XP) vs drift time in ERAM 44
y = 0.16x + 492.29
Energy loss (XP) vs drift time in ERAM 43
y = 0.15x + 464.74
Energy loss (XP) vs drift time in ERAM 39
y = 0.16x + 447.67
Energy loss (XP) vs drift time in ERAM 41
y = 0.15x + 463.18
Energy loss (XP) vs drift time in ERAM 46
y = 0.13x + 451.94
dE/dx (ADC counts/cm)
drift time (timebins)



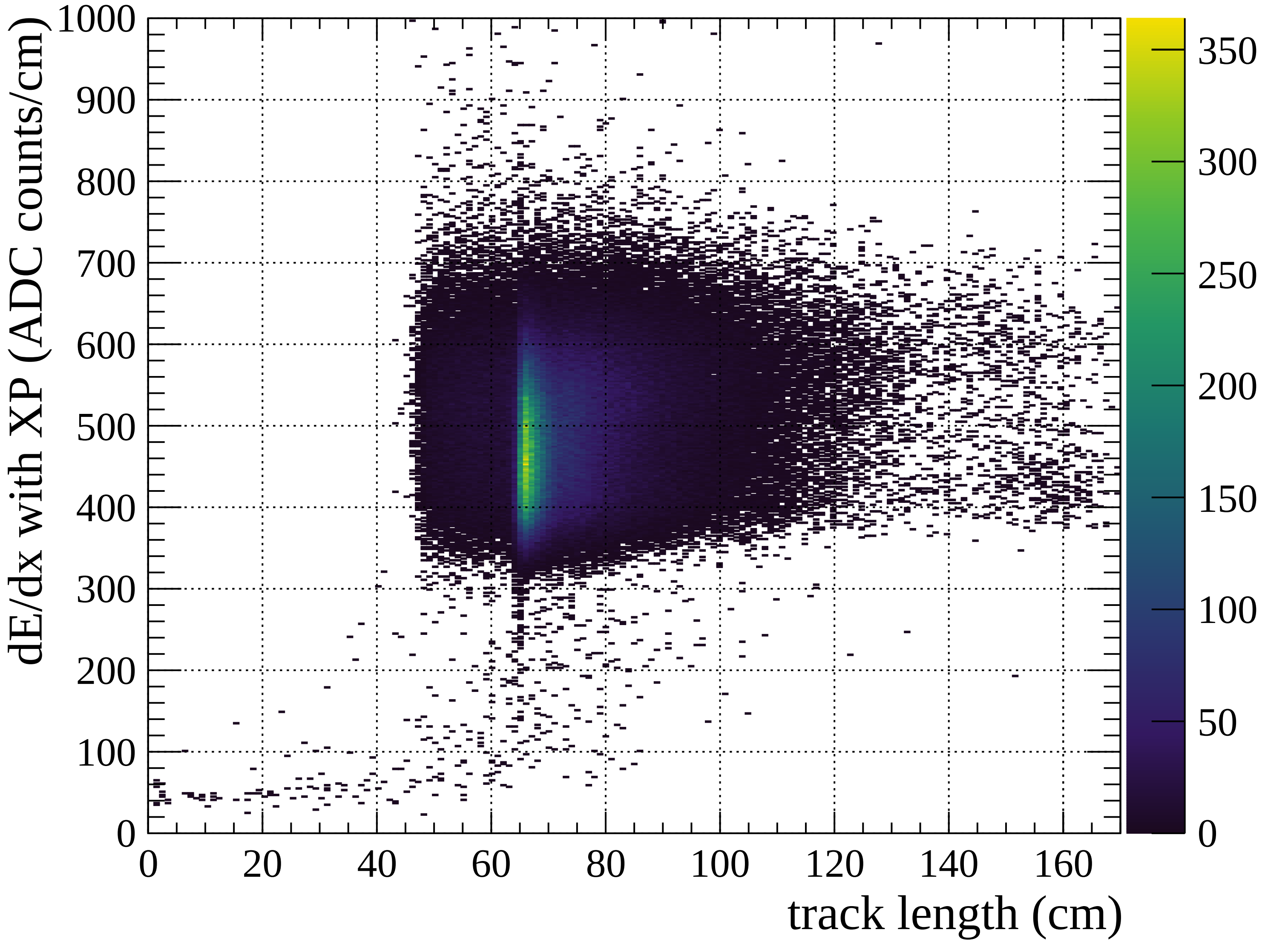
time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
250
200
150
100
50
0

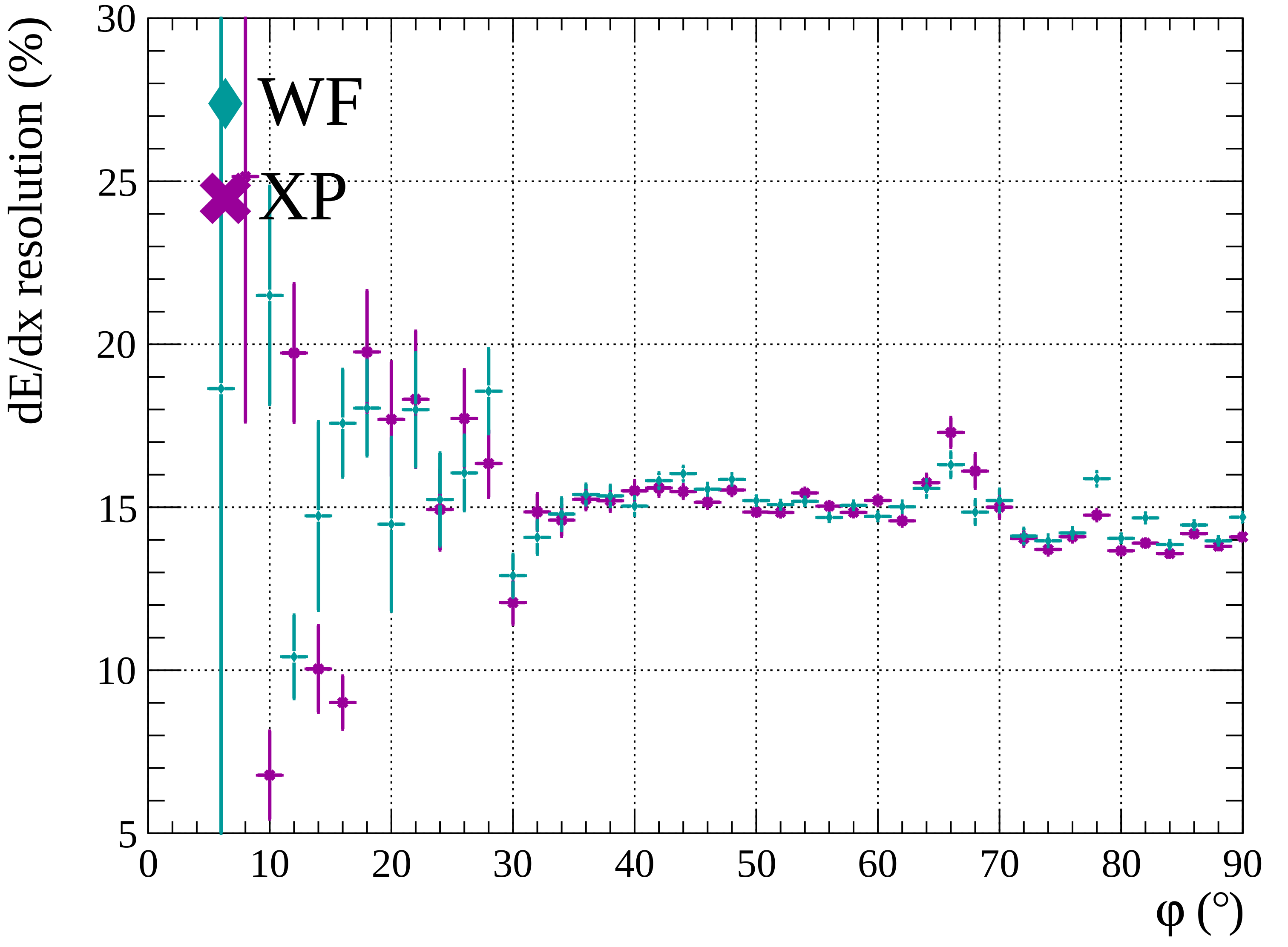


dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
400
600
800
1000
1200
1400
1600
Track length (mm)

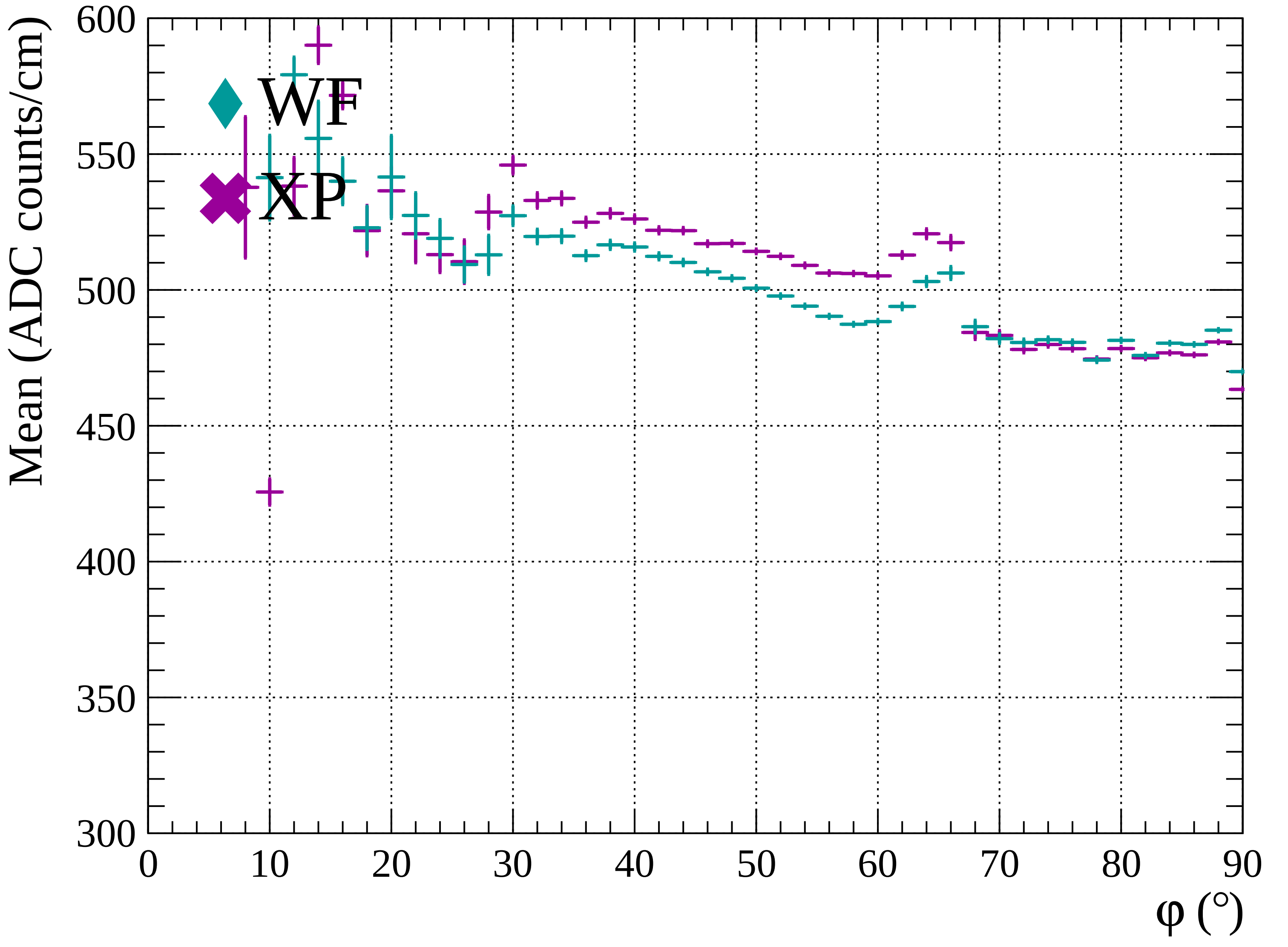


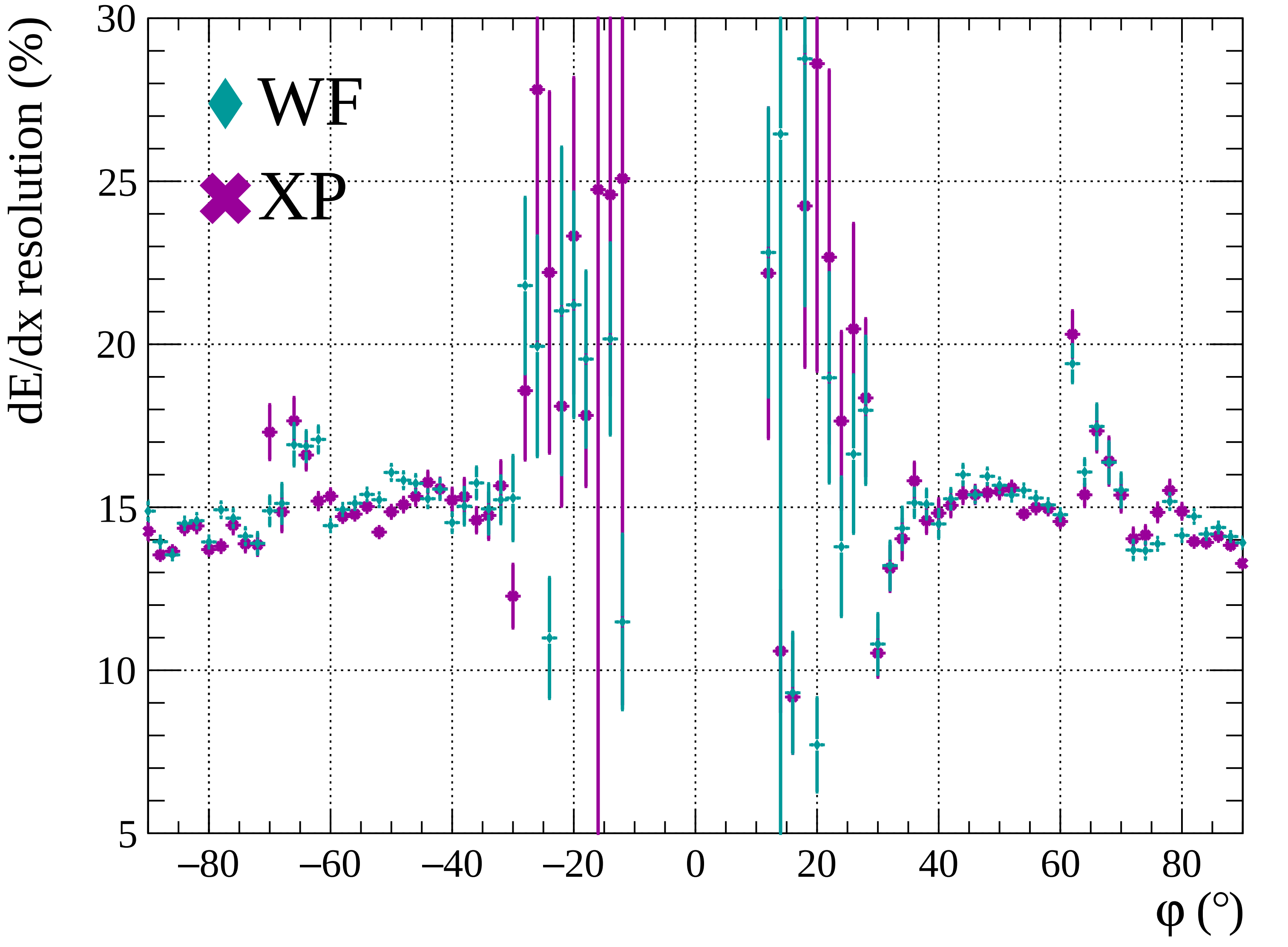
Mean (ADC counts/cm)
600
550
500
450
400
350
300
400
600
800
1000
1200
1400
1600
Track length (mm)
WF
XP


dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600
700
800
900
1000
0
50
100
150
200
250
300
350

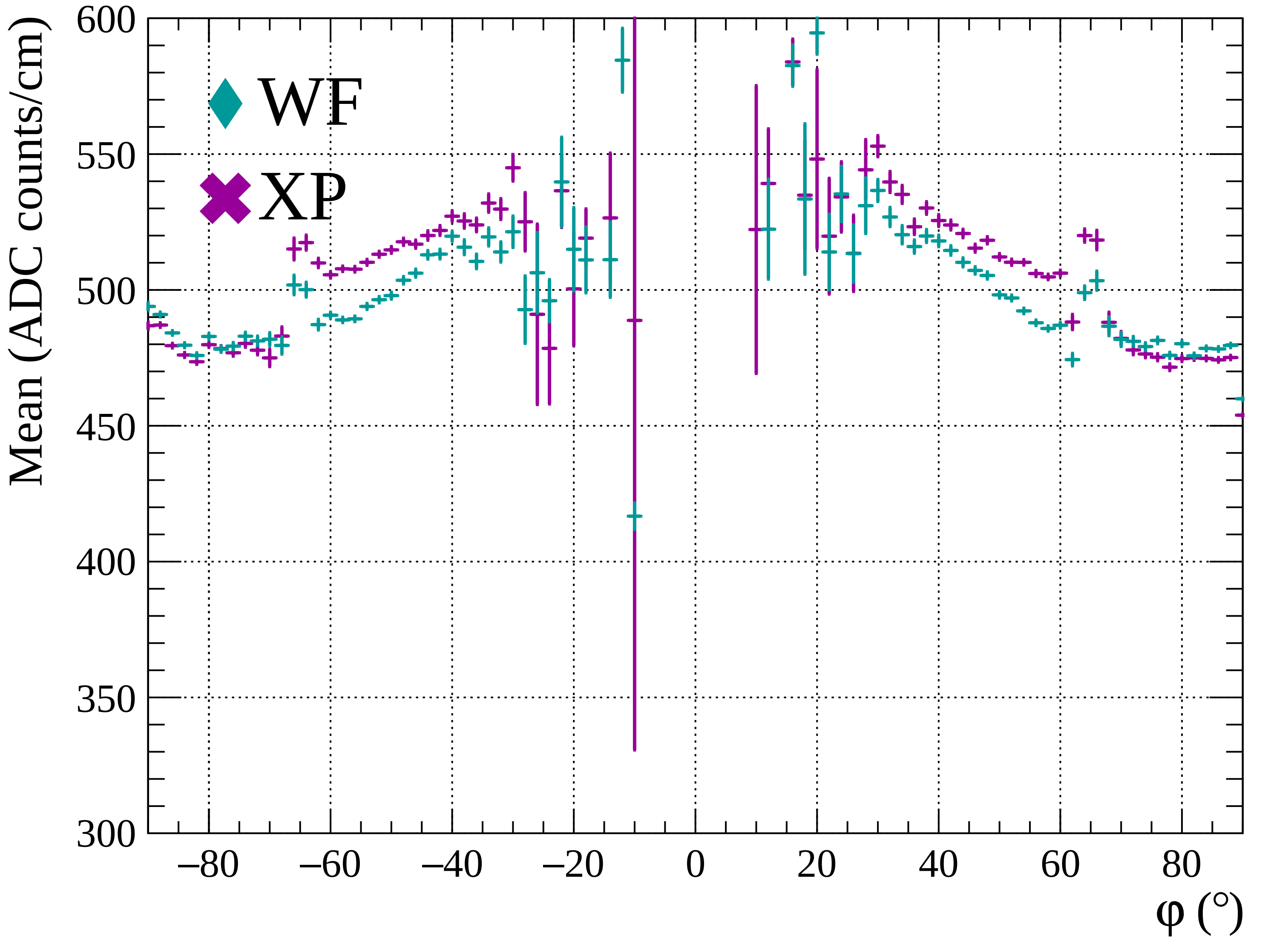


dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
0
10
20
30
40
50
60
70
80
90
φ (°)

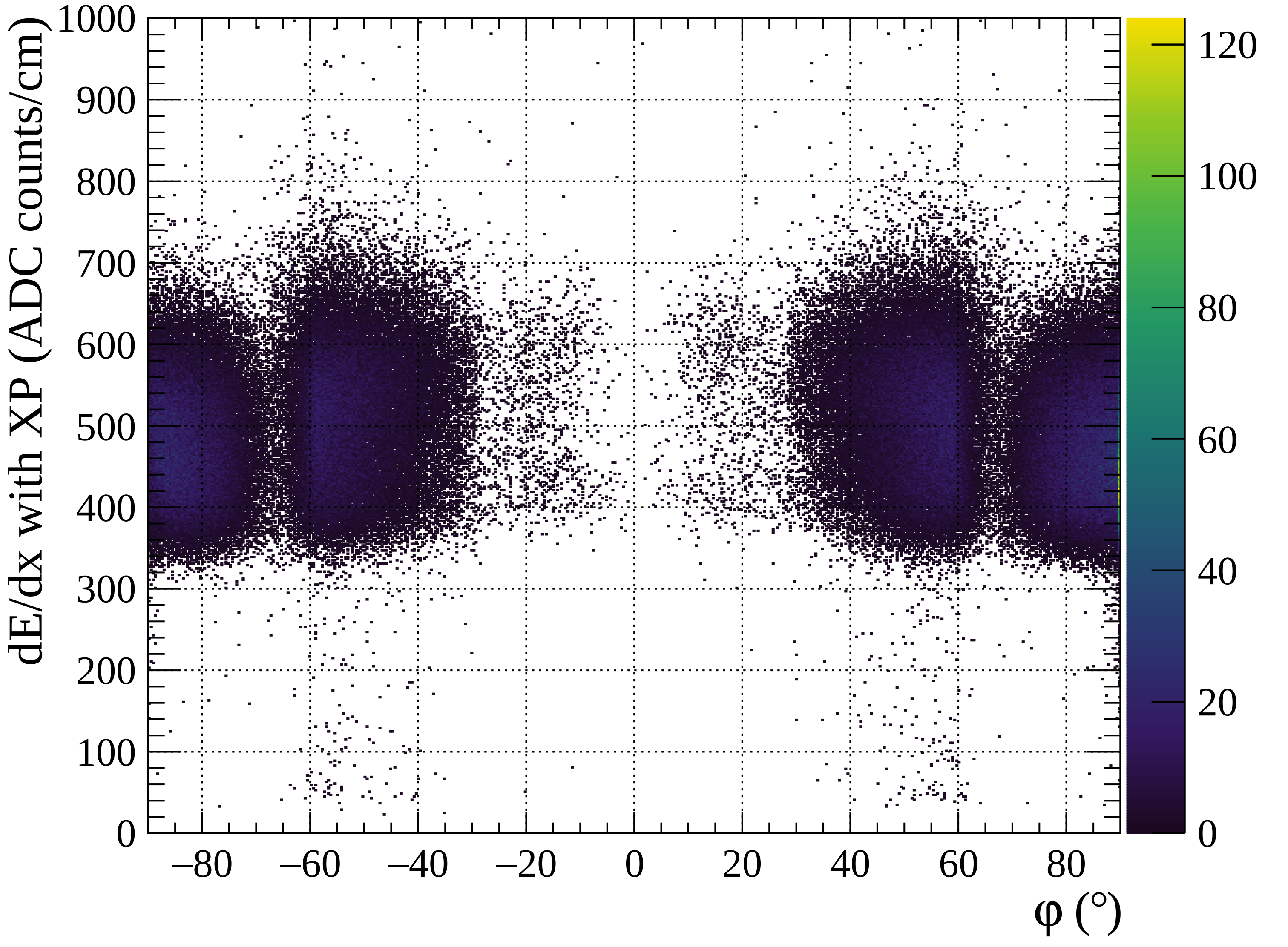

Mean (ADC counts/cm)
φ (°)
WF
XP
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90



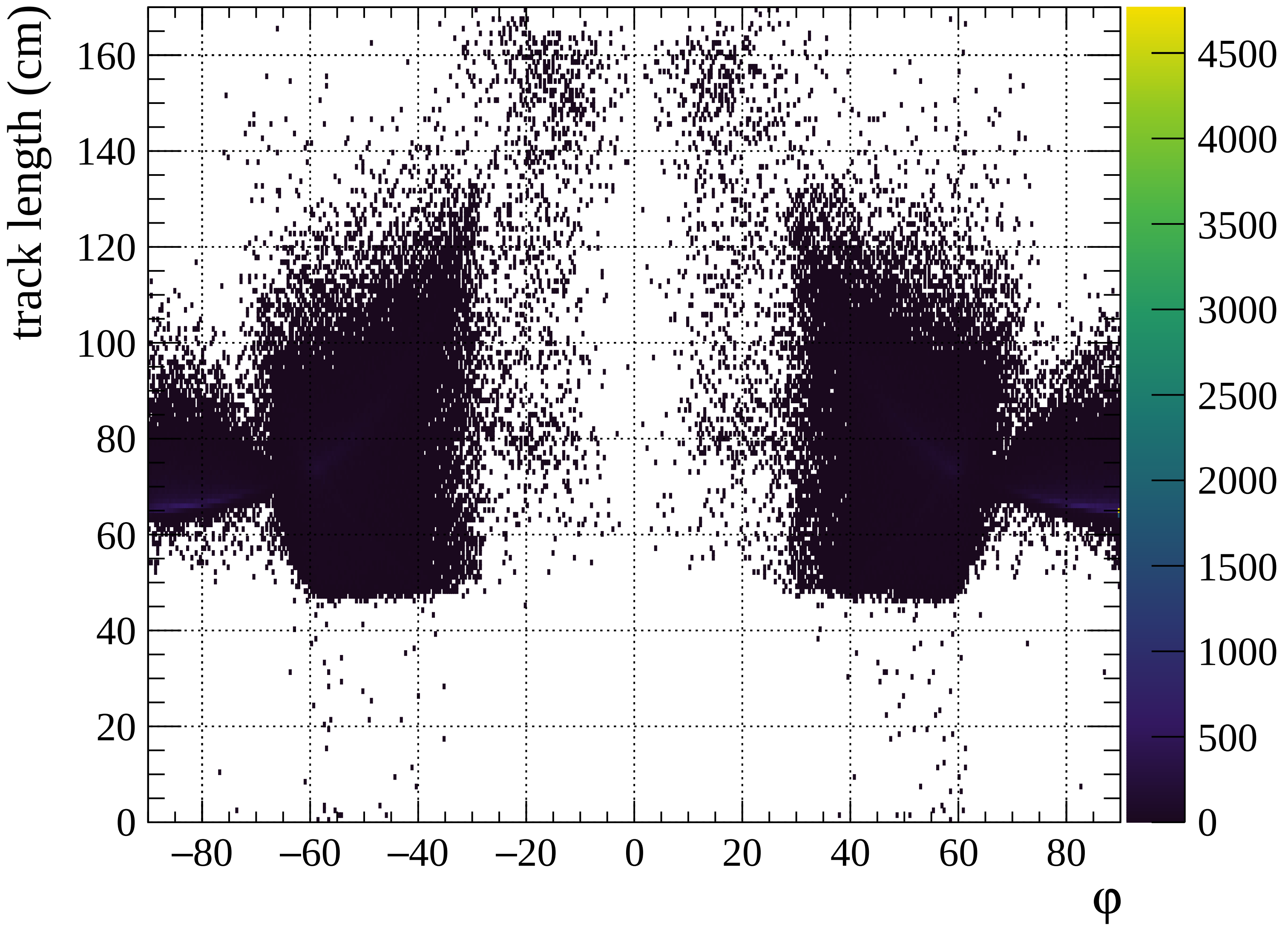
dE/dx resolution (%)
WF
XP
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80
φ (°)



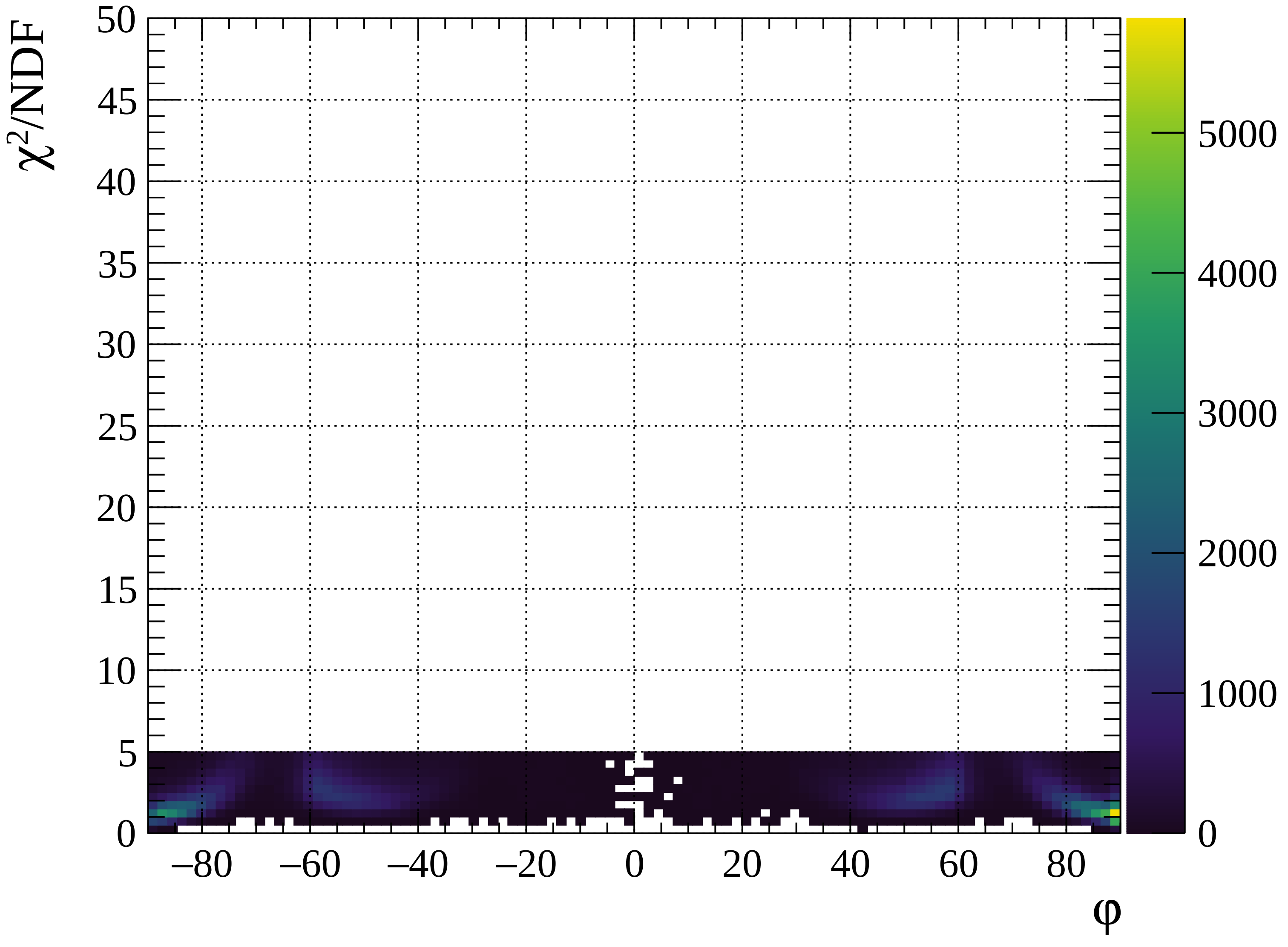
WF
XP
Mean (ADC counts/cm)
φ (°)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80



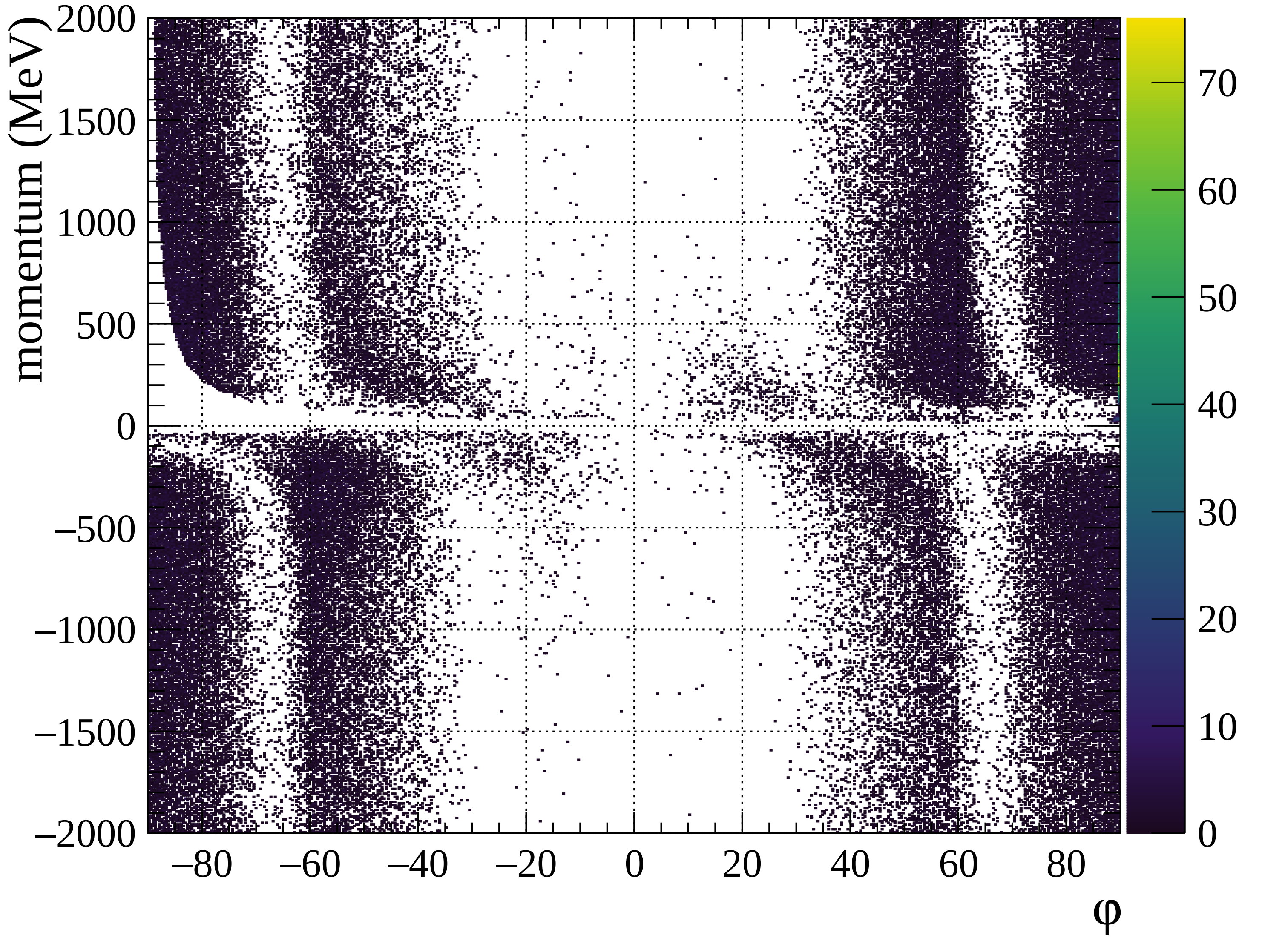
dE/dx with XP (ADC counts/cm)
φ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
20
40
60
80
100
120



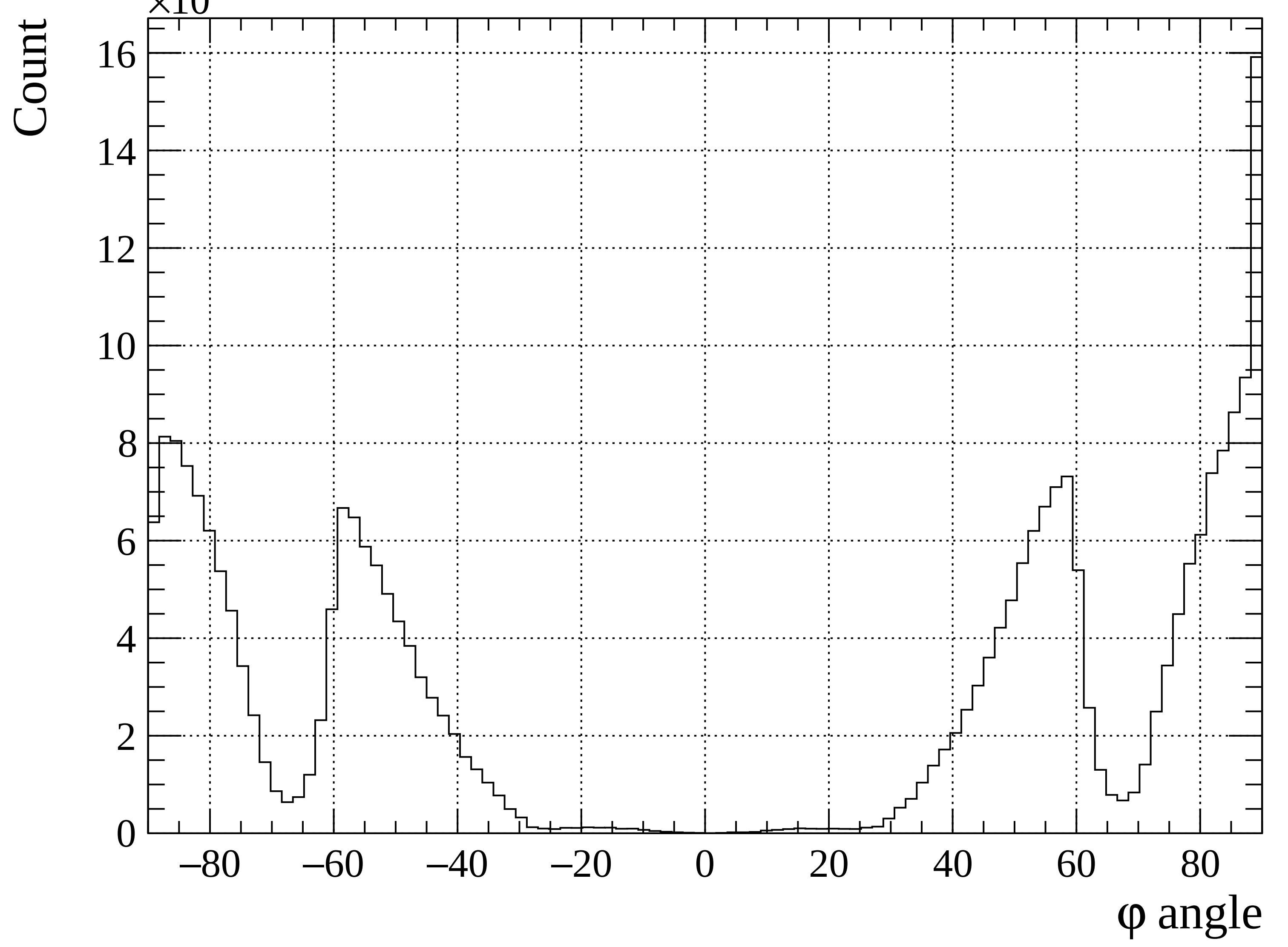
track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
φ
0
500
1000
1500
2000
2500
3000
3500
4000
4500



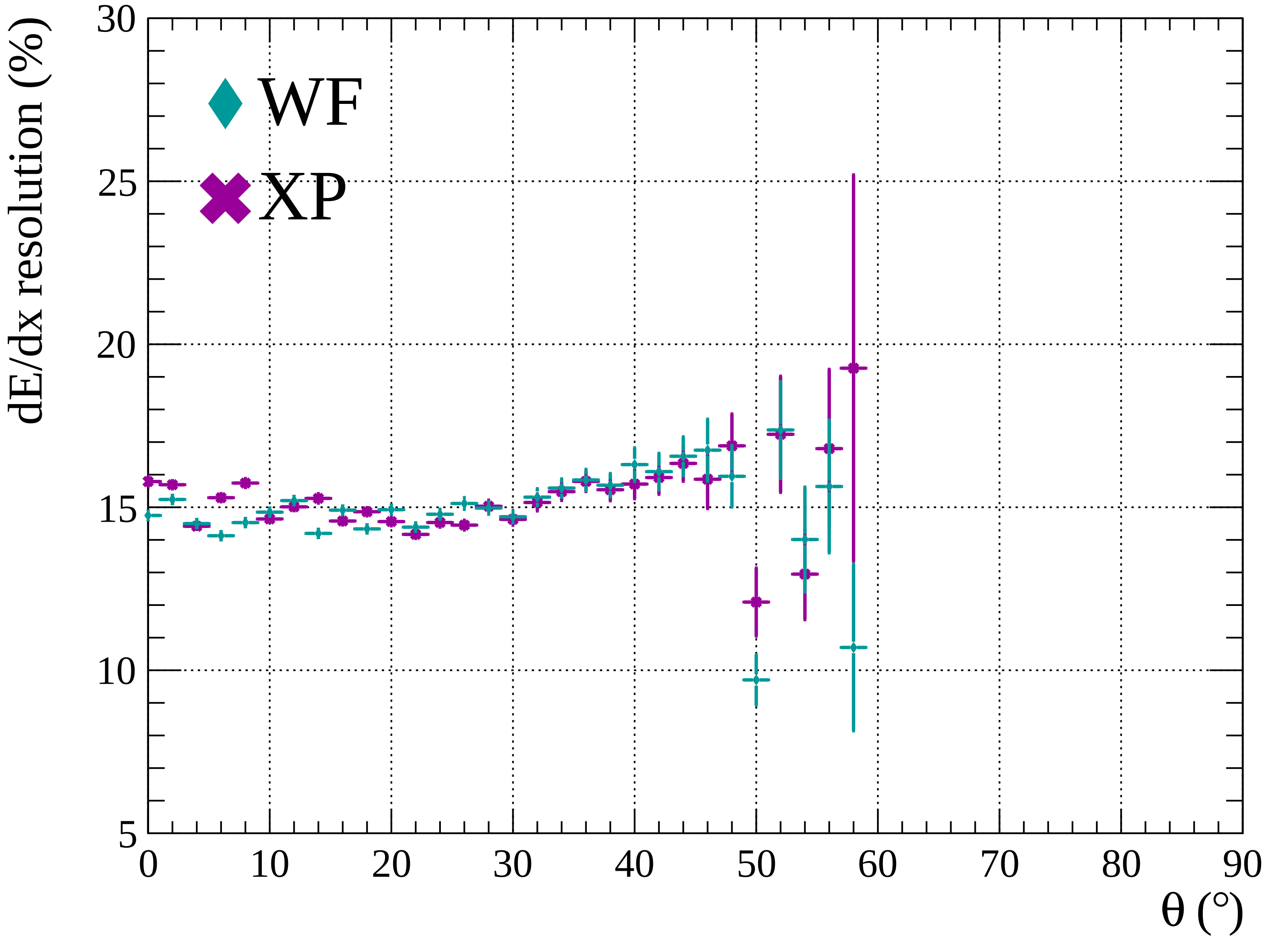
χ^2/NDF
0
5
10
15
20
25
30
35
40
45
50
−80
−60
−40
−20
0
20
40
60
80
φ
0
1000
2000
3000
4000
5000

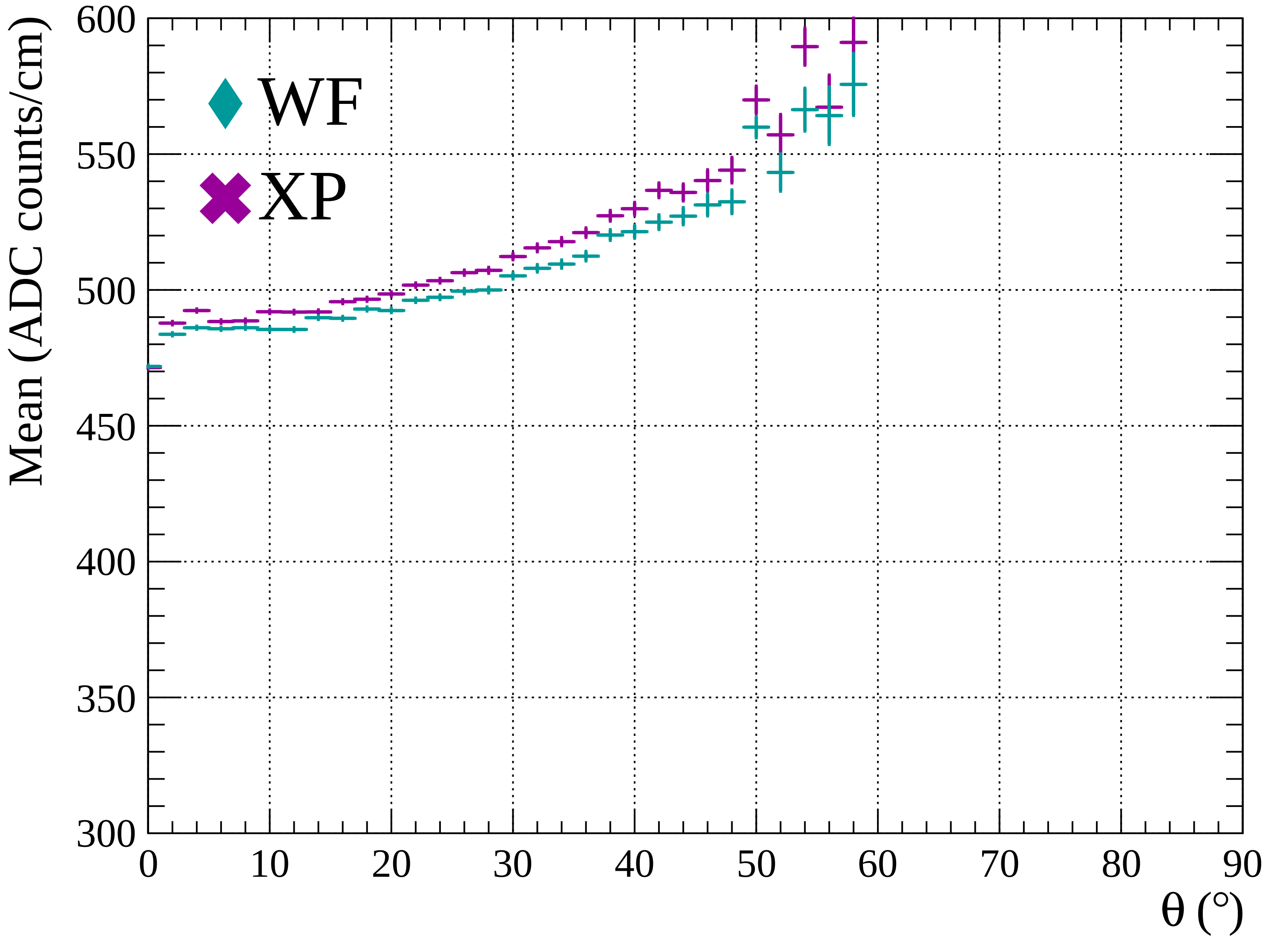


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
70
60
50
40
30
20
10
0

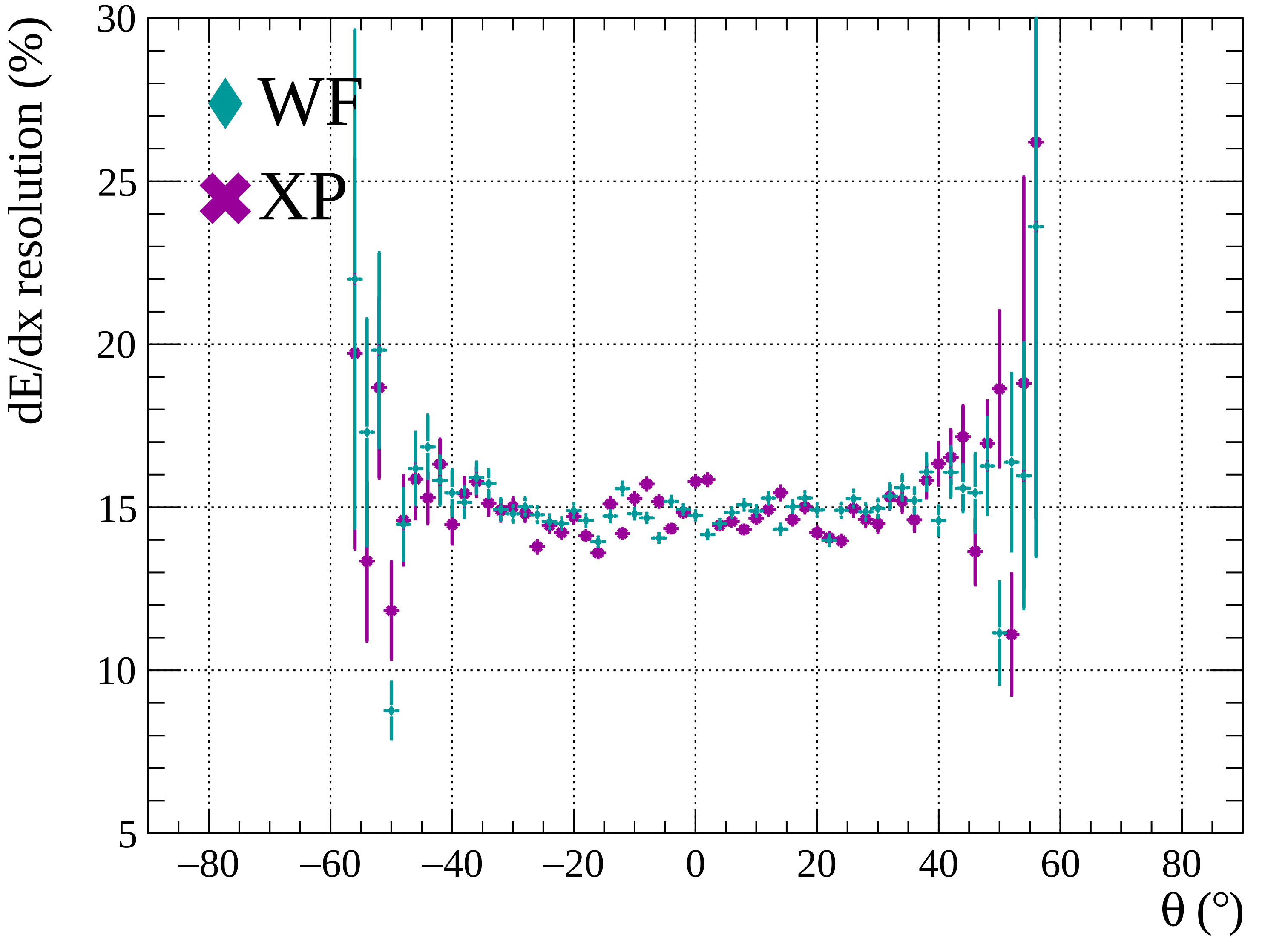


×10
Count
16
14
12
10
8
6
4
2
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

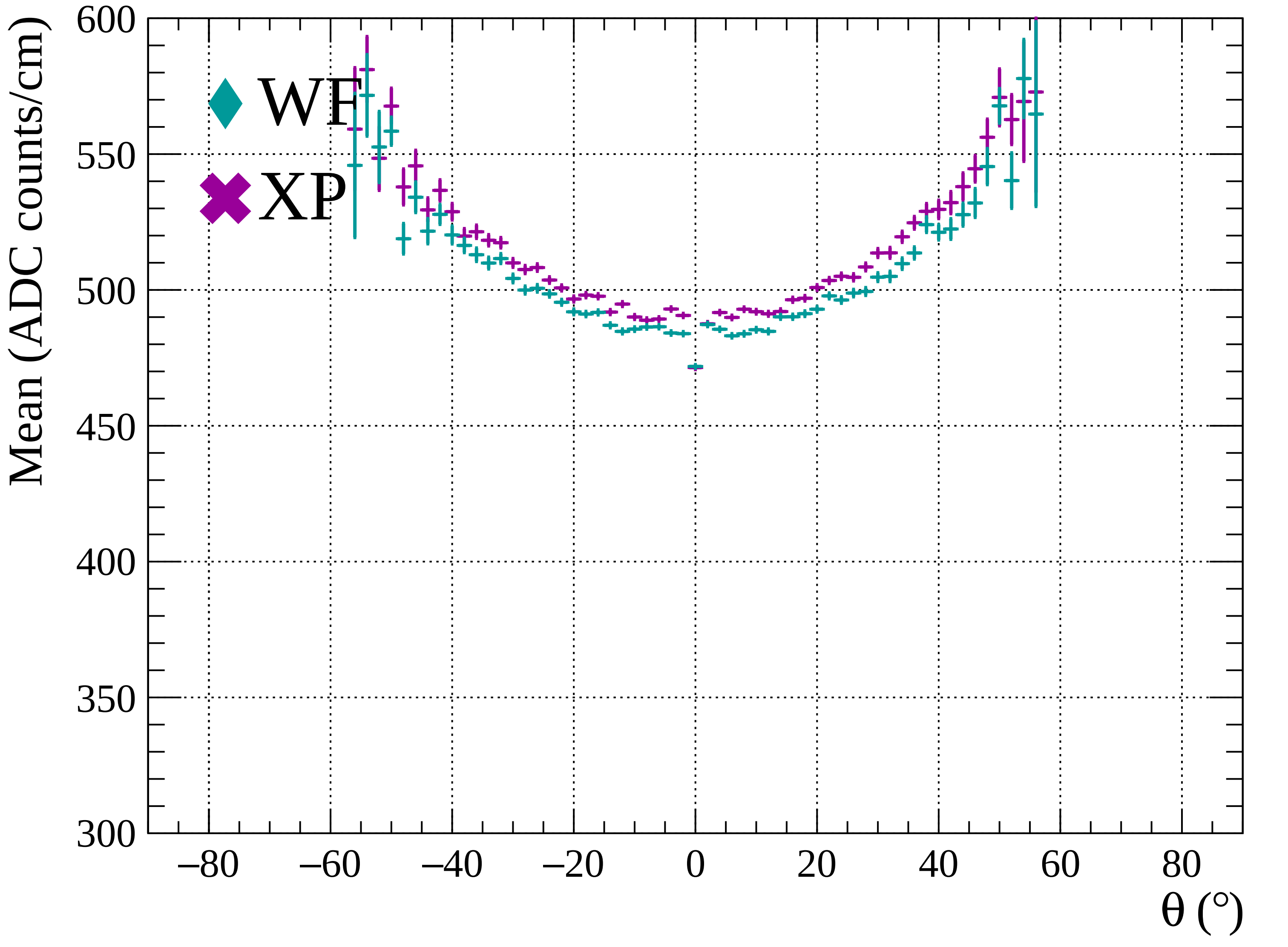

WF
XP
dE/dx resolution (%)
θ (°)
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90



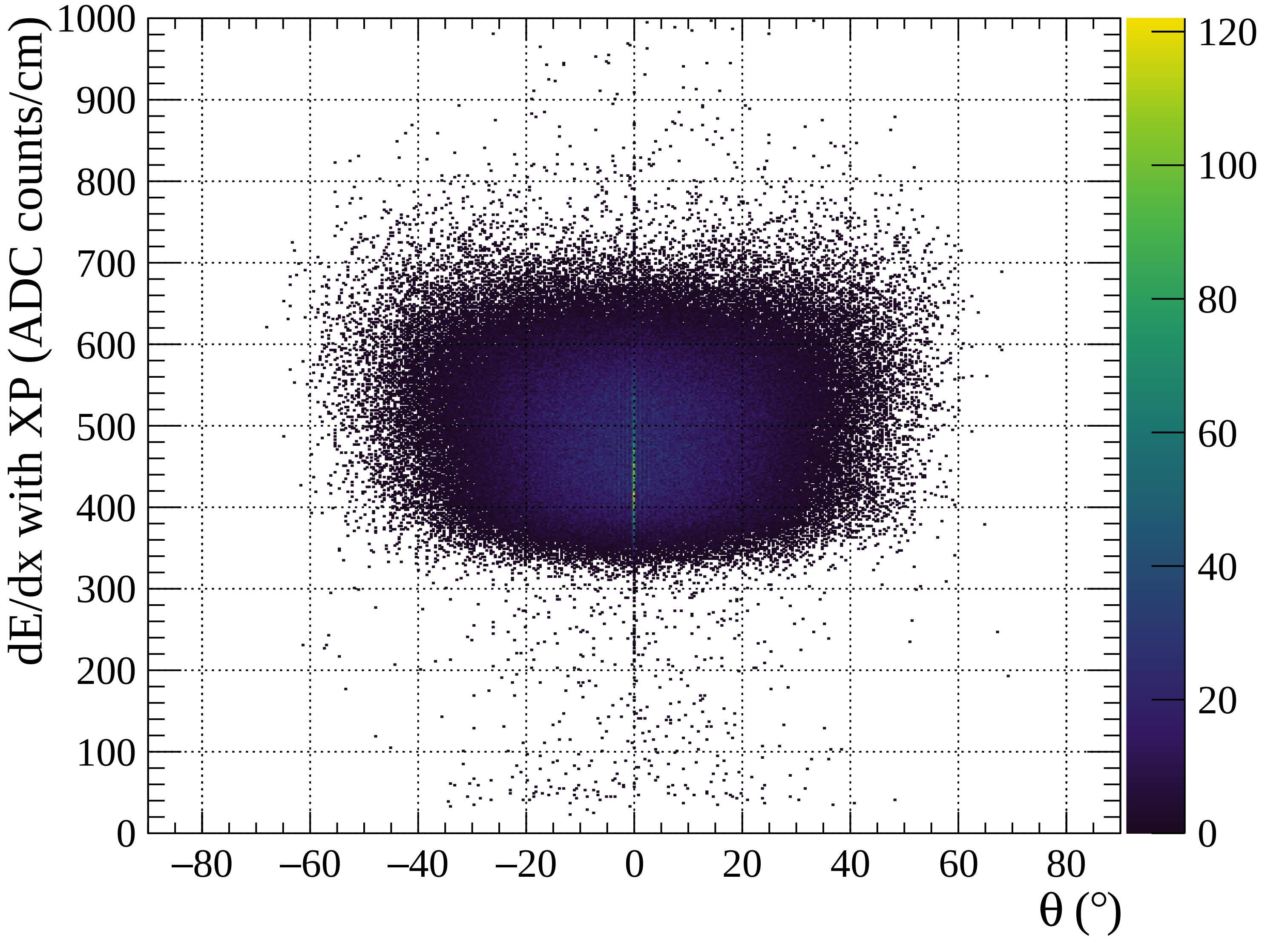
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
0
10
20
30
40
50
60
70
80
90
θ (°)



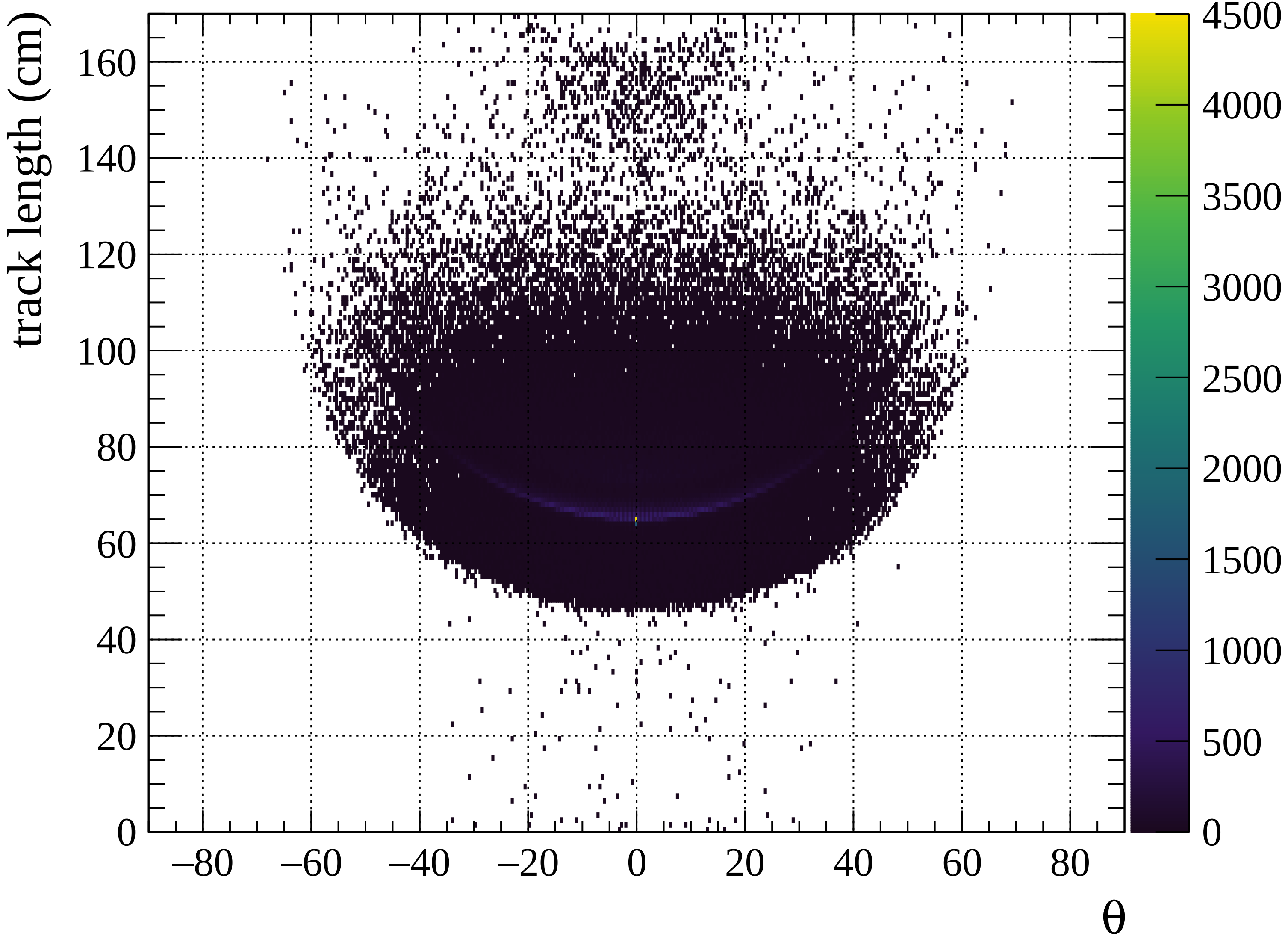
dE/dx resolution (%)
θ (°)
WF
XP
5
10
15
20
25
30
−80
−60
−40
−20
0
20
40
60
80

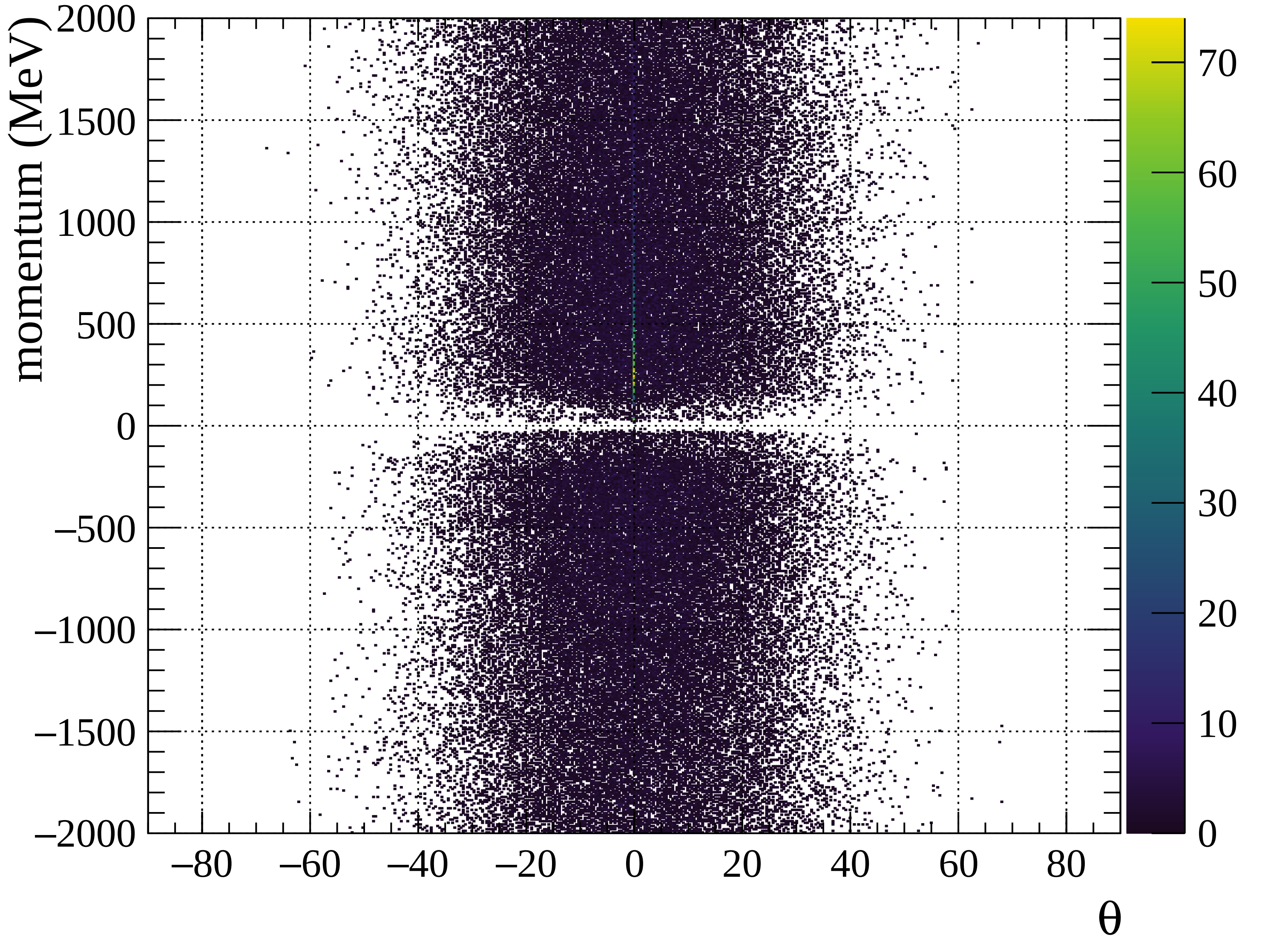


WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
θ (°)

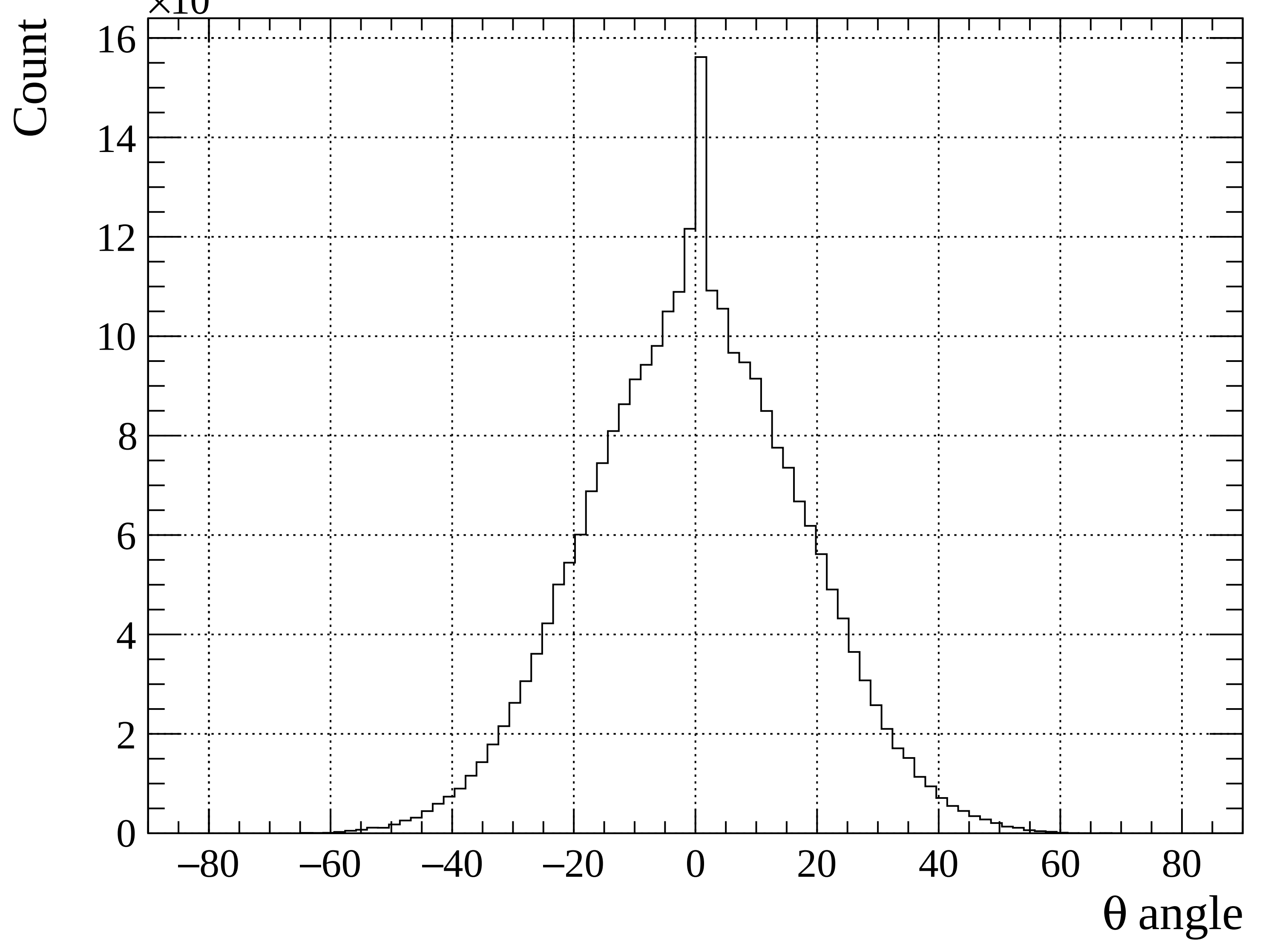


dE/dx with XP (ADC counts/cm)
θ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
120
100
80
60
40
20
0

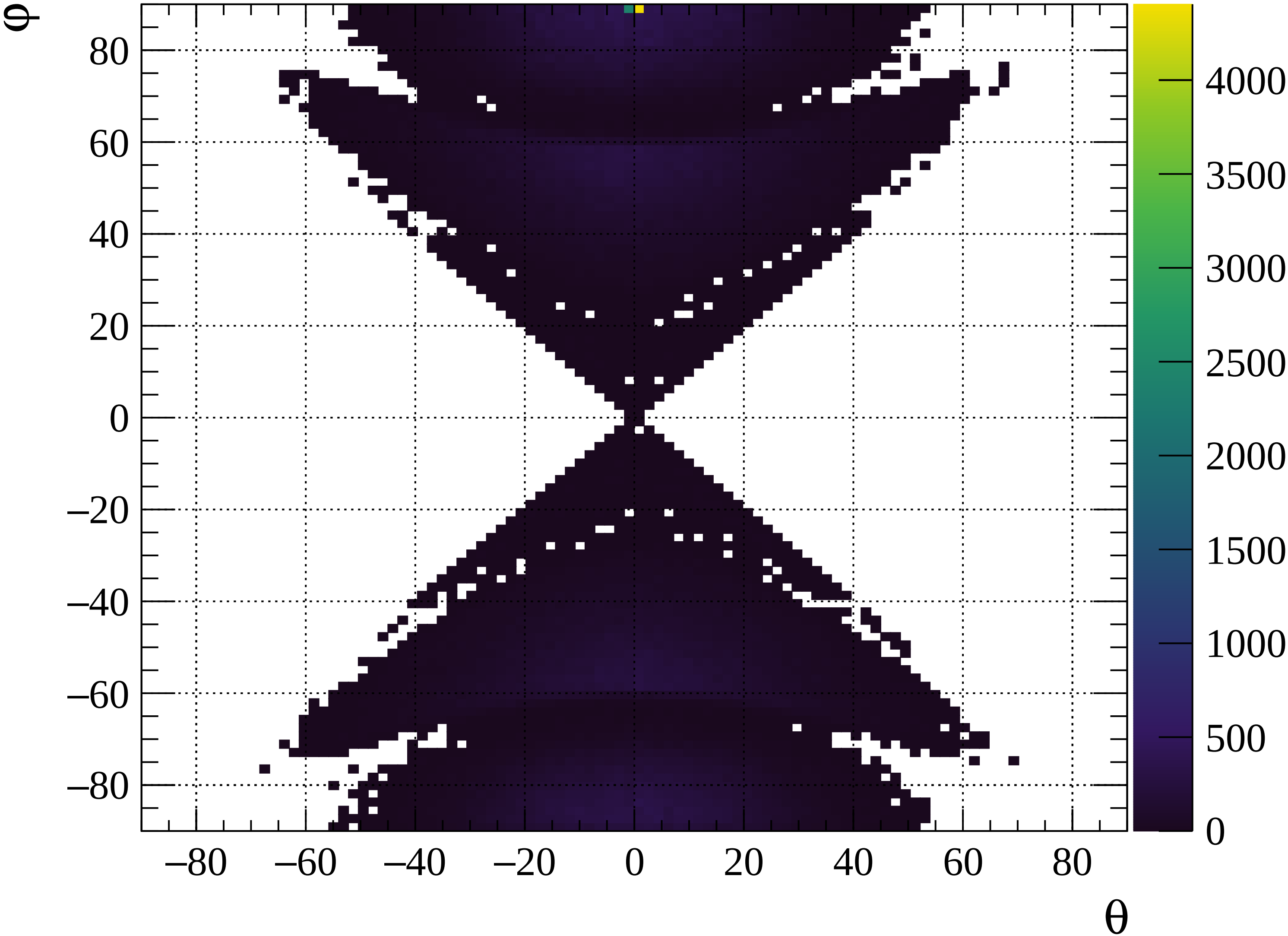

track length (cm)
θ
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
0
500
1000
1500
2000
2500
3000
3500
4000
4500



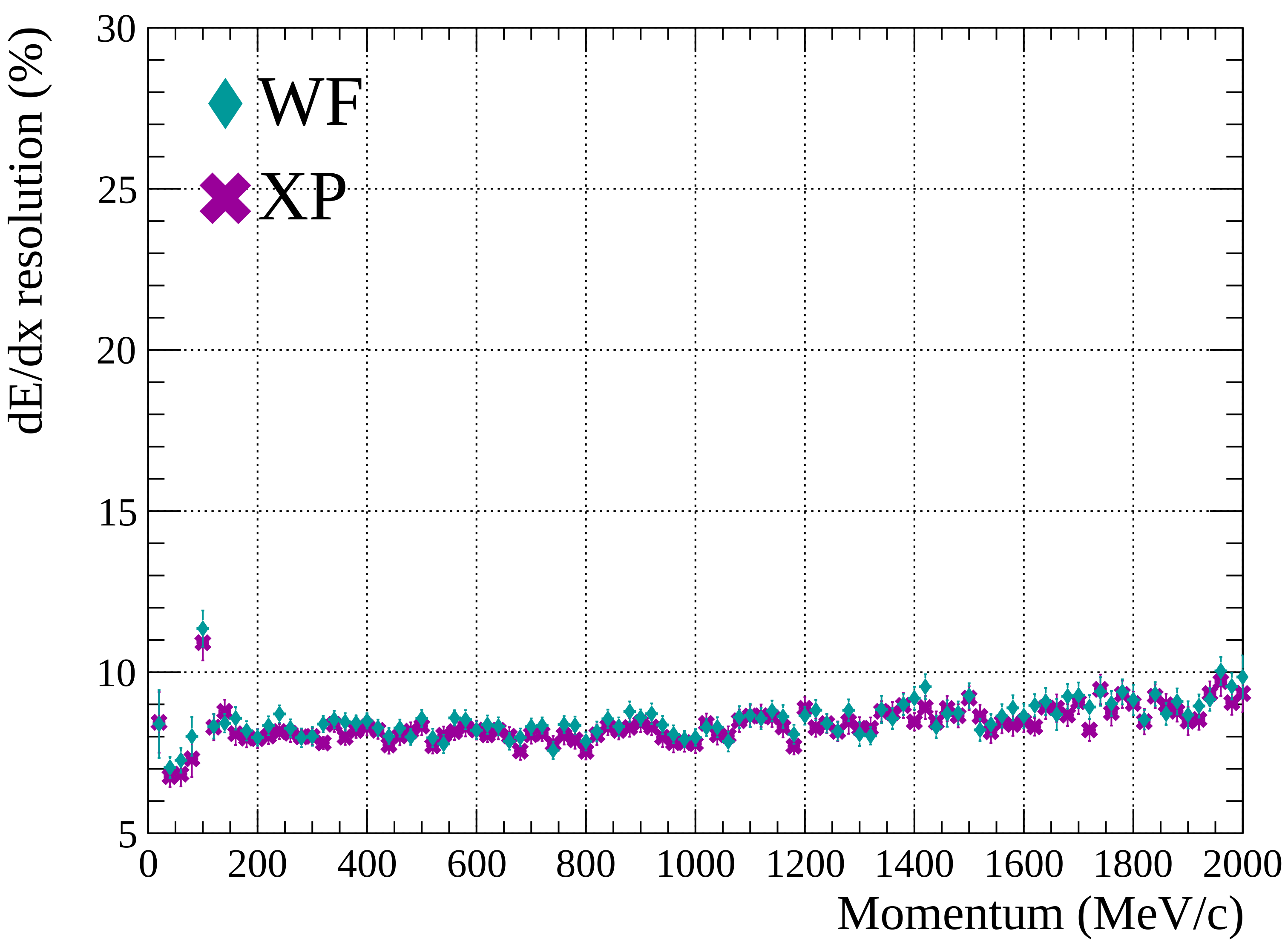
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
70
60
50
40
30
20
10
0

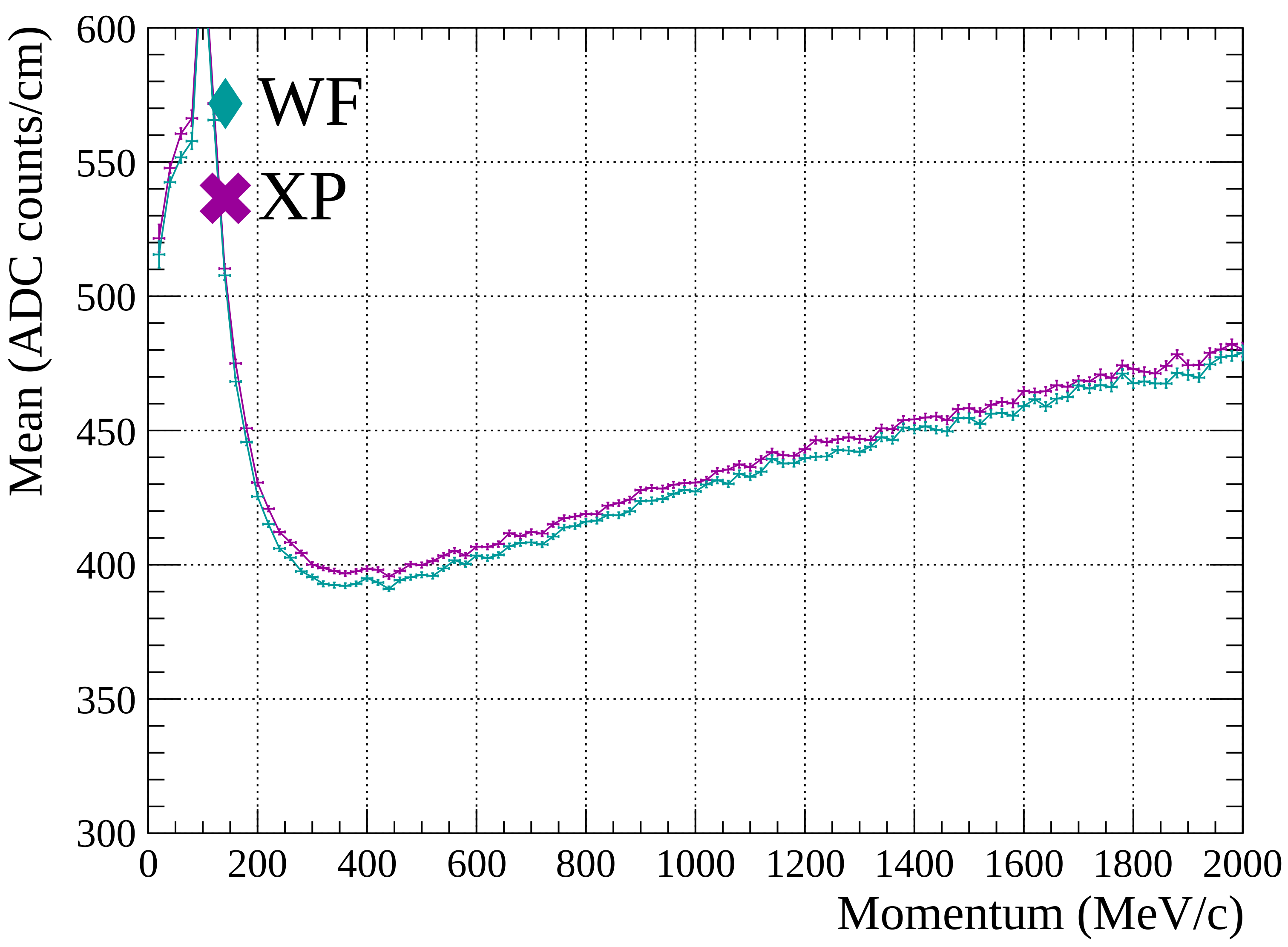


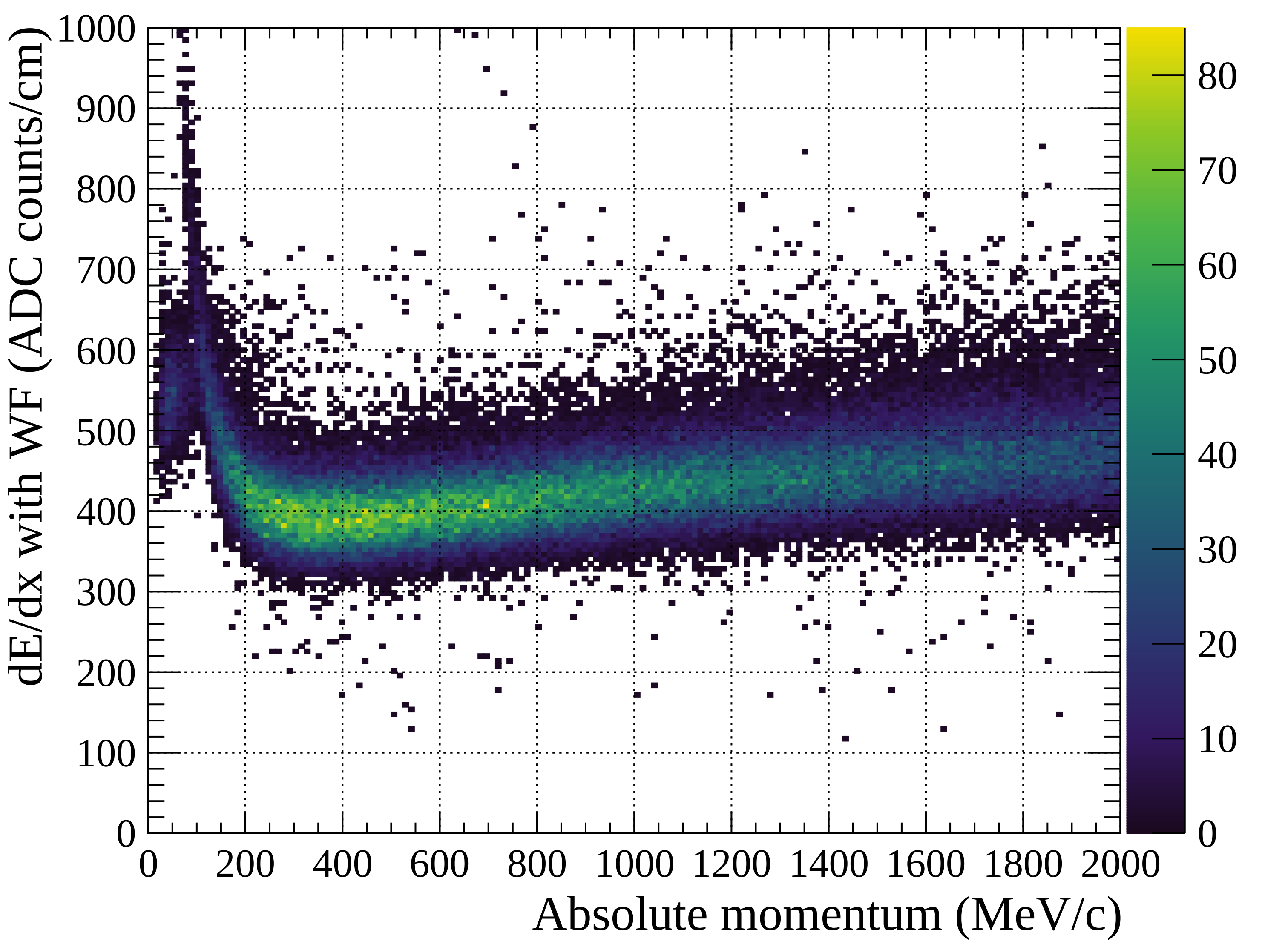
×10
Count
16
14
12
10
8
6
4
2
0
−80
−60
−40
−20
0
20
40
60
80
θ angle



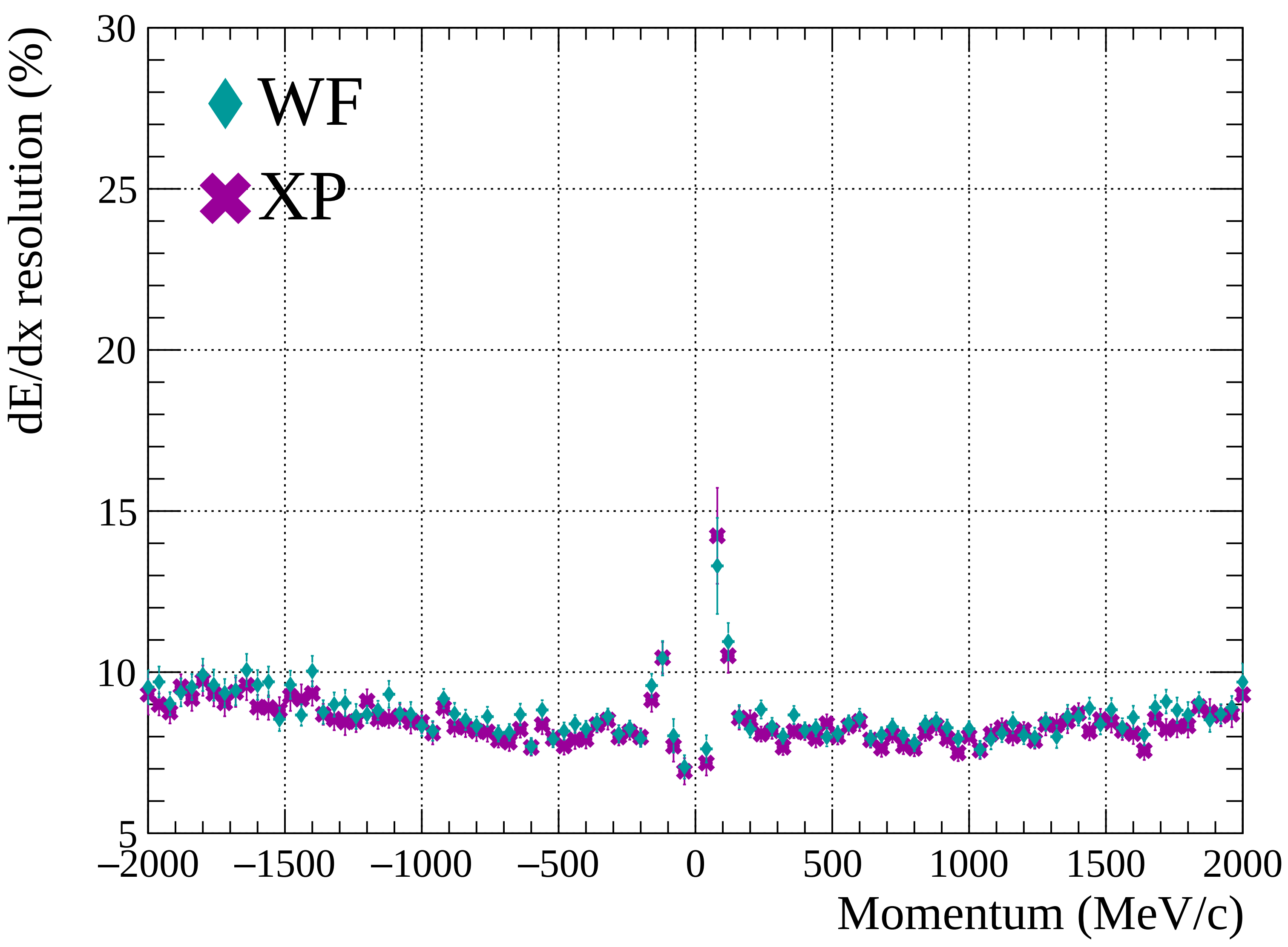
φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
4000
3500
3000
2500
2000
1500
1000
500
0


WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
5
10
15
20
25
30
0
200
400
600
800
1000
1200
1400
1600
1800
2000

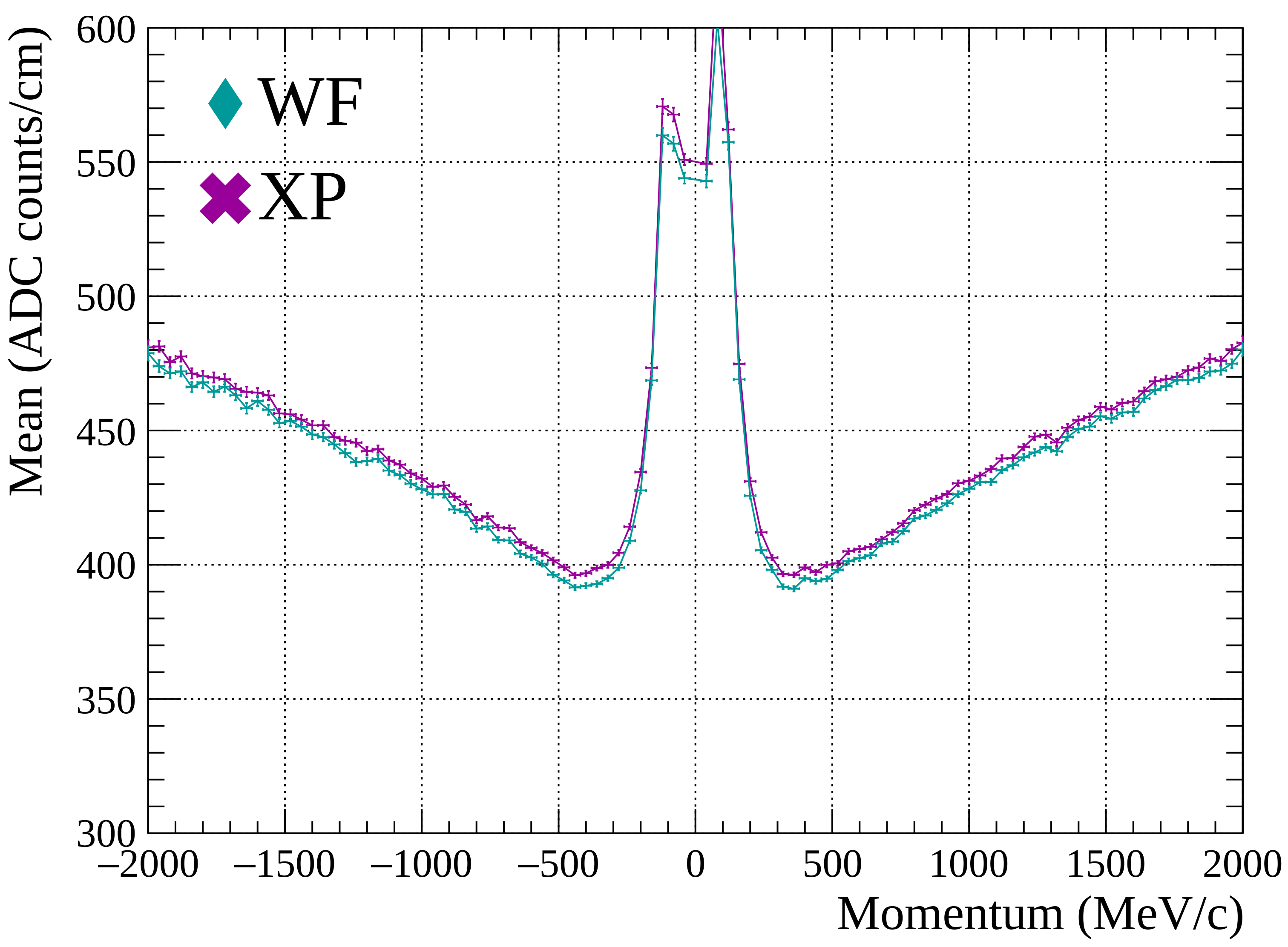

WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000



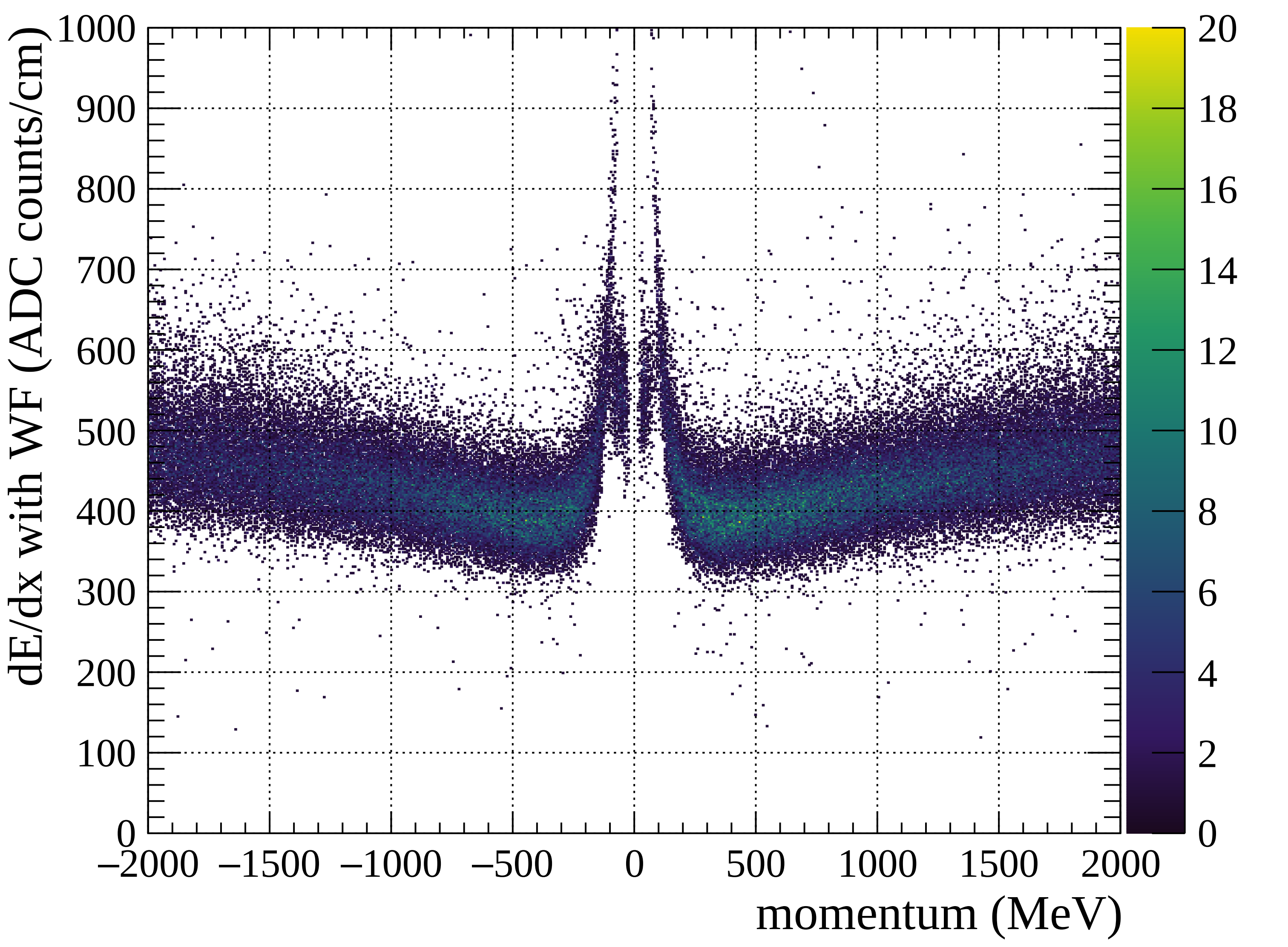
dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
100
300
500
700
900
10
20
30
40
50
60
70
80



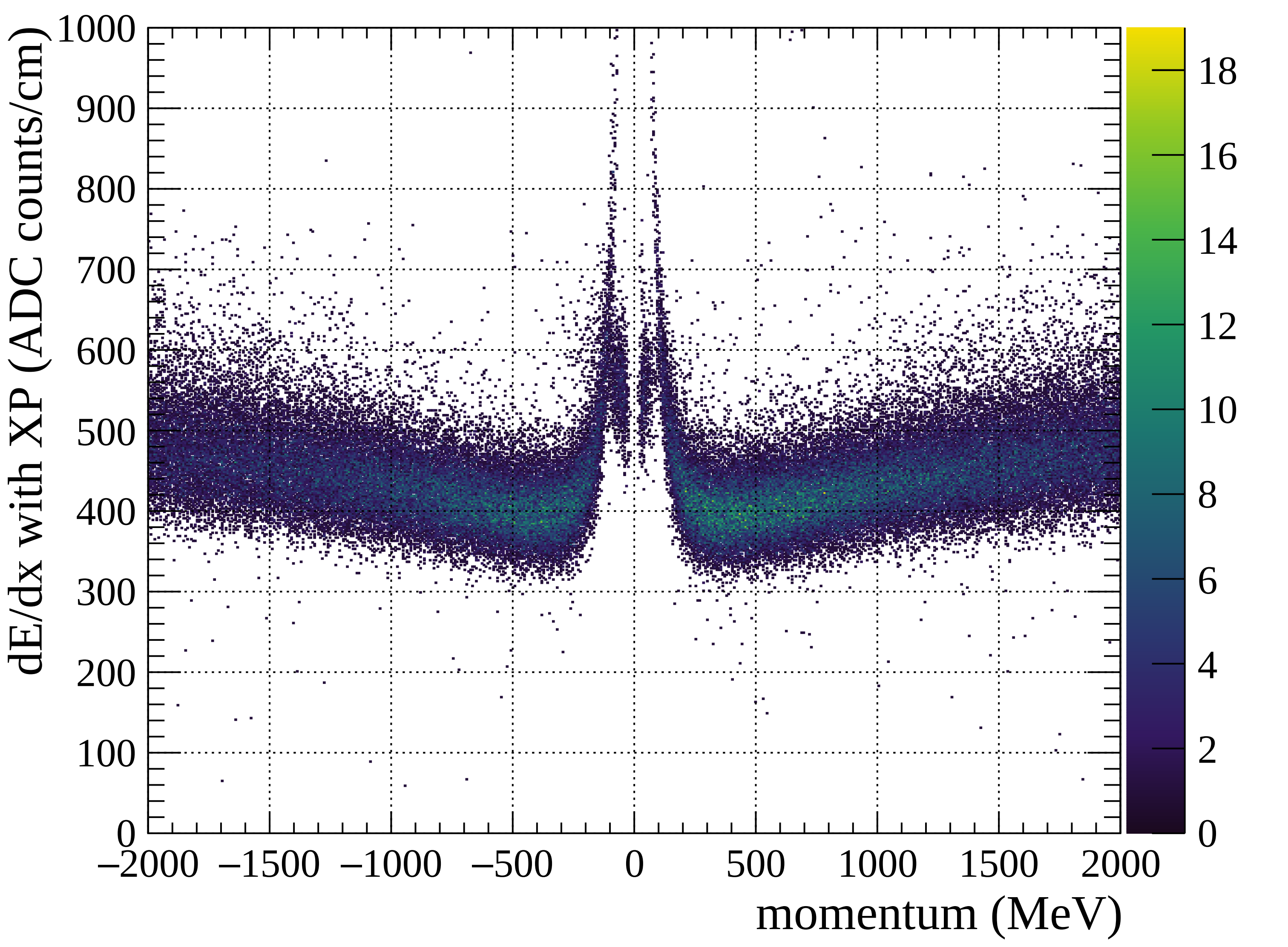
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)

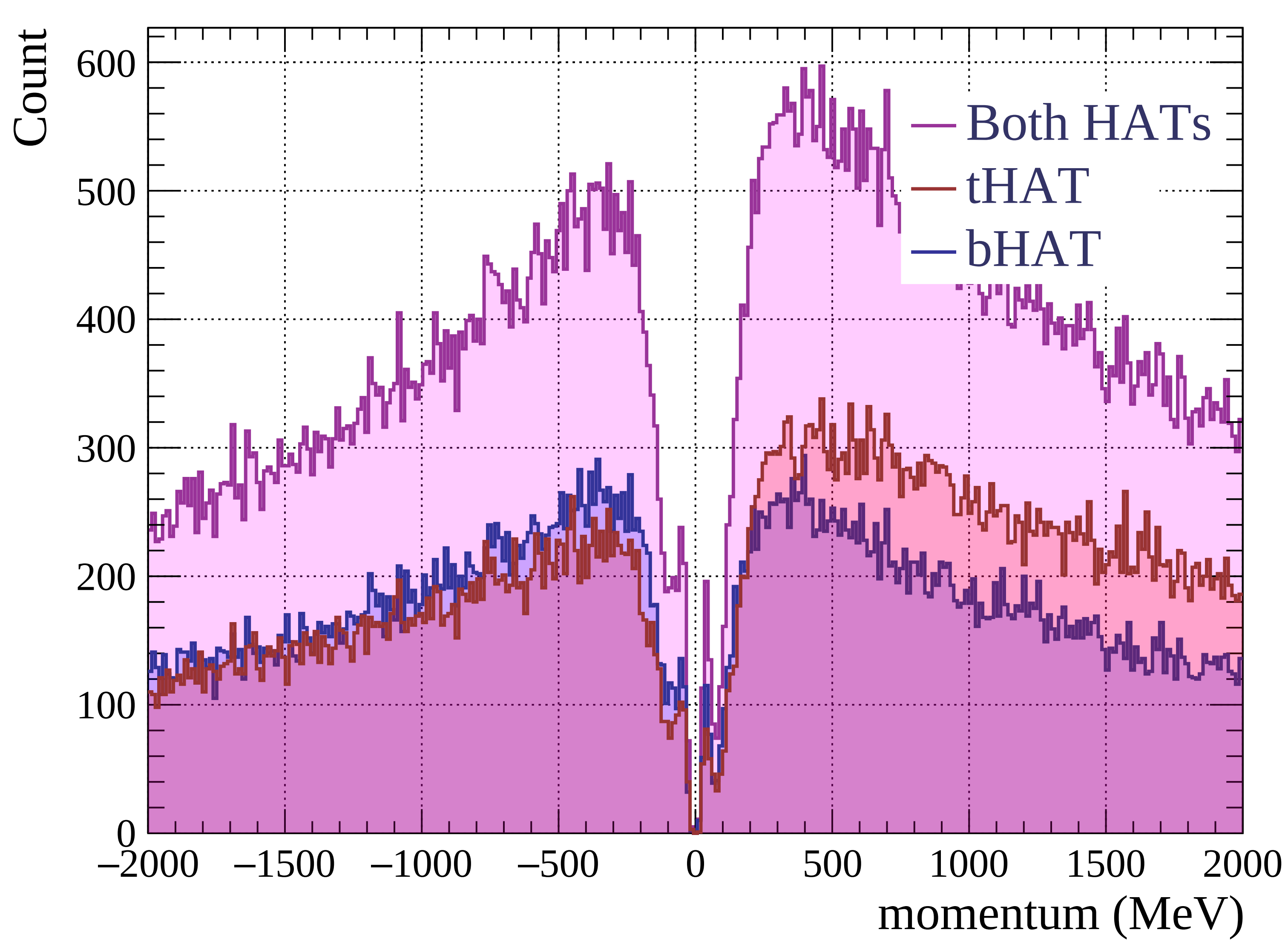


WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000

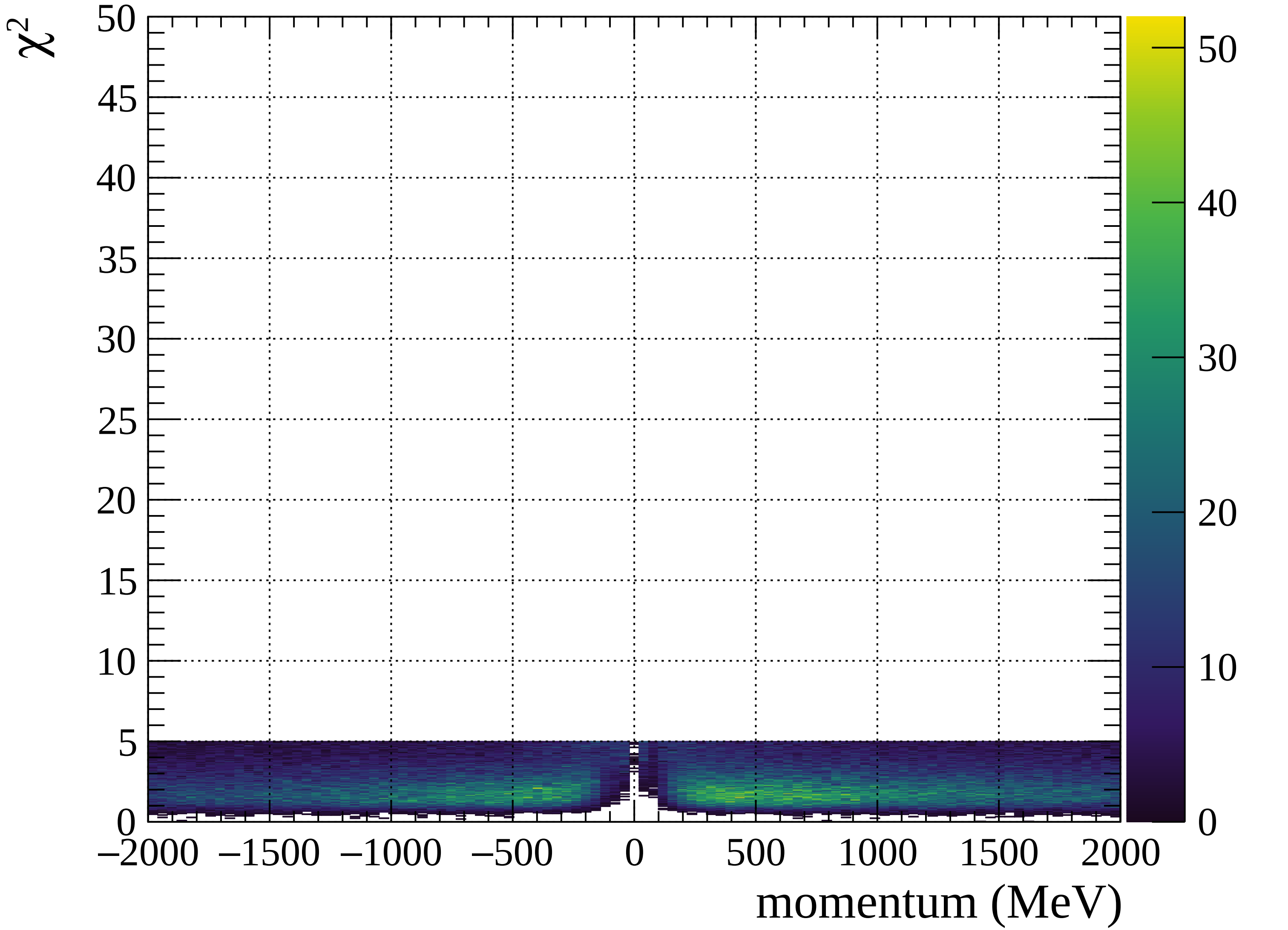


dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
20
18
16
14
12
10
8
6
4
2
0

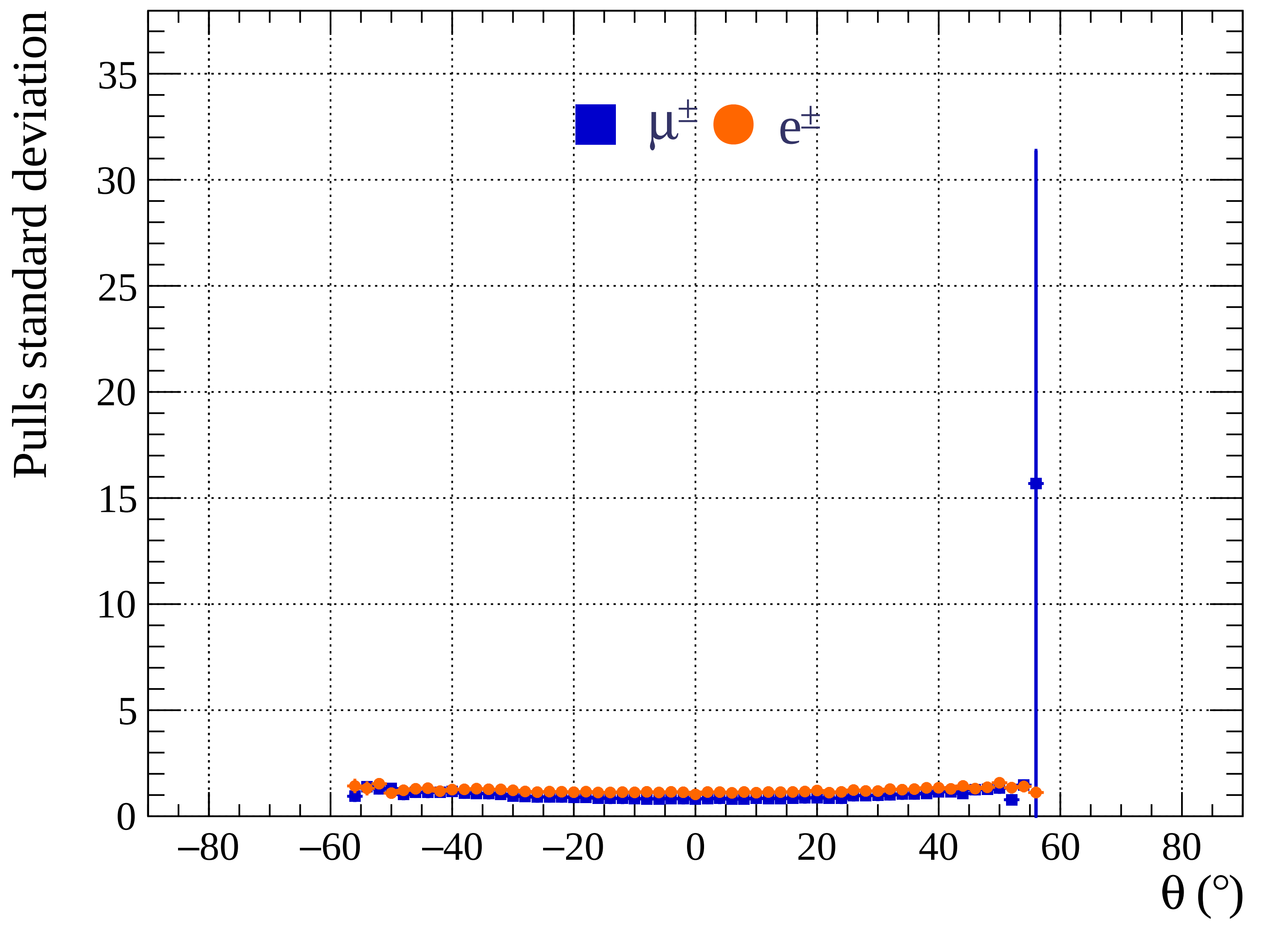

dE/dx with XP (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
2
4
6
8
10
12
14
16
18



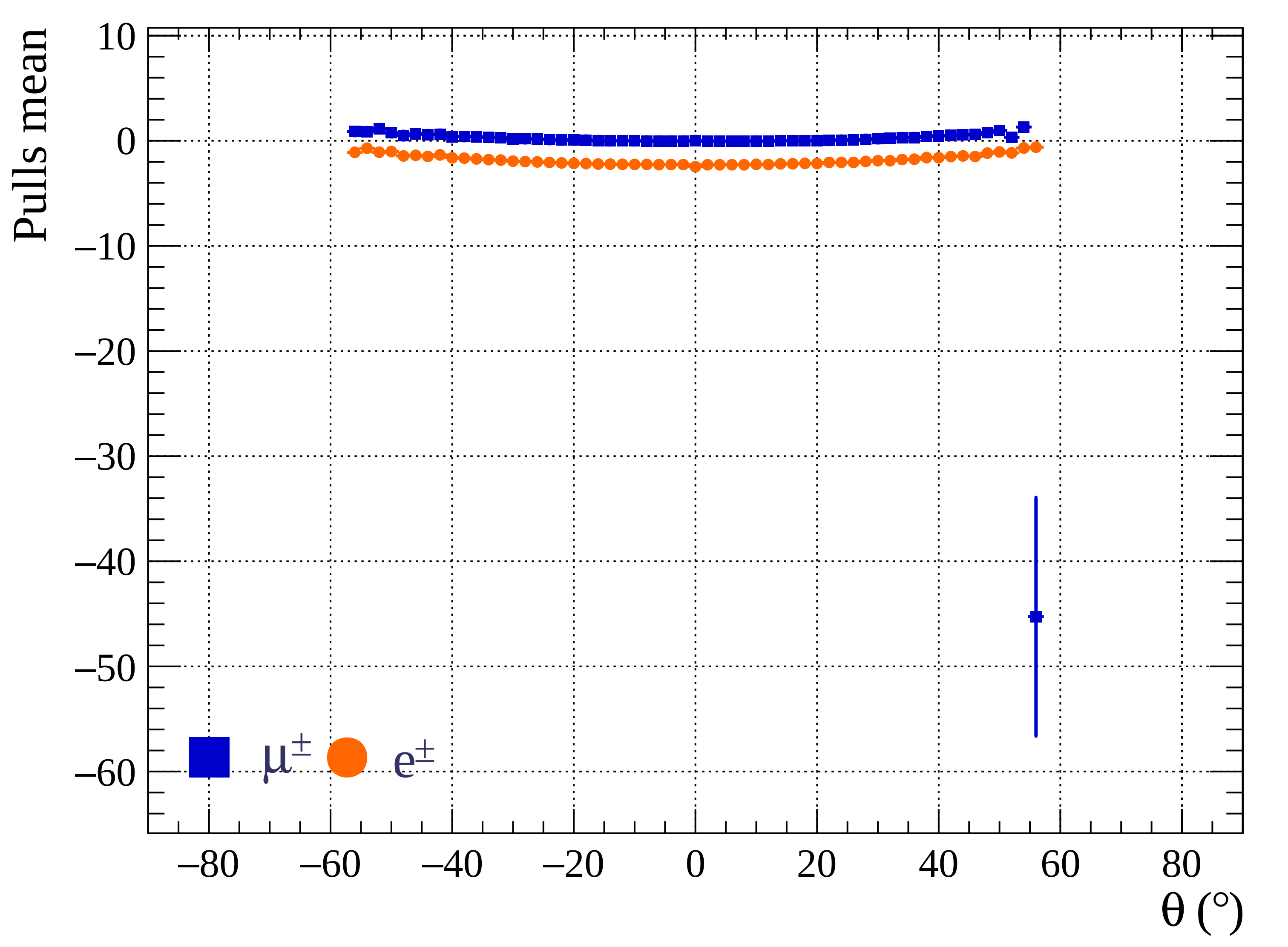
Count
momentum (MeV)
Both HATs
tHAT
bHAT
0
100
200
300
400
500
600
−2000
−1500
−1000
−500
0
500
1000
1500
2000



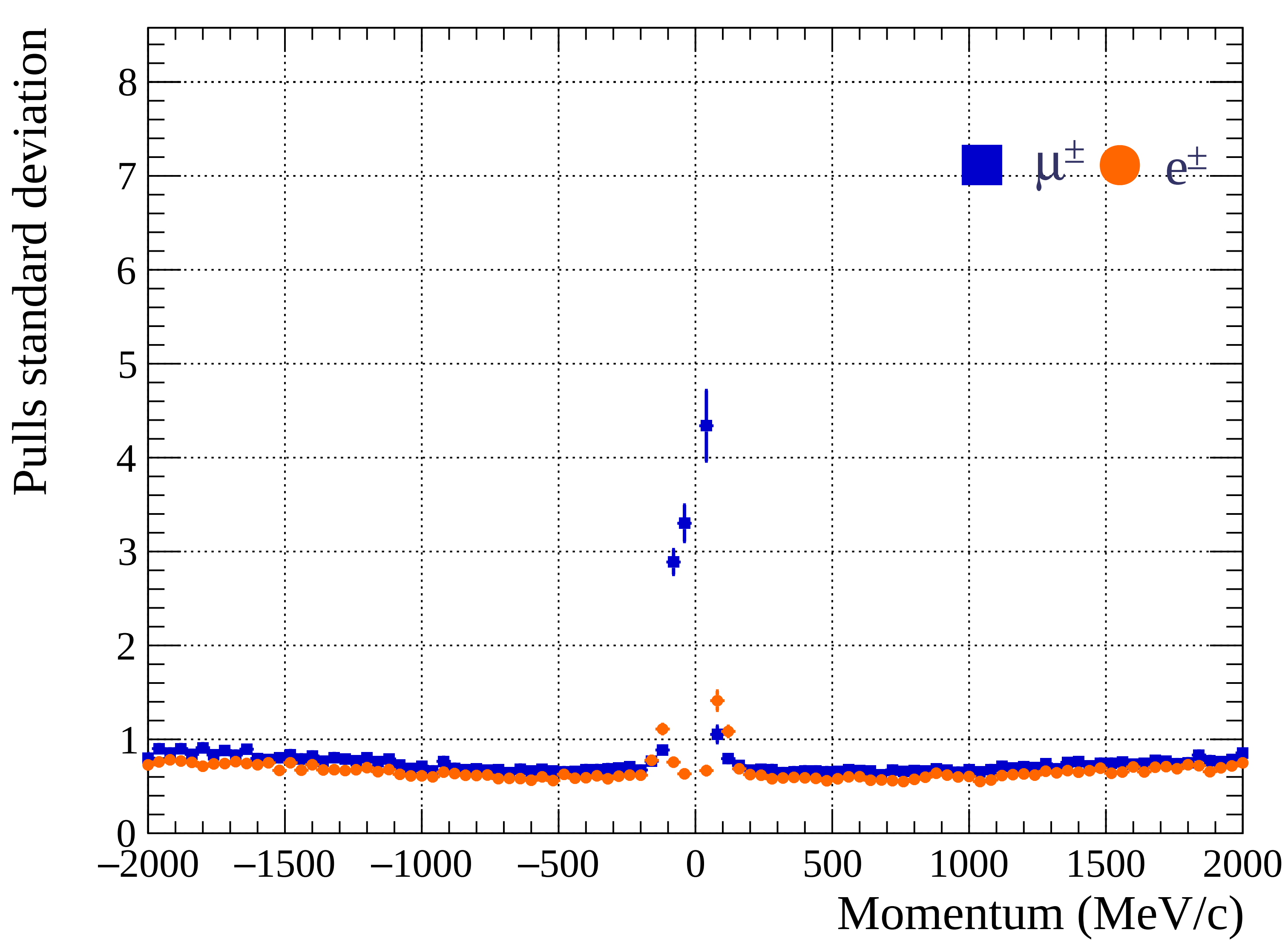
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
50
40
30
20
10
0



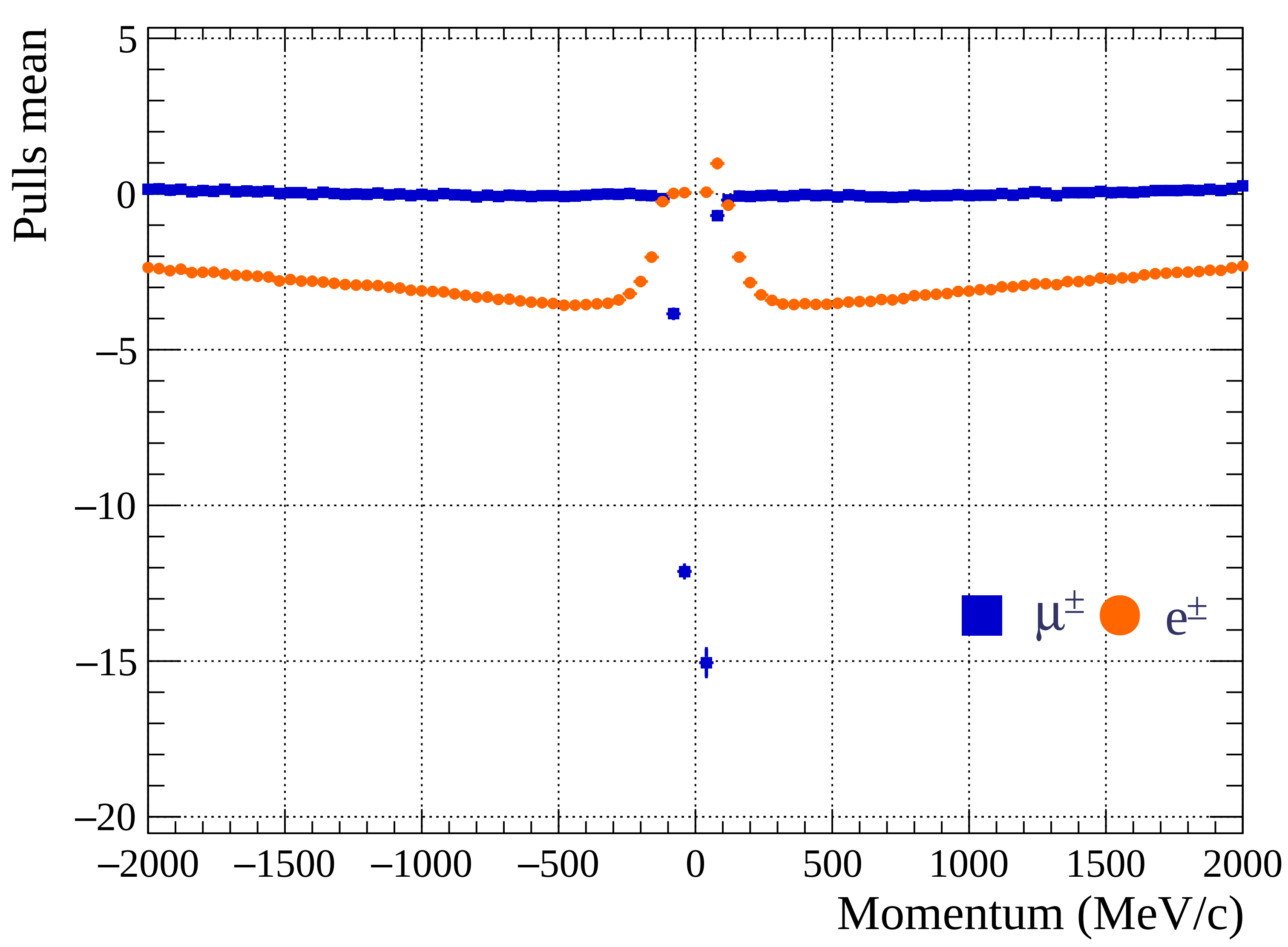
Pulls standard deviation
35
30
25
20
15
10
5
0
μ±
e±
−80
−60
−40
−20
0
20
40
60
80
θ (°)



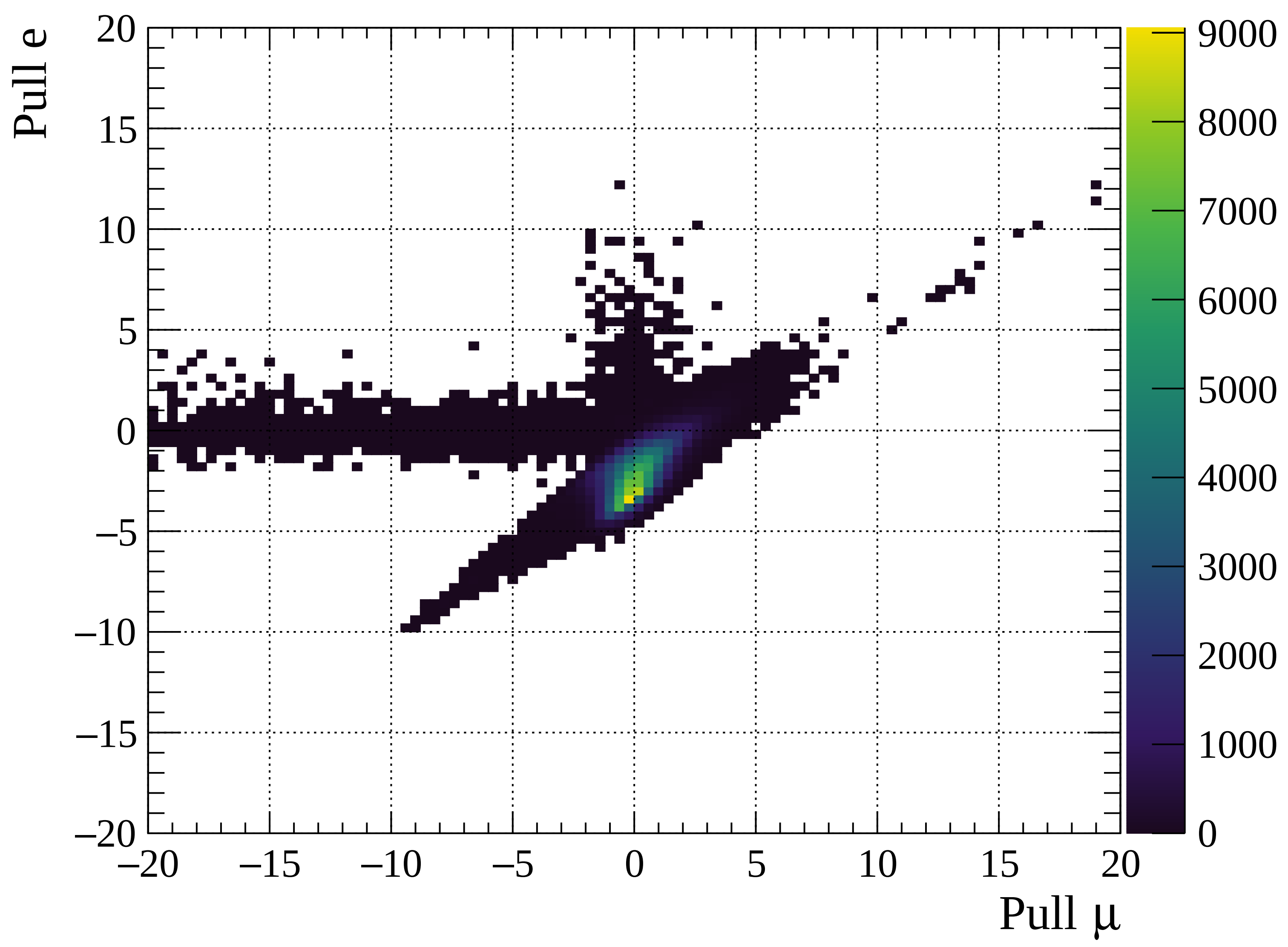
Pulls mean
10
0
−10
−20
−30
−40
−50
−60
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

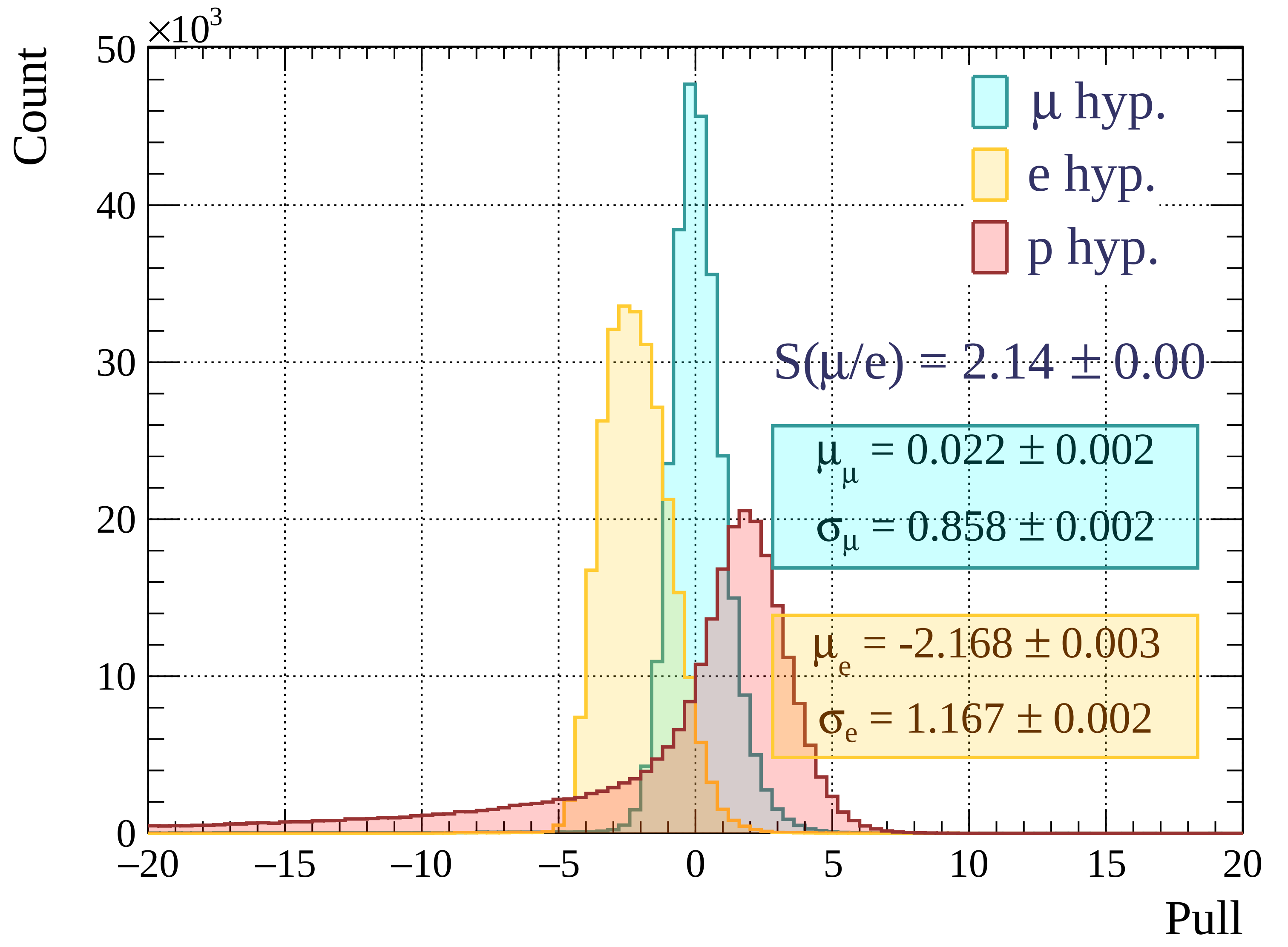


Pulls standard deviation
0
1
2
3
4
5
6
7
8
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

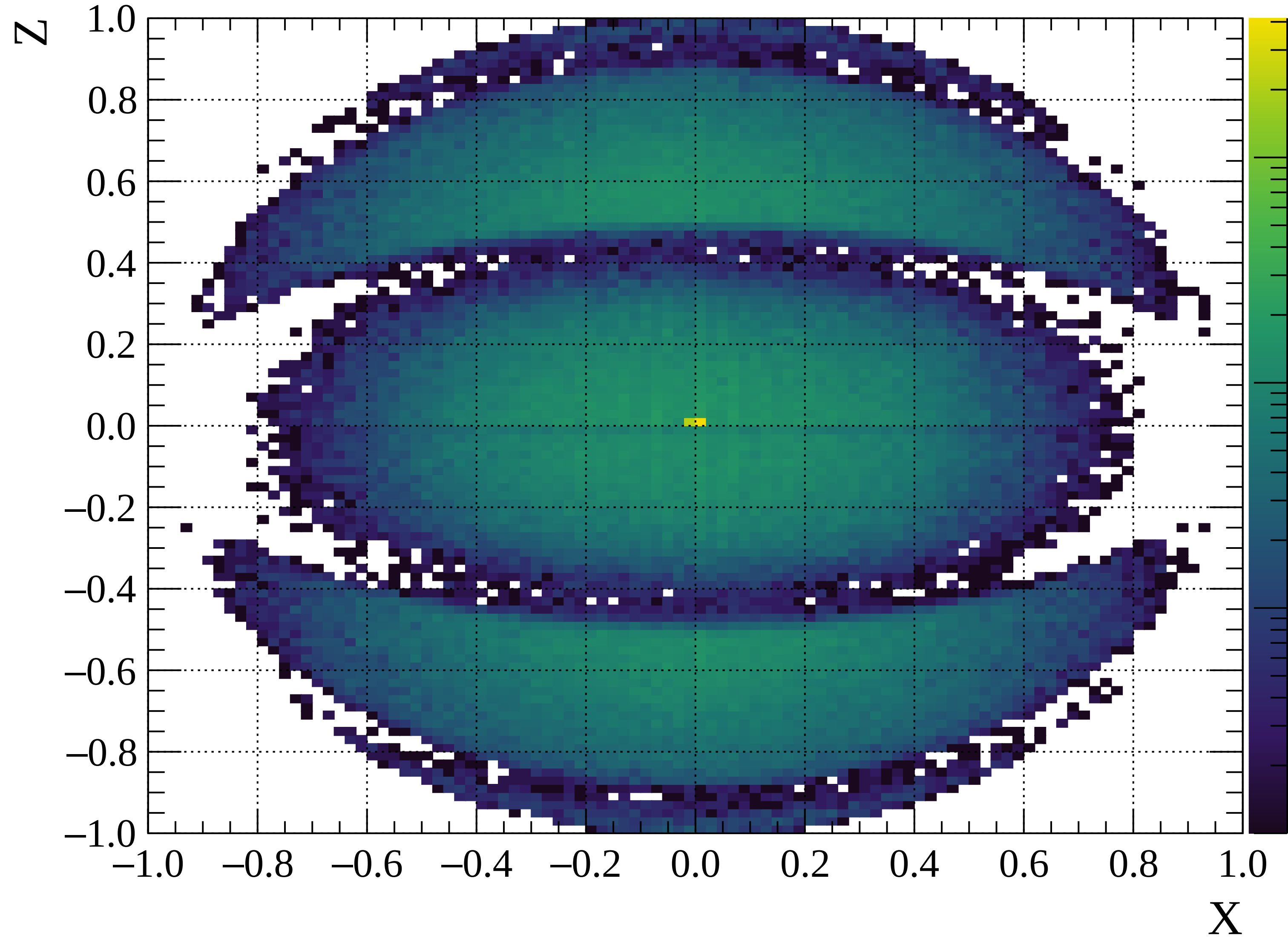


Pulls mean
5
0
−5
−10
−15
−20
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

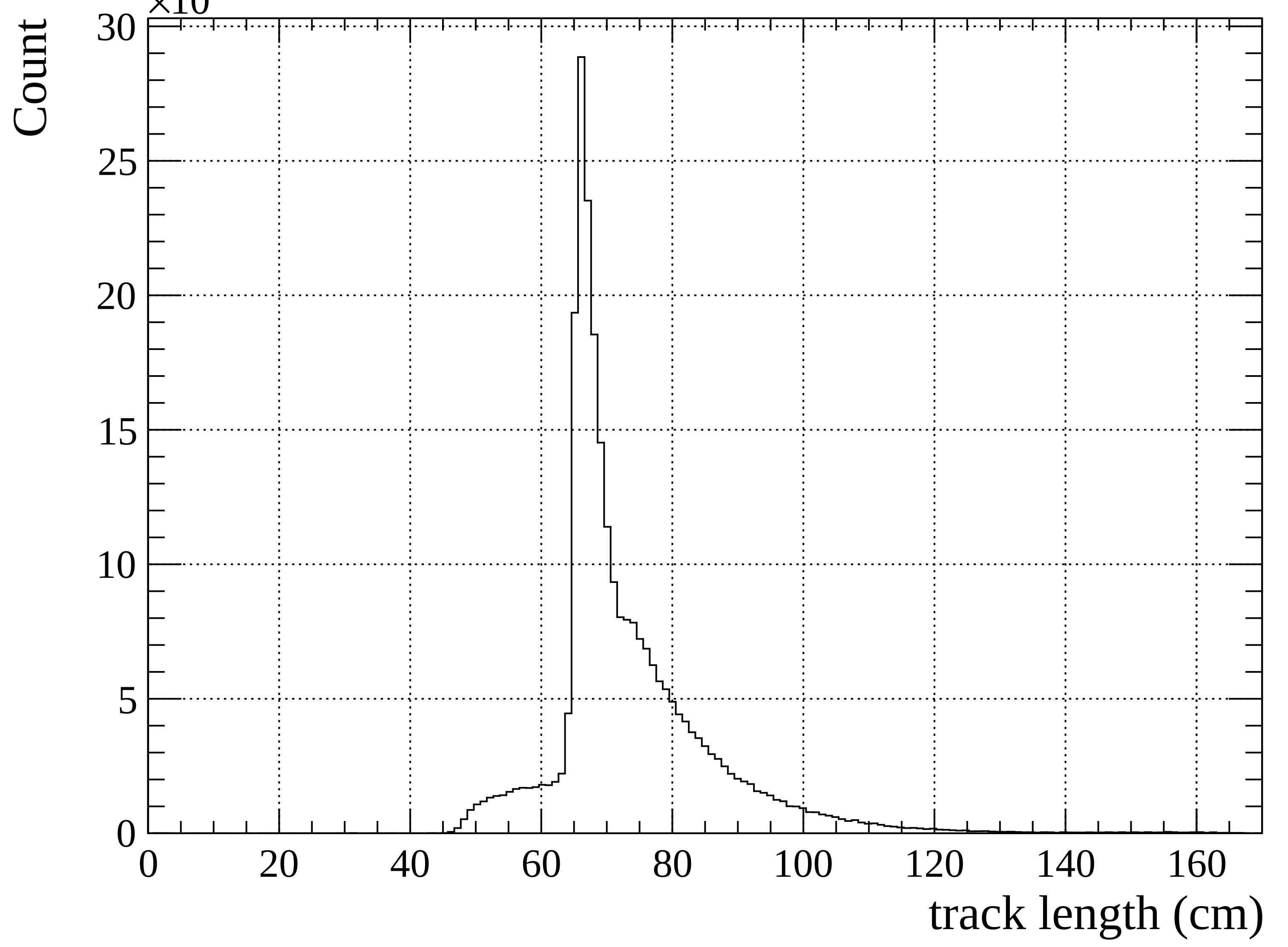

Pull e
Pull μ
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
9000
8000
7000
6000
5000
4000
3000
2000
1000
0

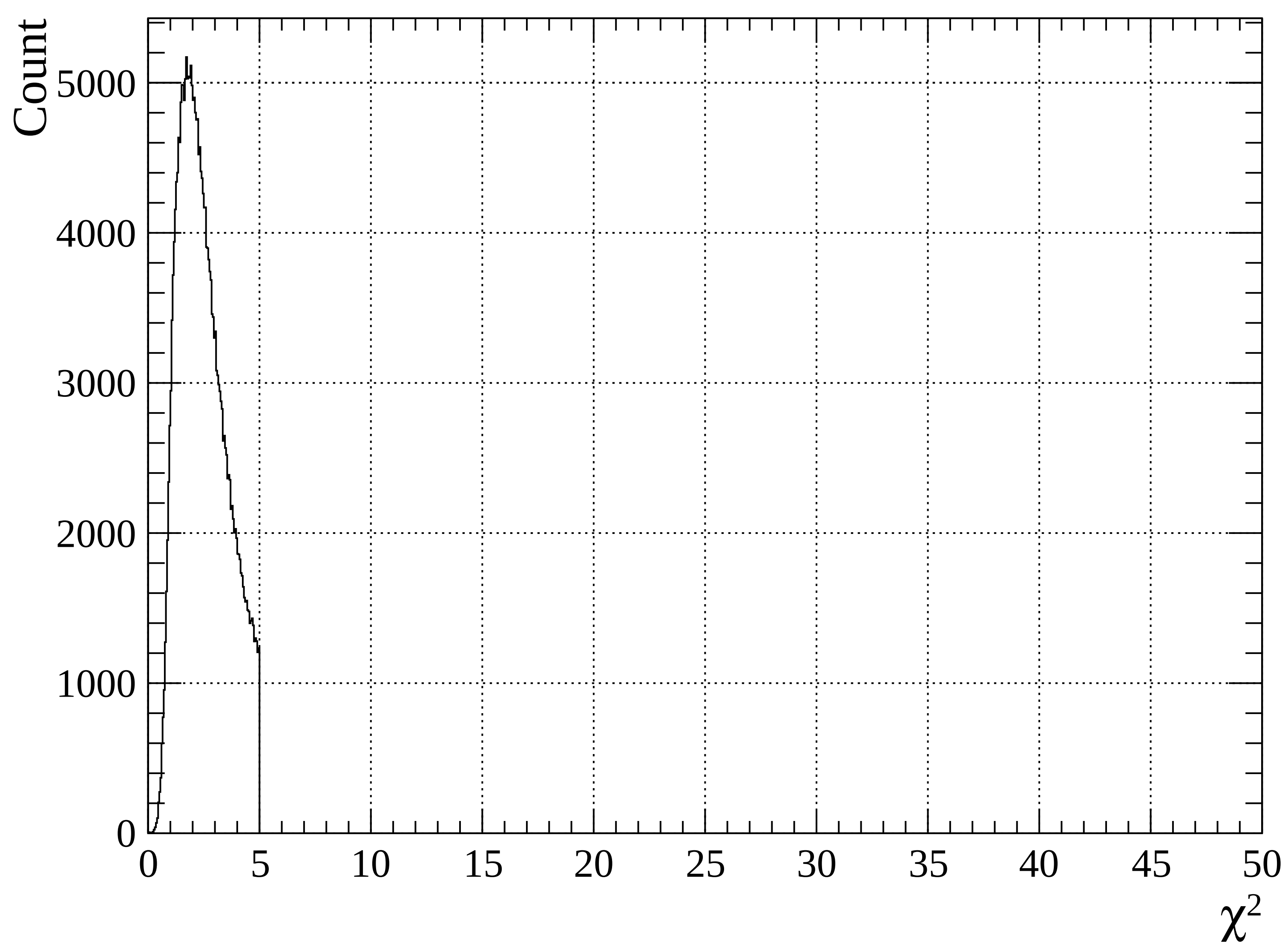


Count
×10^3
50
40
30
20
10
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 2.14 ± 0.00
μ_μ = 0.022 ± 0.002
σ_μ = 0.858 ± 0.002
μ_e = -2.168 ± 0.003
σ_e = 1.167 ± 0.002

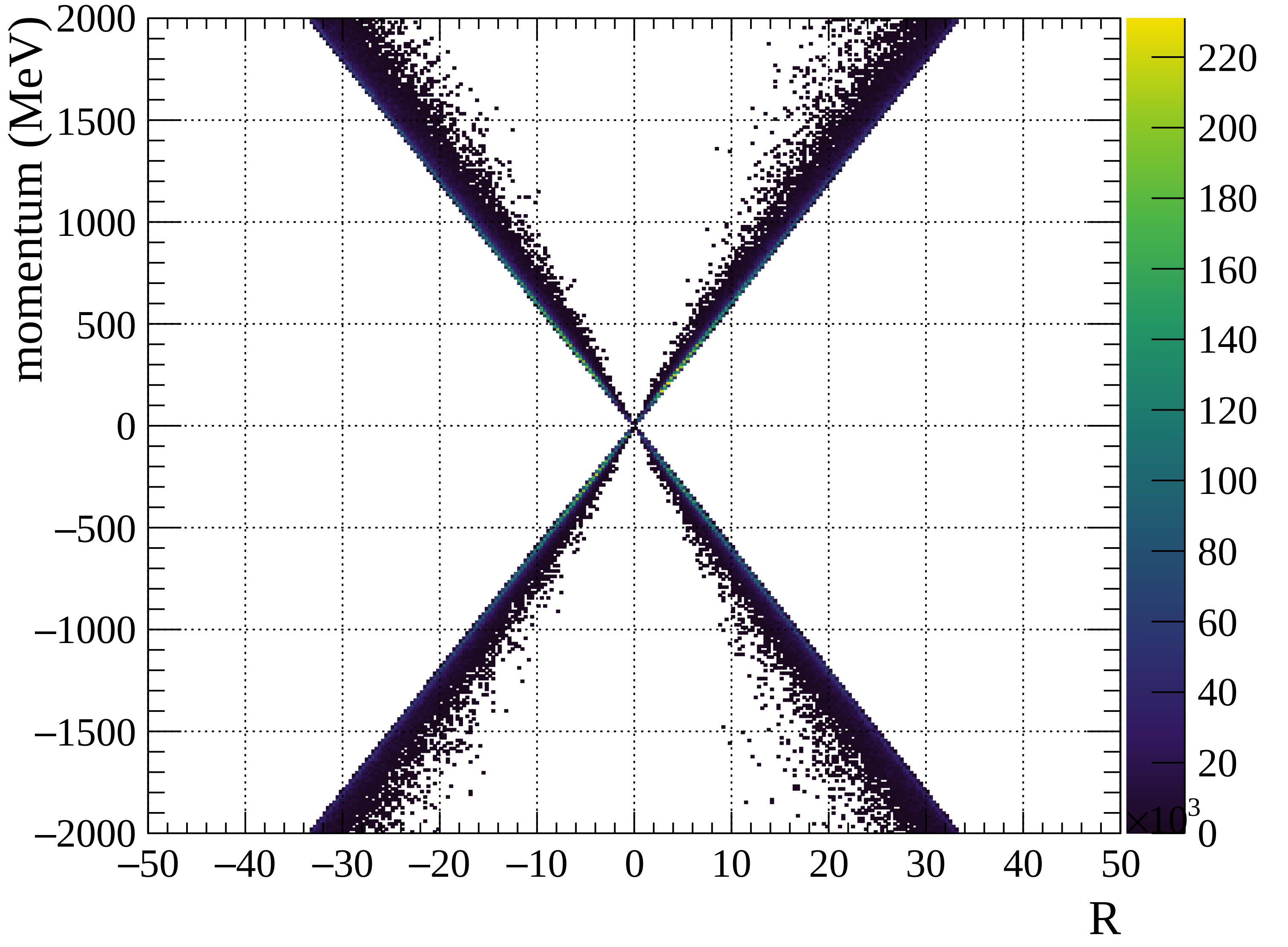


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

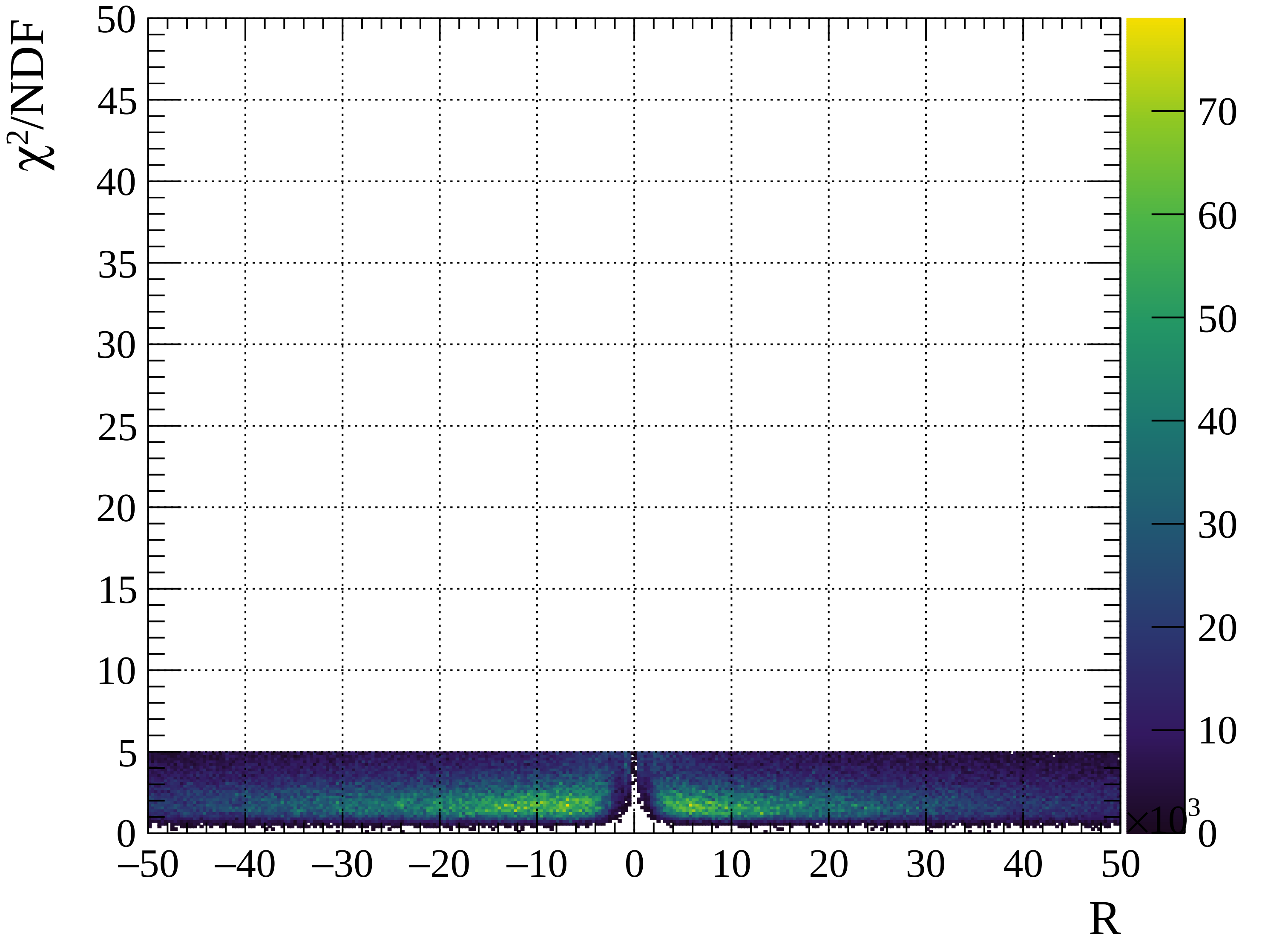

Count
×10
30
25
20
15
10
5
0
0
20
40
60
80
100
120
140
160
track length (cm)



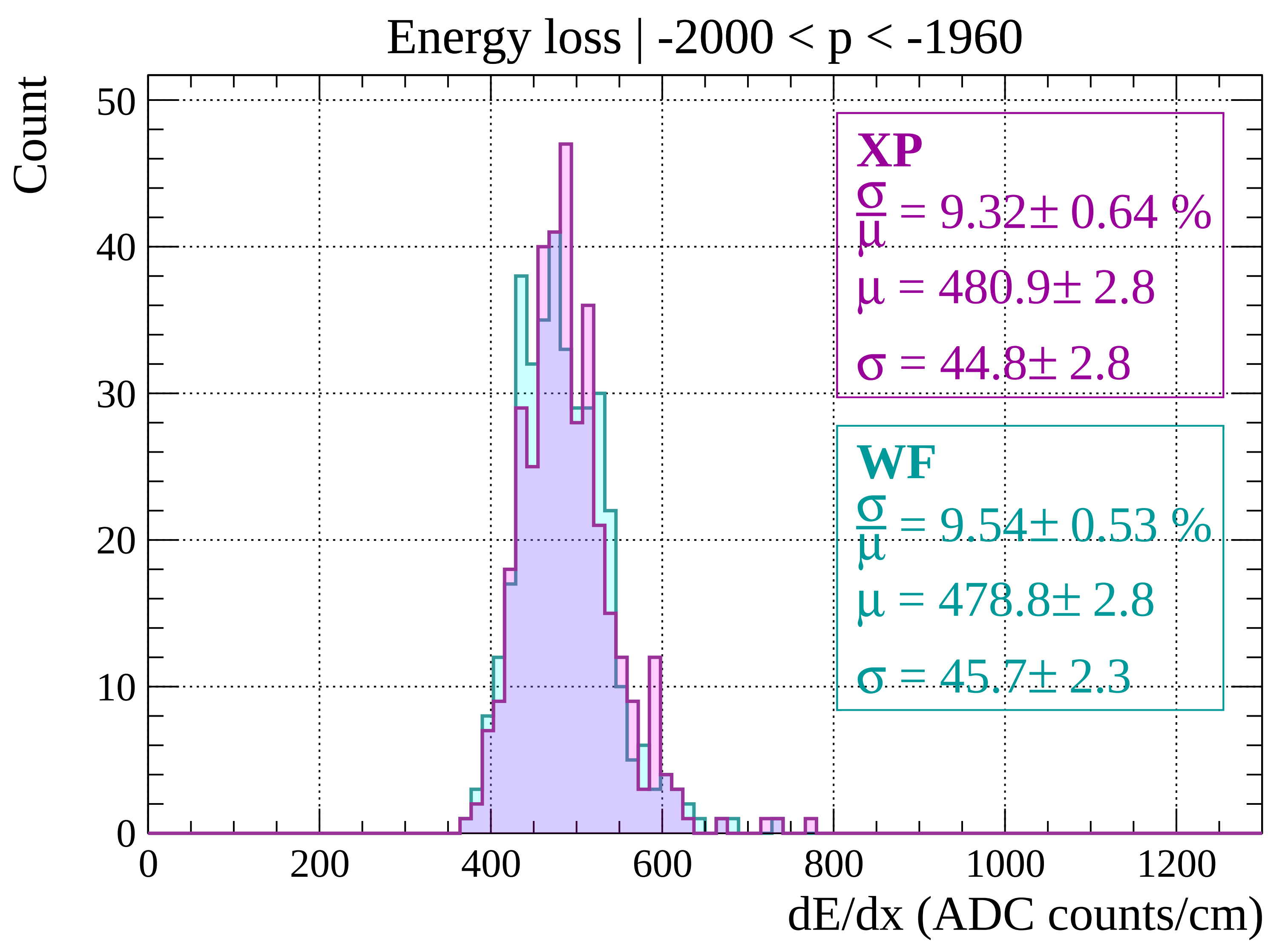
Count
5000
4000
3000
2000
1000
0
0
5
10
15
20
25
30
35
40
45
50
χ^2



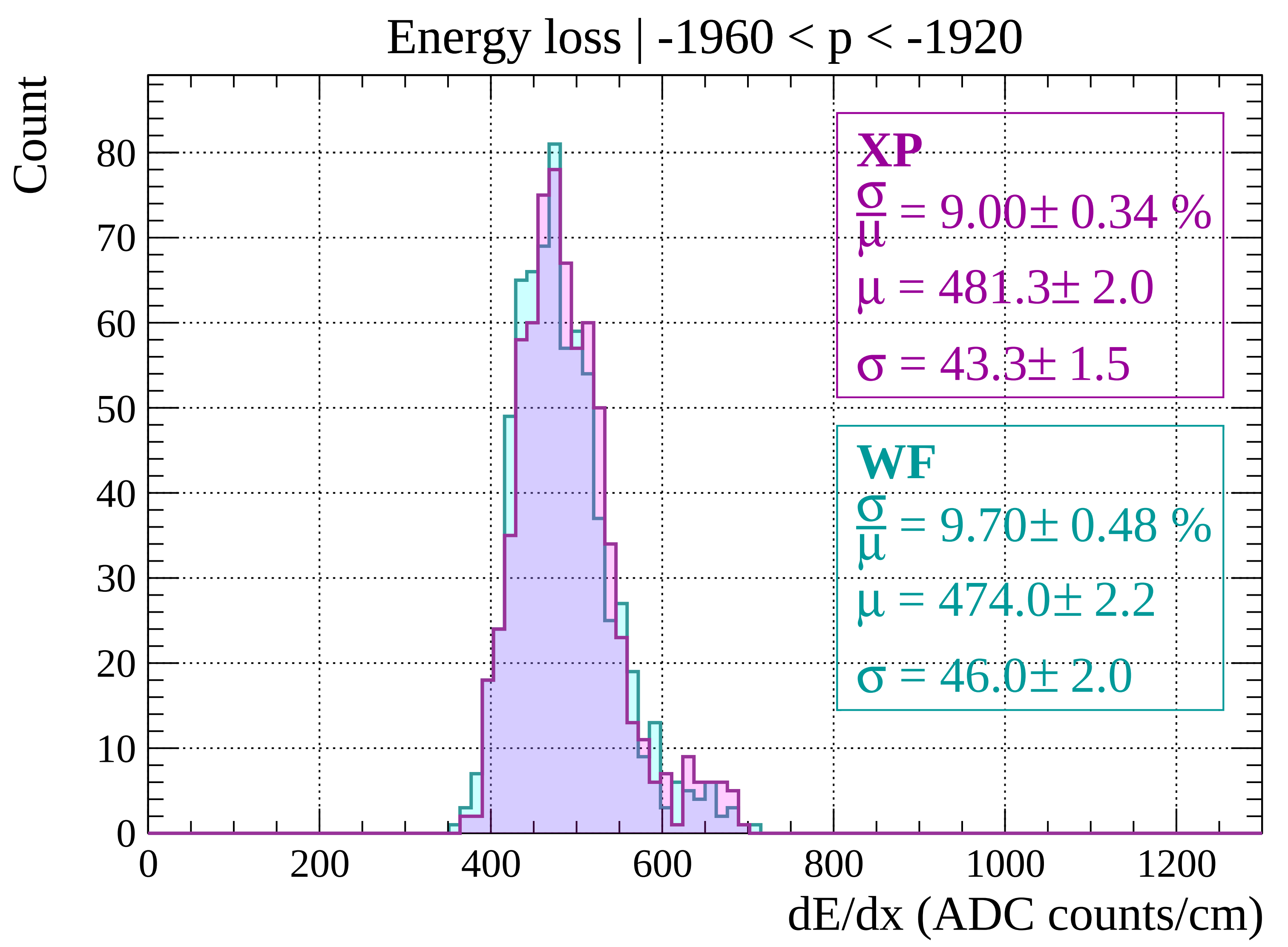
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
220
200
180
160
140
120
100
80
60
40
20
0
×10^3



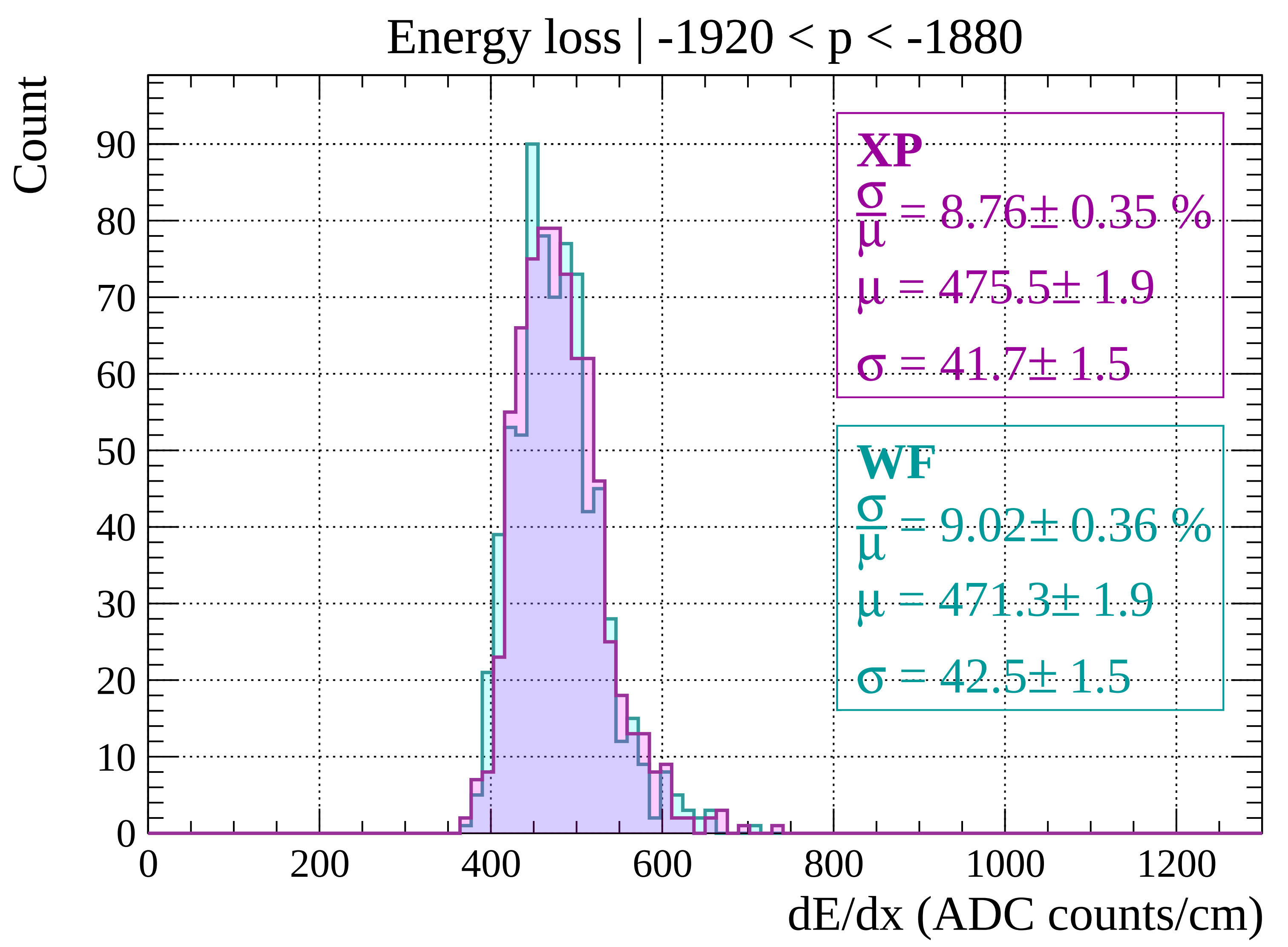
χ^2/NDF
R
$\times 10^3$



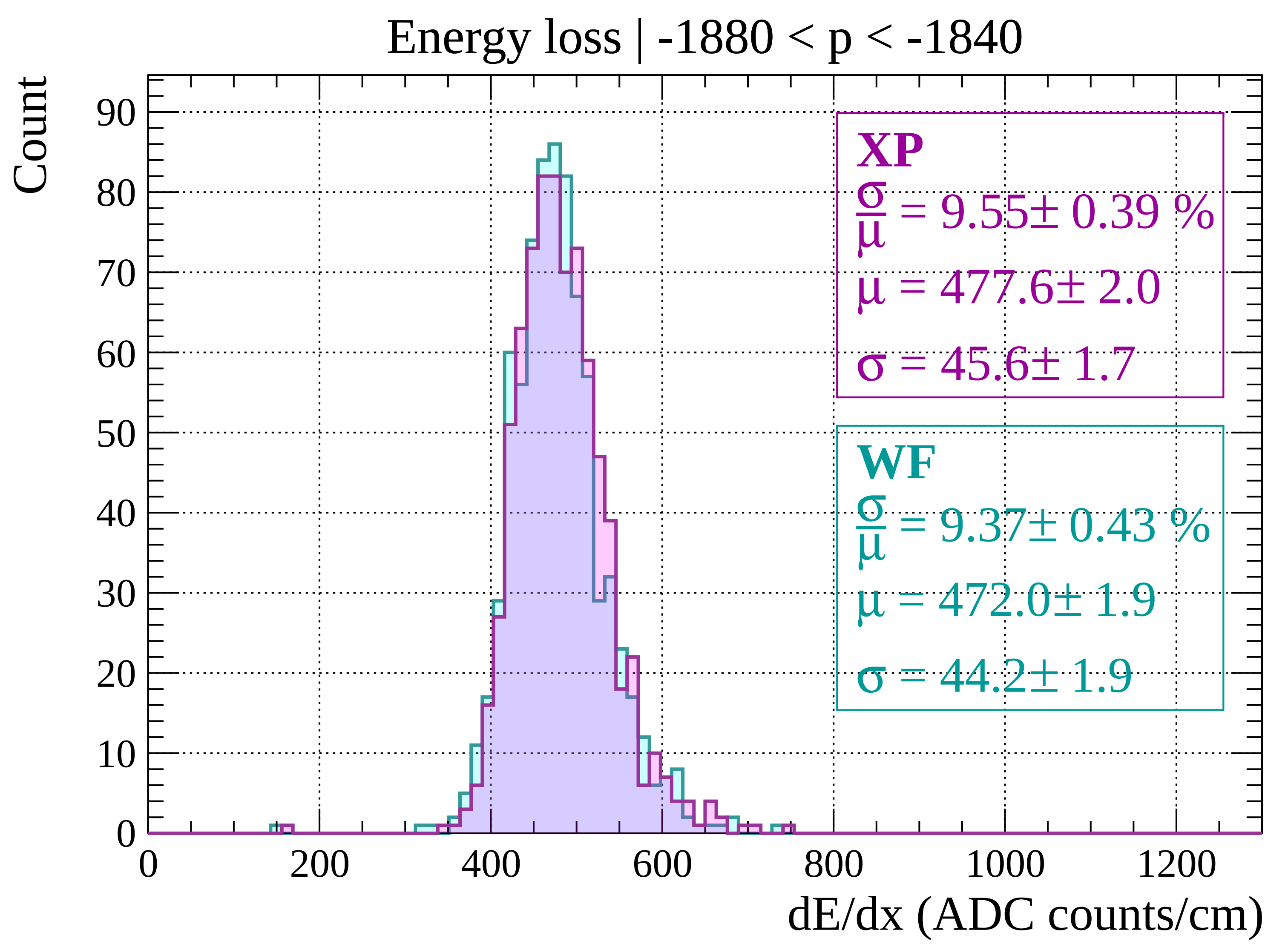
Energy loss | -2000 < p < -1960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP
σ/μ = 9.32± 0.64 %
μ = 480.9± 2.8
σ = 44.8± 2.8
WF
σ/μ = 9.54± 0.53 %
μ = 478.8± 2.8
σ = 45.7± 2.3

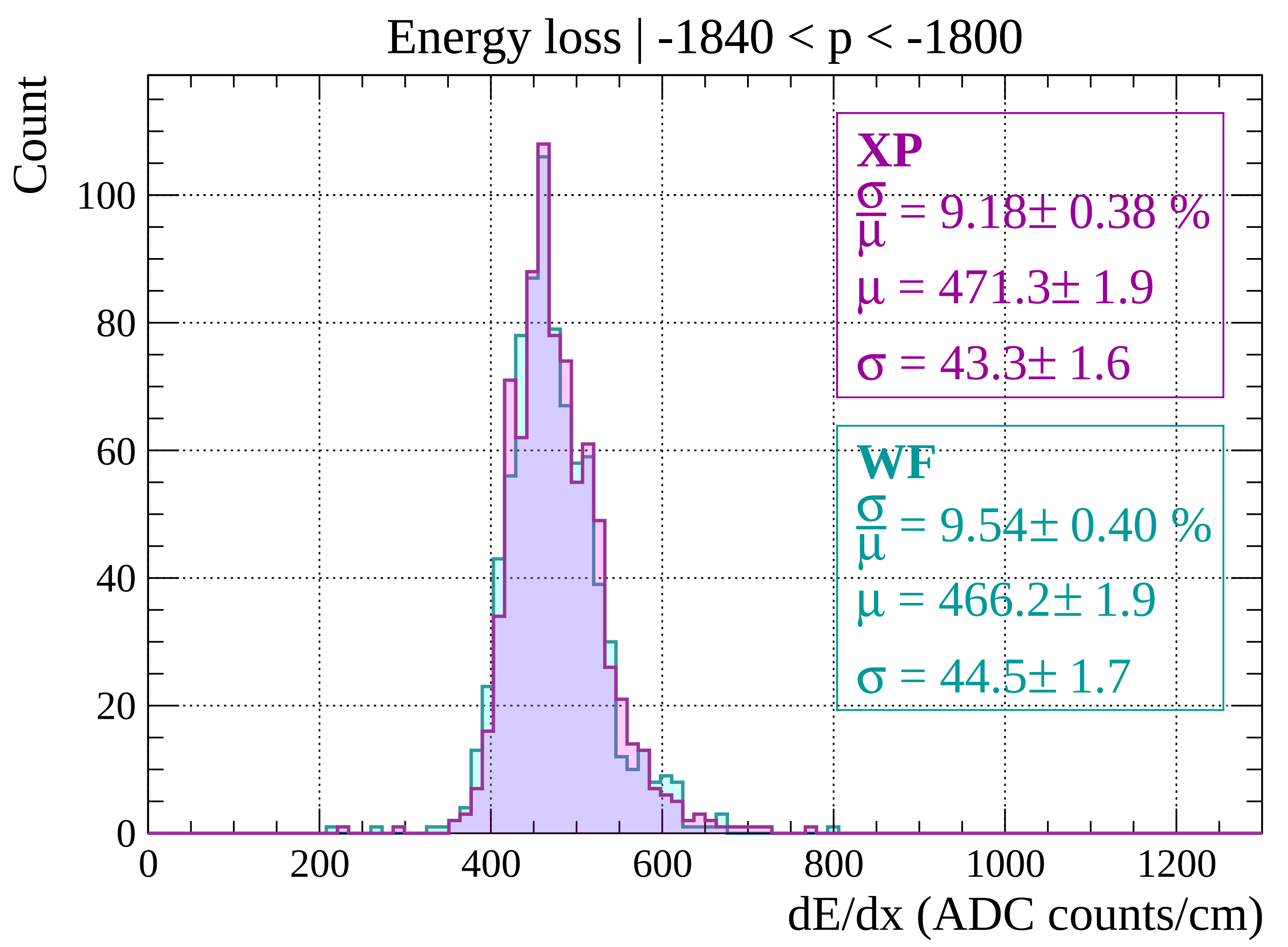


Energy loss | -1960 < p < -1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP
σ/μ = 9.00± 0.34 %
μ = 481.3± 2.0
σ = 43.3± 1.5
WF
σ/μ = 9.70± 0.48 %
μ = 474.0± 2.2
σ = 46.0± 2.0

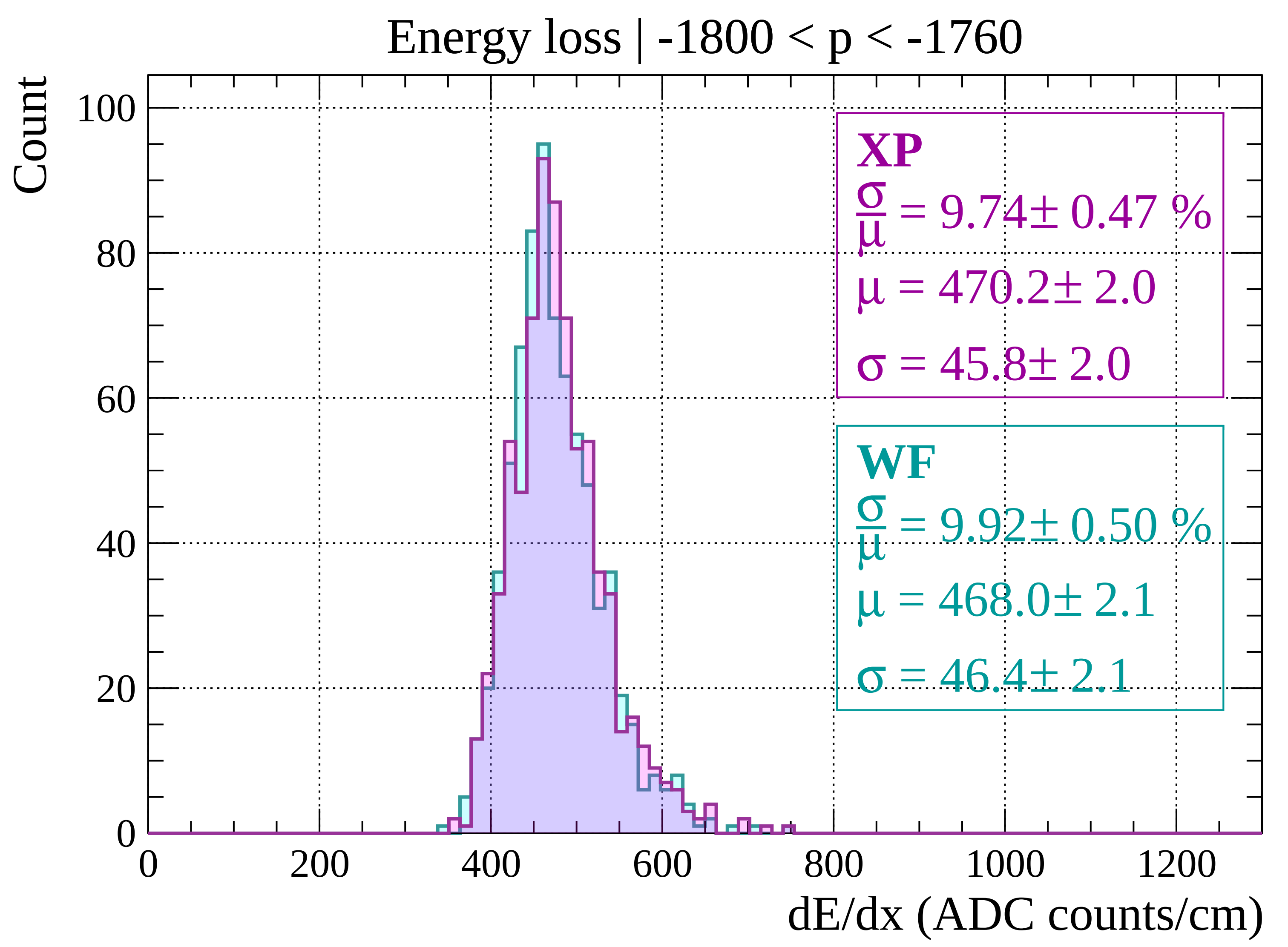


Energy loss | -1920 < p < -1880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
XP
$\frac{\sigma}{\mu}$ = 8.76 ± 0.35 %
μ = 475.5 ± 1.9
σ = 41.7 ± 1.5
WF
$\frac{\sigma}{\mu}$ = 9.02 ± 0.36 %
μ = 471.3 ± 1.9
σ = 42.5 ± 1.5

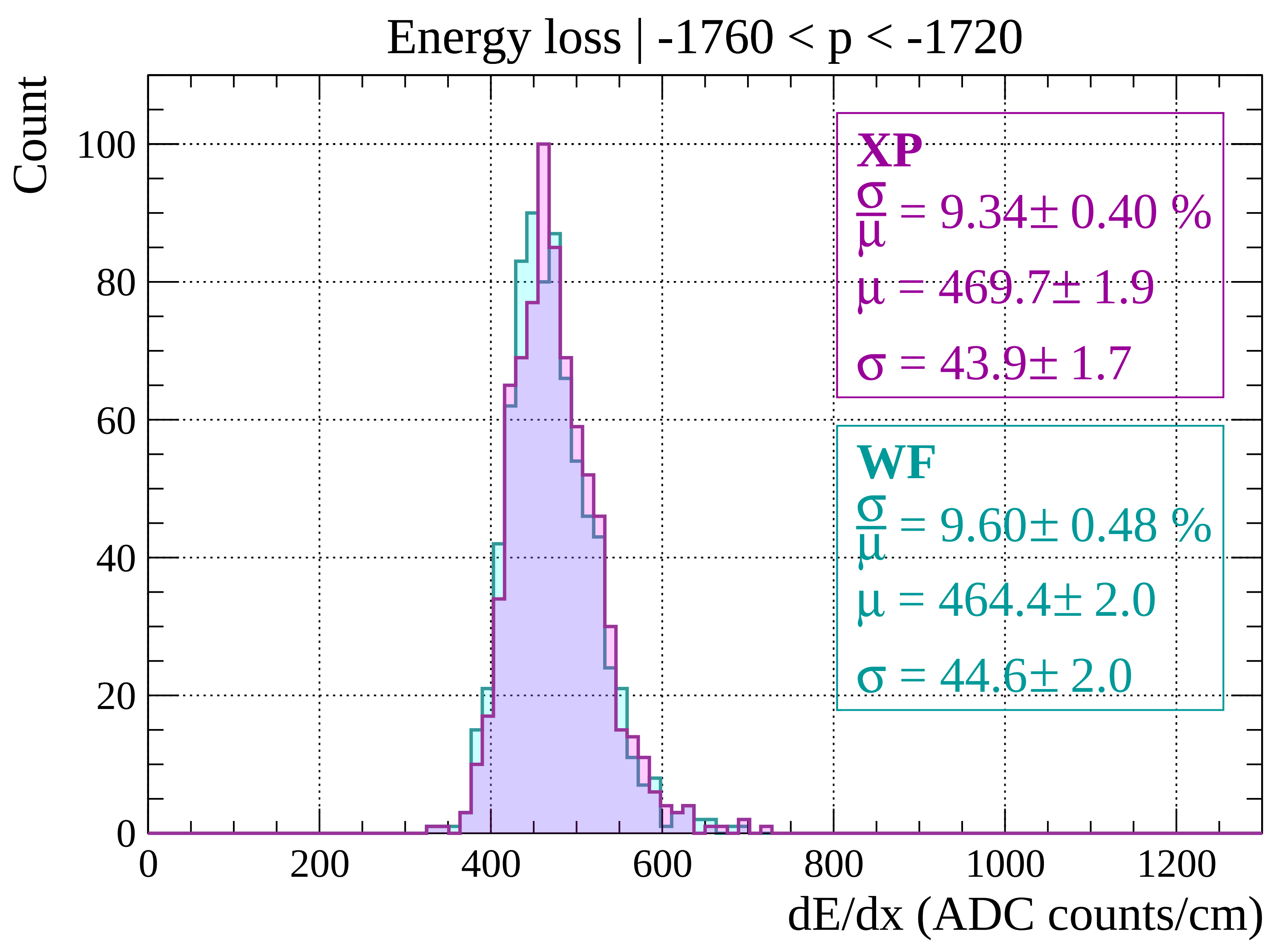

Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
200
400
600
800
1000
1200
XP
σ/μ = 9.55± 0.39 %
μ = 477.6± 2.0
σ = 45.6± 1.7
WF
σ/μ = 9.37± 0.43 %
μ = 472.0± 1.9
σ = 44.2± 1.9



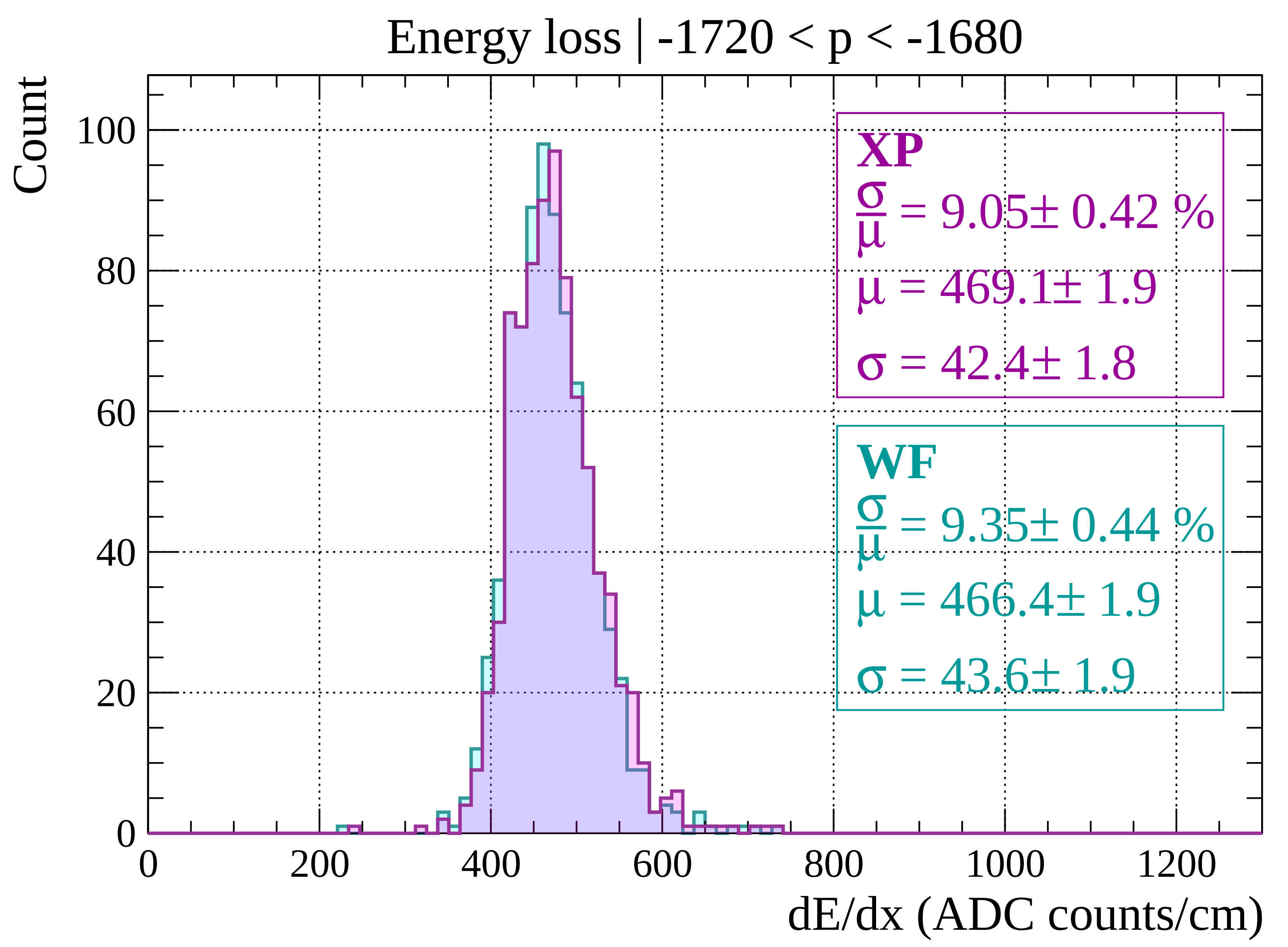
Energy loss | -1840 < p < -1800
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 9.18± 0.38 %
μ = 471.3± 1.9
σ = 43.3± 1.6
WF
σ/μ = 9.54± 0.40 %
μ = 466.2± 1.9
σ = 44.5± 1.7

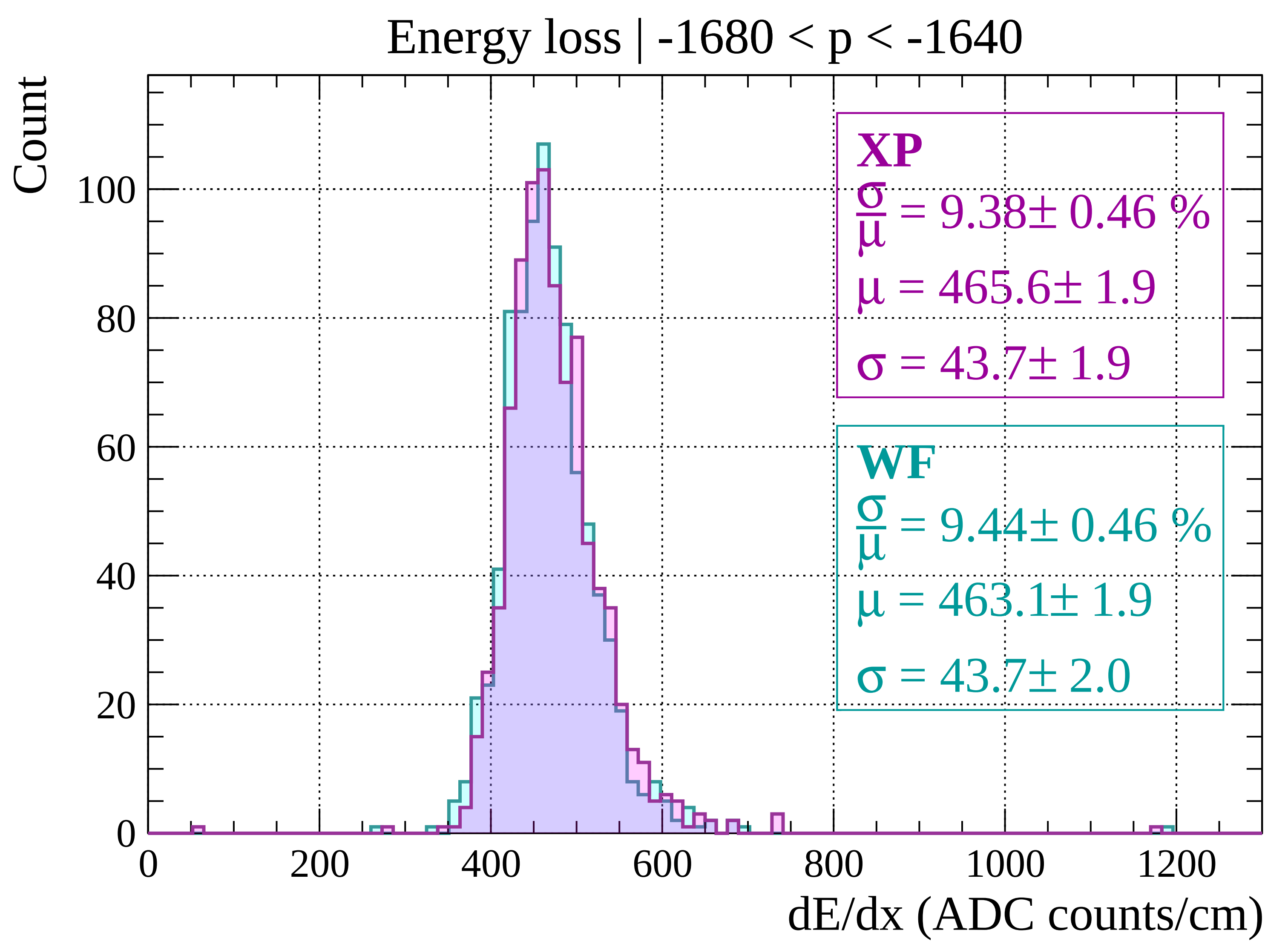


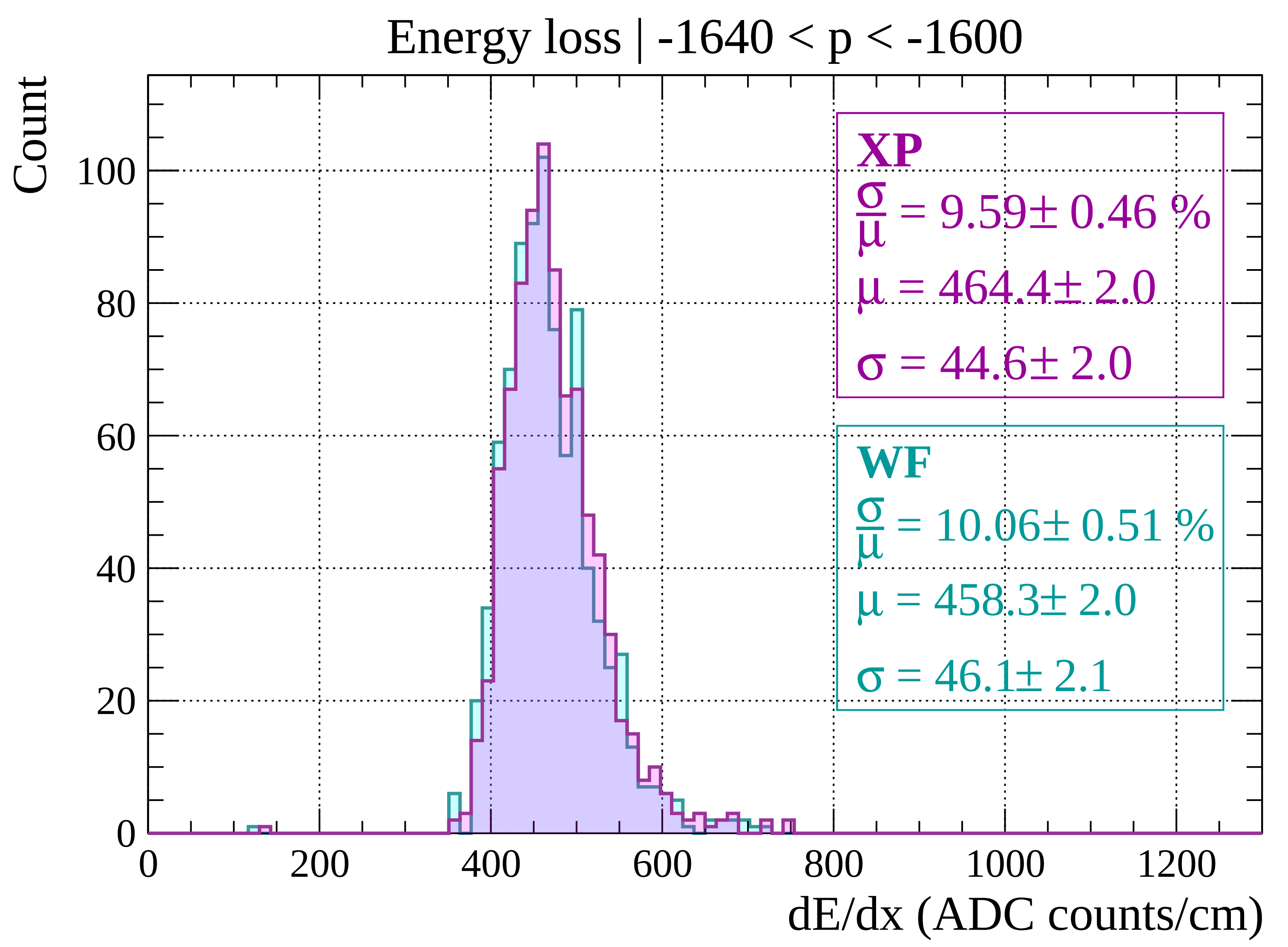
Energy loss | -1800 < p < -1760
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 9.74± 0.47 %
μ = 470.2± 2.0
σ = 45.8± 2.0
WF
σ/μ = 9.92± 0.50 %
μ = 468.0± 2.1
σ = 46.4± 2.1



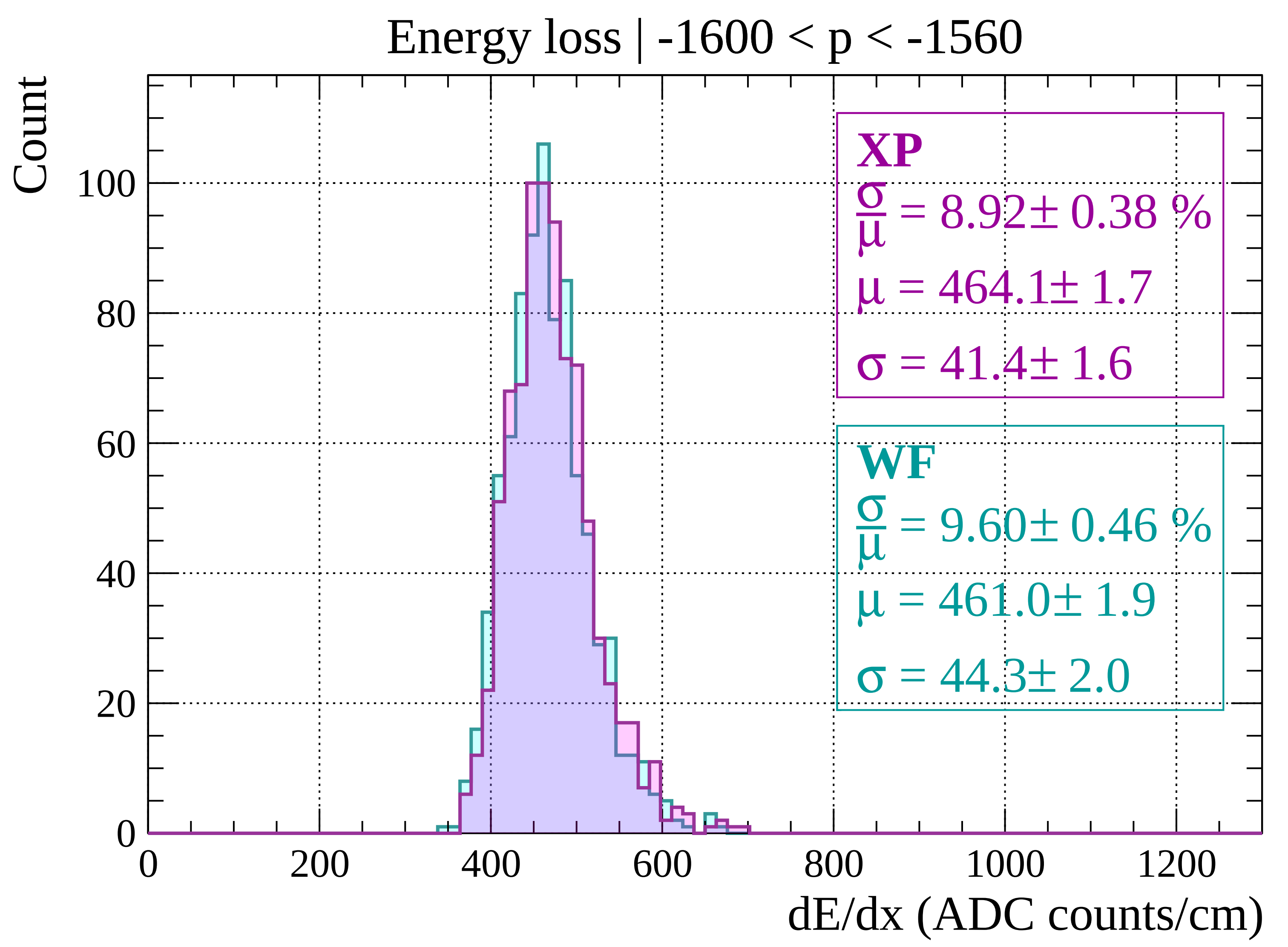
Energy loss | -1760 < p < -1720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
σ/μ = 9.34± 0.40 %
μ = 469.7± 1.9
σ = 43.9± 1.7
WF
σ/μ = 9.60± 0.48 %
μ = 464.4± 2.0
σ = 44.6± 2.0

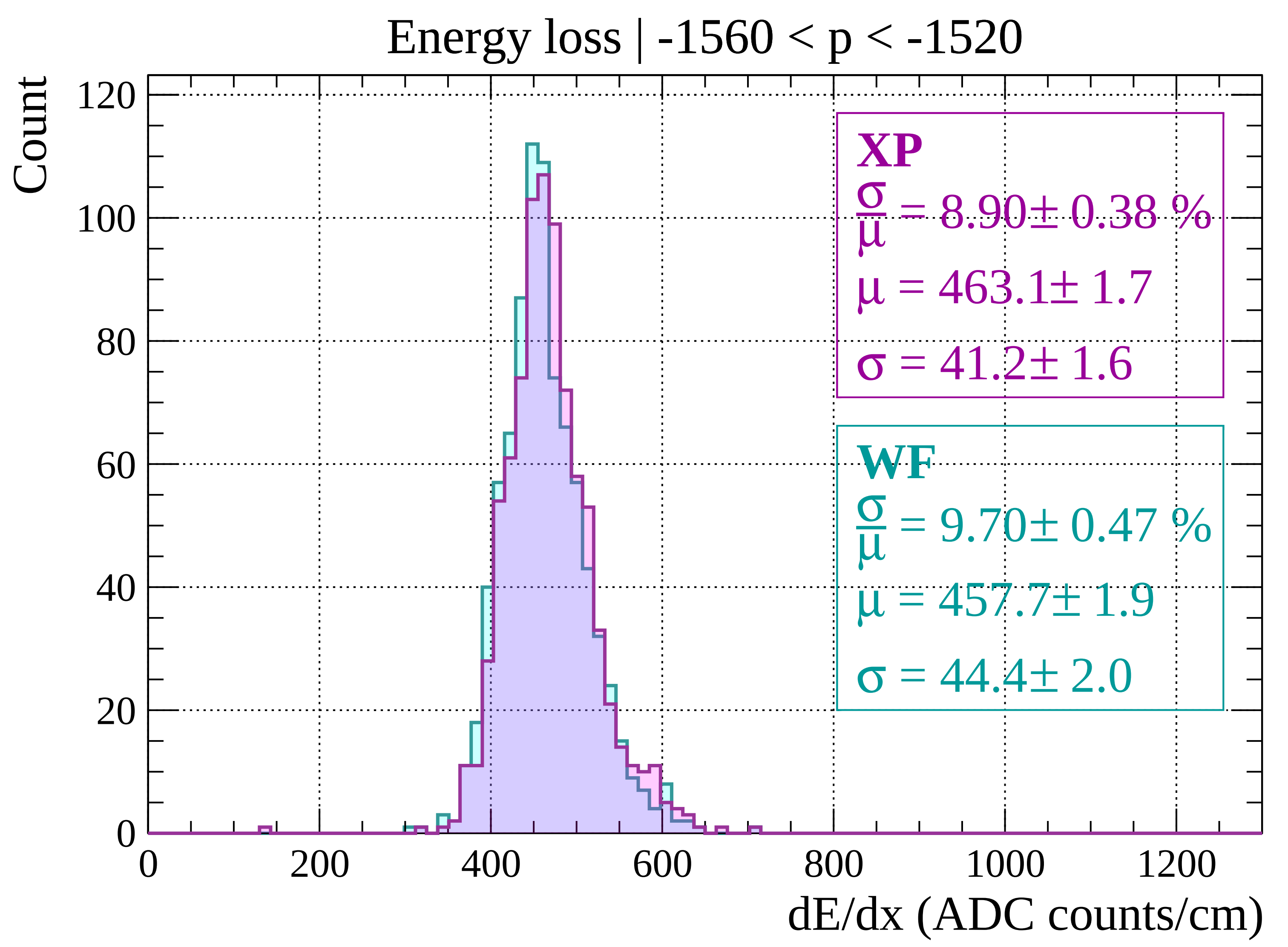

Energy loss | -1720 < p < -1680
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.05± 0.42 %
μ = 469.1± 1.9
σ = 42.4± 1.8
WF
σ/μ = 9.35± 0.44 %
μ = 466.4± 1.9
σ = 43.6± 1.9


Energy loss | -1680 < p < -1640
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.38± 0.46 %
μ = 465.6± 1.9
σ = 43.7± 1.9
WF
σ/μ = 9.44± 0.46 %
μ = 463.1± 1.9
σ = 43.7± 2.0
0
20
40
60
80
100
0
200
400
600
800
1000
1200

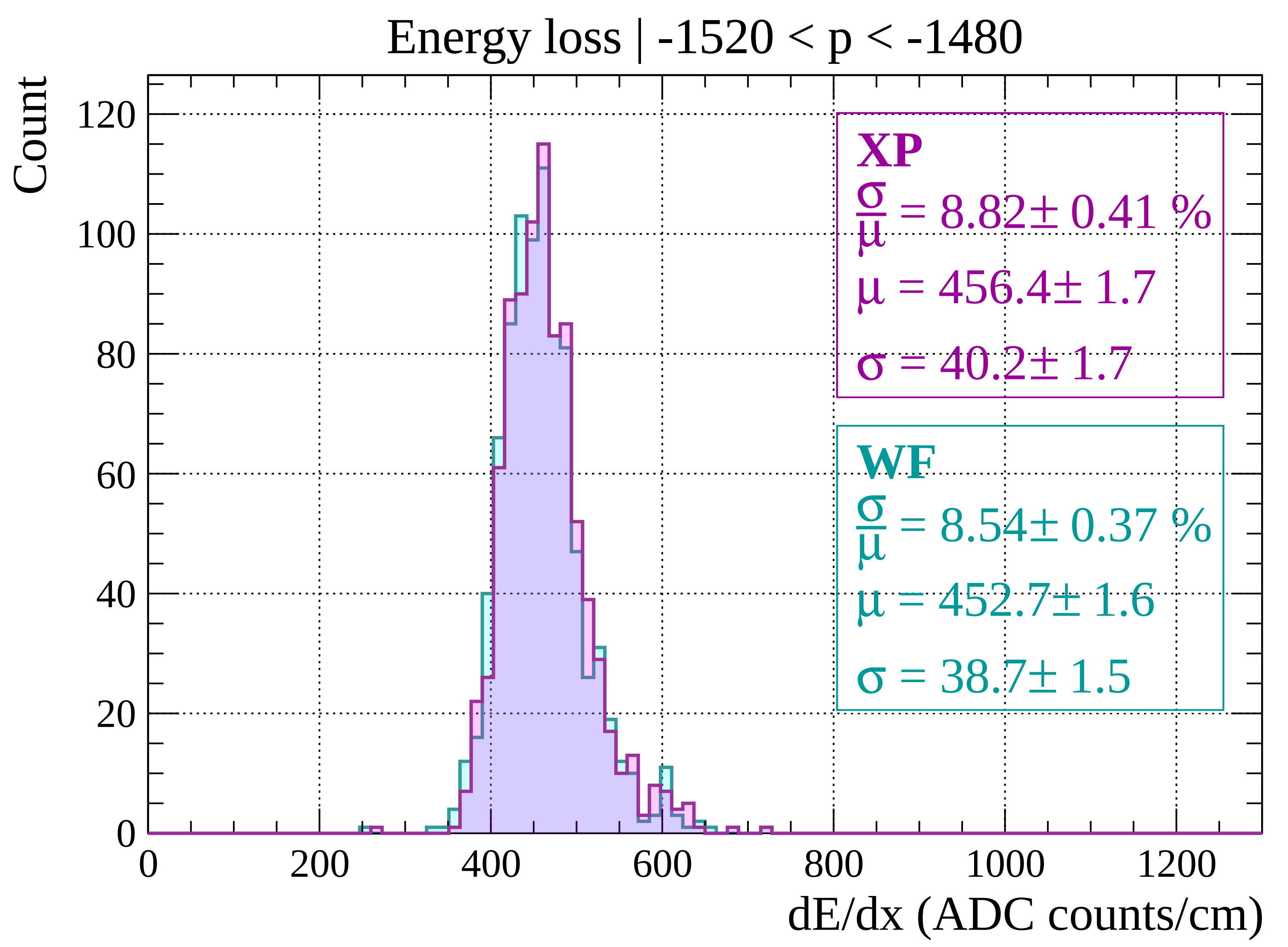


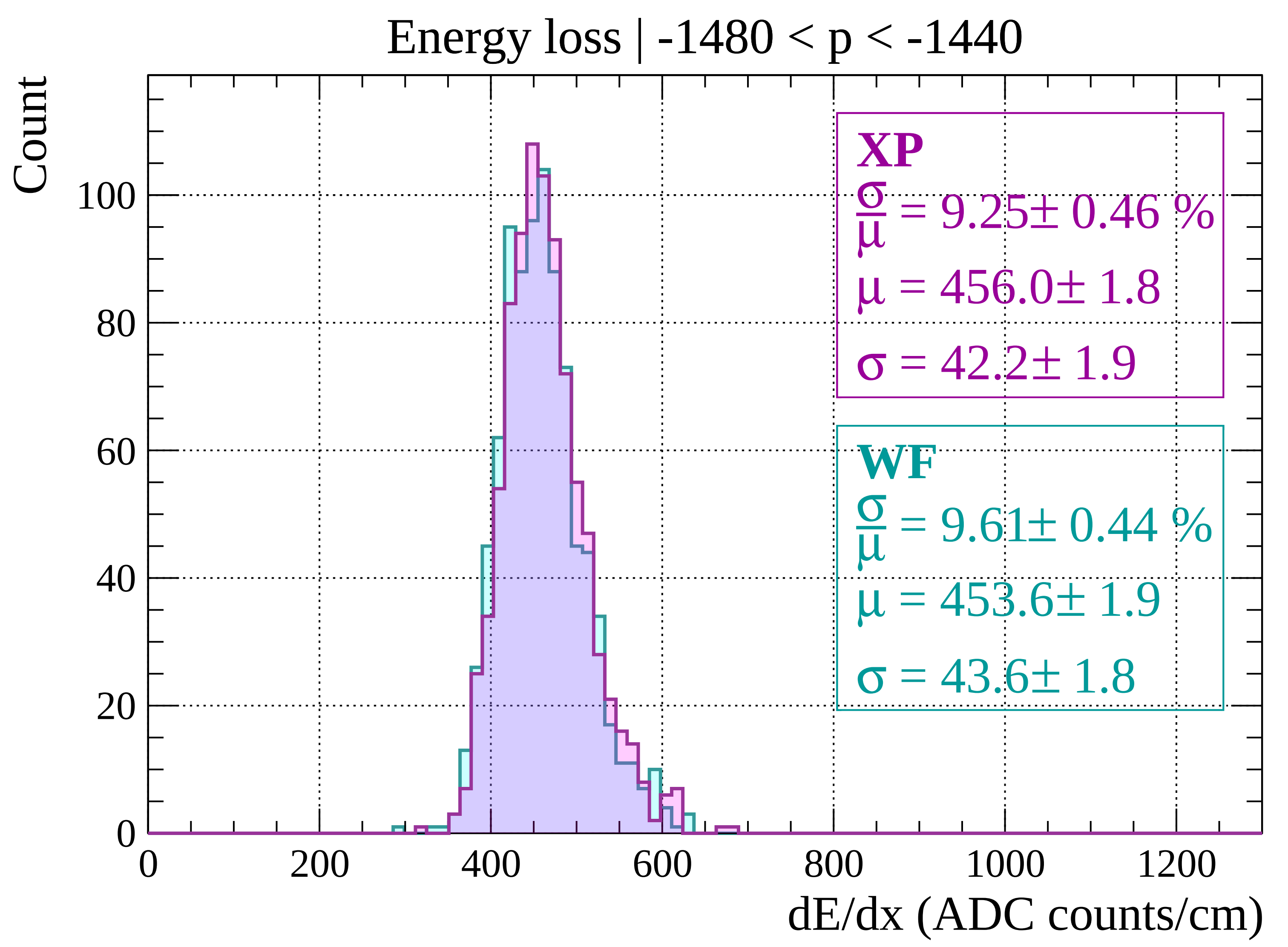
Energy loss | -1640 < p < -1600
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.59± 0.46 %
μ = 464.4± 2.0
σ = 44.6± 2.0
WF
σ/μ = 10.06± 0.51 %
μ = 458.3± 2.0
σ = 46.1± 2.1
0
20
40
60
80
100
0
200
400
600
800
1000
1200


Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
σ/μ = 8.92± 0.38 %
μ = 464.1± 1.7
σ = 41.4± 1.6
WF
σ/μ = 9.60± 0.46 %
μ = 461.0± 1.9
σ = 44.3± 2.0

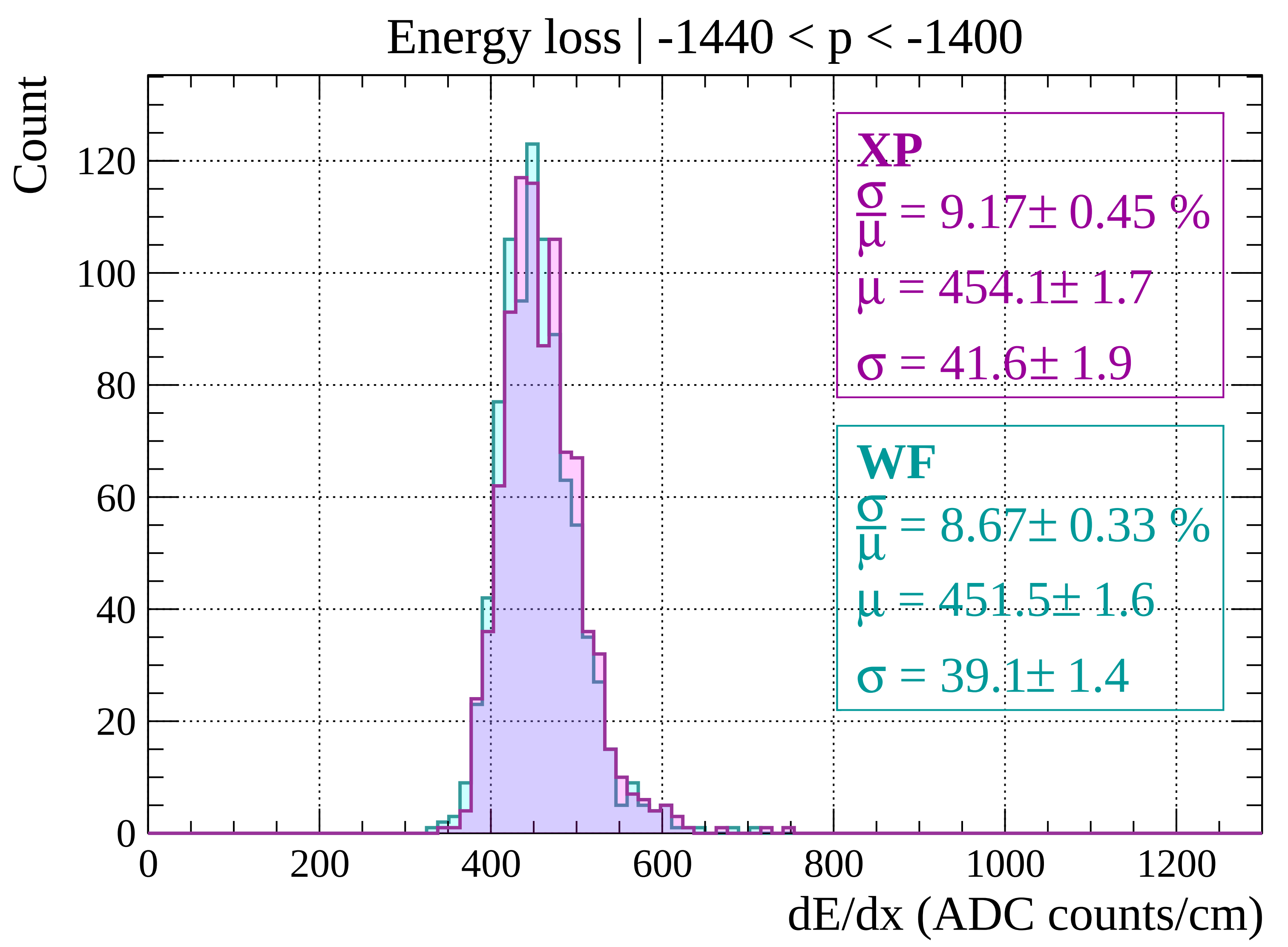


Energy loss | -1560 < p < -1520
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.90± 0.38 %
μ = 463.1± 1.7
σ = 41.2± 1.6
WF
$\frac{\sigma}{\mu}$ = 9.70± 0.47 %
μ = 457.7± 1.9
σ = 44.4± 2.0

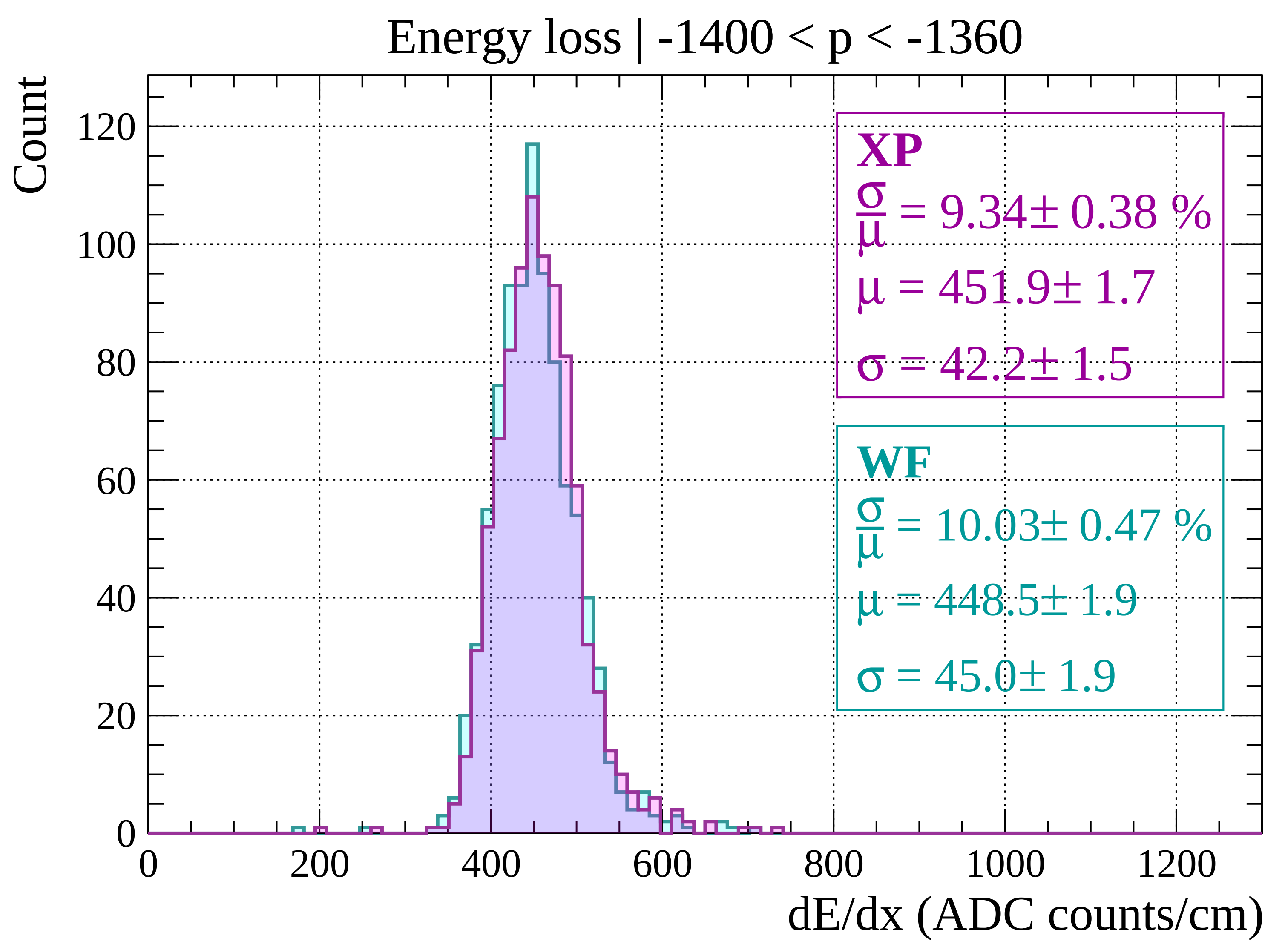

Energy loss | -1520 < p < -1480
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.82± 0.41 %
μ = 456.4± 1.7
σ = 40.2± 1.7
WF
σ/μ = 8.54± 0.37 %
μ = 452.7± 1.6
σ = 38.7± 1.5



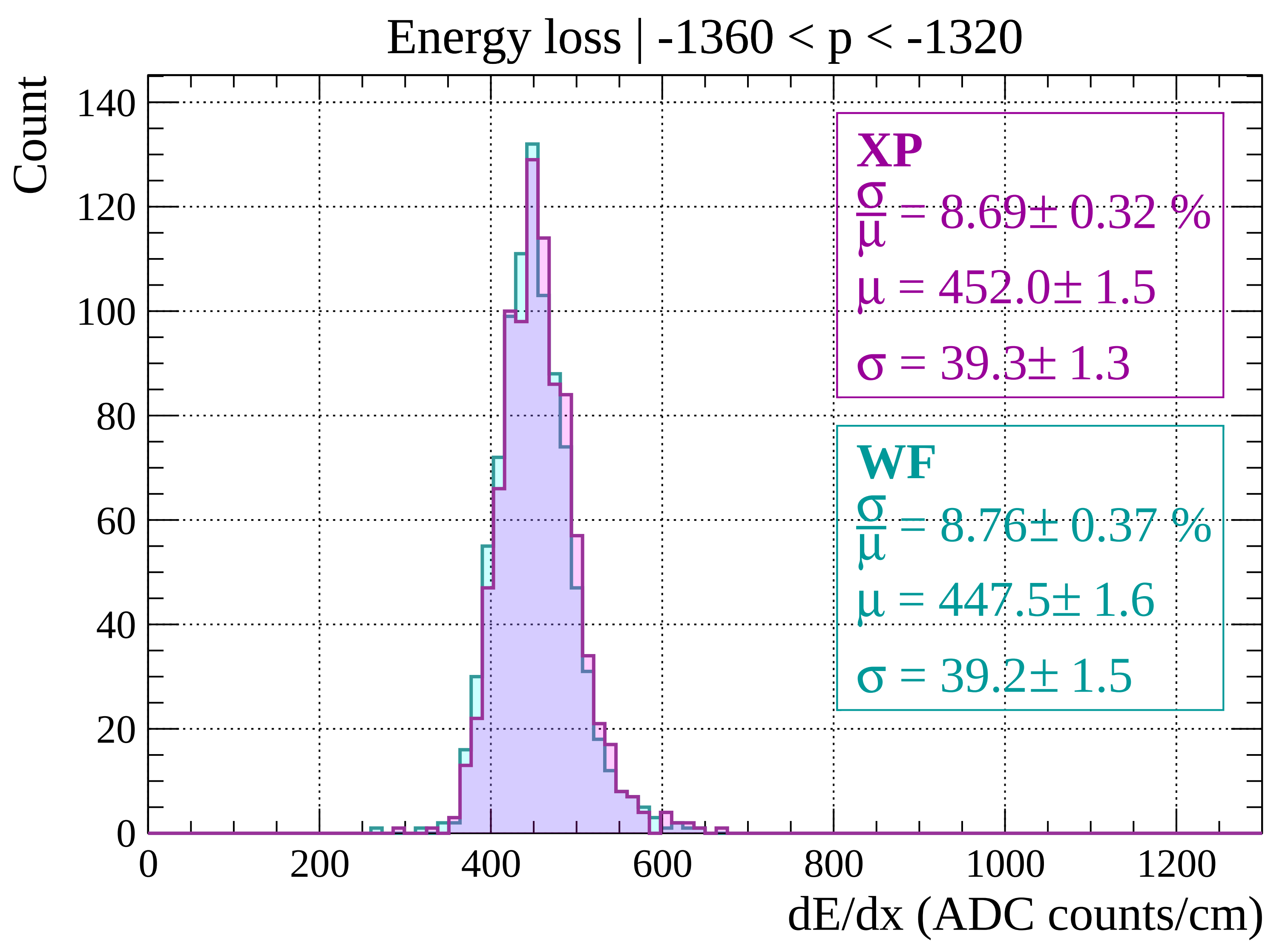
Energy loss | -1480 < p < -1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP
$\frac{\sigma}{\mu}$ = 9.25± 0.46 %
μ = 456.0± 1.8
σ = 42.2± 1.9
WF
$\frac{\sigma}{\mu}$ = 9.61± 0.44 %
μ = 453.6± 1.9
σ = 43.6± 1.8

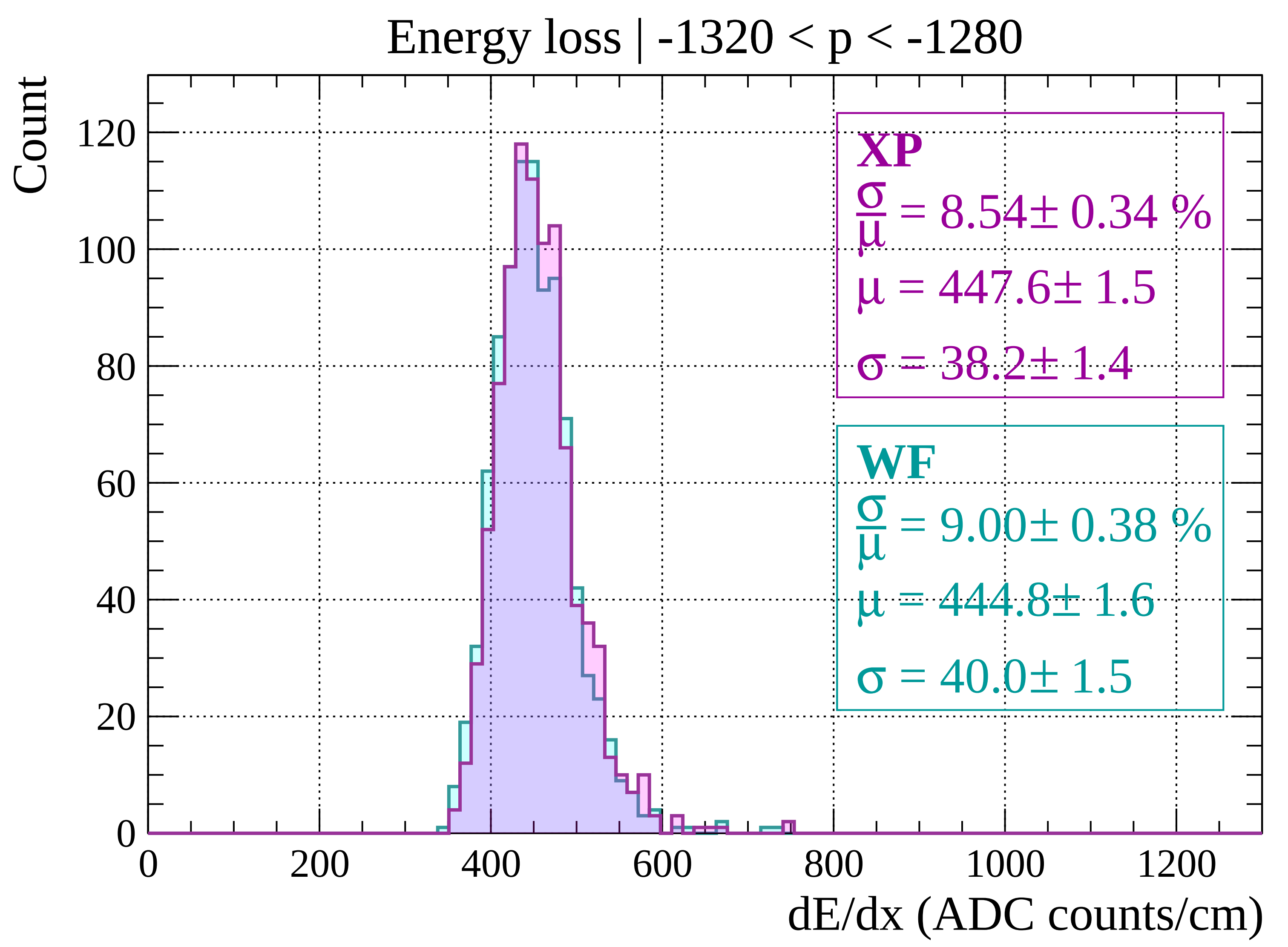


Energy loss | -1440 < p < -1400
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 9.17± 0.45 %
μ = 454.1± 1.7
σ = 41.6± 1.9
WF
σ/μ = 8.67± 0.33 %
μ = 451.5± 1.6
σ = 39.1± 1.4

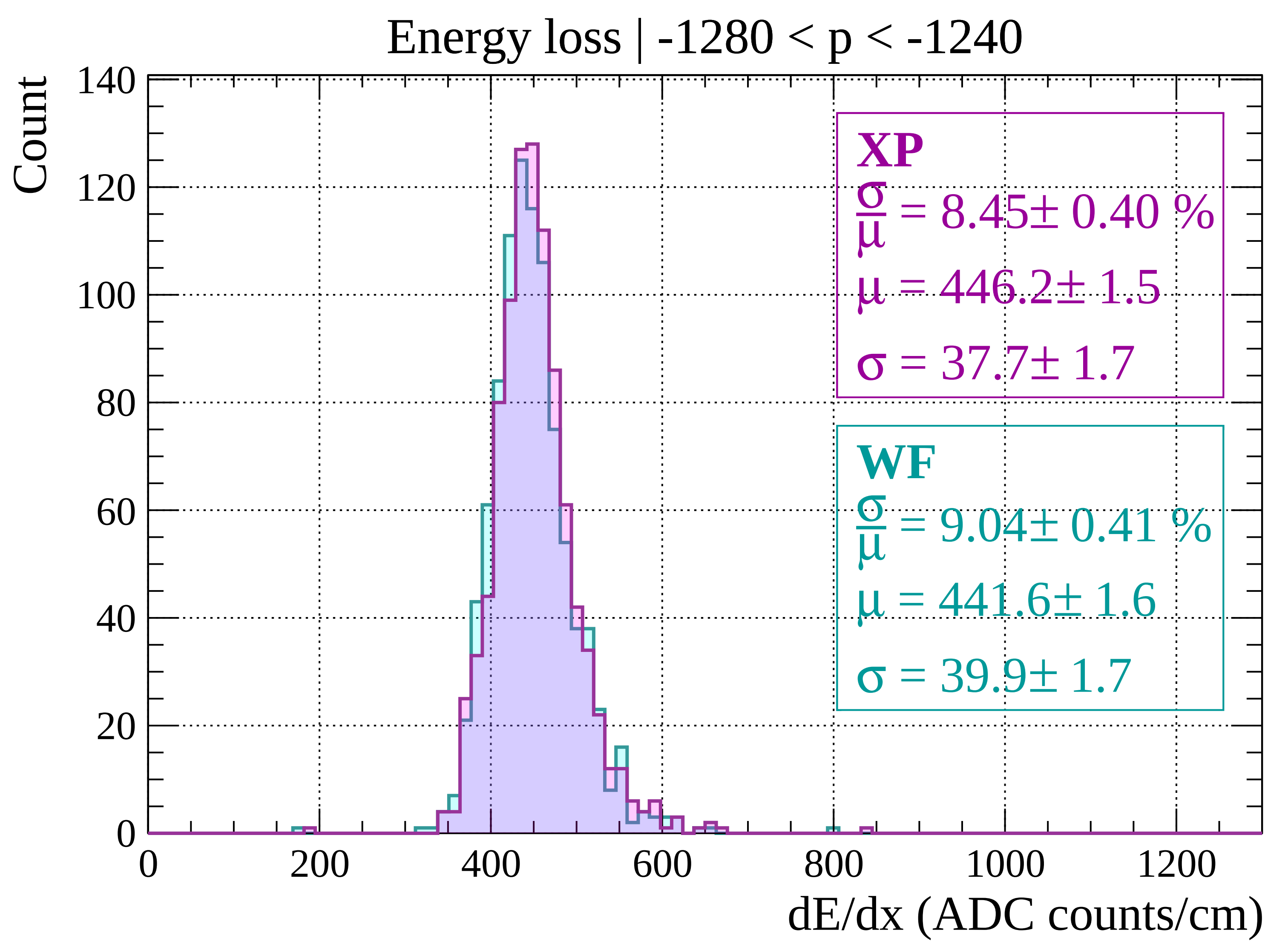


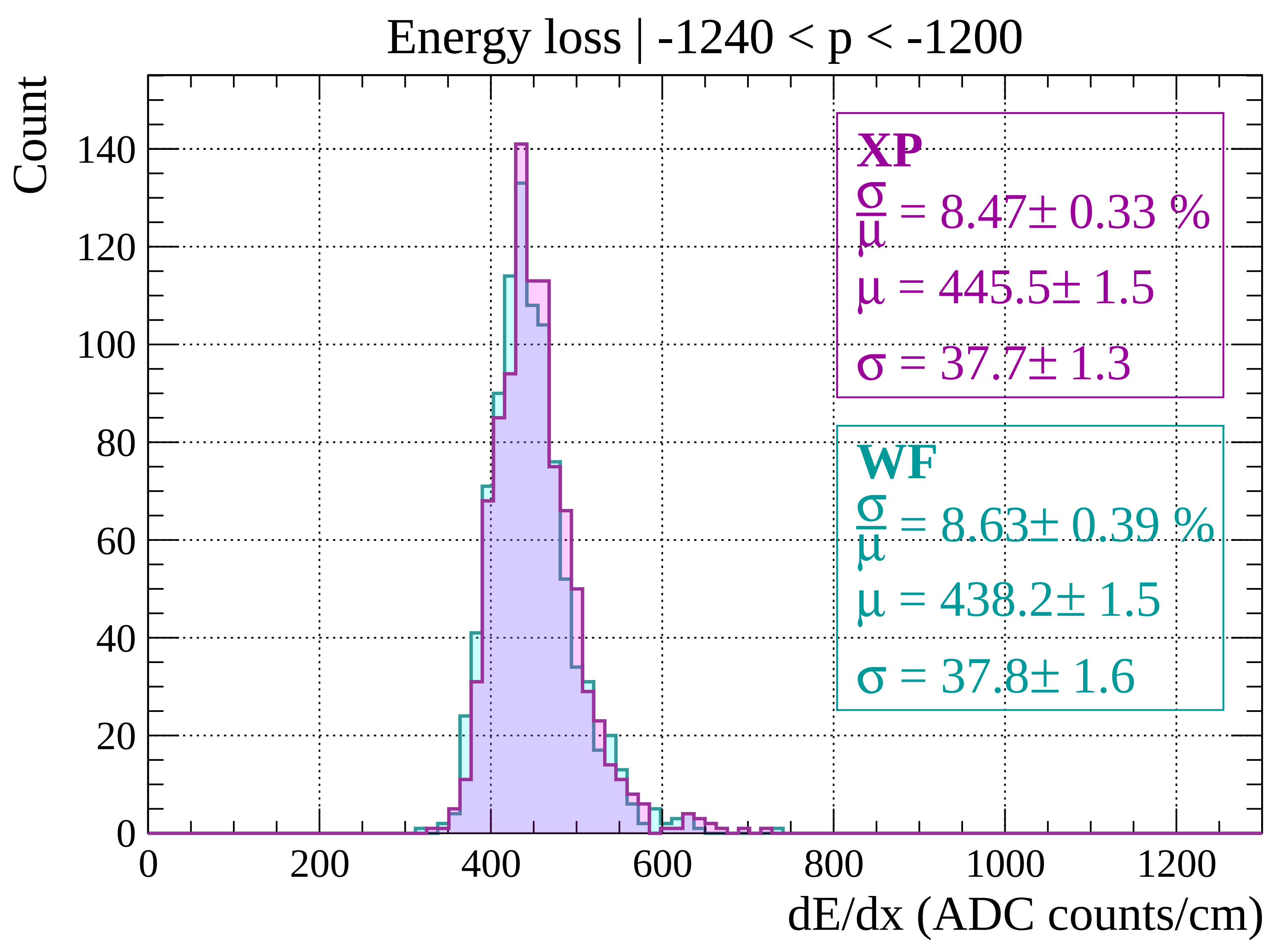
Energy loss | -1400 < p < -1360
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 9.34± 0.38 %
μ = 451.9± 1.7
σ = 42.2± 1.5
WF
σ/μ = 10.03± 0.47 %
μ = 448.5± 1.9
σ = 45.0± 1.9


Energy loss | -1360 < p < -1320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
σ/μ = 8.69± 0.32 %
μ = 452.0± 1.5
σ = 39.3± 1.3
WF
σ/μ = 8.76± 0.37 %
μ = 447.5± 1.6
σ = 39.2± 1.5

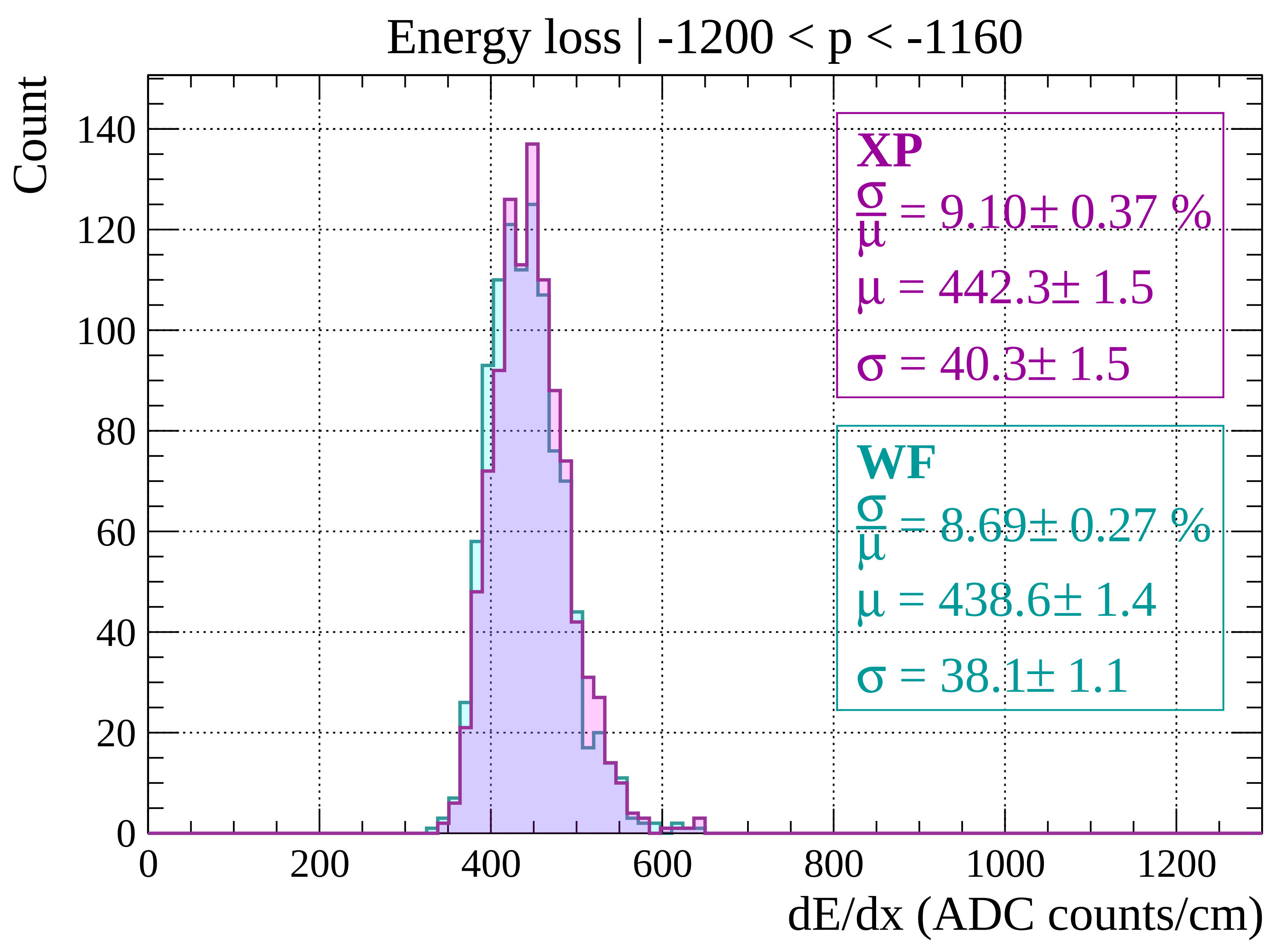


Energy loss | -1320 < p < -1280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
XP
σ/μ = 8.54± 0.34 %
μ = 447.6± 1.5
σ = 38.2± 1.4
WF
σ/μ = 9.00± 0.38 %
μ = 444.8± 1.6
σ = 40.0± 1.5

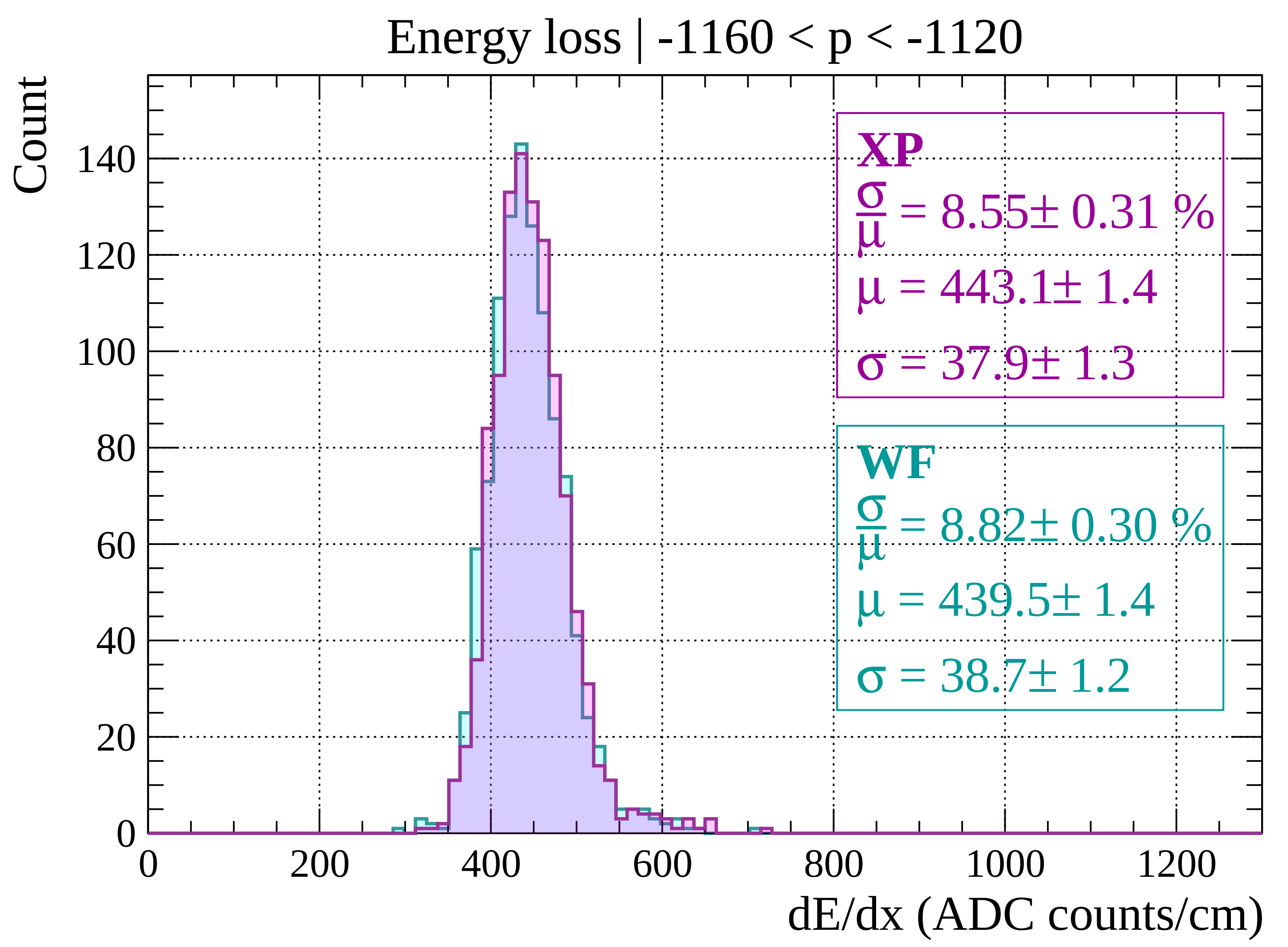

Energy loss | -1280 < p < -1240
Count
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.45± 0.40 %
μ = 446.2± 1.5
σ = 37.7± 1.7
WF
$\frac{\sigma}{\mu}$ = 9.04± 0.41 %
μ = 441.6± 1.6
σ = 39.9± 1.7



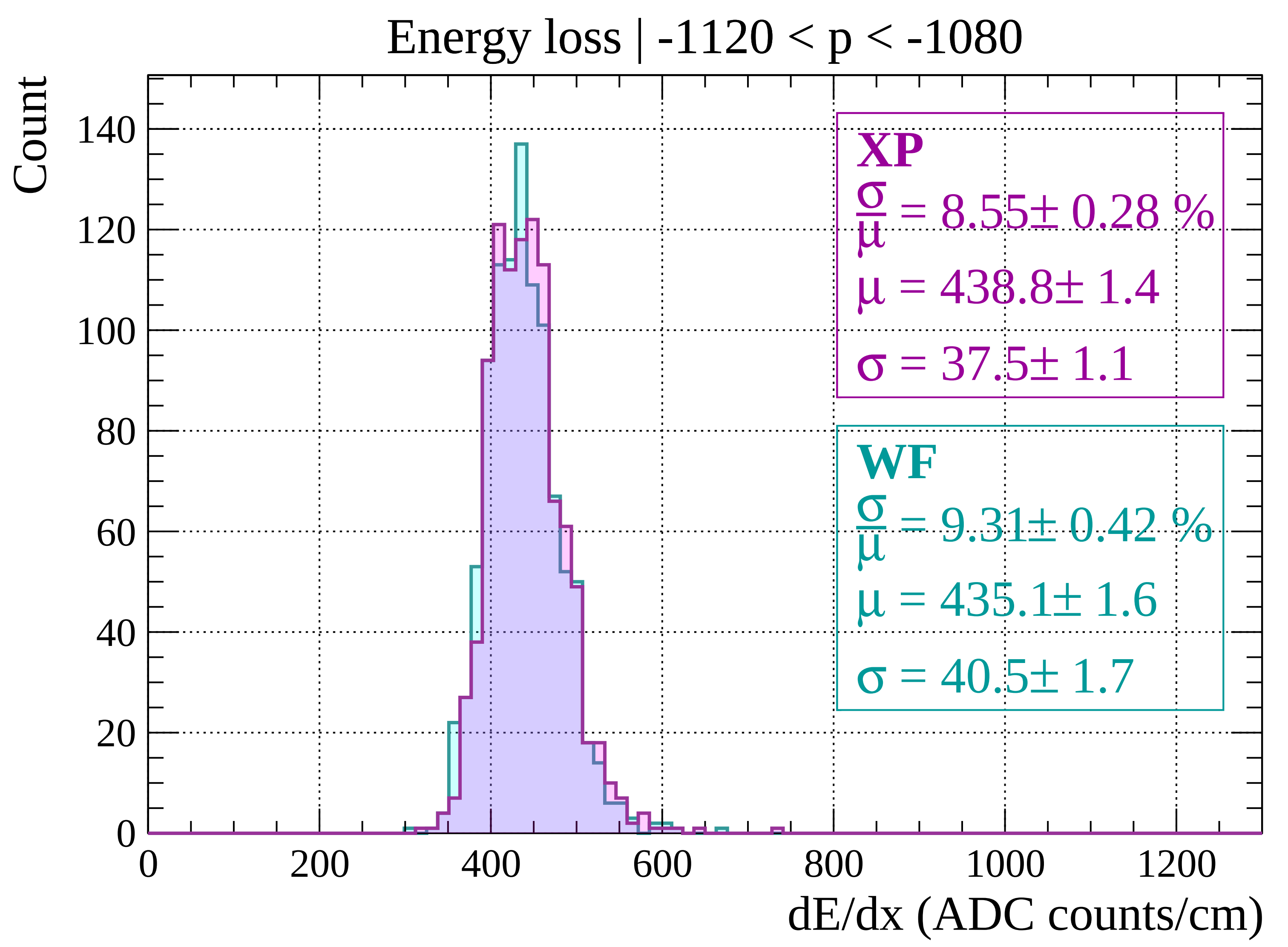
Energy loss | -1240 < p < -1200
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 8.47± 0.33 %
μ = 445.5± 1.5
σ = 37.7± 1.3
WF
σ/μ = 8.63± 0.39 %
μ = 438.2± 1.5
σ = 37.8± 1.6

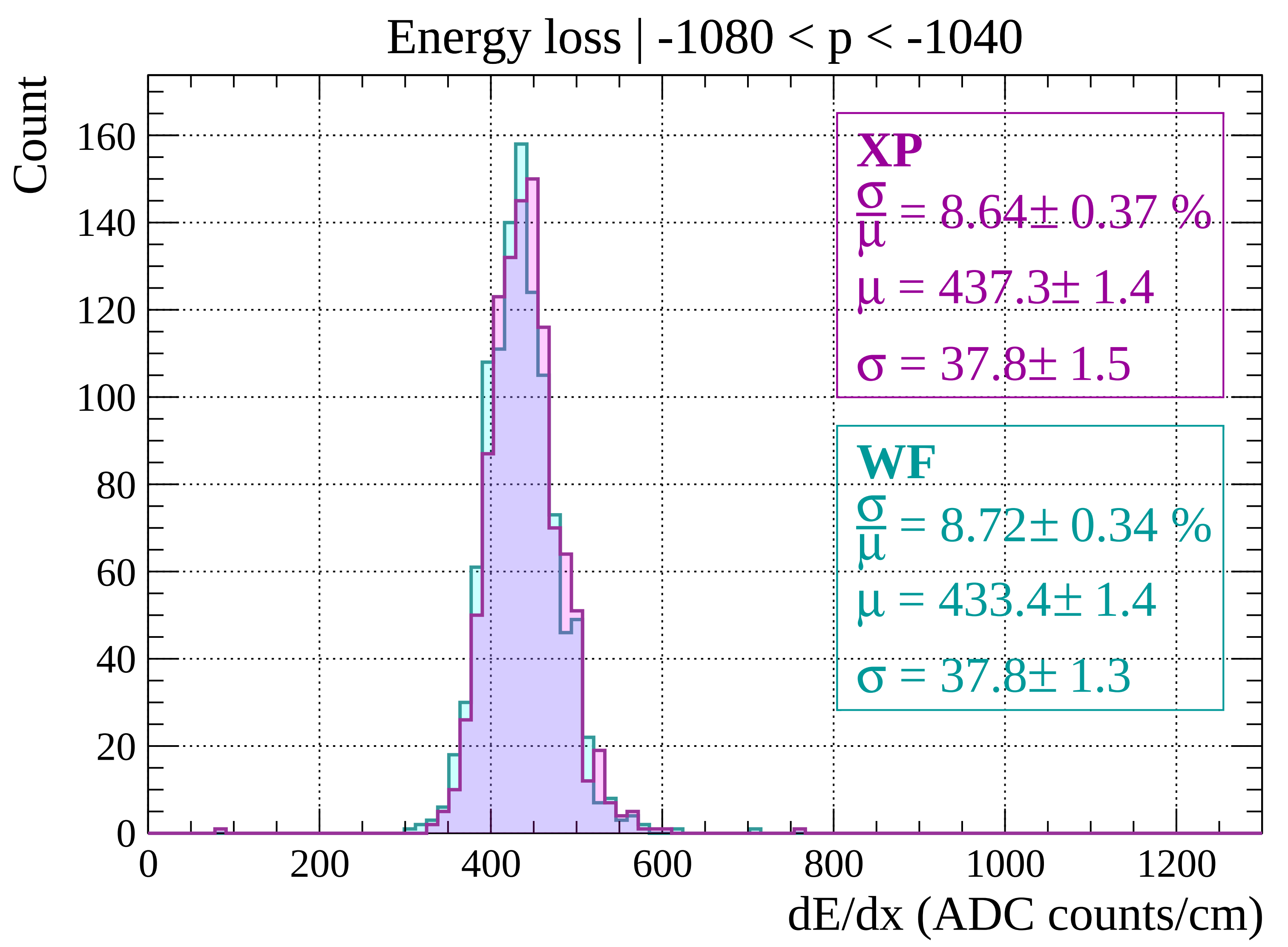


Energy loss | -1200 < p < -1160
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.10± 0.37 %
μ = 442.3± 1.5
σ = 40.3± 1.5
WF
σ/μ = 8.69± 0.27 %
μ = 438.6± 1.4
σ = 38.1± 1.1

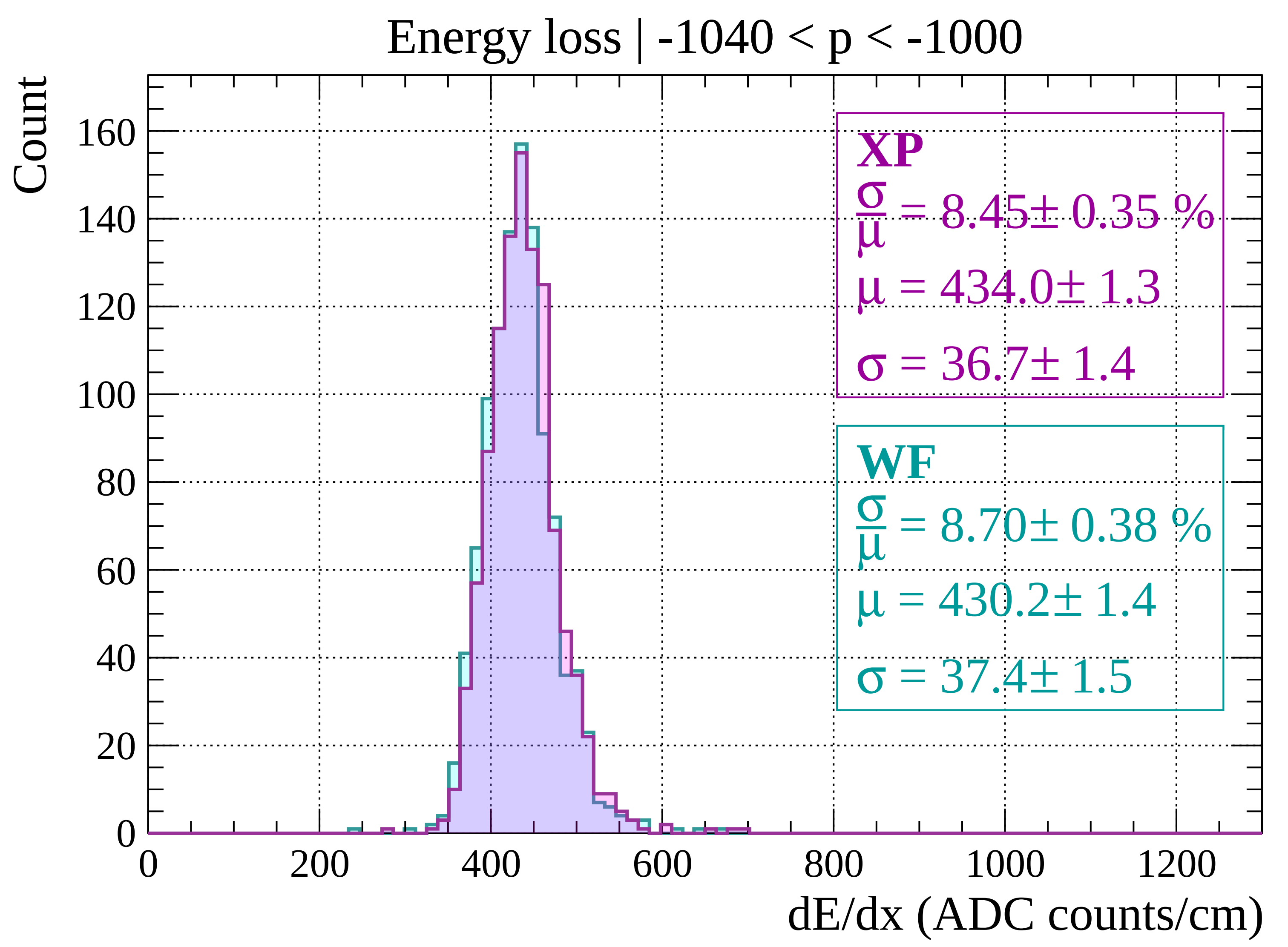


Energy loss | -1160 < p < -1120
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.55± 0.31 %
μ = 443.1± 1.4
σ = 37.9± 1.3
WF
σ/μ = 8.82± 0.30 %
μ = 439.5± 1.4
σ = 38.7± 1.2

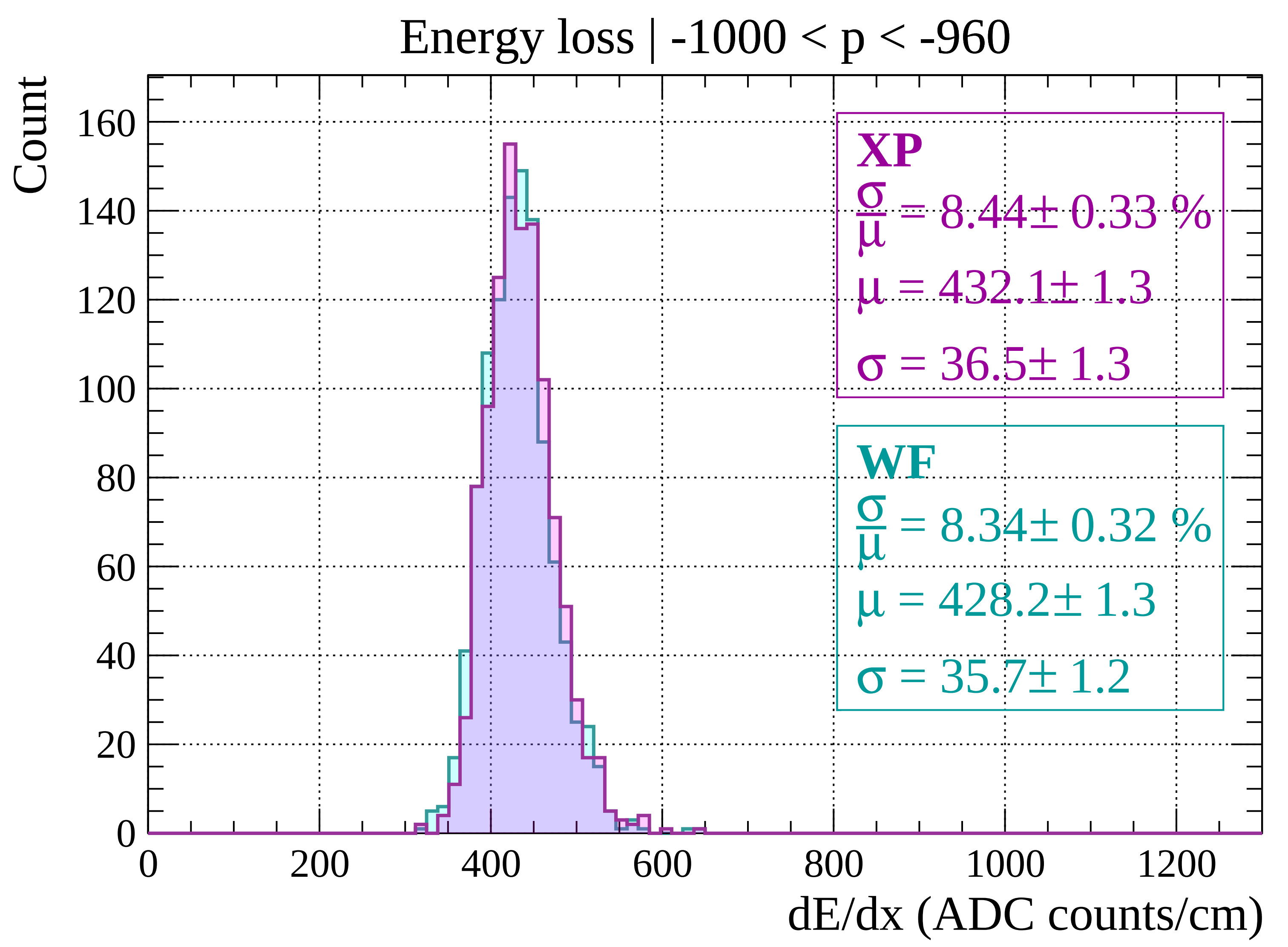

Energy loss | -1120 < p < -1080
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
σ/μ = 8.55± 0.28 %
μ = 438.8± 1.4
σ = 37.5± 1.1
WF
σ/μ = 9.31± 0.42 %
μ = 435.1± 1.6
σ = 40.5± 1.7



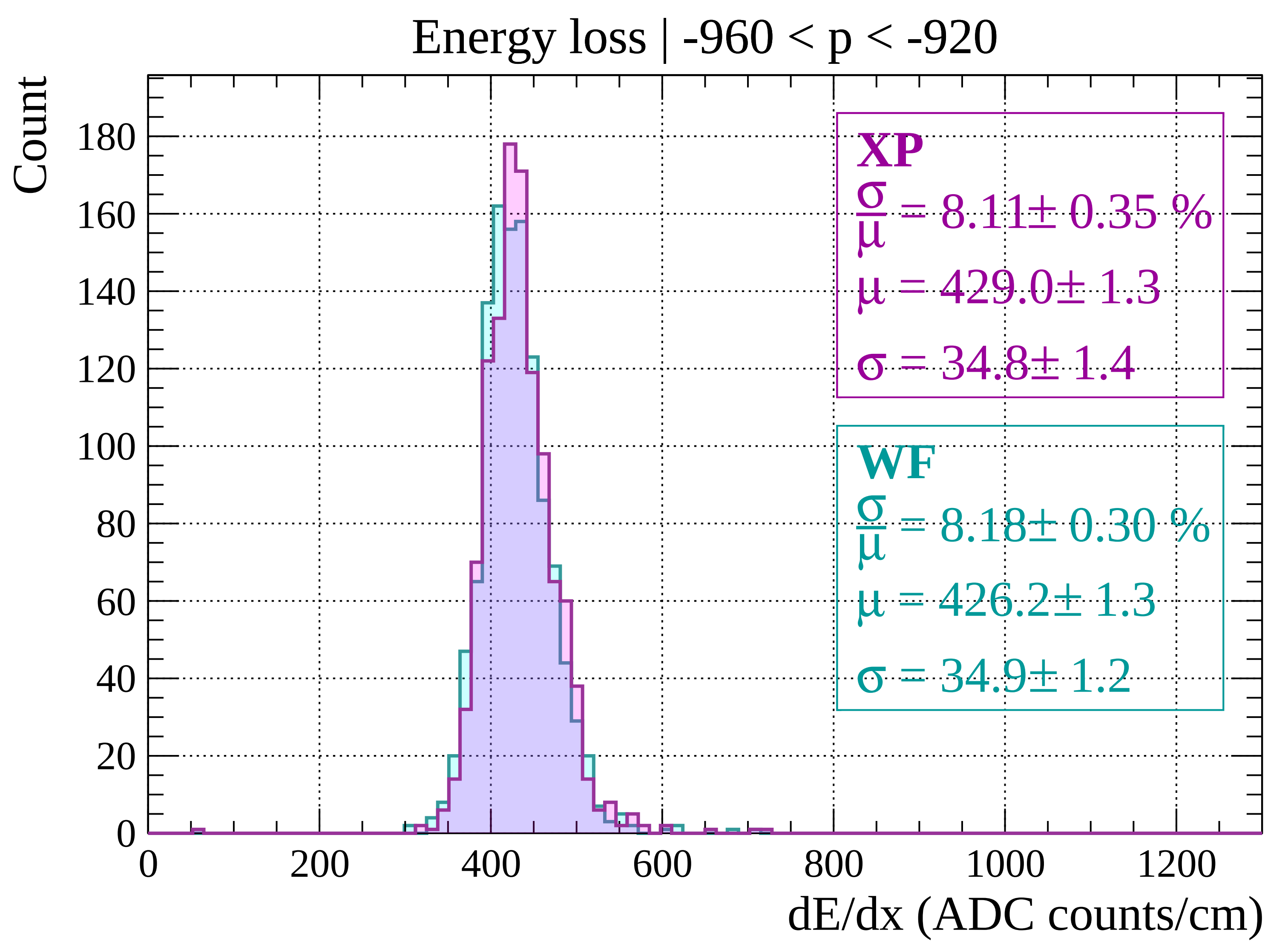
Energy loss | -1080 < p < -1040
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.64± 0.37 %
μ = 437.3± 1.4
σ = 37.8± 1.5
WF
σ/μ = 8.72± 0.34 %
μ = 433.4± 1.4
σ = 37.8± 1.3



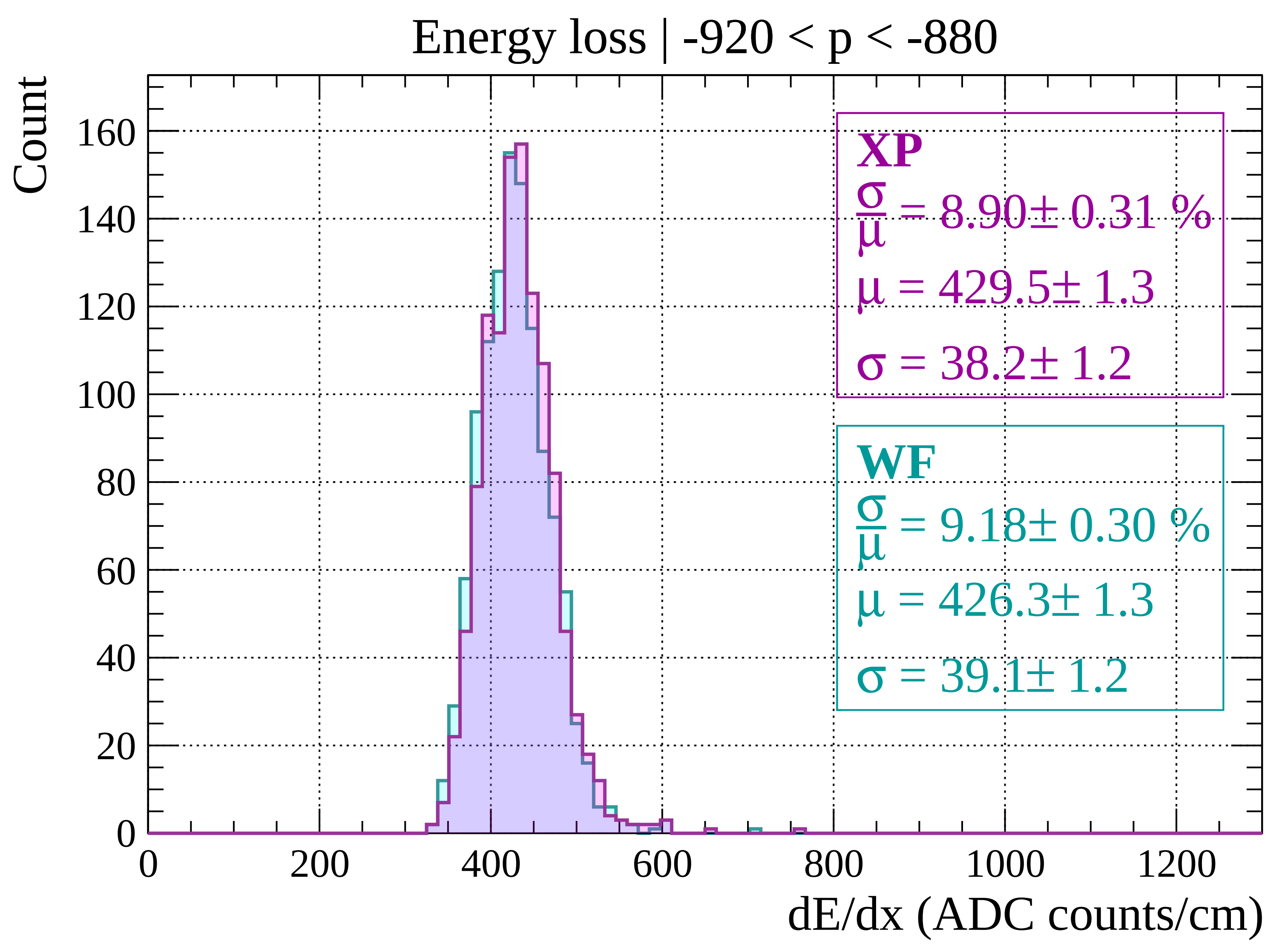
Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 8.45± 0.35 %
μ = 434.0± 1.3
σ = 36.7± 1.4
WF
σ/μ = 8.70± 0.38 %
μ = 430.2± 1.4
σ = 37.4± 1.5



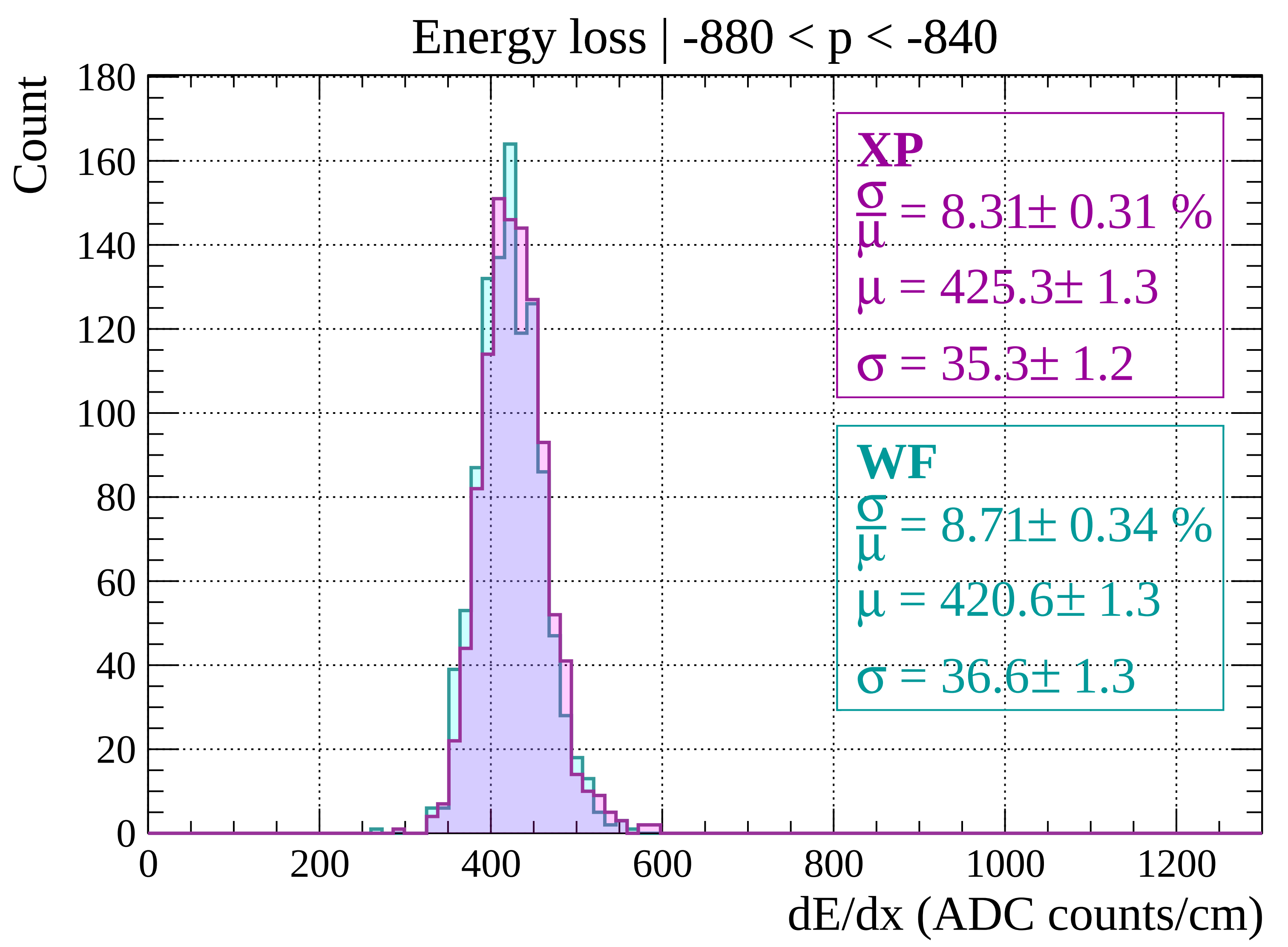
Energy loss | -1000 < p < -960
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.44± 0.33 %
μ = 432.1± 1.3
σ = 36.5± 1.3
WF
$\frac{\sigma}{\mu}$ = 8.34± 0.32 %
μ = 428.2± 1.3
σ = 35.7± 1.2



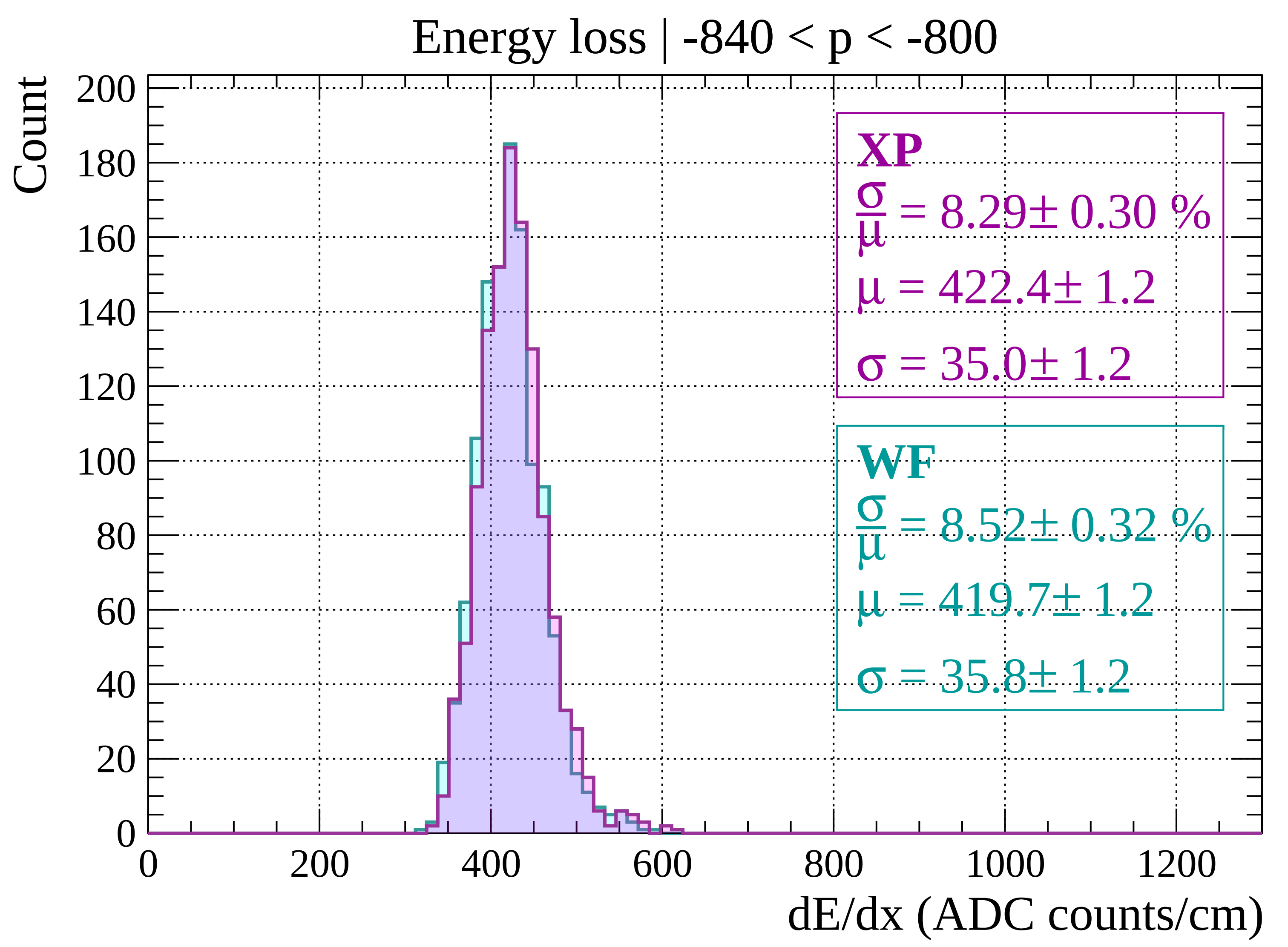
Energy loss | -960 < p < -920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 8.11± 0.35 %
μ = 429.0± 1.3
σ = 34.8± 1.4
WF
σ/μ = 8.18± 0.30 %
μ = 426.2± 1.3
σ = 34.9± 1.2

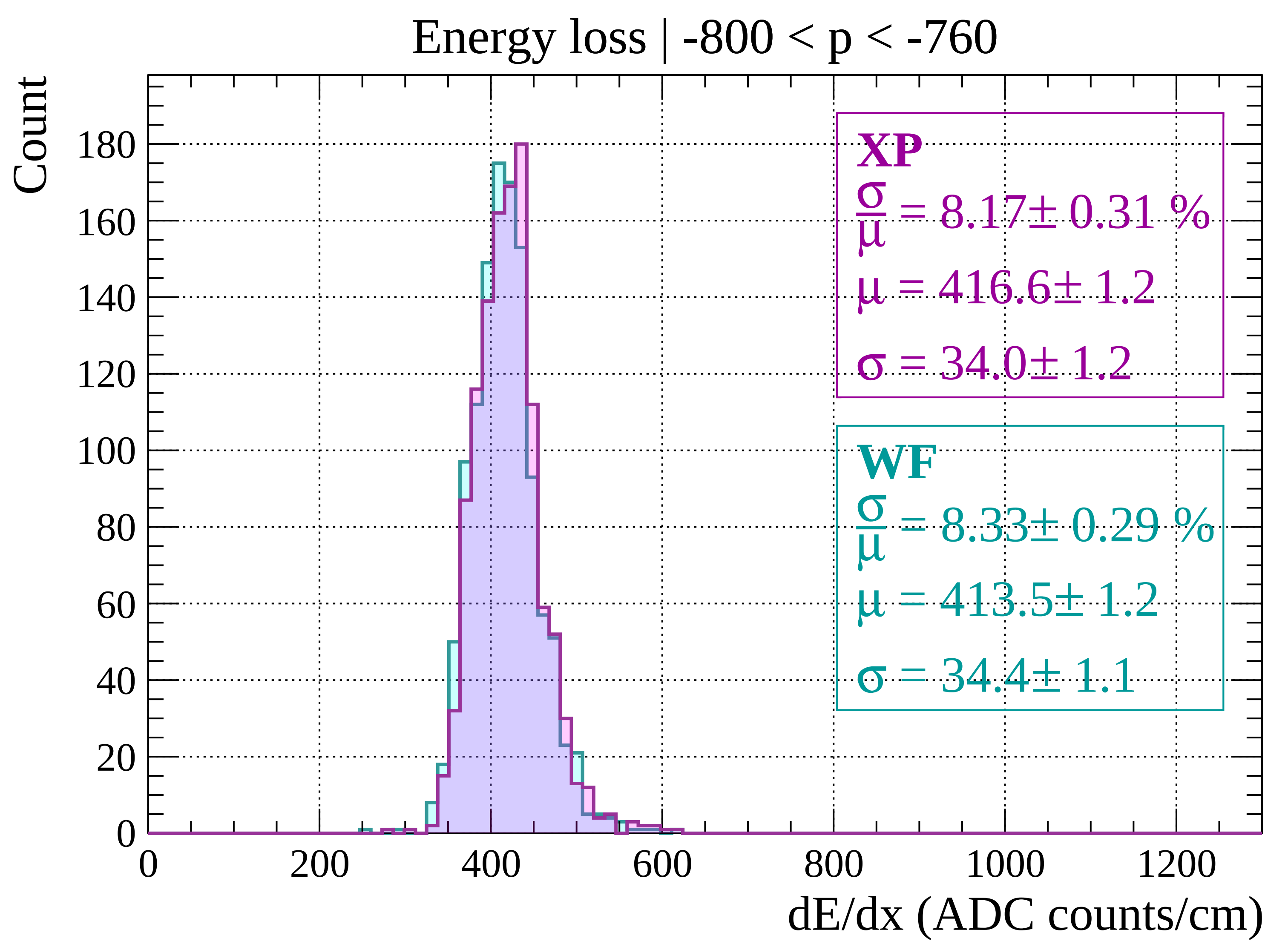


Energy loss | -920 < p < -880
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.90± 0.31 %
μ = 429.5± 1.3
σ = 38.2± 1.2
WF
$\frac{\sigma}{\mu}$ = 9.18± 0.30 %
μ = 426.3± 1.3
σ = 39.1± 1.2

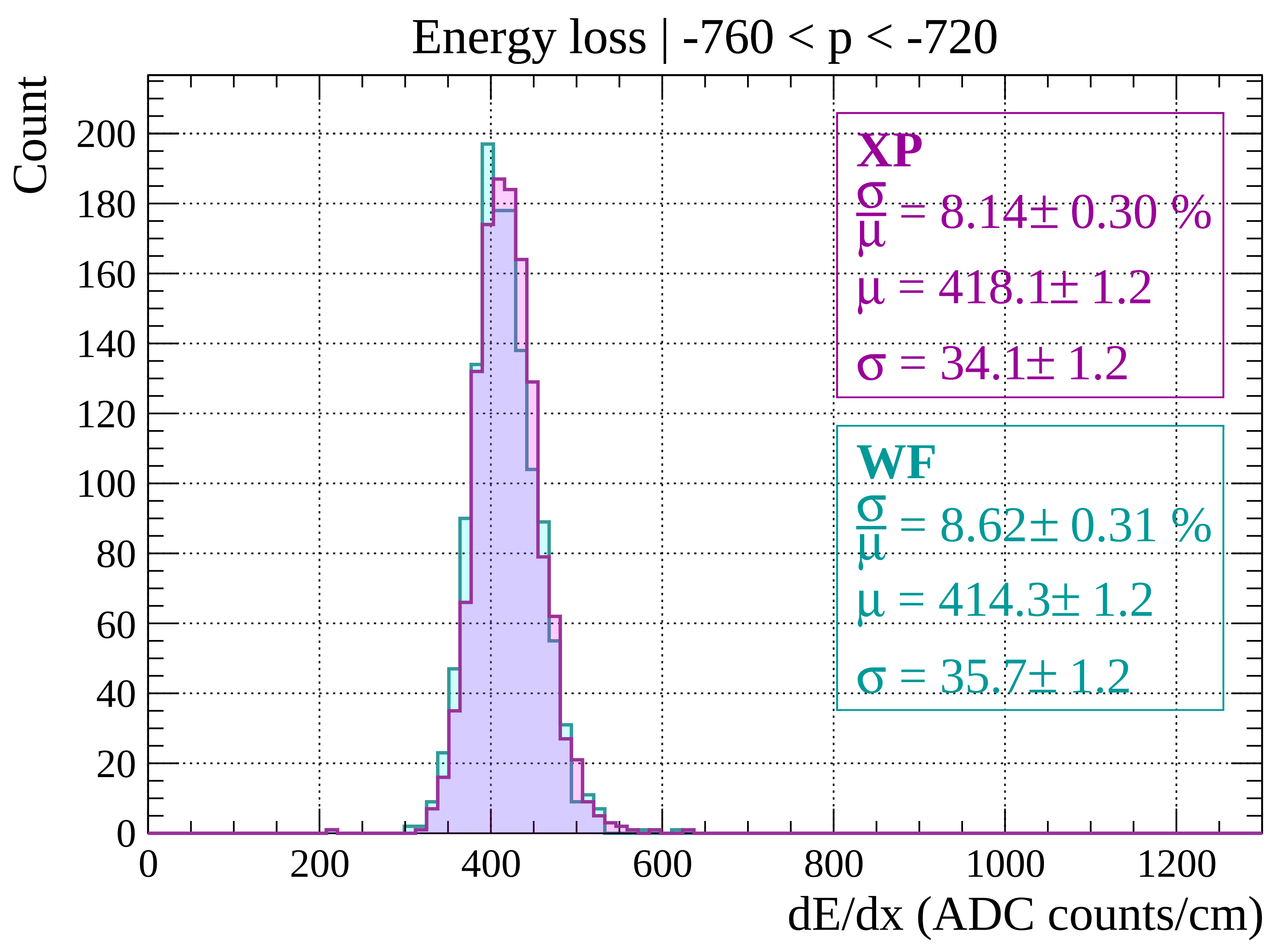


Energy loss | -880 < p < -840
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
σ/μ = 8.31± 0.31 %
μ = 425.3± 1.3
σ = 35.3± 1.2
WF
σ/μ = 8.71± 0.34 %
μ = 420.6± 1.3
σ = 36.6± 1.3

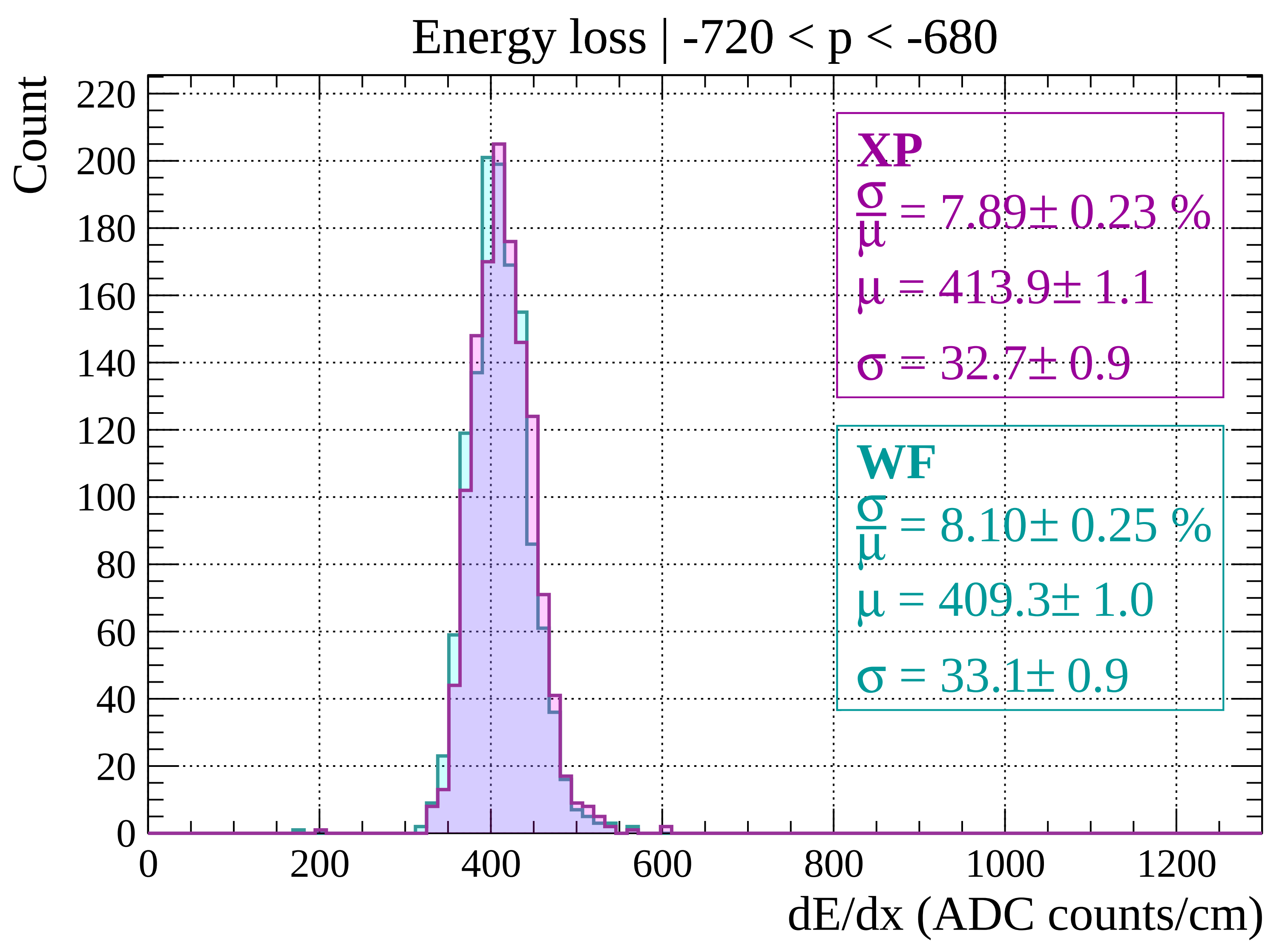

Energy loss | -840 < p < -800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
σ/μ = 8.29± 0.30 %
μ = 422.4± 1.2
σ = 35.0± 1.2
WF
σ/μ = 8.52± 0.32 %
μ = 419.7± 1.2
σ = 35.8± 1.2



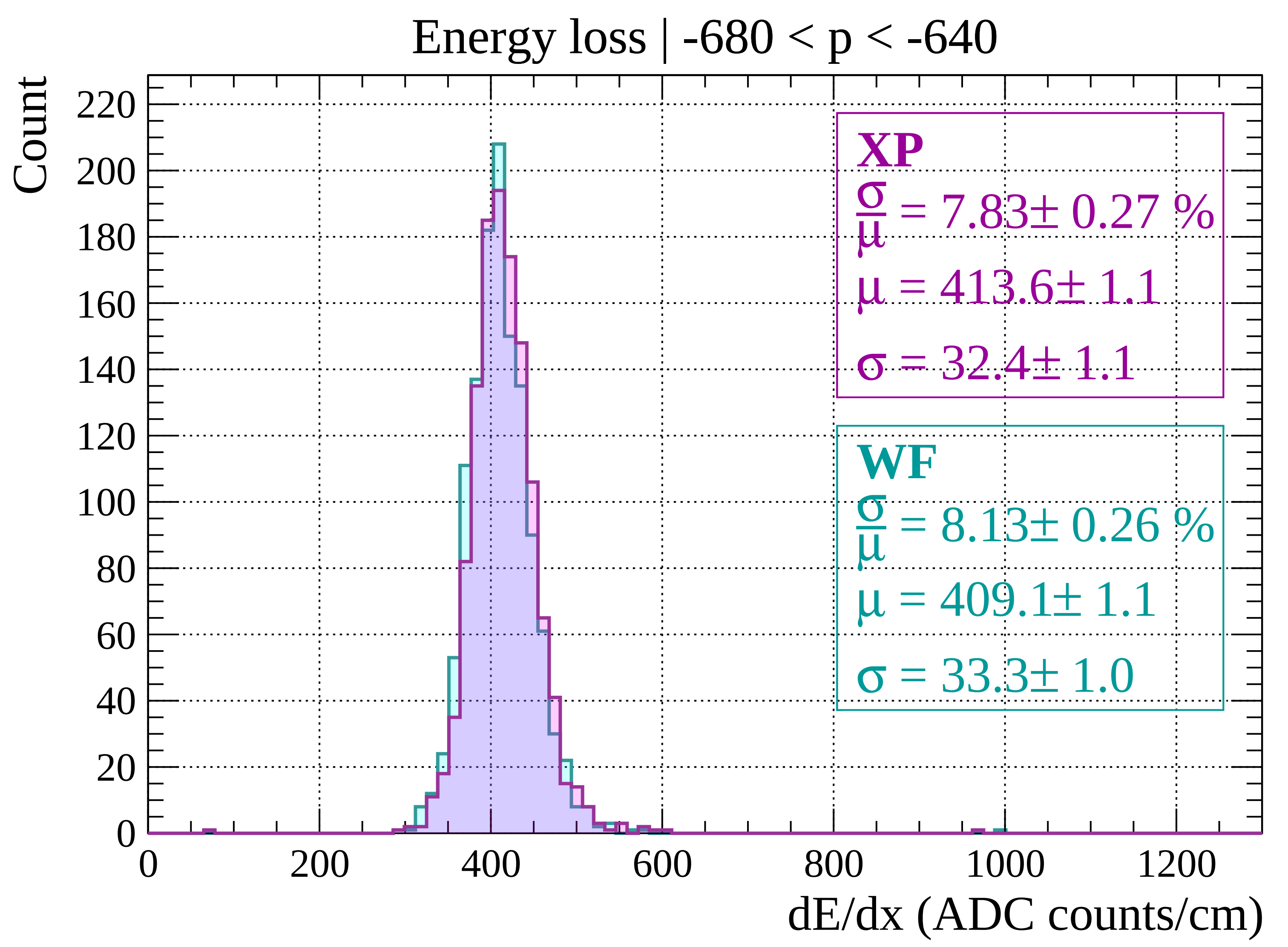
Energy loss | -800 < p < -760
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.17± 0.31 %
μ = 416.6± 1.2
σ = 34.0± 1.2
WF
σ/μ = 8.33± 0.29 %
μ = 413.5± 1.2
σ = 34.4± 1.1



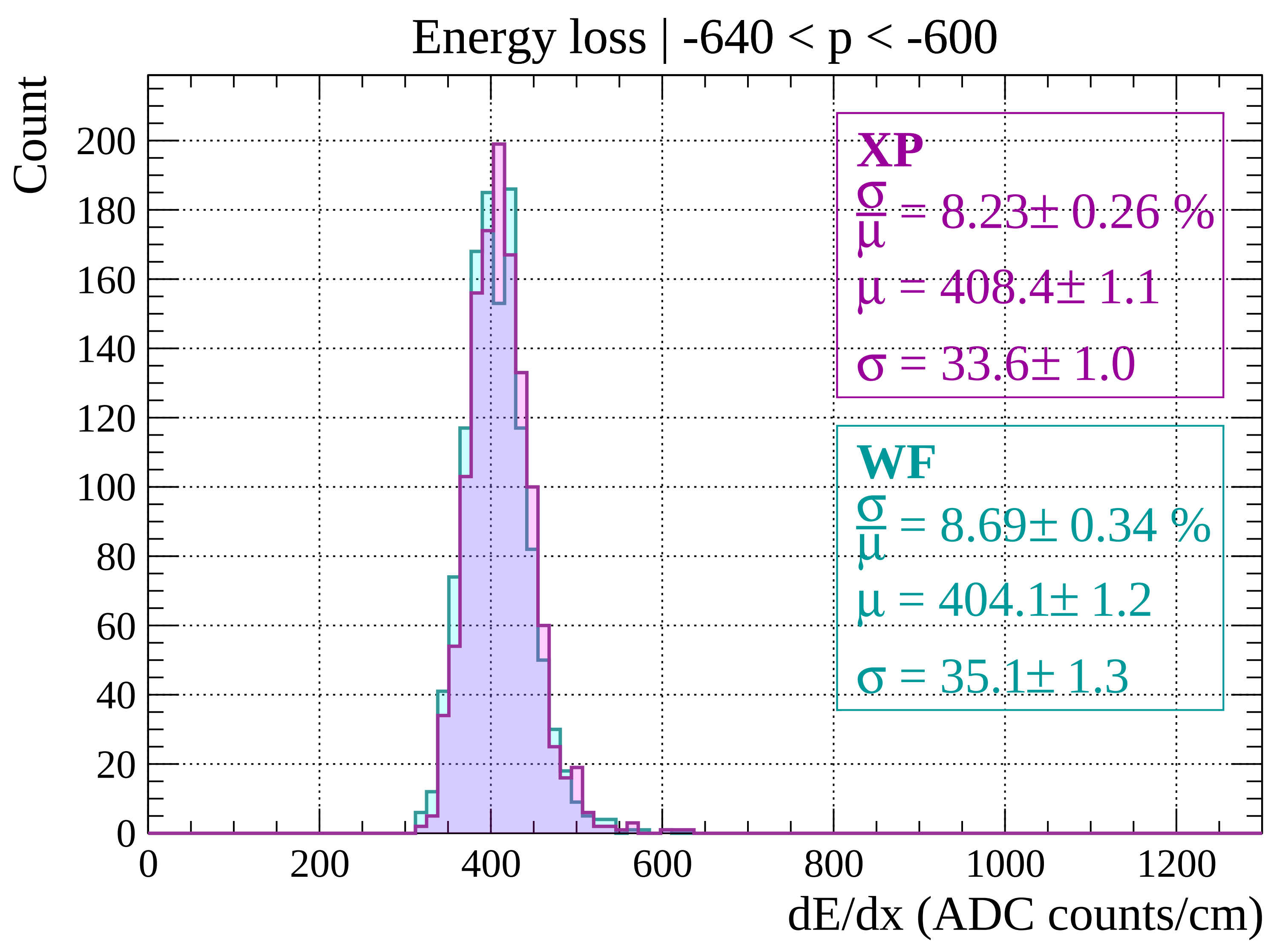
Energy loss | -760 < p < -720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
σ/μ = 8.14± 0.30 %
μ = 418.1± 1.2
σ = 34.1± 1.2
WF
σ/μ = 8.62± 0.31 %
μ = 414.3± 1.2
σ = 35.7± 1.2



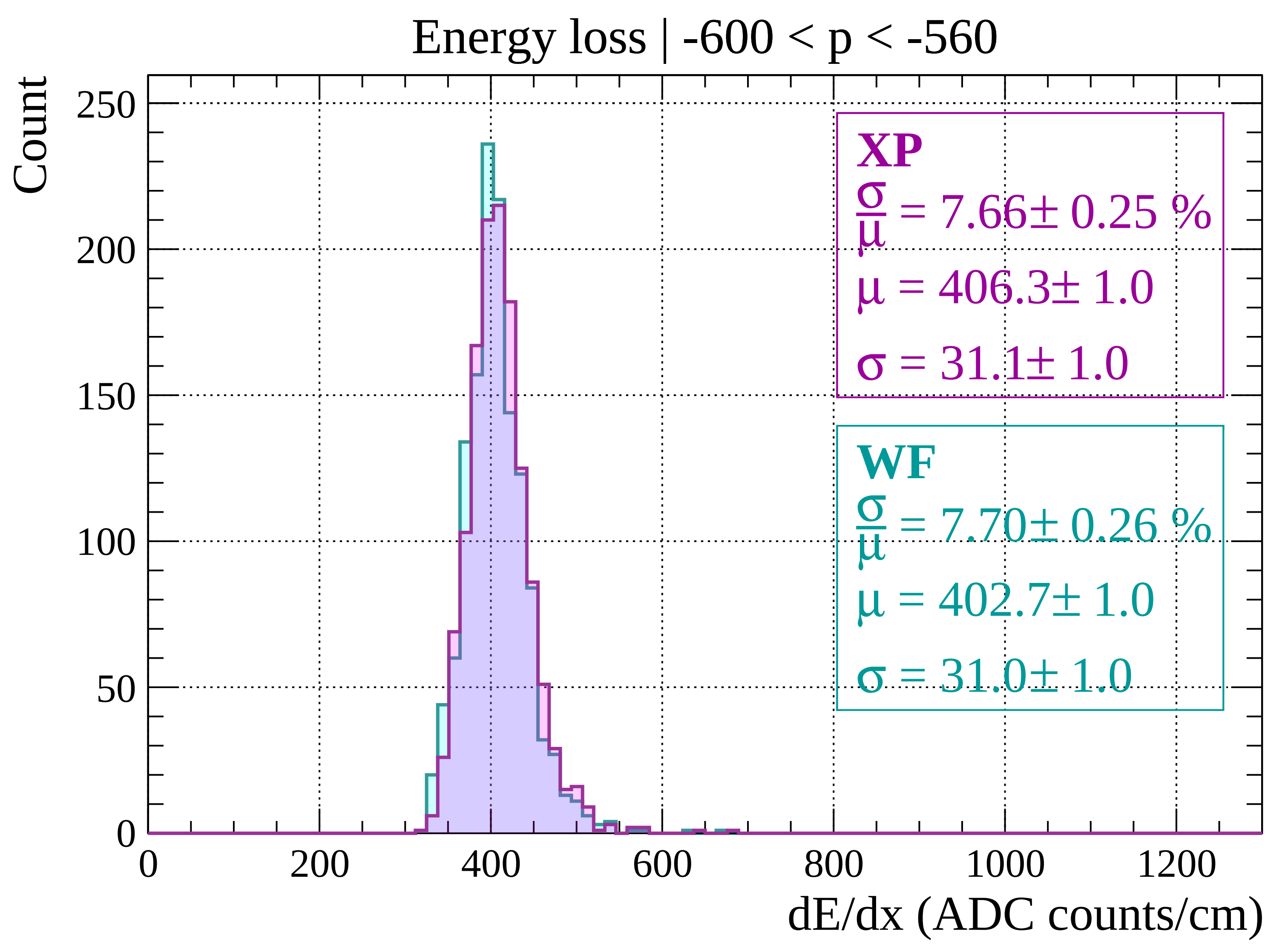
Energy loss | -720 < p < -680
Count
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.89± 0.23 %
μ = 413.9± 1.1
σ = 32.7± 0.9
WF
σ/μ = 8.10± 0.25 %
μ = 409.3± 1.0
σ = 33.1± 0.9



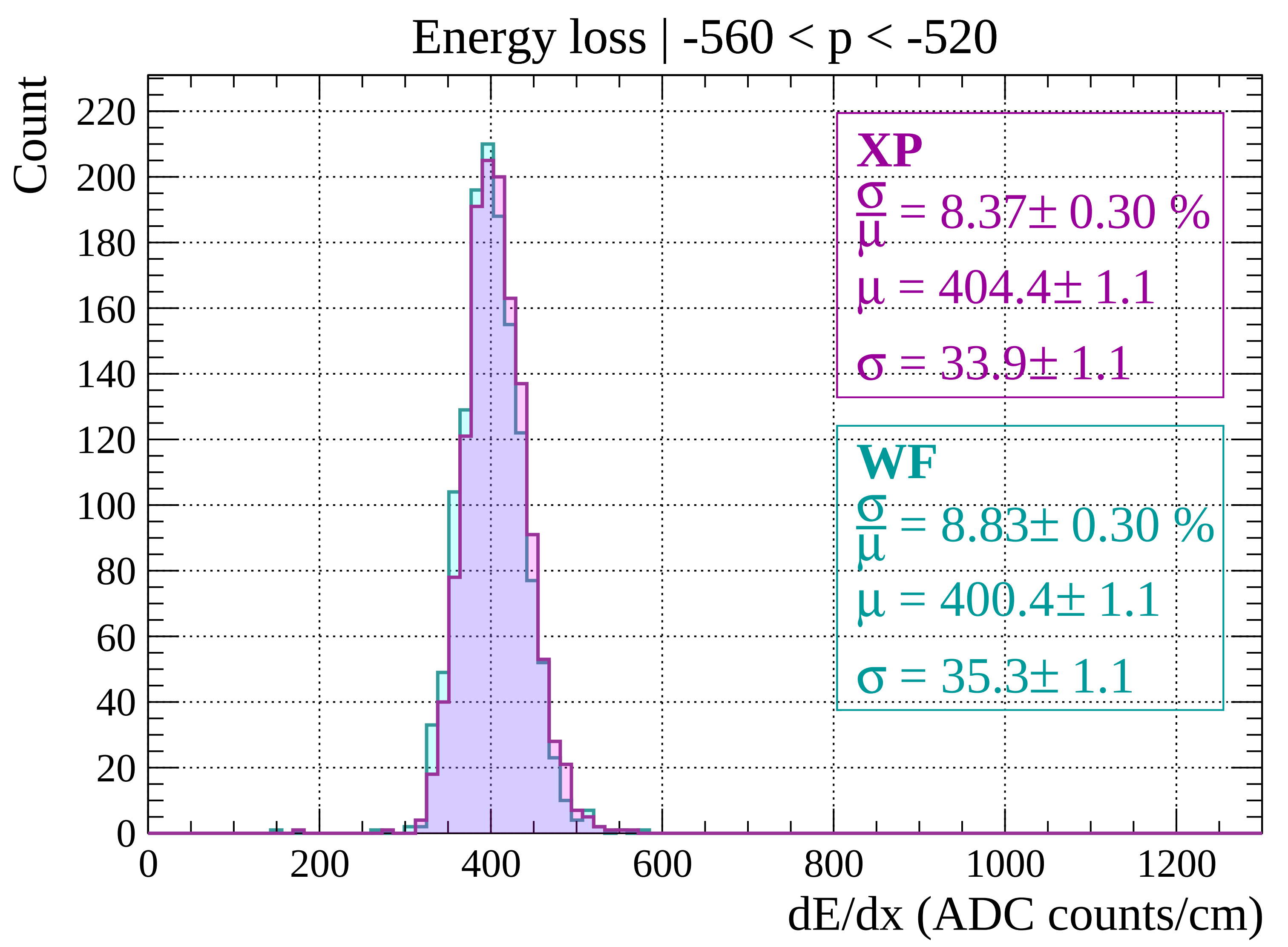
Energy loss | -680 < p < -640
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.83± 0.27 %
μ = 413.6± 1.1
σ = 32.4± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.13± 0.26 %
μ = 409.1± 1.1
σ = 33.3± 1.0



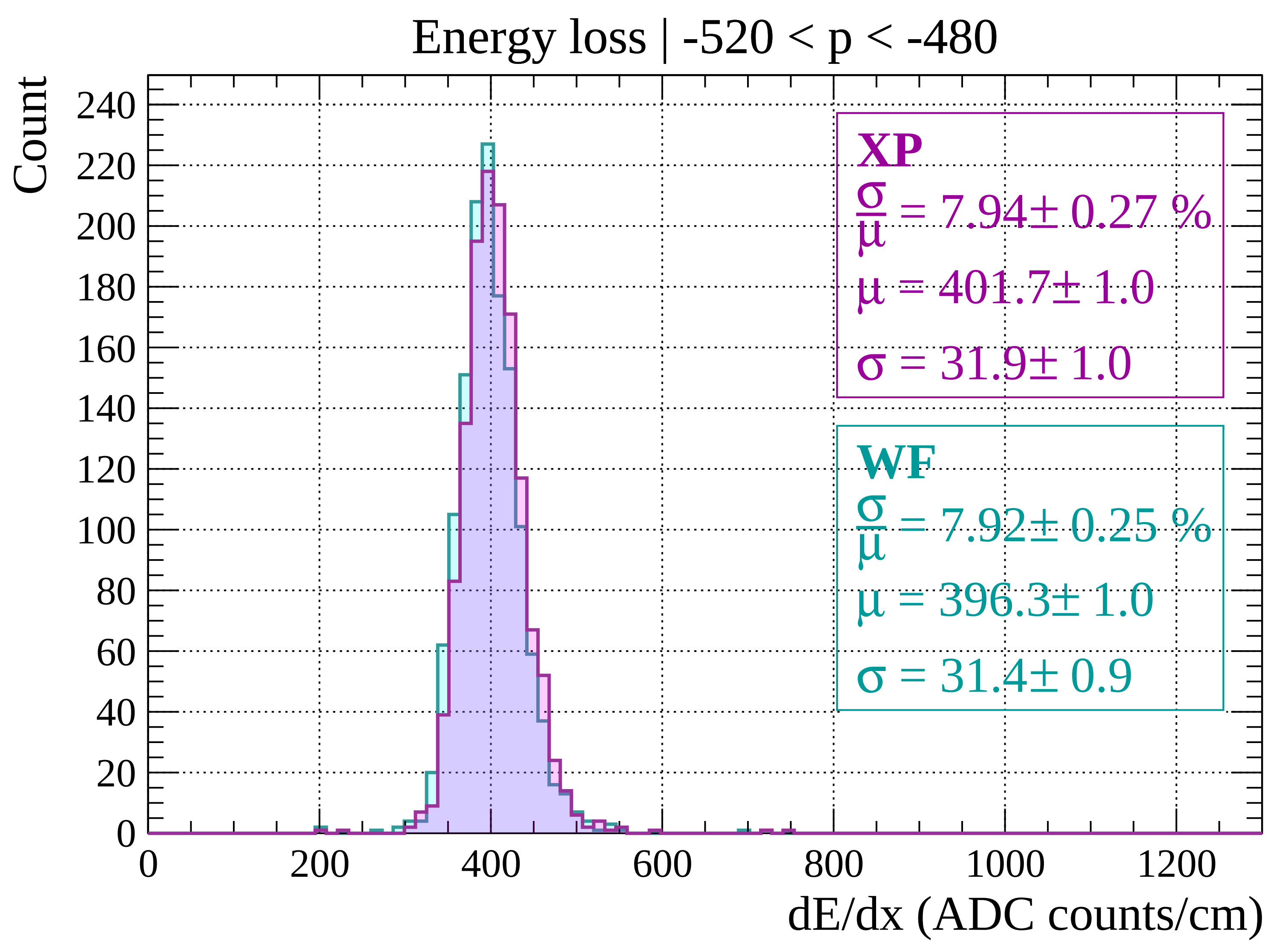
Energy loss | -640 < p < -600
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
σ/μ = 8.23± 0.26 %
μ = 408.4± 1.1
σ = 33.6± 1.0
WF
σ/μ = 8.69± 0.34 %
μ = 404.1± 1.2
σ = 35.1± 1.3



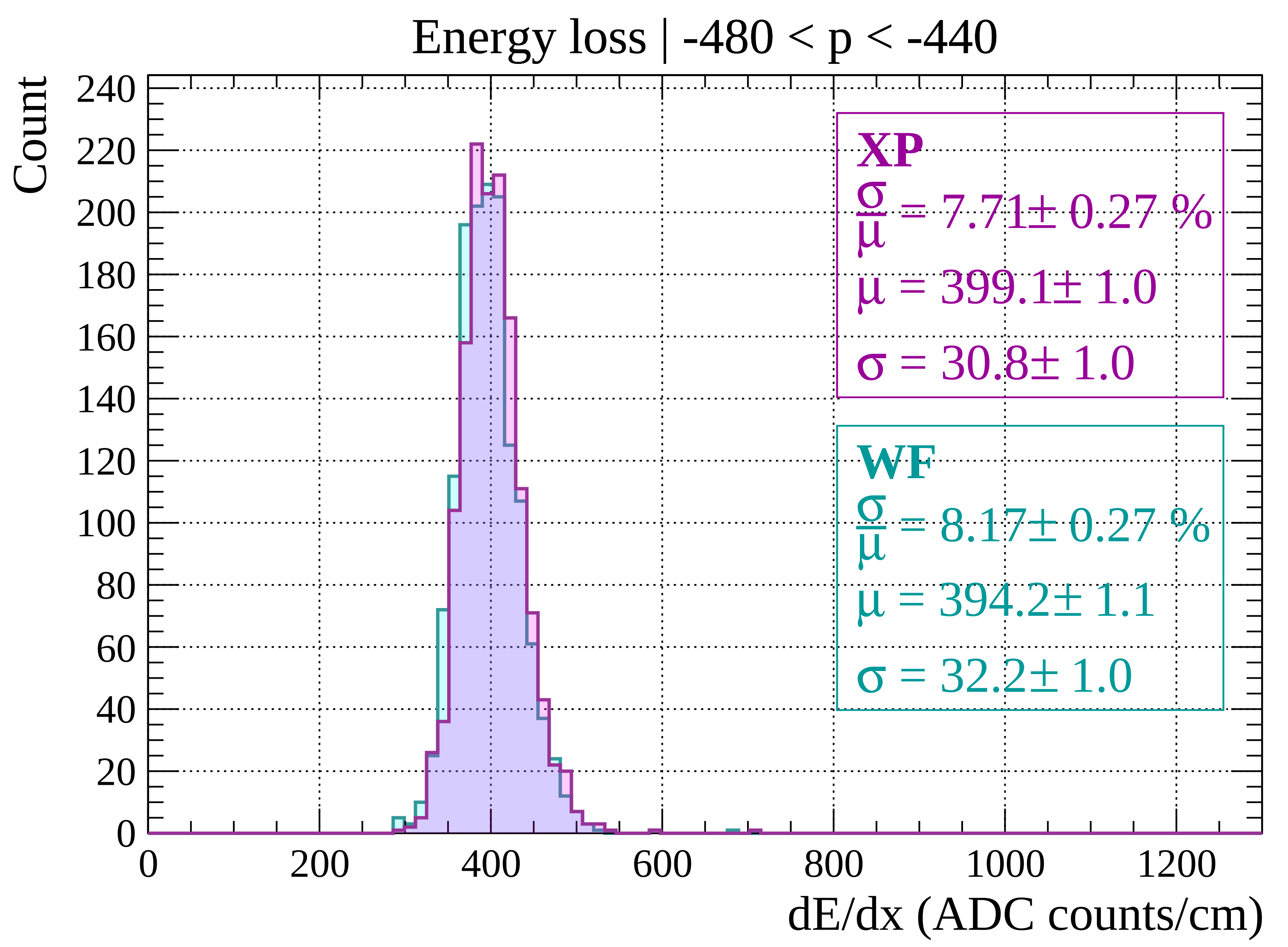
Energy loss | -600 < p < -560
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.66± 0.25 %
μ = 406.3± 1.0
σ = 31.1± 1.0
WF
σ/μ = 7.70± 0.26 %
μ = 402.7± 1.0
σ = 31.0± 1.0

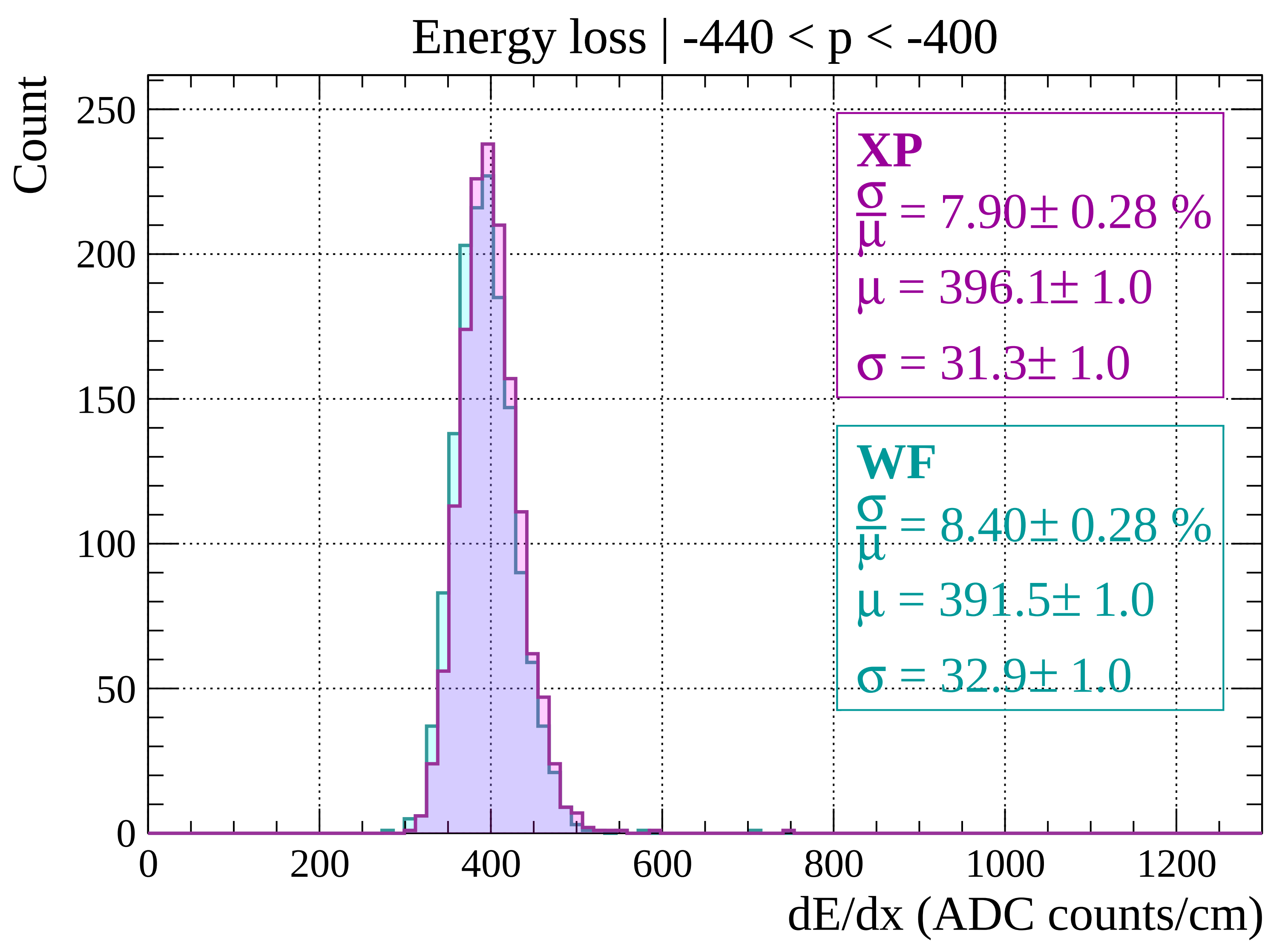


Energy loss | -560 < p < -520
Count
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.37± 0.30 %
μ = 404.4± 1.1
σ = 33.9± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.83± 0.30 %
μ = 400.4± 1.1
σ = 35.3± 1.1

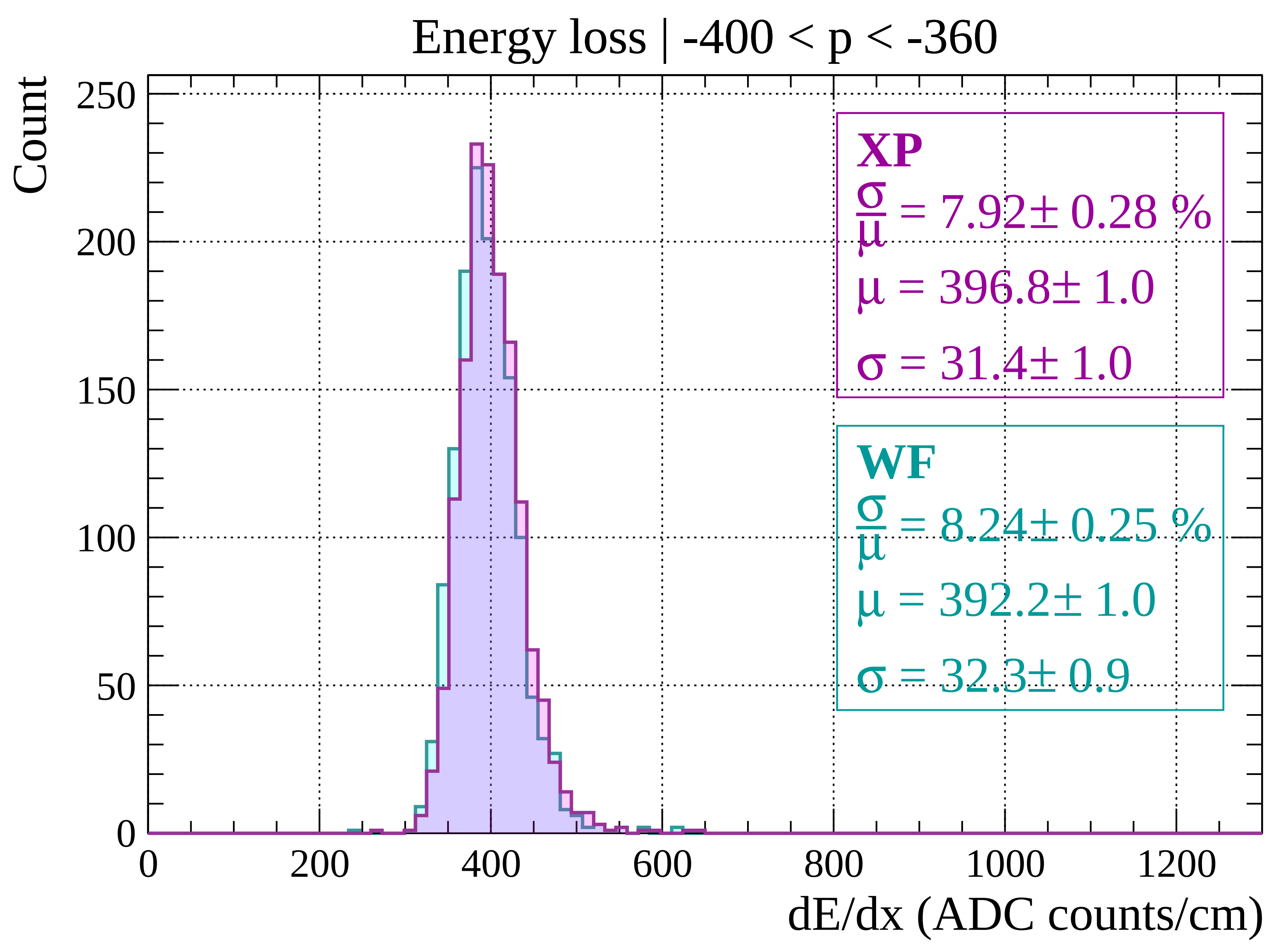


Energy loss | -520 < p < -480
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.94± 0.27 %
μ = 401.7± 1.0
σ = 31.9± 1.0
WF
$\frac{\sigma}{\mu}$ = 7.92± 0.25 %
μ = 396.3± 1.0
σ = 31.4± 0.9

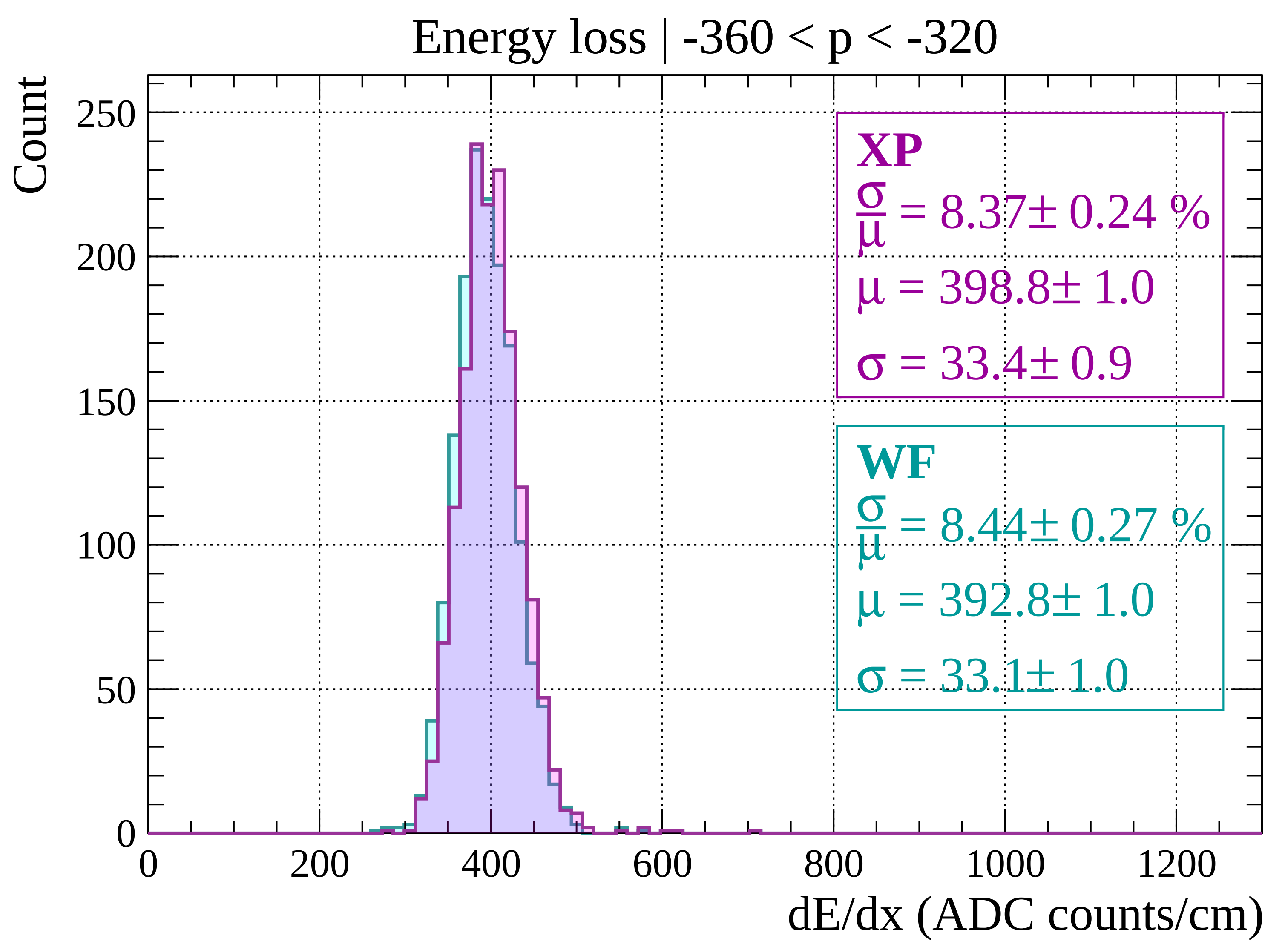

Energy loss | -480 < p < -440
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.71± 0.27 %
μ = 399.1± 1.0
σ = 30.8± 1.0
WF
σ/μ = 8.17± 0.27 %
μ = 394.2± 1.1
σ = 32.2± 1.0



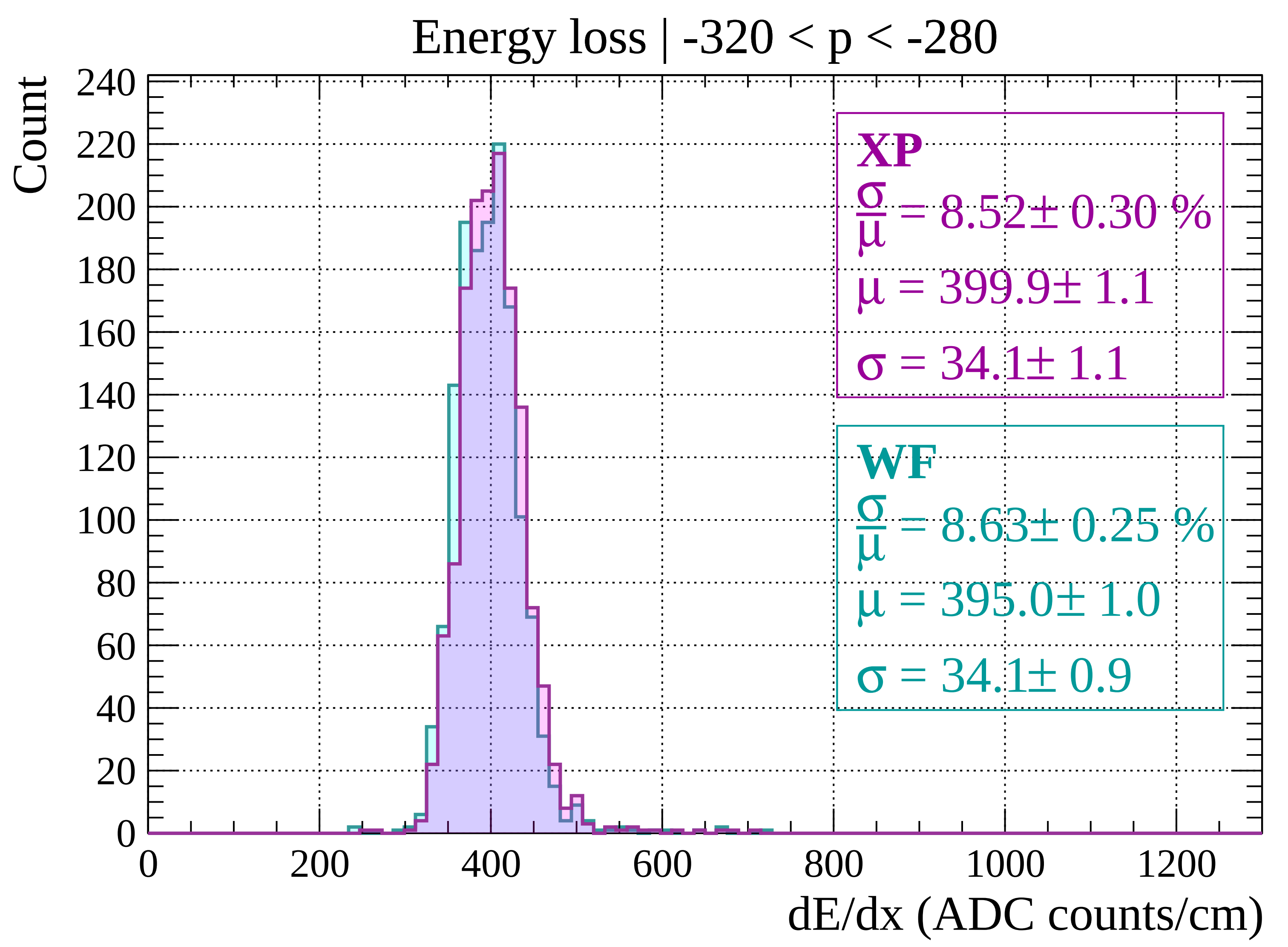
Energy loss | -440 < p < -400
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.90± 0.28 %
μ = 396.1± 1.0
σ = 31.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.40± 0.28 %
μ = 391.5± 1.0
σ = 32.9± 1.0



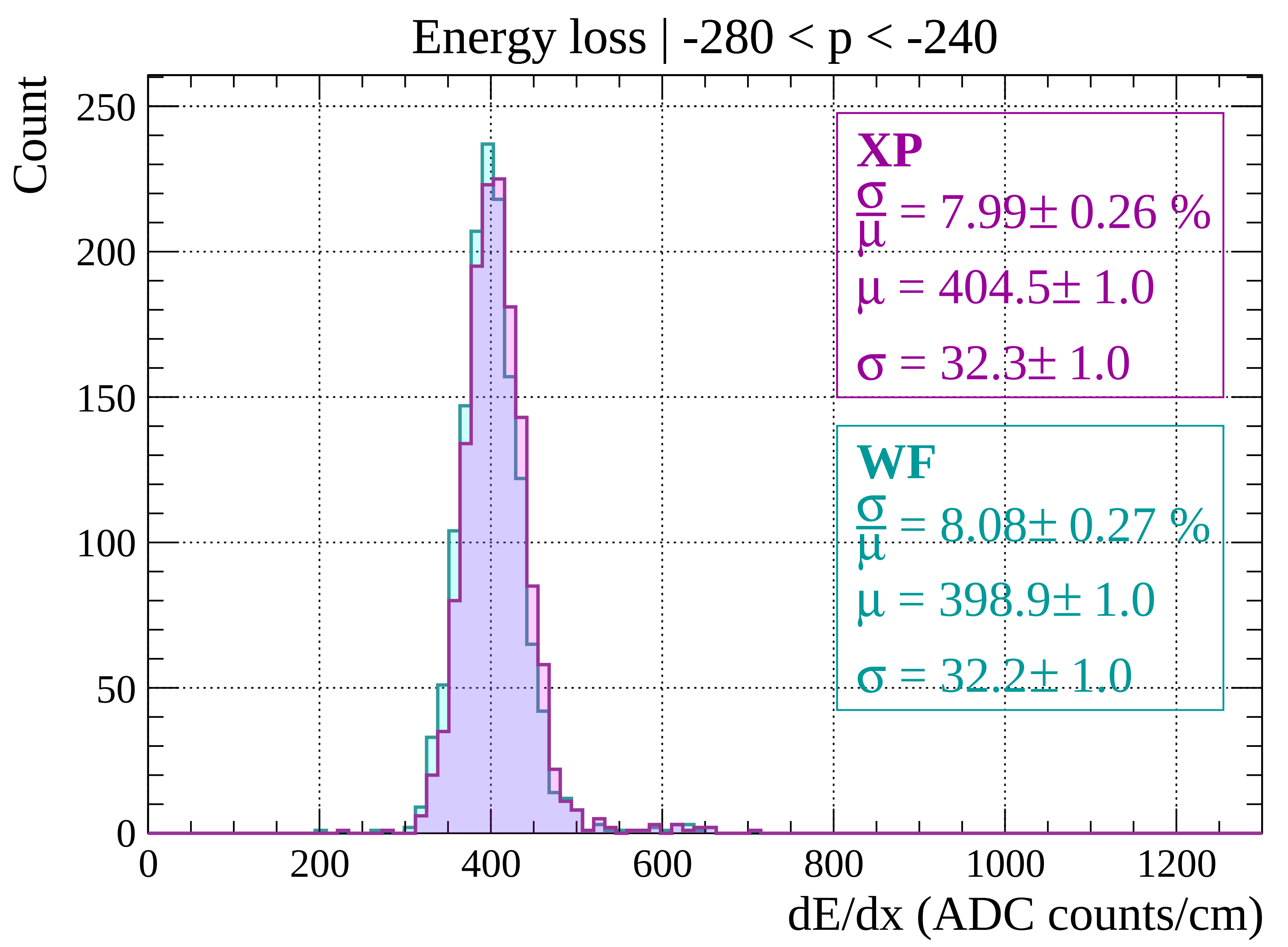
Energy loss | -400 < p < -360
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.92± 0.28 %
μ = 396.8± 1.0
σ = 31.4± 1.0
WF
σ/μ = 8.24± 0.25 %
μ = 392.2± 1.0
σ = 32.3± 0.9



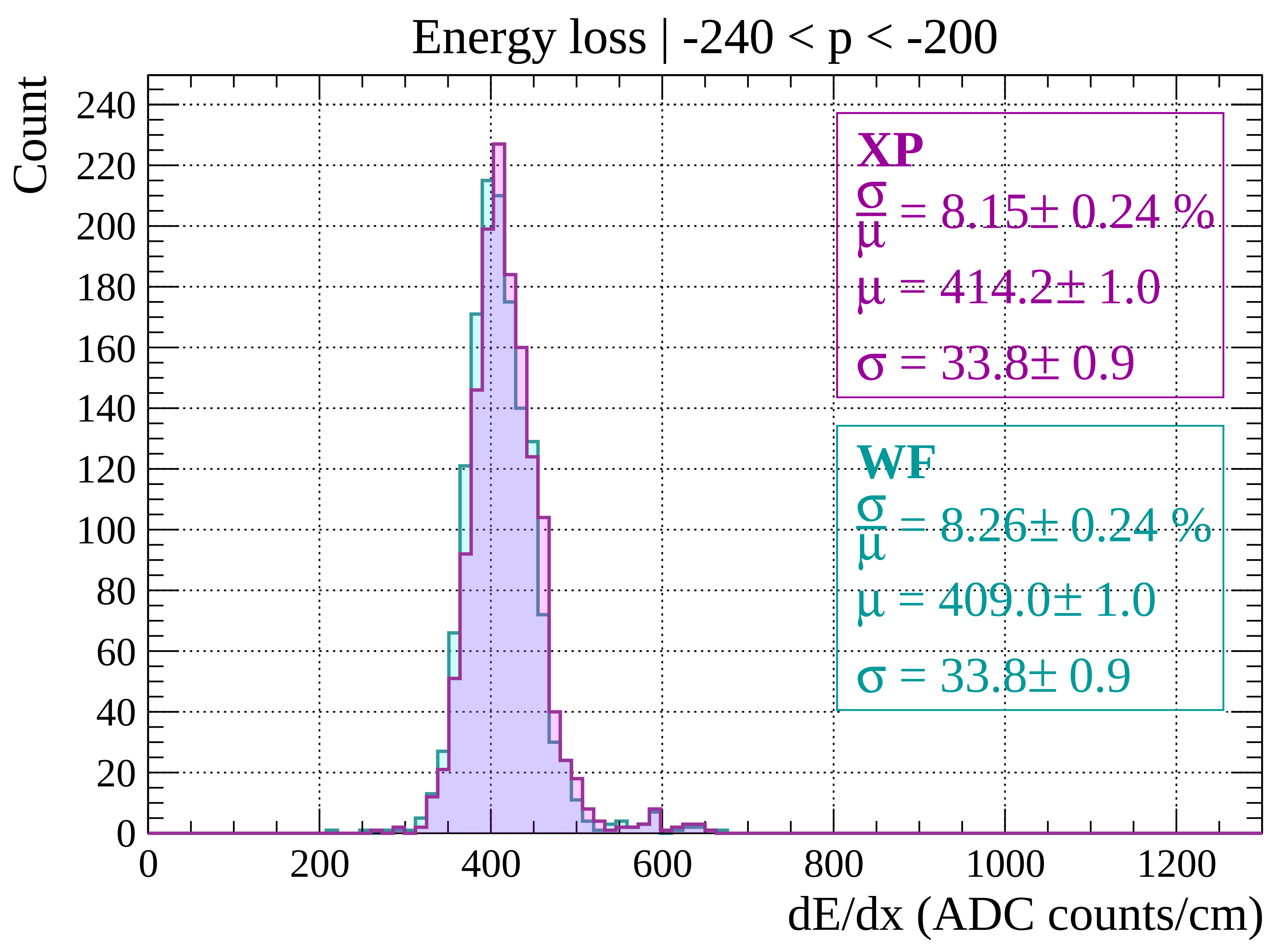
Energy loss | -360 < p < -320
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.37± 0.24 %
μ = 398.8± 1.0
σ = 33.4± 0.9
WF
σ/μ = 8.44± 0.27 %
μ = 392.8± 1.0
σ = 33.1± 1.0



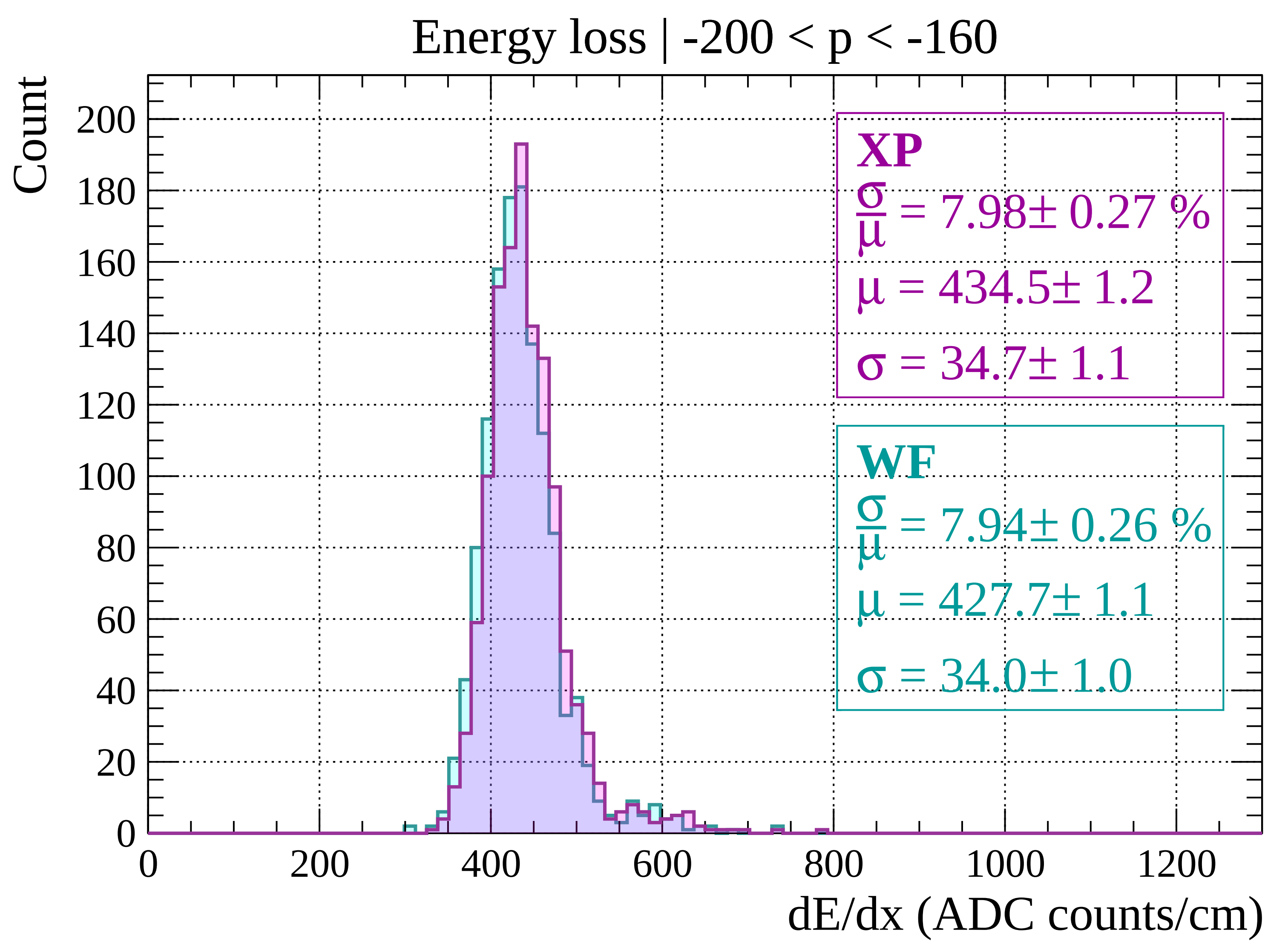
Energy loss | -320 < p < -280
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.52± 0.30 %
μ = 399.9± 1.1
σ = 34.1± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.63± 0.25 %
μ = 395.0± 1.0
σ = 34.1± 0.9

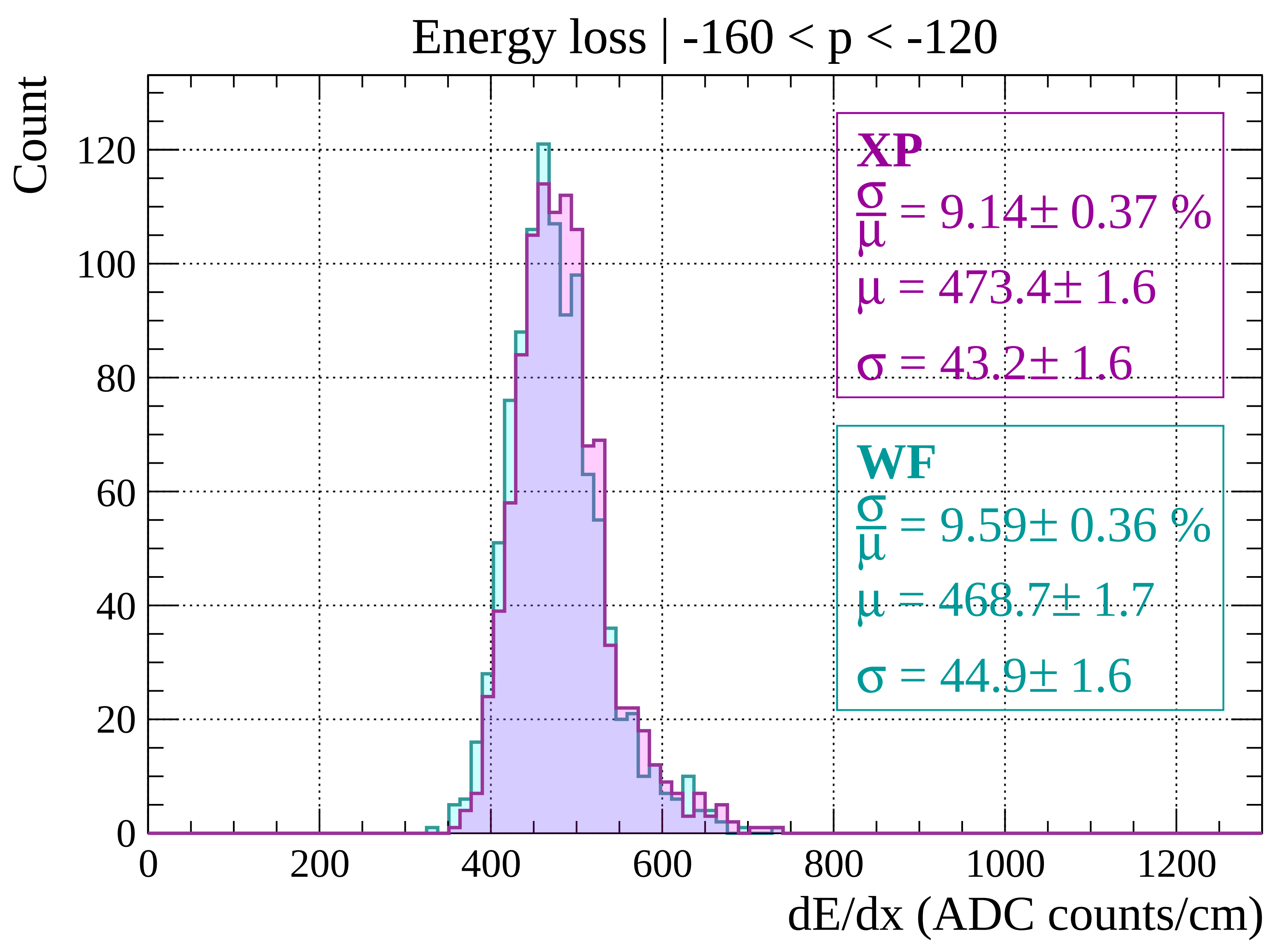


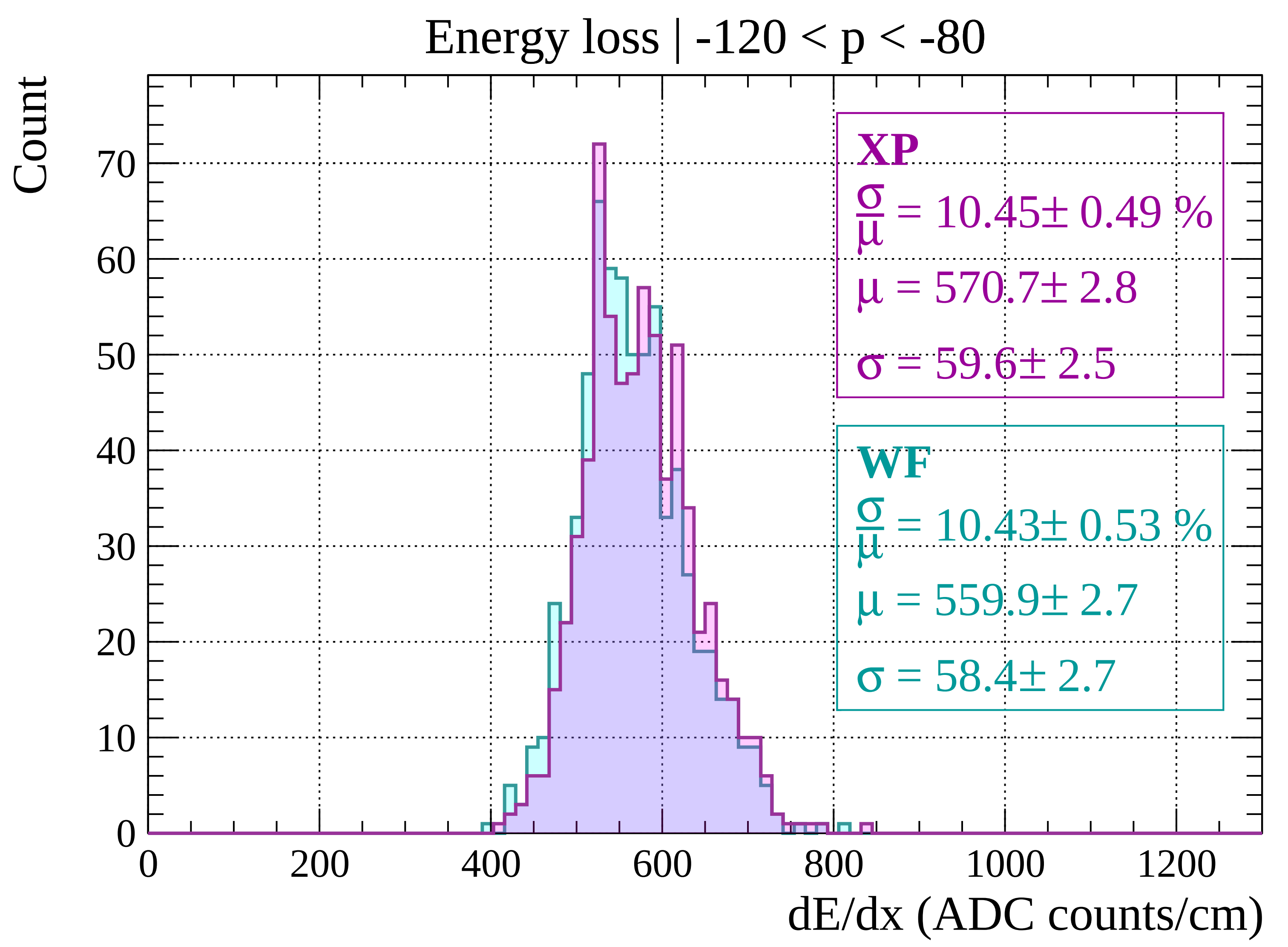
Energy loss | -280 < p < -240
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.99± 0.26 %
μ = 404.5± 1.0
σ = 32.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.08± 0.27 %
μ = 398.9± 1.0
σ = 32.2± 1.0



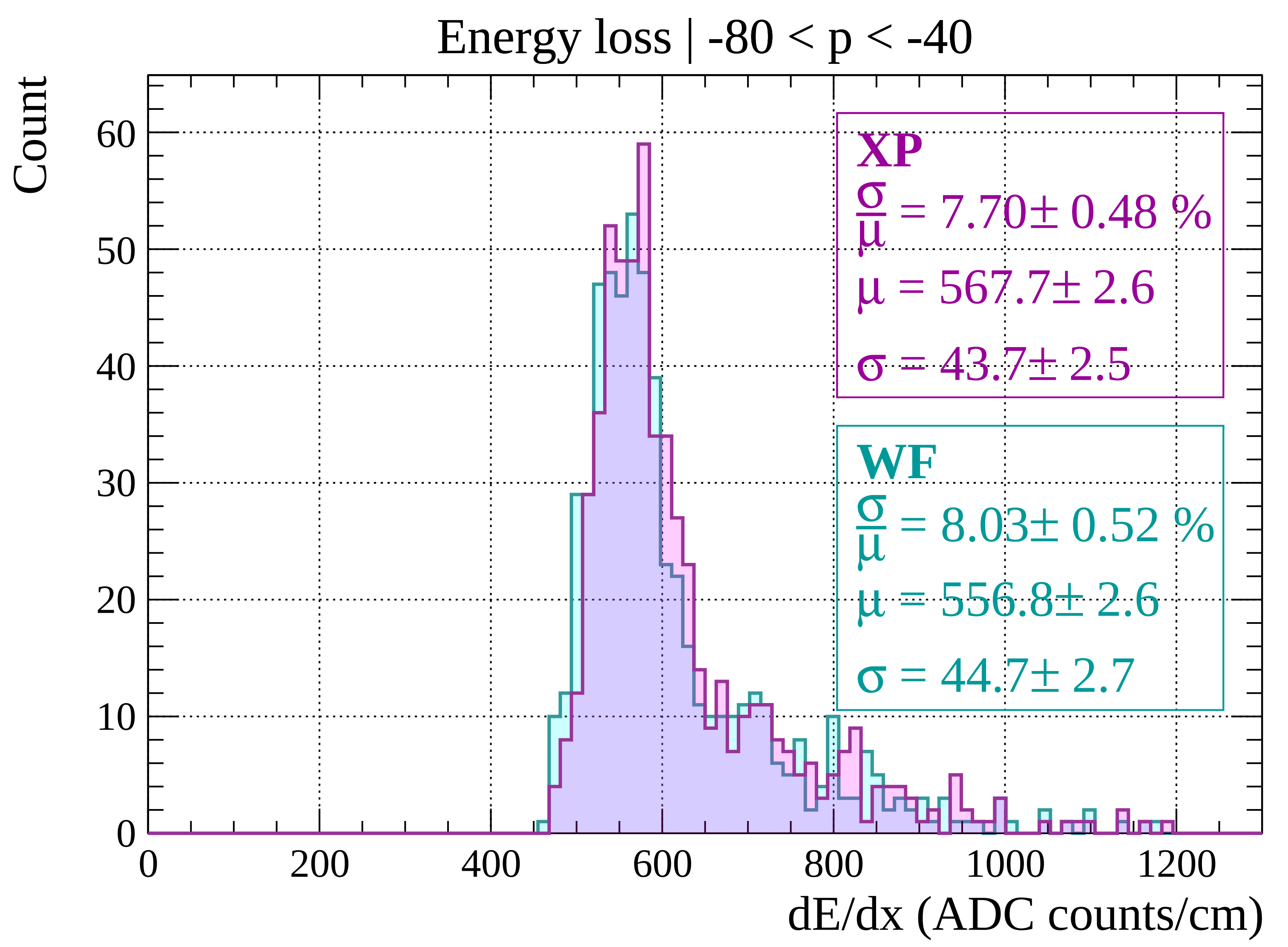
Energy loss | -240 < p < -200
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.15± 0.24 %
μ = 414.2± 1.0
σ = 33.8± 0.9
WF
σ/μ = 8.26± 0.24 %
μ = 409.0± 1.0
σ = 33.8± 0.9
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200


Energy loss | -200 < p < -160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
σ/μ = 7.98± 0.27 %
μ = 434.5± 1.2
σ = 34.7± 1.1
WF
σ/μ = 7.94± 0.26 %
μ = 427.7± 1.1
σ = 34.0± 1.0

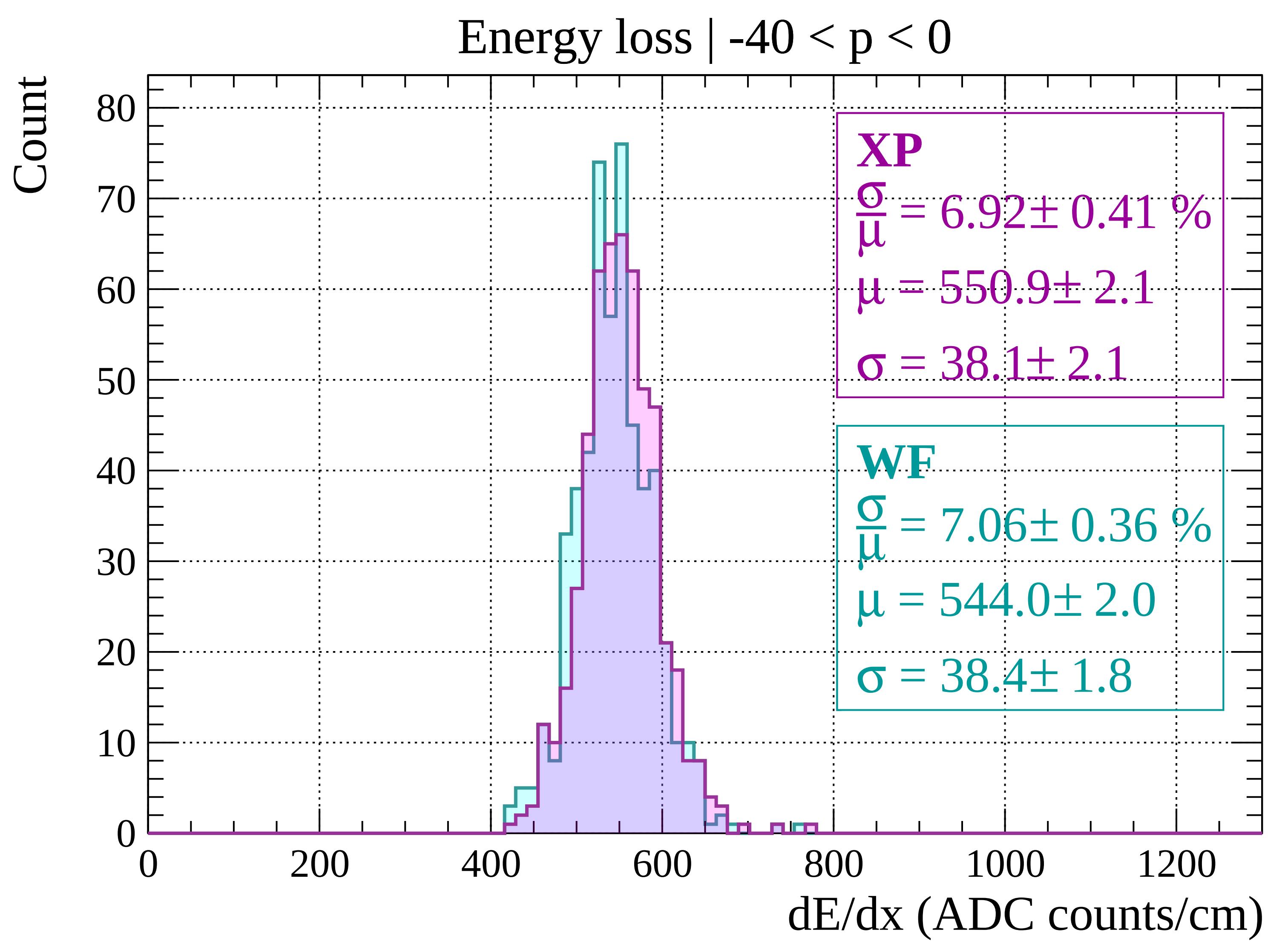

Energy loss | -160 < p < -120
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.14± 0.37 %
μ = 473.4± 1.6
σ = 43.2± 1.6
WF
$\frac{\sigma}{\mu}$ = 9.59± 0.36 %
μ = 468.7± 1.7
σ = 44.9± 1.6
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

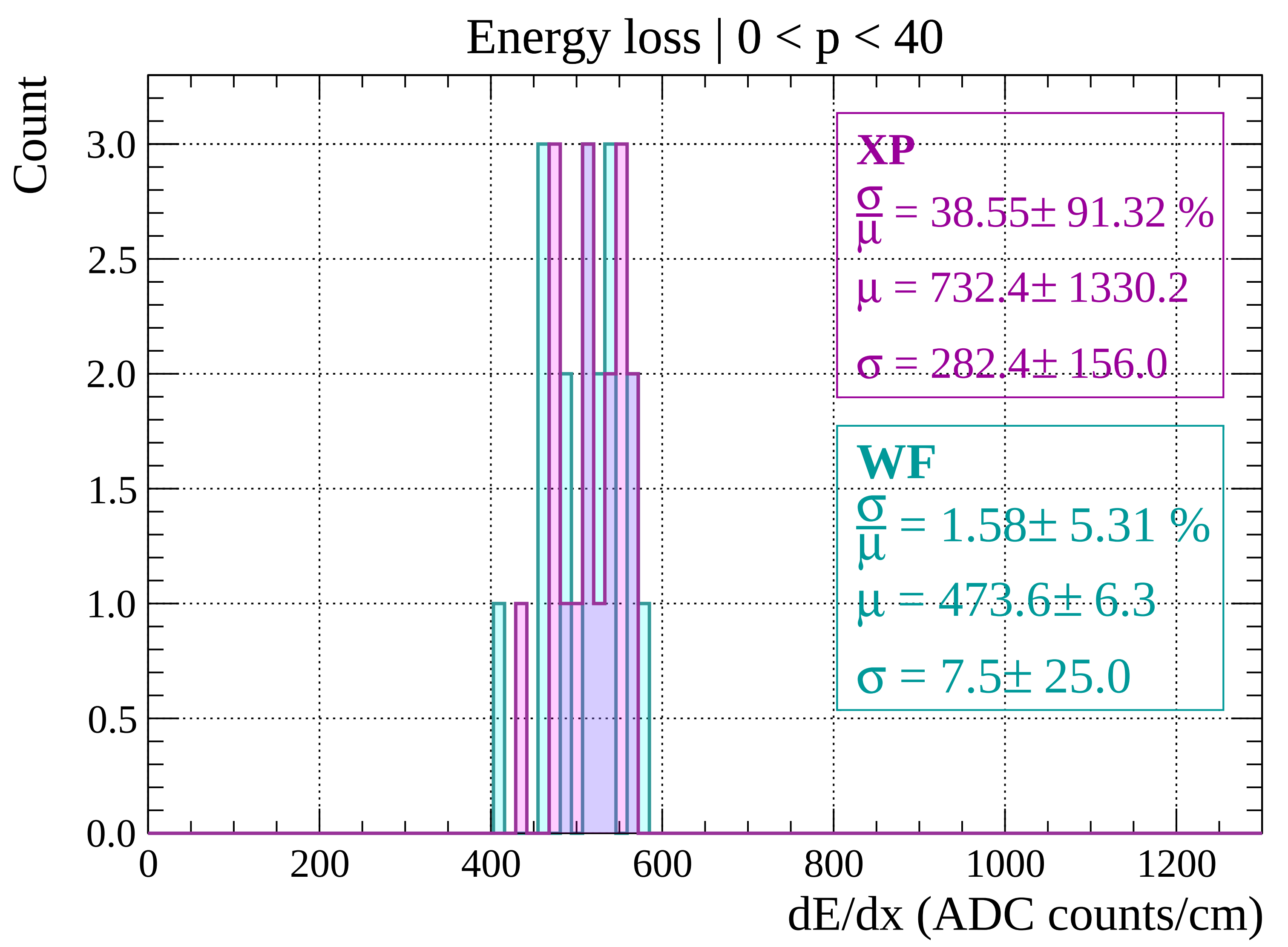


Energy loss | -120 < p < -80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP
σ/μ = 10.45± 0.49 %
μ = 570.7± 2.8
σ = 59.6± 2.5
WF
σ/μ = 10.43± 0.53 %
μ = 559.9± 2.7
σ = 58.4± 2.7

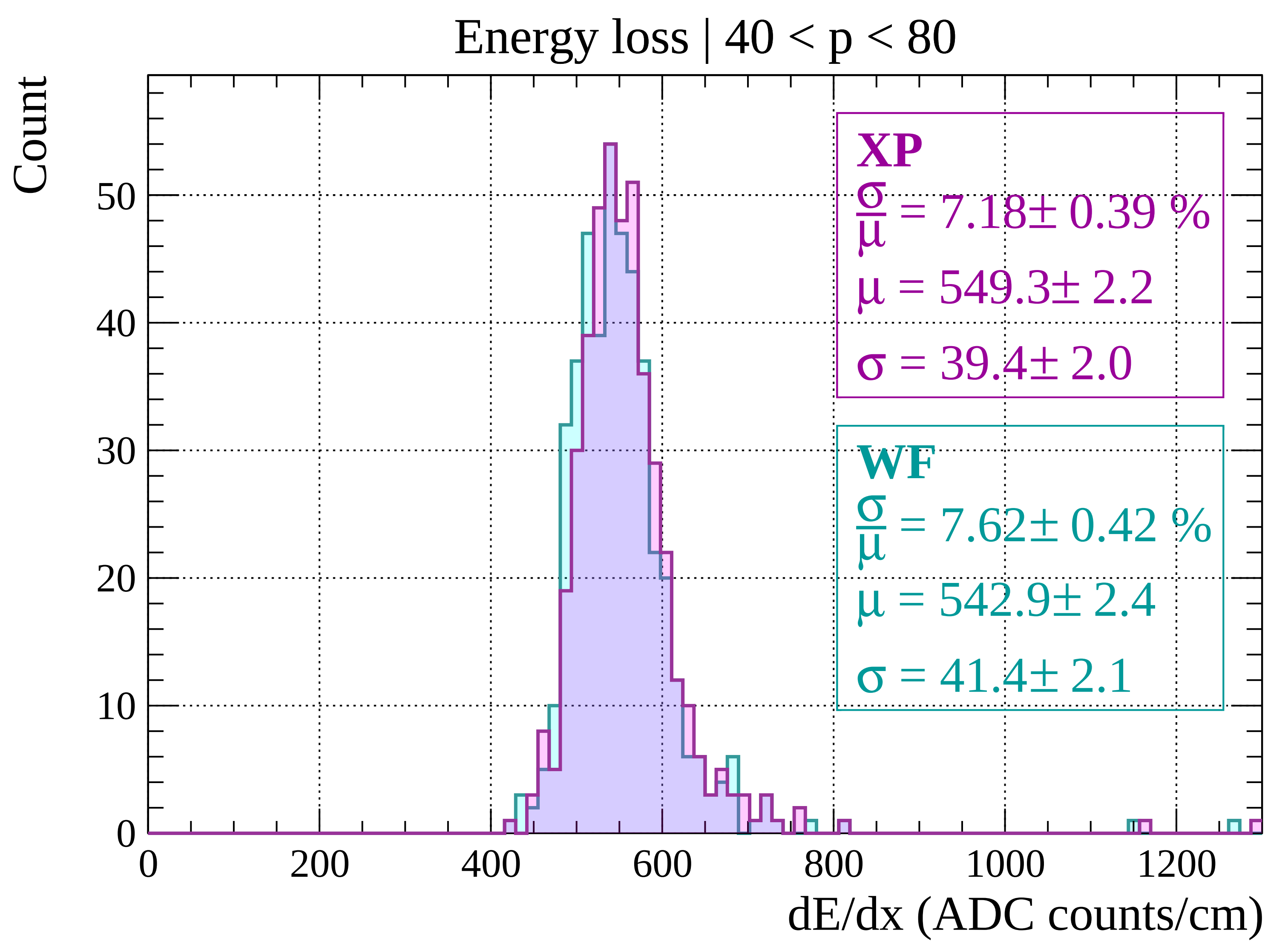


Energy loss | -80 < p < -40
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 7.70± 0.48 %
μ = 567.7± 2.6
σ = 43.7± 2.5
WF
σ/μ = 8.03± 0.52 %
μ = 556.8± 2.6
σ = 44.7± 2.7

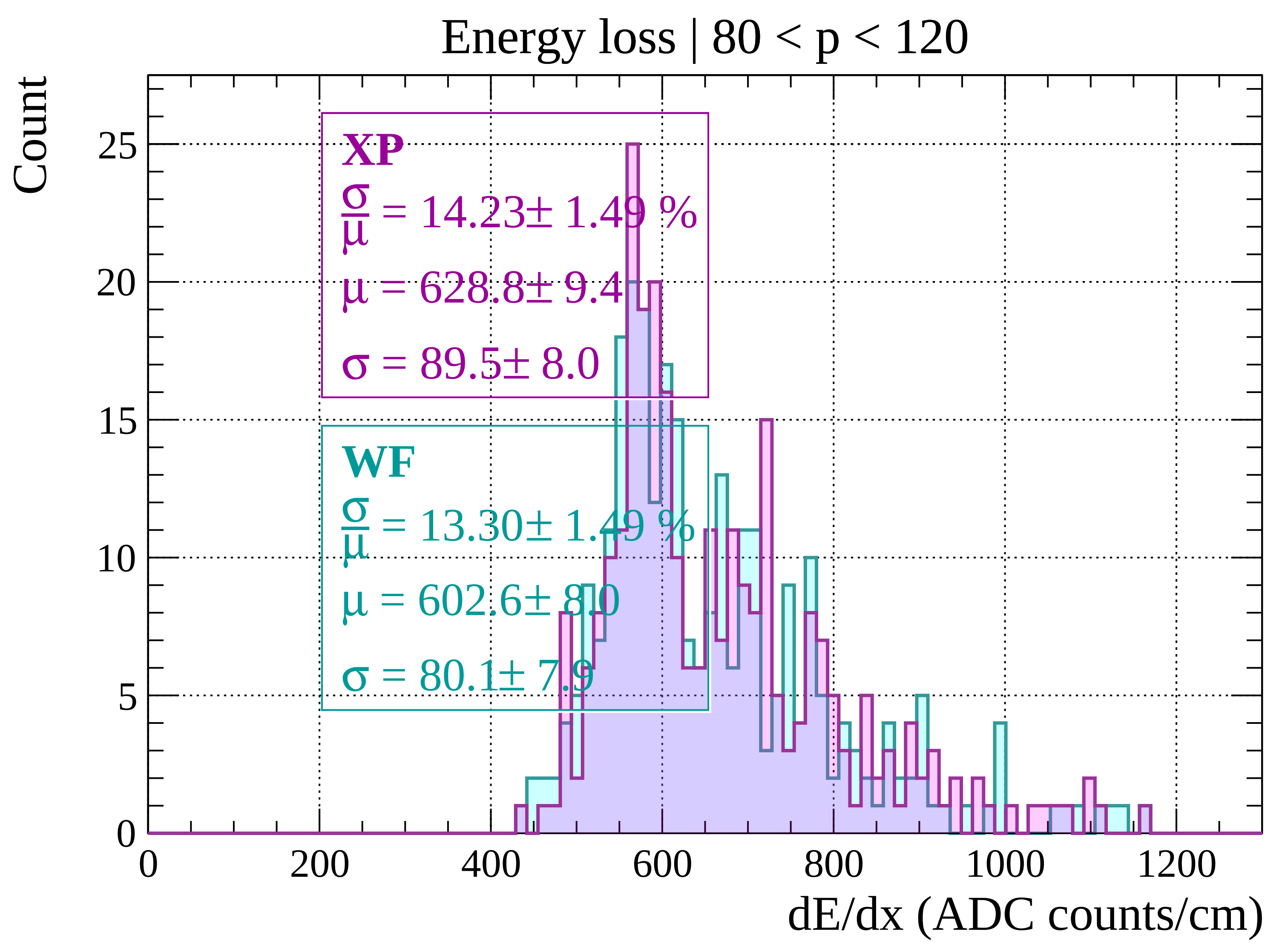

Energy loss | -40 < p < 0
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP
σ/μ = 6.92± 0.41 %
μ = 550.9± 2.1
σ = 38.1± 2.1
WF
σ/μ = 7.06± 0.36 %
μ = 544.0± 2.0
σ = 38.4± 1.8



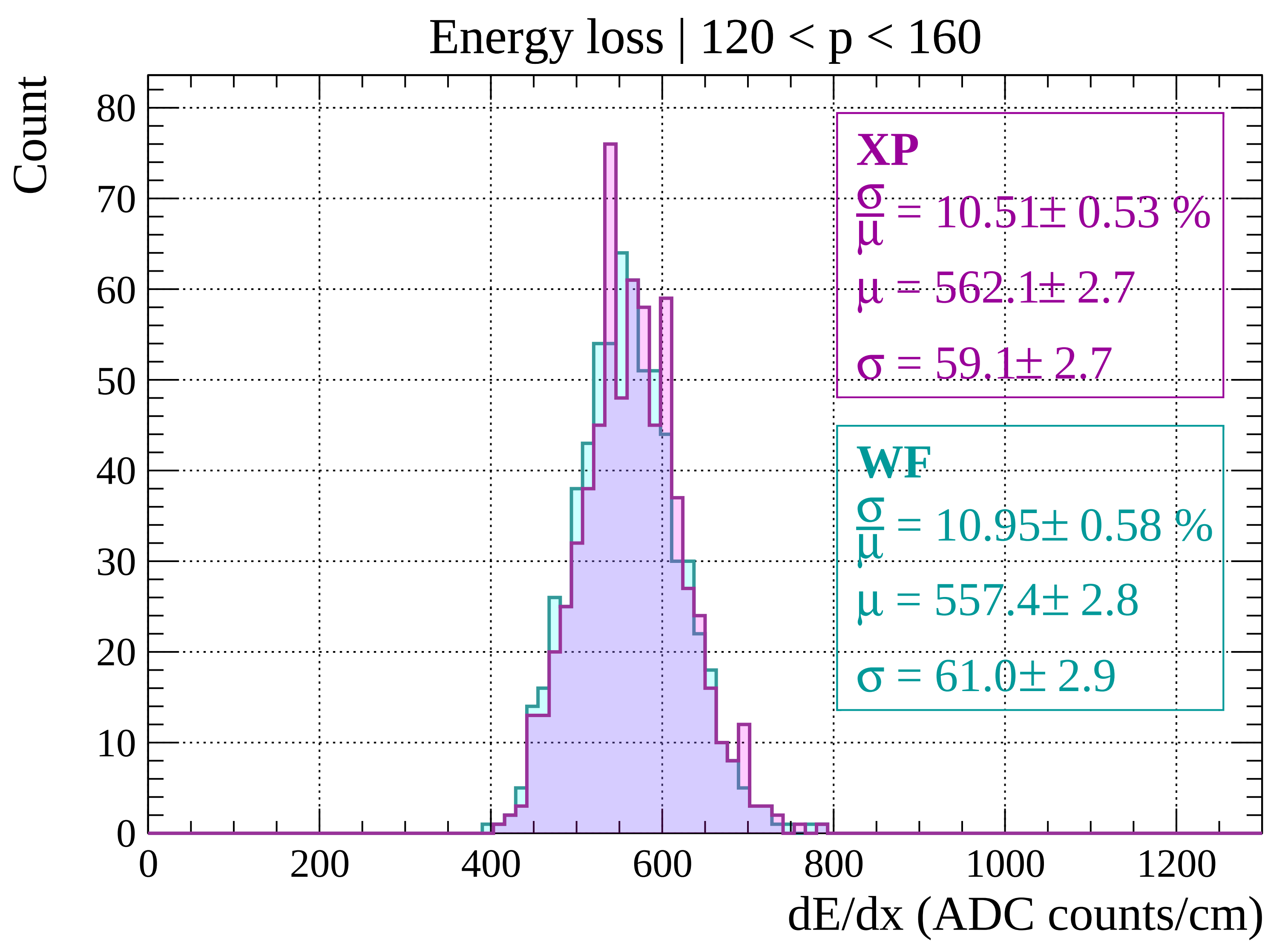
Energy loss | 0 < p < 40
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
σ/μ = 38.55± 91.32 %
μ = 732.4± 1330.2
σ = 282.4± 156.0
WF
σ/μ = 1.58± 5.31 %
μ = 473.6± 6.3
σ = 7.5± 25.0



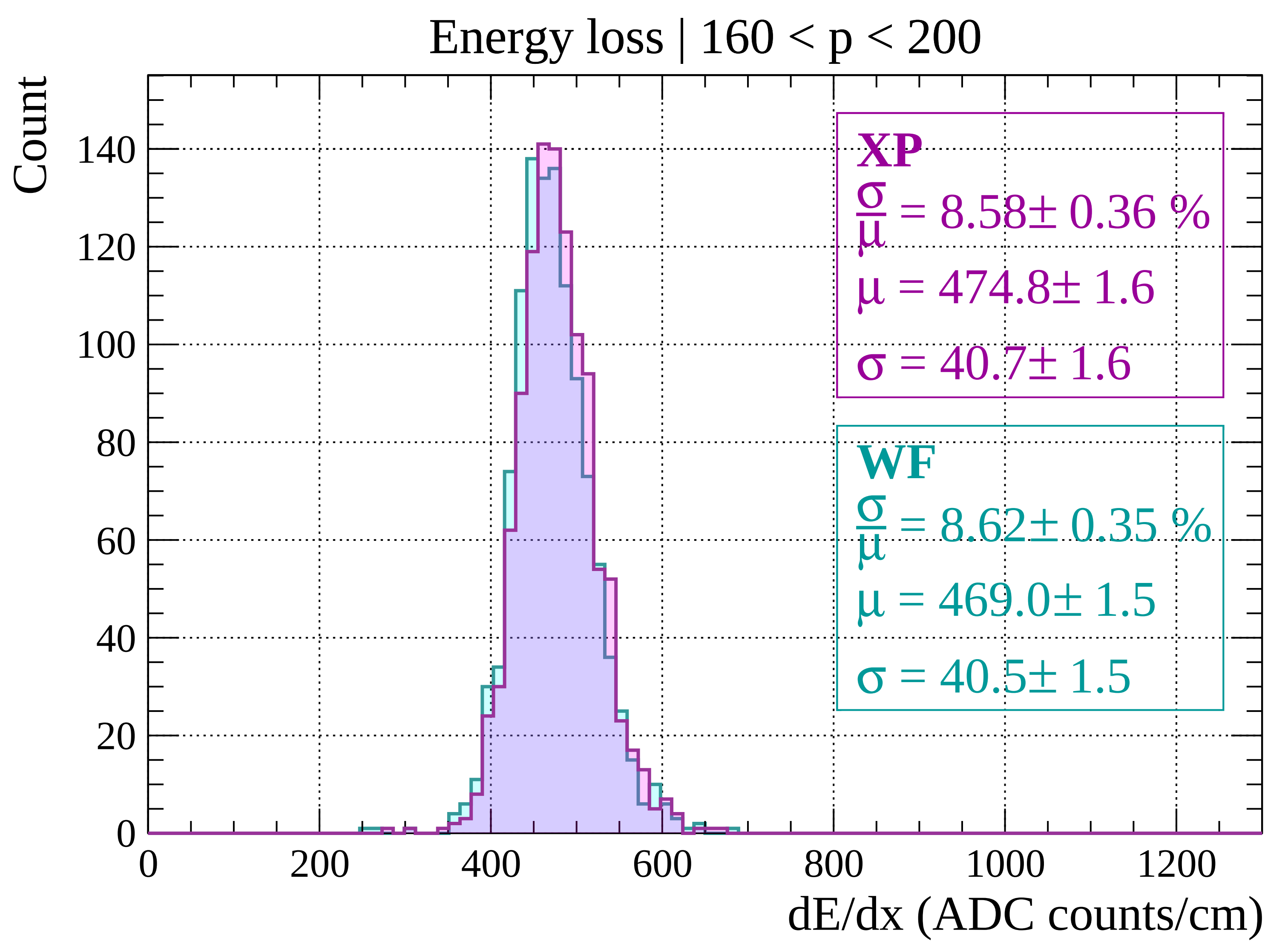
Energy loss | 40 < p < 80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP
$\frac{\sigma}{\mu}$ = 7.18± 0.39 %
μ = 549.3± 2.2
σ = 39.4± 2.0
WF
$\frac{\sigma}{\mu}$ = 7.62± 0.42 %
μ = 542.9± 2.4
σ = 41.4± 2.1



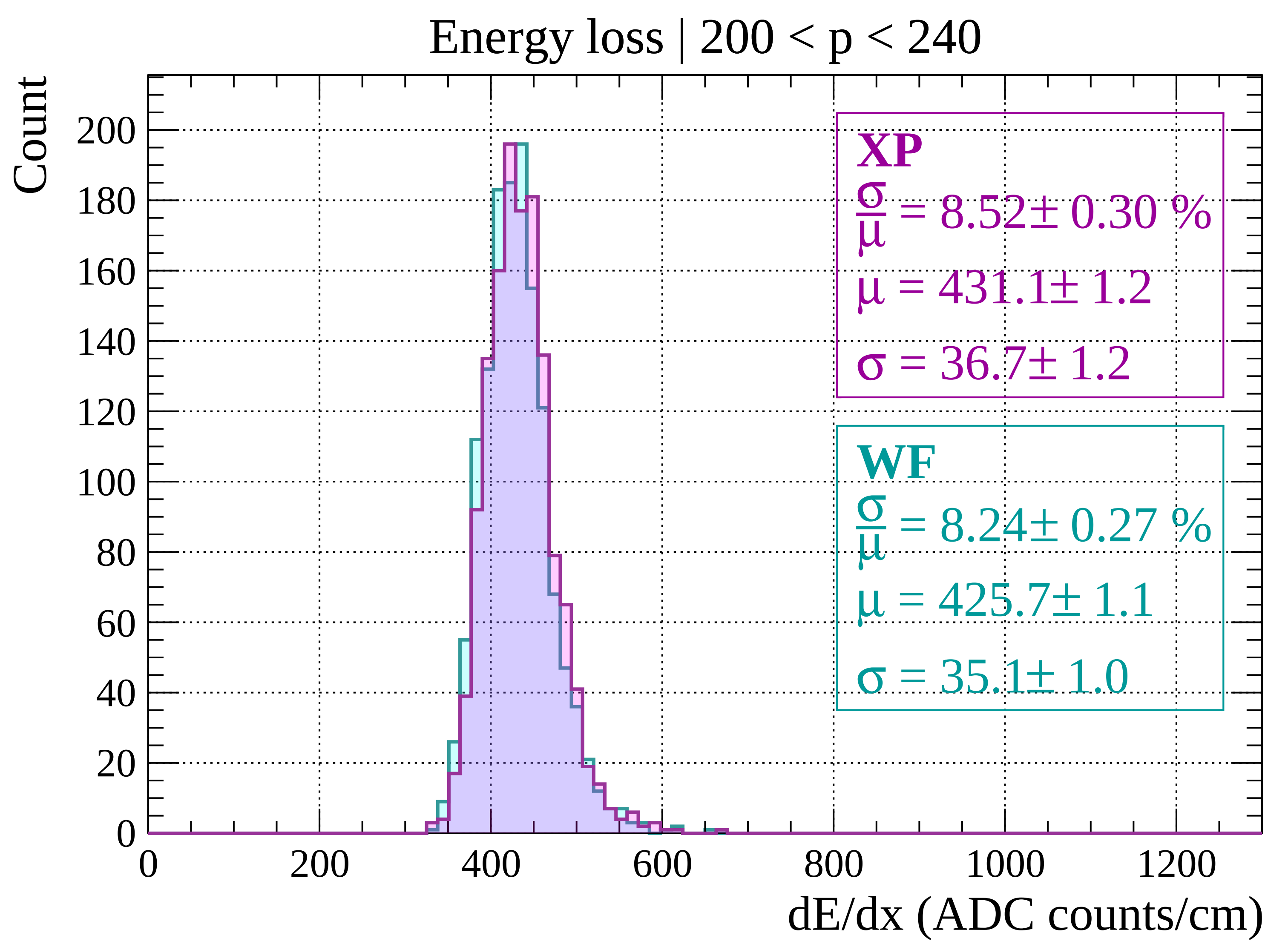
Energy loss | 80 < p < 120
Count
XP
σ/μ = 14.23± 1.49 %
μ = 628.8± 9.4
σ = 89.5± 8.0
WF
σ/μ = 13.30± 1.49 %
μ = 602.6± 8.0
σ = 80.1± 7.9
0
5
10
15
20
25
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

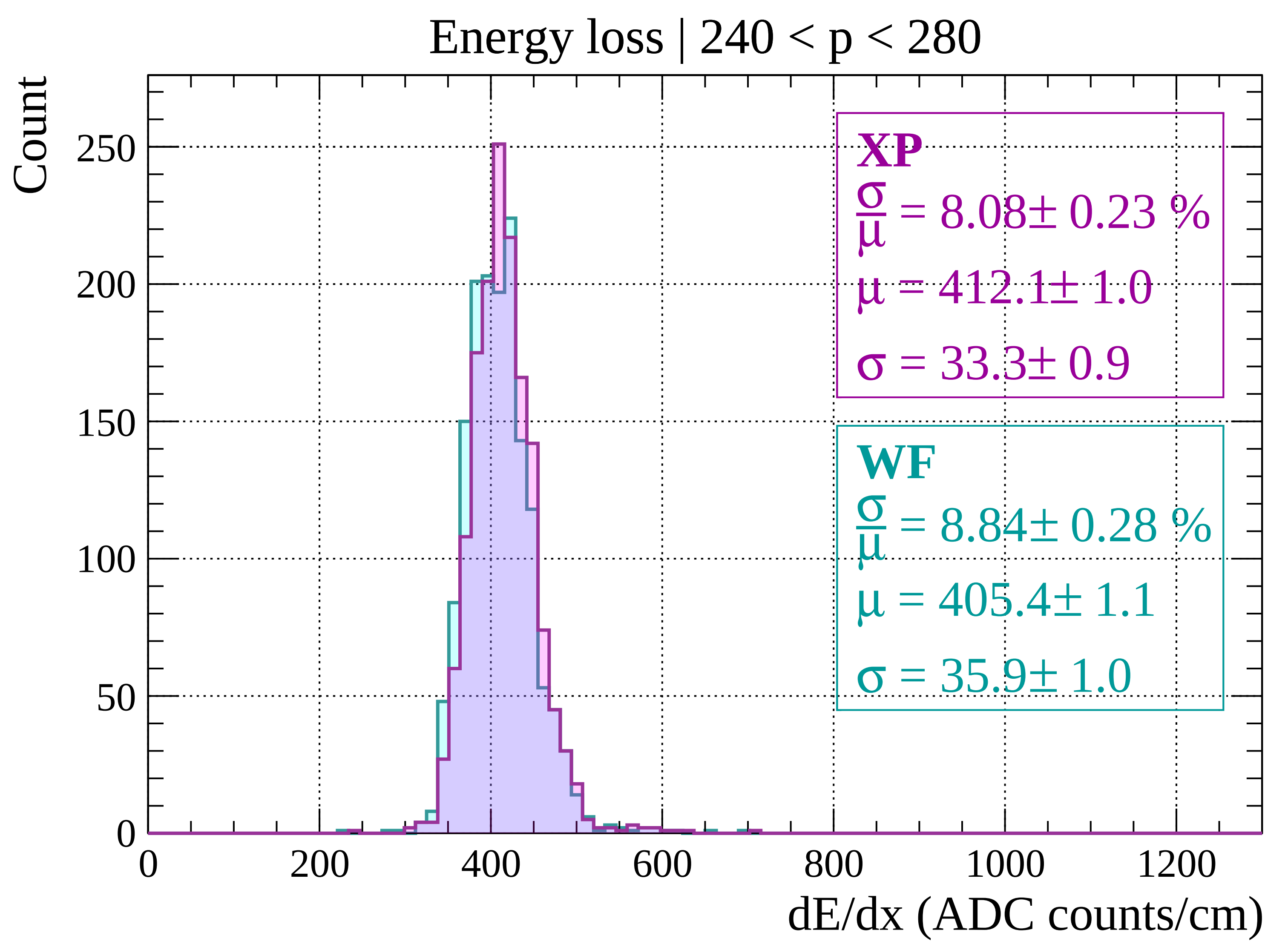


Energy loss | 120 < p < 160
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP
σ/μ = 10.51± 0.53 %
μ = 562.1± 2.7
σ = 59.1± 2.7
WF
σ/μ = 10.95± 0.58 %
μ = 557.4± 2.8
σ = 61.0± 2.9

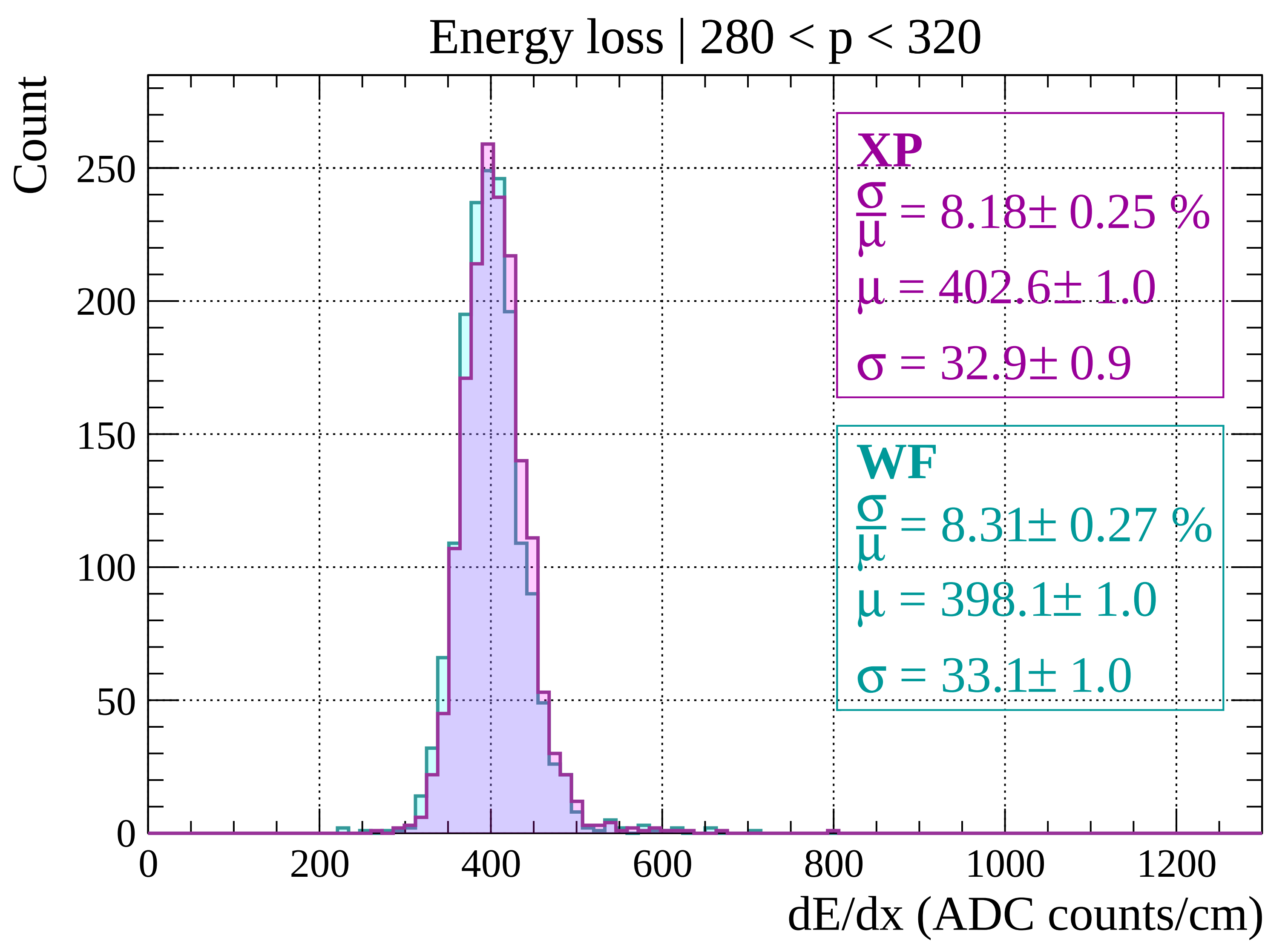


Energy loss | 160 < p < 200
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.58± 0.36 %
μ = 474.8± 1.6
σ = 40.7± 1.6
WF
σ/μ = 8.62± 0.35 %
μ = 469.0± 1.5
σ = 40.5± 1.5

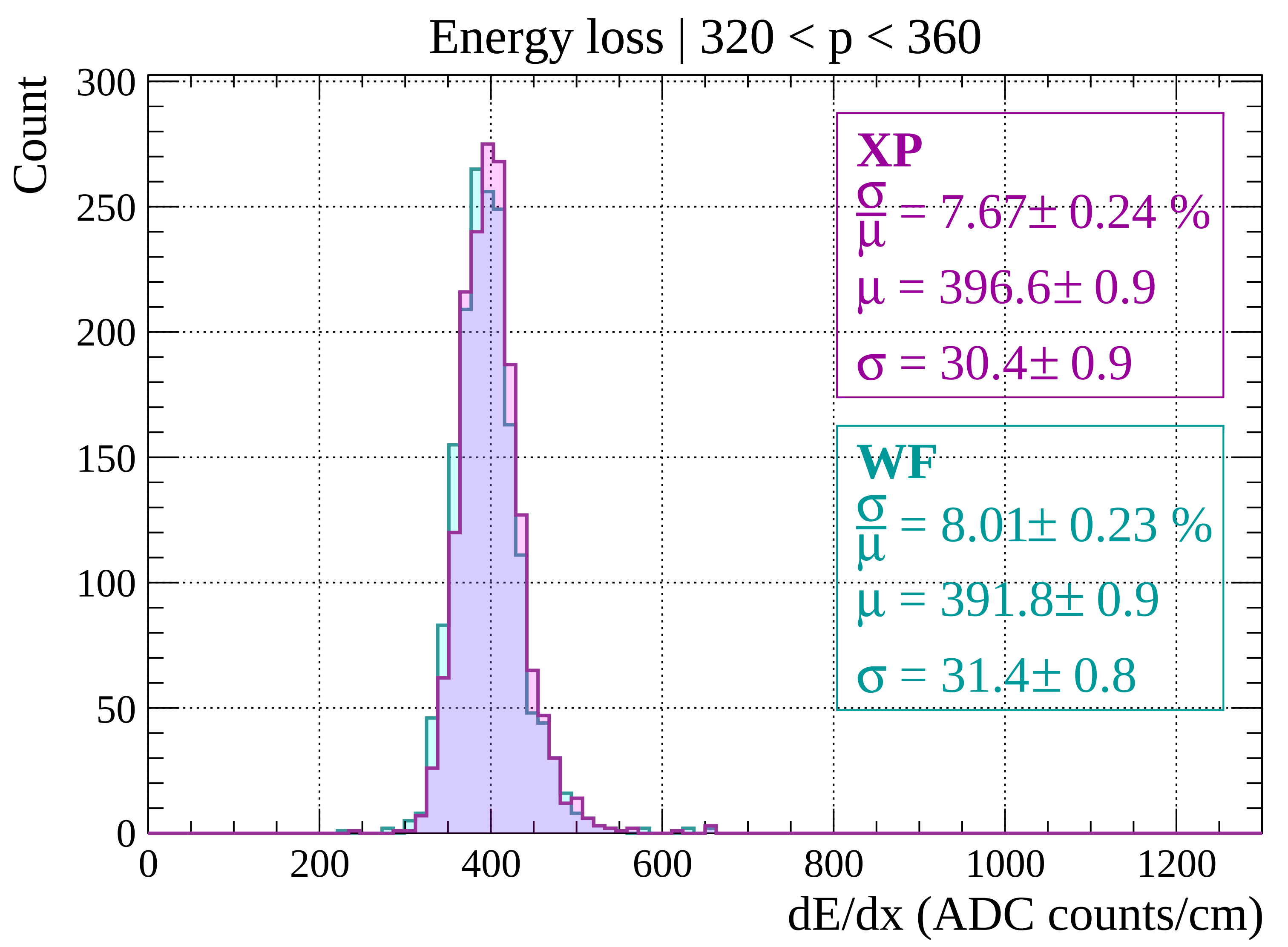

Energy loss | 200 < p < 240
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.52± 0.30 %
μ = 431.1± 1.2
σ = 36.7± 1.2
WF
σ/μ = 8.24± 0.27 %
μ = 425.7± 1.1
σ = 35.1± 1.0
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200



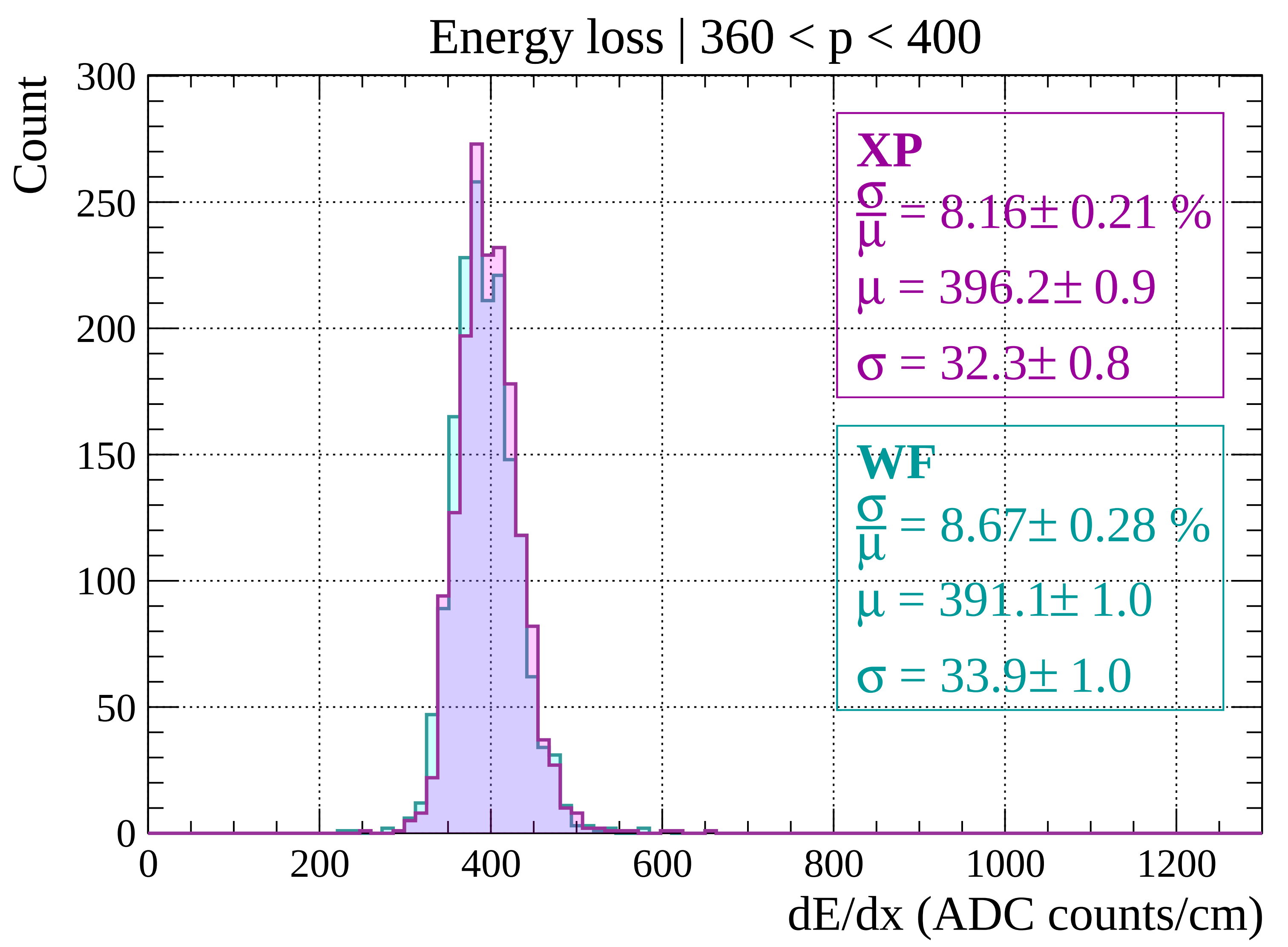
Energy loss | 240 < p < 280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 8.08± 0.23 %
μ = 412.1± 1.0
σ = 33.3± 0.9
WF
σ/μ = 8.84± 0.28 %
μ = 405.4± 1.1
σ = 35.9± 1.0

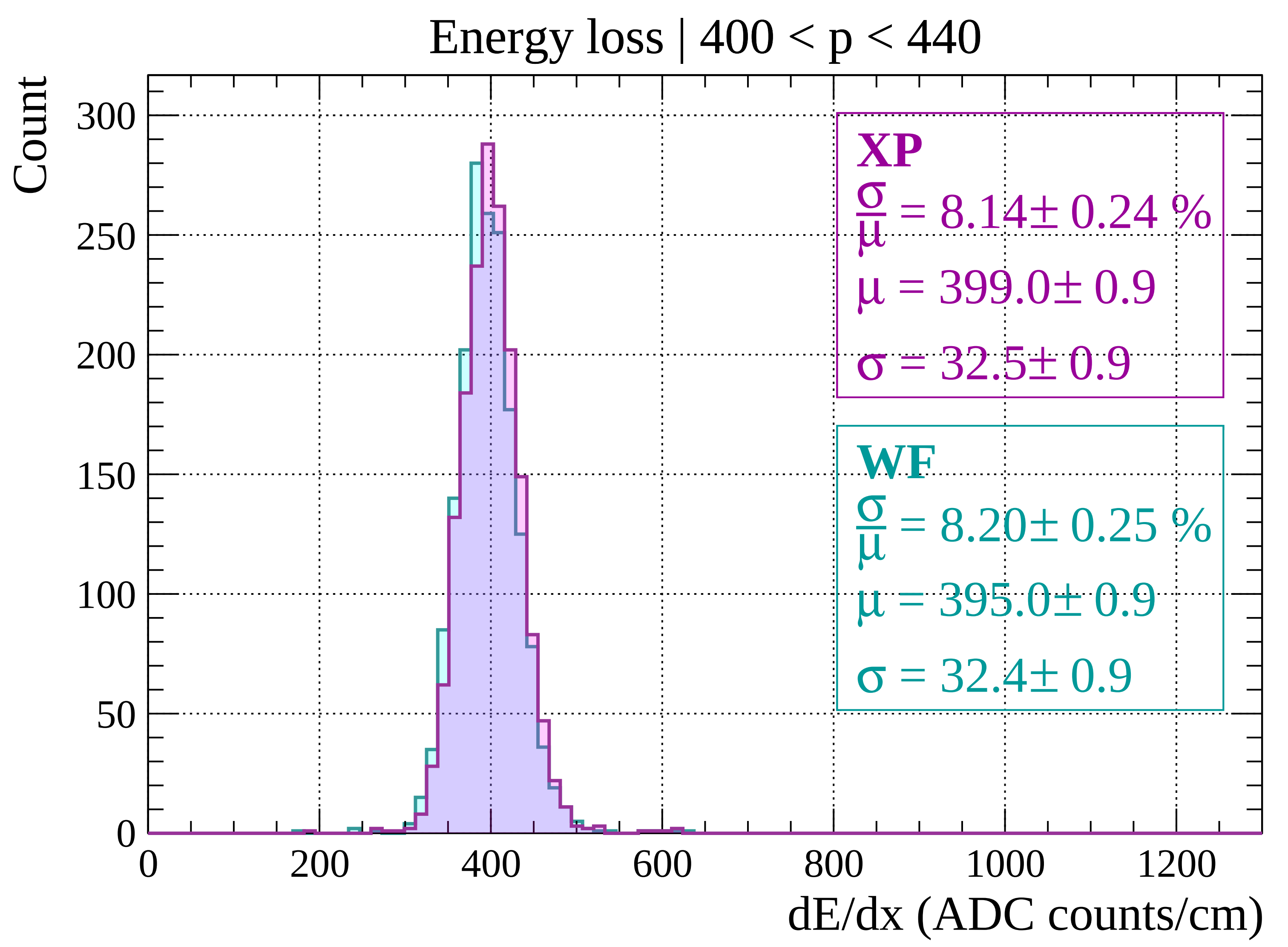


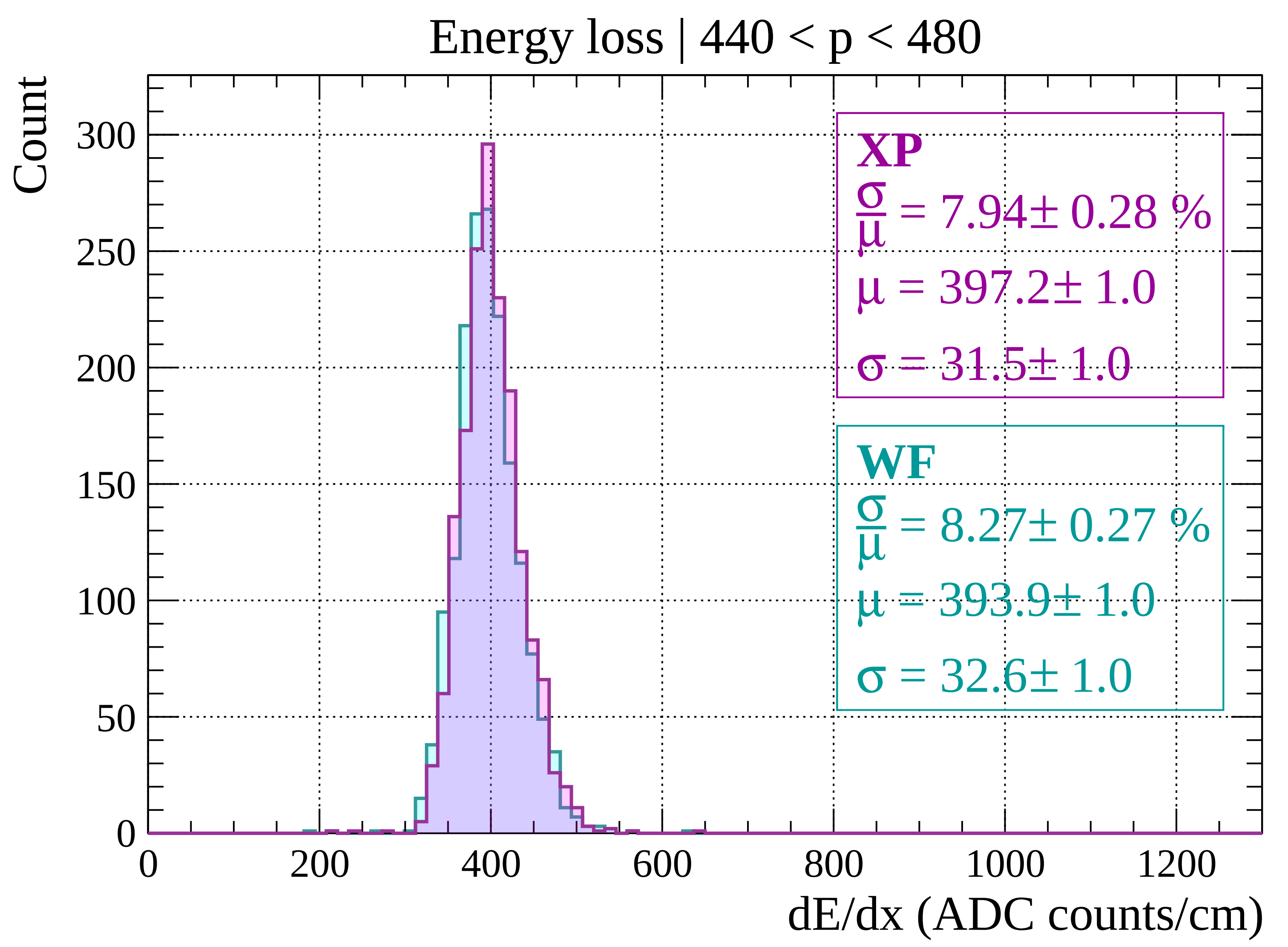
Energy loss | 280 < p < 320
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 8.18± 0.25 %
μ = 402.6± 1.0
σ = 32.9± 0.9
WF
σ/μ = 8.31± 0.27 %
μ = 398.1± 1.0
σ = 33.1± 1.0

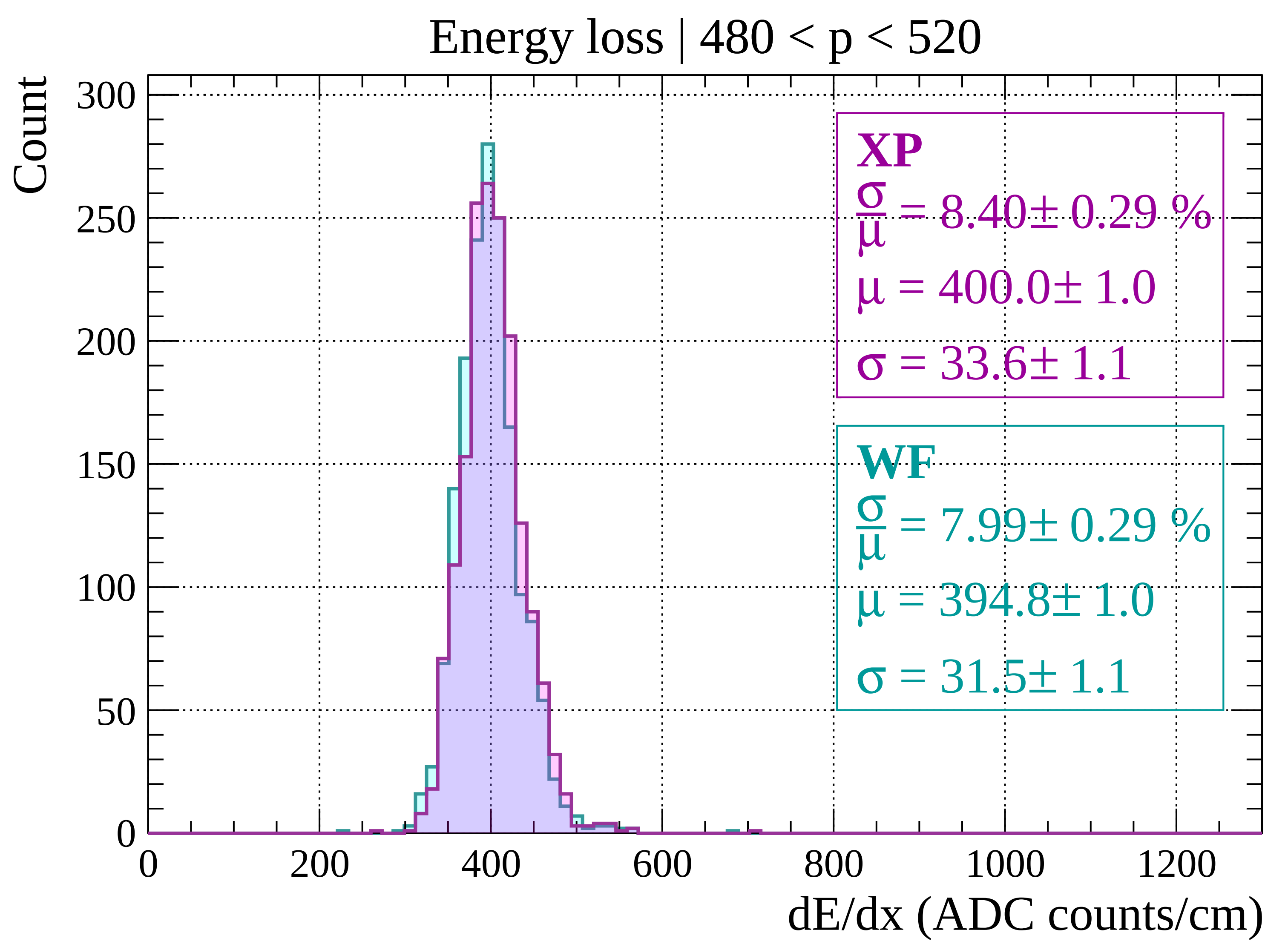


Energy loss | 320 < p < 360
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 7.67 \pm 0.24$ %
$\mu = 396.6 \pm 0.9$
$\sigma = 30.4 \pm 0.9$
WF
$\frac{\sigma}{\mu} = 8.01 \pm 0.23$ %
$\mu = 391.8 \pm 0.9$
$\sigma = 31.4 \pm 0.8$

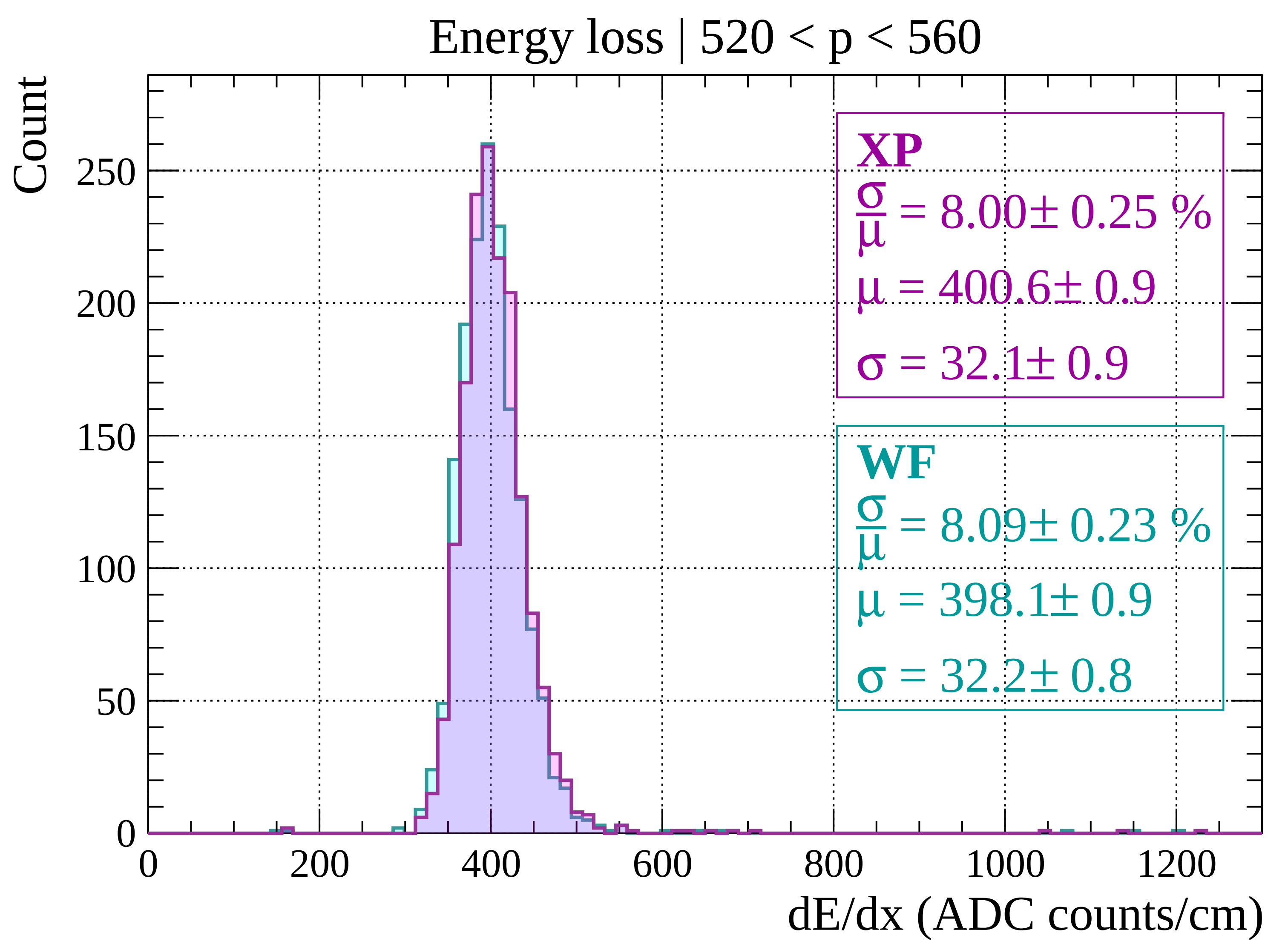

Energy loss | 360 < p < 400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 8.16± 0.21 %
μ = 396.2± 0.9
σ = 32.3± 0.8
WF
σ/μ = 8.67± 0.28 %
μ = 391.1± 1.0
σ = 33.9± 1.0


Energy loss | 400 < p < 440
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.14± 0.24 %
μ = 399.0± 0.9
σ = 32.5± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.20± 0.25 %
μ = 395.0± 0.9
σ = 32.4± 0.9
0
50
100
150
200
250
300
400
600
800
1000
1200

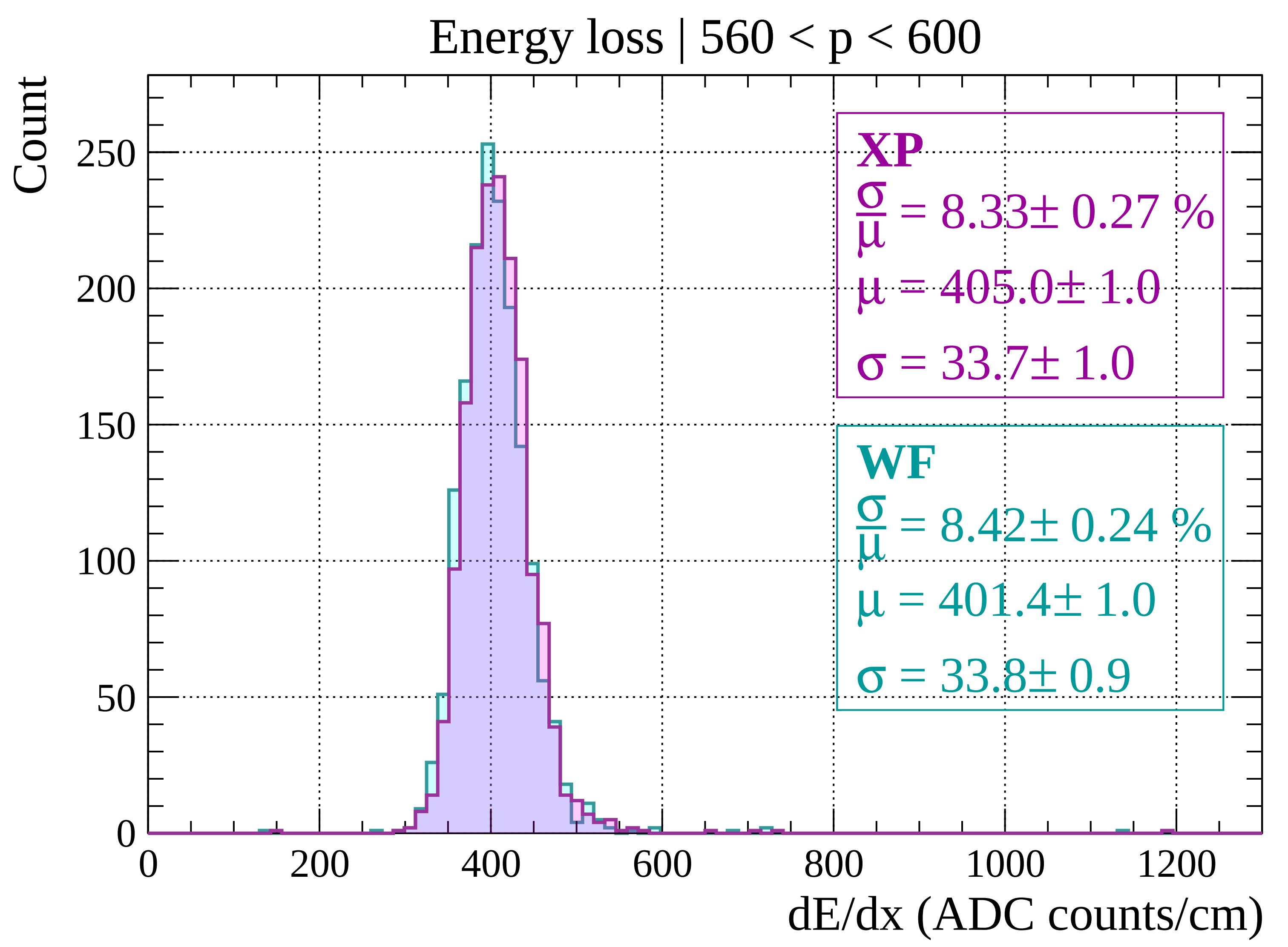

Energy loss | 440 < p < 480
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 7.94± 0.28 %
μ = 397.2± 1.0
σ = 31.5± 1.0
WF
σ/μ = 8.27± 0.27 %
μ = 393.9± 1.0
σ = 32.6± 1.0



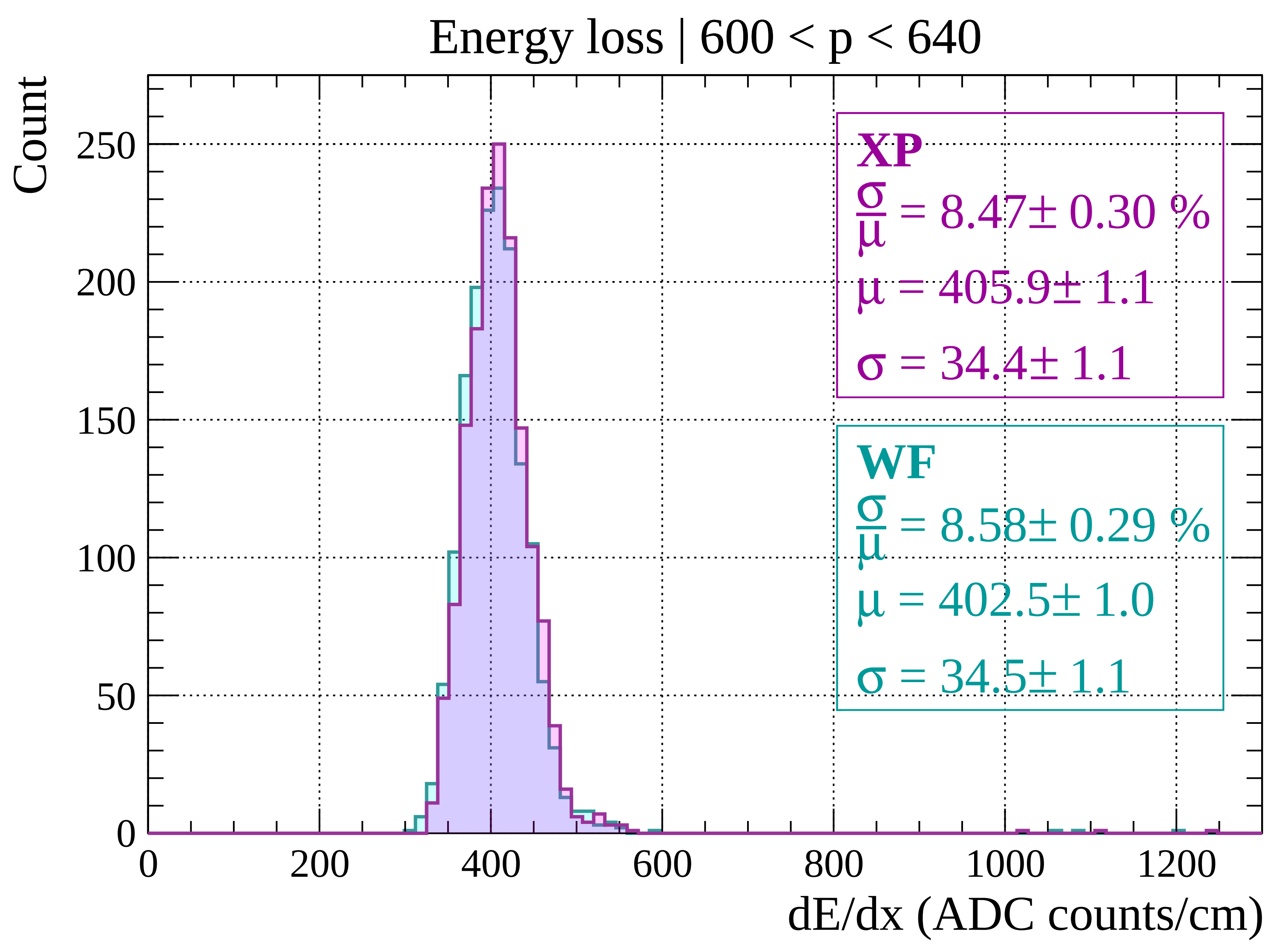
Energy loss | 480 < p < 520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 8.40 ± 0.29 %
μ = 400.0 ± 1.0
σ = 33.6 ± 1.1
WF
σ/μ = 7.99 ± 0.29 %
μ = 394.8 ± 1.0
σ = 31.5 ± 1.1



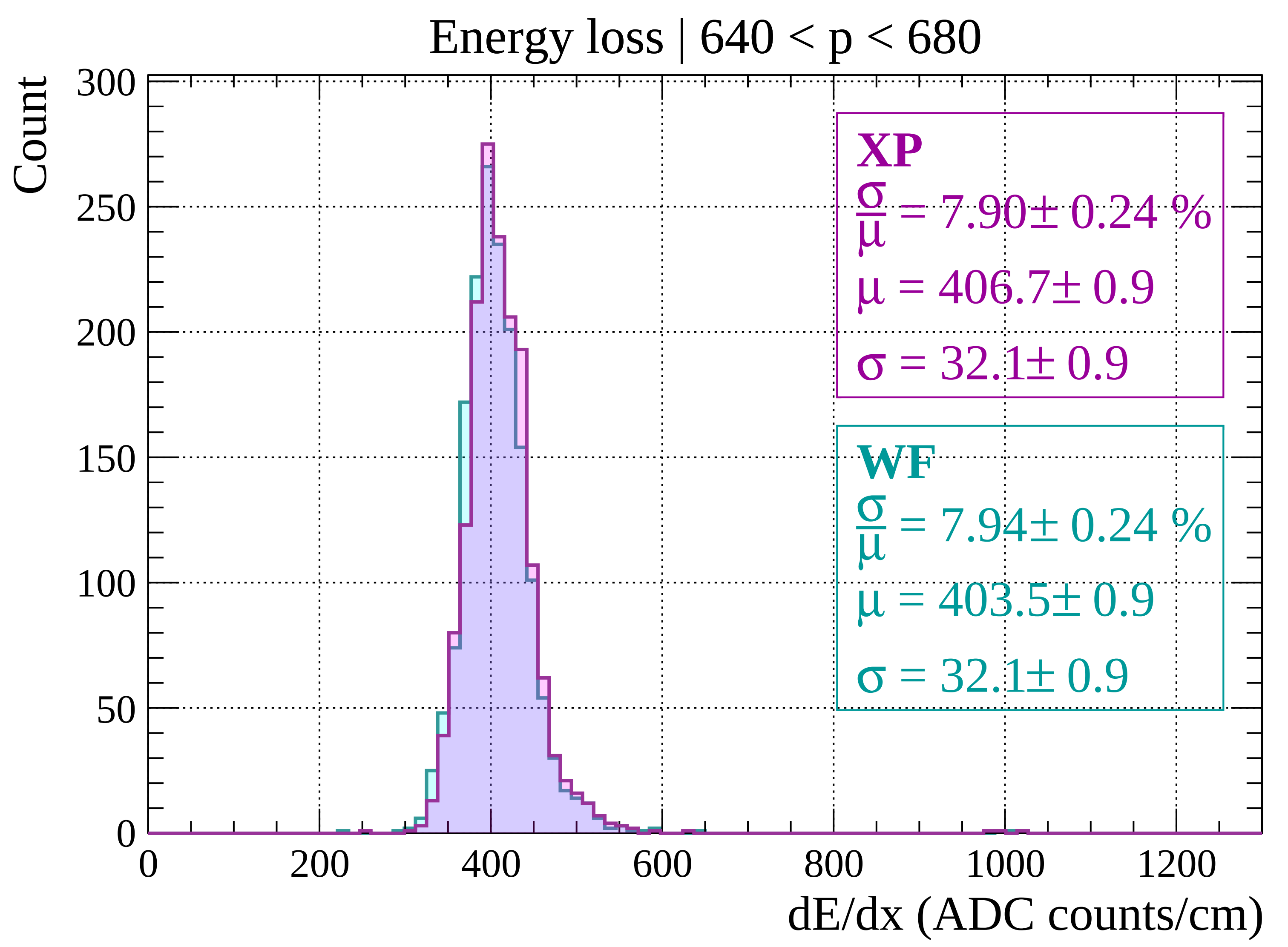
Energy loss | 520 < p < 560
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 8.00± 0.25 %
μ = 400.6± 0.9
σ = 32.1± 0.9
WF
σ/μ = 8.09± 0.23 %
μ = 398.1± 0.9
σ = 32.2± 0.8



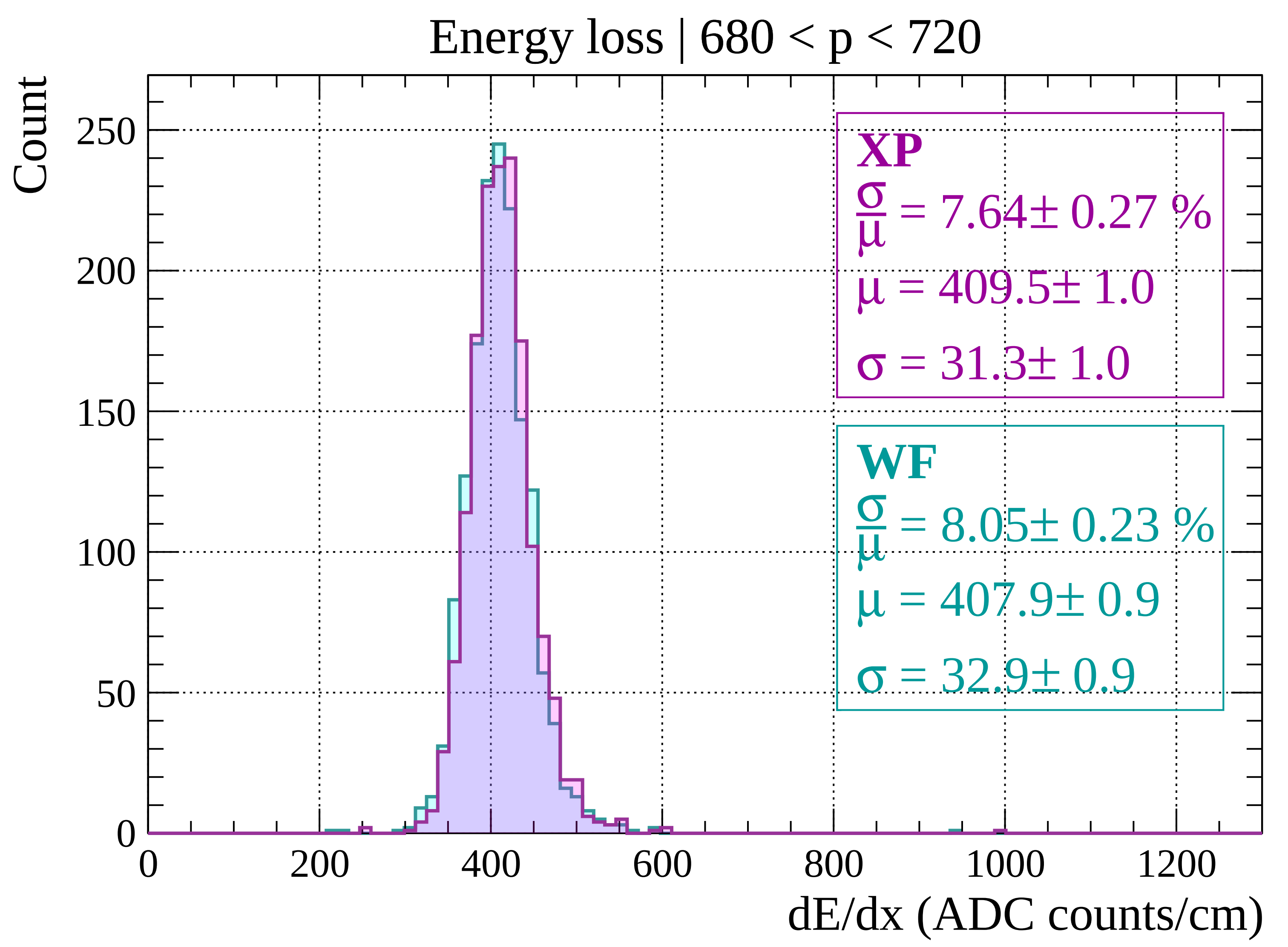
Energy loss | 560 < p < 600
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.33± 0.27 %
μ = 405.0± 1.0
σ = 33.7± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.42± 0.24 %
μ = 401.4± 1.0
σ = 33.8± 0.9



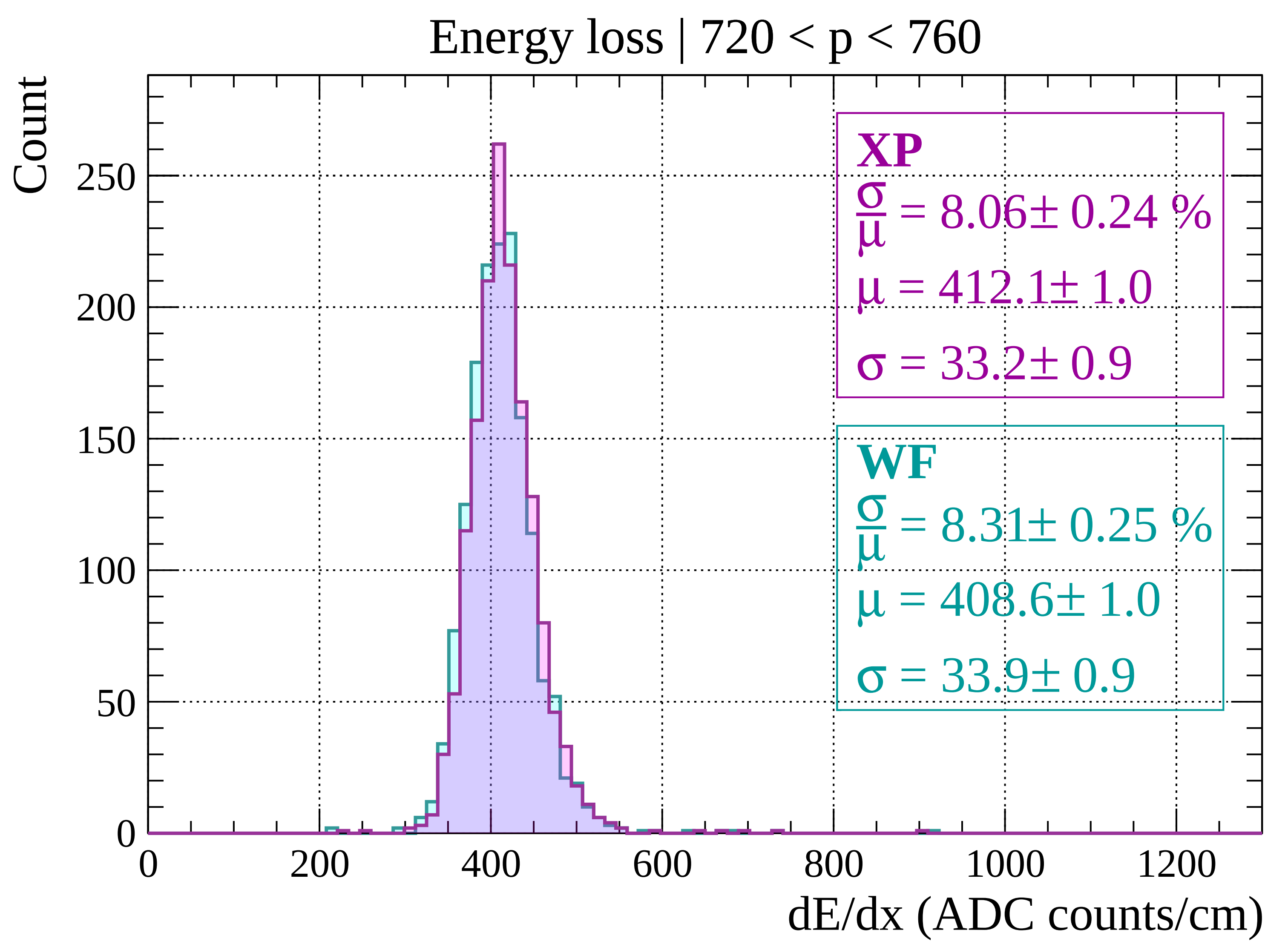
Energy loss | 600 < p < 640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.47± 0.30 %
μ = 405.9± 1.1
σ = 34.4± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.58± 0.29 %
μ = 402.5± 1.0
σ = 34.5± 1.1

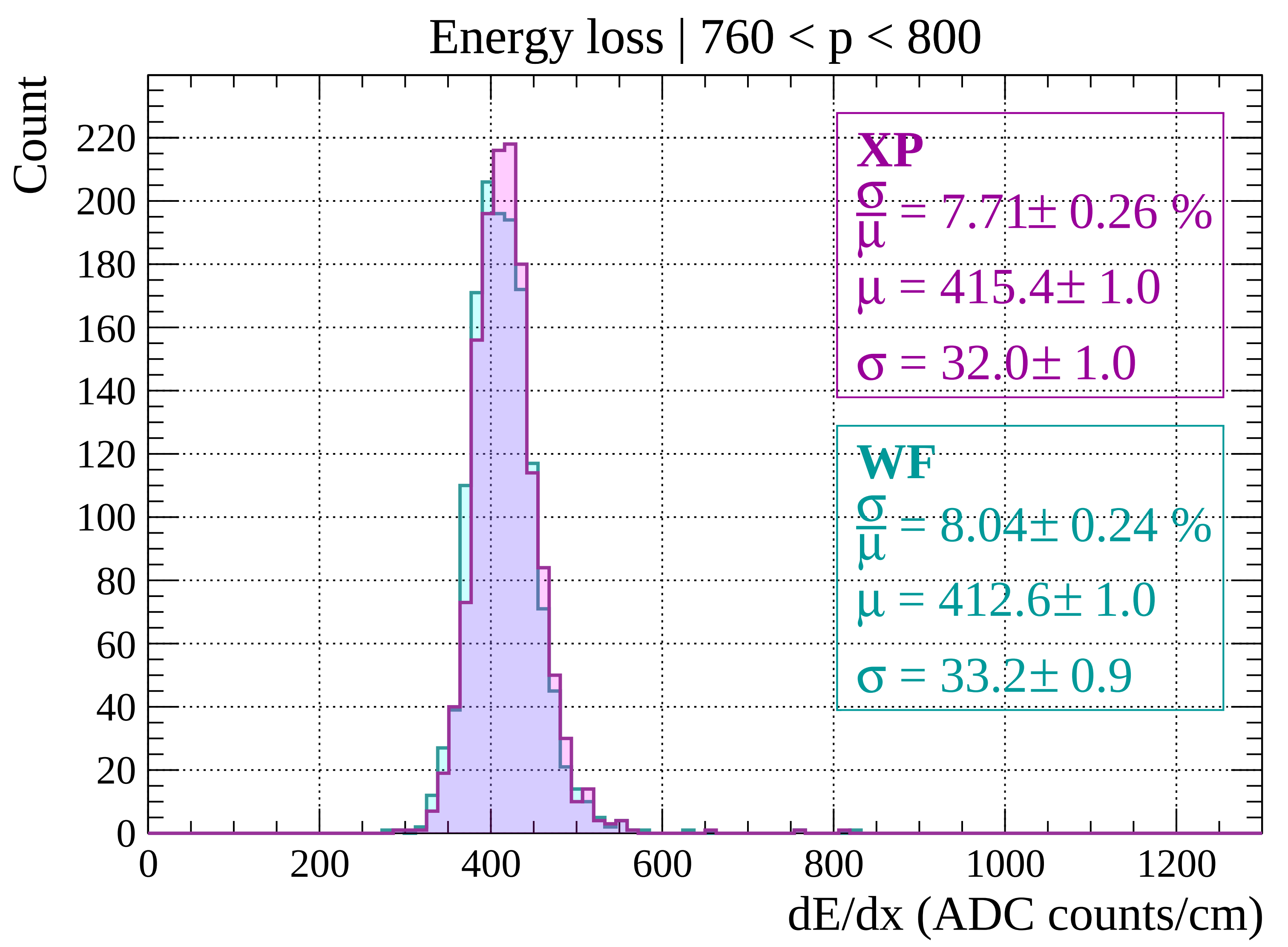


Energy loss | 640 < p < 680
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.90± 0.24 %
μ = 406.7± 0.9
σ = 32.1± 0.9
WF
$\frac{\sigma}{\mu}$ = 7.94± 0.24 %
μ = 403.5± 0.9
σ = 32.1± 0.9

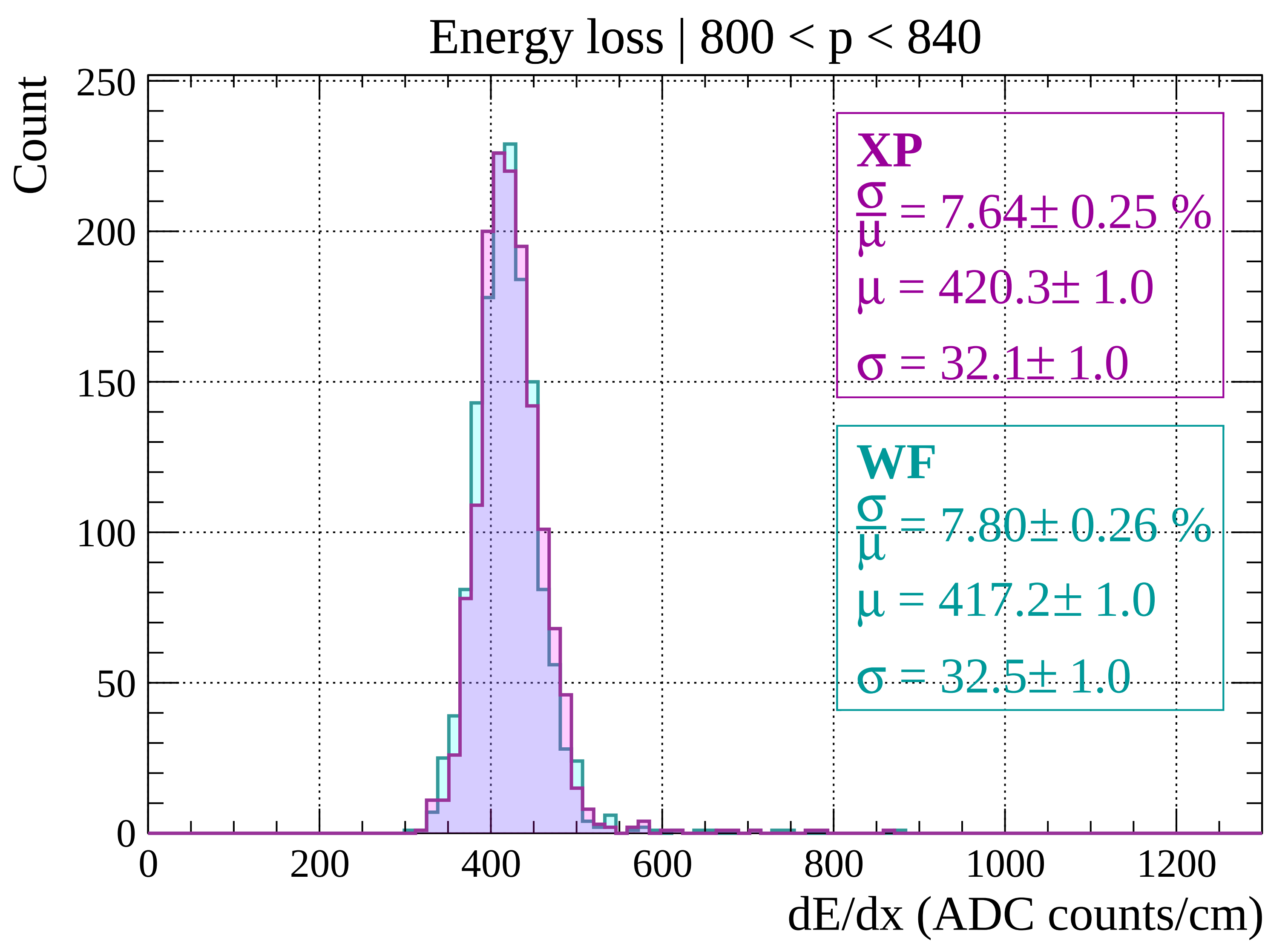


Energy loss | 680 < p < 720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
250
XP
σ/μ = 7.64± 0.27 %
μ = 409.5± 1.0
σ = 31.3± 1.0
WF
σ/μ = 8.05± 0.23 %
μ = 407.9± 0.9
σ = 32.9± 0.9

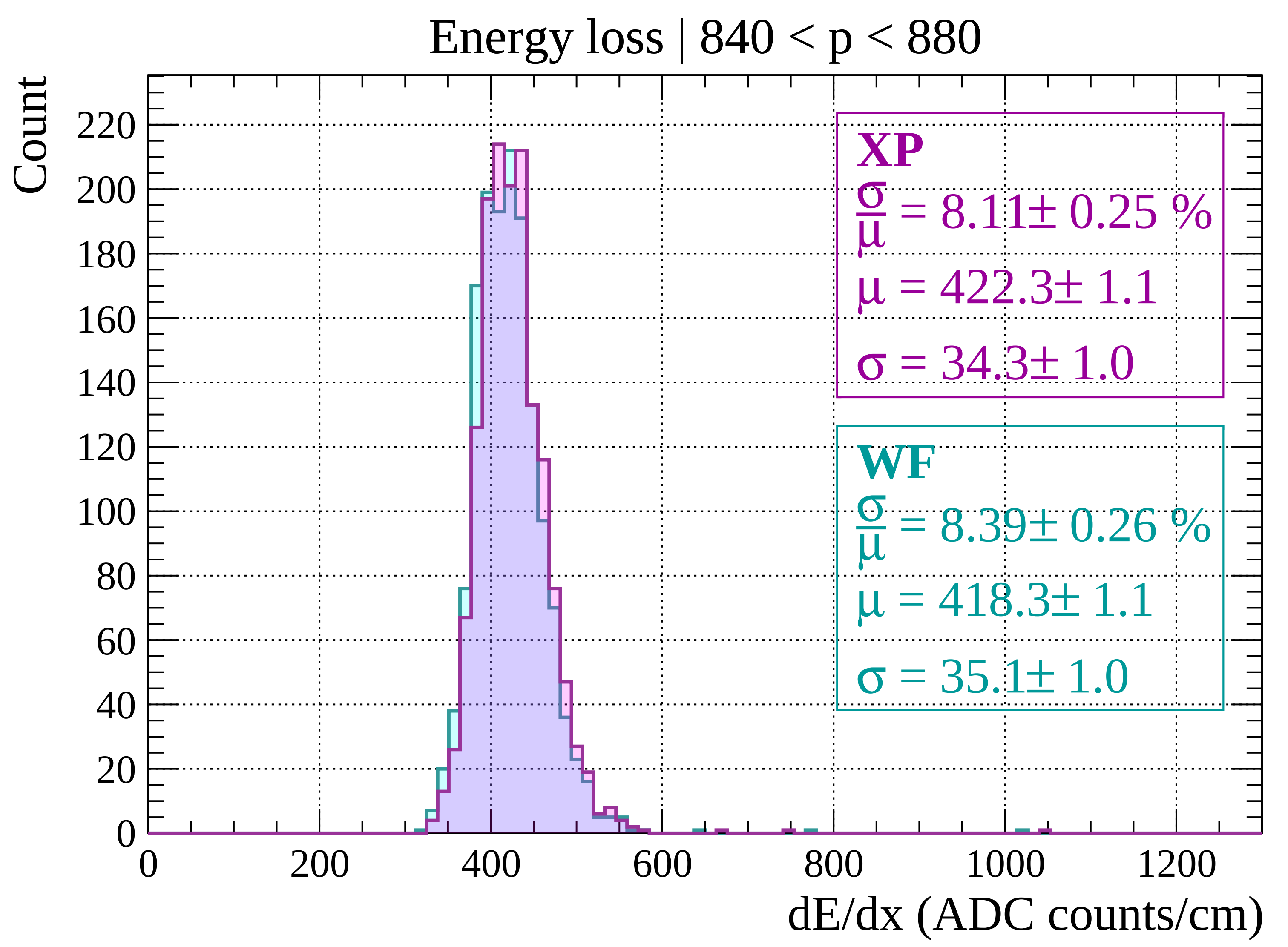

Energy loss | 720 < p < 760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 8.06± 0.24 %
μ = 412.1± 1.0
σ = 33.2± 0.9
WF
σ/μ = 8.31± 0.25 %
μ = 408.6± 1.0
σ = 33.9± 0.9



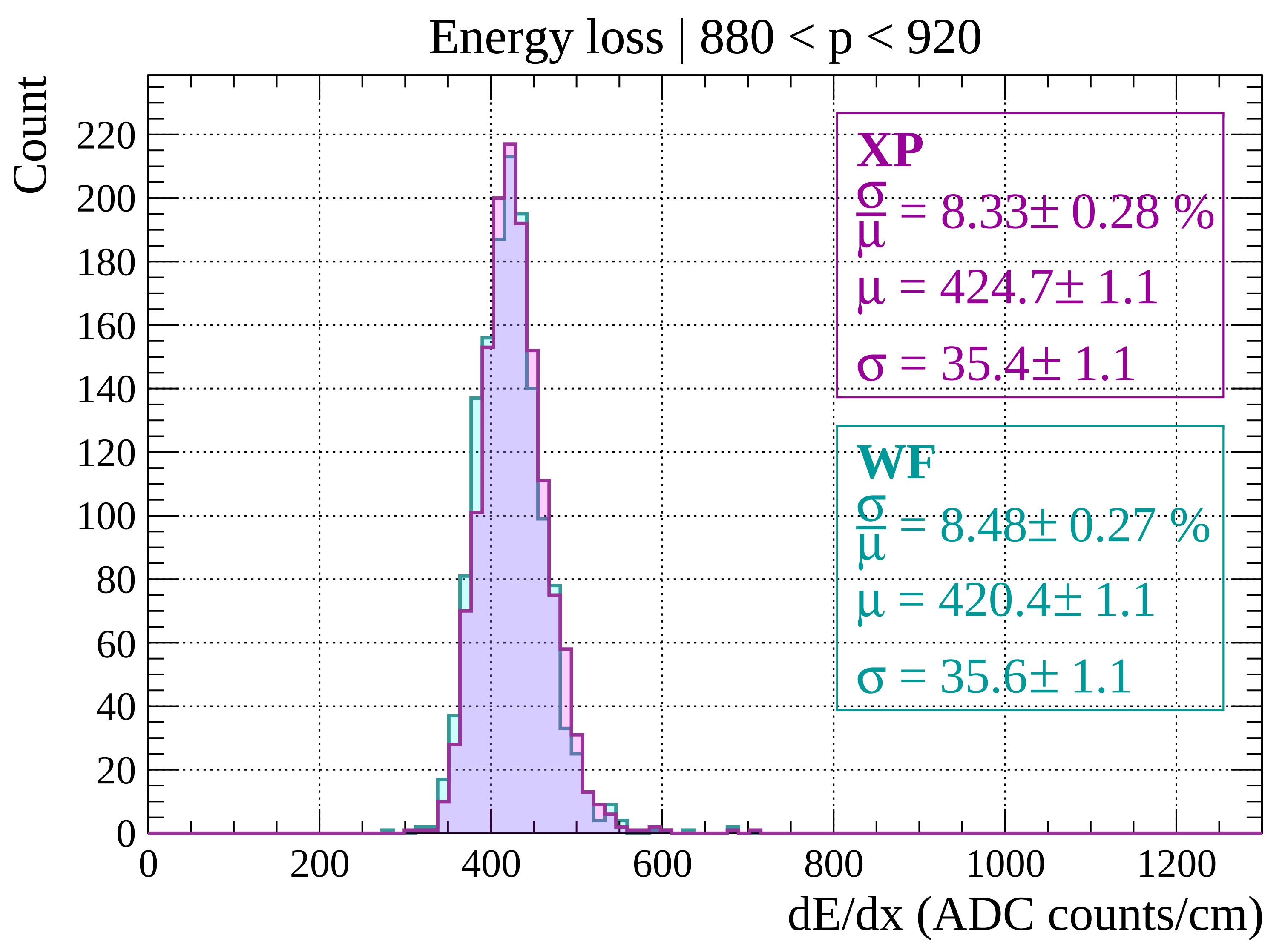
Energy loss | 760 < p < 800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
σ/μ = 7.71 ± 0.26 %
μ = 415.4 ± 1.0
σ = 32.0 ± 1.0
WF
σ/μ = 8.04 ± 0.24 %
μ = 412.6 ± 1.0
σ = 33.2 ± 0.9



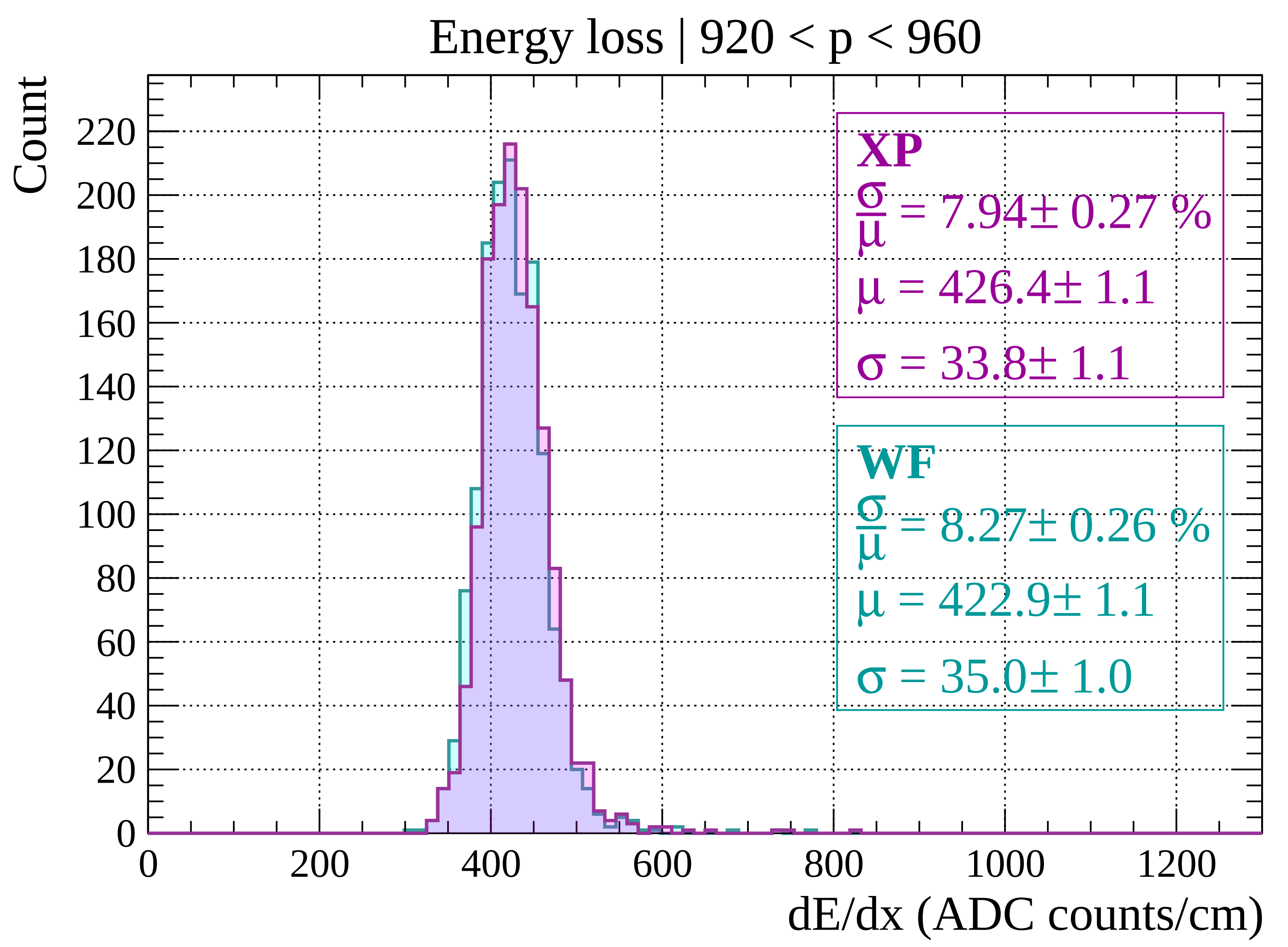
Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP
$\frac{\sigma}{\mu} = 7.64 \pm 0.25$ %
$\mu = 420.3 \pm 1.0$
$\sigma = 32.1 \pm 1.0$
WF
$\frac{\sigma}{\mu} = 7.80 \pm 0.26$ %
$\mu = 417.2 \pm 1.0$
$\sigma = 32.5 \pm 1.0$



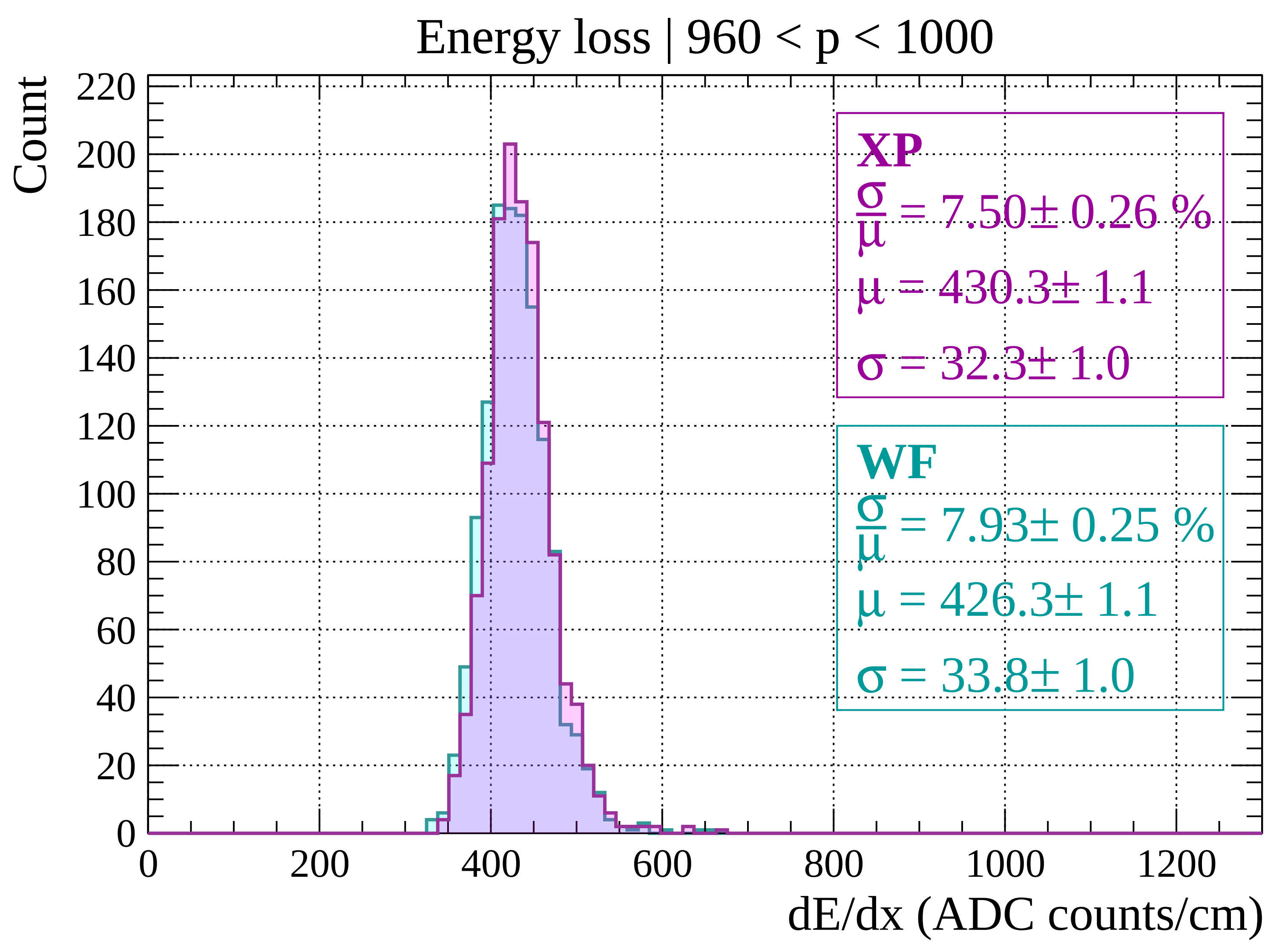
Energy loss | 840 < p < 880
Count
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.11± 0.25 %
μ = 422.3± 1.1
σ = 34.3± 1.0
WF
σ/μ = 8.39± 0.26 %
μ = 418.3± 1.1
σ = 35.1± 1.0

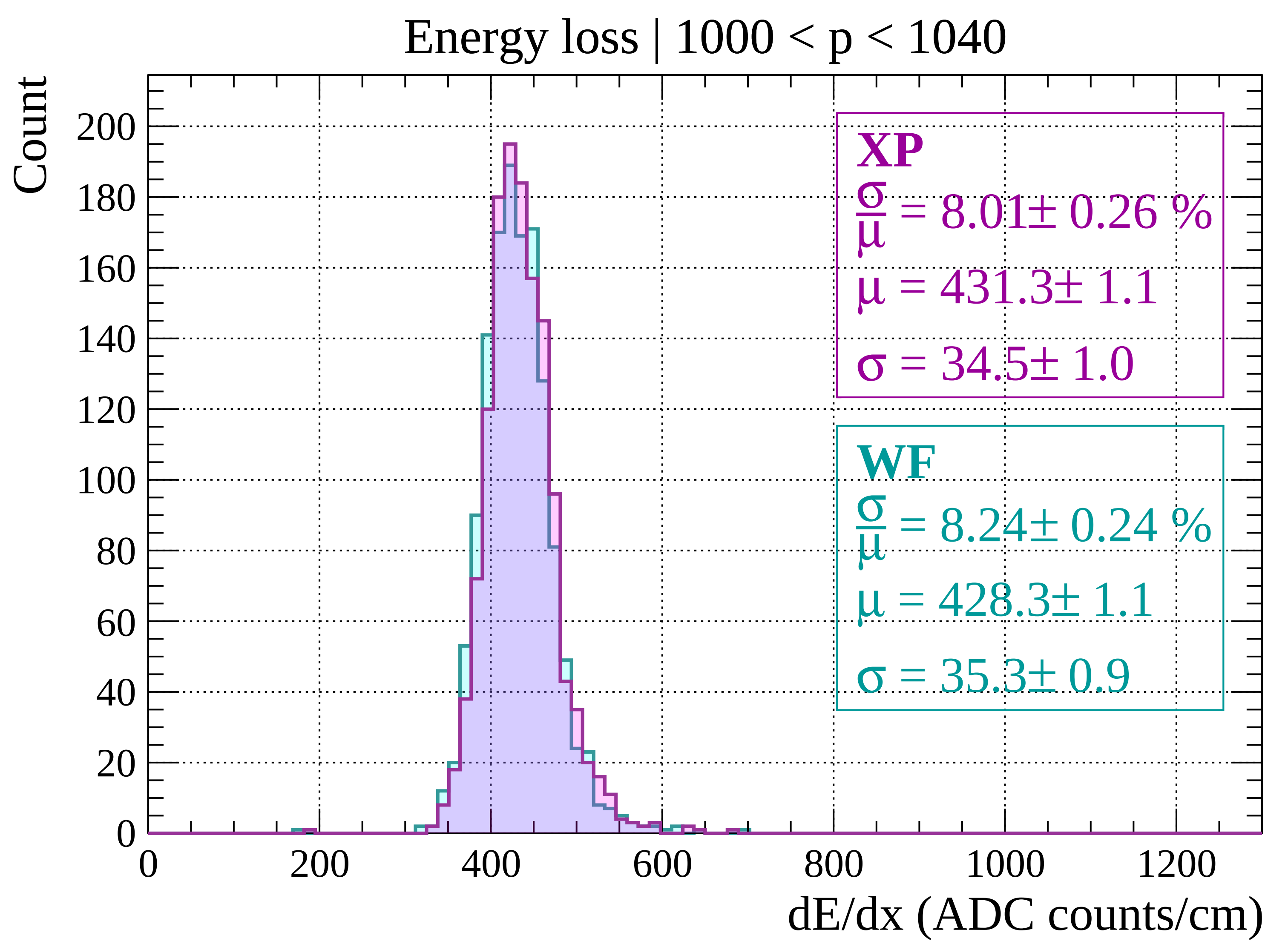


Energy loss | 880 < p < 920
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.33± 0.28 %
μ = 424.7± 1.1
σ = 35.4± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.48± 0.27 %
μ = 420.4± 1.1
σ = 35.6± 1.1

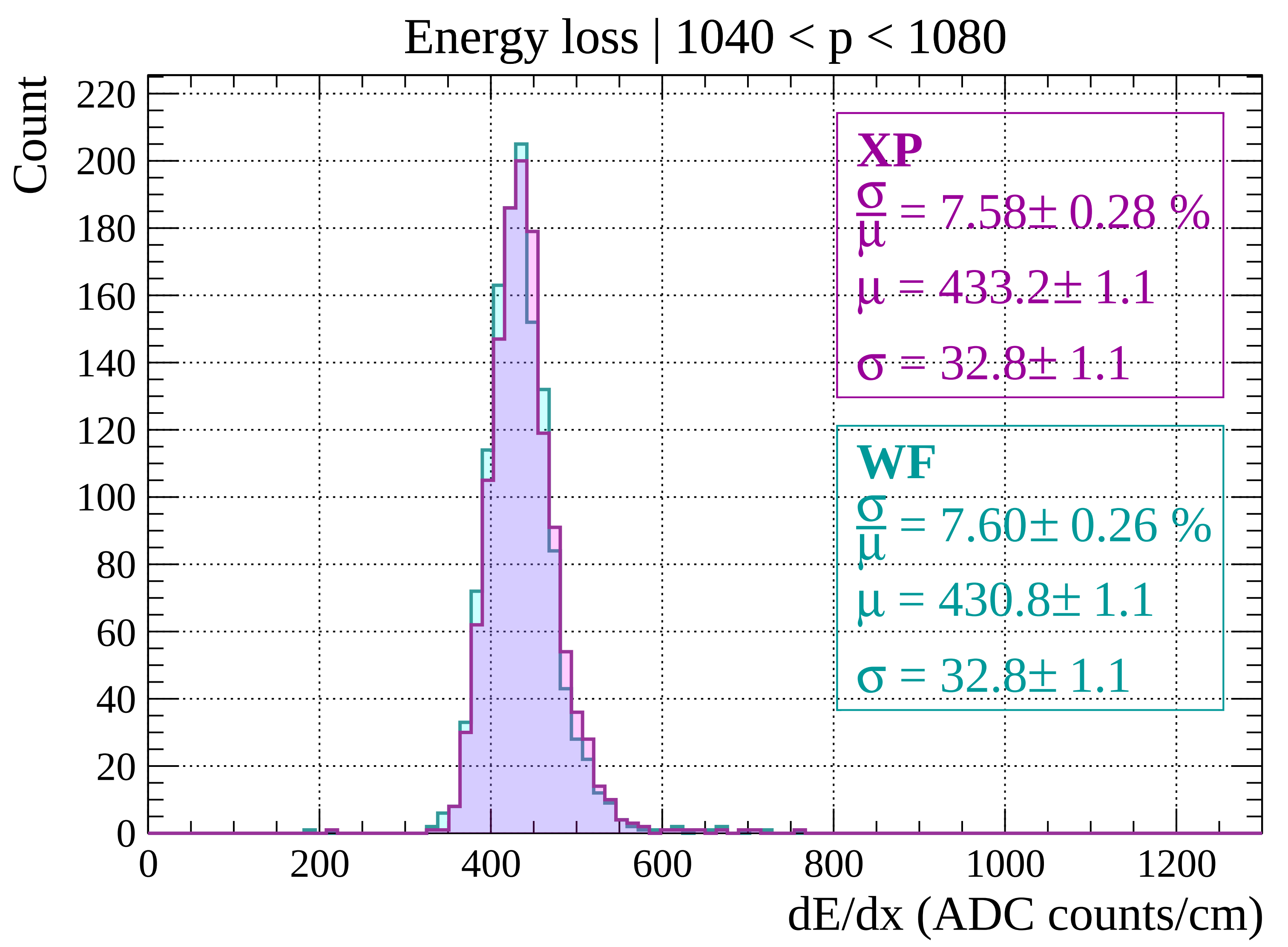


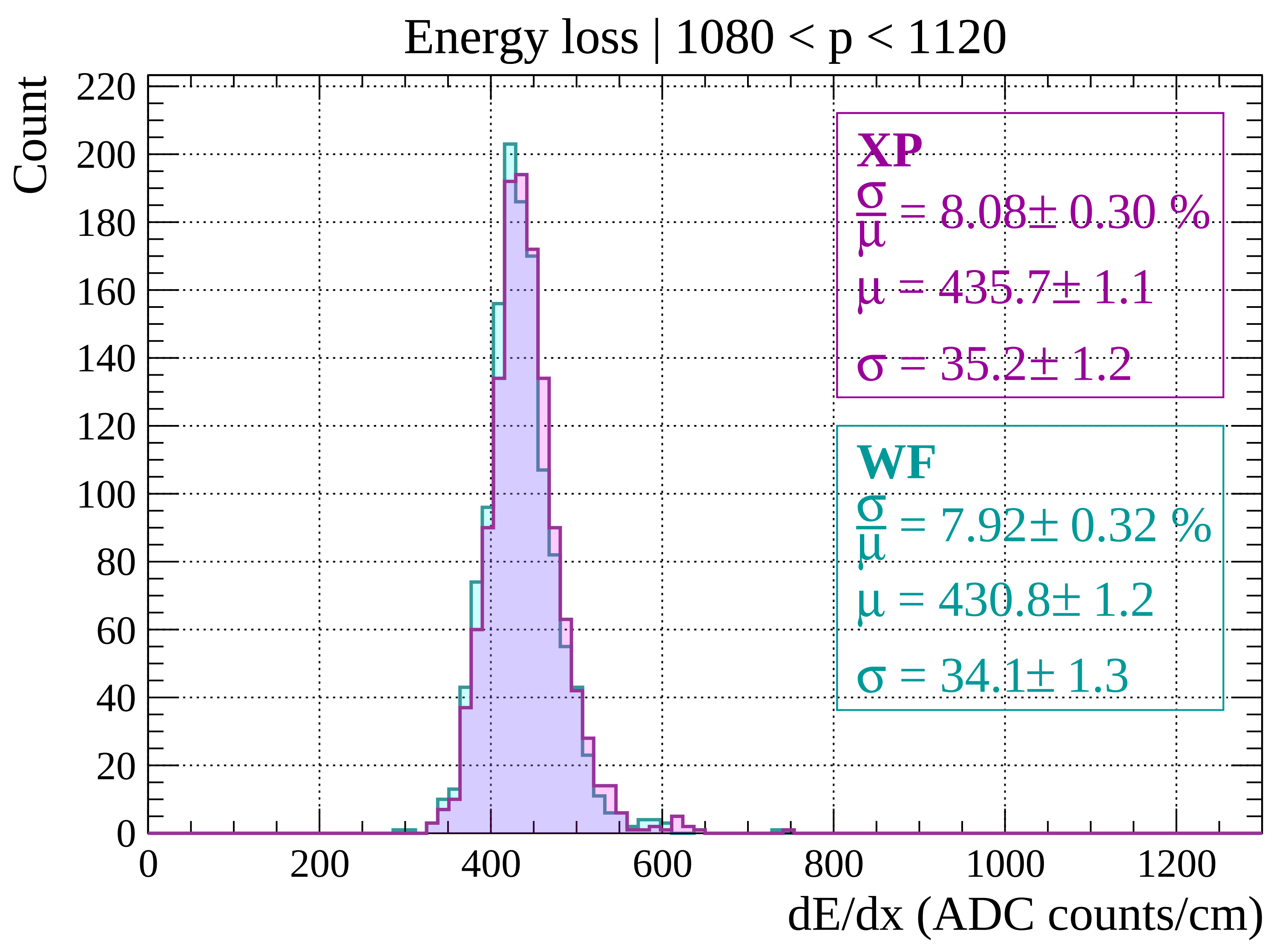
Energy loss | 920 < p < 960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
σ/μ = 7.94± 0.27 %
μ = 426.4± 1.1
σ = 33.8± 1.1
WF
σ/μ = 8.27± 0.26 %
μ = 422.9± 1.1
σ = 35.0± 1.0

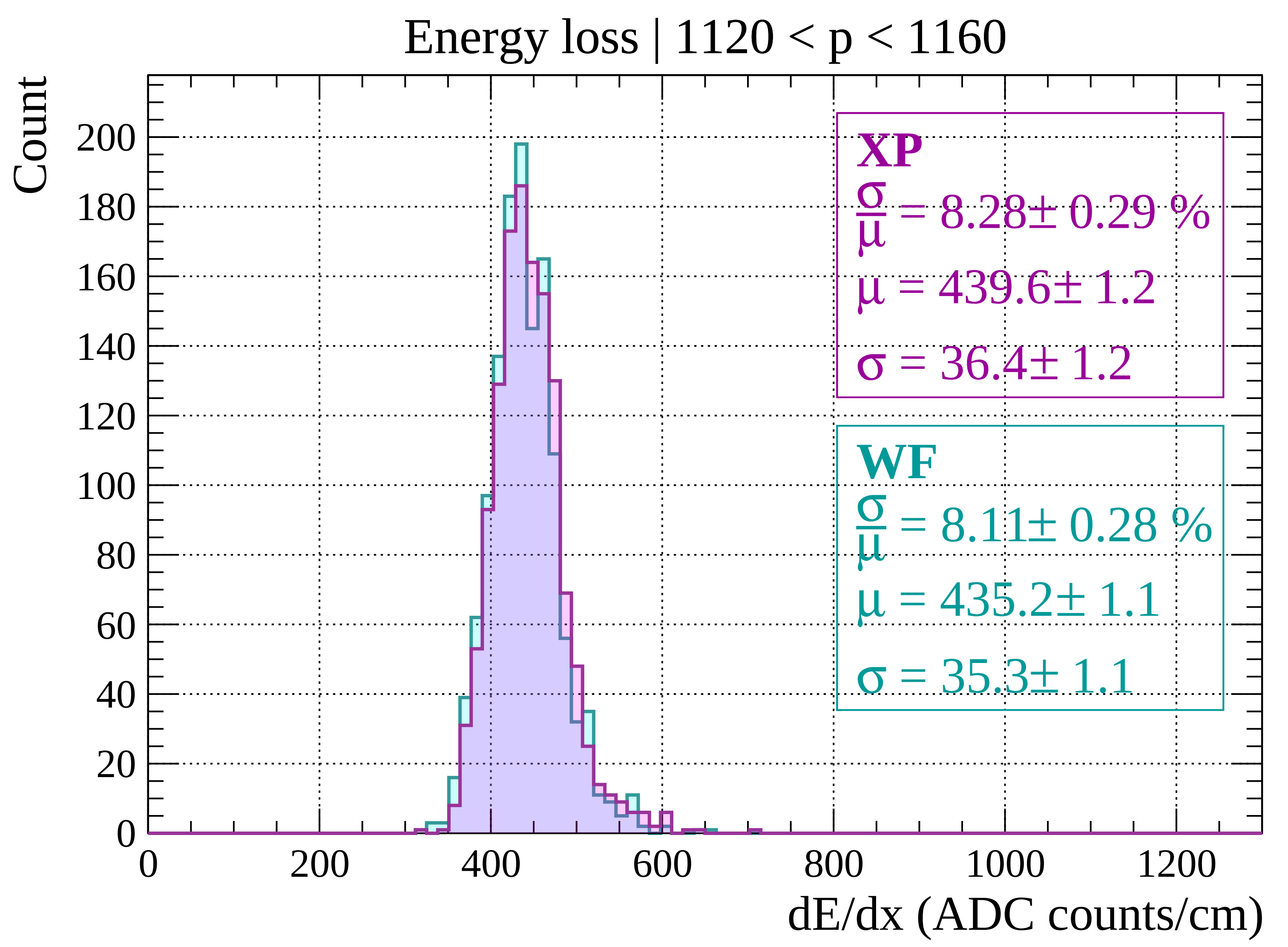

Energy loss | 960 < p < 1000
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.50± 0.26 %
μ = 430.3± 1.1
σ = 32.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 7.93± 0.25 %
μ = 426.3± 1.1
σ = 33.8± 1.0



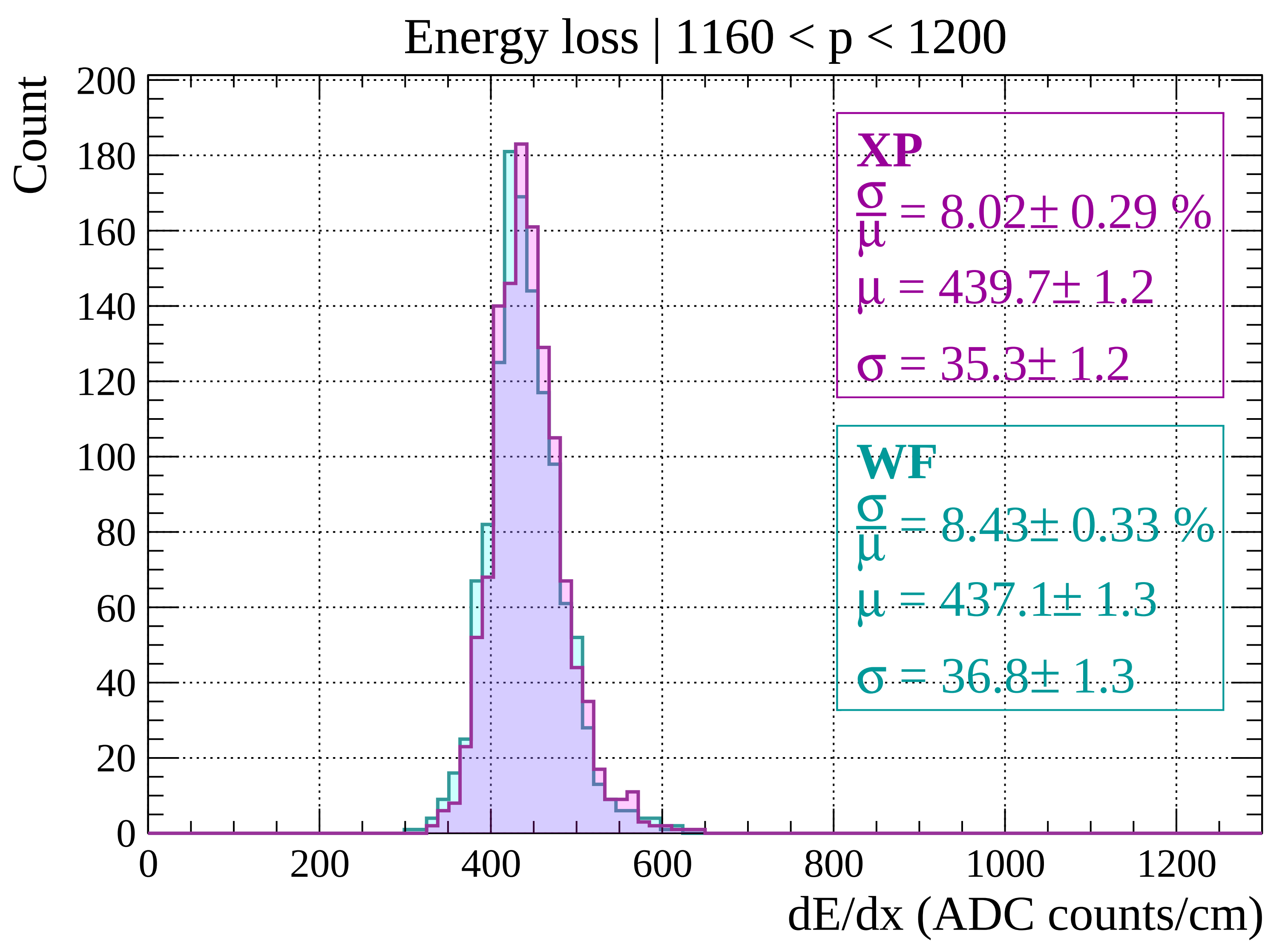
Energy loss | 1000 < p < 1040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
σ/μ = 8.01± 0.26 %
μ = 431.3± 1.1
σ = 34.5± 1.0
WF
σ/μ = 8.24± 0.24 %
μ = 428.3± 1.1
σ = 35.3± 0.9


Energy loss | 1040 < p < 1080
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.58± 0.28 %
μ = 433.2± 1.1
σ = 32.8± 1.1
WF
$\frac{\sigma}{\mu}$ = 7.60± 0.26 %
μ = 430.8± 1.1
σ = 32.8± 1.1
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200

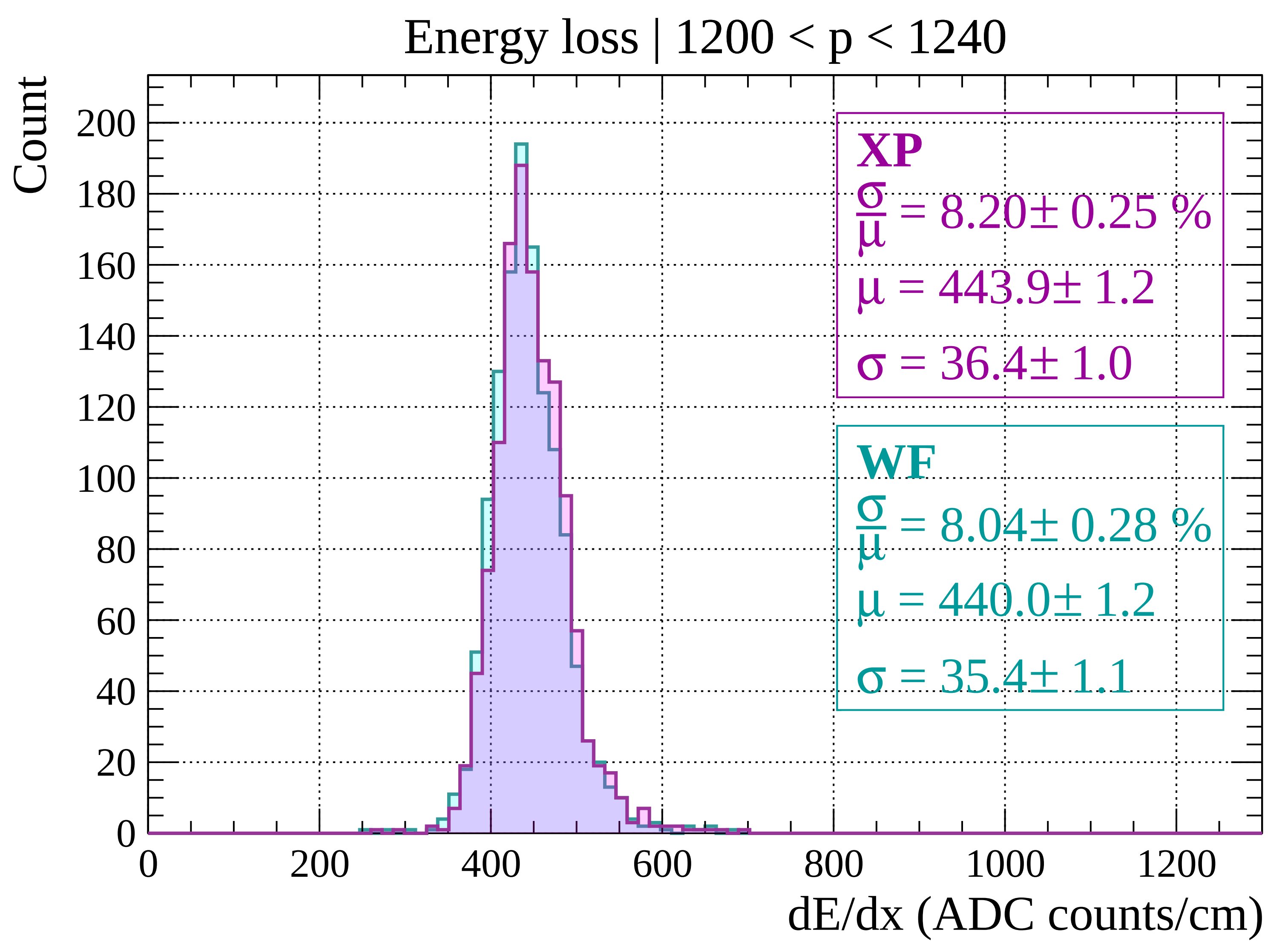

Energy loss | 1080 < p < 1120
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
σ/μ = 8.08± 0.30 %
μ = 435.7± 1.1
σ = 35.2± 1.2
WF
σ/μ = 7.92± 0.32 %
μ = 430.8± 1.2
σ = 34.1± 1.3



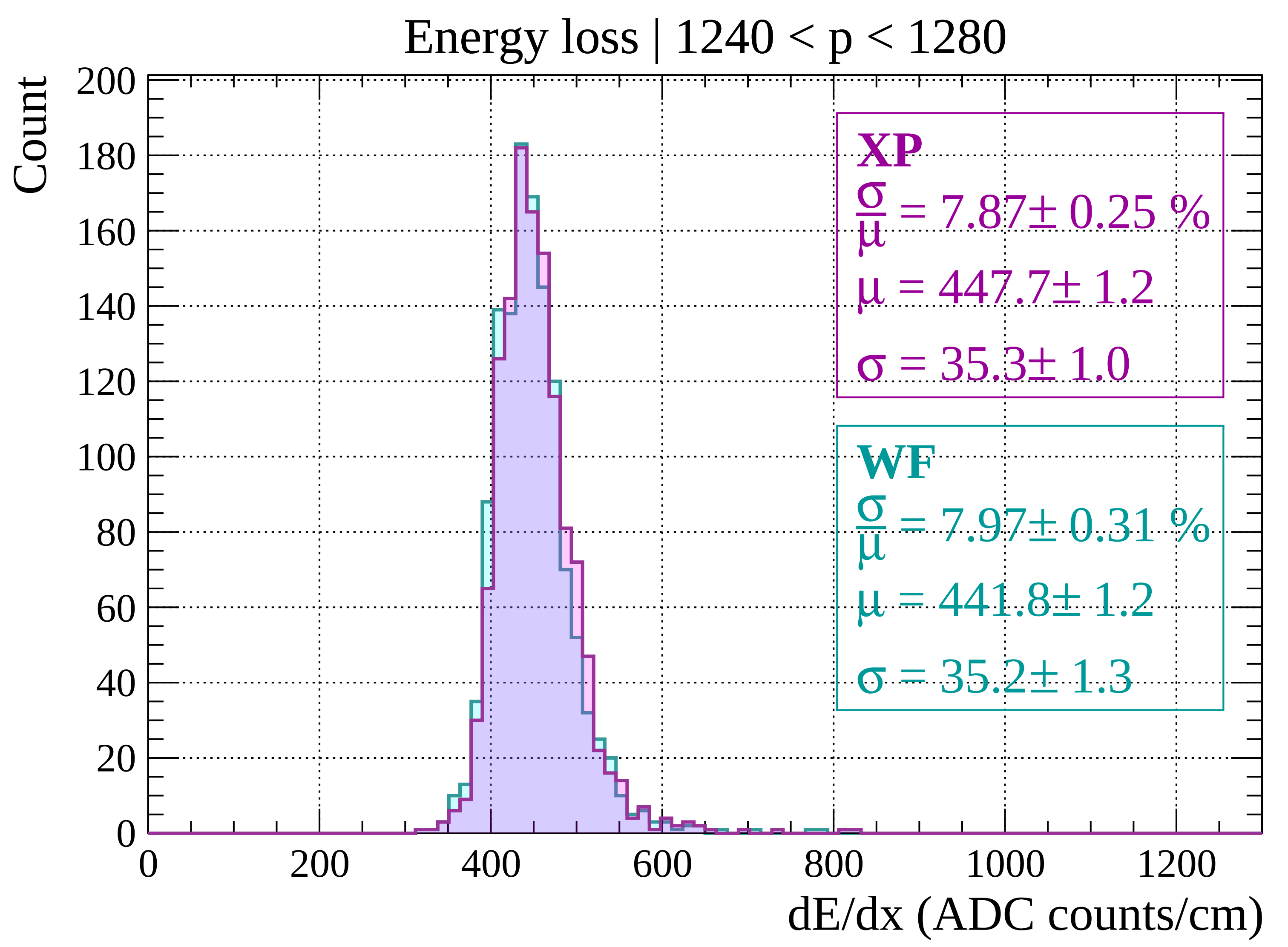
Energy loss | 1120 < p < 1160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
$\frac{\sigma}{\mu} = 8.28 \pm 0.29$ %
$\mu = 439.6 \pm 1.2$
$\sigma = 36.4 \pm 1.2$
WF
$\frac{\sigma}{\mu} = 8.11 \pm 0.28$ %
$\mu = 435.2 \pm 1.1$
$\sigma = 35.3 \pm 1.1$



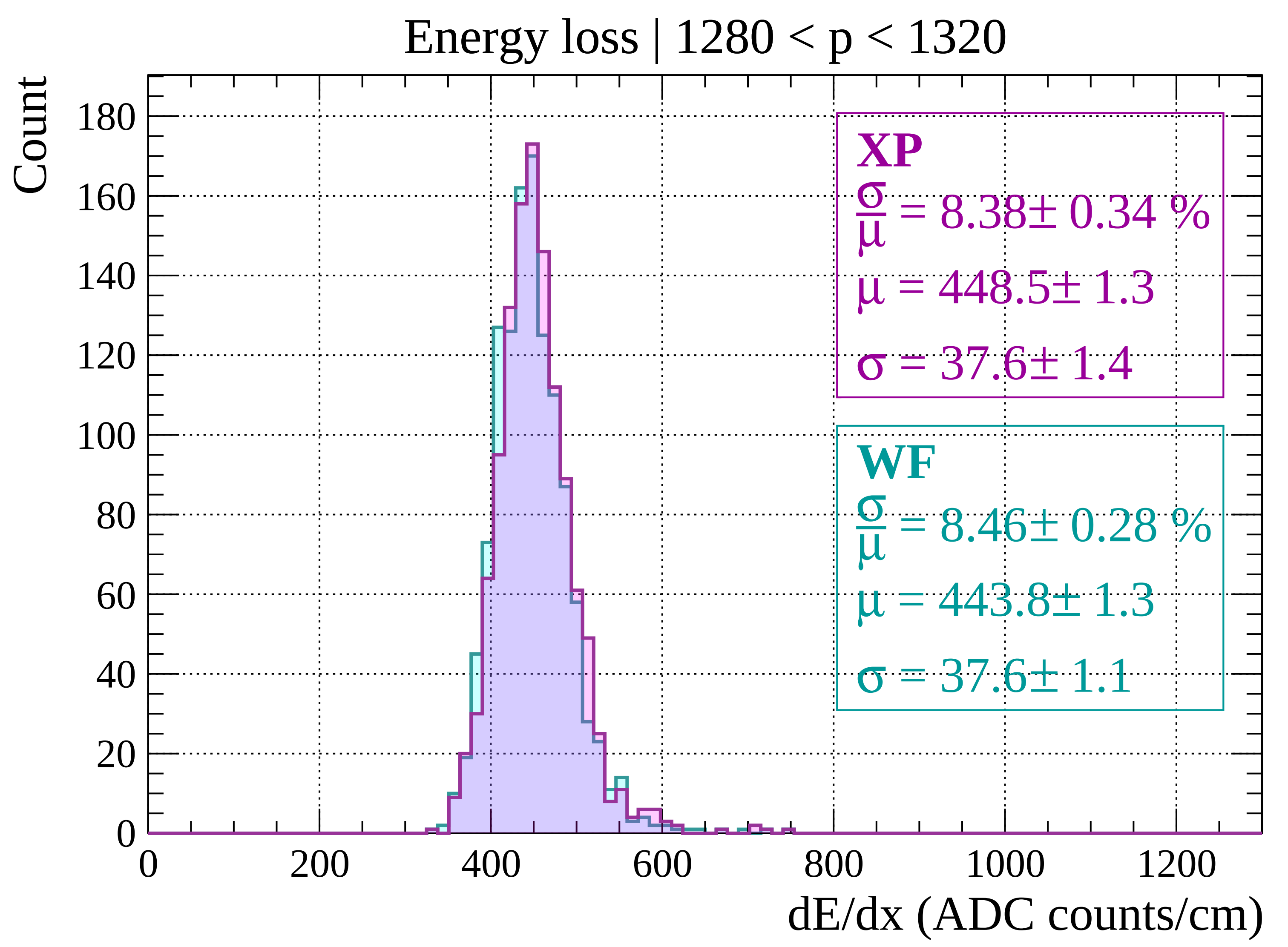
Energy loss | 1160 < p < 1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
σ/μ = 8.02± 0.29 %
μ = 439.7± 1.2
σ = 35.3± 1.2
WF
σ/μ = 8.43± 0.33 %
μ = 437.1± 1.3
σ = 36.8± 1.3



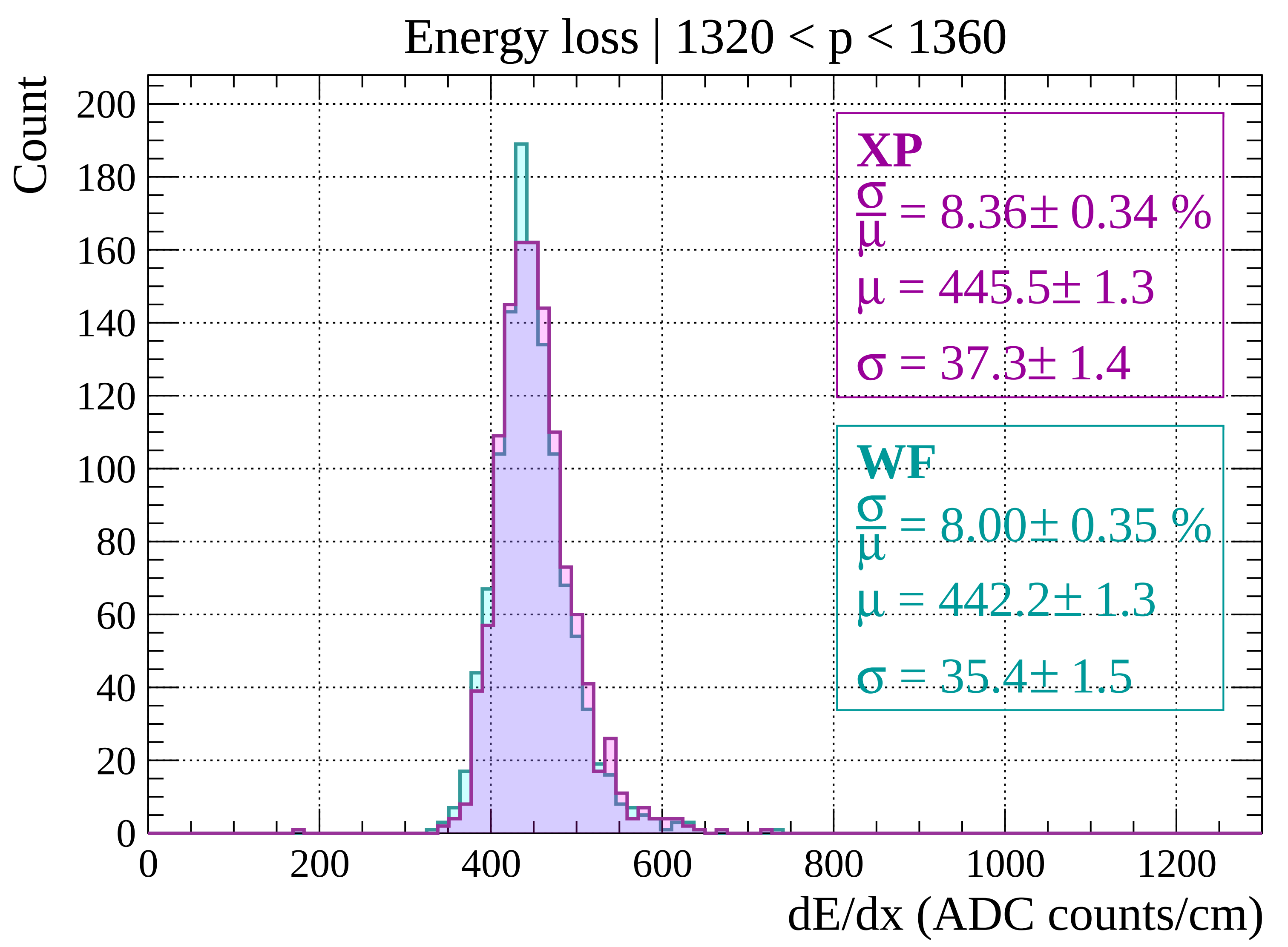
Energy loss | 1200 < p < 1240
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
σ/μ = 8.20± 0.25 %
μ = 443.9± 1.2
σ = 36.4± 1.0
WF
σ/μ = 8.04± 0.28 %
μ = 440.0± 1.2
σ = 35.4± 1.1

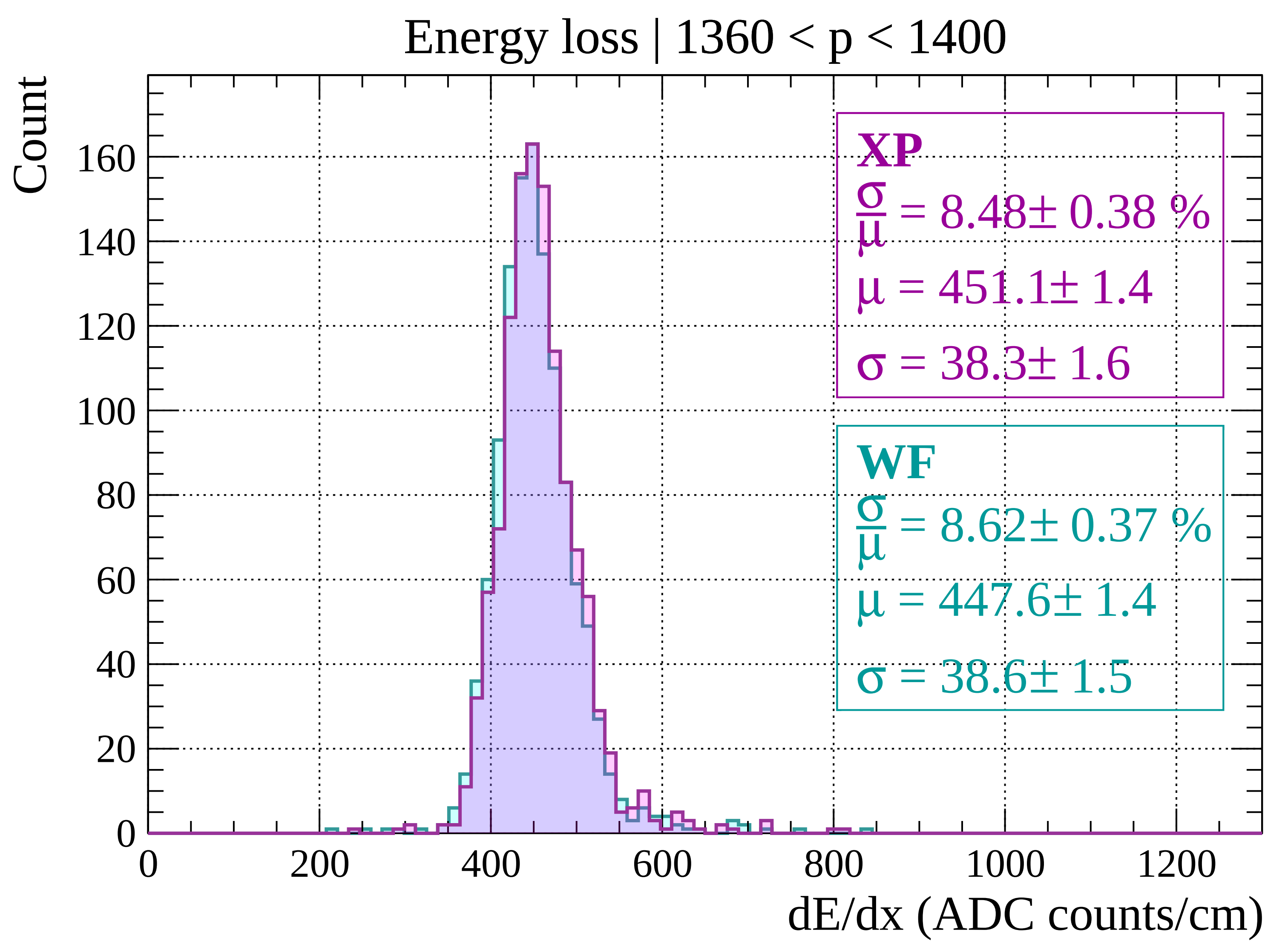


Energy loss | 1240 < p < 1280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
$\frac{\sigma}{\mu}$ = 7.87 ± 0.25 %
μ = 447.7 ± 1.2
σ = 35.3 ± 1.0
WF
$\frac{\sigma}{\mu}$ = 7.97 ± 0.31 %
μ = 441.8 ± 1.2
σ = 35.2 ± 1.3

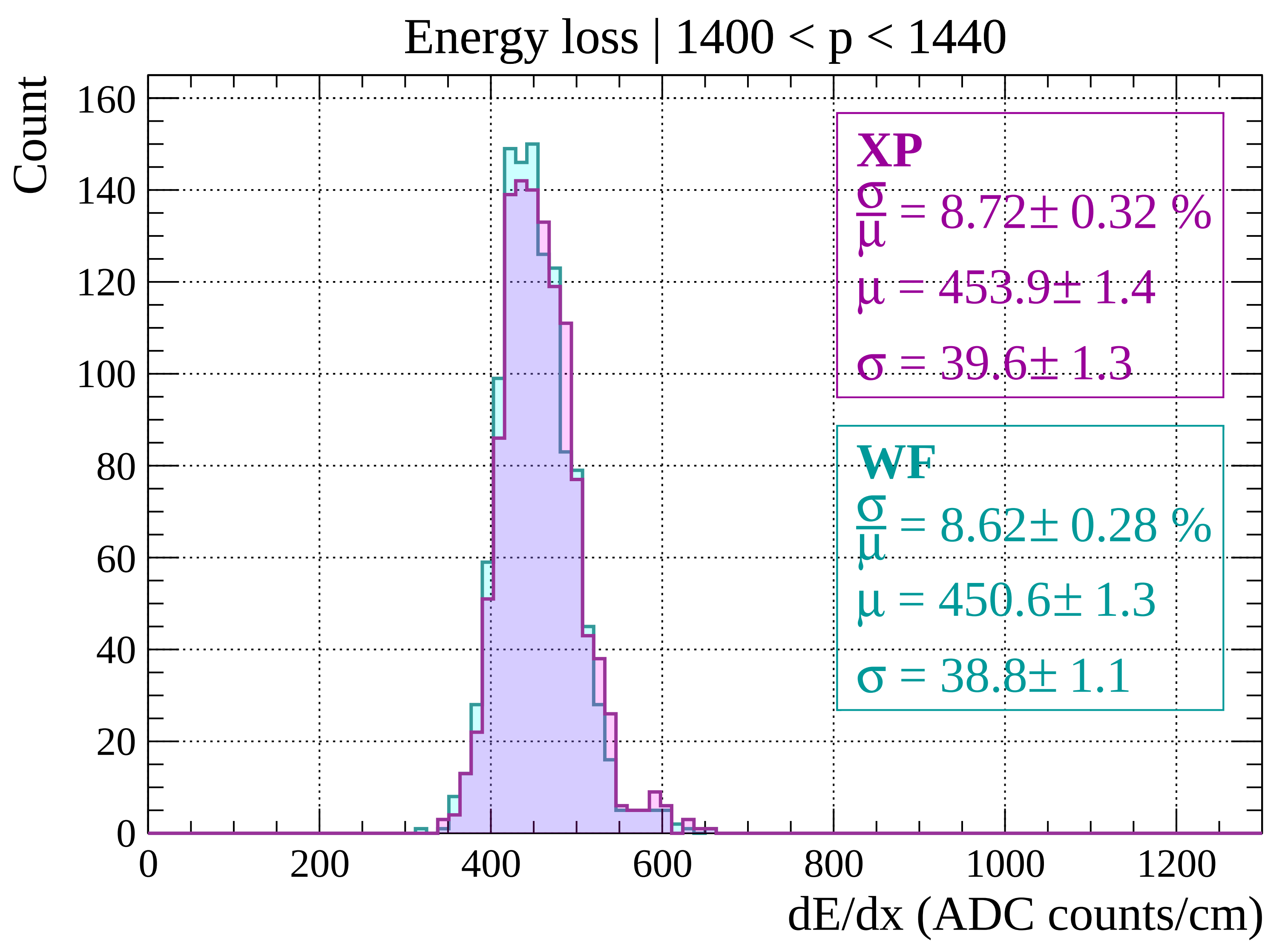


Energy loss | 1280 < p < 1320
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.38± 0.34 %
μ = 448.5± 1.3
σ = 37.6± 1.4
WF
$\frac{\sigma}{\mu}$ = 8.46± 0.28 %
μ = 443.8± 1.3
σ = 37.6± 1.1

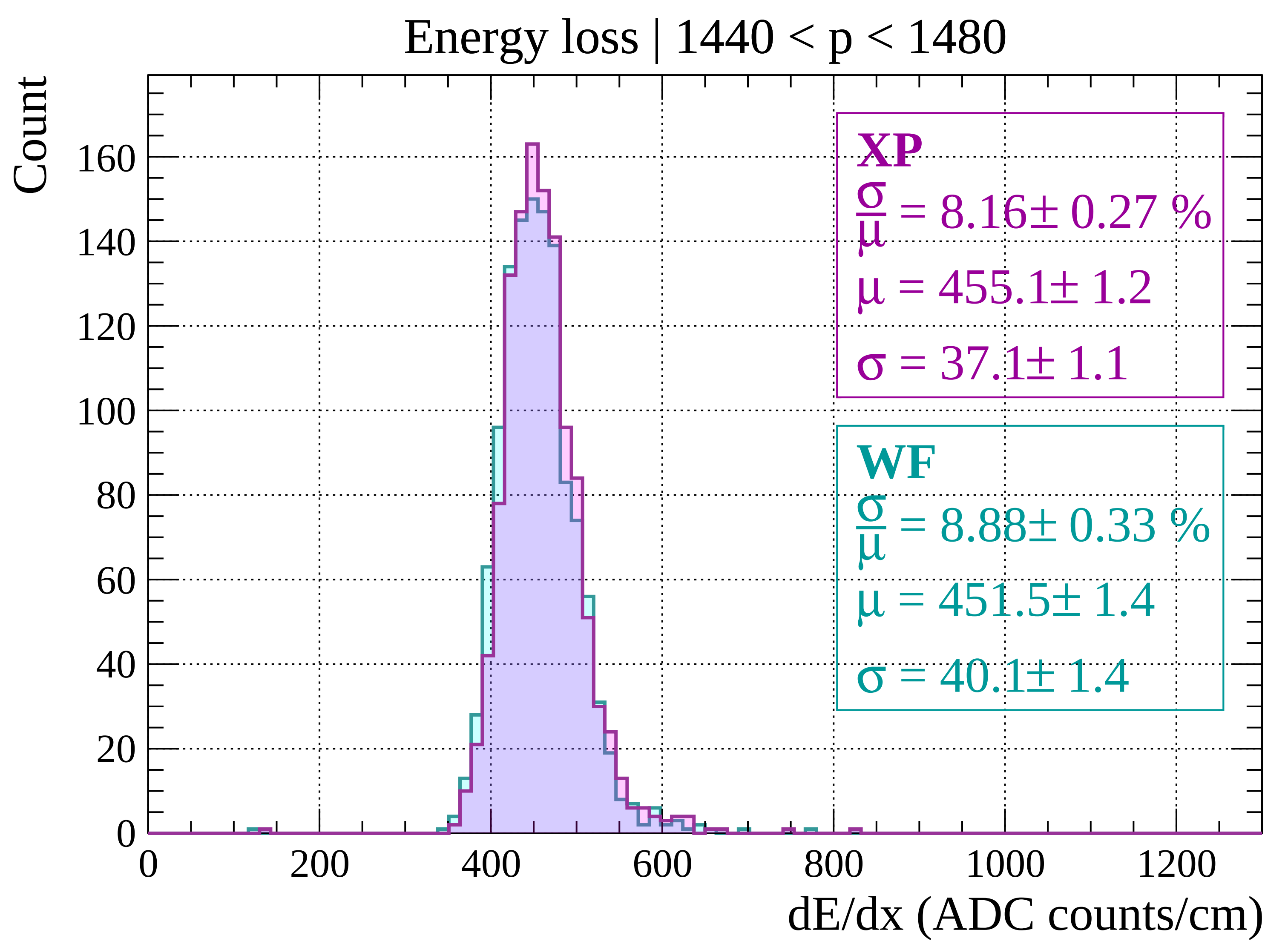

Energy loss | 1320 < p < 1360
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
σ/μ = 8.36± 0.34 %
μ = 445.5± 1.3
σ = 37.3± 1.4
WF
σ/μ = 8.00± 0.35 %
μ = 442.2± 1.3
σ = 35.4± 1.5



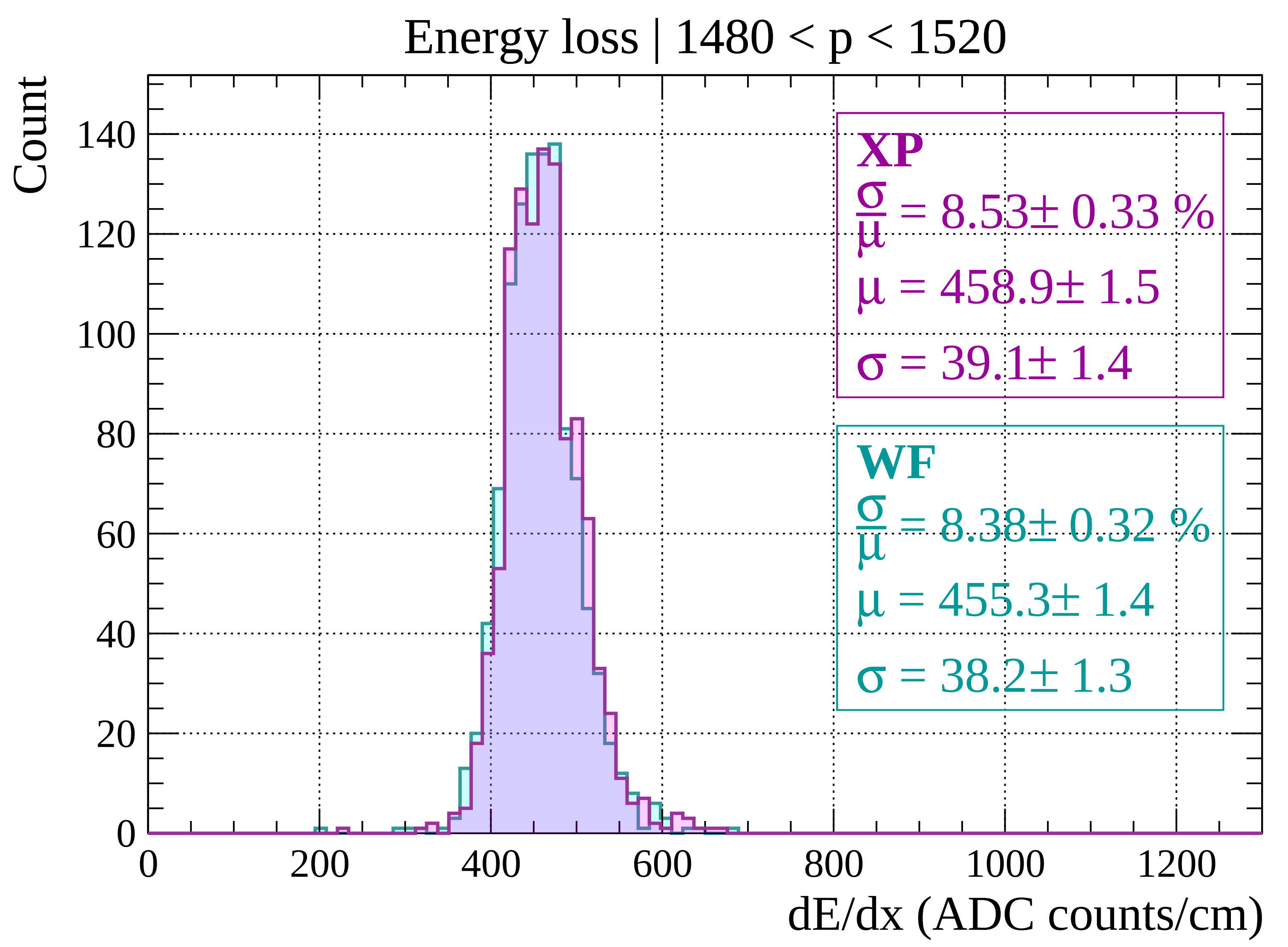
Energy loss | 1360 < p < 1400
Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.48± 0.38 %
μ = 451.1± 1.4
σ = 38.3± 1.6
WF
σ/μ = 8.62± 0.37 %
μ = 447.6± 1.4
σ = 38.6± 1.5



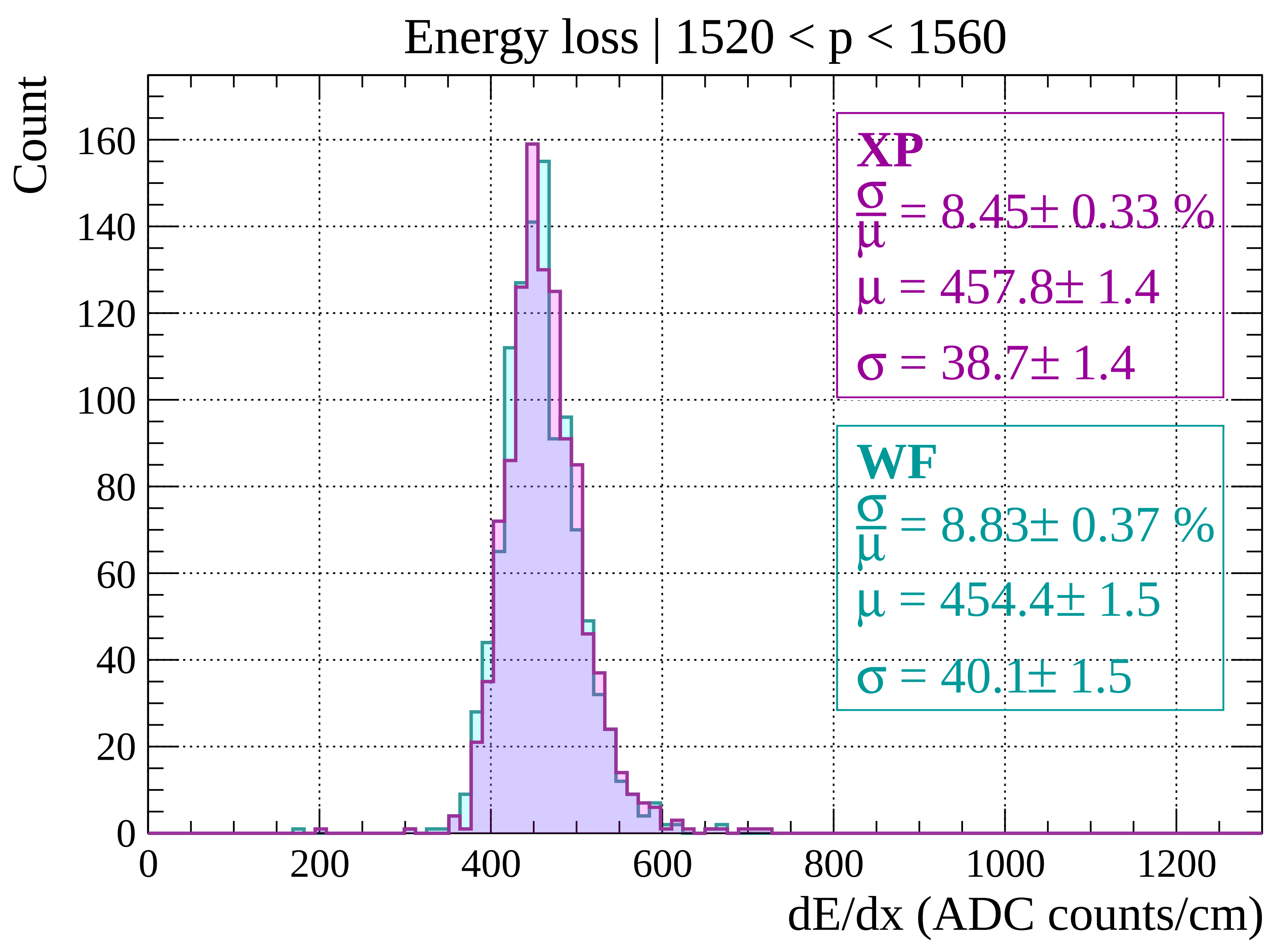
Energy loss | 1400 < p < 1440
Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.72± 0.32 %
μ = 453.9± 1.4
σ = 39.6± 1.3
WF
$\frac{\sigma}{\mu}$ = 8.62± 0.28 %
μ = 450.6± 1.3
σ = 38.8± 1.1



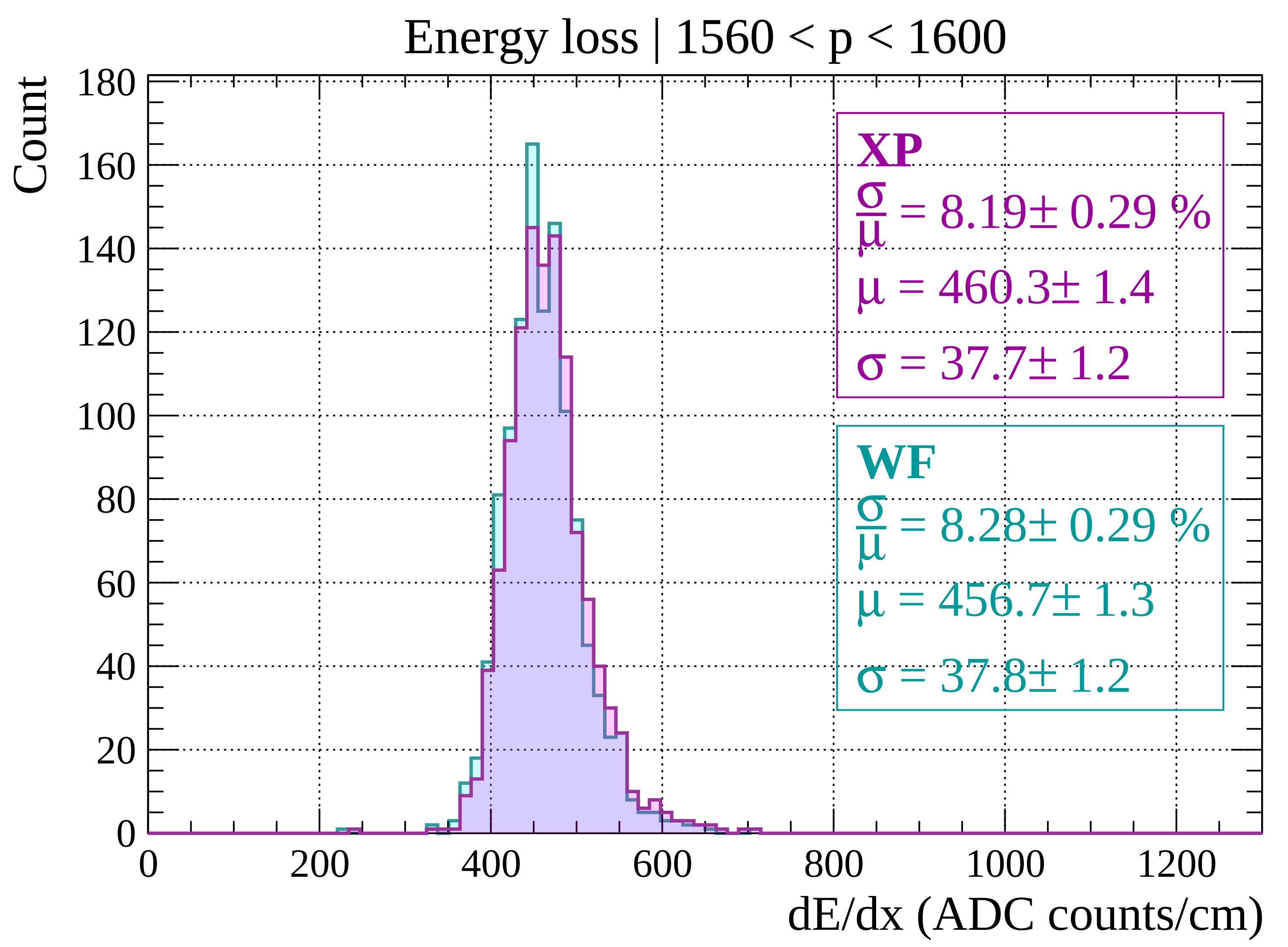
Energy loss | 1440 < p < 1480
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.16± 0.27 %
μ = 455.1± 1.2
σ = 37.1± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.88± 0.33 %
μ = 451.5± 1.4
σ = 40.1± 1.4



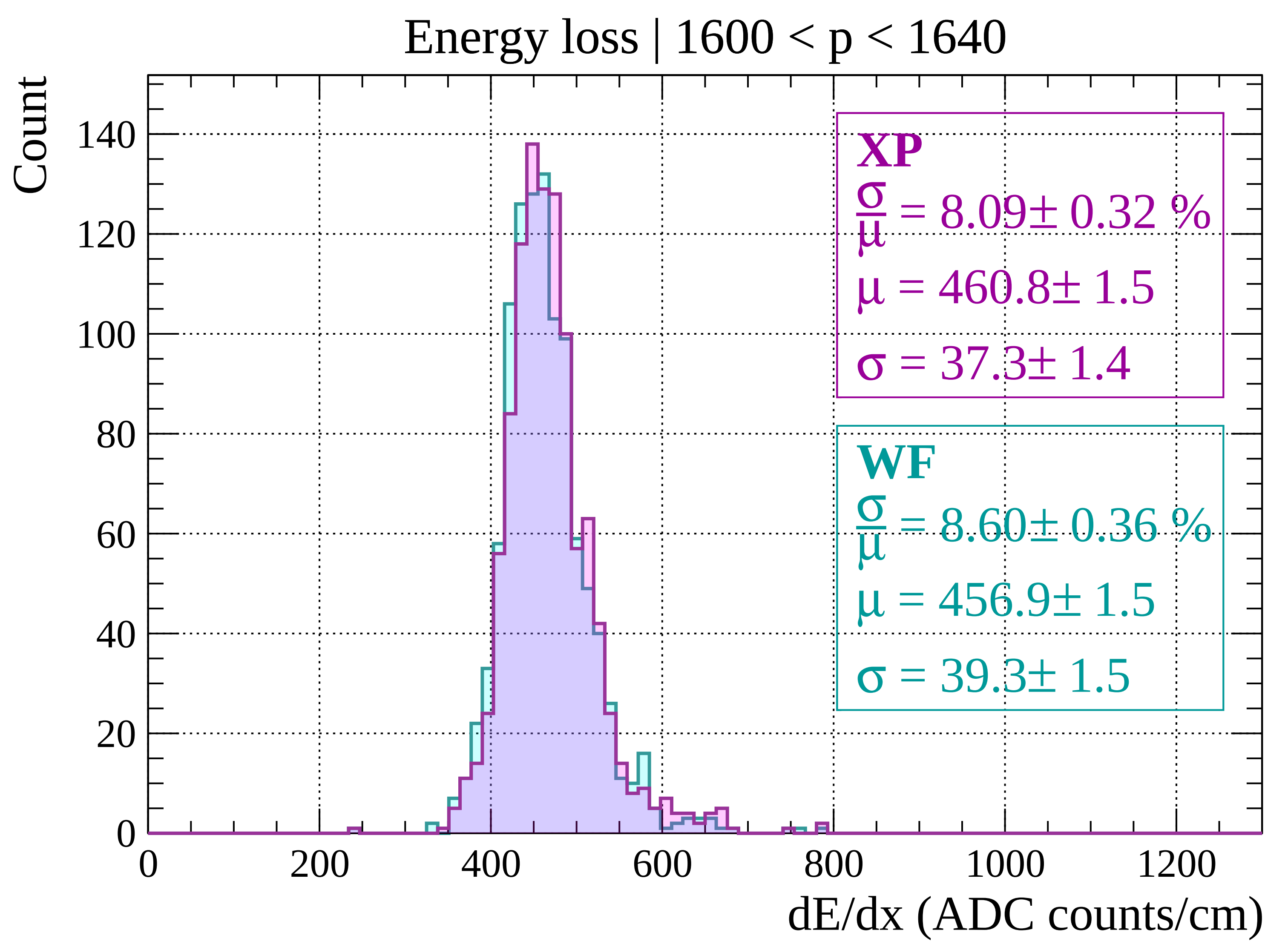
Energy loss | 1480 < p < 1520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
XP
σ/μ = 8.53± 0.33 %
μ = 458.9± 1.5
σ = 39.1± 1.4
WF
σ/μ = 8.38± 0.32 %
μ = 455.3± 1.4
σ = 38.2± 1.3



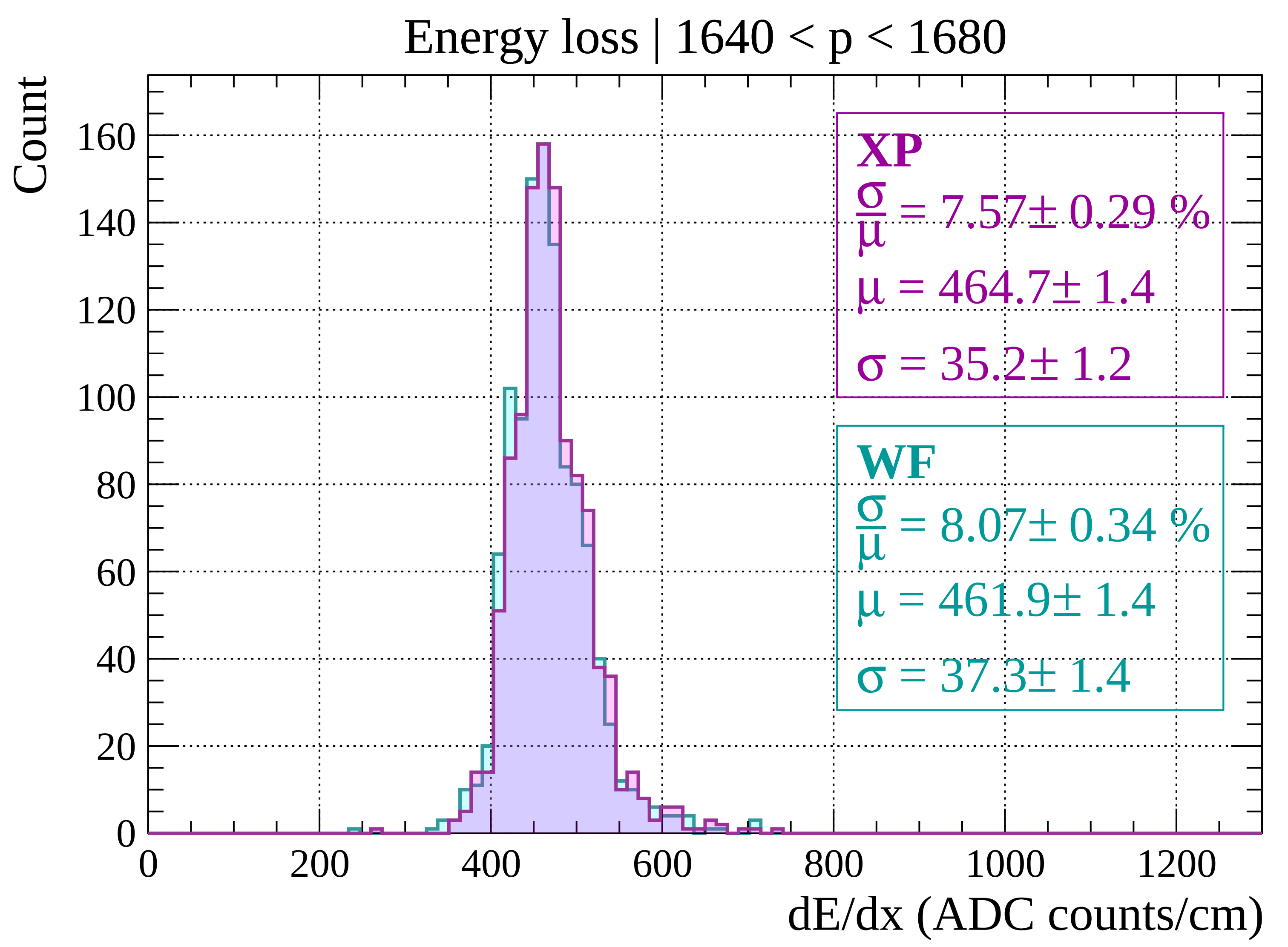
Energy loss | 1520 < p < 1560
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.45± 0.33 %
μ = 457.8± 1.4
σ = 38.7± 1.4
WF
$\frac{\sigma}{\mu}$ = 8.83± 0.37 %
μ = 454.4± 1.5
σ = 40.1± 1.5

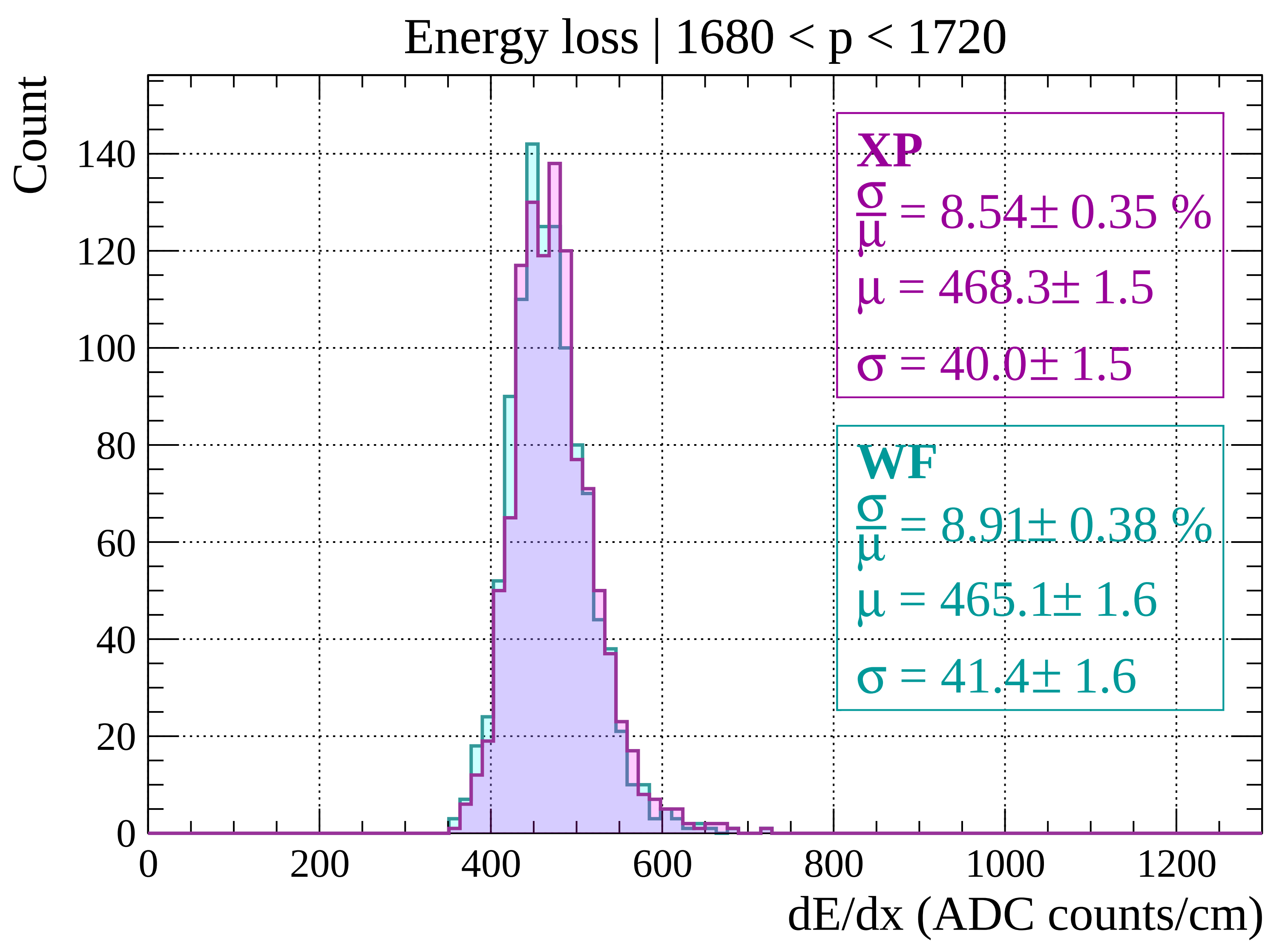


Energy loss | 1560 < p < 1600
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.19± 0.29 %
μ = 460.3± 1.4
σ = 37.7± 1.2
WF
$\frac{\sigma}{\mu}$ = 8.28± 0.29 %
μ = 456.7± 1.3
σ = 37.8± 1.2

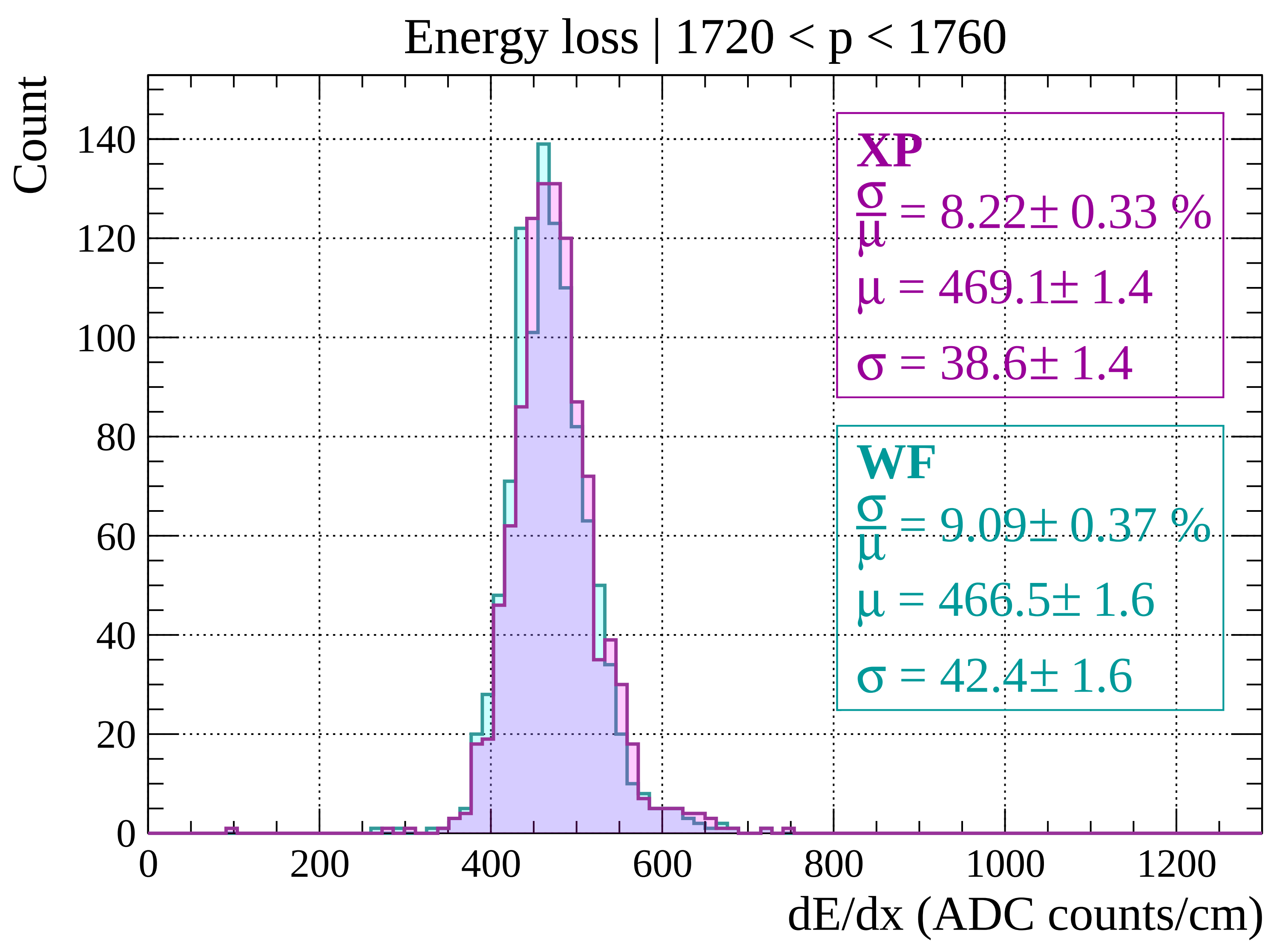


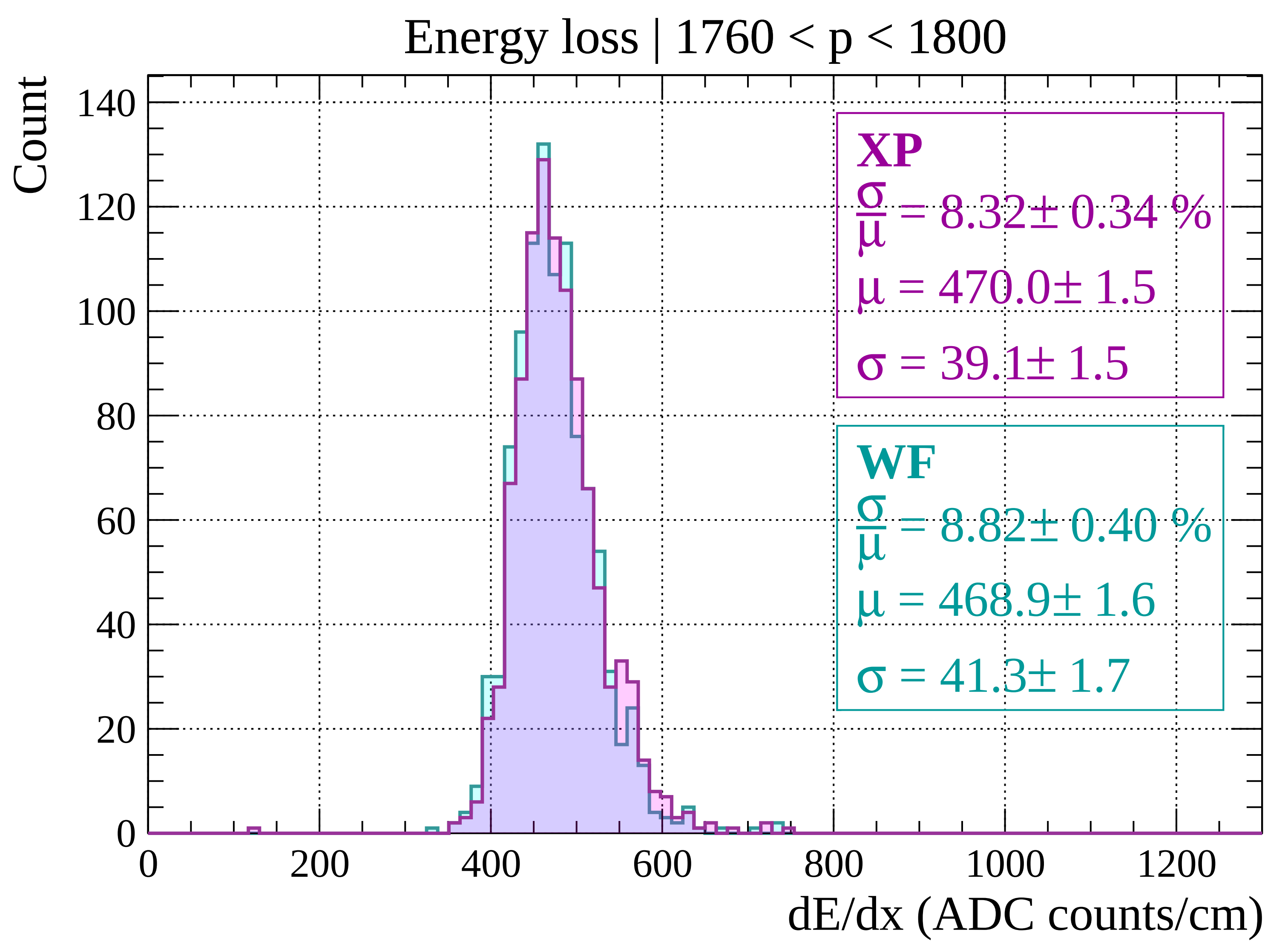
Energy loss | 1600 < p < 1640
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.09± 0.32 %
μ = 460.8± 1.5
σ = 37.3± 1.4
WF
σ/μ = 8.60± 0.36 %
μ = 456.9± 1.5
σ = 39.3± 1.5


Energy loss | 1640 < p < 1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
σ/μ = 7.57± 0.29 %
μ = 464.7± 1.4
σ = 35.2± 1.2
WF
σ/μ = 8.07± 0.34 %
μ = 461.9± 1.4
σ = 37.3± 1.4

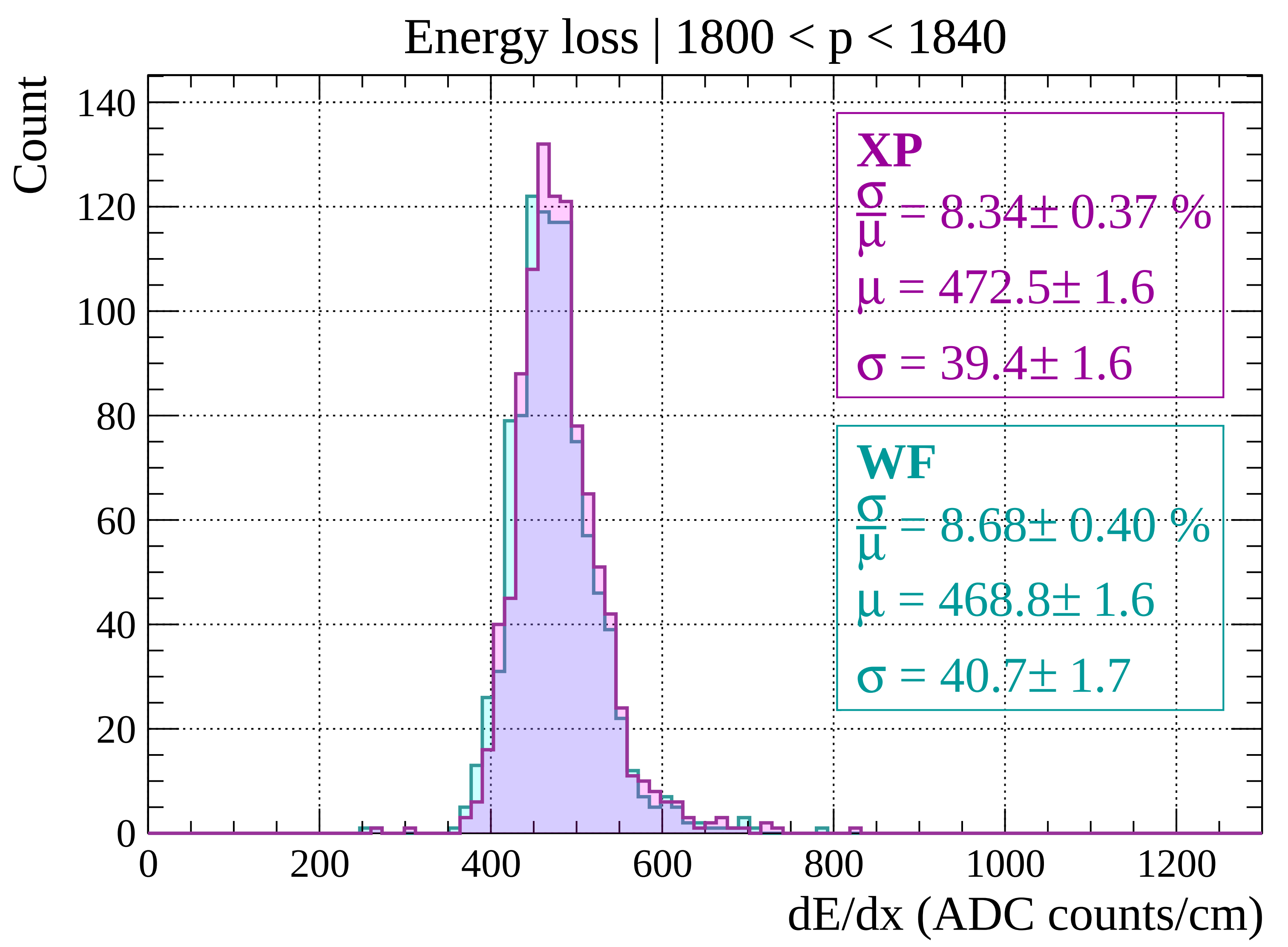


Energy loss | 1680 < p < 1720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
$\frac{\sigma}{\mu}$ = 8.54± 0.35 %
μ = 468.3± 1.5
σ = 40.0± 1.5
WF
$\frac{\sigma}{\mu}$ = 8.91± 0.38 %
μ = 465.1± 1.6
σ = 41.4± 1.6

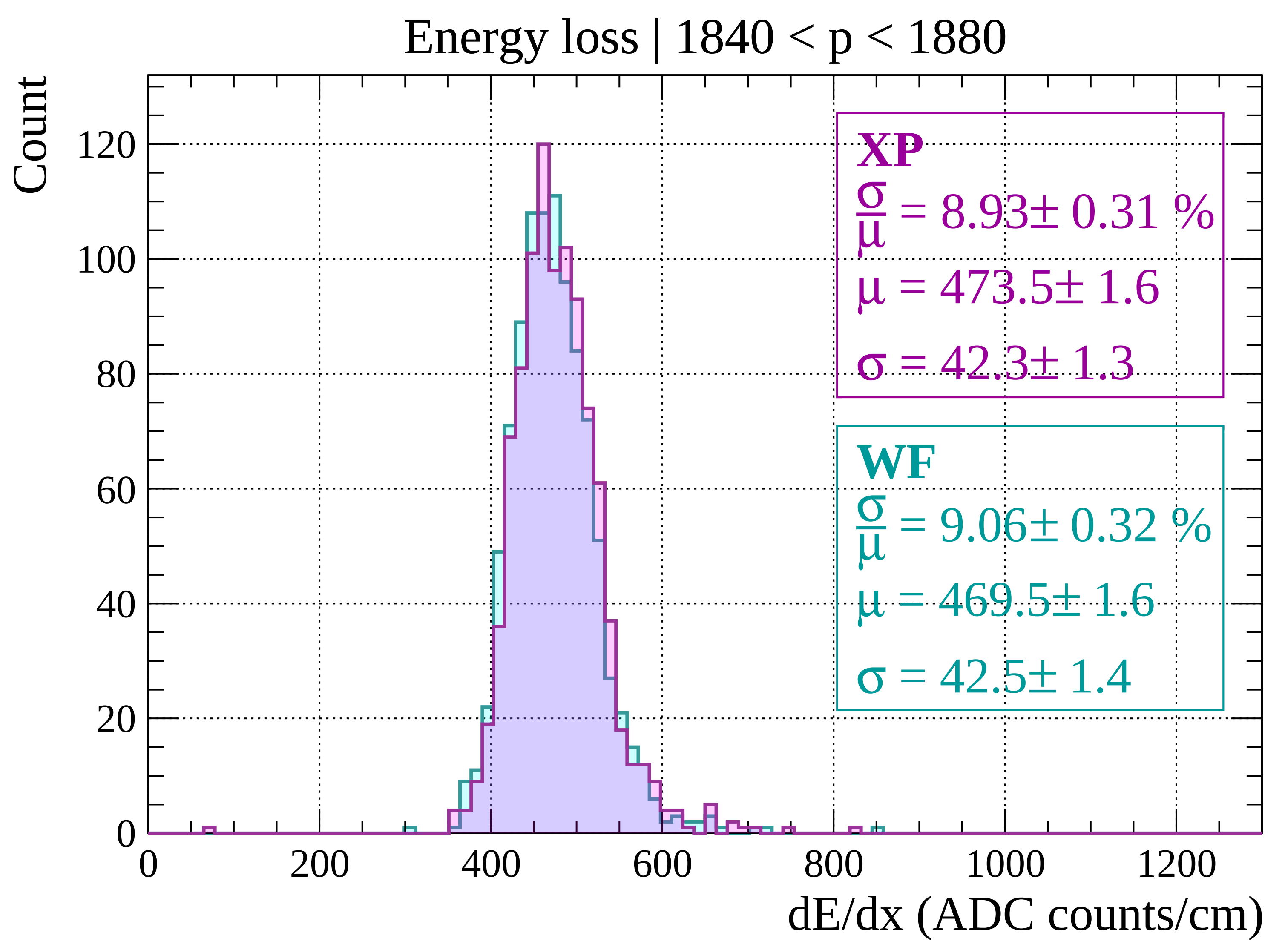

Energy loss | 1720 < p < 1760
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 8.22± 0.33 %
μ = 469.1± 1.4
σ = 38.6± 1.4
WF
σ/μ = 9.09± 0.37 %
μ = 466.5± 1.6
σ = 42.4± 1.6



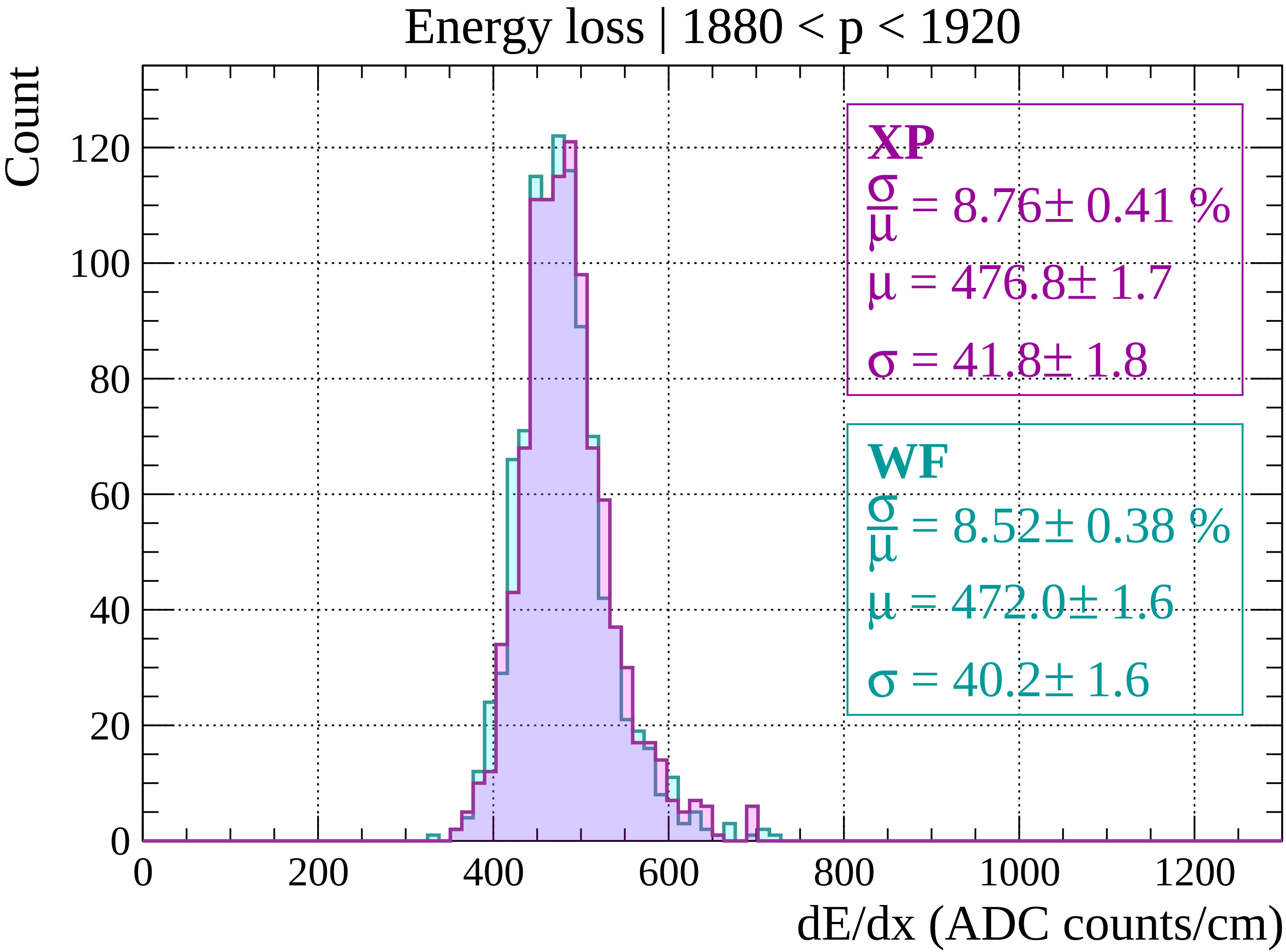
Energy loss | 1760 < p < 1800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
$\frac{\sigma}{\mu}$ = 8.32± 0.34 %
μ = 470.0± 1.5
σ = 39.1± 1.5
WF
$\frac{\sigma}{\mu}$ = 8.82± 0.40 %
μ = 468.9± 1.6
σ = 41.3± 1.7



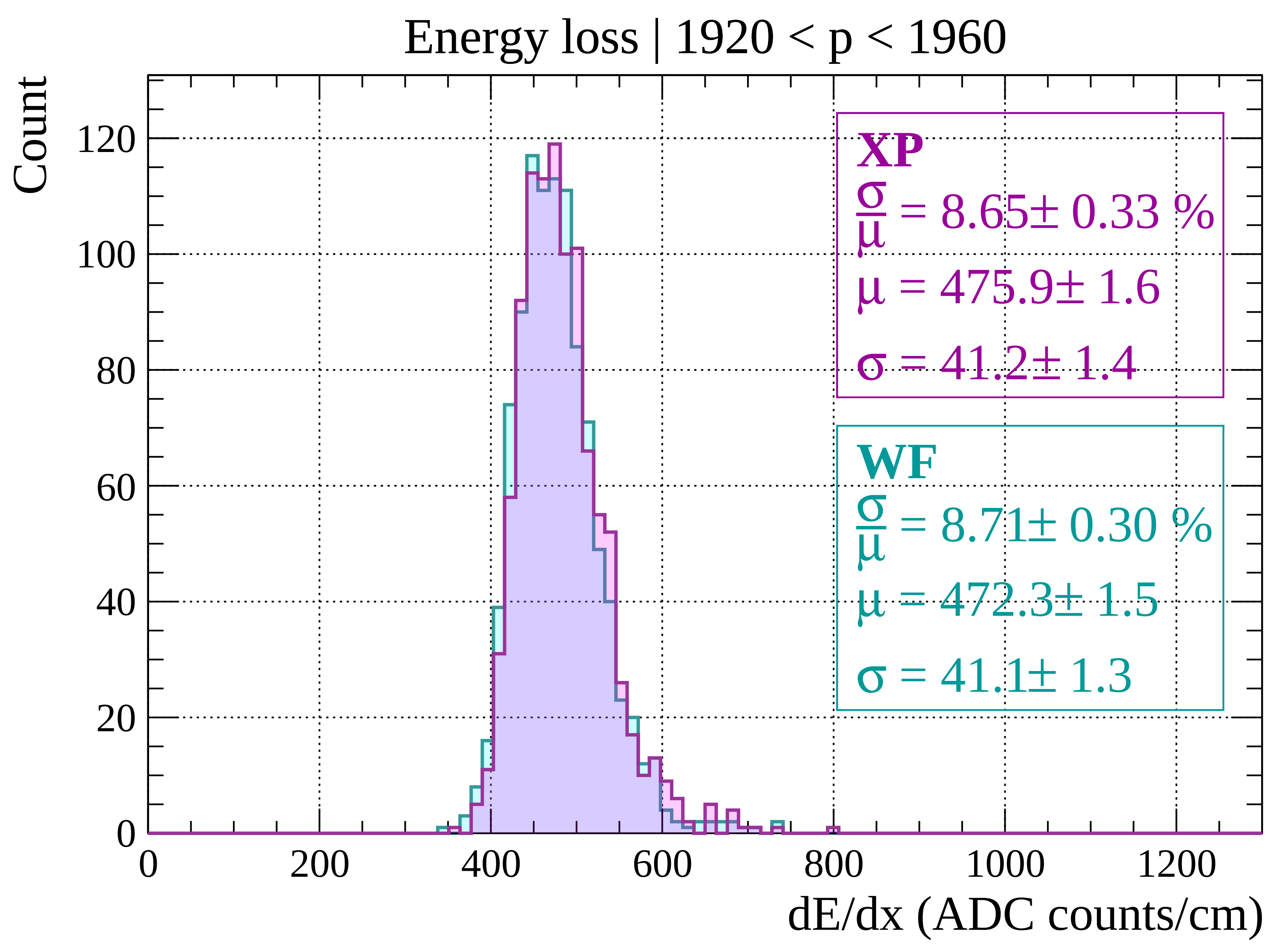
Energy loss | 1800 < p < 1840
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 8.34± 0.37 %
μ = 472.5± 1.6
σ = 39.4± 1.6
WF
σ/μ = 8.68± 0.40 %
μ = 468.8± 1.6
σ = 40.7± 1.7



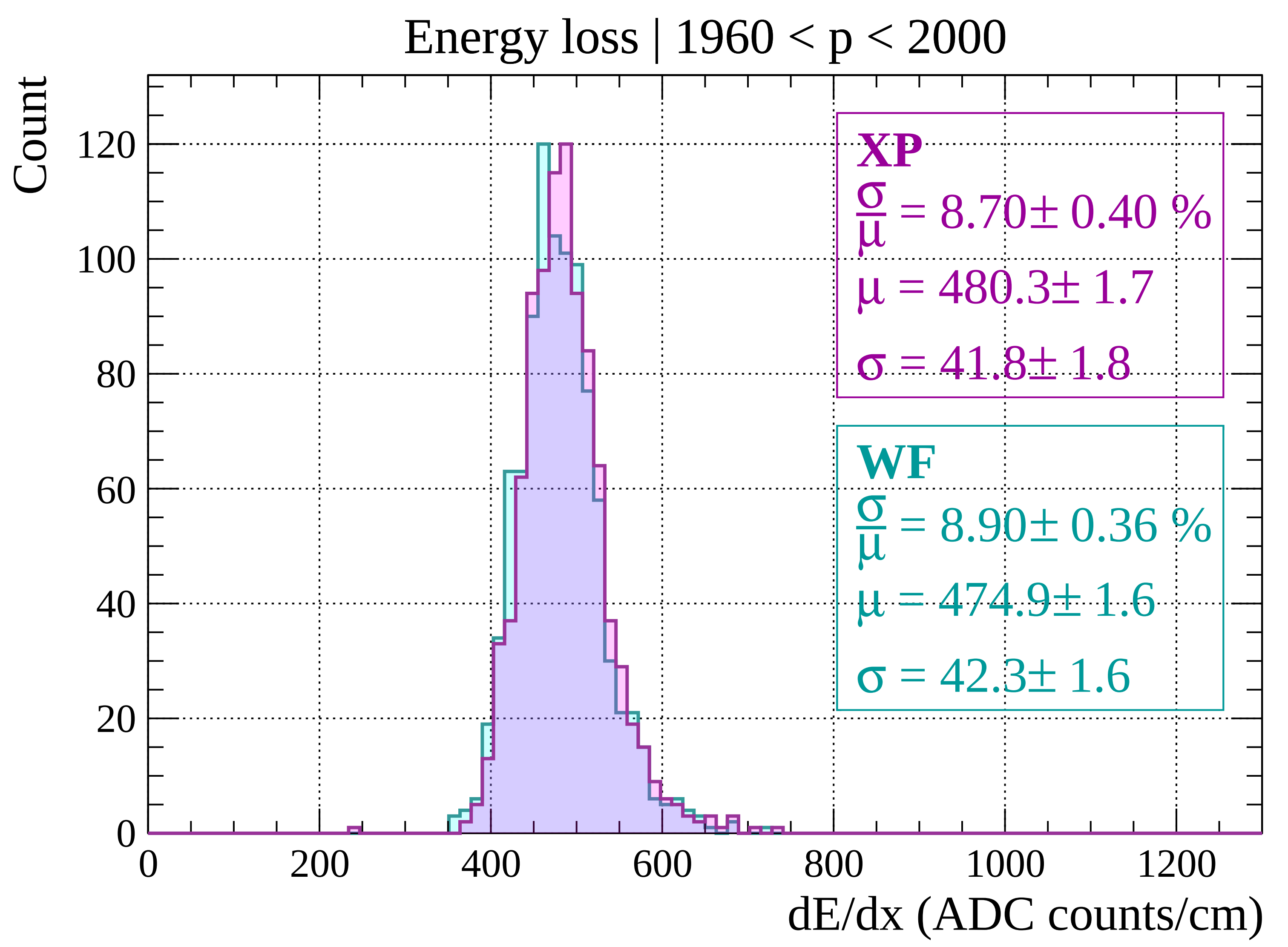
Energy loss | 1840 < p < 1880
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.93± 0.31 %
μ = 473.5± 1.6
σ = 42.3± 1.3
WF
σ/μ = 9.06± 0.32 %
μ = 469.5± 1.6
σ = 42.5± 1.4



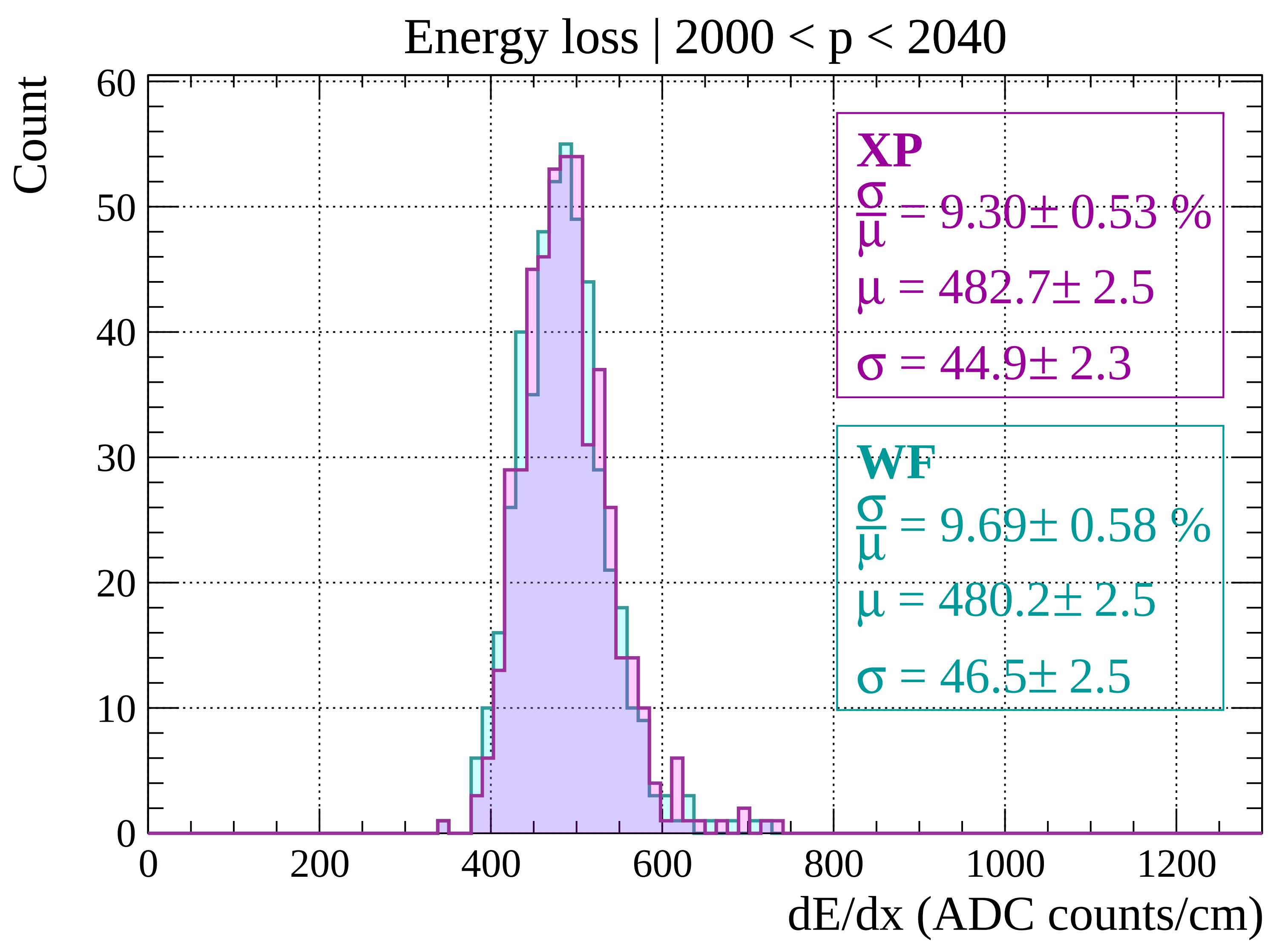
Energy loss | 1880 < p < 1920
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.76± 0.41 %
μ = 476.8± 1.7
σ = 41.8± 1.8
WF
$\frac{\sigma}{\mu}$ = 8.52± 0.38 %
μ = 472.0± 1.6
σ = 40.2± 1.6



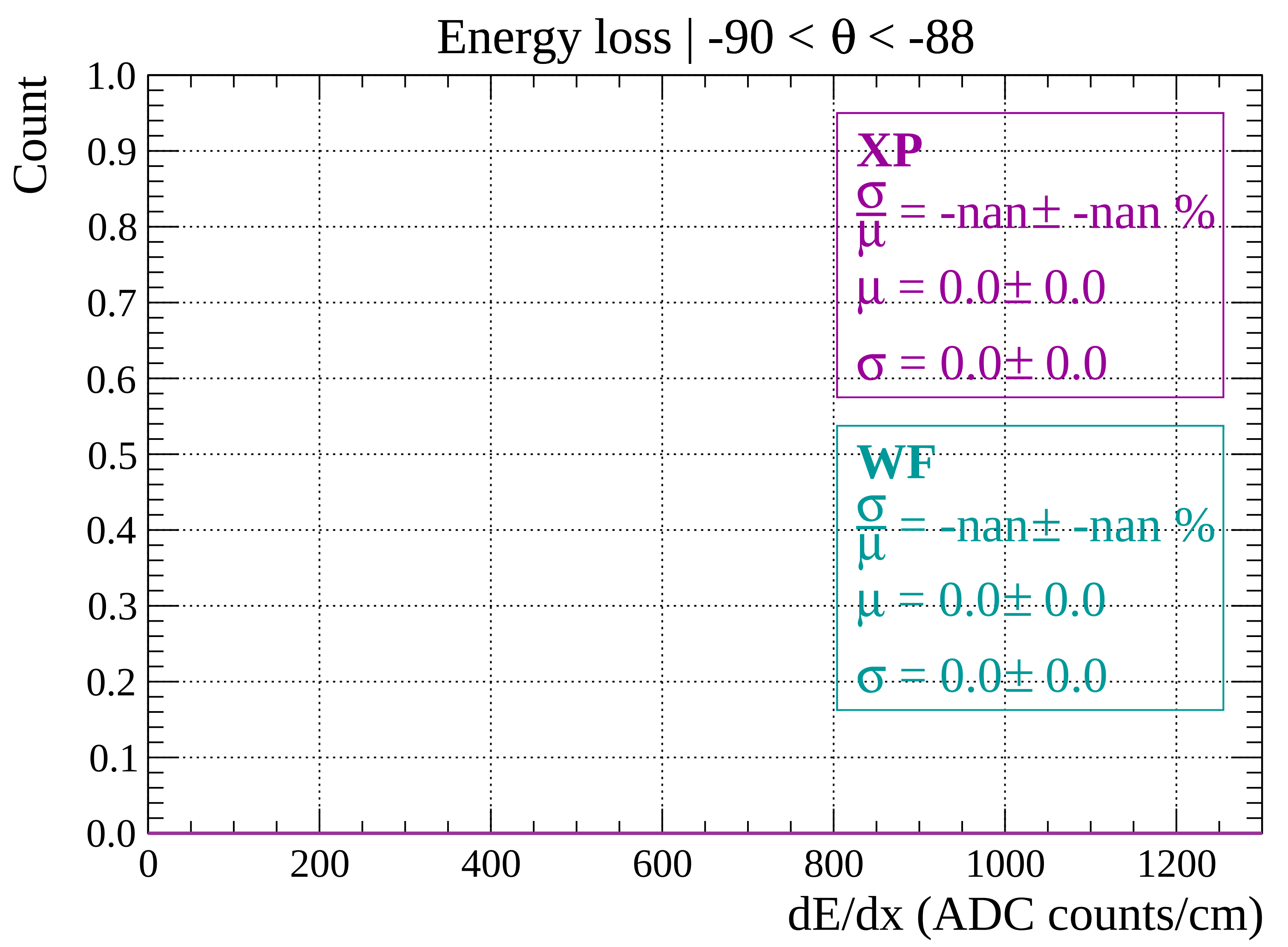
Energy loss | 1920 < p < 1960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 8.65± 0.33 %
μ = 475.9± 1.6
σ = 41.2± 1.4
WF
σ/μ = 8.71± 0.30 %
μ = 472.3± 1.5
σ = 41.1± 1.3



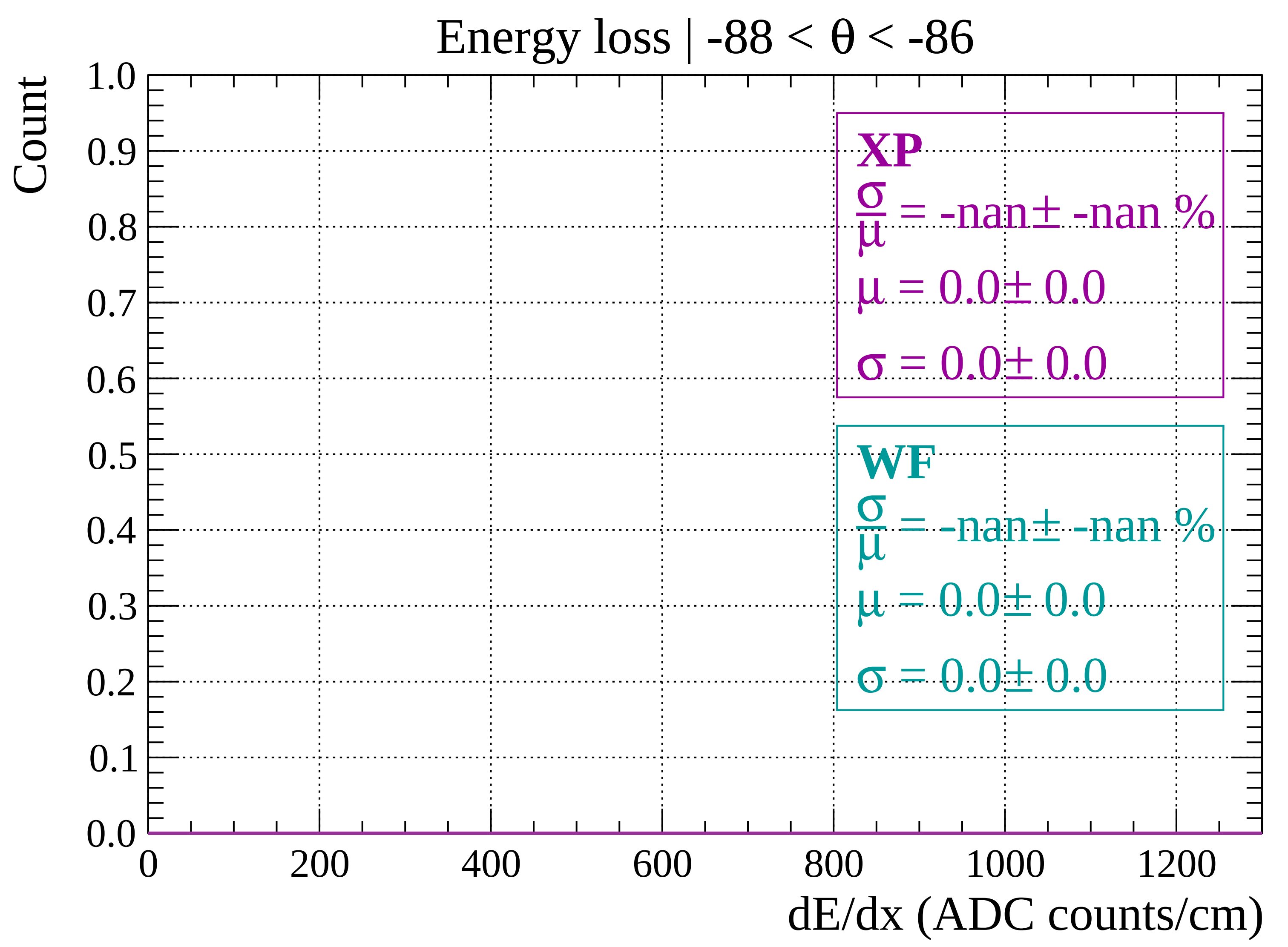
Energy loss | 1960 < p < 2000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 8.70± 0.40 %
μ = 480.3± 1.7
σ = 41.8± 1.8
WF
σ/μ = 8.90± 0.36 %
μ = 474.9± 1.6
σ = 42.3± 1.6

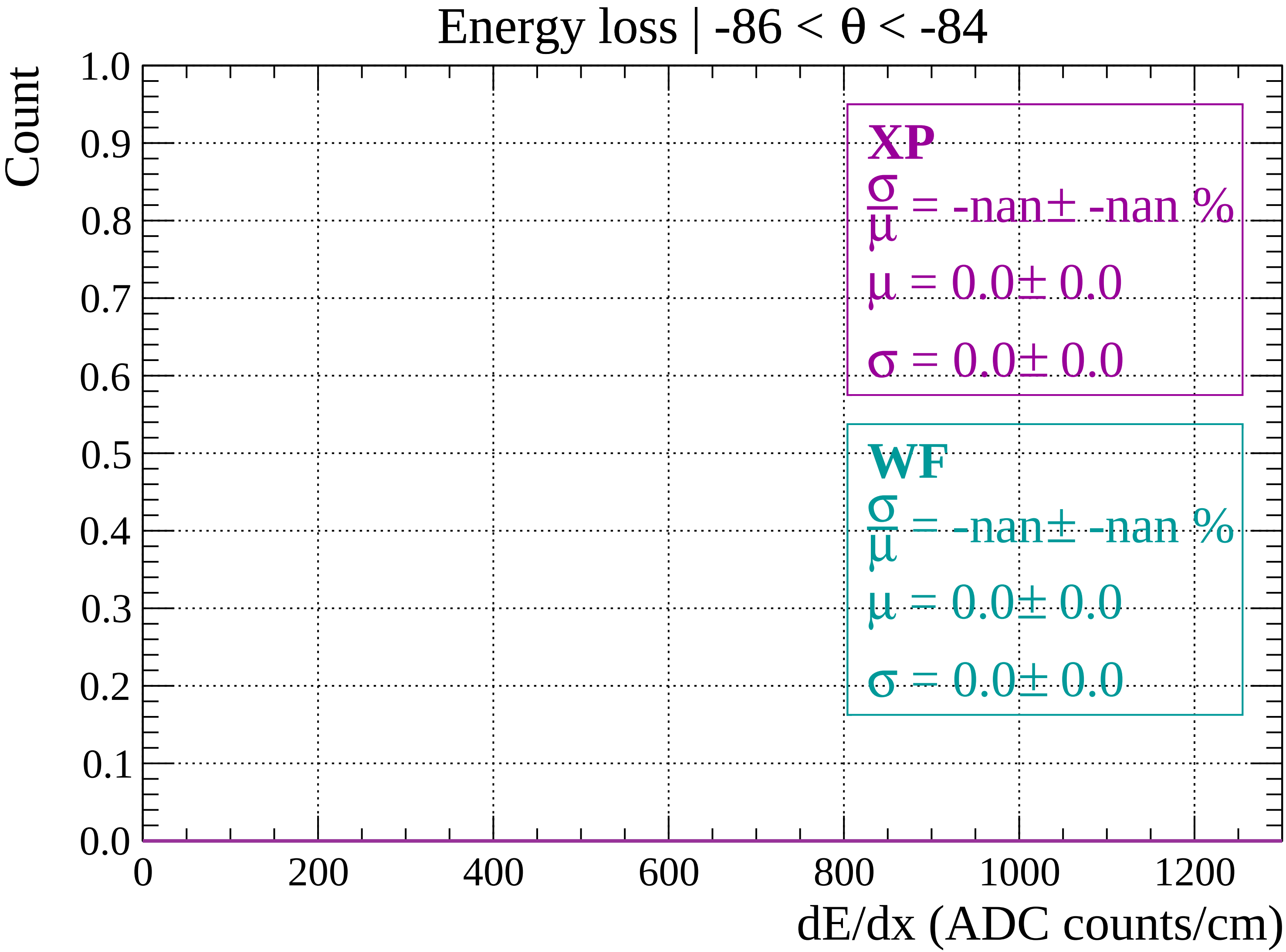


Energy loss | 2000 < p < 2040
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 9.30± 0.53 %
μ = 482.7± 2.5
σ = 44.9± 2.3
WF
σ/μ = 9.69± 0.58 %
μ = 480.2± 2.5
σ = 46.5± 2.5

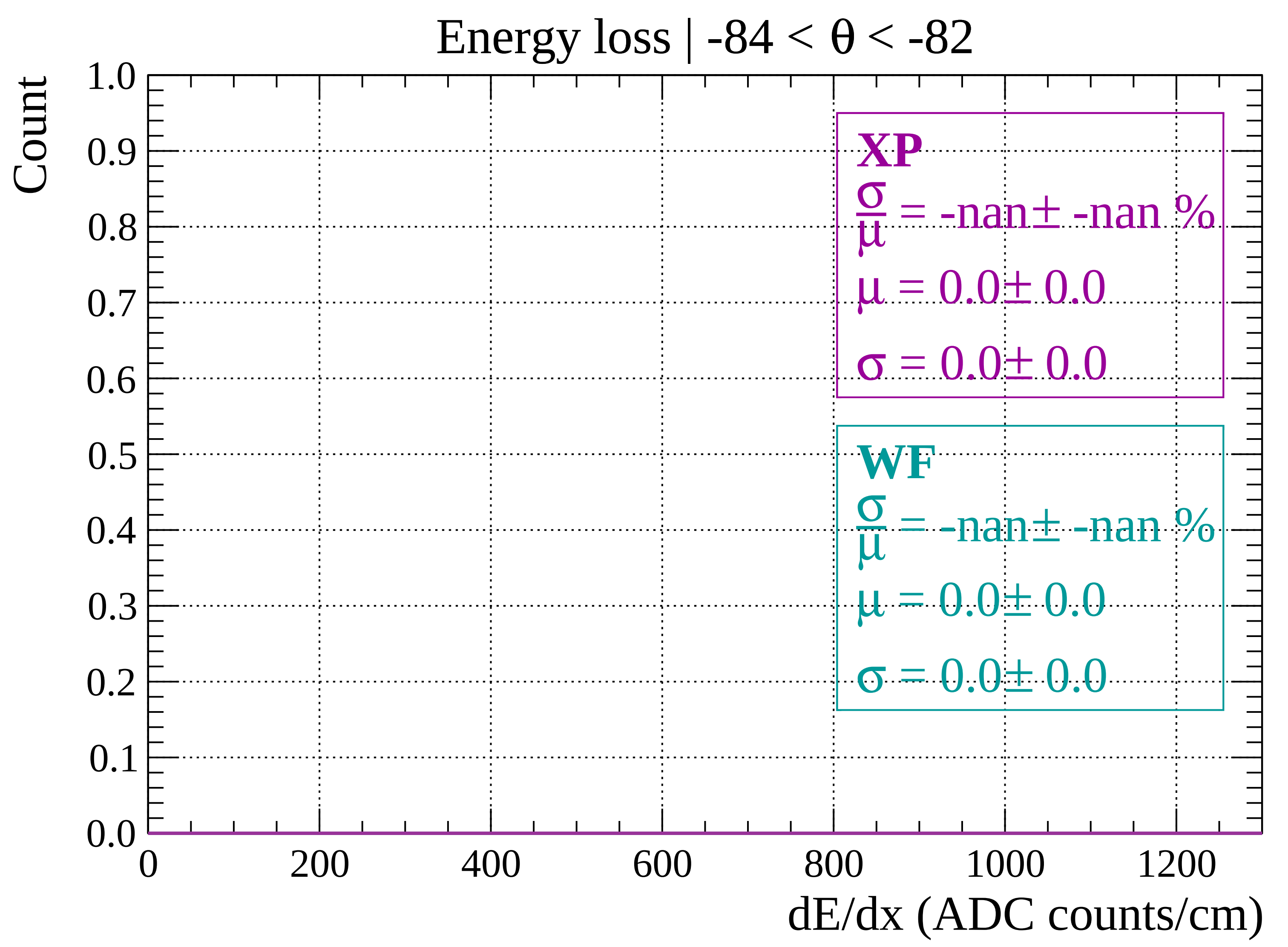


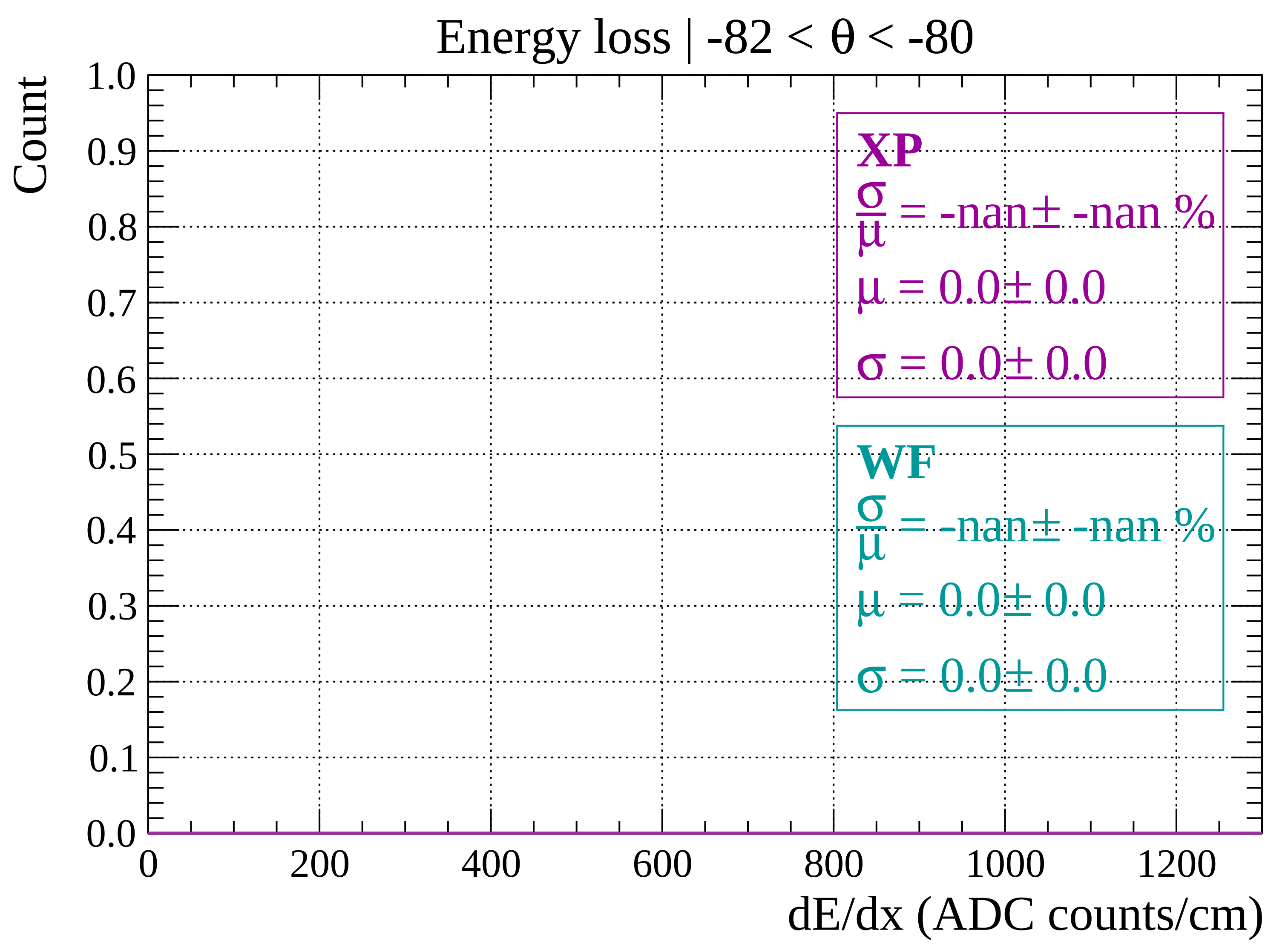
Energy loss | -90 < θ < -88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -88 < θ < -86
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

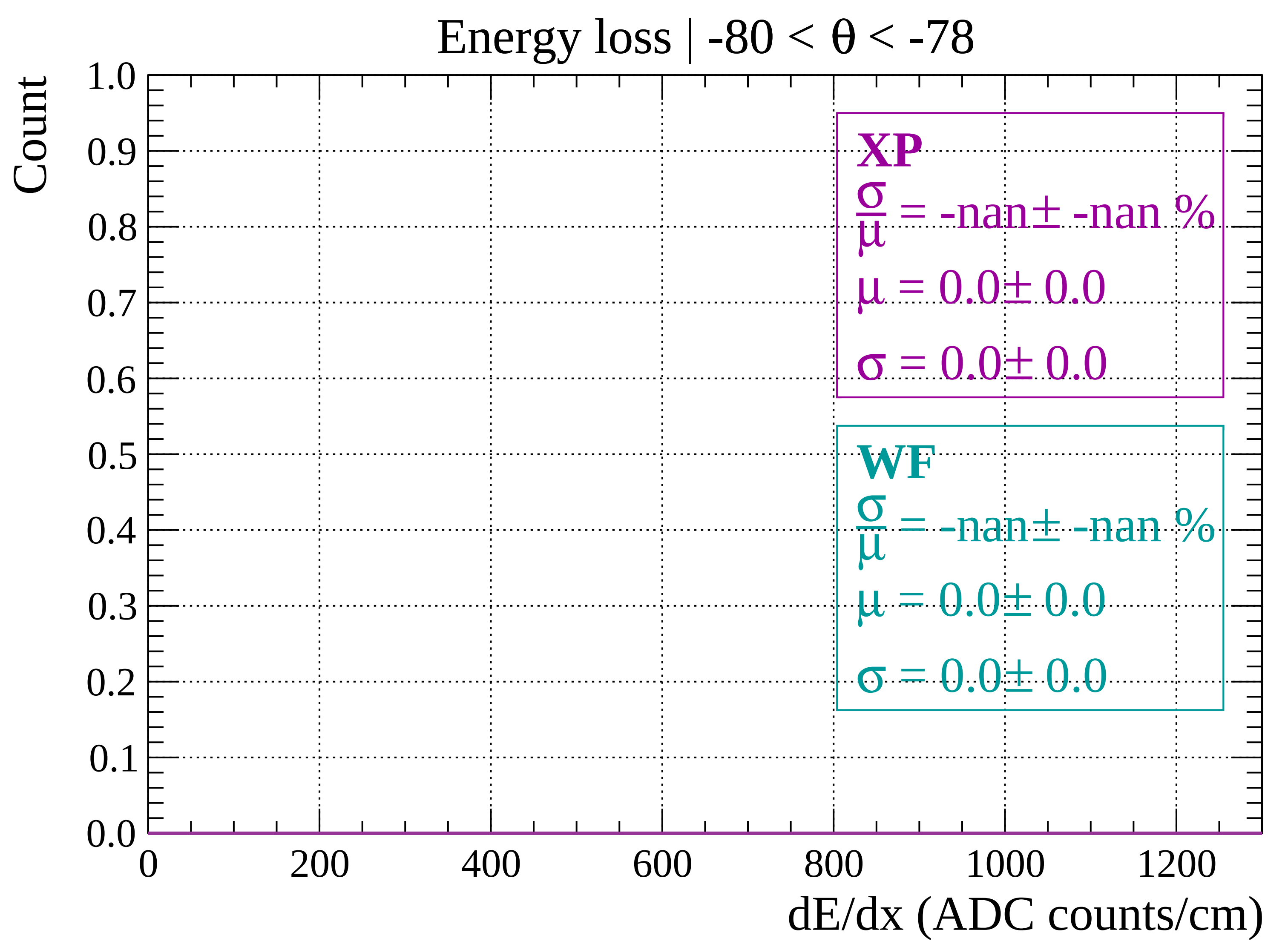


Energy loss | -86 < θ < -84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

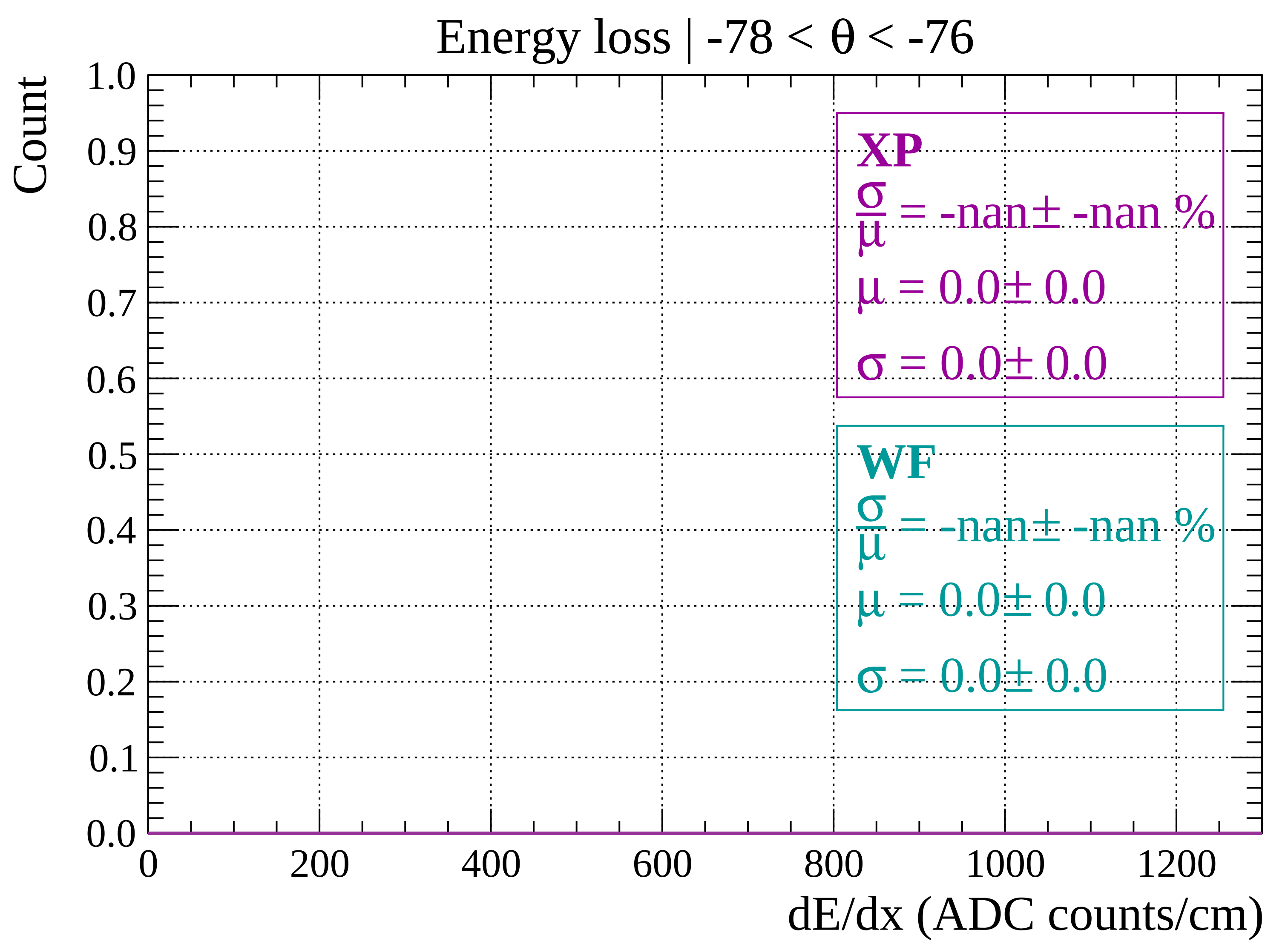

Energy loss | -84 < θ < -82
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



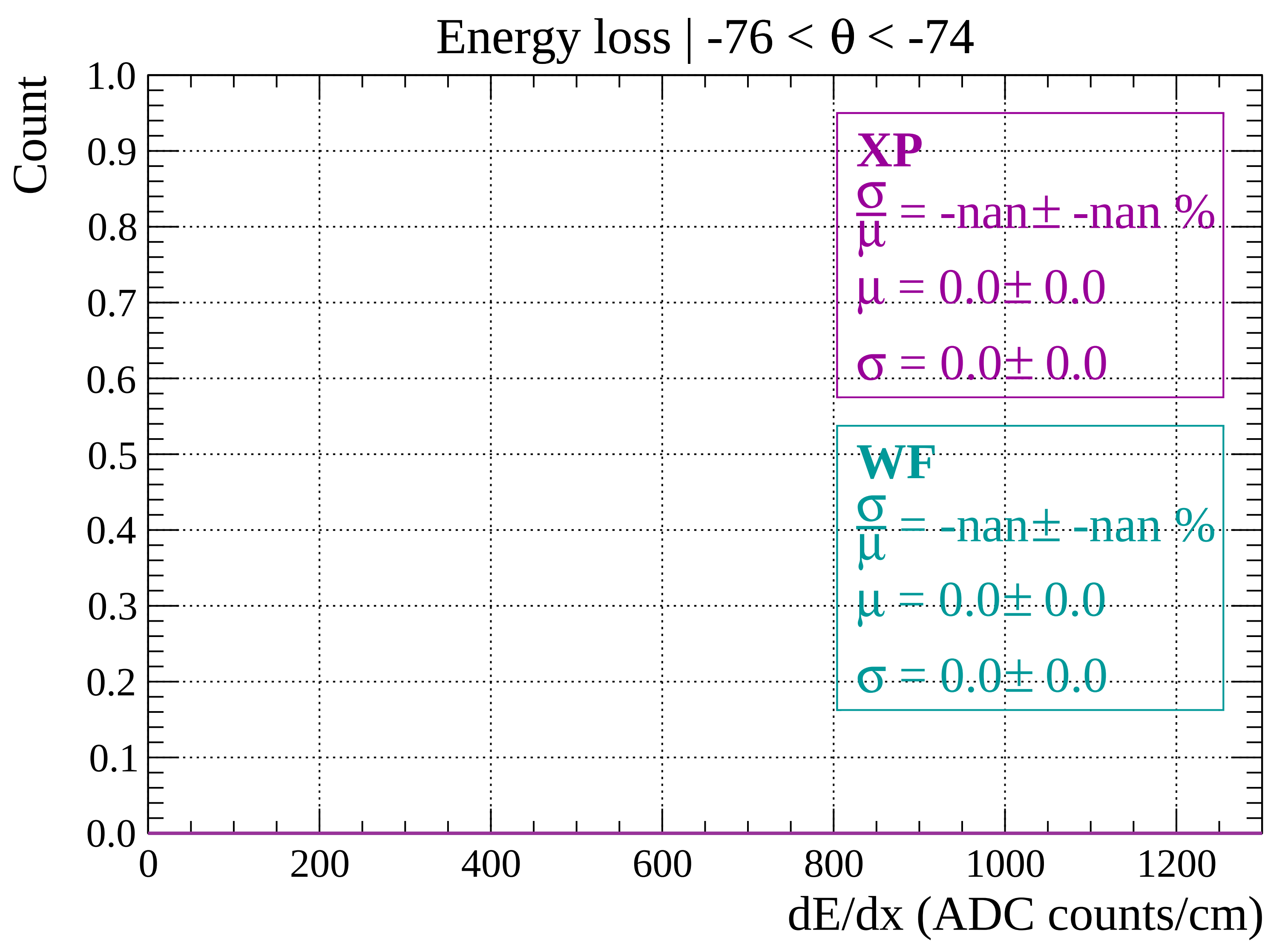
Energy loss | -82 < θ < -80
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



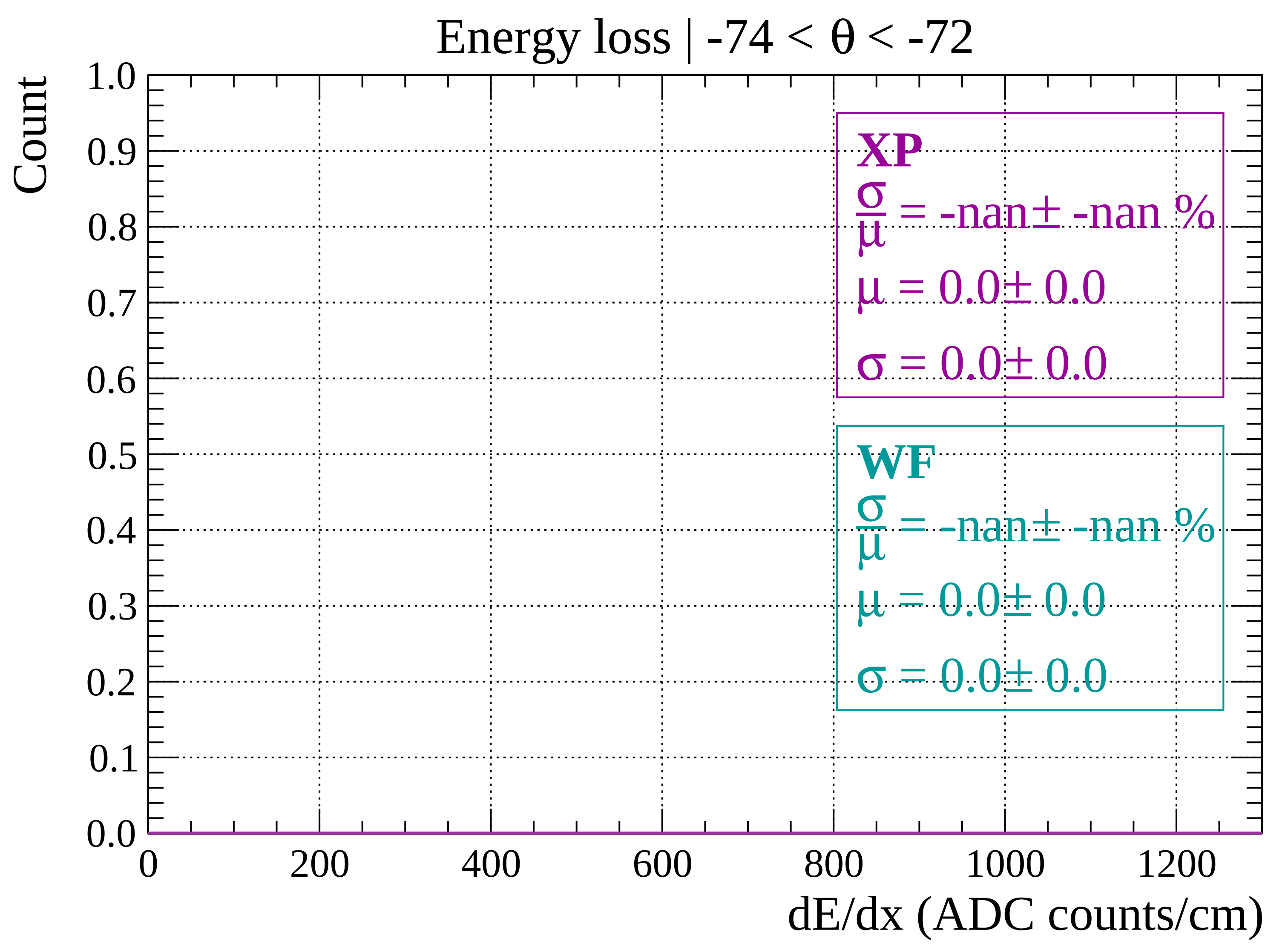
Energy loss | -80 < θ < -78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



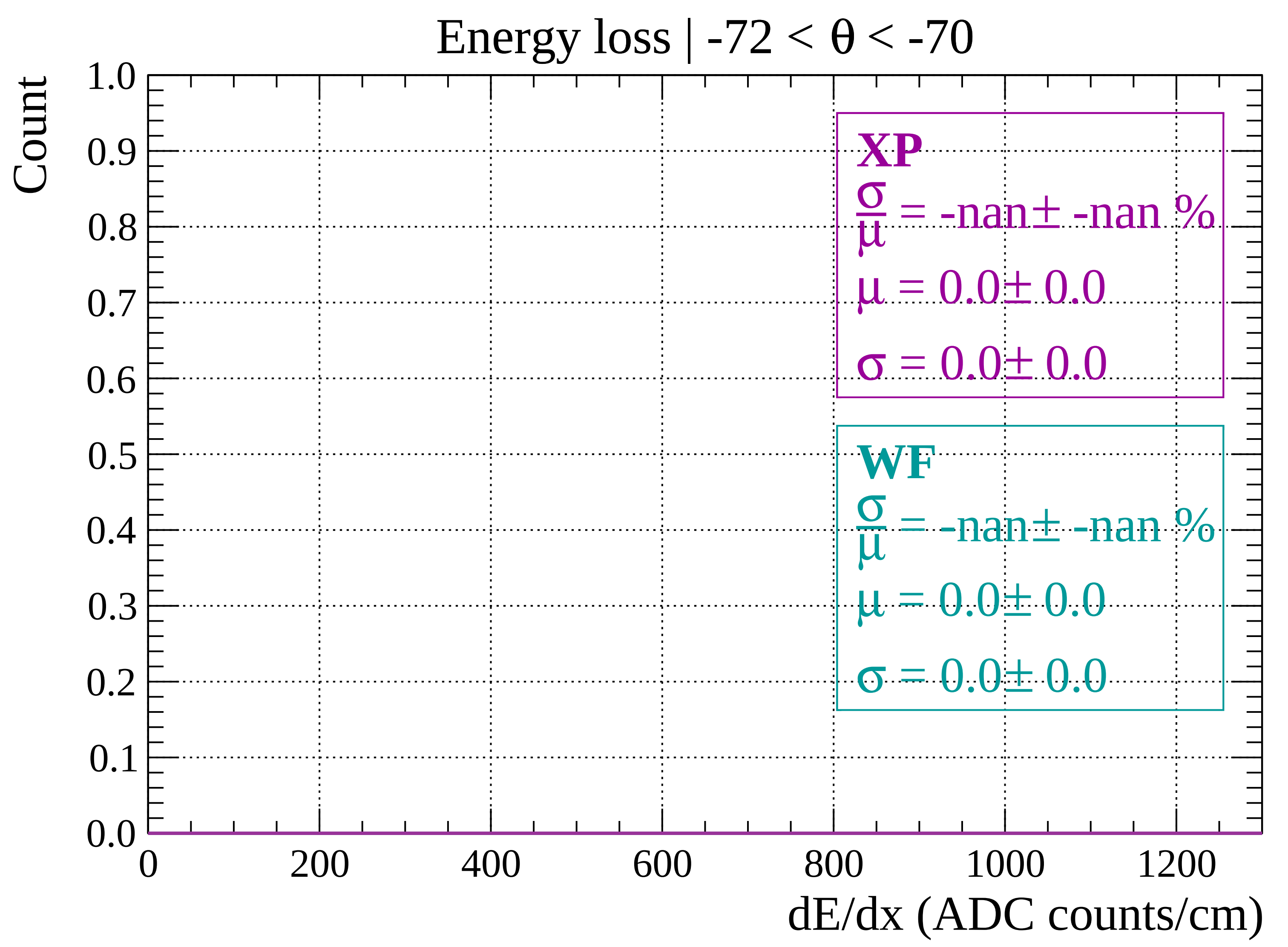
Energy loss | -78 < θ < -76
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

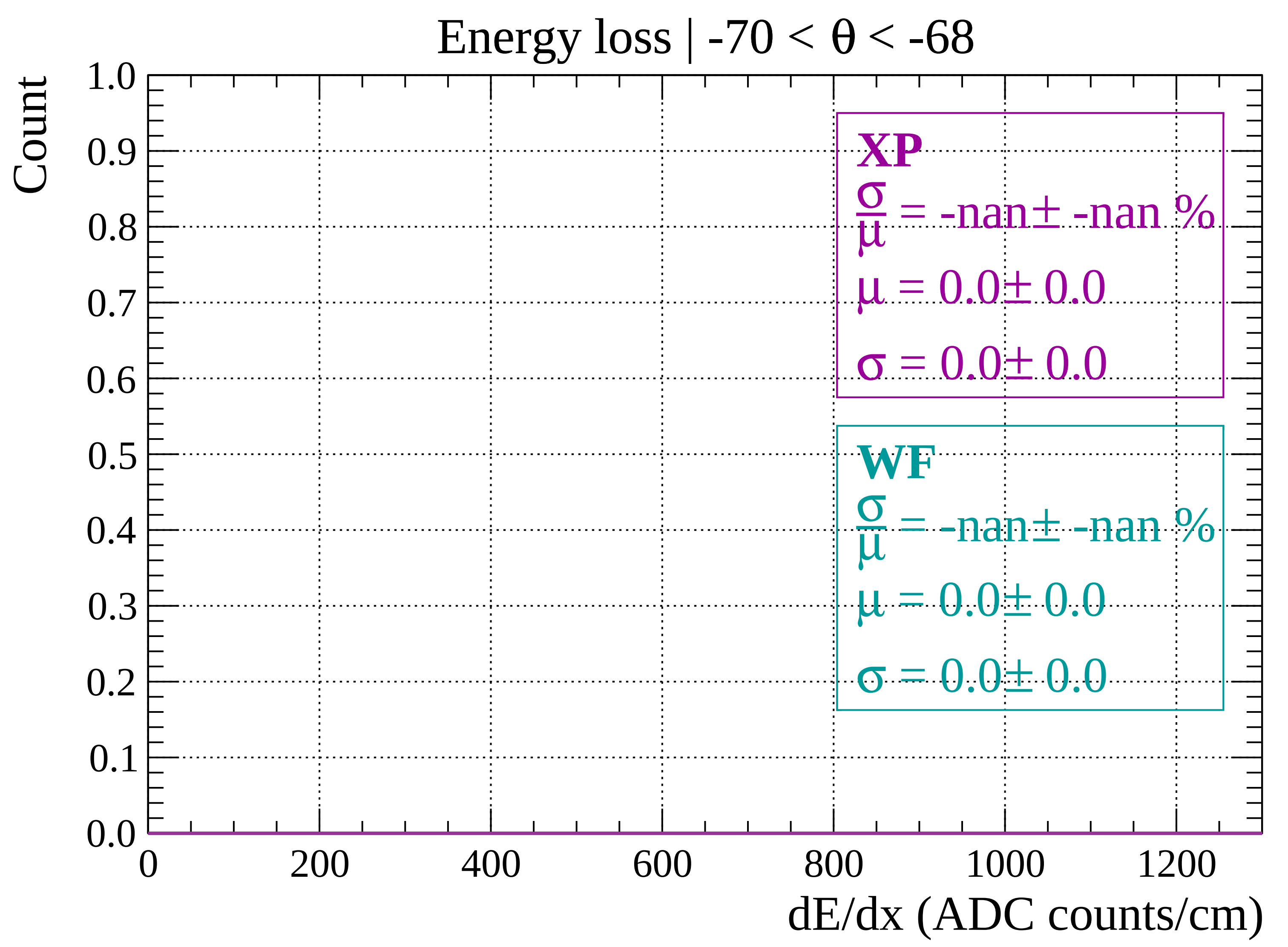


Energy loss | -76 < θ < -74
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

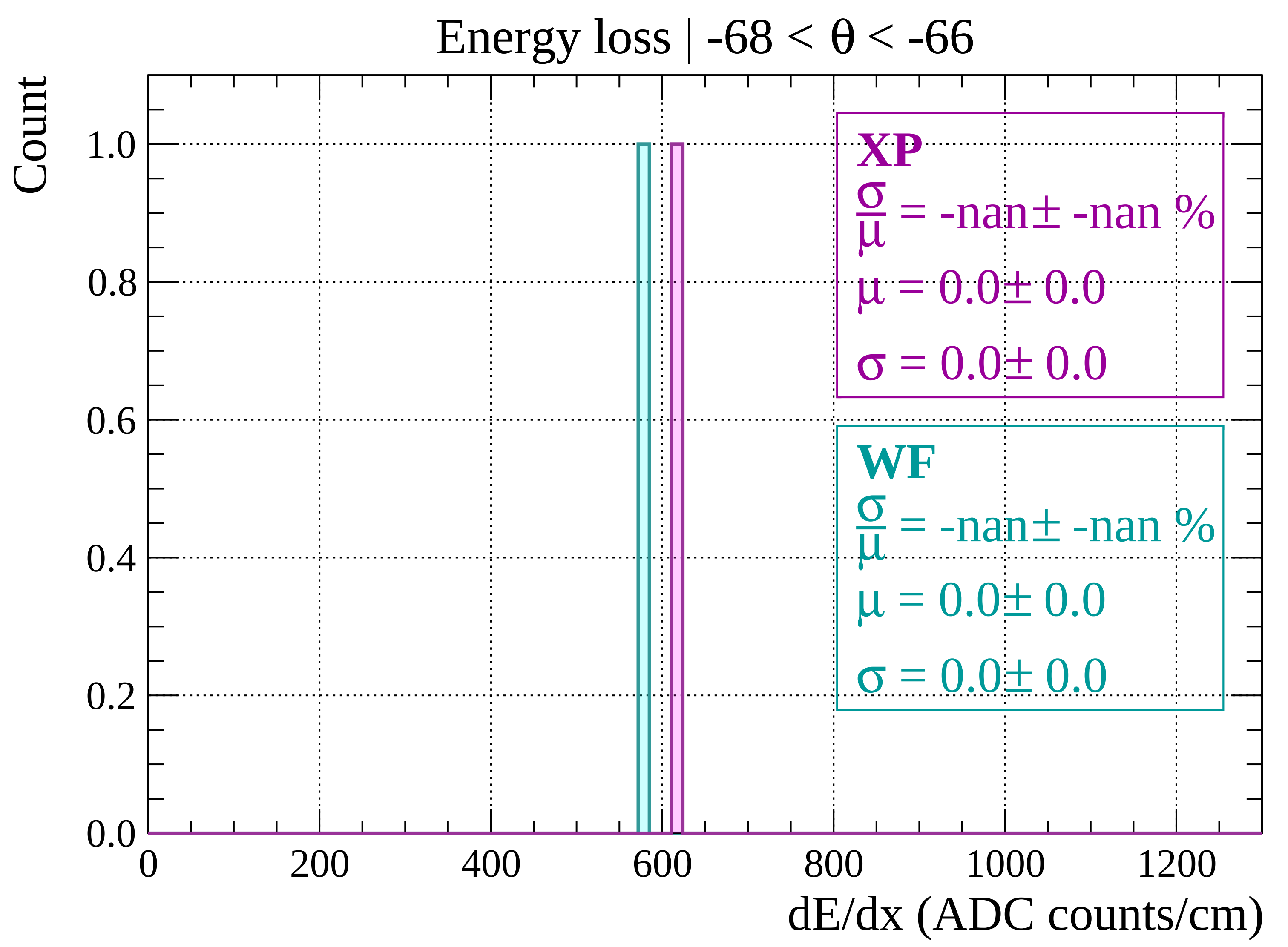


Energy loss | -74 < θ < -72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

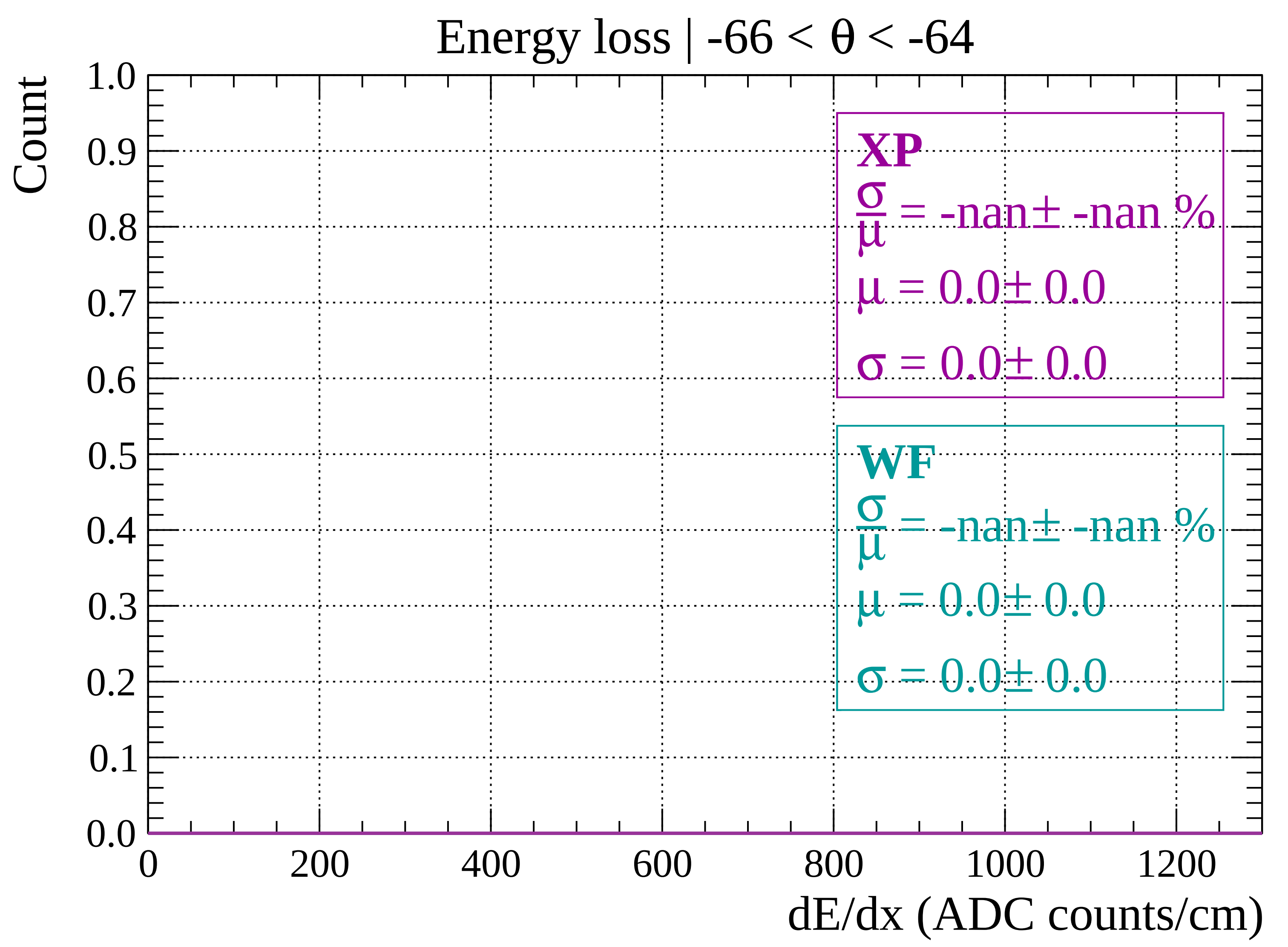

Energy loss | -72 < θ < -70
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200

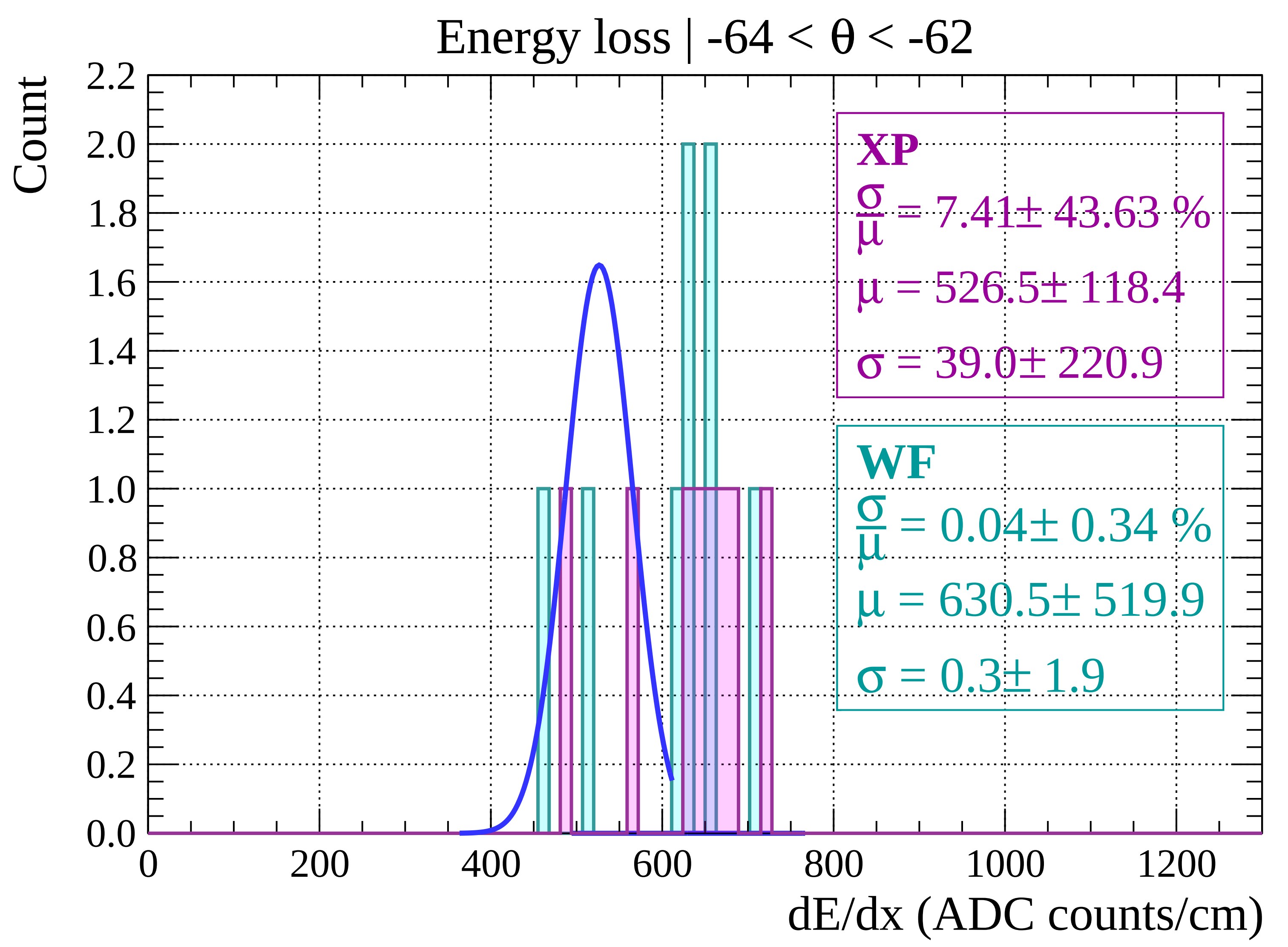


Energy loss | -70 < θ < -68
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

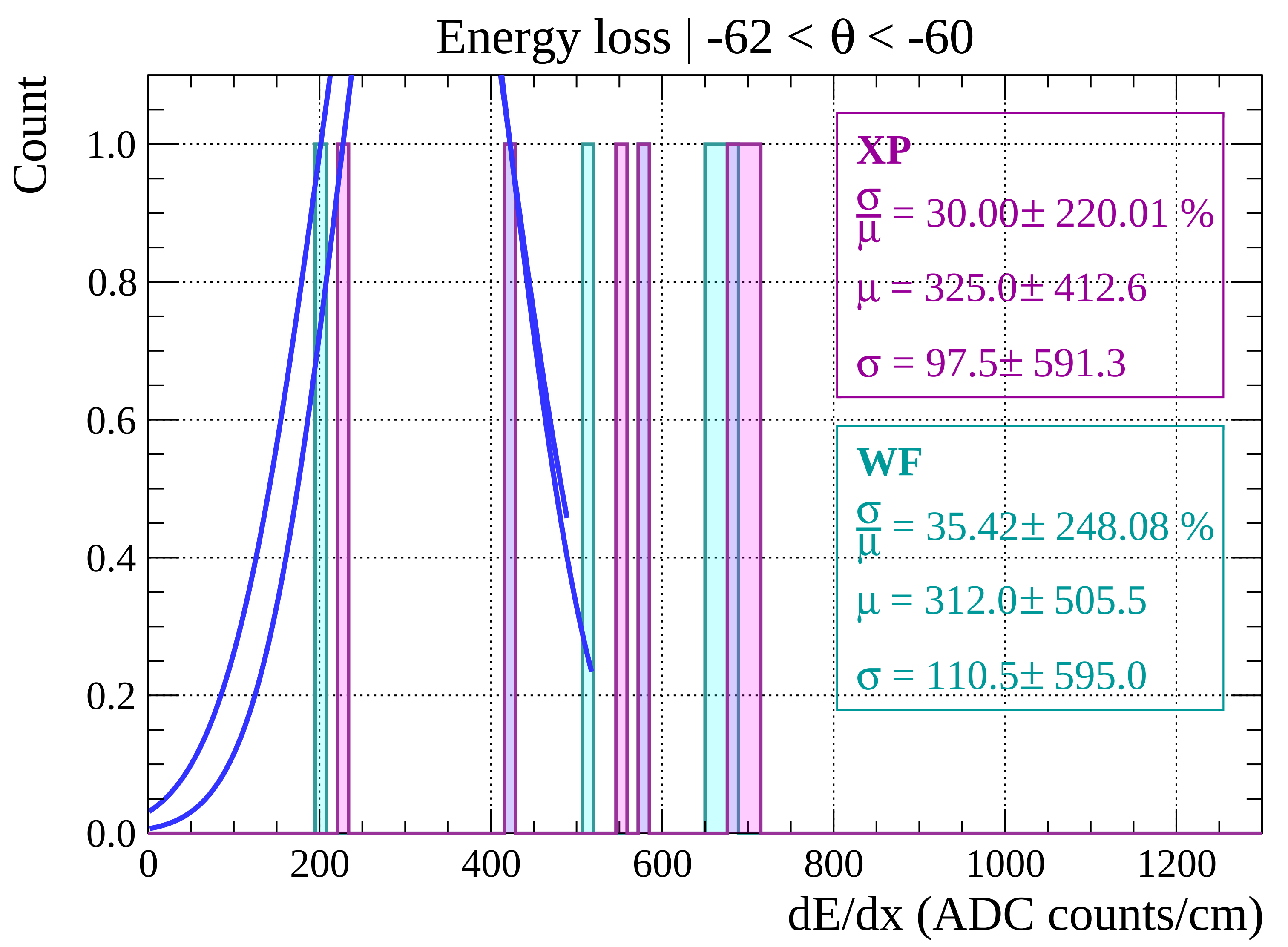


Energy loss | -68 < θ < -66
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

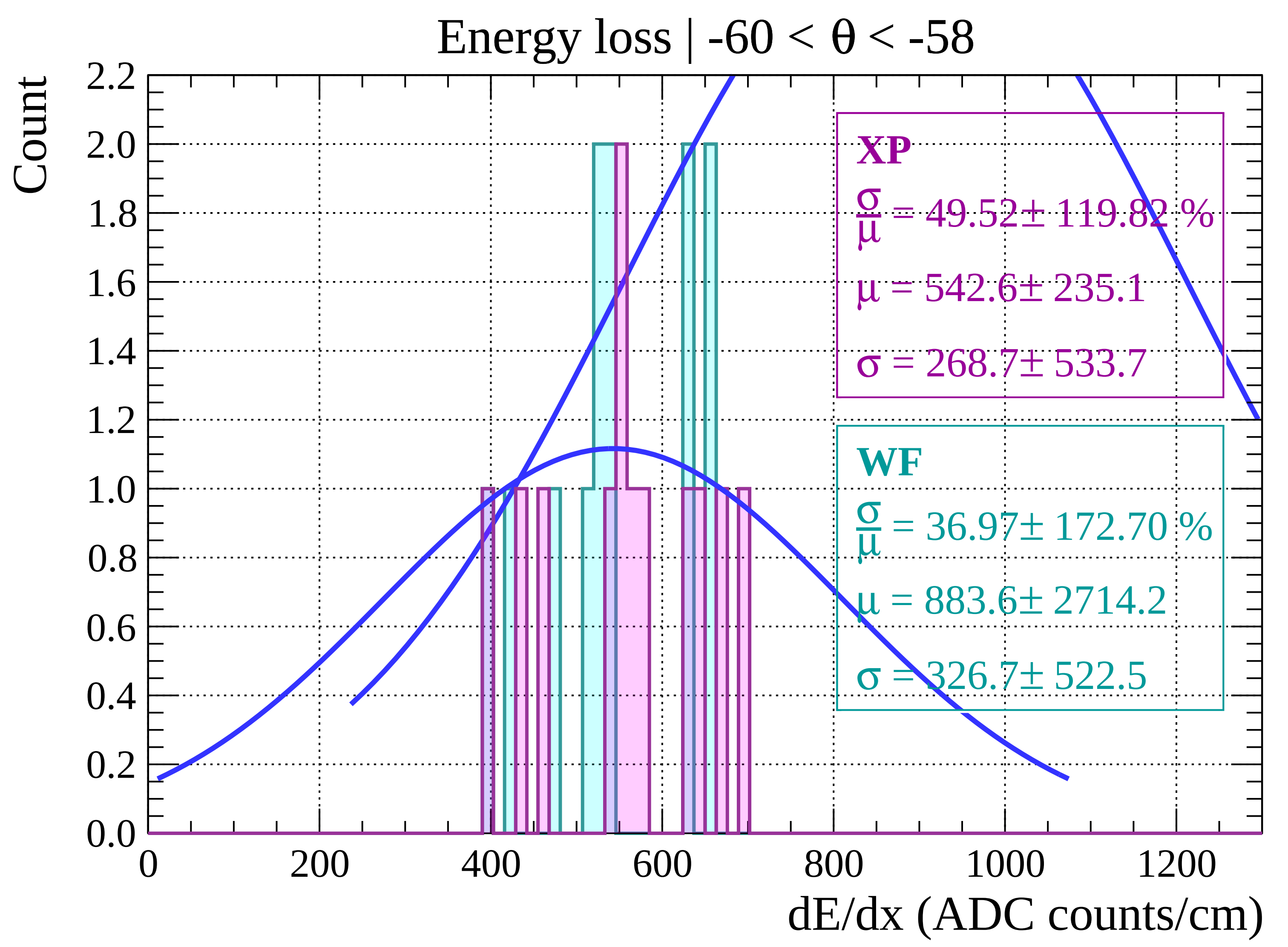

Energy loss | -66 < θ < -64
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



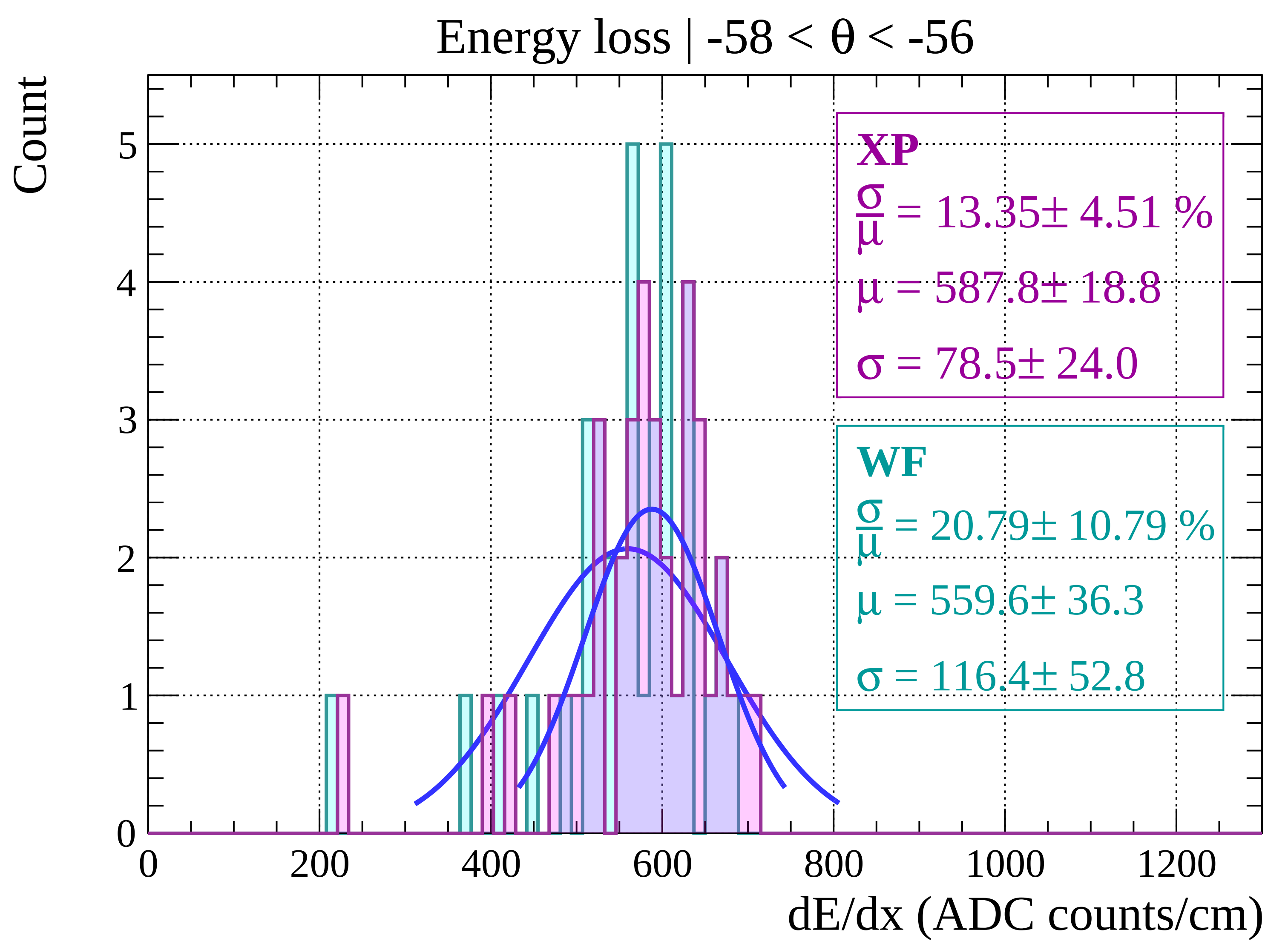
Energy loss | -64 < θ < -62
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.41± 43.63 %
μ = 526.5± 118.4
σ = 39.0± 220.9
WF
σ/μ = 0.04± 0.34 %
μ = 630.5± 519.9
σ = 0.3± 1.9



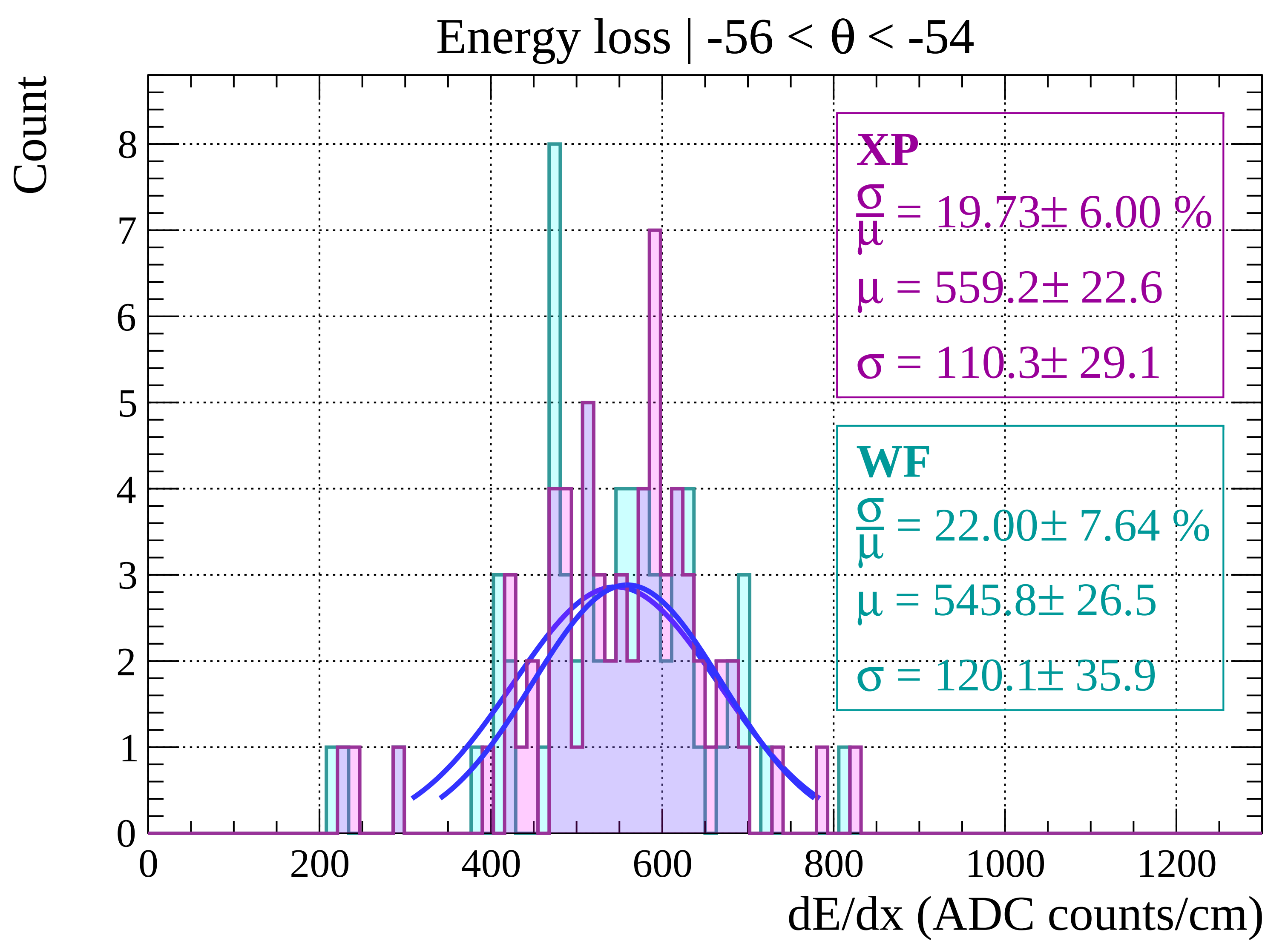
Energy loss | -62 < θ < -60
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 30.00± 220.01 %
μ = 325.0± 412.6
σ = 97.5± 591.3
WF
σ/μ = 35.42± 248.08 %
μ = 312.0± 505.5
σ = 110.5± 595.0



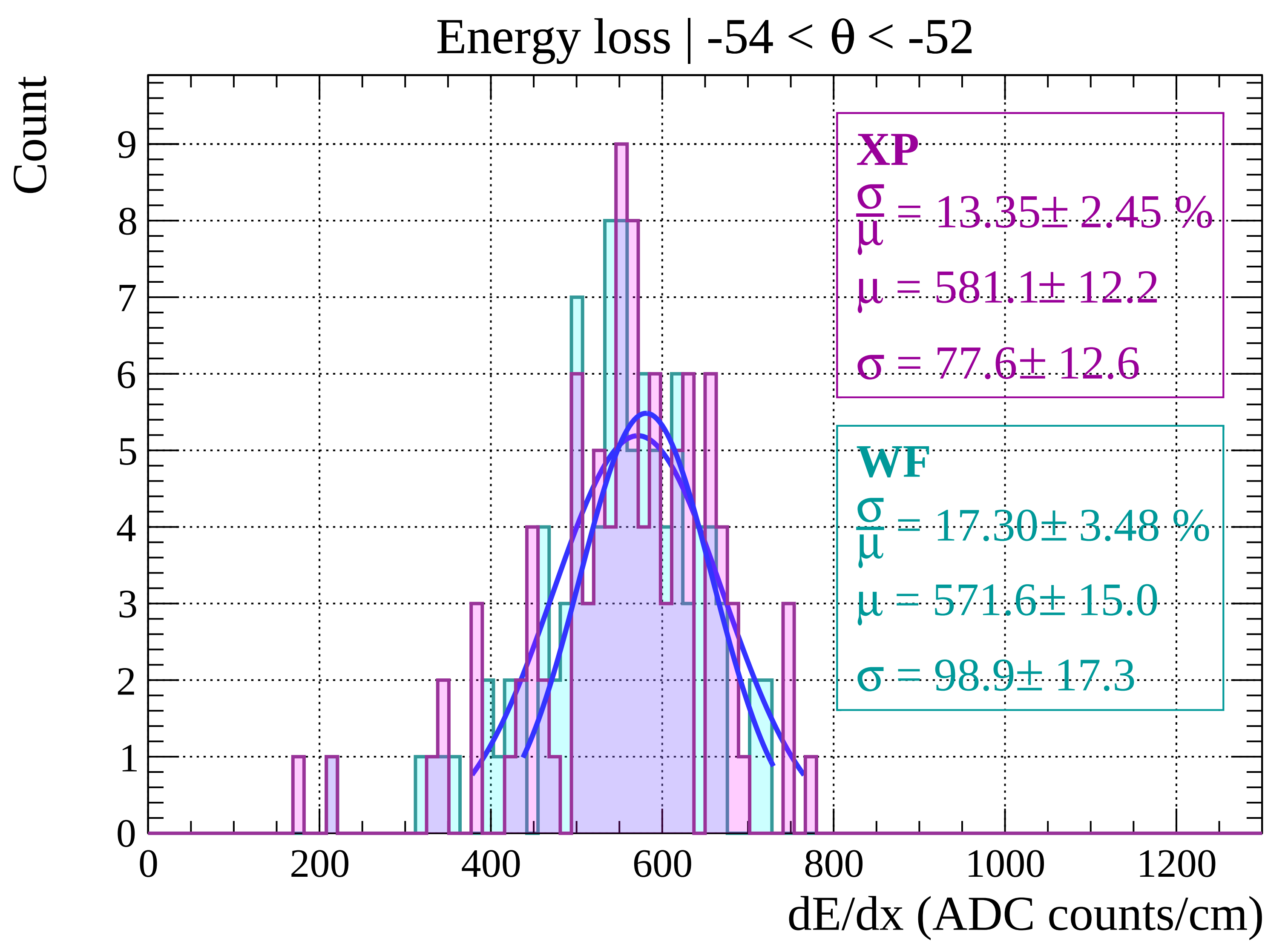
Energy loss | -60 < θ < -58
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 49.52± 119.82 %
μ = 542.6± 235.1
σ = 268.7± 533.7
WF
σ/μ = 36.97± 172.70 %
μ = 883.6± 2714.2
σ = 326.7± 522.5



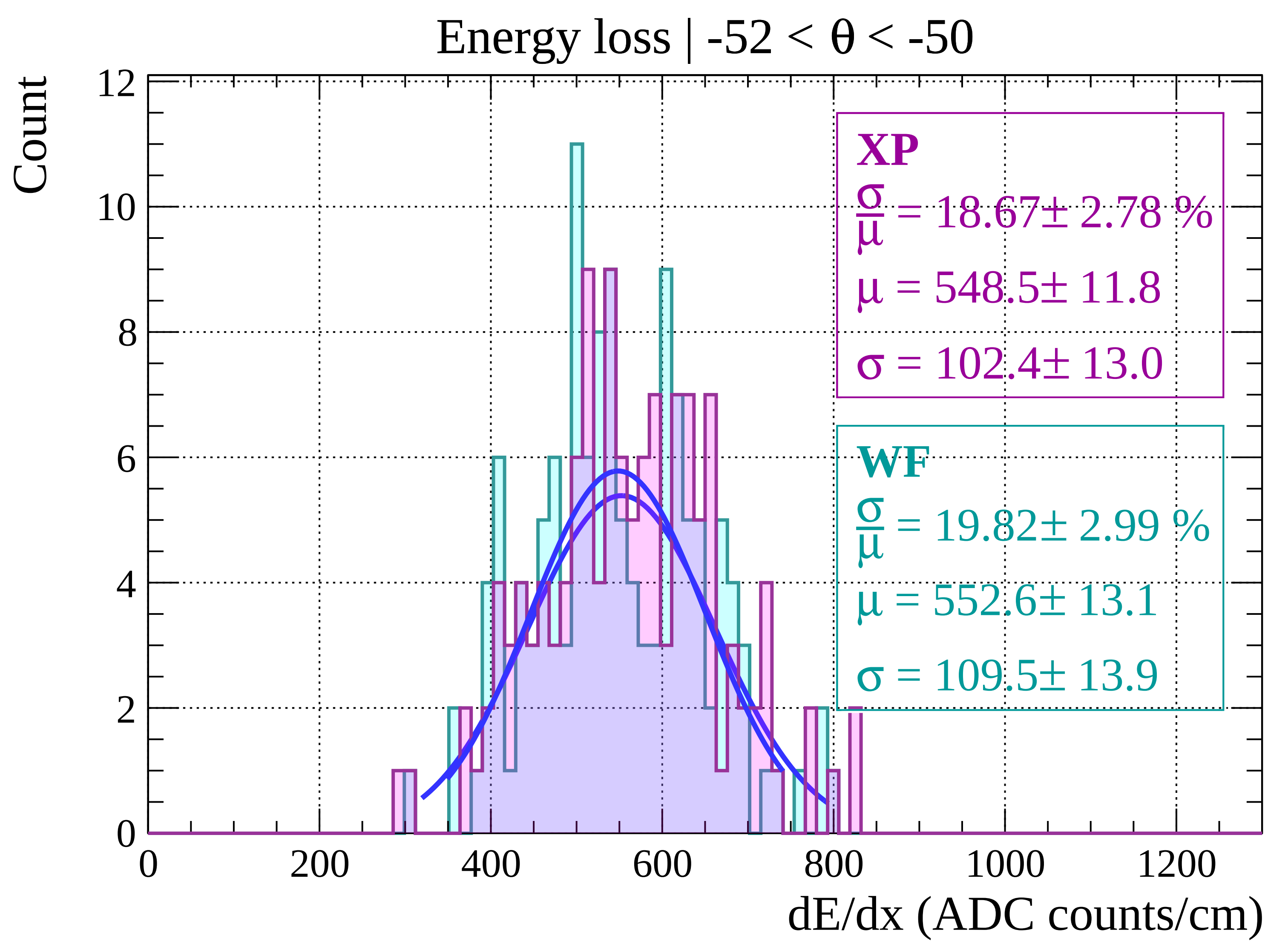
Energy loss | -58 < θ < -56
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
σ/μ = 13.35± 4.51 %
μ = 587.8± 18.8
σ = 78.5± 24.0
WF
σ/μ = 20.79± 10.79 %
μ = 559.6± 36.3
σ = 116.4± 52.8

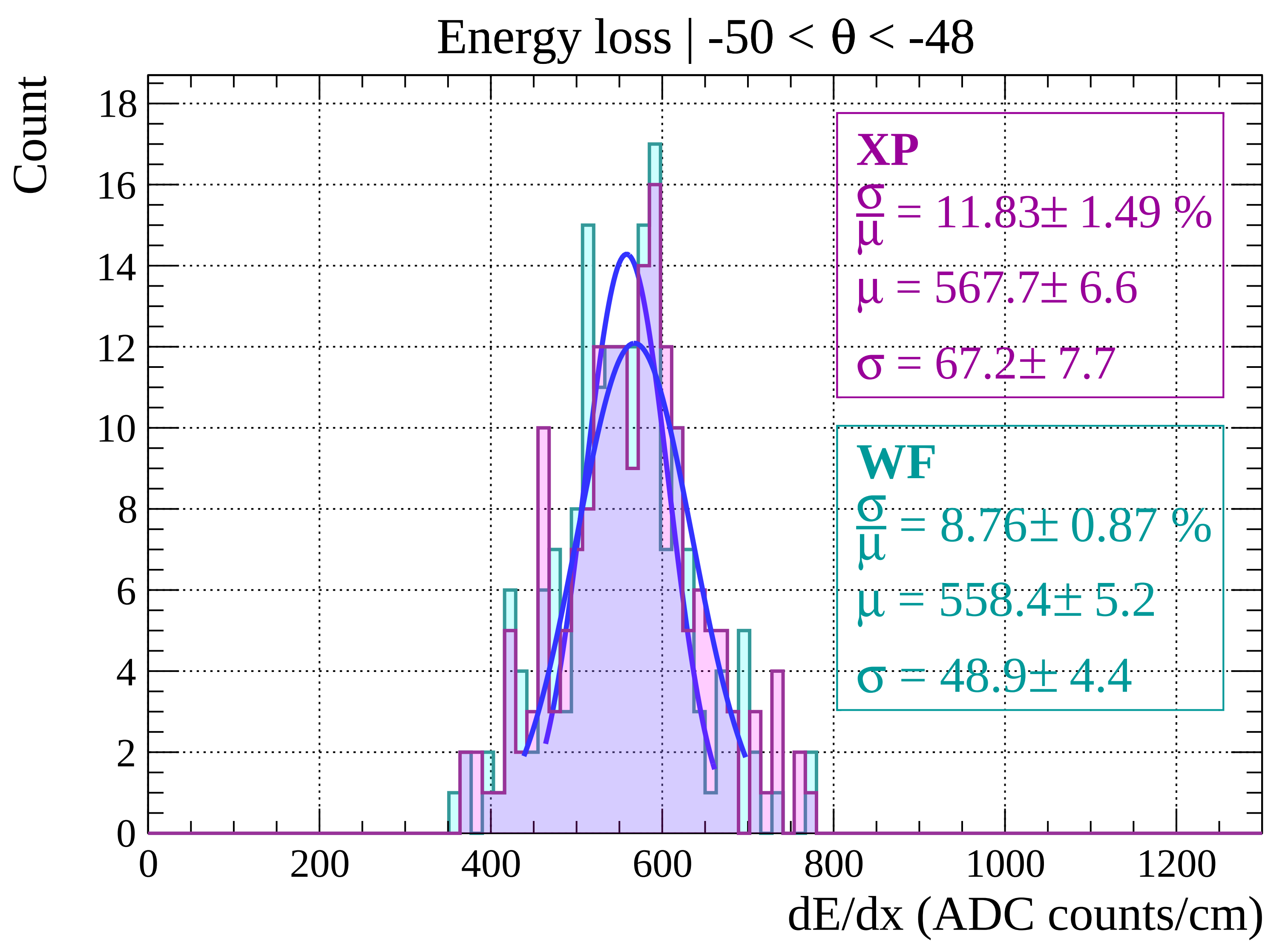


Energy loss | -56 < θ < -54
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 19.73± 6.00 %
μ = 559.2± 22.6
σ = 110.3± 29.1
WF
σ/μ = 22.00± 7.64 %
μ = 545.8± 26.5
σ = 120.1± 35.9
0
1
2
3
4
5
6
7
8
0
200
400
600
800
1000
1200

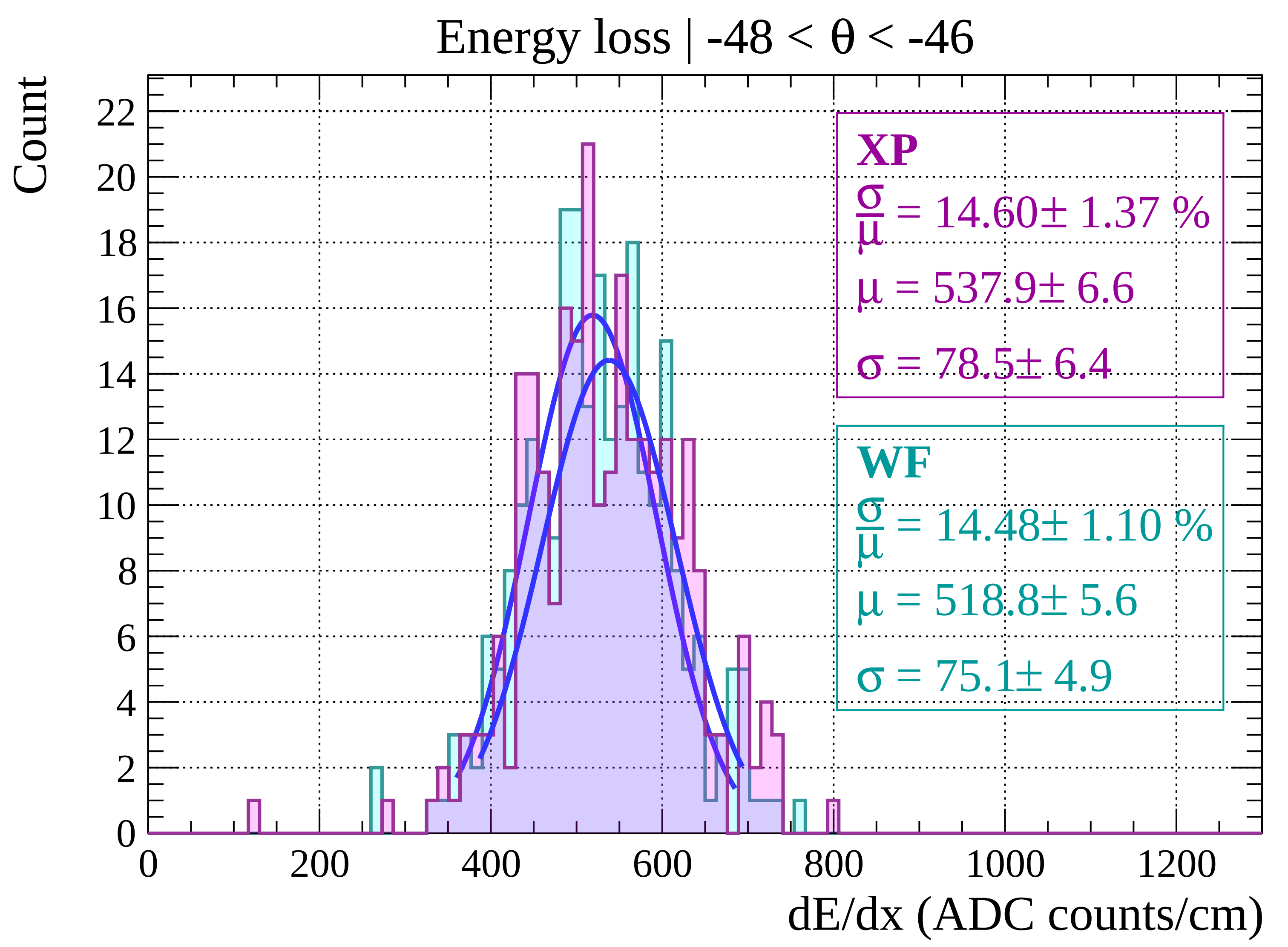


Energy loss | -54 < θ < -52
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 13.35± 2.45 %
μ = 581.1± 12.2
σ = 77.6± 12.6
WF
σ/μ = 17.30± 3.48 %
μ = 571.6± 15.0
σ = 98.9± 17.3
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
9

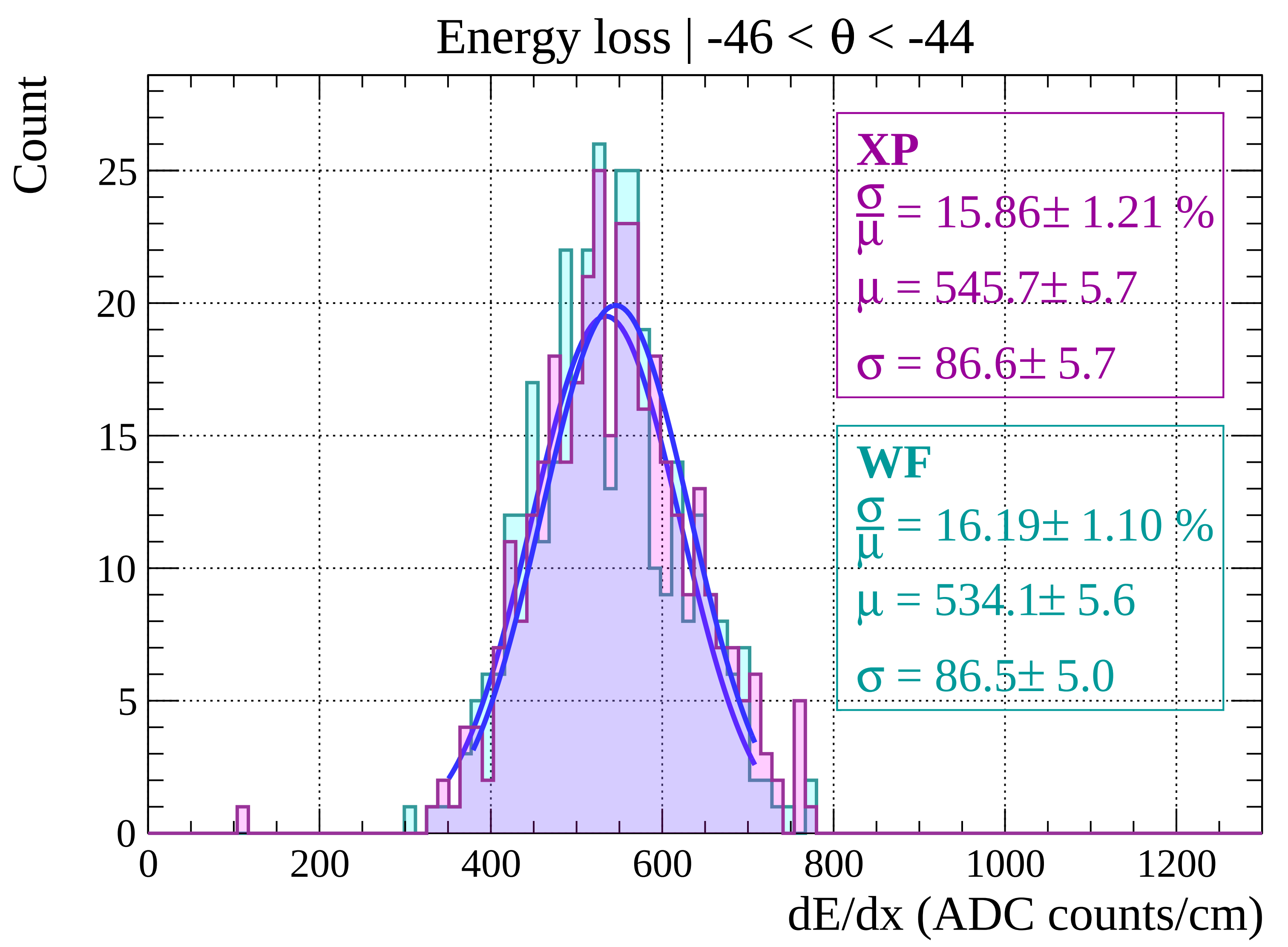

Energy loss | -52 < θ < -50
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 18.67± 2.78 %
μ = 548.5± 11.8
σ = 102.4± 13.0
WF
σ/μ = 19.82± 2.99 %
μ = 552.6± 13.1
σ = 109.5± 13.9



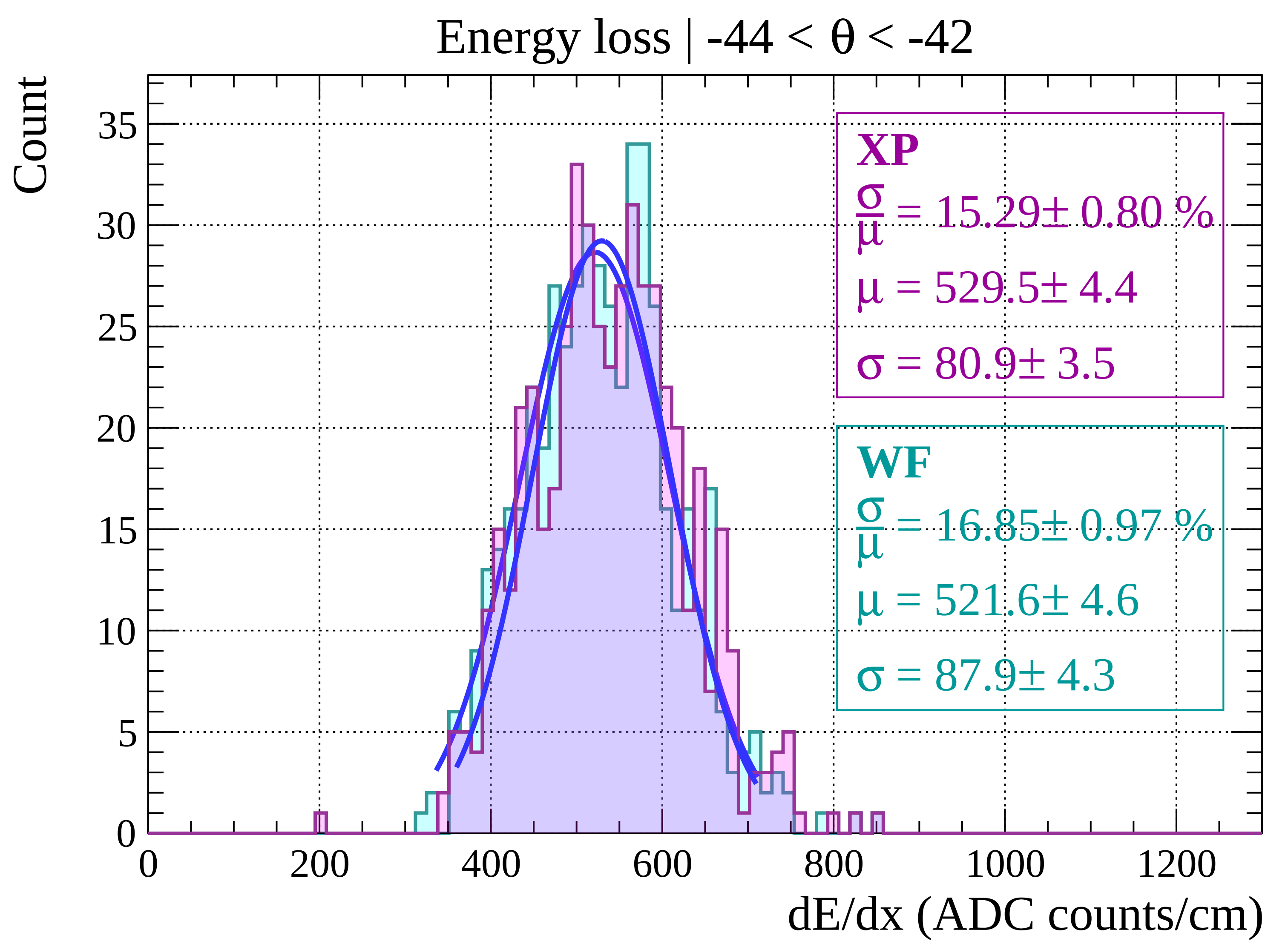
Energy loss | -50 < θ < -48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
XP
σ/μ = 11.83± 1.49 %
μ = 567.7± 6.6
σ = 67.2± 7.7
WF
σ/μ = 8.76± 0.87 %
μ = 558.4± 5.2
σ = 48.9± 4.4



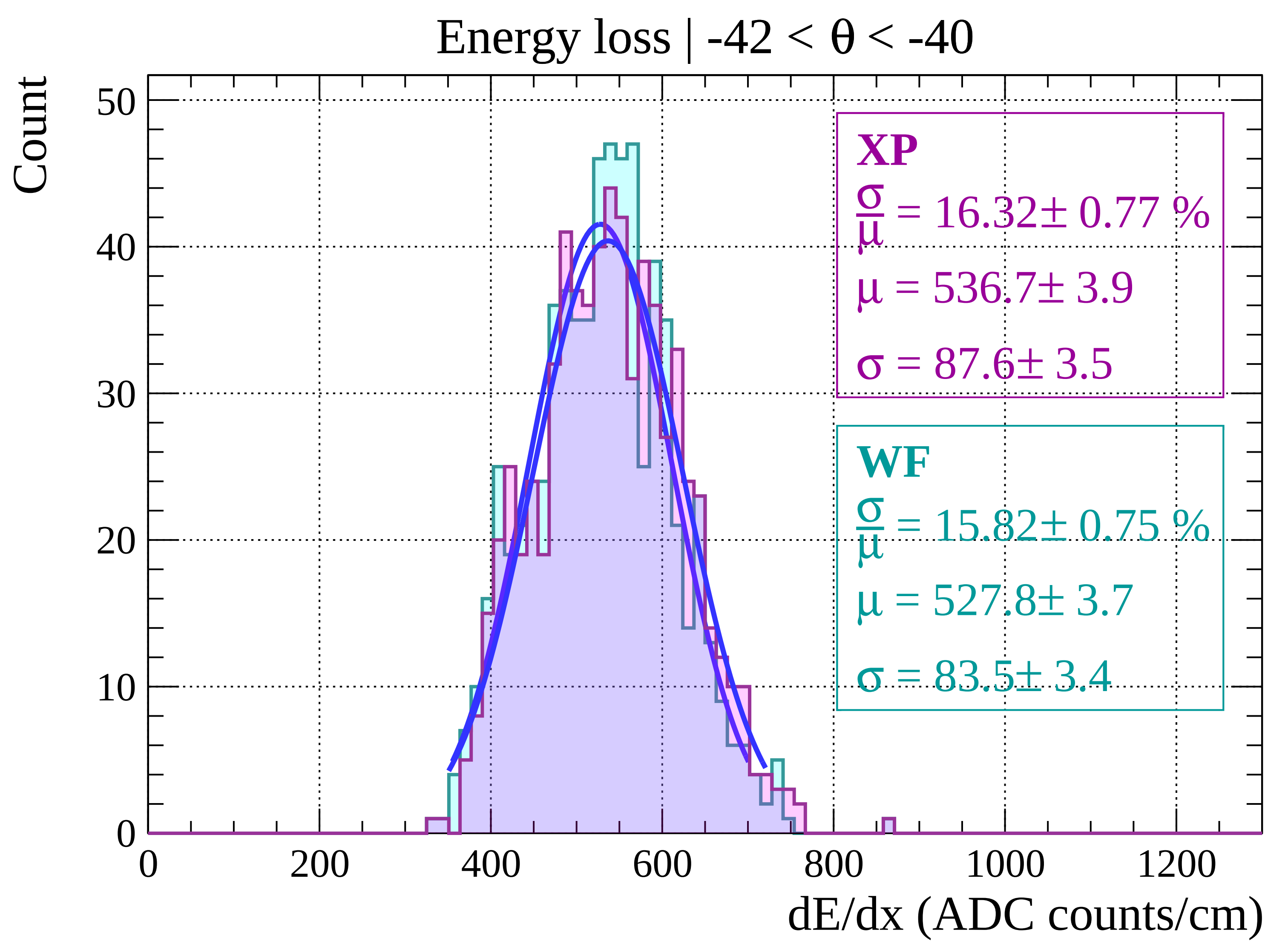
Energy loss | -48 < θ < -46
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
20
22
XP
σ/μ = 14.60± 1.37 %
μ = 537.9± 6.6
σ = 78.5± 6.4
WF
σ/μ = 14.48± 1.10 %
μ = 518.8± 5.6
σ = 75.1± 4.9



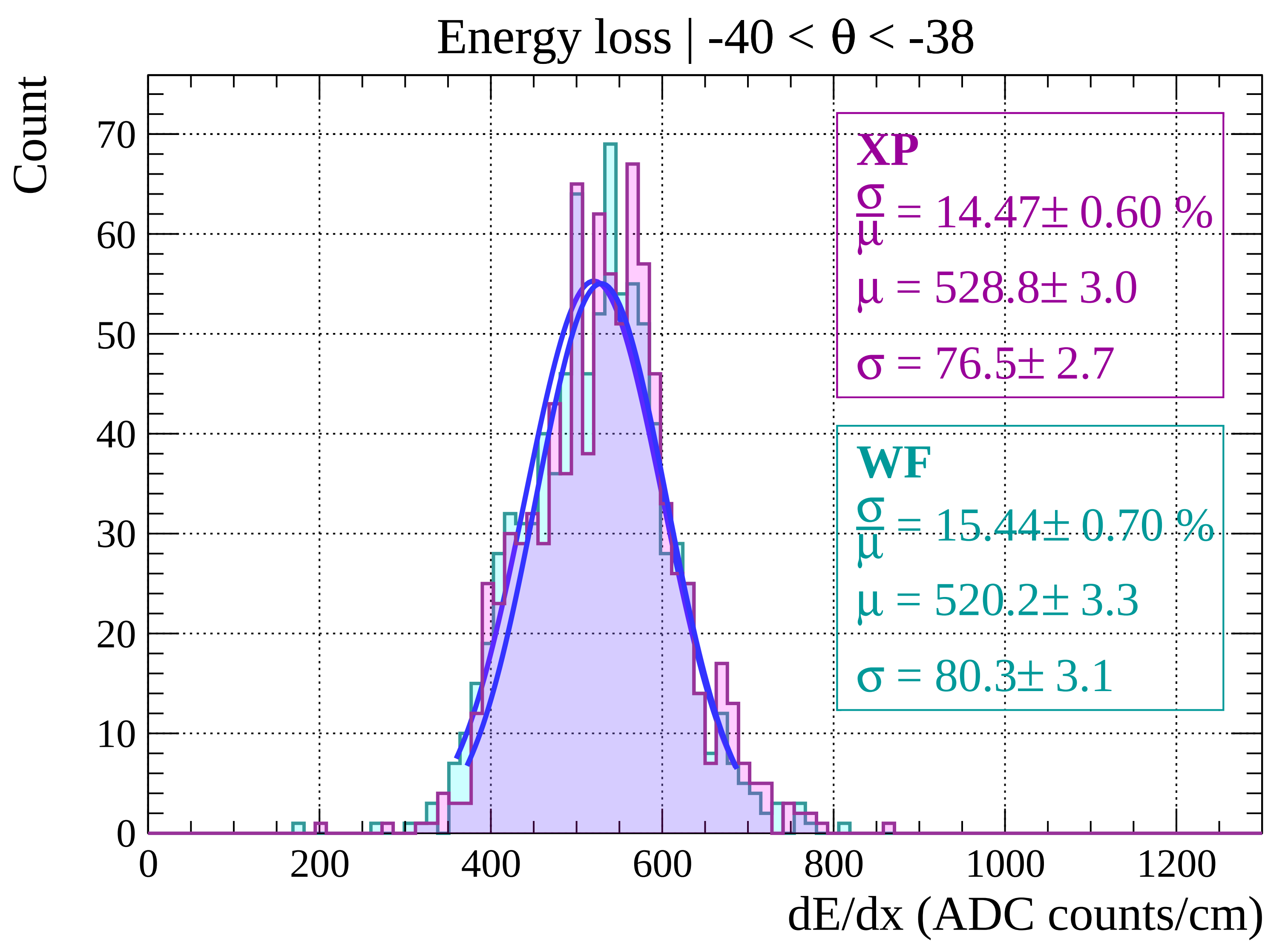
Energy loss | -46 < θ < -44
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 15.86± 1.21 %
μ = 545.7± 5.7
σ = 86.6± 5.7
WF
σ/μ = 16.19± 1.10 %
μ = 534.1± 5.6
σ = 86.5± 5.0



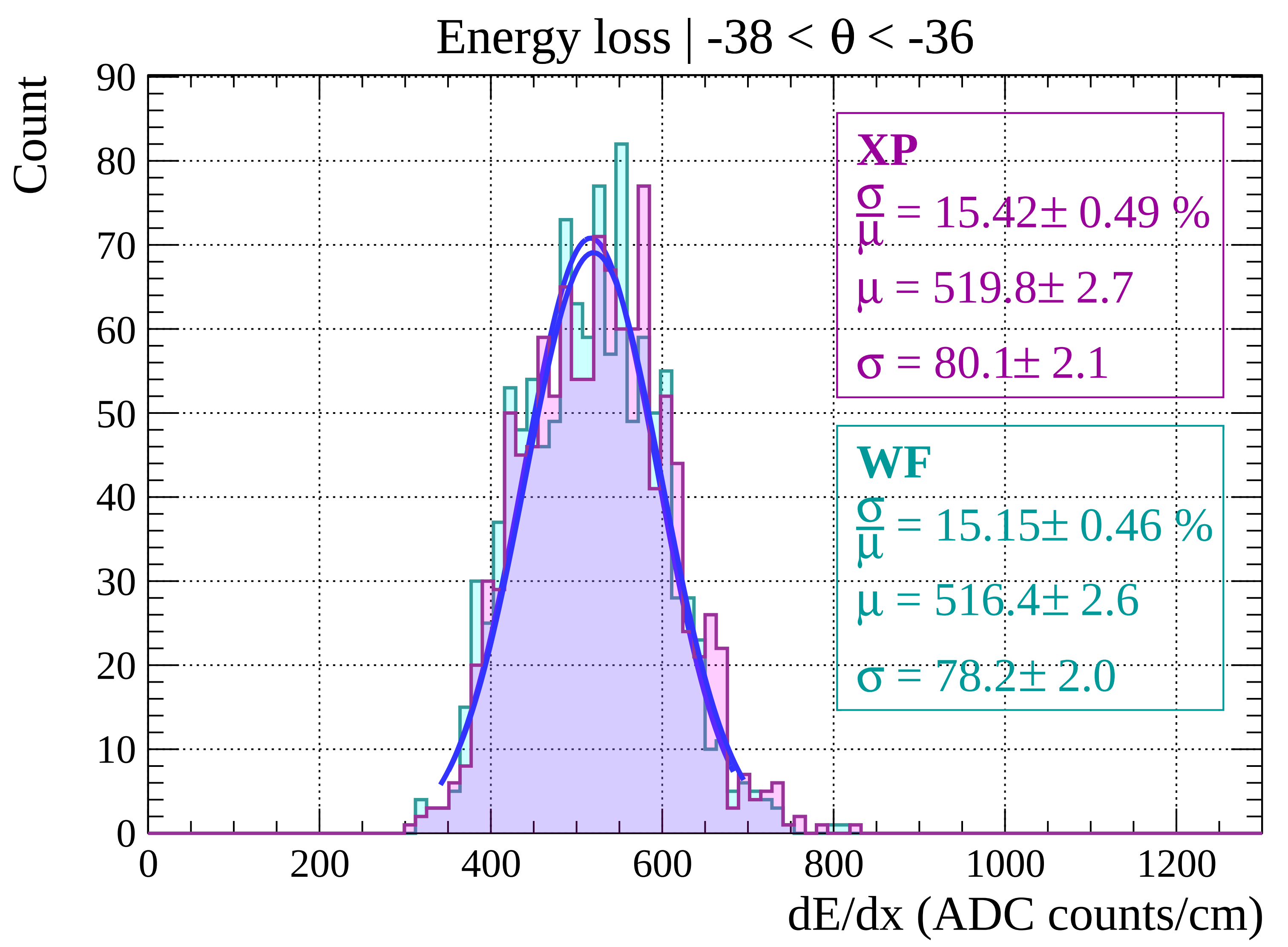
Energy loss | -44 < θ < -42
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP
σ/μ = 15.29± 0.80 %
μ = 529.5± 4.4
σ = 80.9± 3.5
WF
σ/μ = 16.85± 0.97 %
μ = 521.6± 4.6
σ = 87.9± 4.3



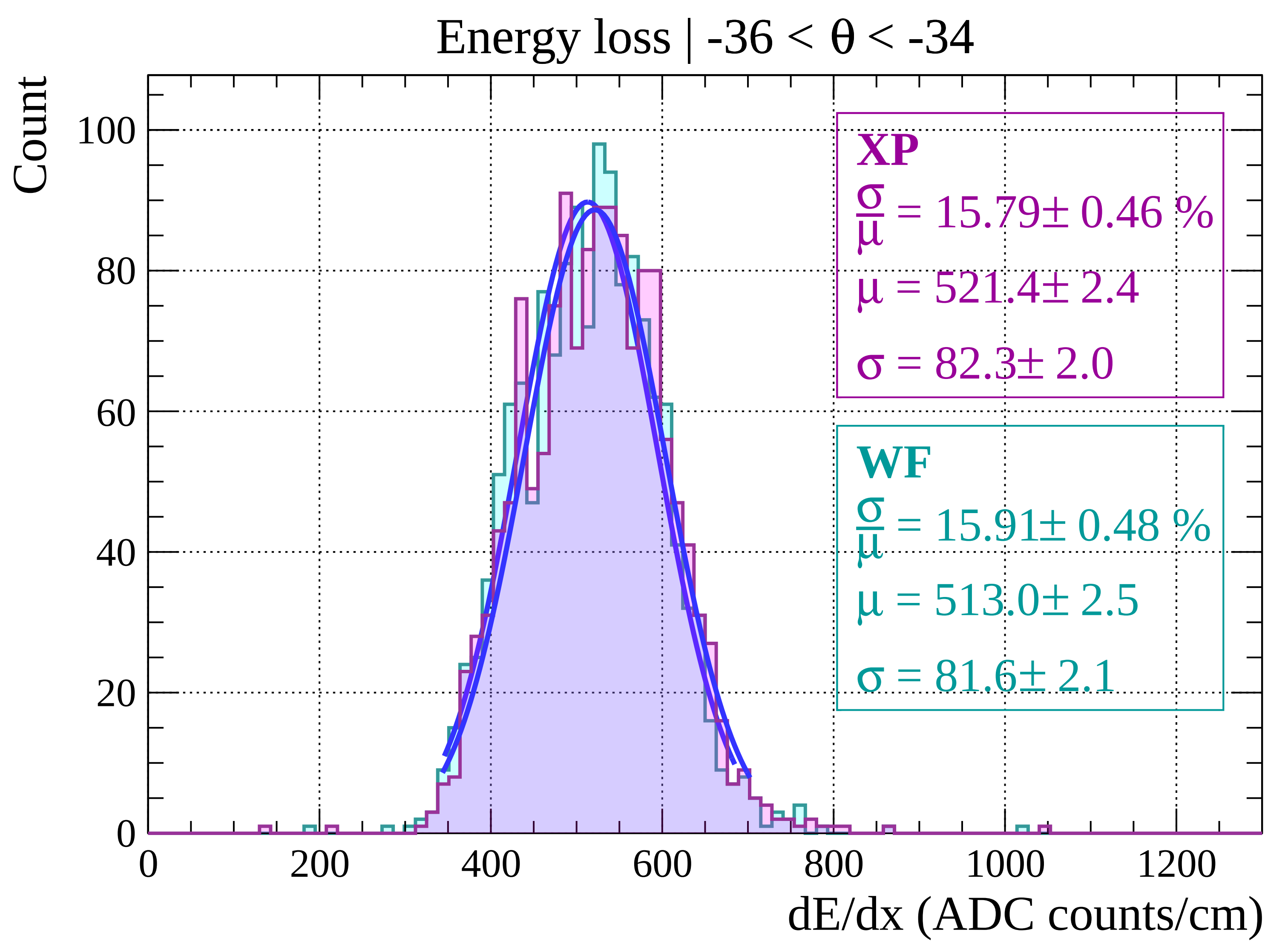
Energy loss | -42 < θ < -40
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP
$\frac{\sigma}{\mu}$ = 16.32± 0.77 %
μ = 536.7± 3.9
σ = 87.6± 3.5
WF
$\frac{\sigma}{\mu}$ = 15.82± 0.75 %
μ = 527.8± 3.7
σ = 83.5± 3.4



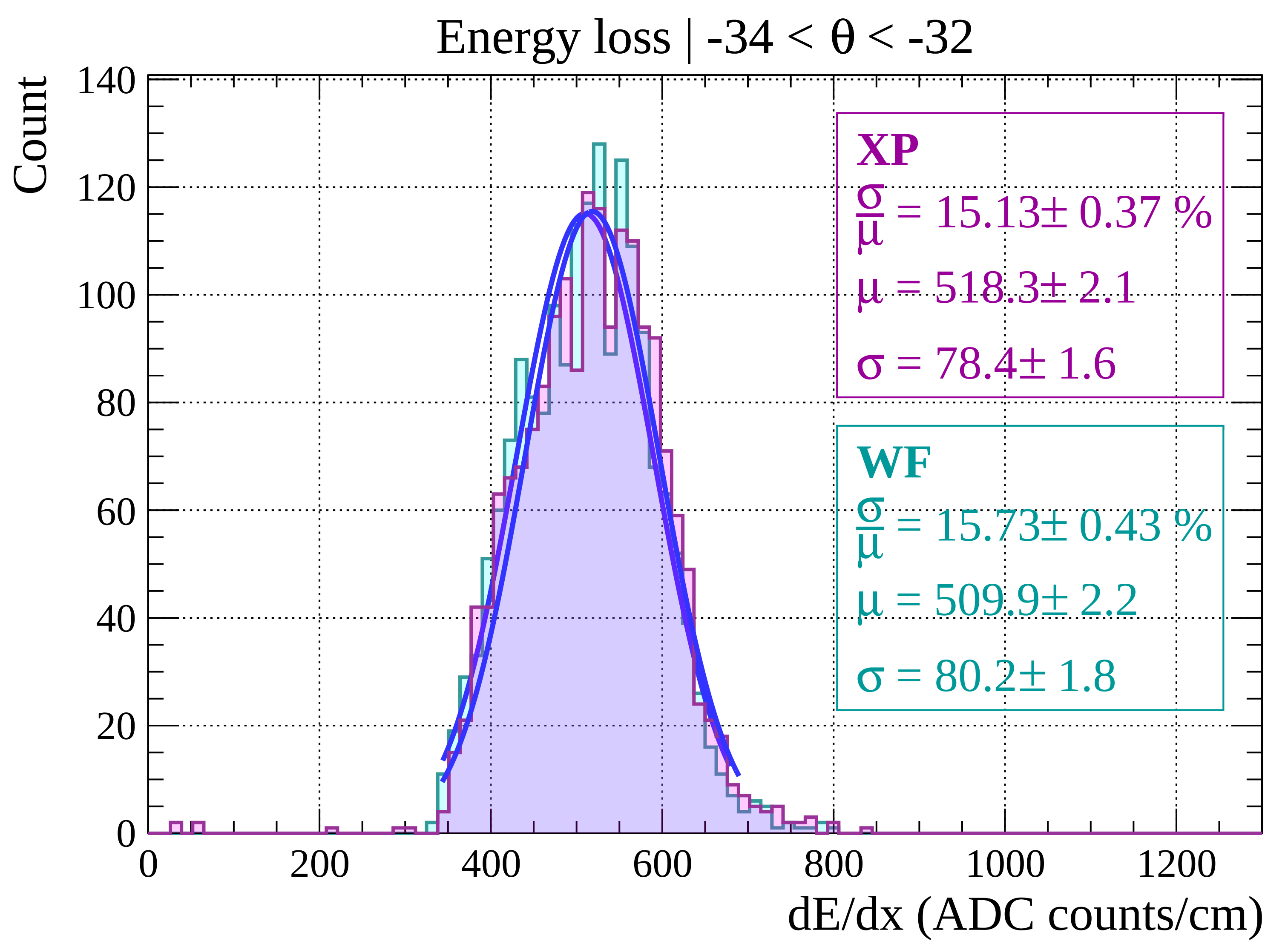
Energy loss | -40 < θ < -38
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
σ/μ = 14.47± 0.60 %
μ = 528.8± 3.0
σ = 76.5± 2.7
WF
σ/μ = 15.44± 0.70 %
μ = 520.2± 3.3
σ = 80.3± 3.1



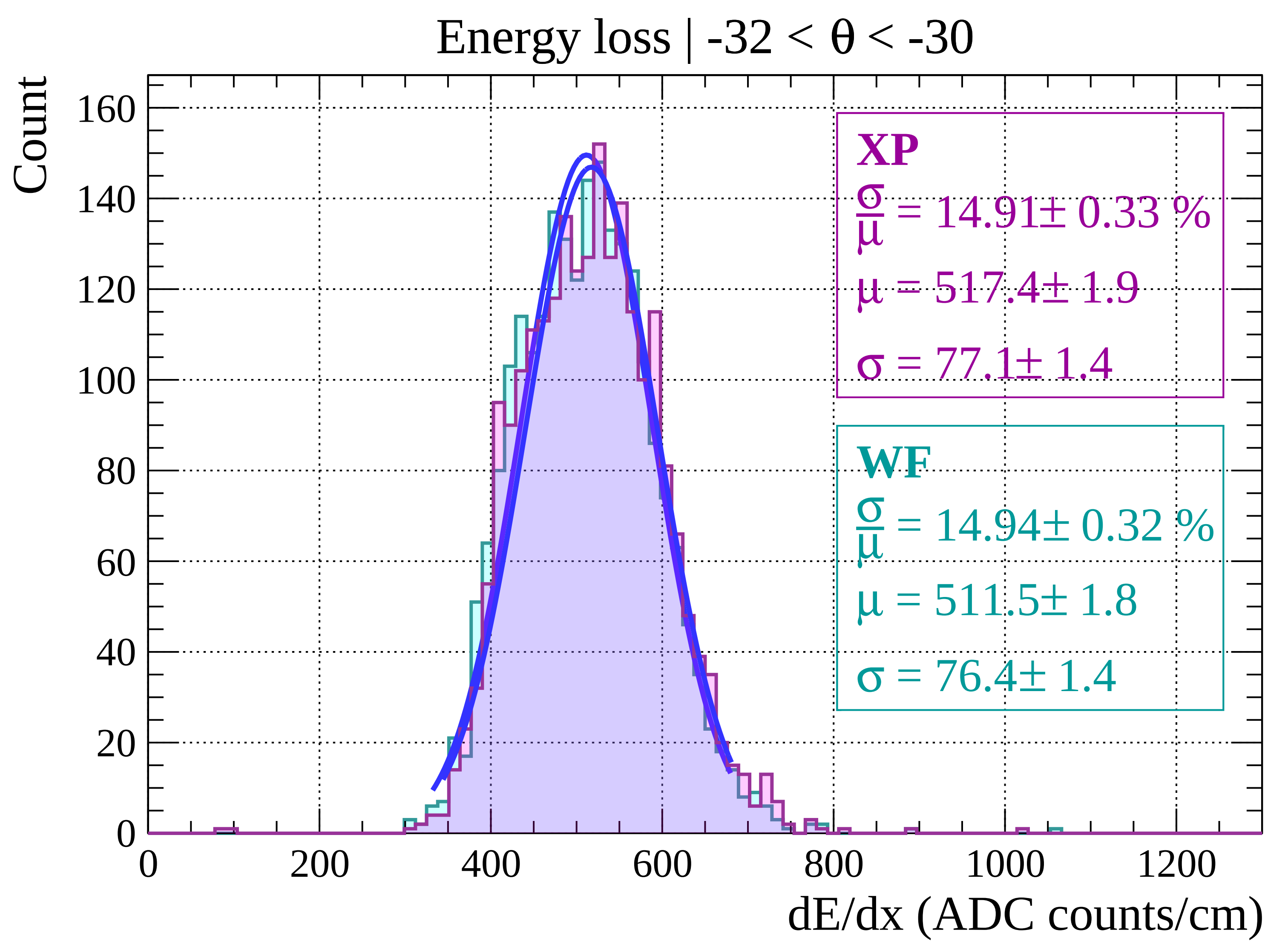
Energy loss | -38 < θ < -36
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.42± 0.49 %
μ = 519.8± 2.7
σ = 80.1± 2.1
WF
σ/μ = 15.15± 0.46 %
μ = 516.4± 2.6
σ = 78.2± 2.0
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200



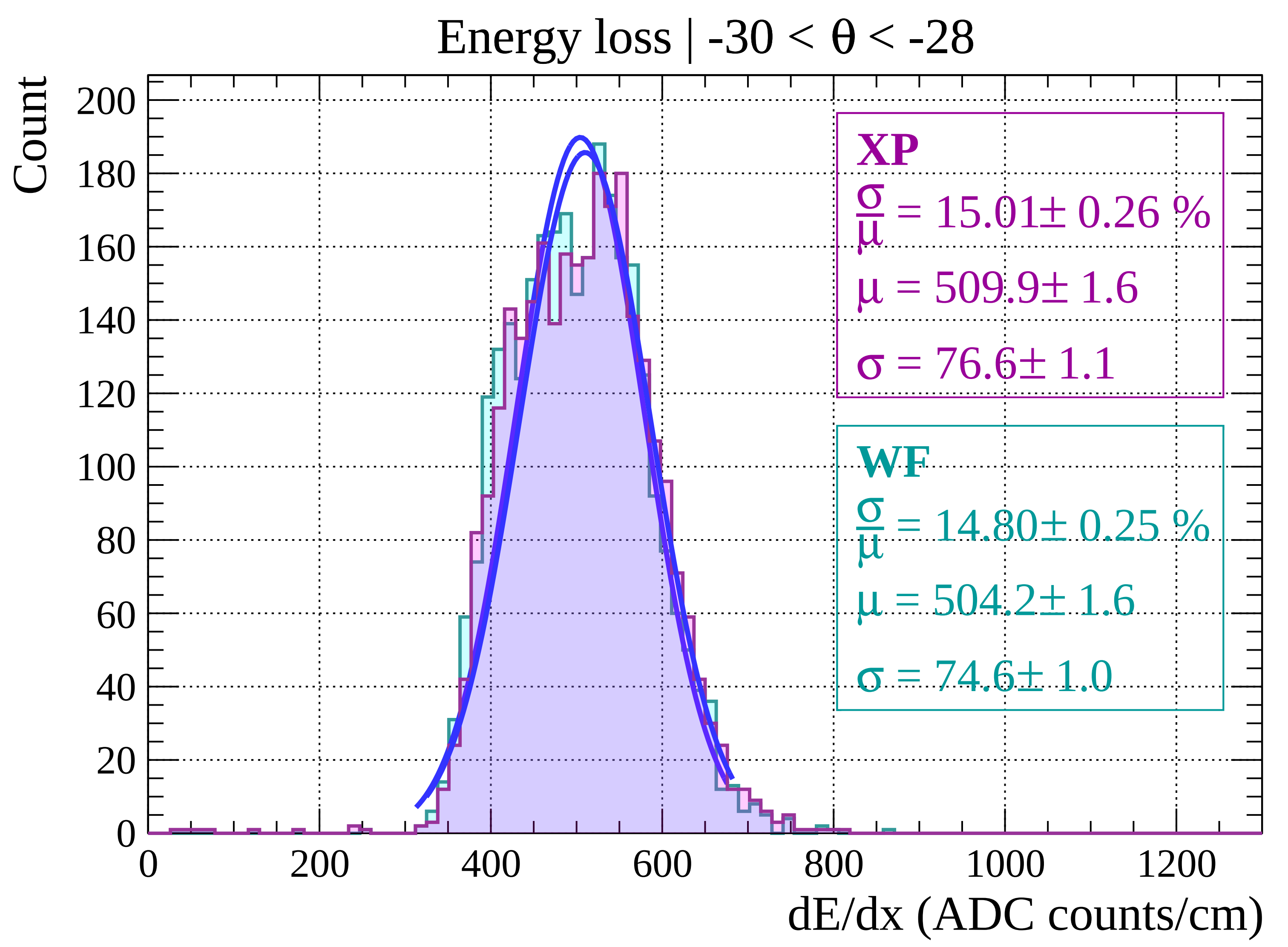
Energy loss | -36 < θ < -34
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.79± 0.46 %
μ = 521.4± 2.4
σ = 82.3± 2.0
WF
σ/μ = 15.91± 0.48 %
μ = 513.0± 2.5
σ = 81.6± 2.1



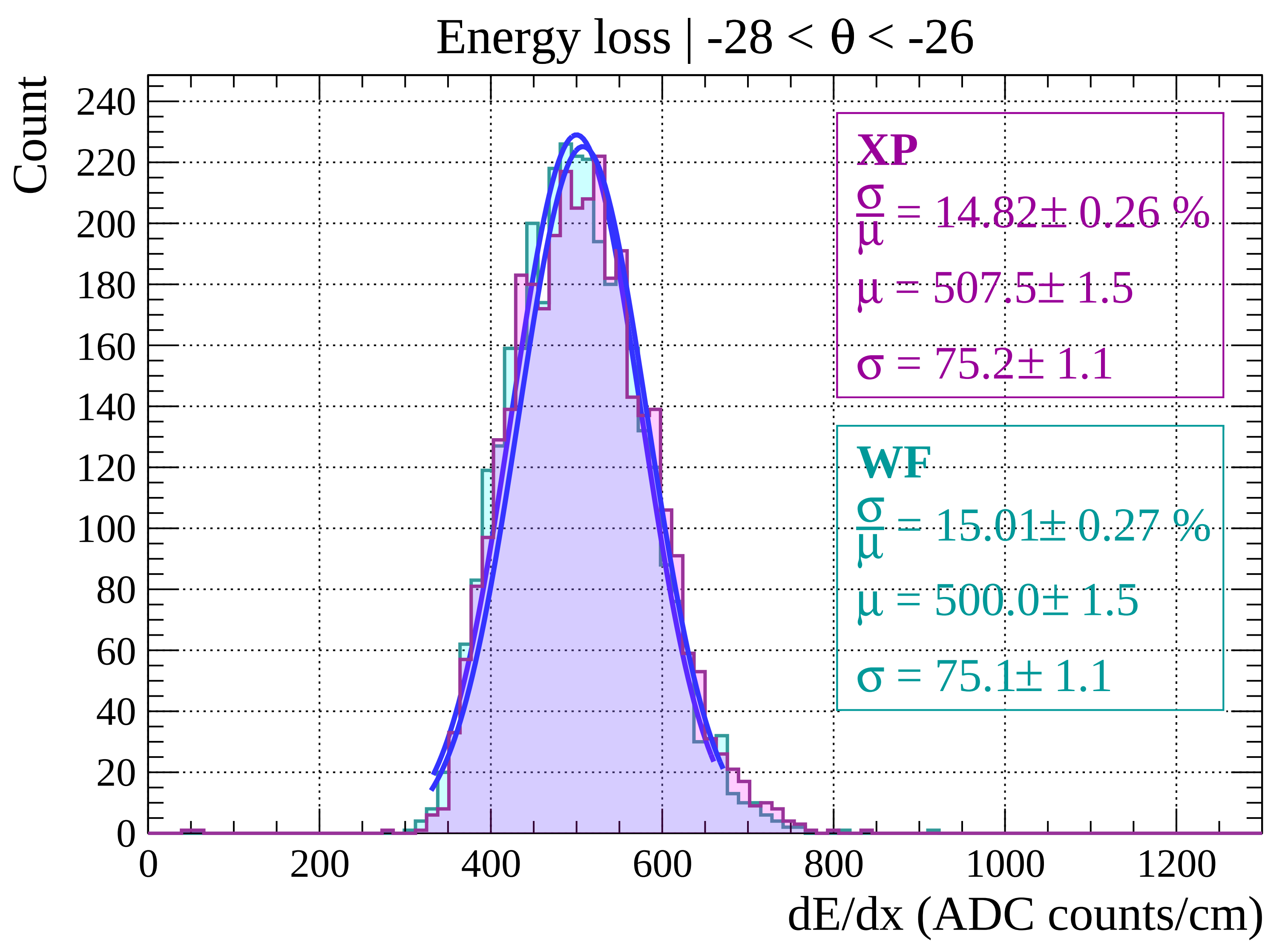
Energy loss | -34 < θ < -32
Count
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.13± 0.37 %
μ = 518.3± 2.1
σ = 78.4± 1.6
WF
σ/μ = 15.73± 0.43 %
μ = 509.9± 2.2
σ = 80.2± 1.8



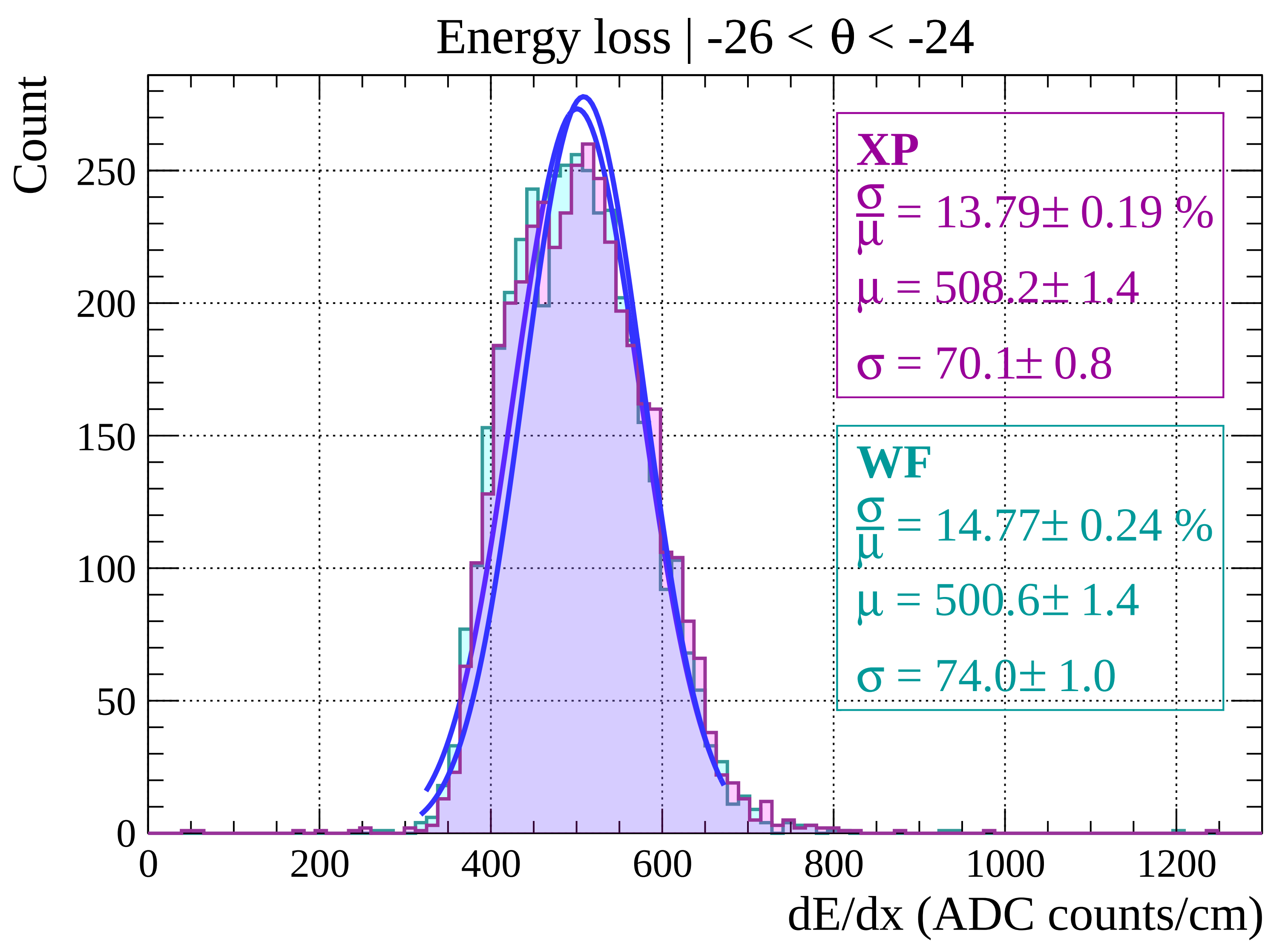
Energy loss | -32 < θ < -30
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 14.91± 0.33 %
μ = 517.4± 1.9
σ = 77.1± 1.4
WF
σ/μ = 14.94± 0.32 %
μ = 511.5± 1.8
σ = 76.4± 1.4



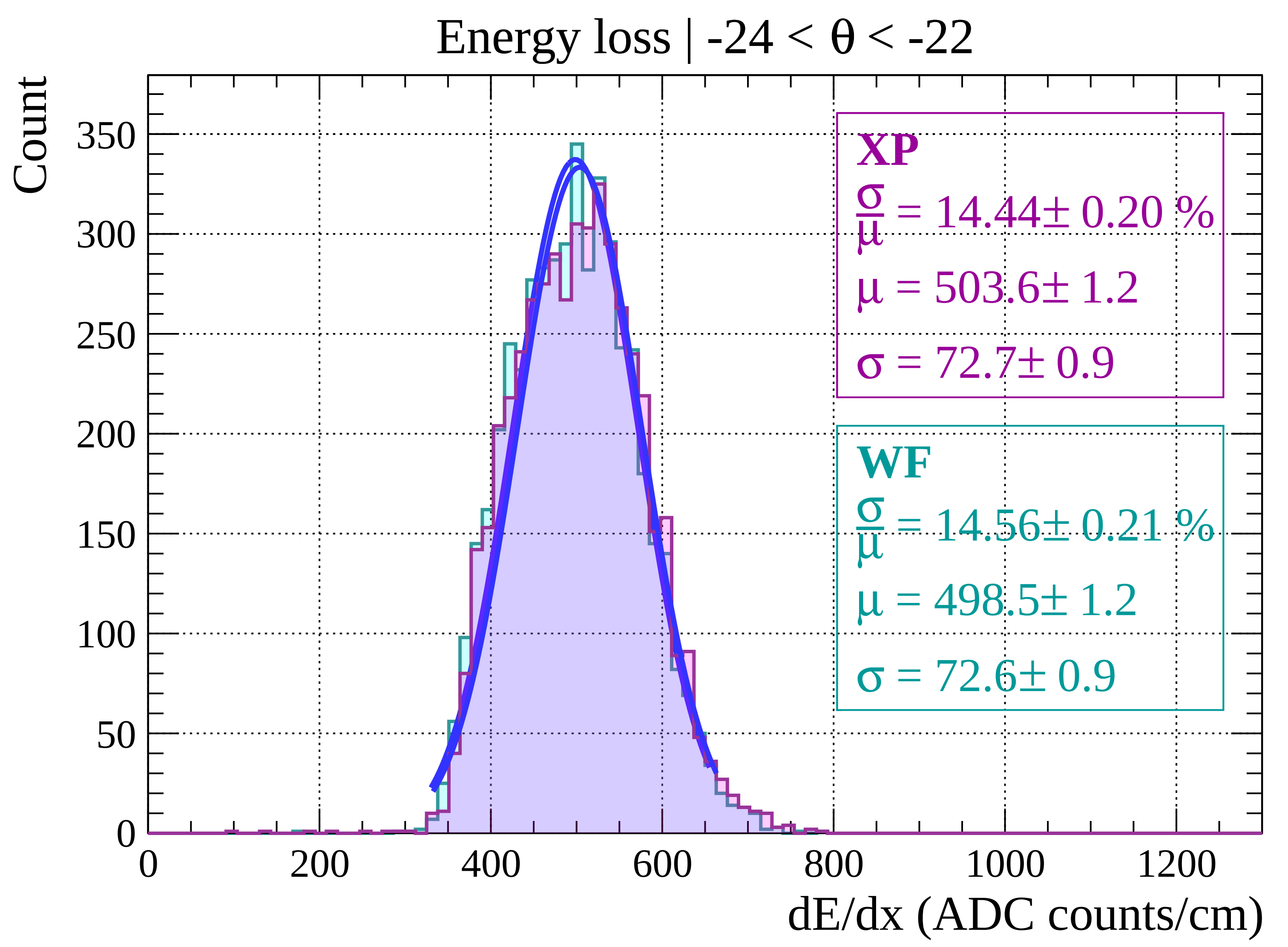
Energy loss | -30 < θ < -28
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
σ/μ = 15.01± 0.26 %
μ = 509.9± 1.6
σ = 76.6± 1.1
WF
σ/μ = 14.80± 0.25 %
μ = 504.2± 1.6
σ = 74.6± 1.0



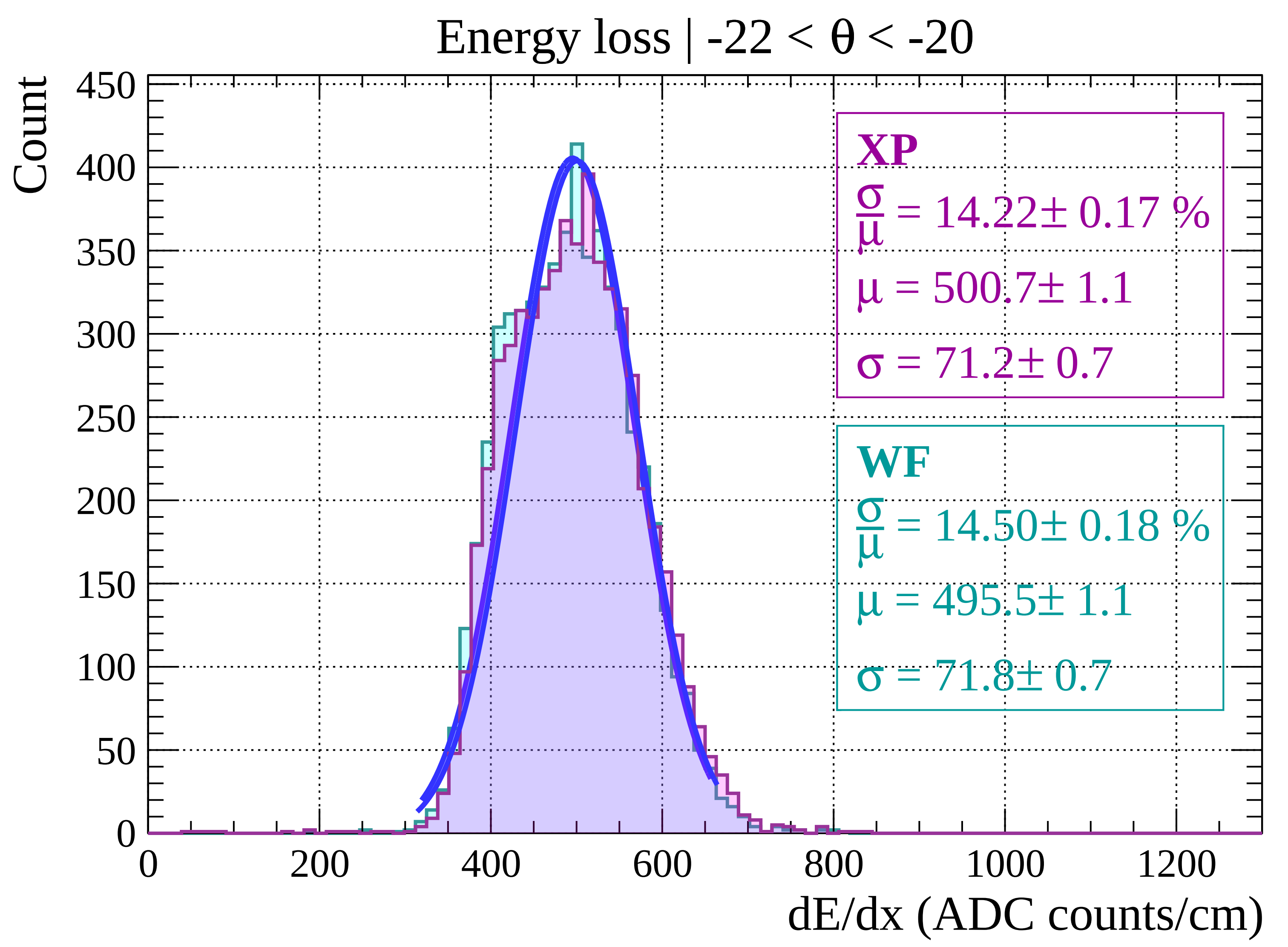
Energy loss | -28 < θ < -26
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.82± 0.26 %
μ = 507.5± 1.5
σ = 75.2± 1.1
WF
σ/μ = 15.01± 0.27 %
μ = 500.0± 1.5
σ = 75.1± 1.1



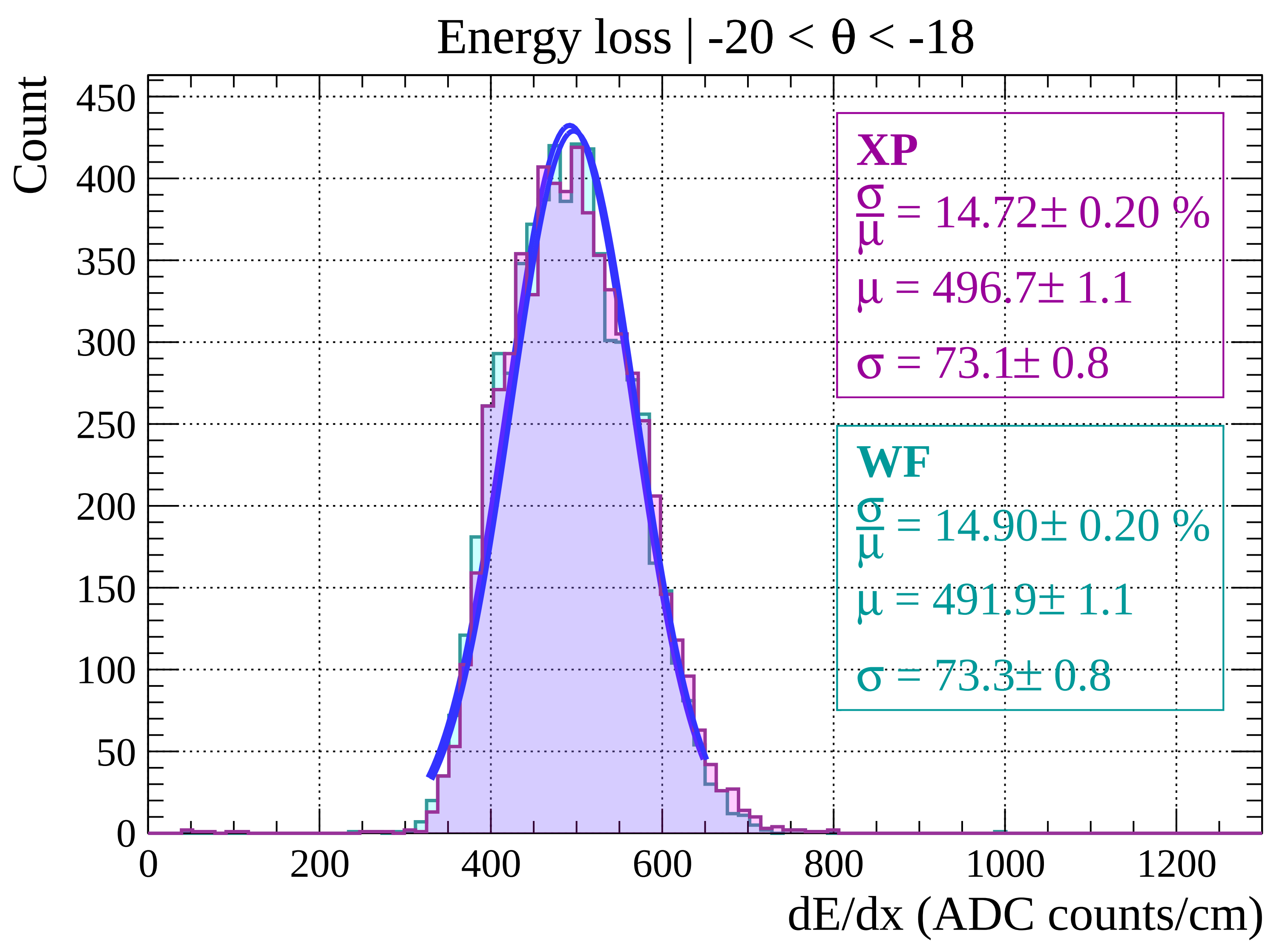
Energy loss | -26 < θ < -24
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.79± 0.19 %
μ = 508.2± 1.4
σ = 70.1± 0.8
WF
σ/μ = 14.77± 0.24 %
μ = 500.6± 1.4
σ = 74.0± 1.0



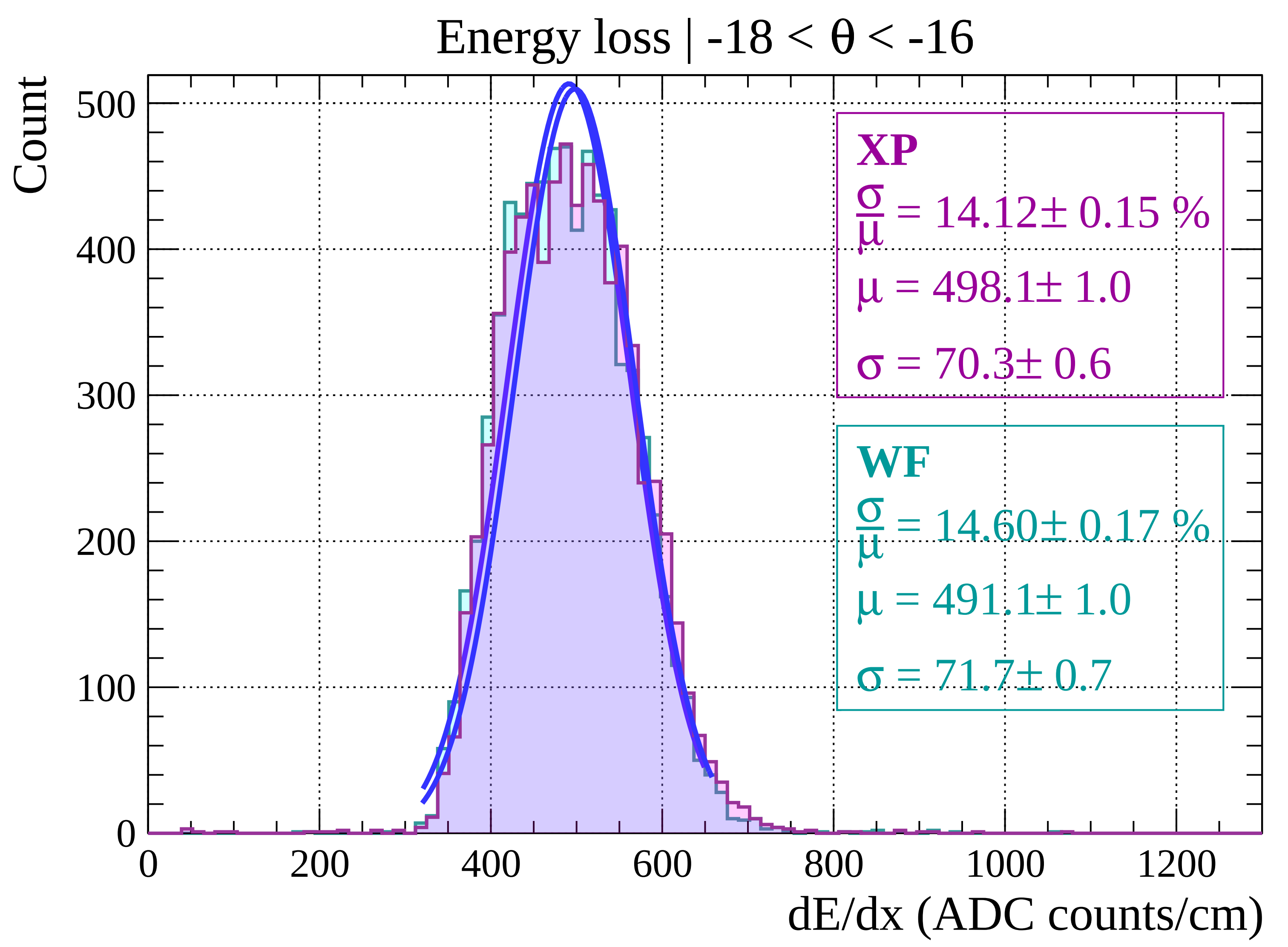
Energy loss | -24 < θ < -22
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 14.44± 0.20 %
μ = 503.6± 1.2
σ = 72.7± 0.9
WF
σ/μ = 14.56± 0.21 %
μ = 498.5± 1.2
σ = 72.6± 0.9



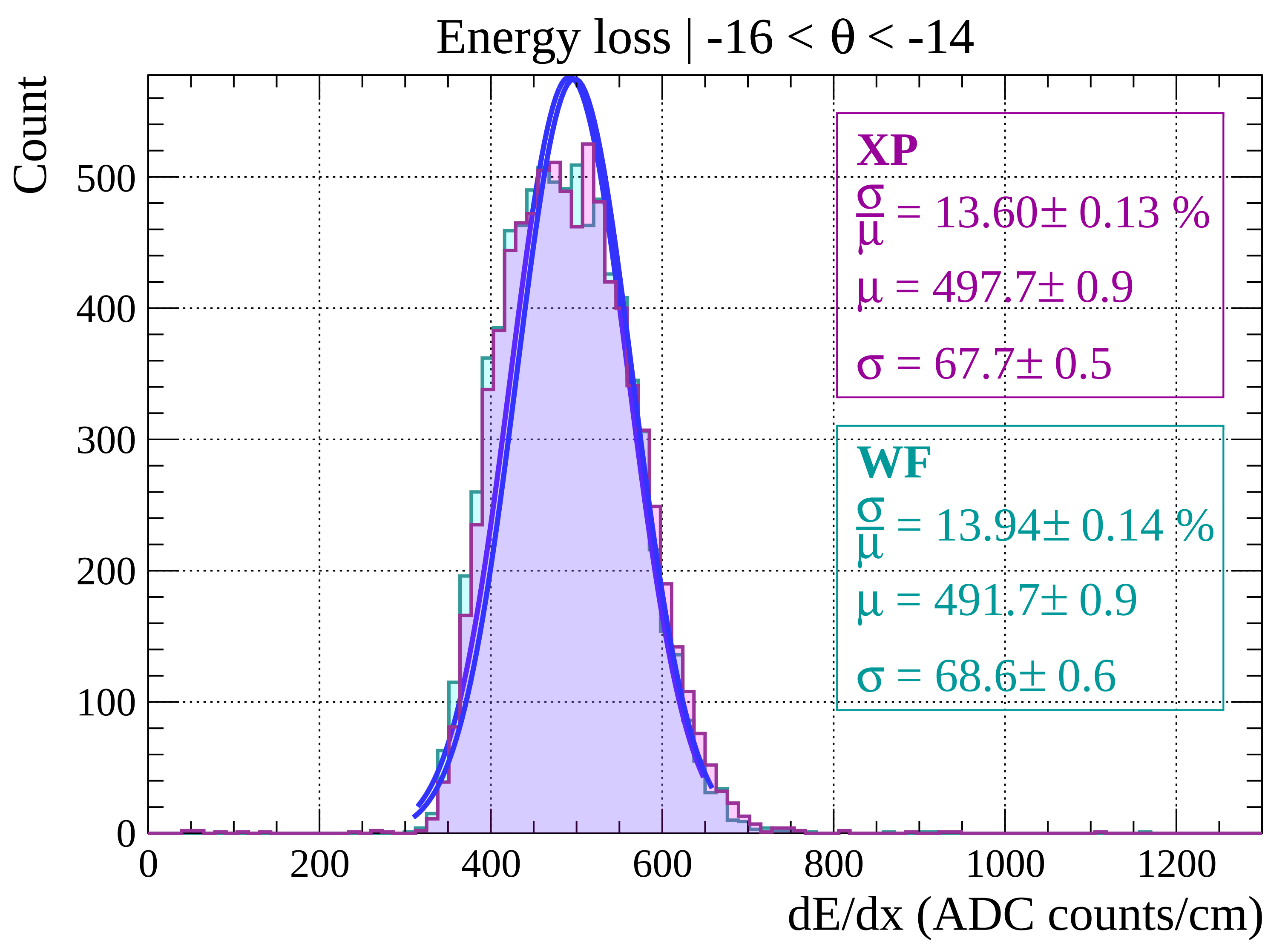
Energy loss | -22 < θ < -20
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 14.22± 0.17 %
μ = 500.7± 1.1
σ = 71.2± 0.7
WF
σ/μ = 14.50± 0.18 %
μ = 495.5± 1.1
σ = 71.8± 0.7



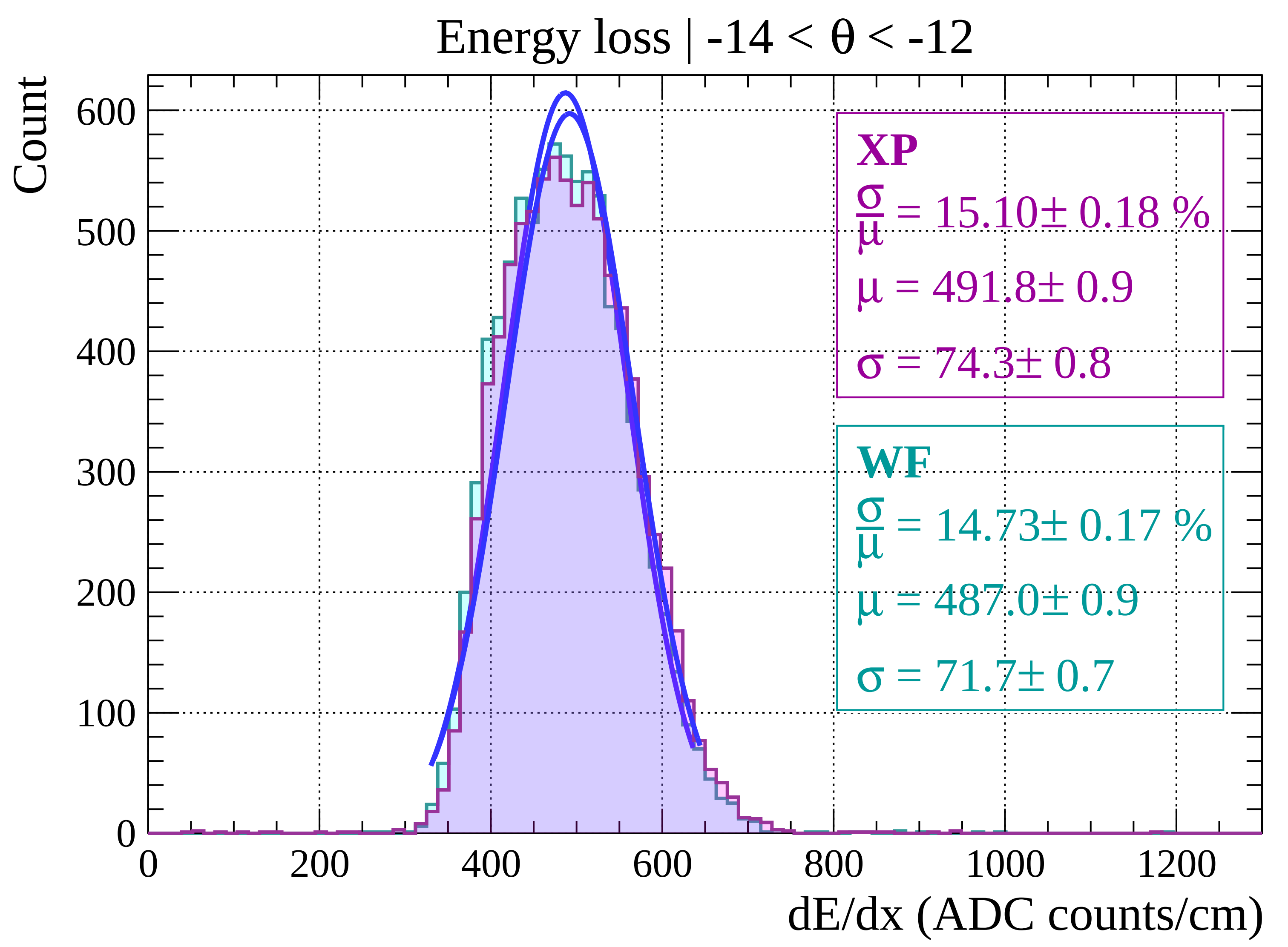
Energy loss | -20 < θ < -18
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.72± 0.20 %
μ = 496.7± 1.1
σ = 73.1± 0.8
WF
σ/μ = 14.90± 0.20 %
μ = 491.9± 1.1
σ = 73.3± 0.8



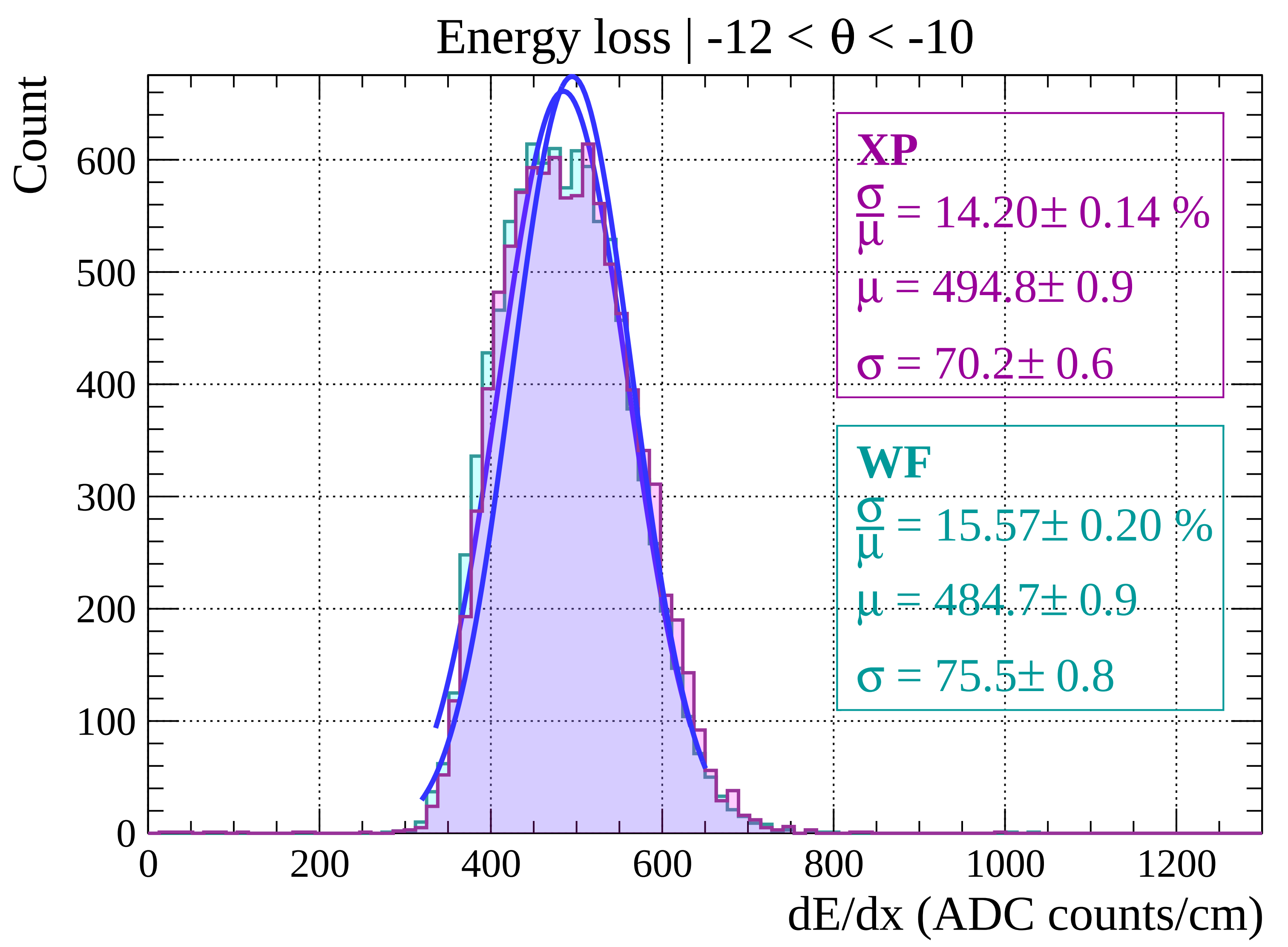
Energy loss | -18 < θ < -16
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 14.12± 0.15 %
μ = 498.1± 1.0
σ = 70.3± 0.6
WF
σ/μ = 14.60± 0.17 %
μ = 491.1± 1.0
σ = 71.7± 0.7

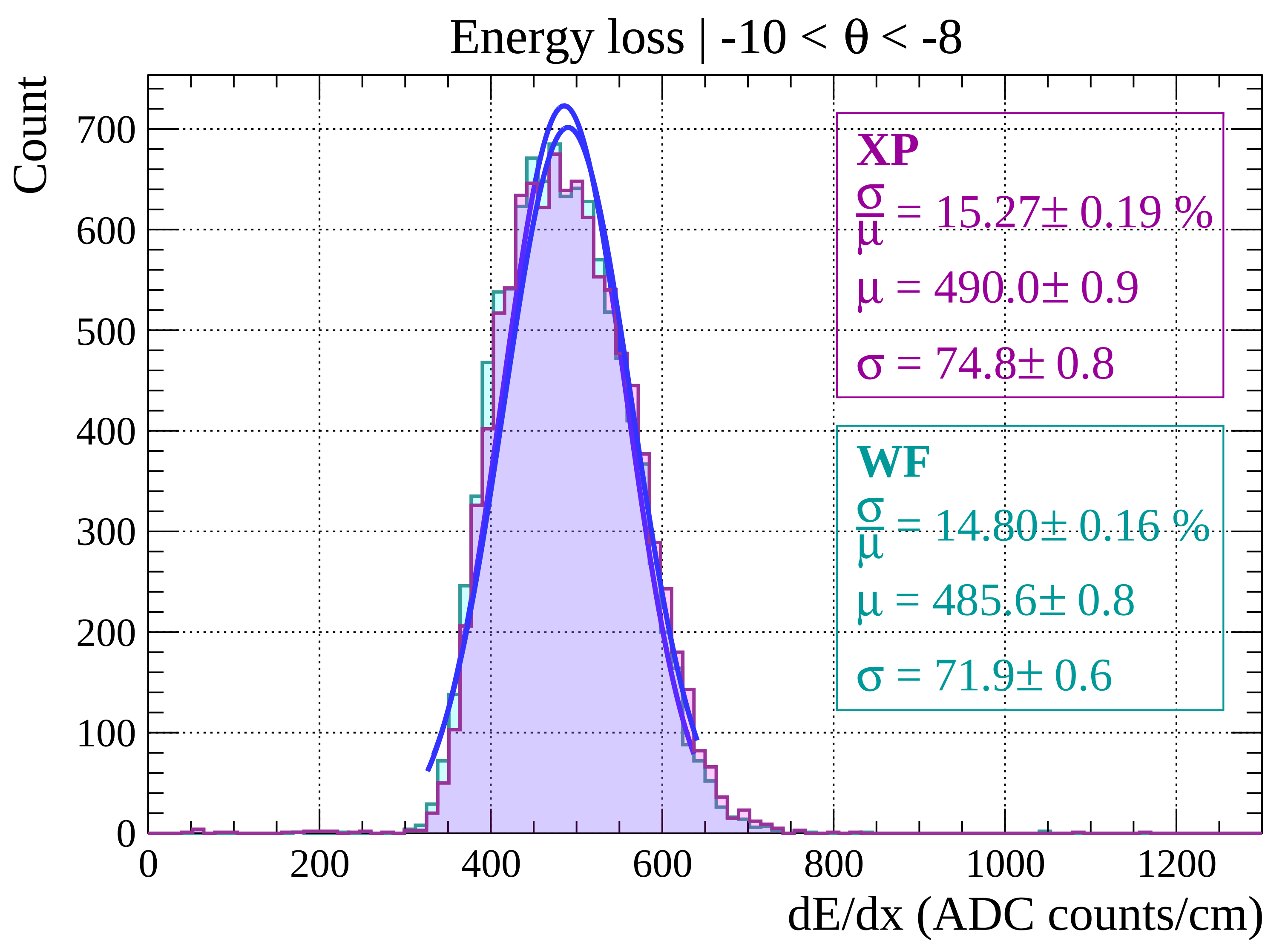


Energy loss | -16 < θ < -14
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.60± 0.13 %
μ = 497.7± 0.9
σ = 67.7± 0.5
WF
σ/μ = 13.94± 0.14 %
μ = 491.7± 0.9
σ = 68.6± 0.6

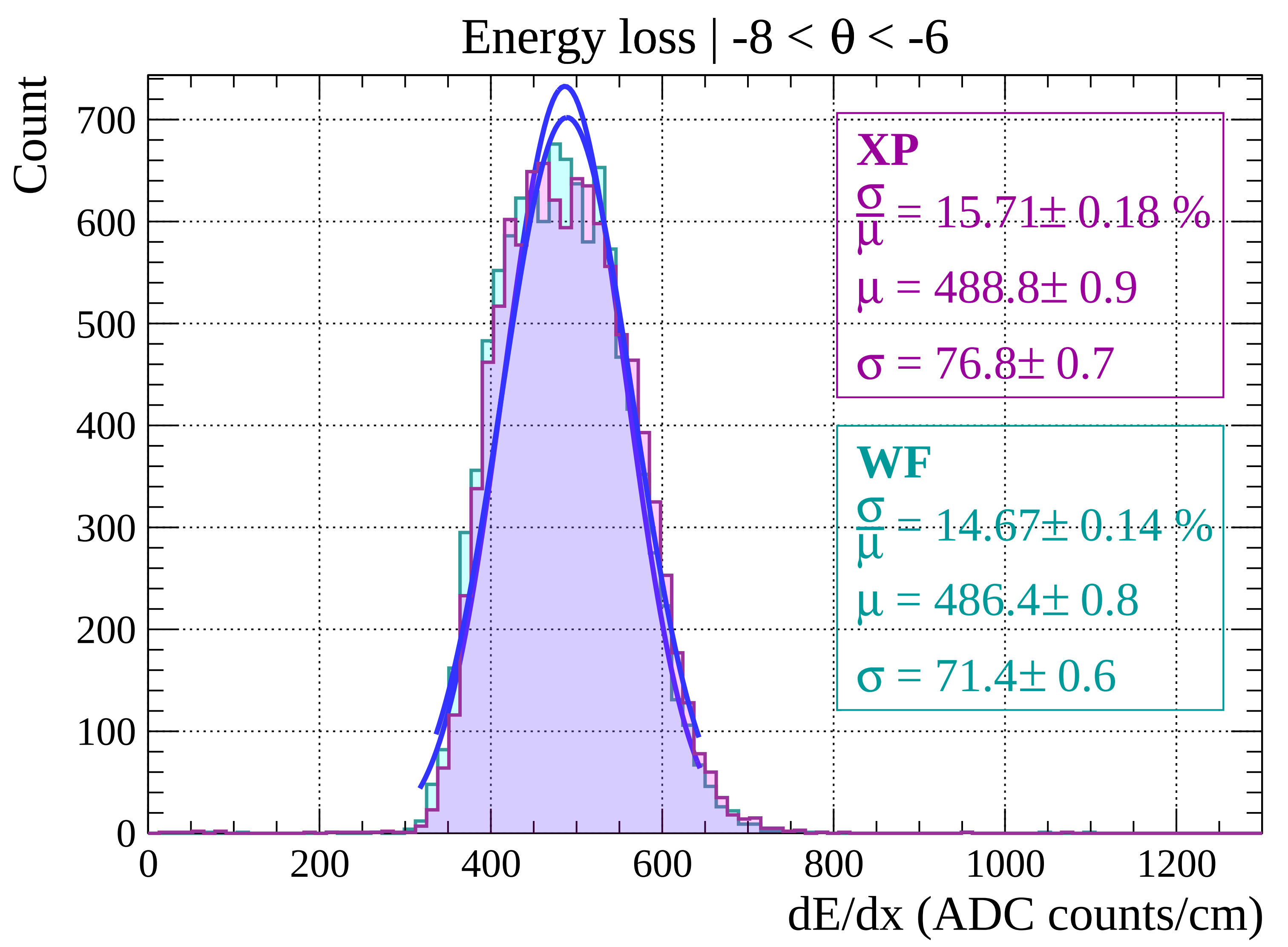


Energy loss | -14 < θ < -12
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.10± 0.18 %
μ = 491.8± 0.9
σ = 74.3± 0.8
WF
σ/μ = 14.73± 0.17 %
μ = 487.0± 0.9
σ = 71.7± 0.7

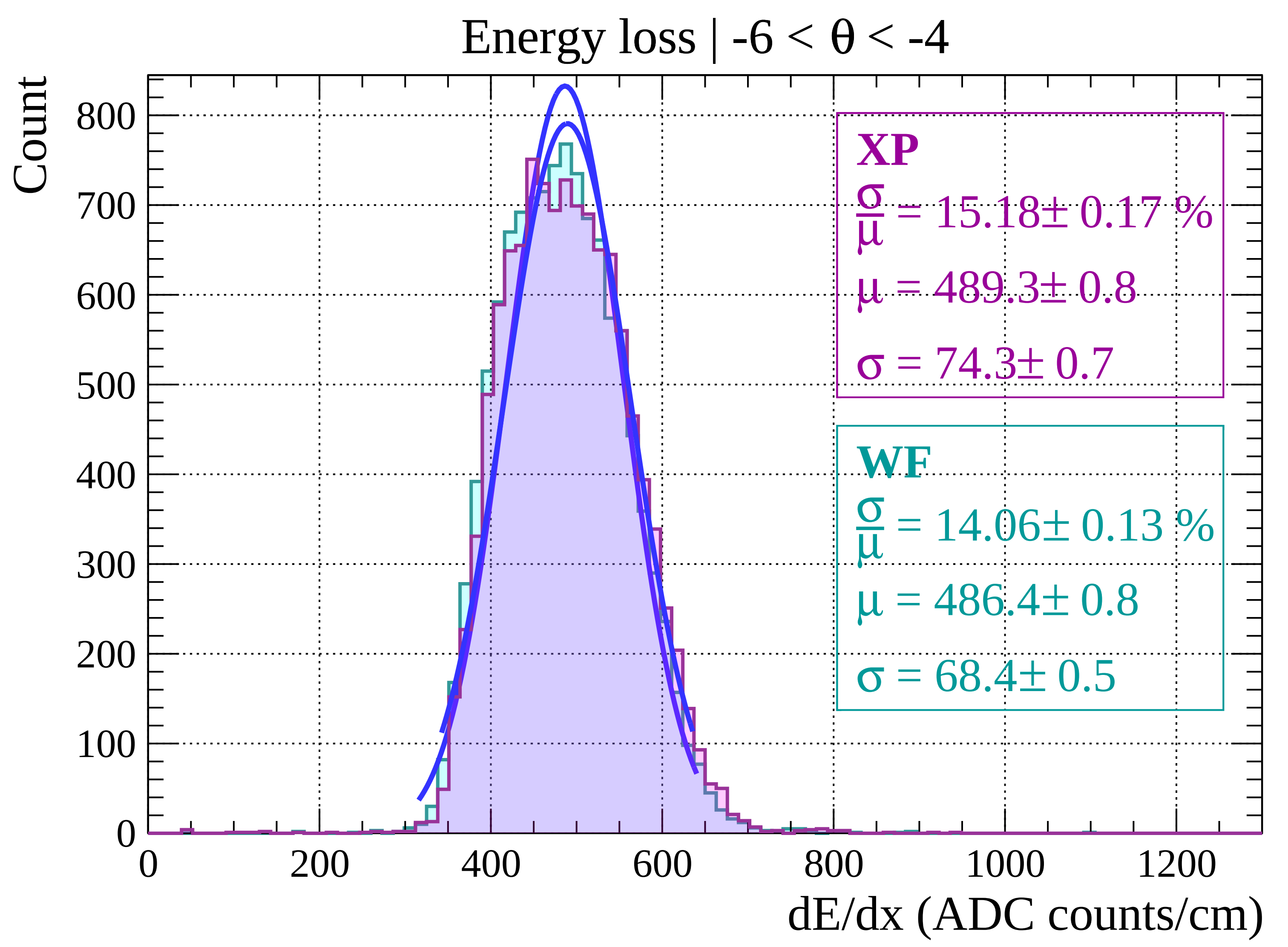

Energy loss | -12 < θ < -10
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.20± 0.14 %
μ = 494.8± 0.9
σ = 70.2± 0.6
WF
σ/μ = 15.57± 0.20 %
μ = 484.7± 0.9
σ = 75.5± 0.8
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200



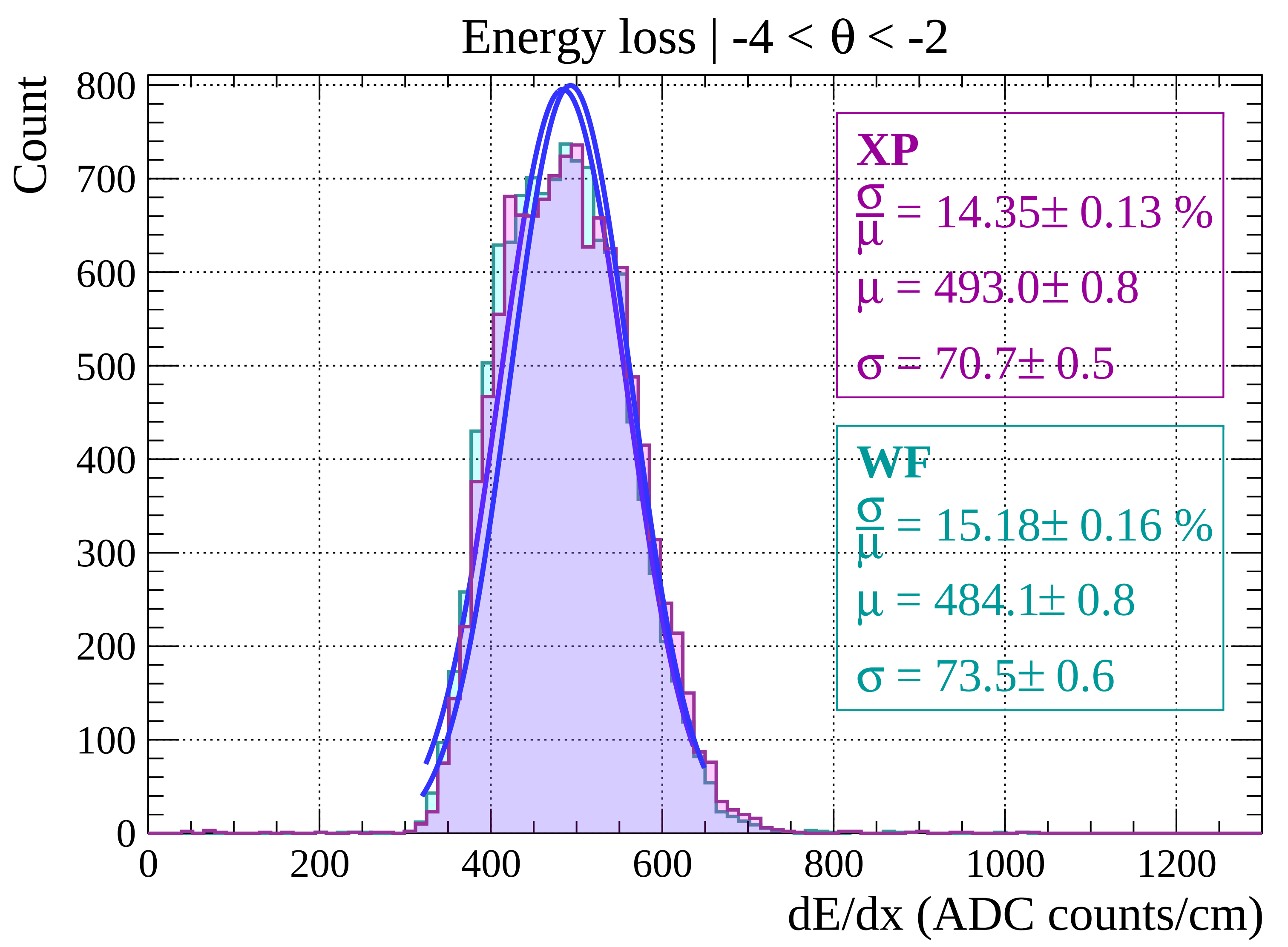
Energy loss | -10 < θ < -8
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 15.27± 0.19 %
μ = 490.0± 0.9
σ = 74.8± 0.8
WF
σ/μ = 14.80± 0.16 %
μ = 485.6± 0.8
σ = 71.9± 0.6



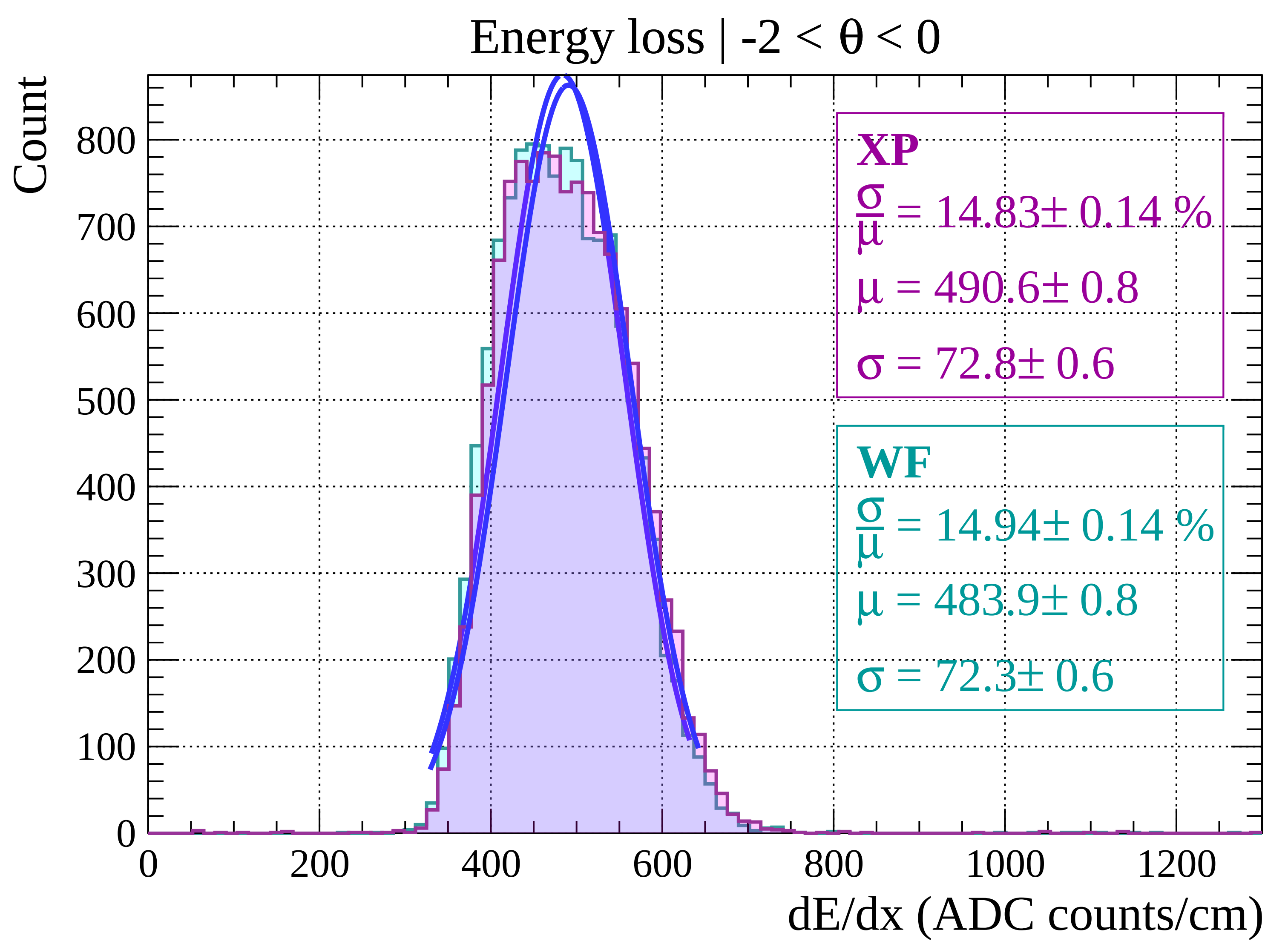
Energy loss | -8 < θ < -6
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 15.71 ± 0.18 %
μ = 488.8 ± 0.9
σ = 76.8 ± 0.7
WF
σ/μ = 14.67 ± 0.14 %
μ = 486.4 ± 0.8
σ = 71.4 ± 0.6



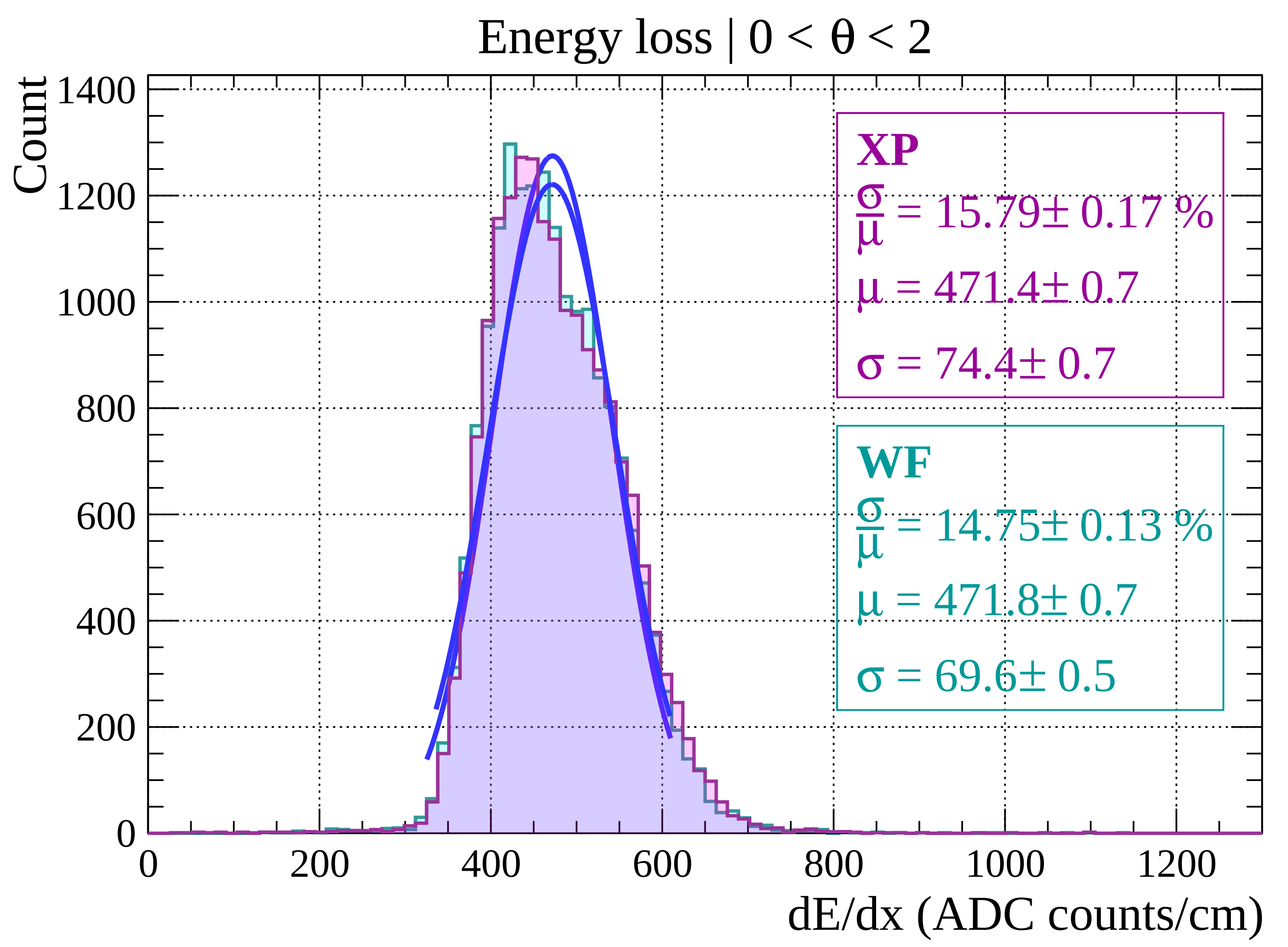
Energy loss | -6 < θ < -4
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.18± 0.17 %
μ = 489.3± 0.8
σ = 74.3± 0.7
WF
σ/μ = 14.06± 0.13 %
μ = 486.4± 0.8
σ = 68.4± 0.5
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

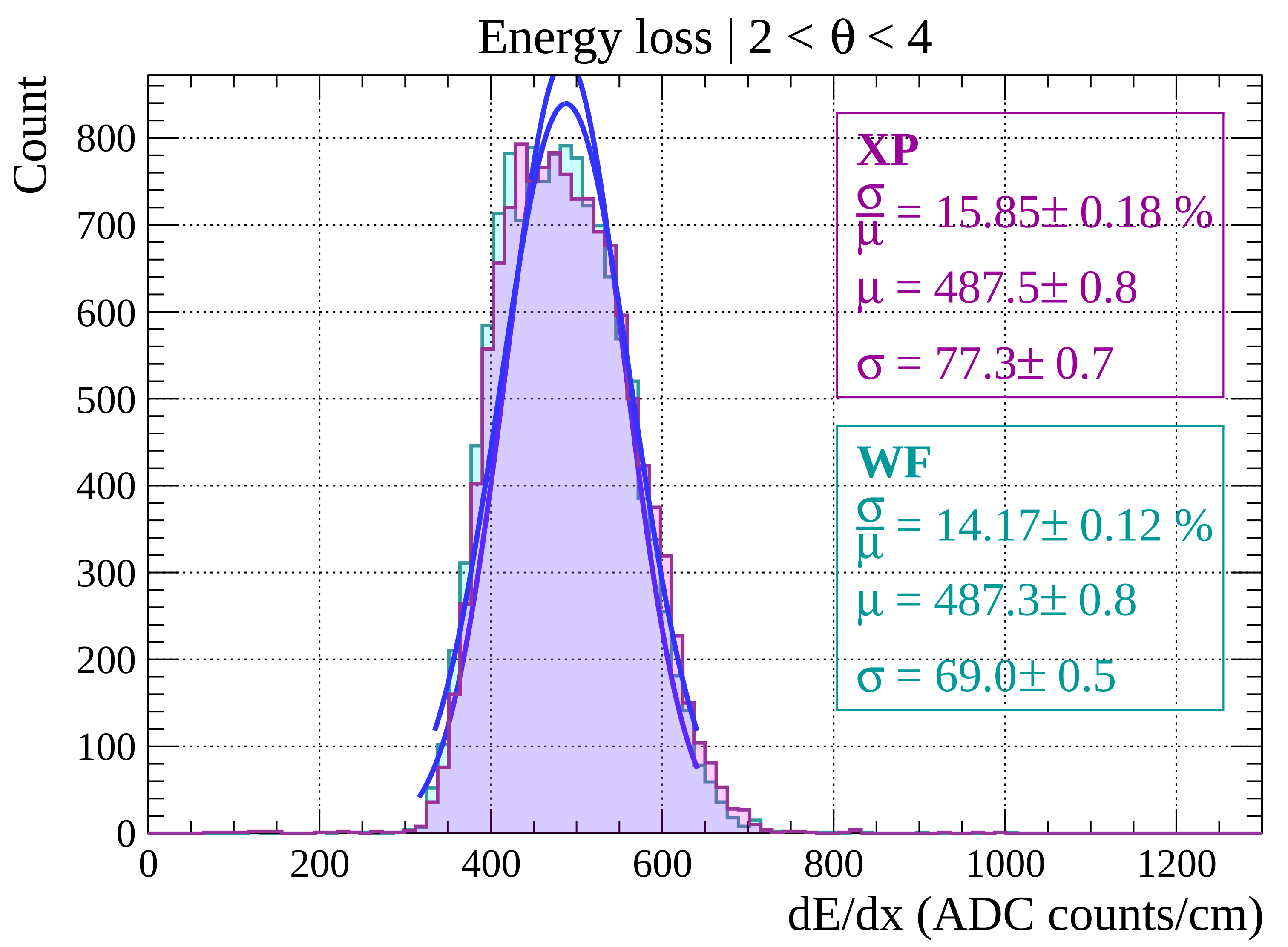


Energy loss | -4 < θ < -2
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 14.35± 0.13 %
μ = 493.0± 0.8
σ = 70.7± 0.5
WF
σ/μ = 15.18± 0.16 %
μ = 484.1± 0.8
σ = 73.5± 0.6

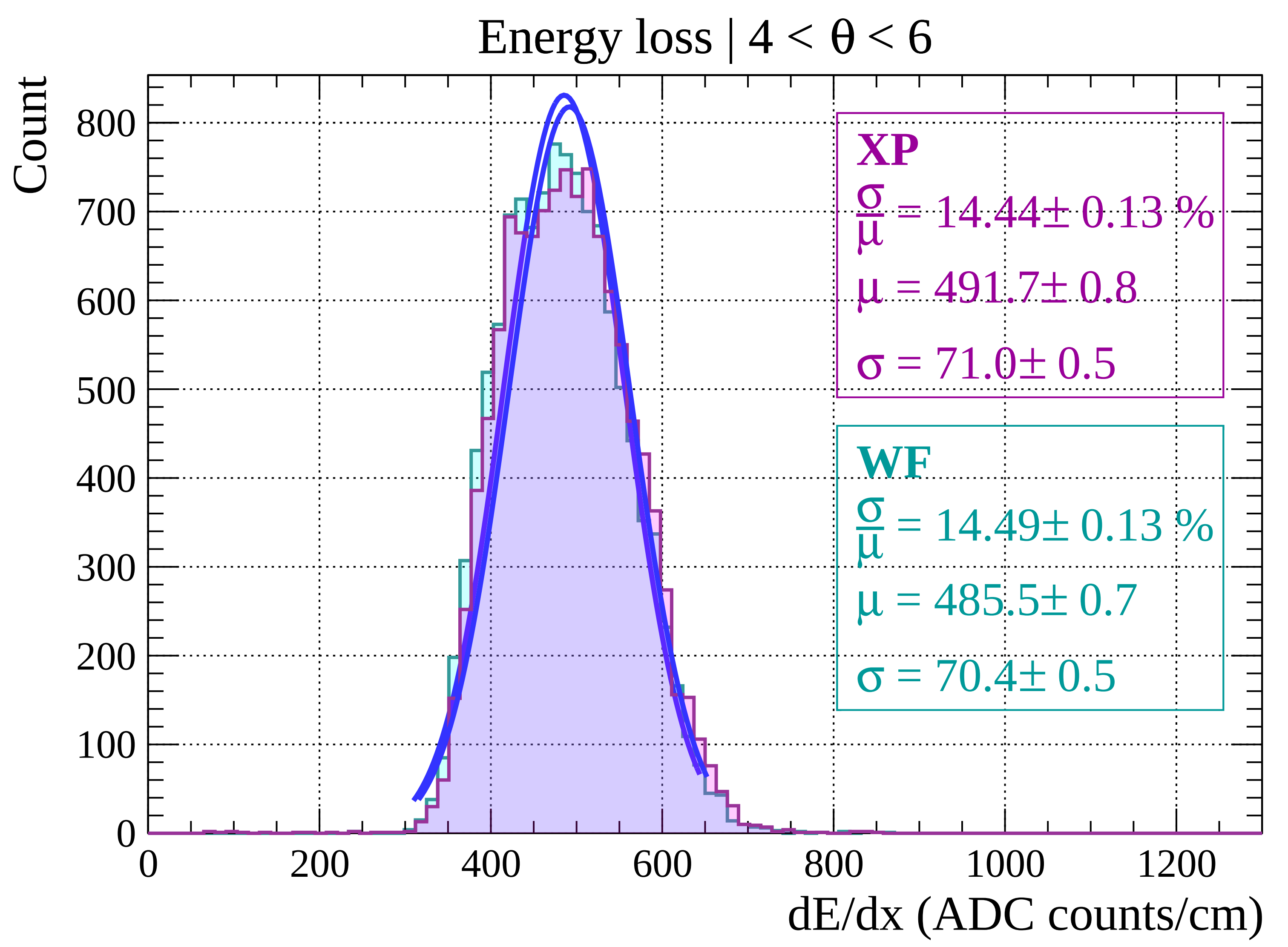


Energy loss | -2 < θ < 0
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 14.83± 0.14 %
μ = 490.6± 0.8
σ = 72.8± 0.6
WF
σ/μ = 14.94± 0.14 %
μ = 483.9± 0.8
σ = 72.3± 0.6

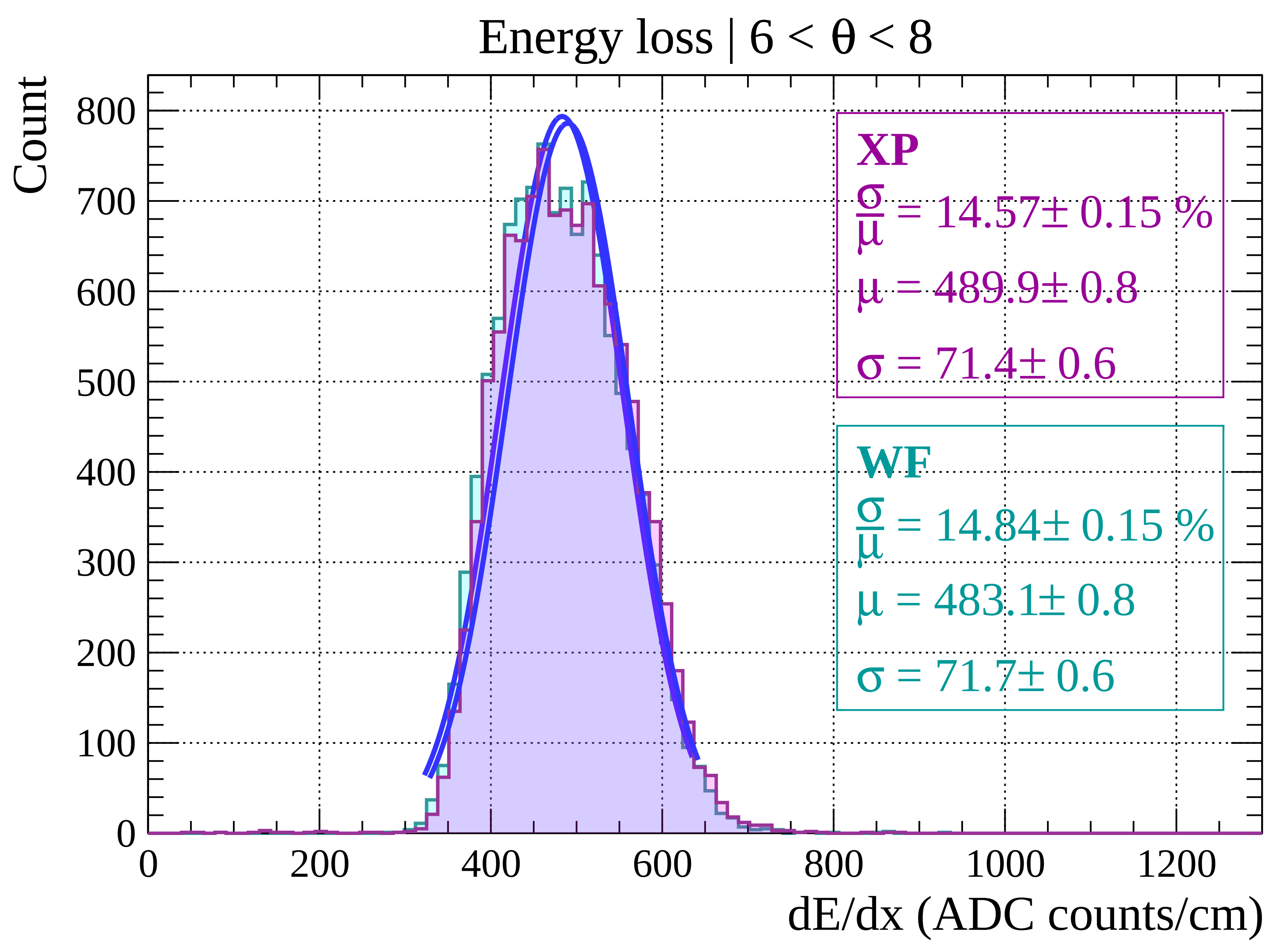

Energy loss | 0 < θ < 2
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.79± 0.17 %
μ = 471.4± 0.7
σ = 74.4± 0.7
WF
σ/μ = 14.75± 0.13 %
μ = 471.8± 0.7
σ = 69.6± 0.5



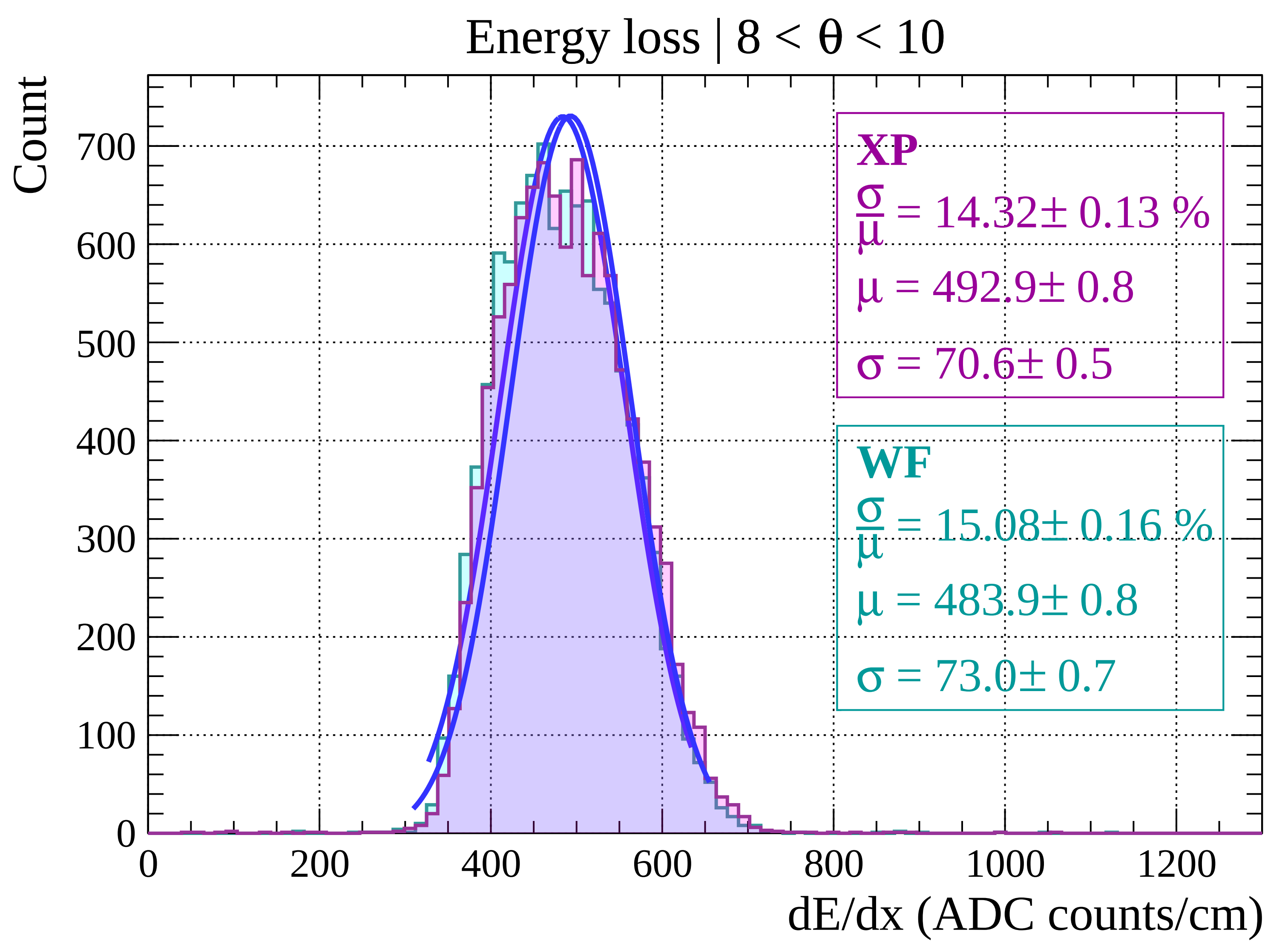
Energy loss | 2 < θ < 4
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.85± 0.18 %
μ = 487.5± 0.8
σ = 77.3± 0.7
WF
σ/μ = 14.17± 0.12 %
μ = 487.3± 0.8
σ = 69.0± 0.5
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200



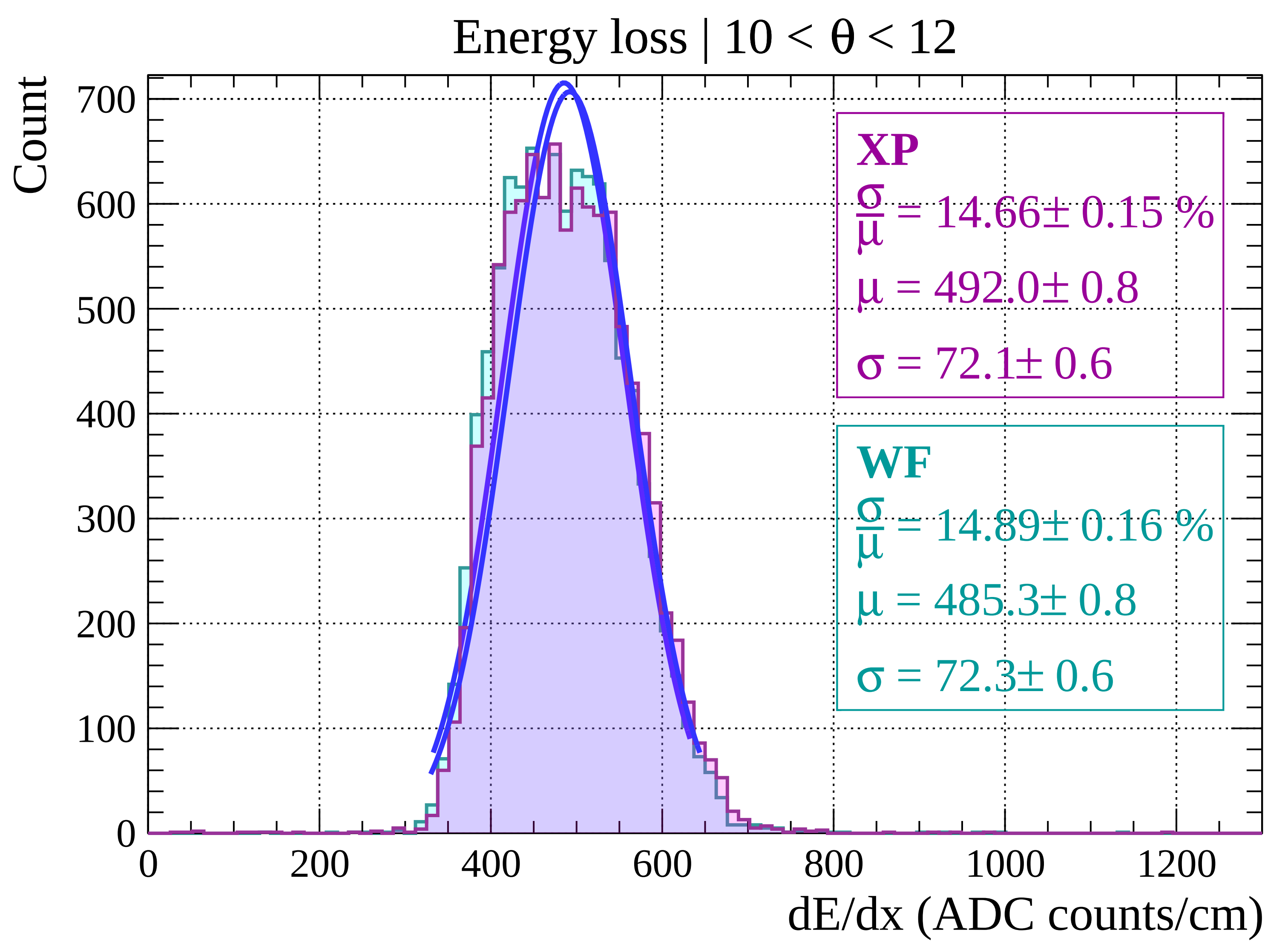
Energy loss | 4 < θ < 6
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP
σ/μ = 14.44± 0.13 %
μ = 491.7± 0.8
σ = 71.0± 0.5
WF
σ/μ = 14.49± 0.13 %
μ = 485.5± 0.7
σ = 70.4± 0.5



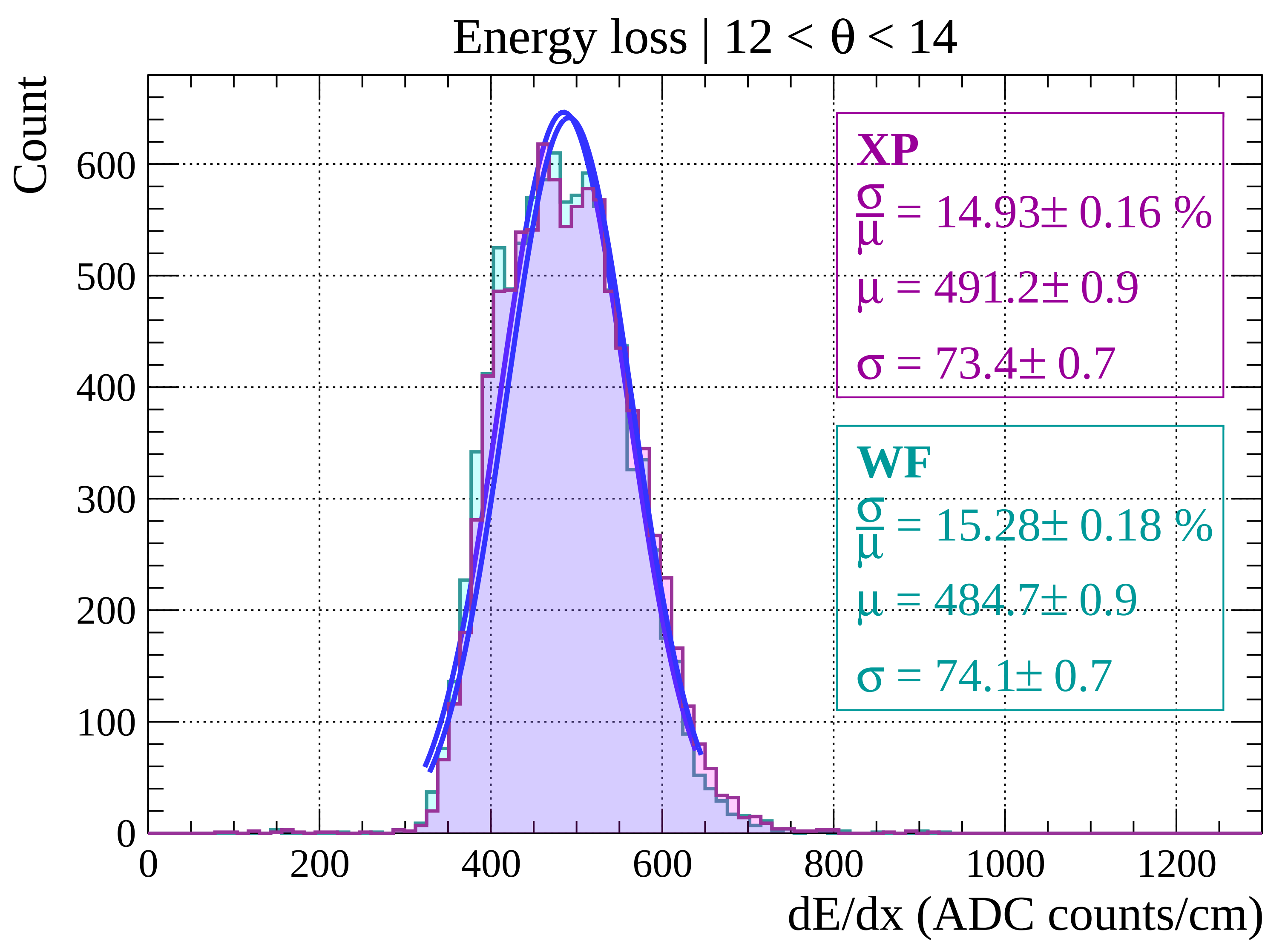
Energy loss | 6 < θ < 8
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 14.57± 0.15 %
μ = 489.9± 0.8
σ = 71.4± 0.6
WF
σ/μ = 14.84± 0.15 %
μ = 483.1± 0.8
σ = 71.7± 0.6



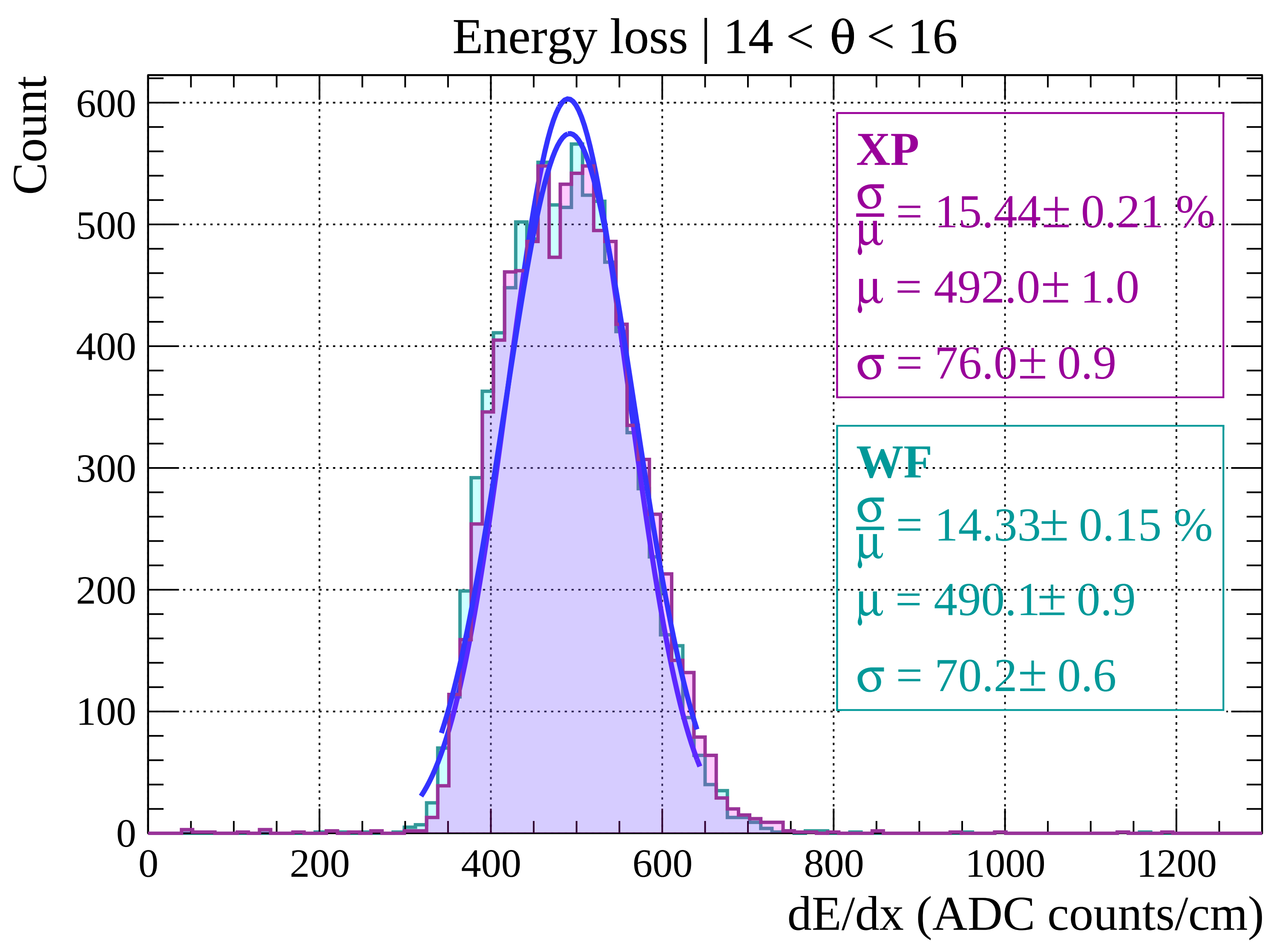
Energy loss | 8 < θ < 10
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 14.32± 0.13 %
μ = 492.9± 0.8
σ = 70.6± 0.5
WF
σ/μ = 15.08± 0.16 %
μ = 483.9± 0.8
σ = 73.0± 0.7



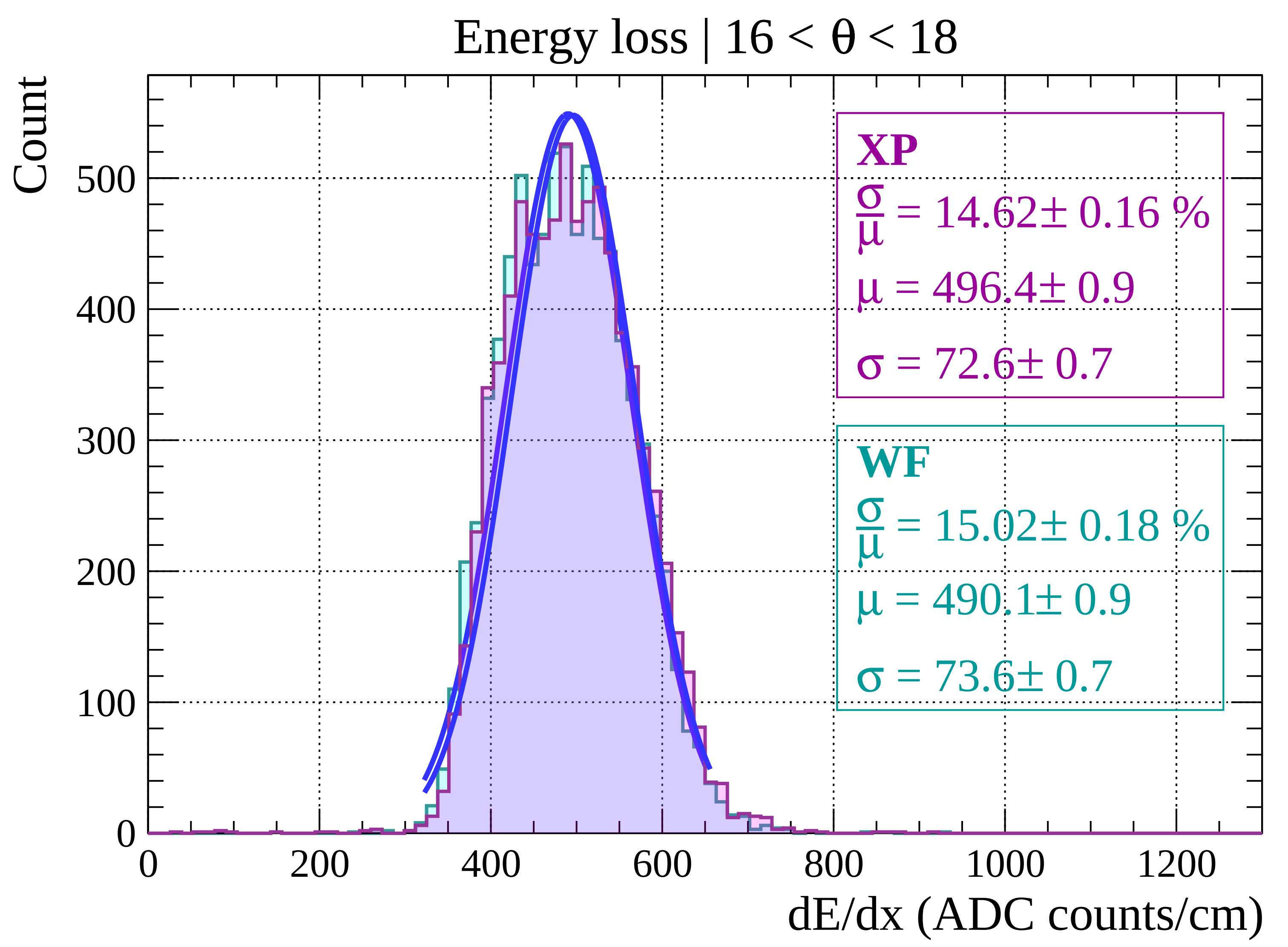
Energy loss | 10 < θ < 12
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.66 ± 0.15 %
μ = 492.0 ± 0.8
σ = 72.1 ± 0.6
WF
σ/μ = 14.89 ± 0.16 %
μ = 485.3 ± 0.8
σ = 72.3 ± 0.6



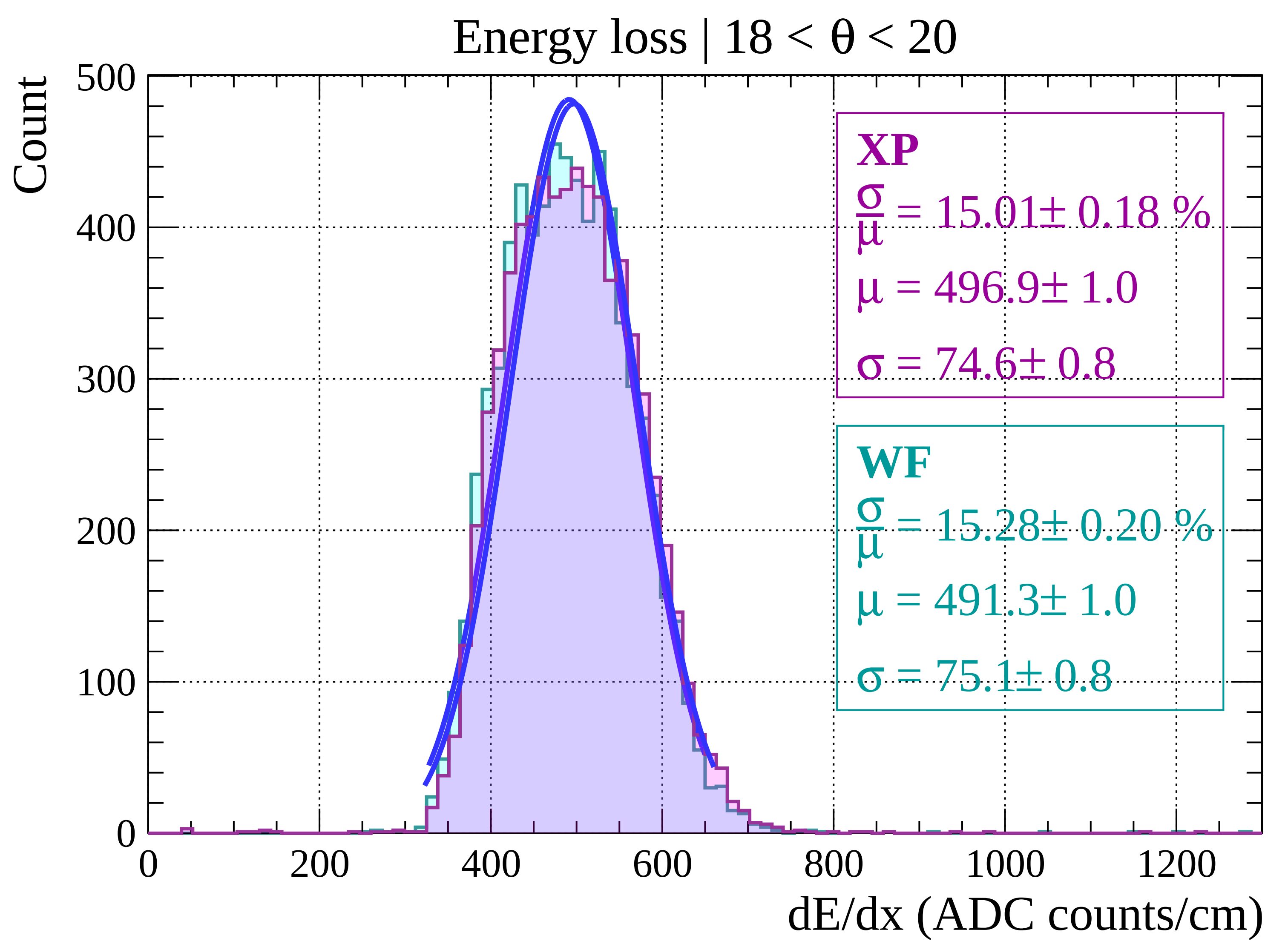
Energy loss | 12 < θ < 14
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 14.93± 0.16 %
μ = 491.2± 0.9
σ = 73.4± 0.7
WF
σ/μ = 15.28± 0.18 %
μ = 484.7± 0.9
σ = 74.1± 0.7



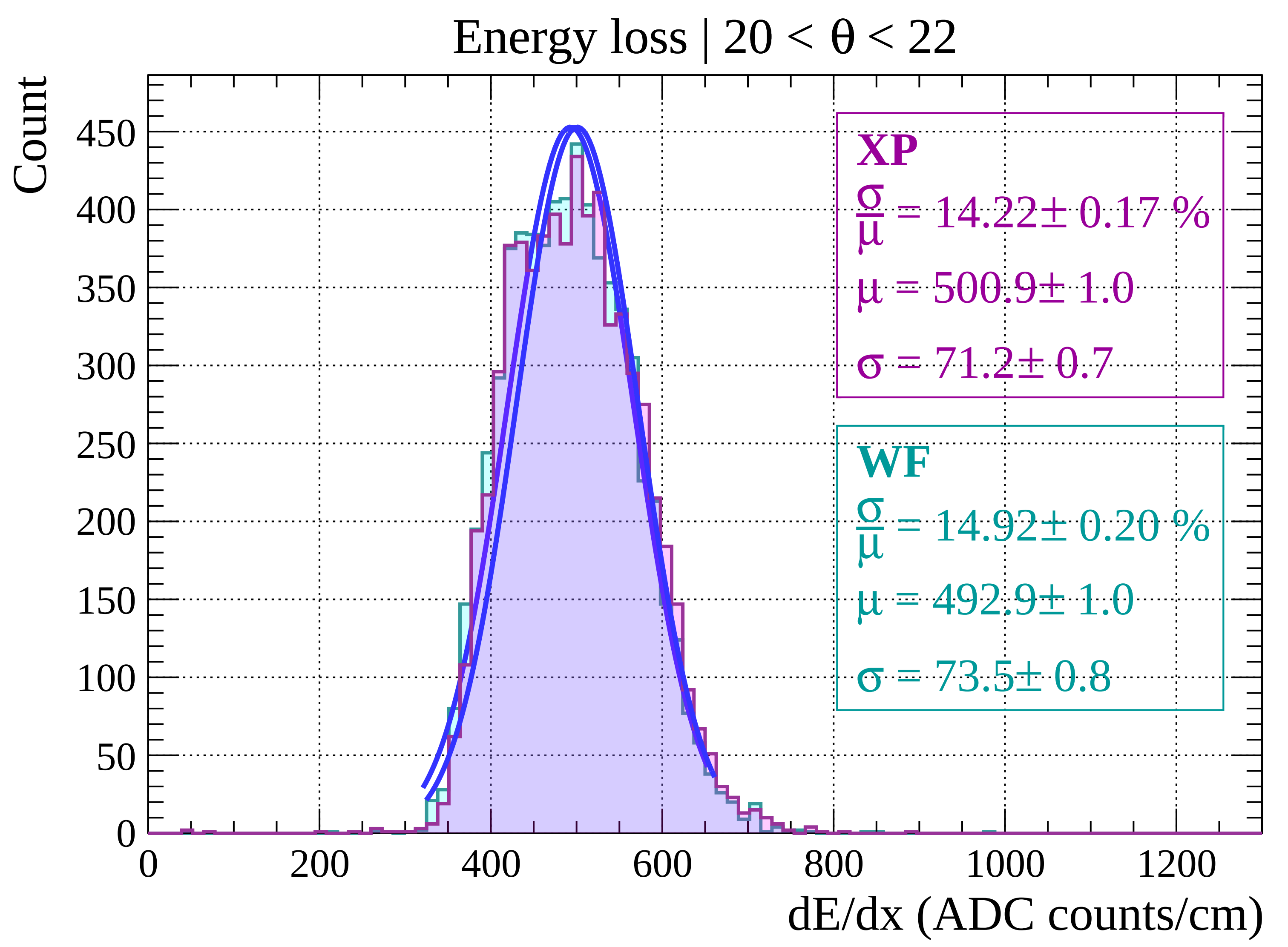
Energy loss | 14 < θ < 16
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.44± 0.21 %
μ = 492.0± 1.0
σ = 76.0± 0.9
WF
σ/μ = 14.33± 0.15 %
μ = 490.1± 0.9
σ = 70.2± 0.6



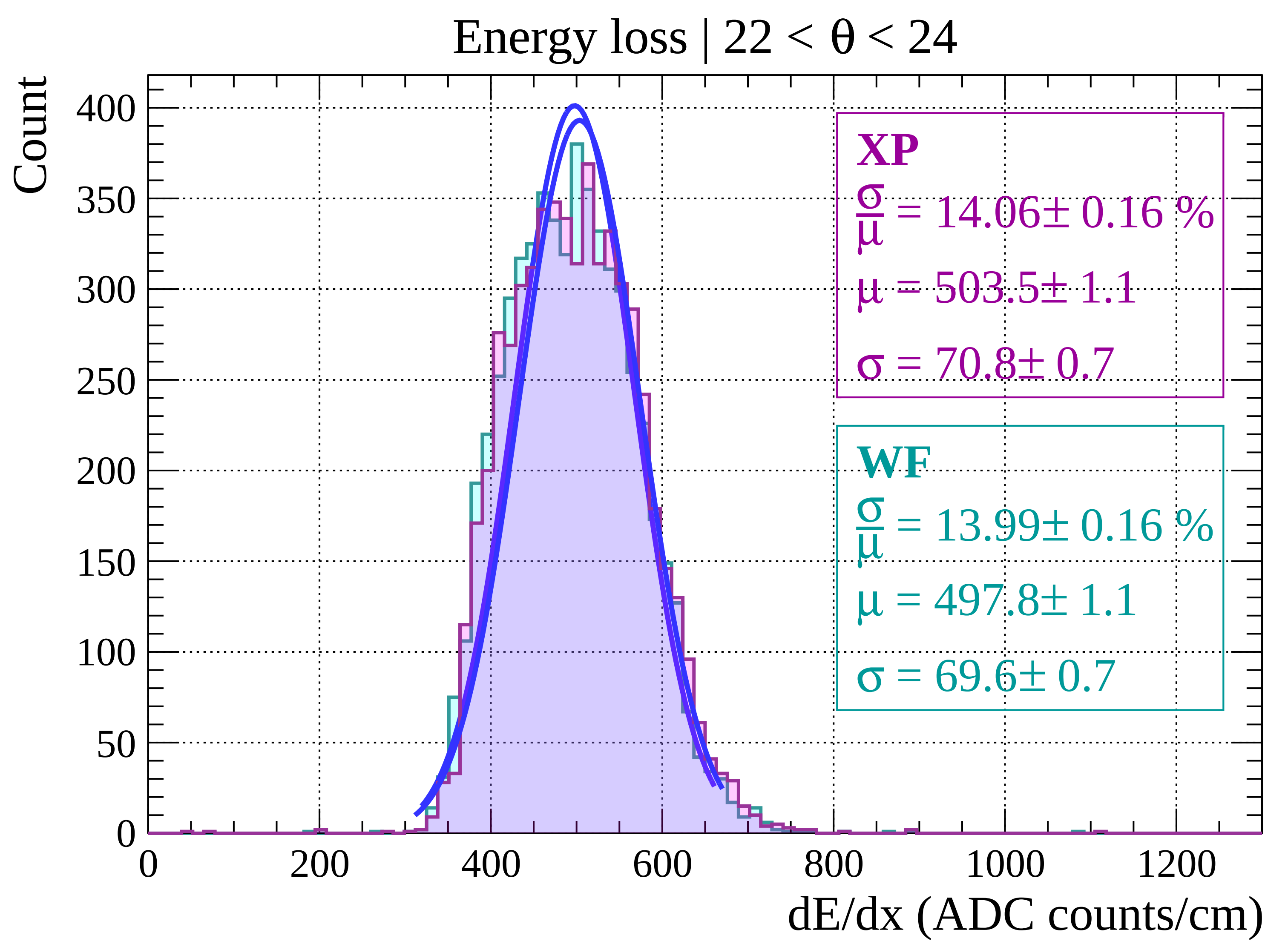
Energy loss | 16 < θ < 18
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 14.62± 0.16 %
μ = 496.4± 0.9
σ = 72.6± 0.7
WF
σ/μ = 15.02± 0.18 %
μ = 490.1± 0.9
σ = 73.6± 0.7



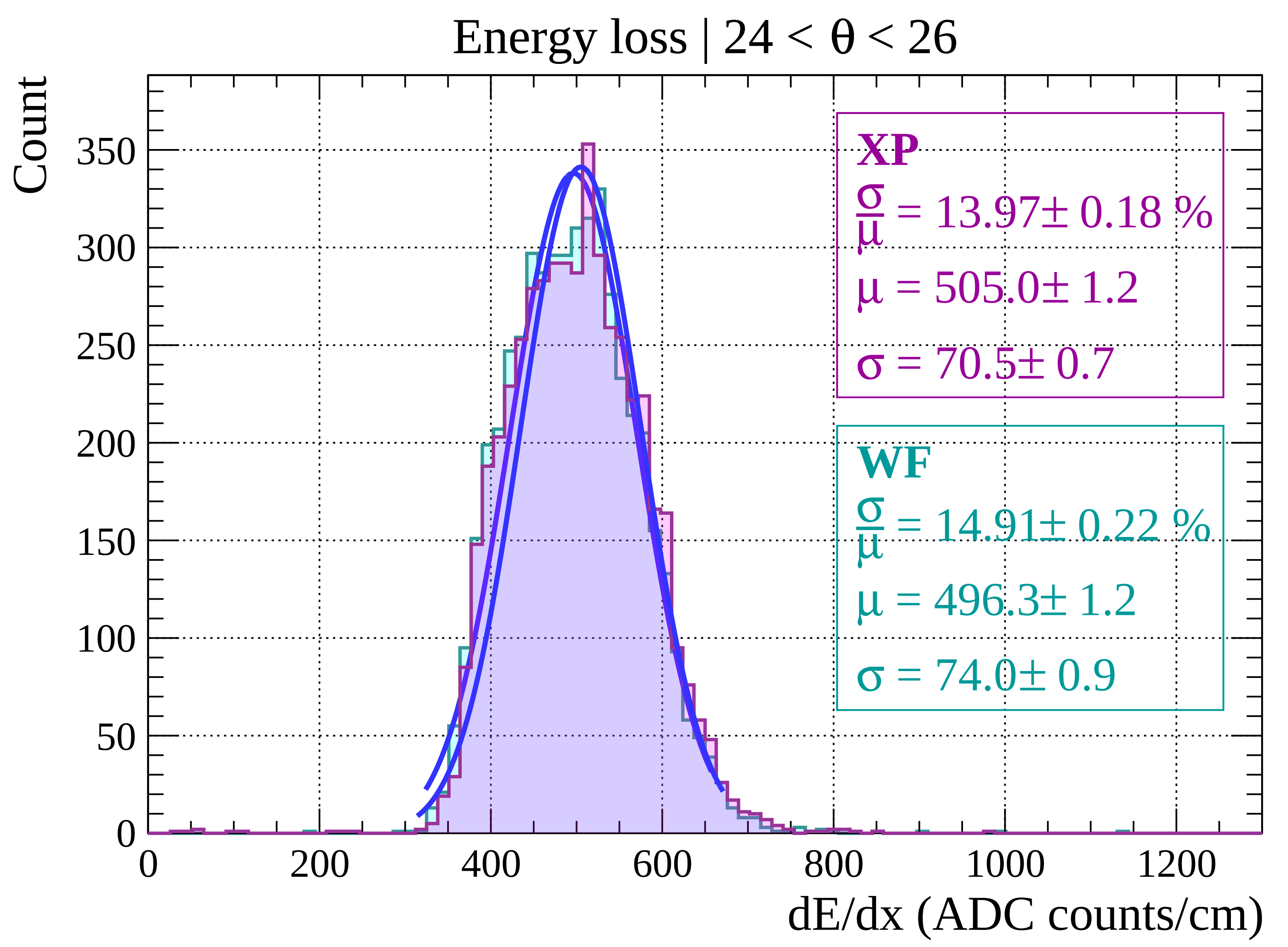
Energy loss | 18 < θ < 20
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 15.01± 0.18 %
μ = 496.9± 1.0
σ = 74.6± 0.8
WF
$\frac{\sigma}{\mu}$ = 15.28± 0.20 %
μ = 491.3± 1.0
σ = 75.1± 0.8
0
100
200
300
400
500
0
200
400
600
800
1000
1200



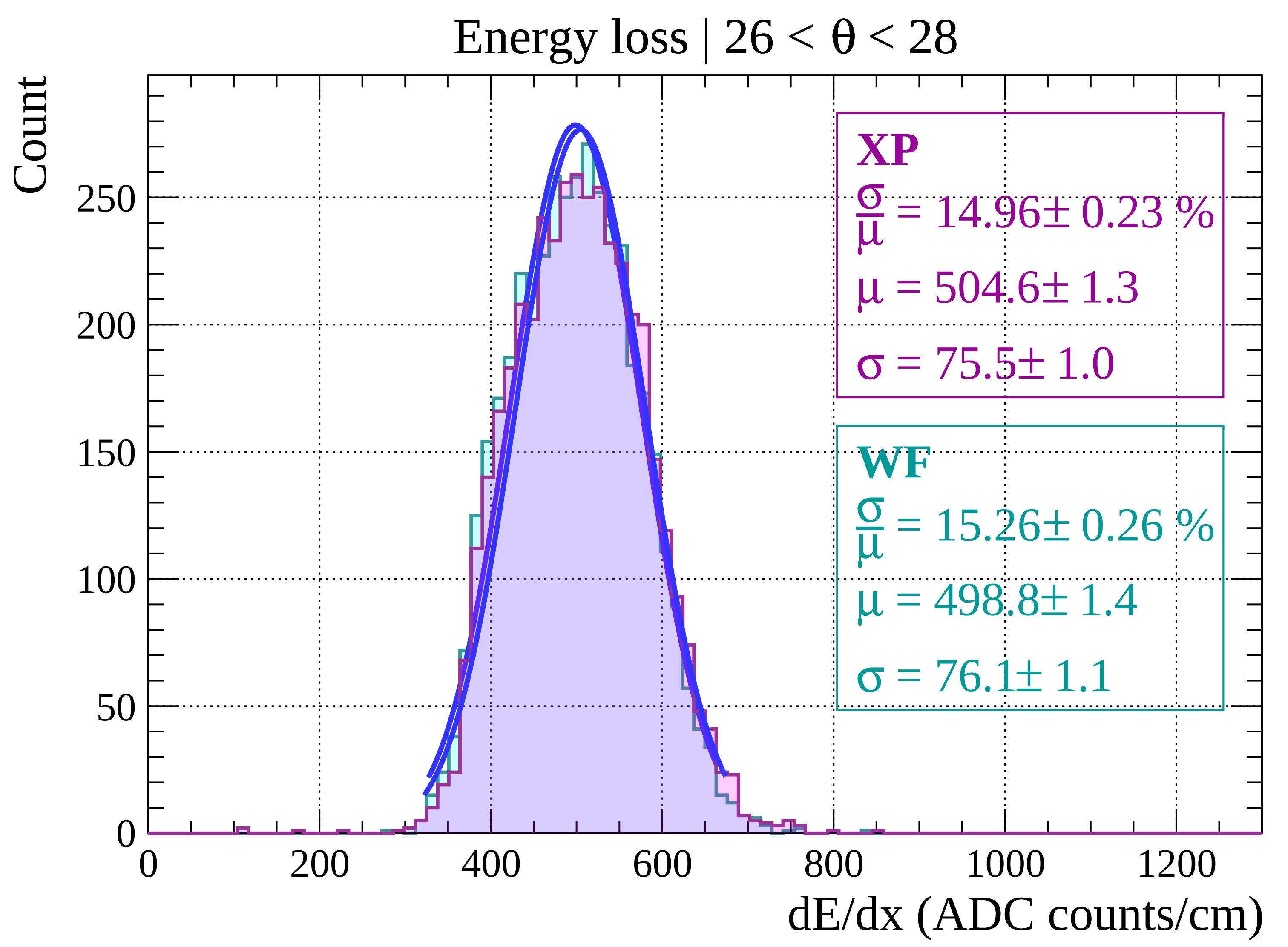
Energy loss | 20 < θ < 22
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.22± 0.17 %
μ = 500.9± 1.0
σ = 71.2± 0.7
WF
σ/μ = 14.92± 0.20 %
μ = 492.9± 1.0
σ = 73.5± 0.8



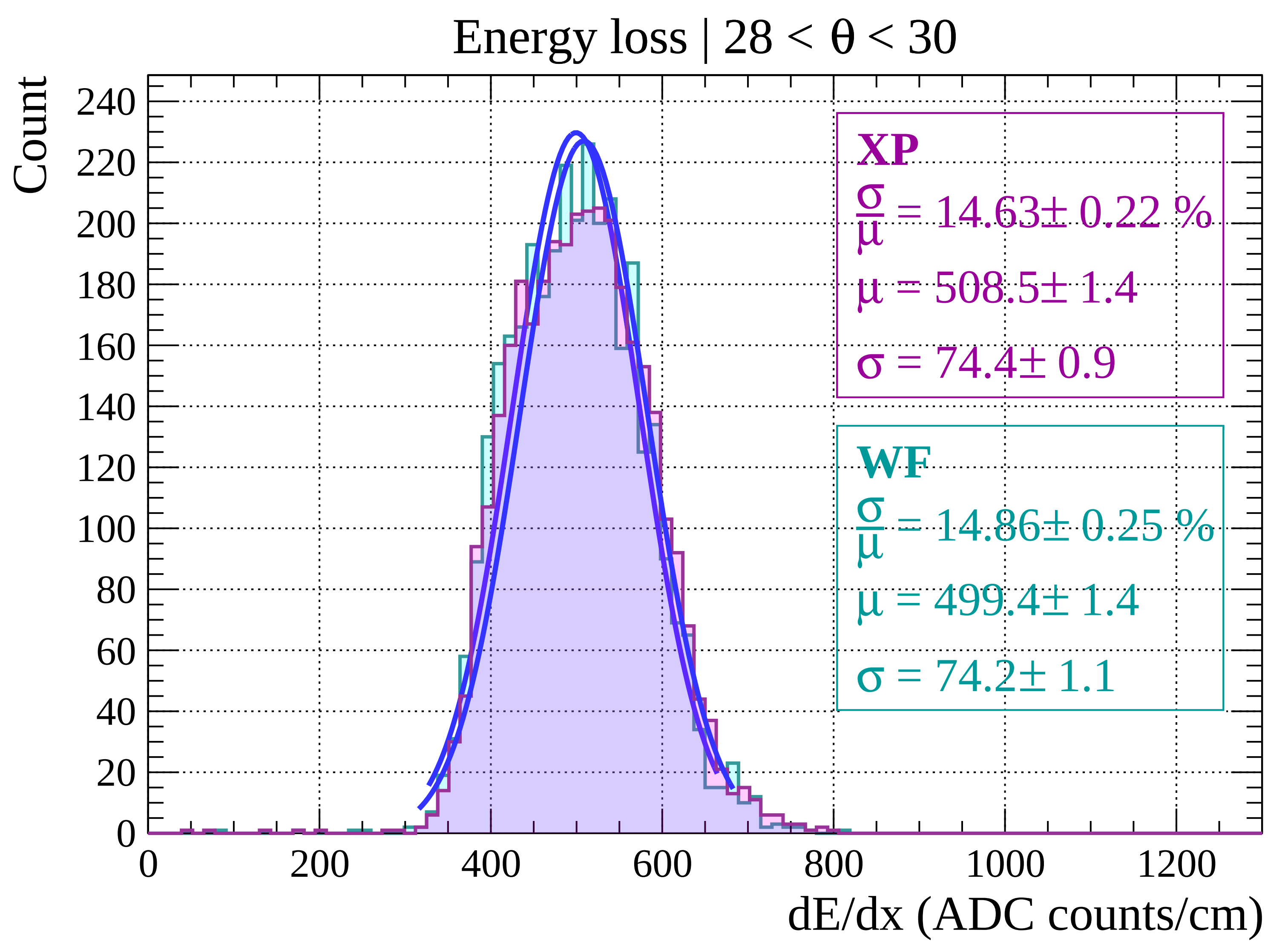
Energy loss | 22 < θ < 24
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 14.06± 0.16 %
μ = 503.5± 1.1
σ = 70.8± 0.7
WF
σ/μ = 13.99± 0.16 %
μ = 497.8± 1.1
σ = 69.6± 0.7



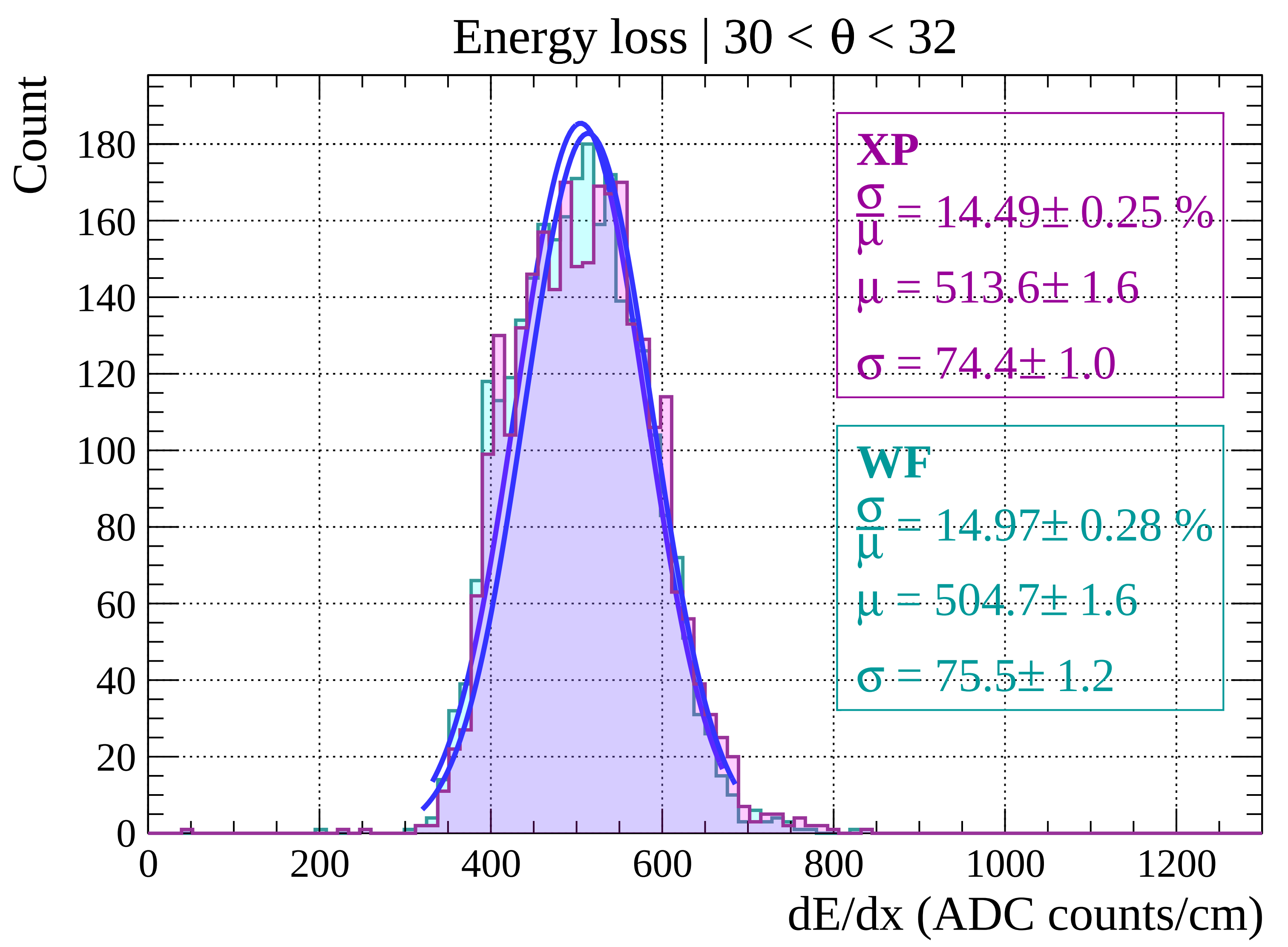
Energy loss | 24 < θ < 26
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 13.97± 0.18 %
μ = 505.0± 1.2
σ = 70.5± 0.7
WF
σ/μ = 14.91± 0.22 %
μ = 496.3± 1.2
σ = 74.0± 0.9



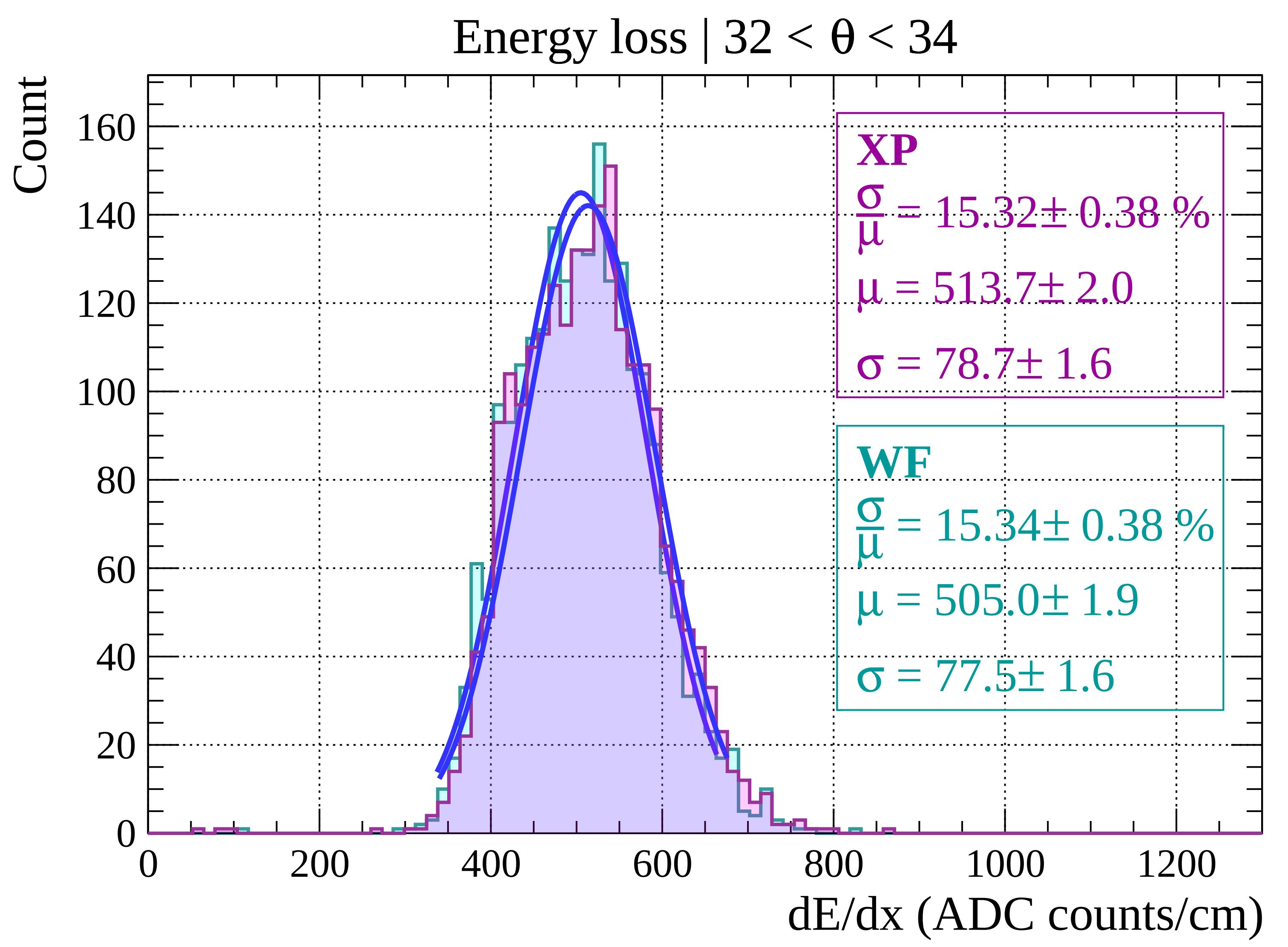
Energy loss | 26 < θ < 28
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 14.96± 0.23 %
μ = 504.6± 1.3
σ = 75.5± 1.0
WF
$\frac{\sigma}{\mu}$ = 15.26± 0.26 %
μ = 498.8± 1.4
σ = 76.1± 1.1
0
50
100
150
200
250
0
200
400
600
800
1000
1200



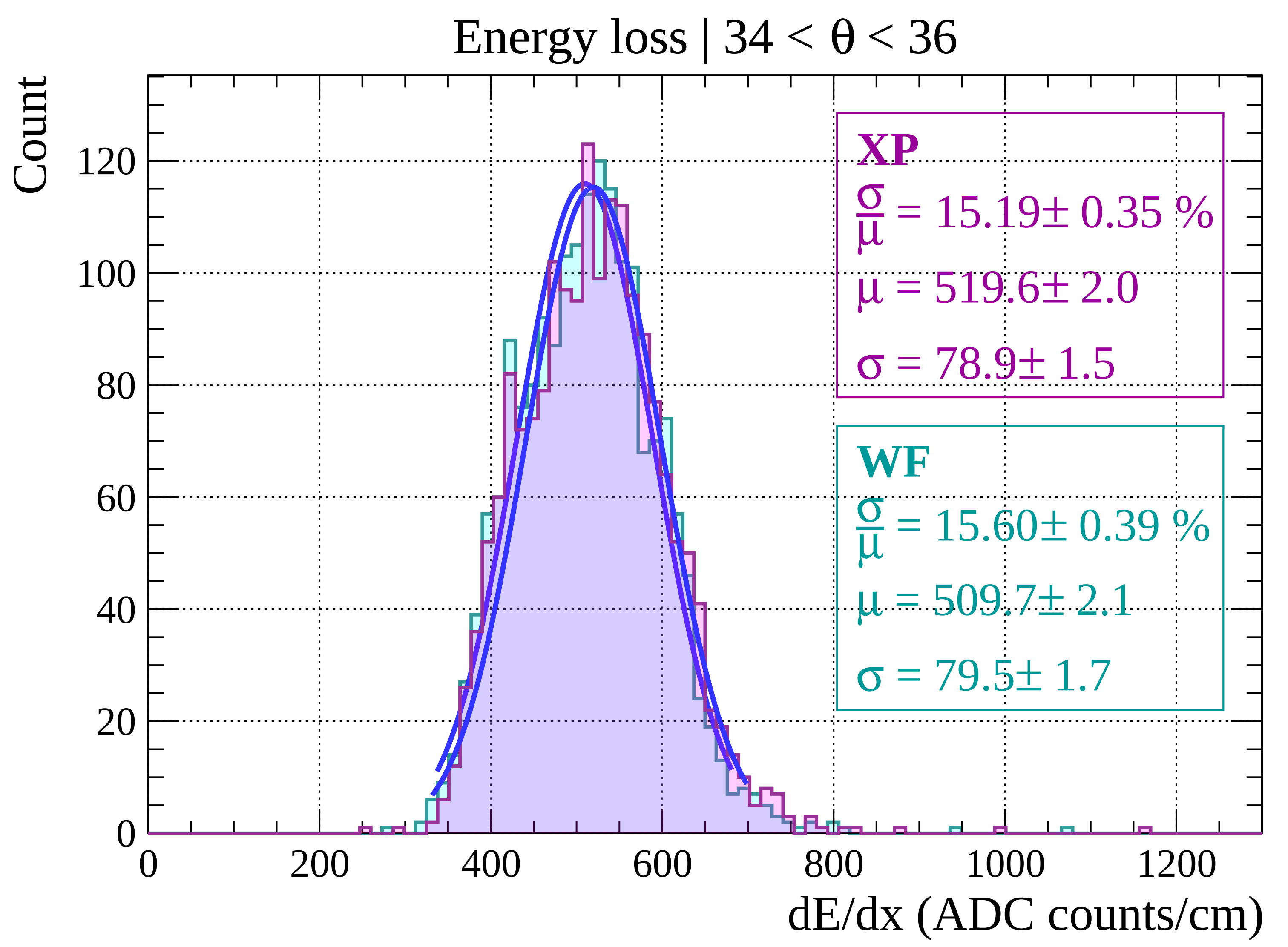
Energy loss | 28 < θ < 30
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
XP
σ/μ = 14.63± 0.22 %
μ = 508.5± 1.4
σ = 74.4± 0.9
WF
σ/μ = 14.86± 0.25 %
μ = 499.4± 1.4
σ = 74.2± 1.1



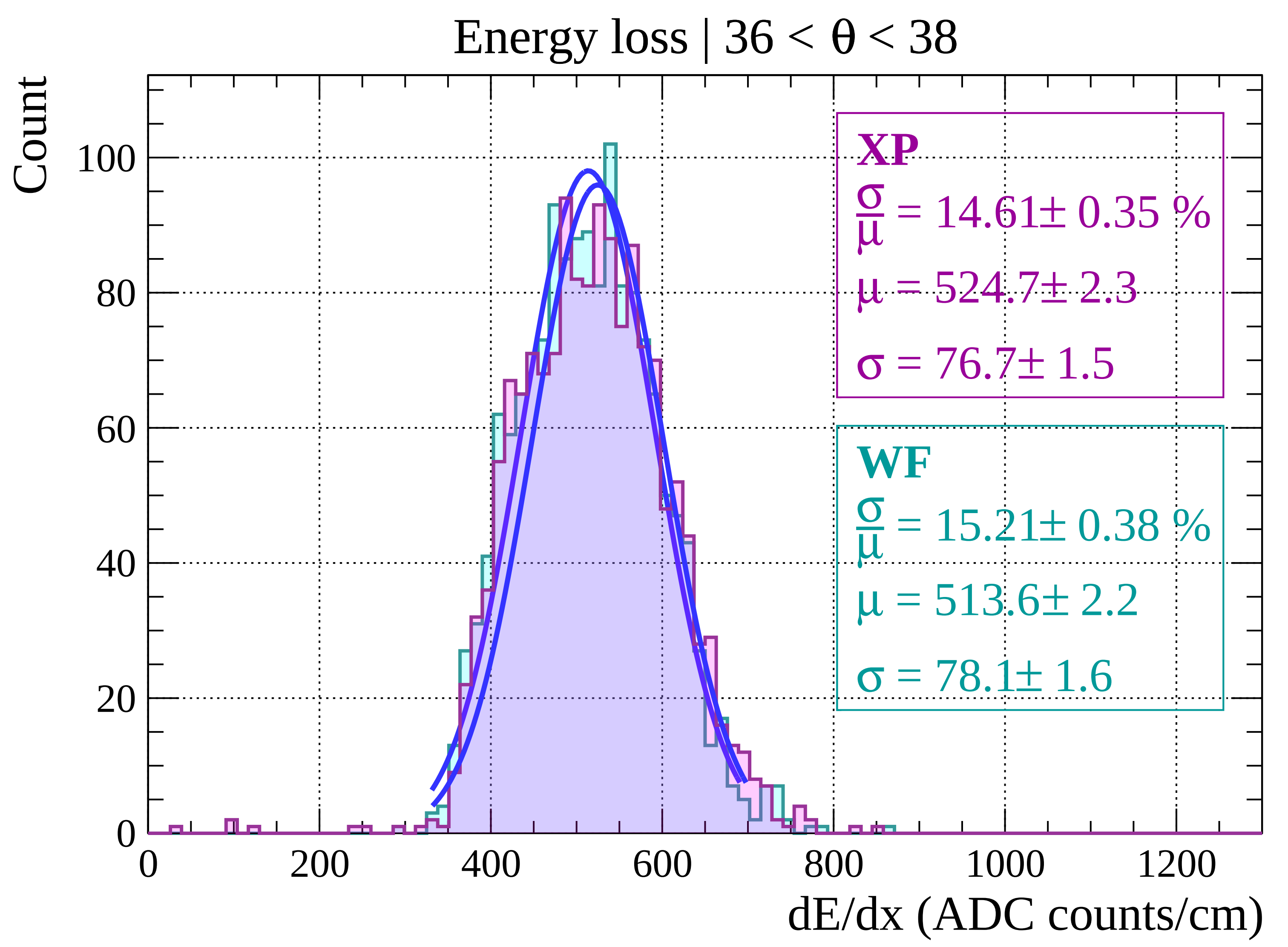
Energy loss | 30 < θ < 32
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 14.49± 0.25 %
μ = 513.6± 1.6
σ = 74.4± 1.0
WF
σ/μ = 14.97± 0.28 %
μ = 504.7± 1.6
σ = 75.5± 1.2



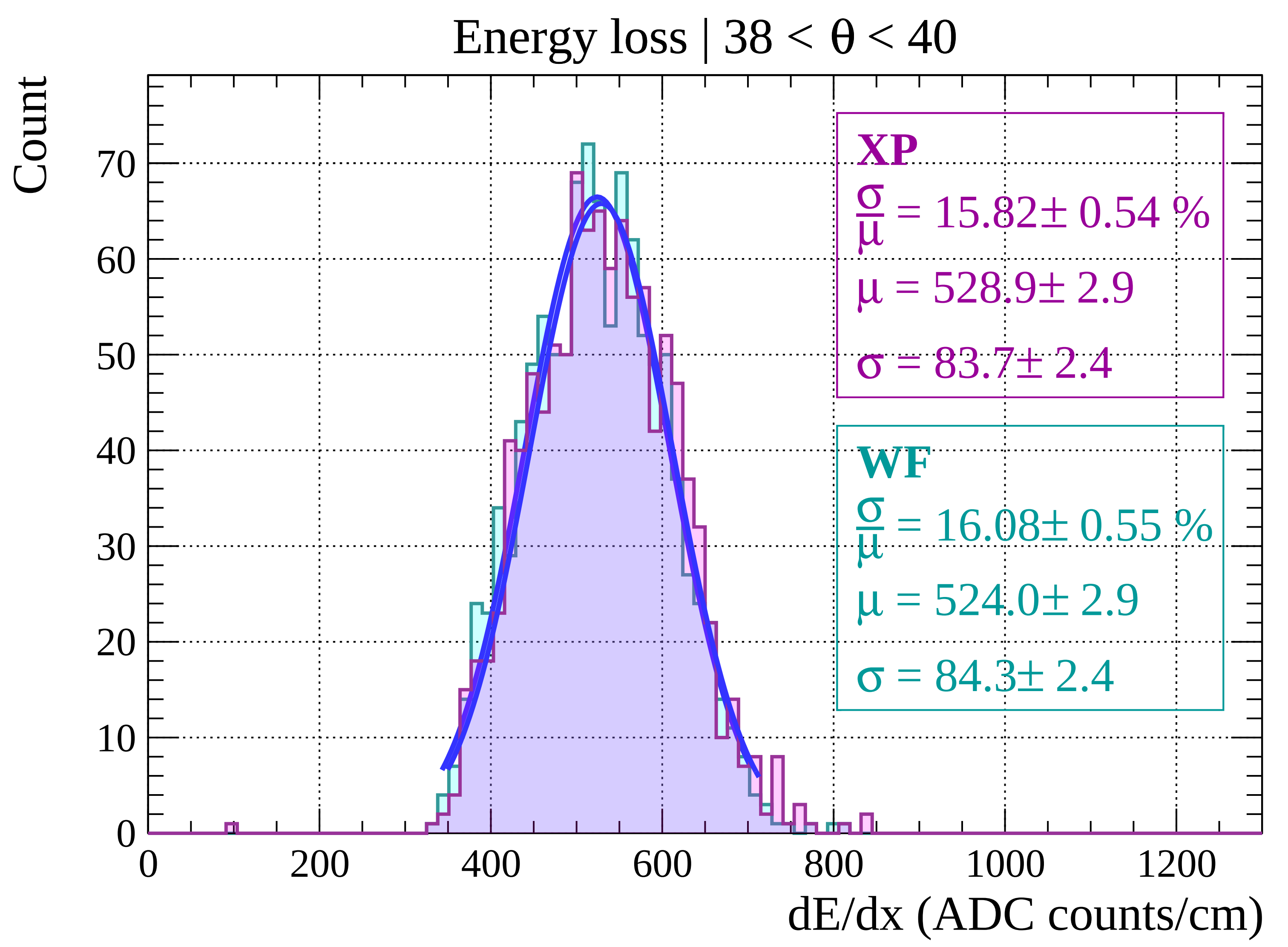
Energy loss | 32 < θ < 34
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
σ/μ = 15.32± 0.38 %
μ = 513.7± 2.0
σ = 78.7± 1.6
WF
σ/μ = 15.34± 0.38 %
μ = 505.0± 1.9
σ = 77.5± 1.6



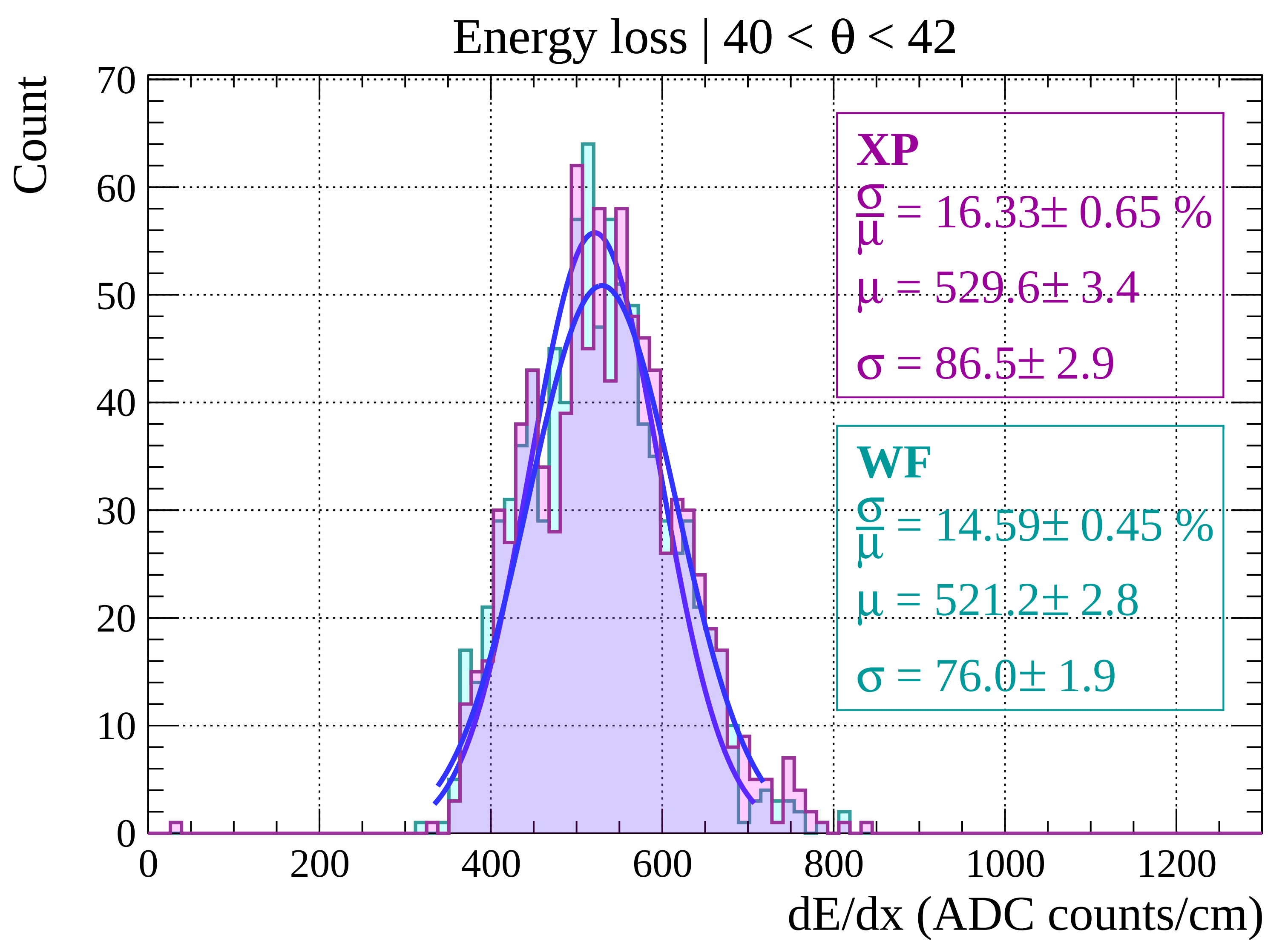
Energy loss | 34 < θ < 36
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 15.19± 0.35 %
μ = 519.6± 2.0
σ = 78.9± 1.5
WF
σ/μ = 15.60± 0.39 %
μ = 509.7± 2.1
σ = 79.5± 1.7



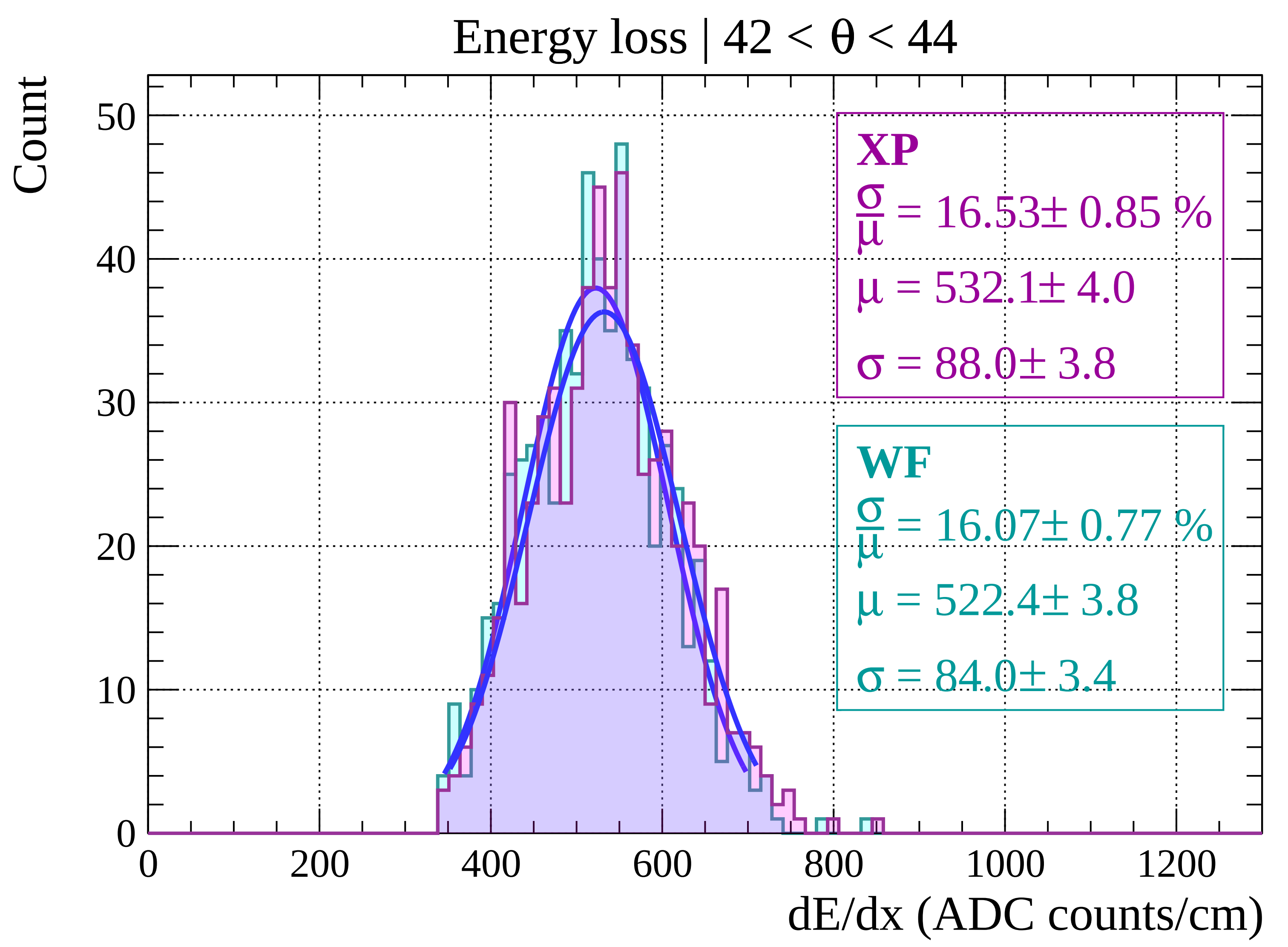
Energy loss | 36 < θ < 38
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 14.61± 0.35 %
μ = 524.7± 2.3
σ = 76.7± 1.5
WF
σ/μ = 15.21± 0.38 %
μ = 513.6± 2.2
σ = 78.1± 1.6



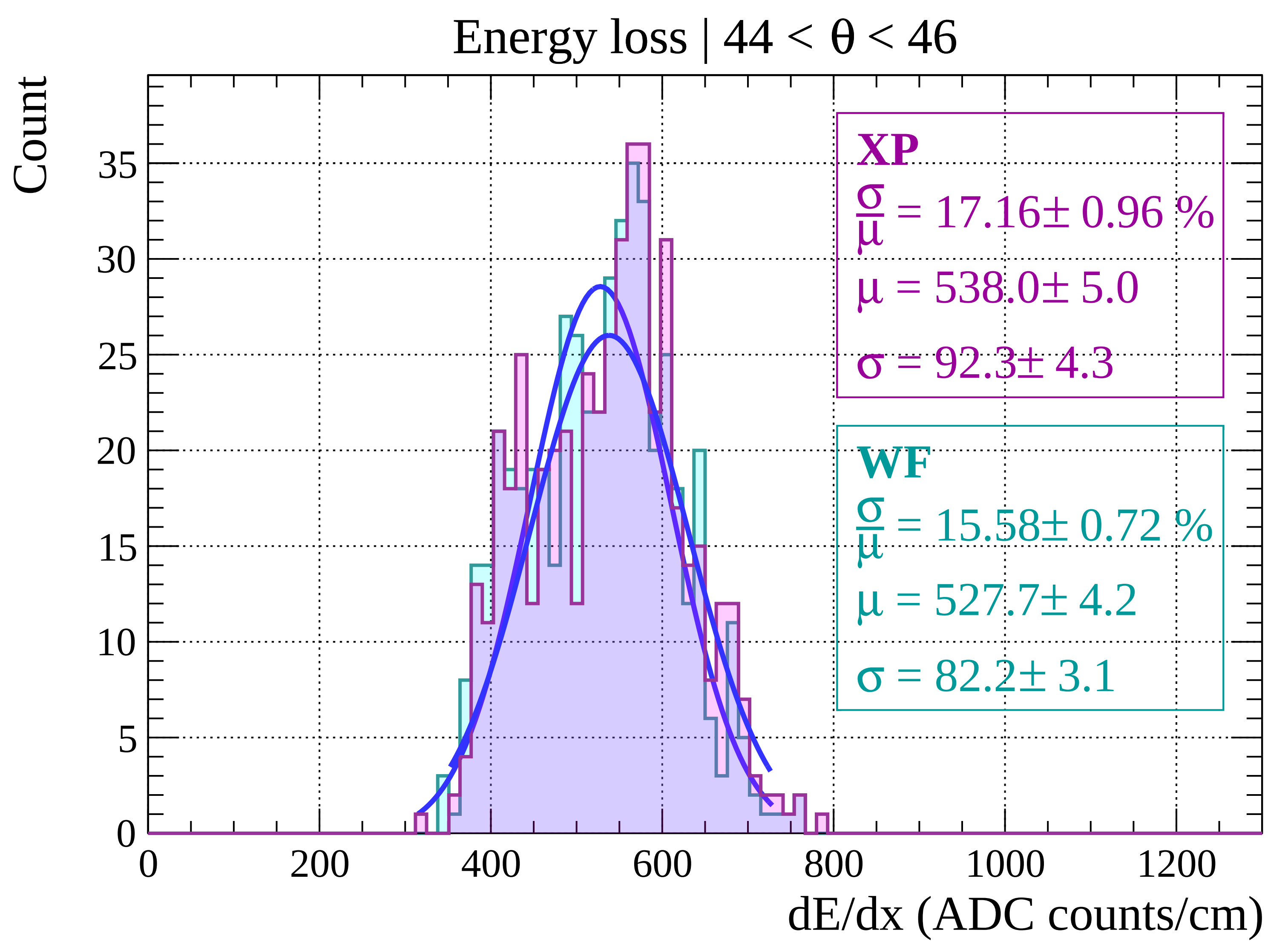
Energy loss | 38 < θ < 40
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
σ/μ = 15.82± 0.54 %
μ = 528.9± 2.9
σ = 83.7± 2.4
WF
σ/μ = 16.08± 0.55 %
μ = 524.0± 2.9
σ = 84.3± 2.4



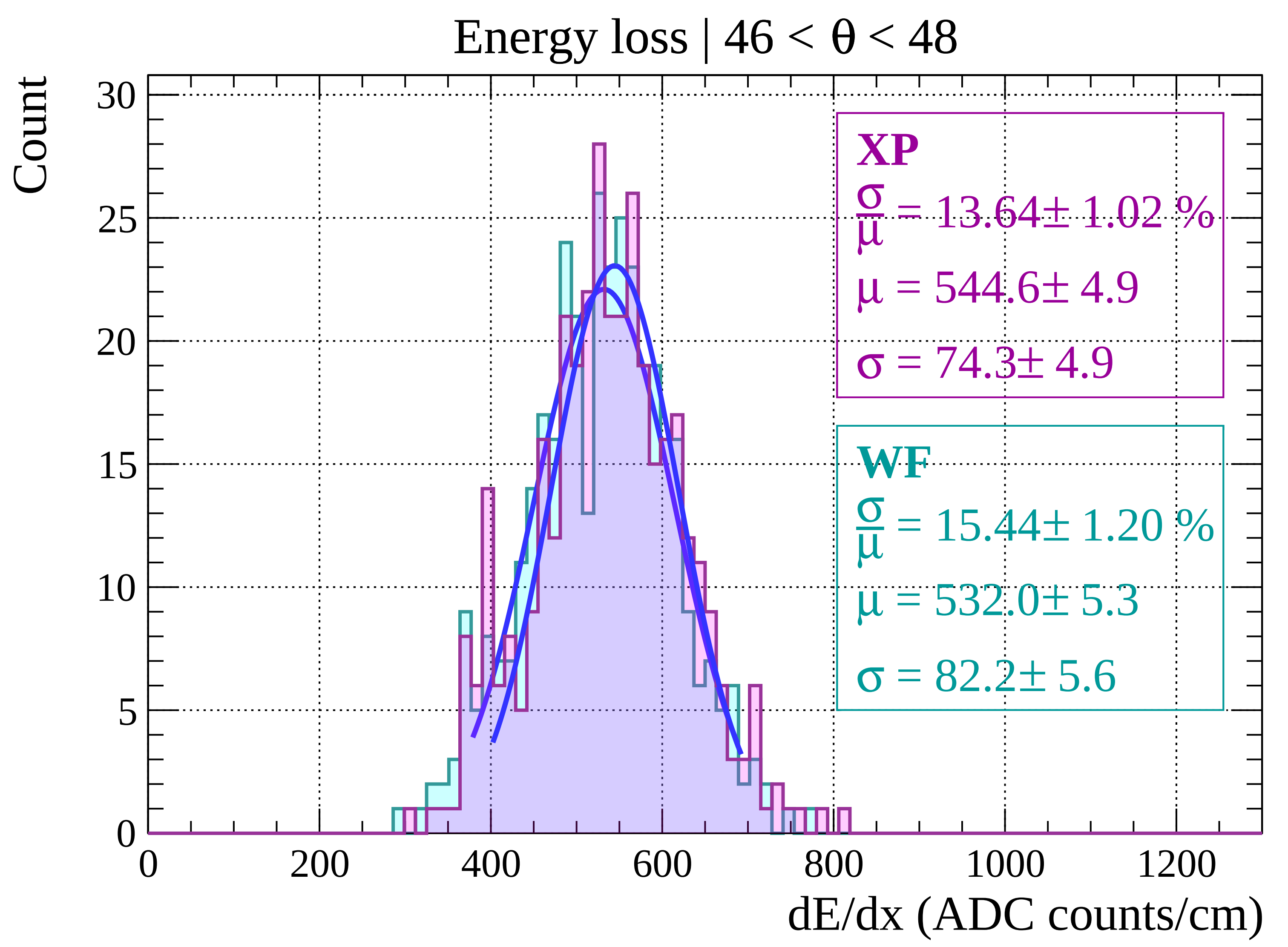
Energy loss | 40 < θ < 42
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
XP
σ/μ = 16.33± 0.65 %
μ = 529.6± 3.4
σ = 86.5± 2.9
WF
σ/μ = 14.59± 0.45 %
μ = 521.2± 2.8
σ = 76.0± 1.9



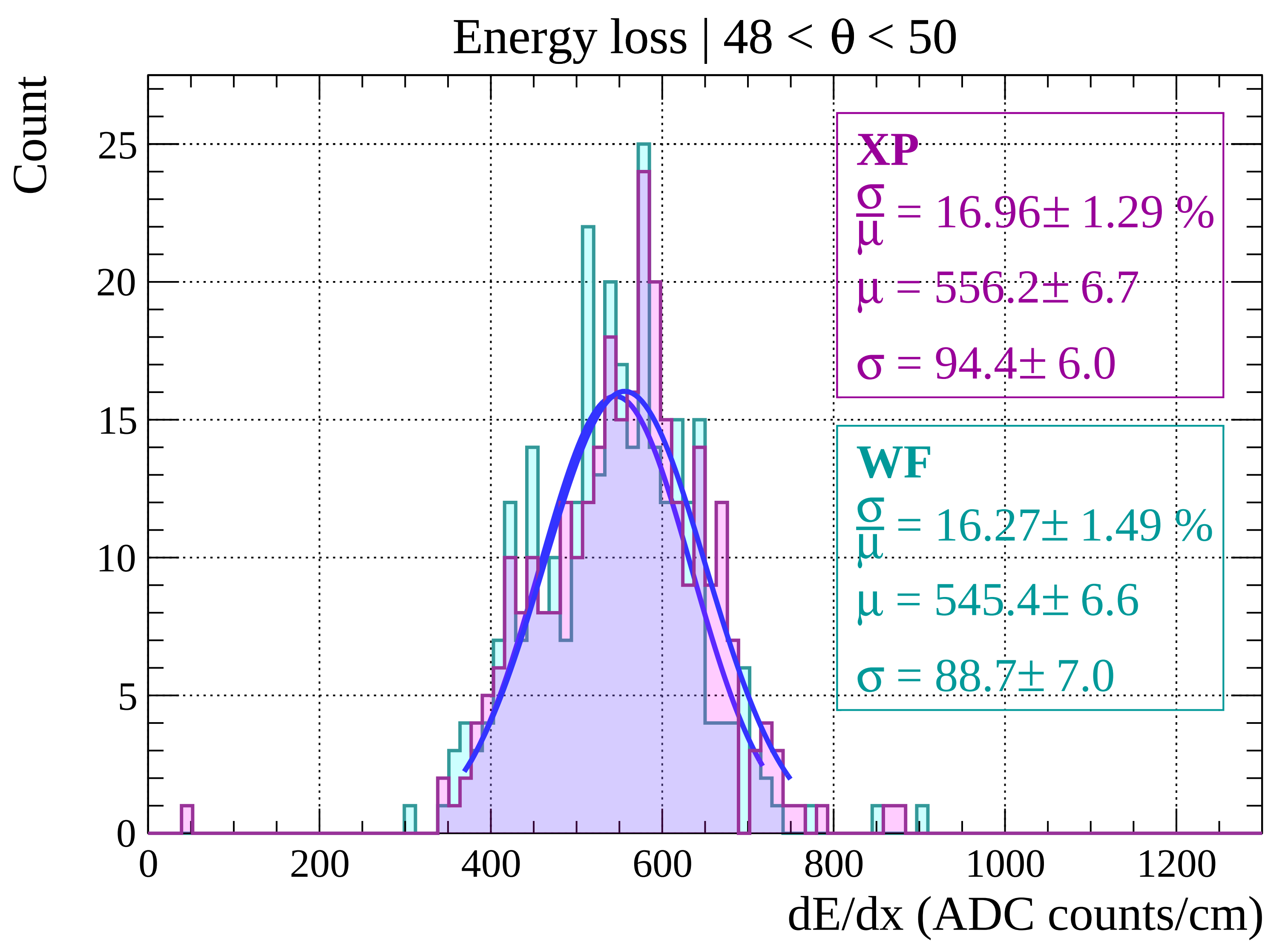
Energy loss | 42 < θ < 44
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 16.53± 0.85 %
μ = 532.1± 4.0
σ = 88.0± 3.8
WF
σ/μ = 16.07± 0.77 %
μ = 522.4± 3.8
σ = 84.0± 3.4

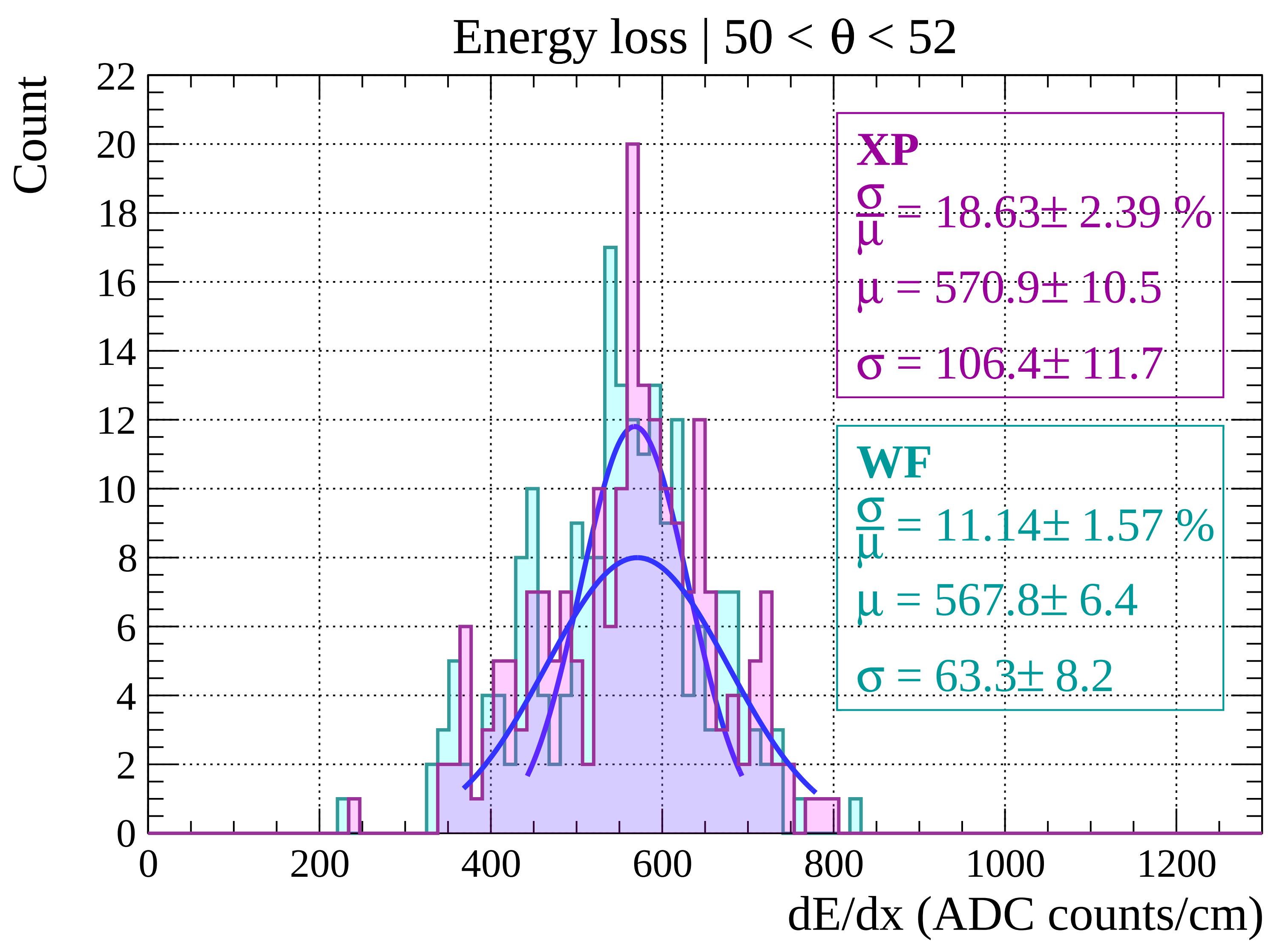


Energy loss | 44 < θ < 46
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP
σ/μ = 17.16± 0.96 %
μ = 538.0± 5.0
σ = 92.3± 4.3
WF
σ/μ = 15.58± 0.72 %
μ = 527.7± 4.2
σ = 82.2± 3.1

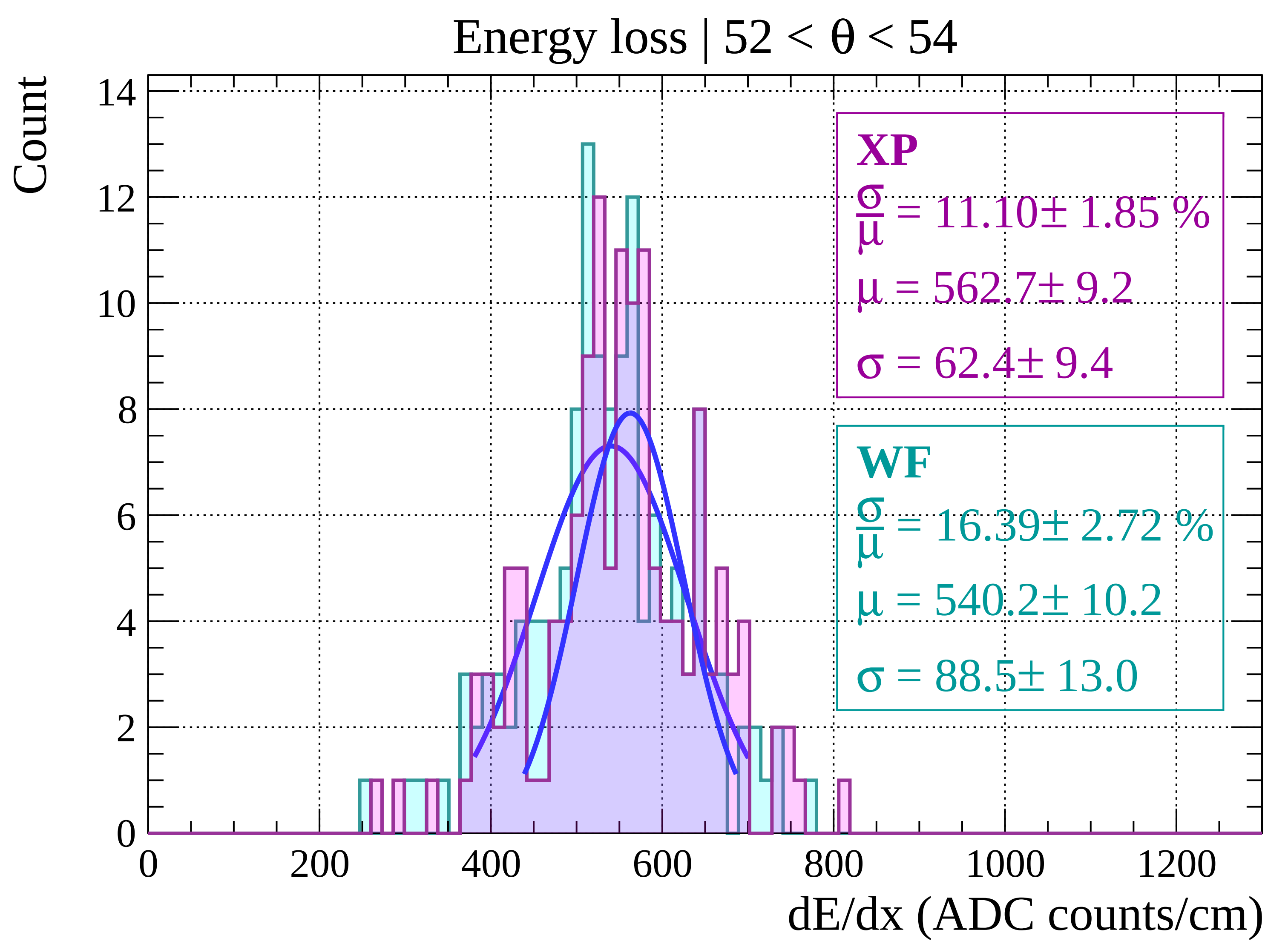


Energy loss | 46 < θ < 48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
XP
σ/μ = 13.64± 1.02 %
μ = 544.6± 4.9
σ = 74.3± 4.9
WF
σ/μ = 15.44± 1.20 %
μ = 532.0± 5.3
σ = 82.2± 5.6

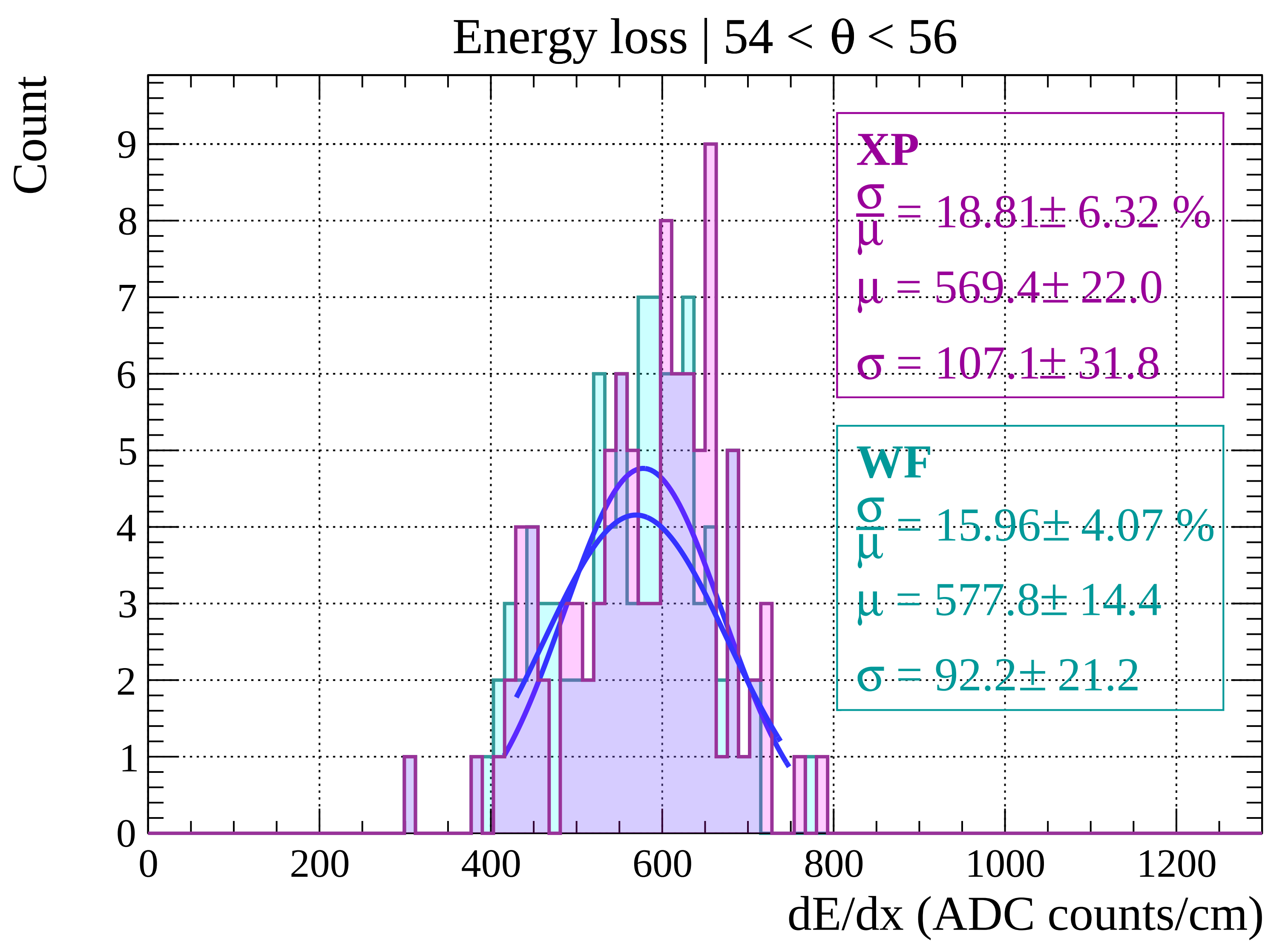

Energy loss | 48 < θ < 50
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
XP
σ/μ = 16.96± 1.29 %
μ = 556.2± 6.7
σ = 94.4± 6.0
WF
σ/μ = 16.27± 1.49 %
μ = 545.4± 6.6
σ = 88.7± 7.0

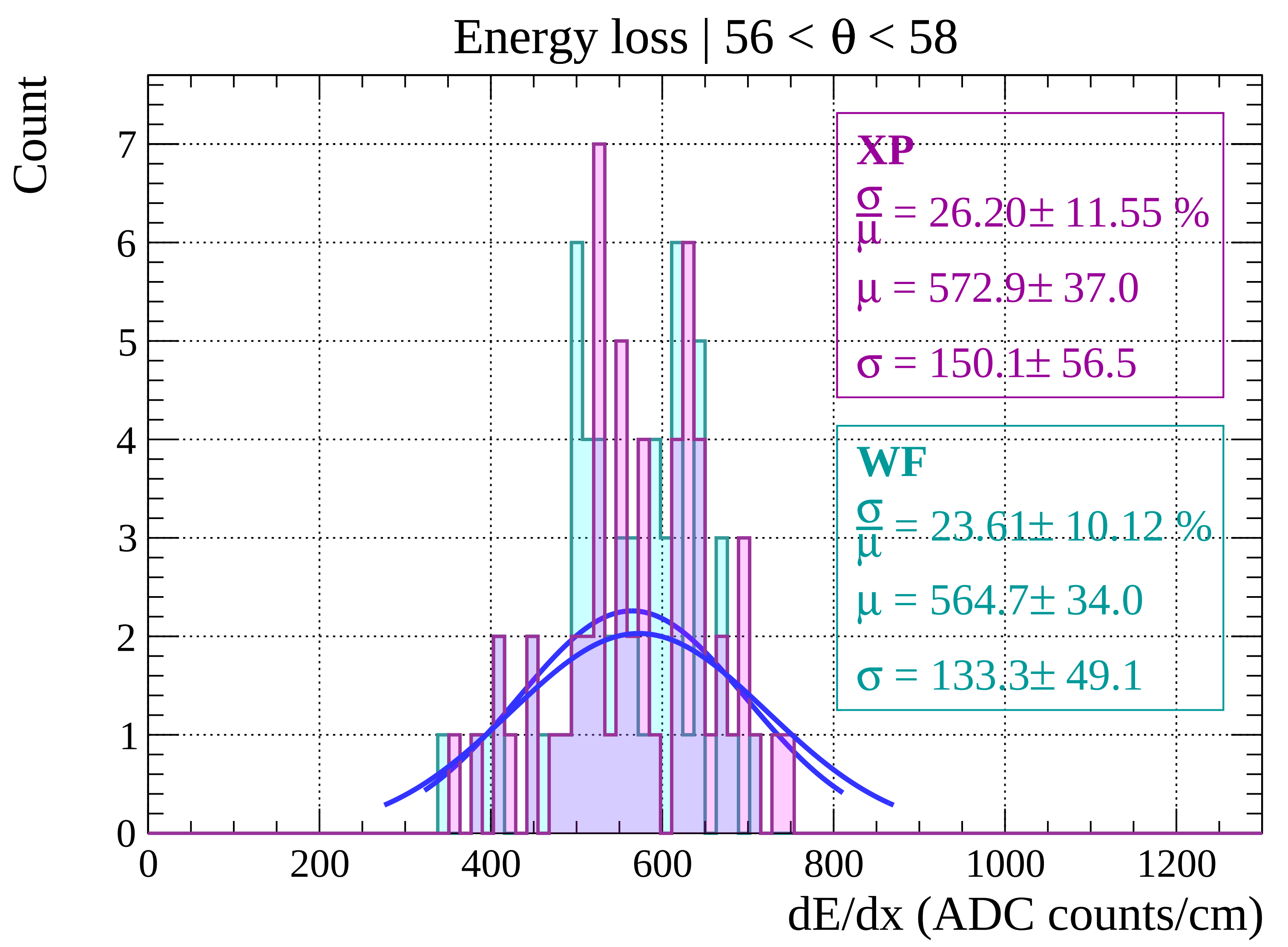


Energy loss | 50 < θ < 52
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
20
22
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 18.63± 2.39 %
μ = 570.9± 10.5
σ = 106.4± 11.7
WF
$\frac{\sigma}{\mu}$ = 11.14± 1.57 %
μ = 567.8± 6.4
σ = 63.3± 8.2

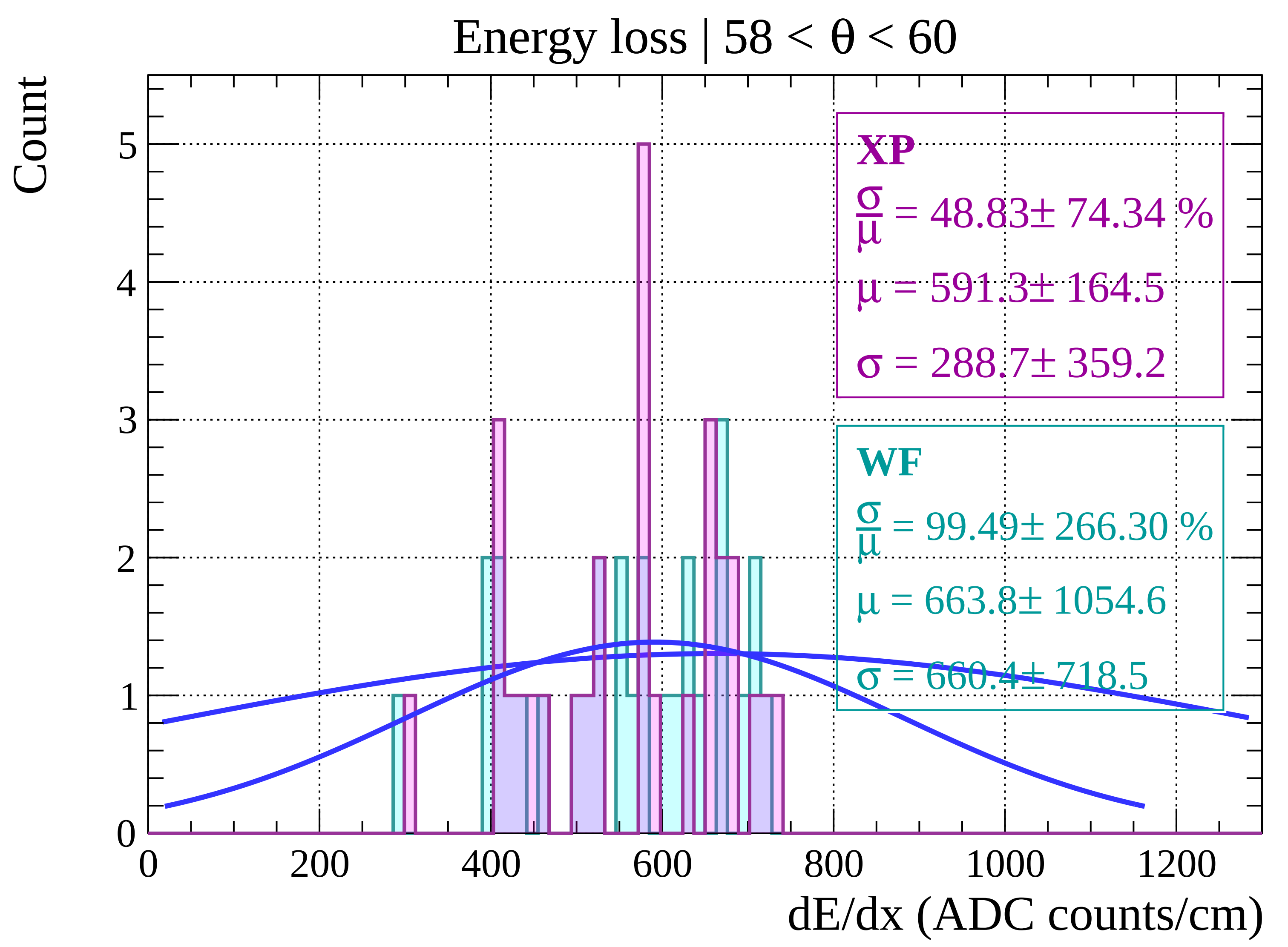


Energy loss | 52 < θ < 54
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.10± 1.85 %
μ = 562.7± 9.2
σ = 62.4± 9.4
WF
σ/μ = 16.39± 2.72 %
μ = 540.2± 10.2
σ = 88.5± 13.0

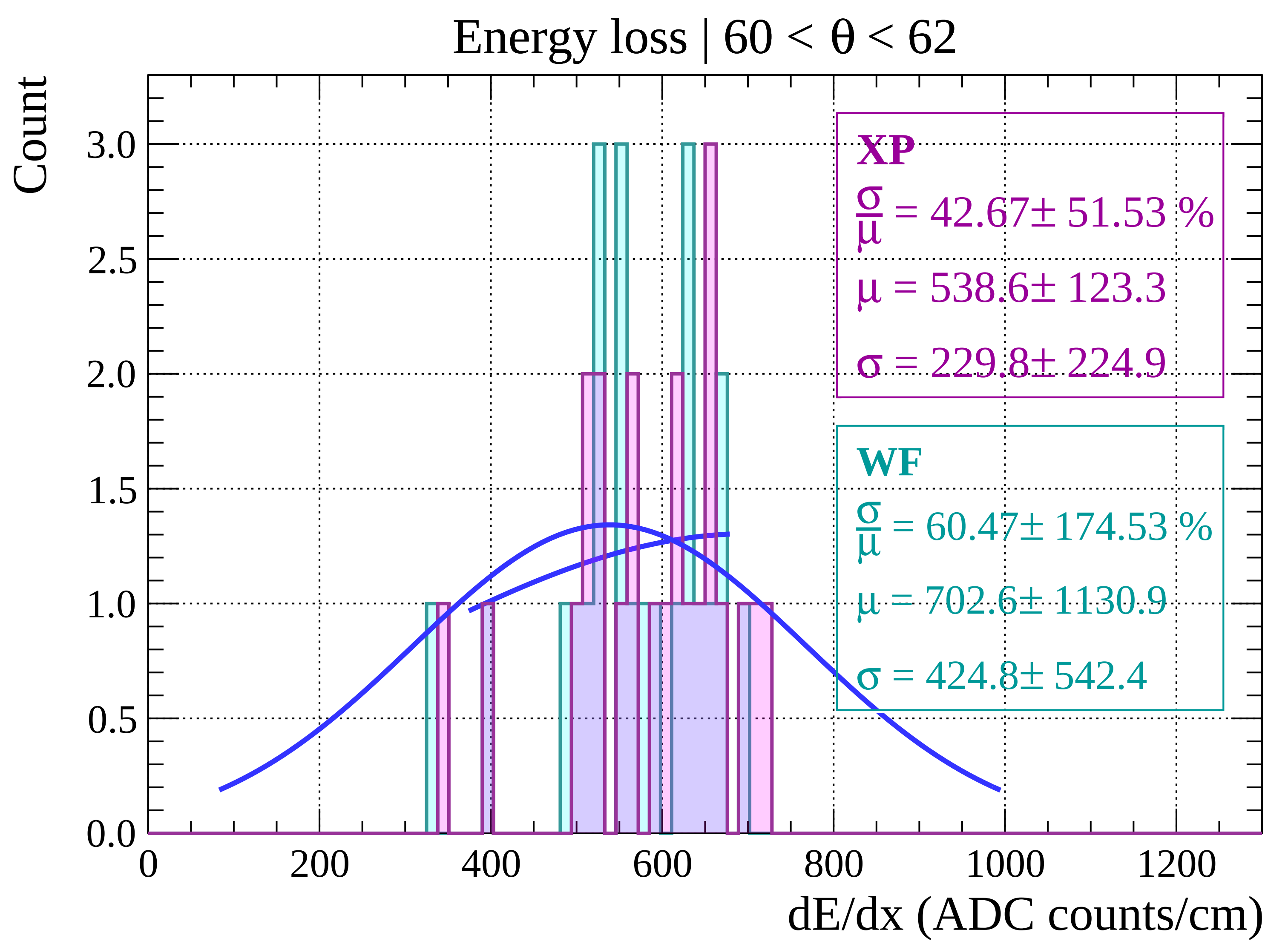

Energy loss | 54 < θ < 56
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 18.81± 6.32 %
μ = 569.4± 22.0
σ = 107.1± 31.8
WF
σ/μ = 15.96± 4.07 %
μ = 577.8± 14.4
σ = 92.2± 21.2
0
1
2
3
4
5
6
7
8
9
0
200
400
600
800
1000
1200



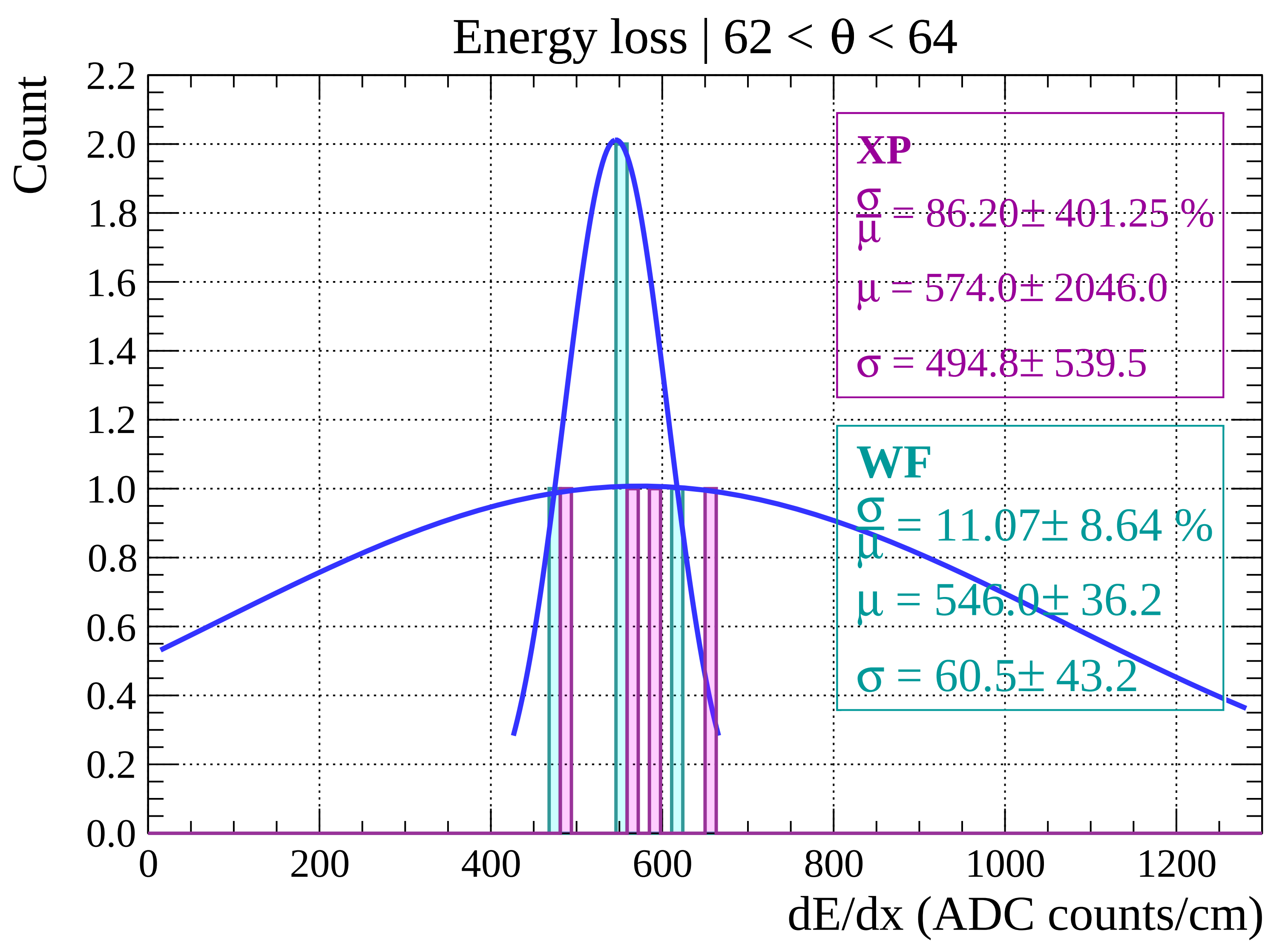
Energy loss | 56 < θ < 58
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
XP
σ/μ = 26.20± 11.55 %
μ = 572.9± 37.0
σ = 150.1± 56.5
WF
σ/μ = 23.61± 10.12 %
μ = 564.7± 34.0
σ = 133.3± 49.1



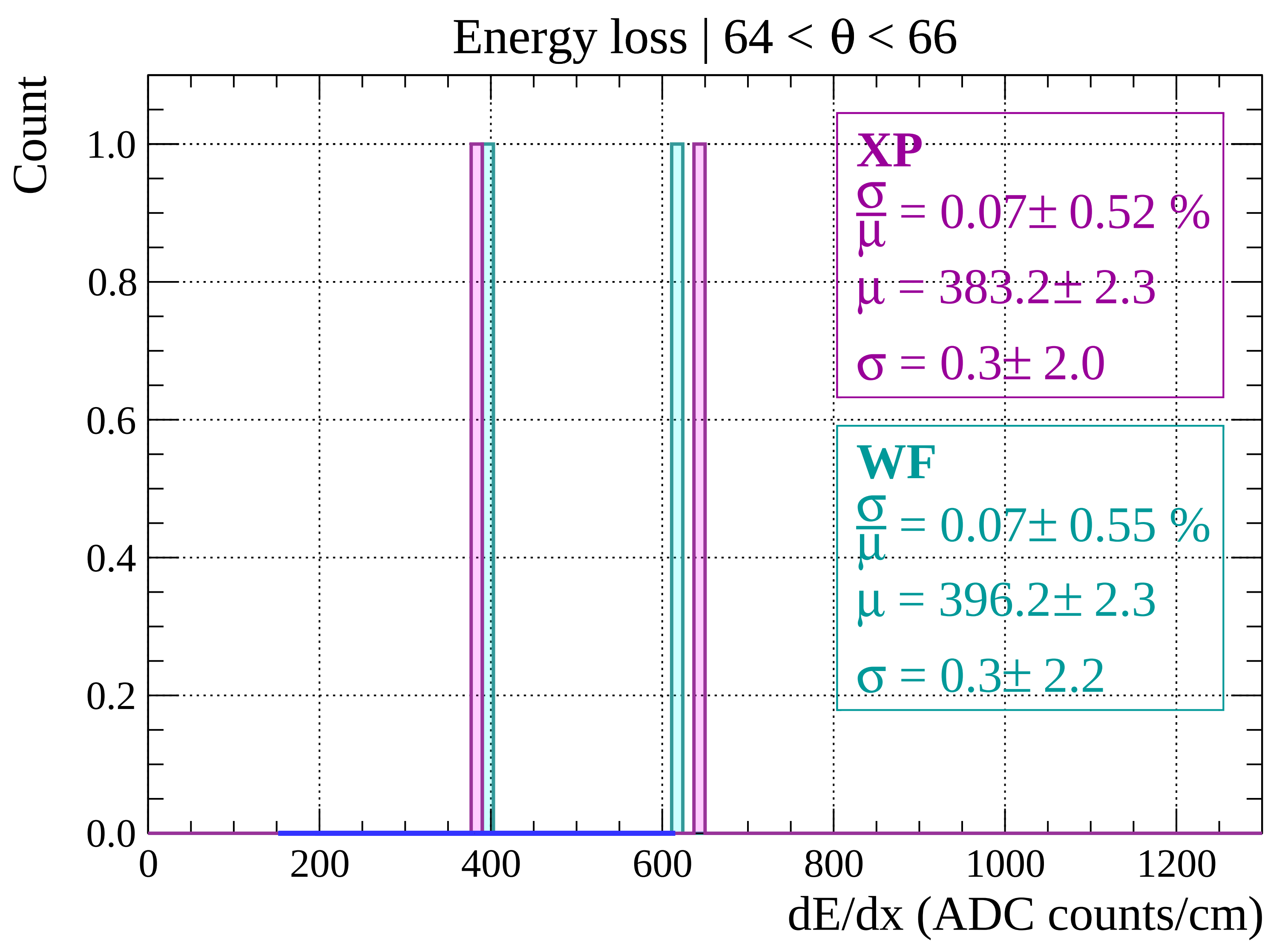
Energy loss | 58 < θ < 60
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
σ/μ = 48.83± 74.34 %
μ = 591.3± 164.5
σ = 288.7± 359.2
WF
σ/μ = 99.49± 266.30 %
μ = 663.8± 1054.6
σ = 660.4± 718.5



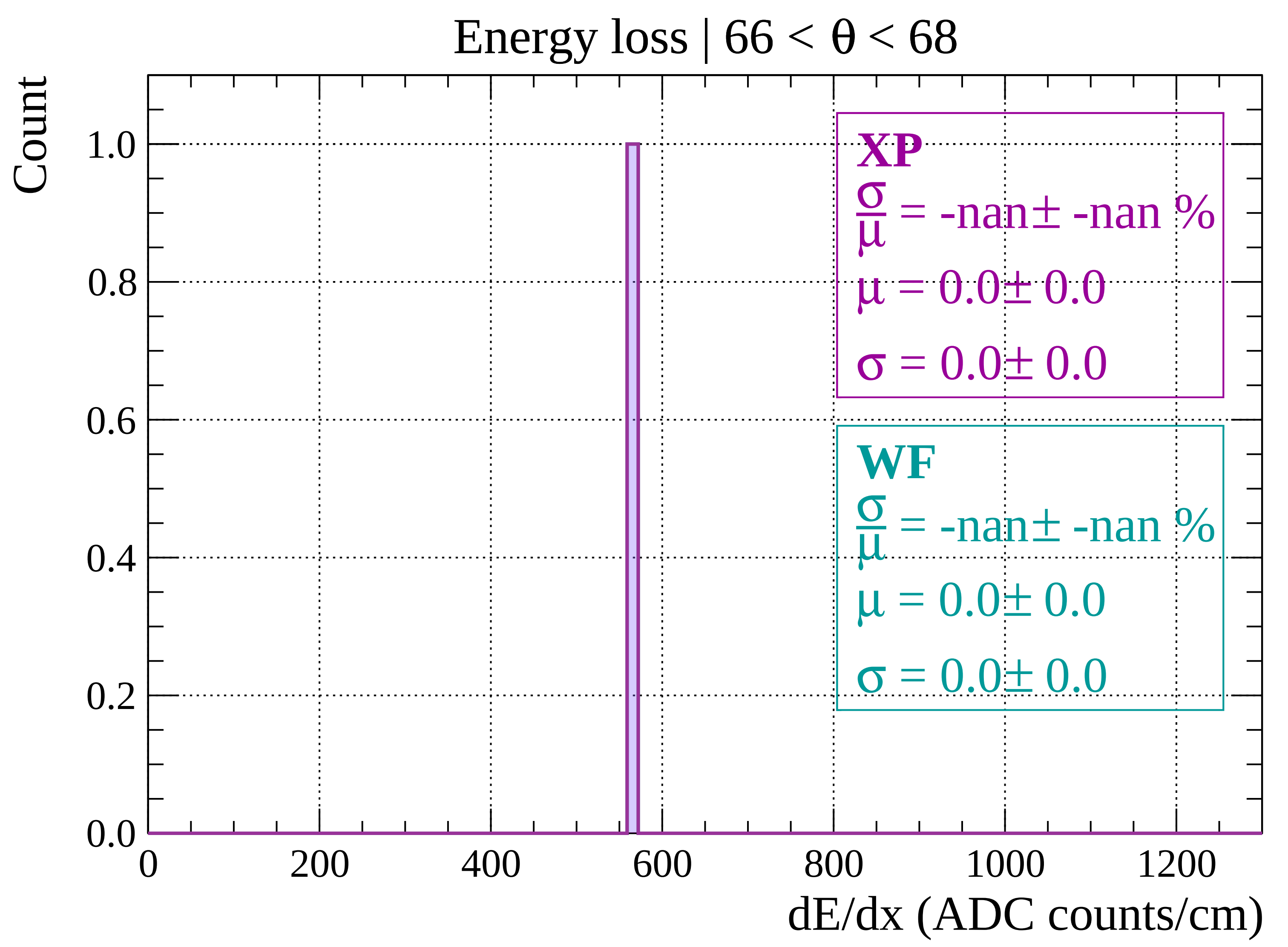
Energy loss | 60 < θ < 62
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.5
1.0
1.5
2.0
2.5
3.0
XP
σ/μ = 42.67± 51.53 %
μ = 538.6± 123.3
σ = 229.8± 224.9
WF
σ/μ = 60.47± 174.53 %
μ = 702.6± 1130.9
σ = 424.8± 542.4



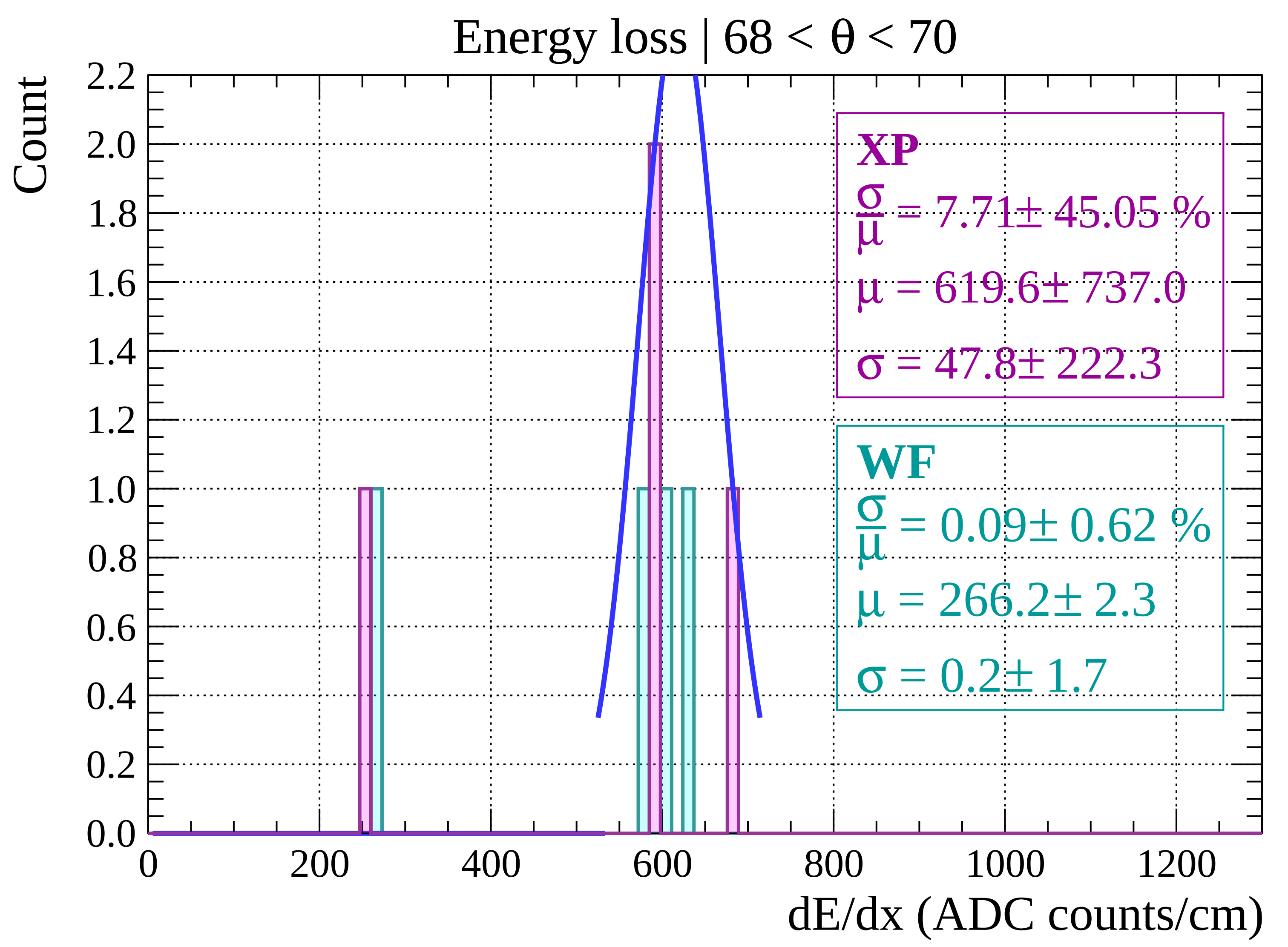
Energy loss | 62 < θ < 64
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
XP
σ/μ = 86.20± 401.25 %
μ = 574.0± 2046.0
σ = 494.8± 539.5
WF
σ/μ = 11.07± 8.64 %
μ = 546.0± 36.2
σ = 60.5± 43.2



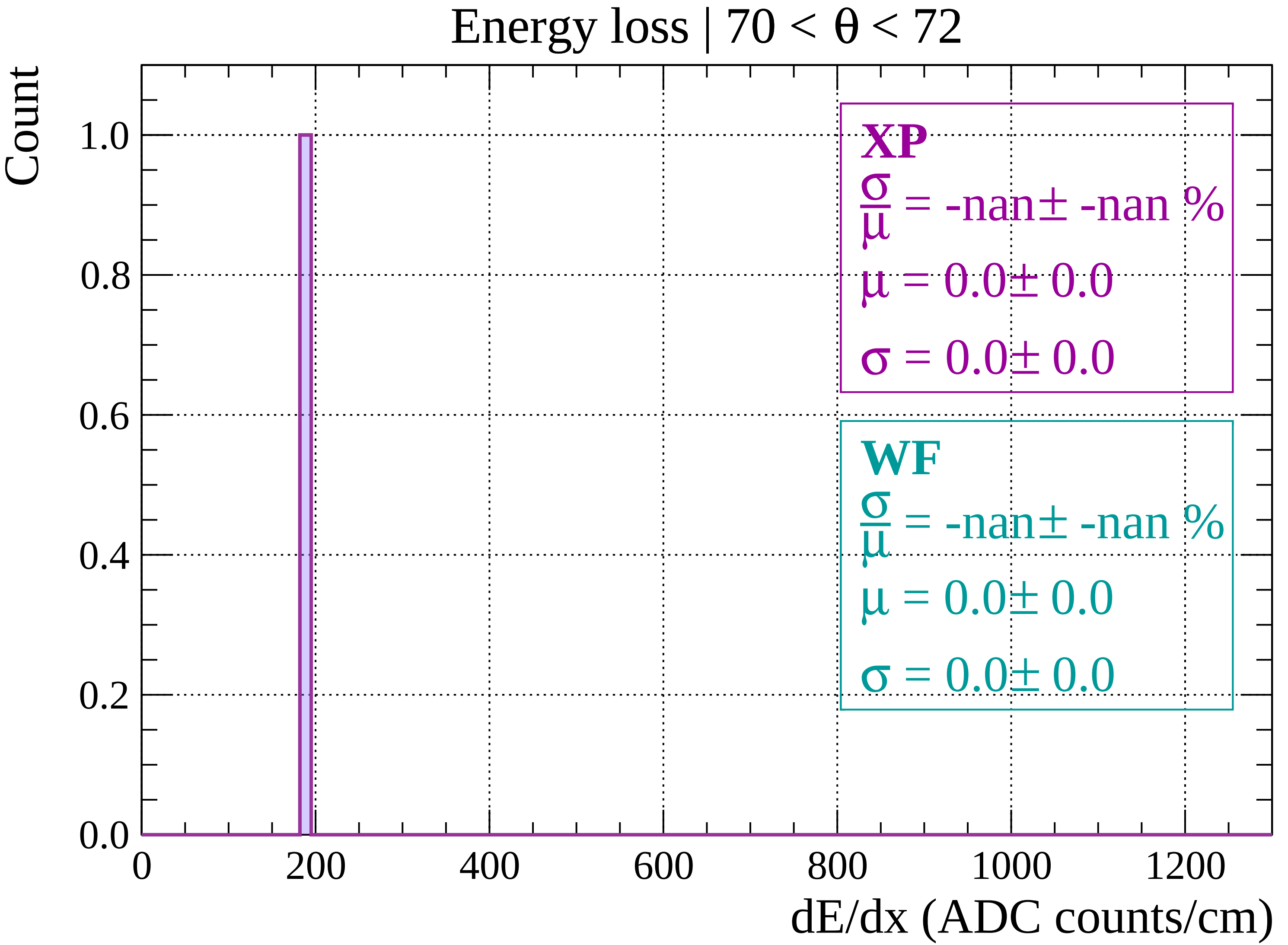
Energy loss | 64 < θ < 66
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 0.07 \pm 0.52$ %
$\mu = 383.2 \pm 2.3$
$\sigma = 0.3 \pm 2.0$
WF
$\frac{\sigma}{\mu} = 0.07 \pm 0.55$ %
$\mu = 396.2 \pm 2.3$
$\sigma = 0.3 \pm 2.2$

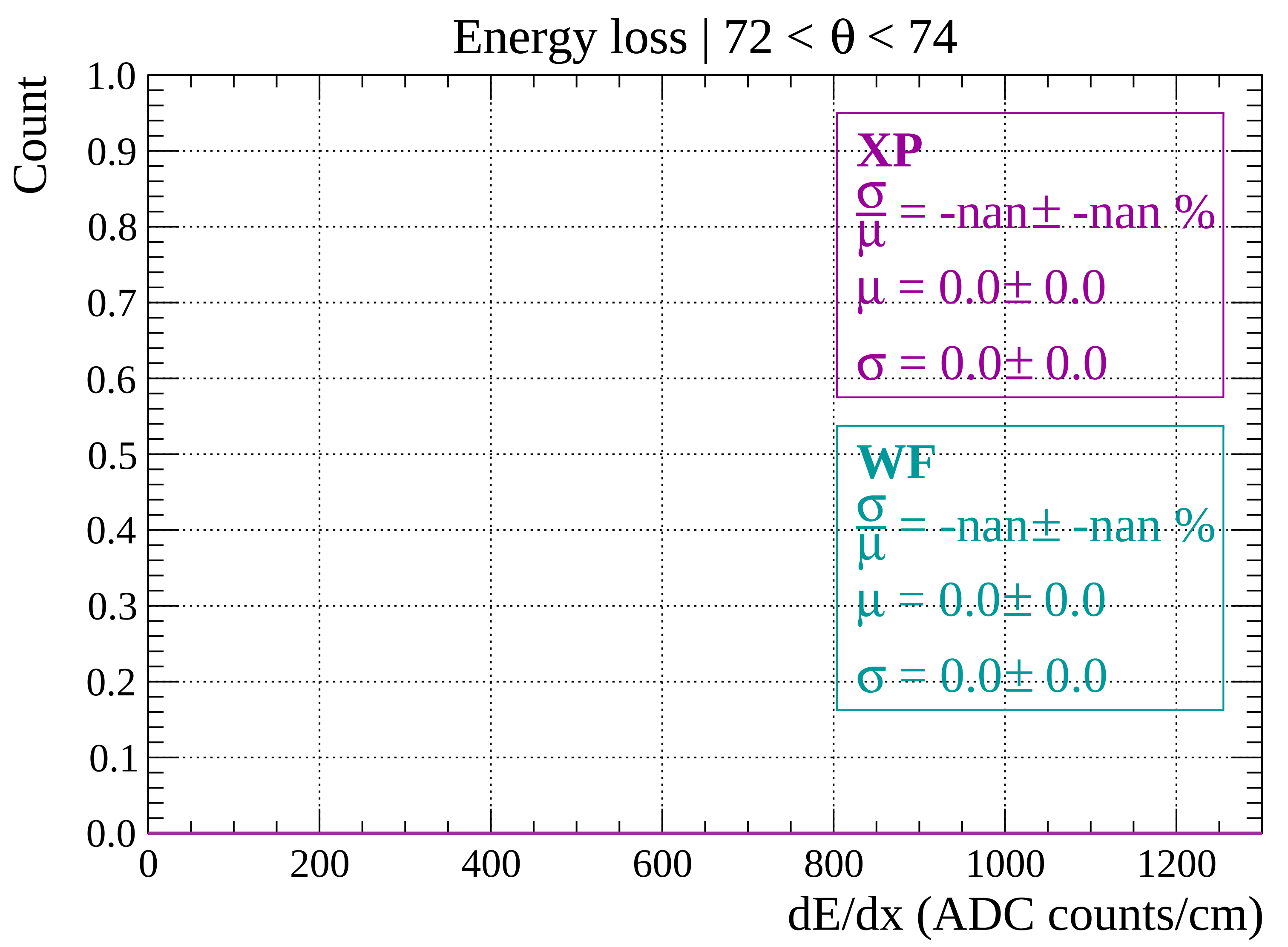


Energy loss | 66 < θ < 68
Count
dE/dx (ADC counts/cm)
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

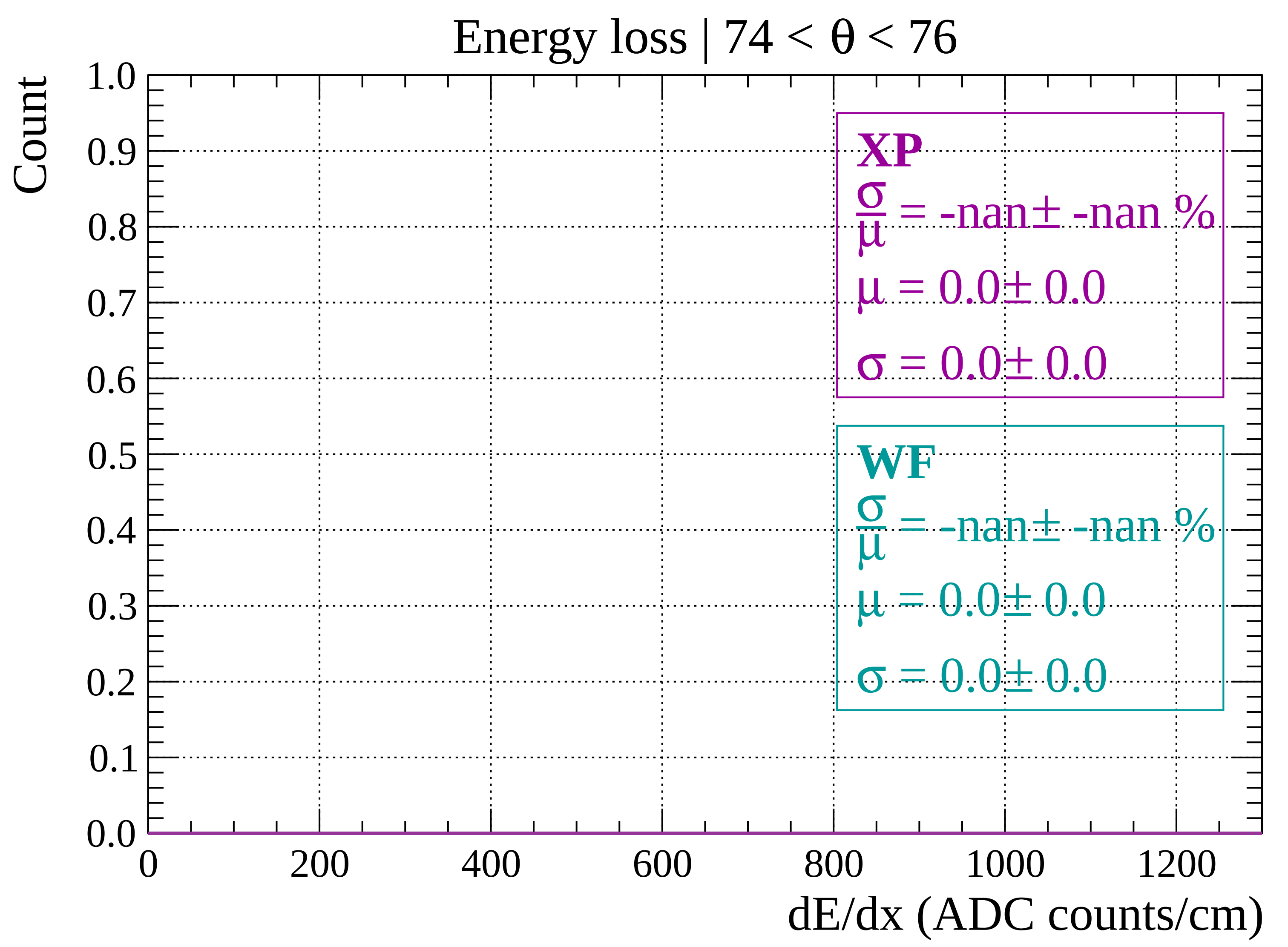


Energy loss | 68 < θ < 70
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
XP
σ/μ = 7.71± 45.05 %
μ = 619.6± 737.0
σ = 47.8± 222.3
WF
σ/μ = 0.09± 0.62 %
μ = 266.2± 2.3
σ = 0.2± 1.7

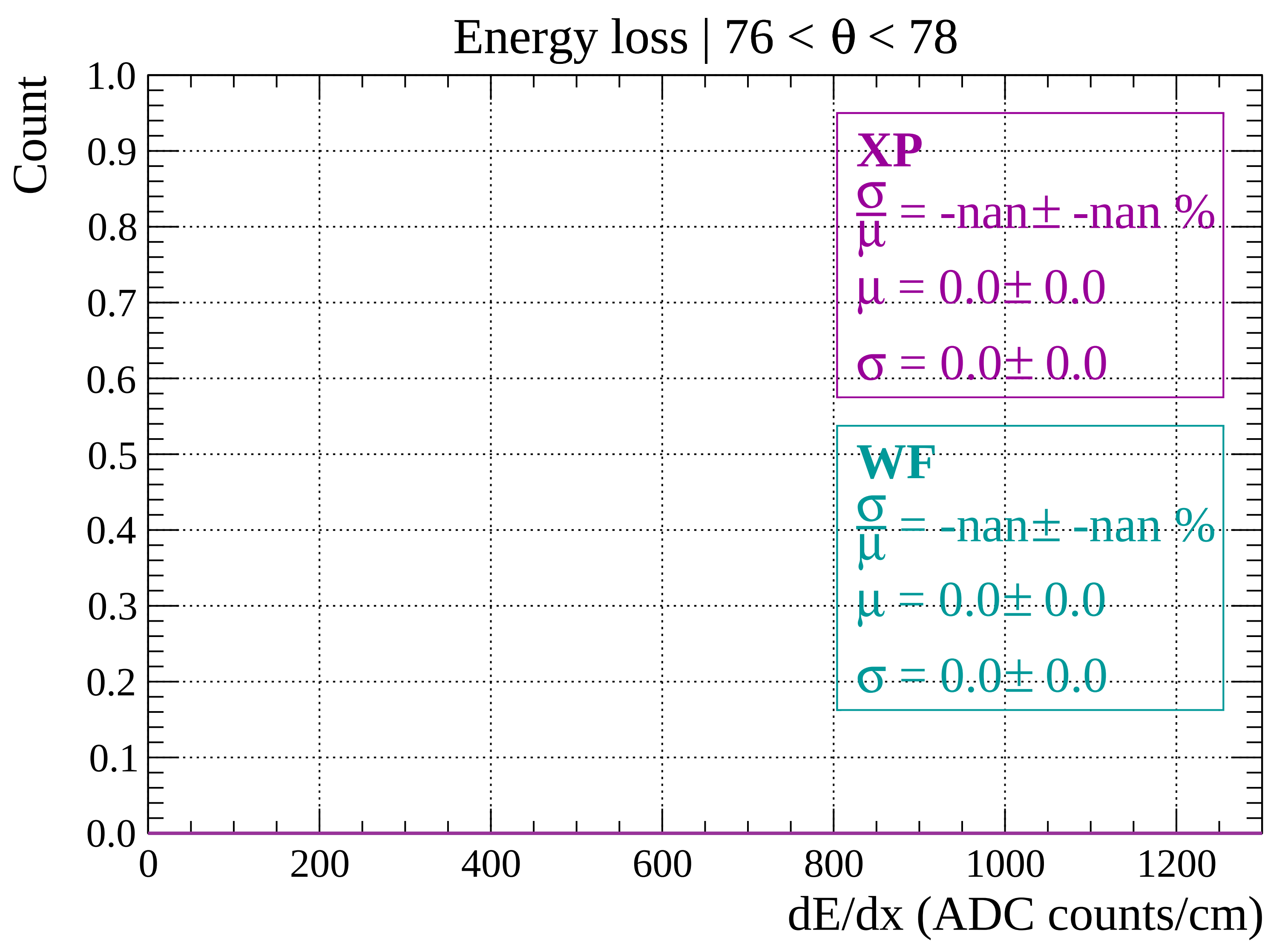

Energy loss | 70 < θ < 72
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

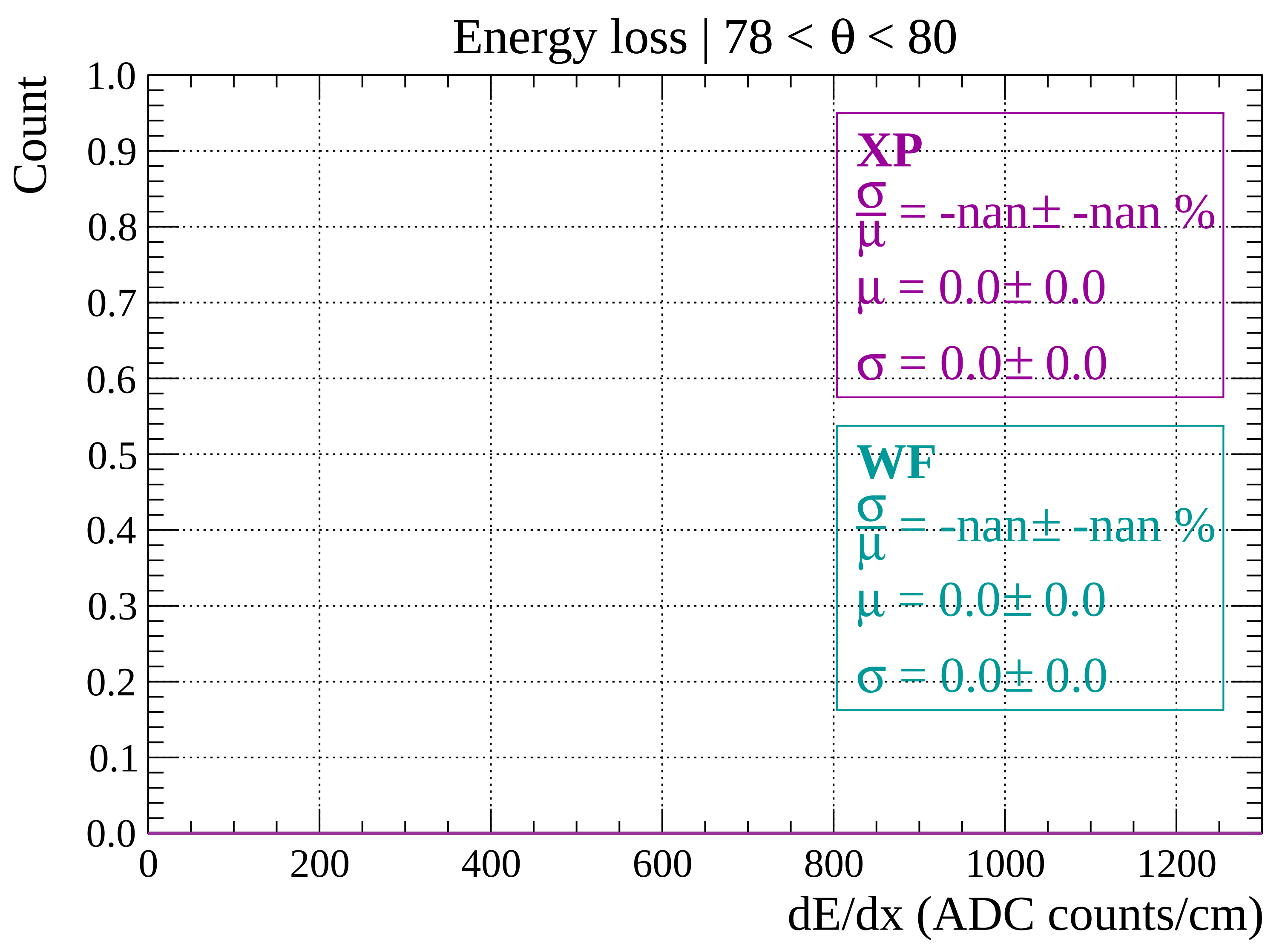


Energy loss | 72 < θ < 74
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

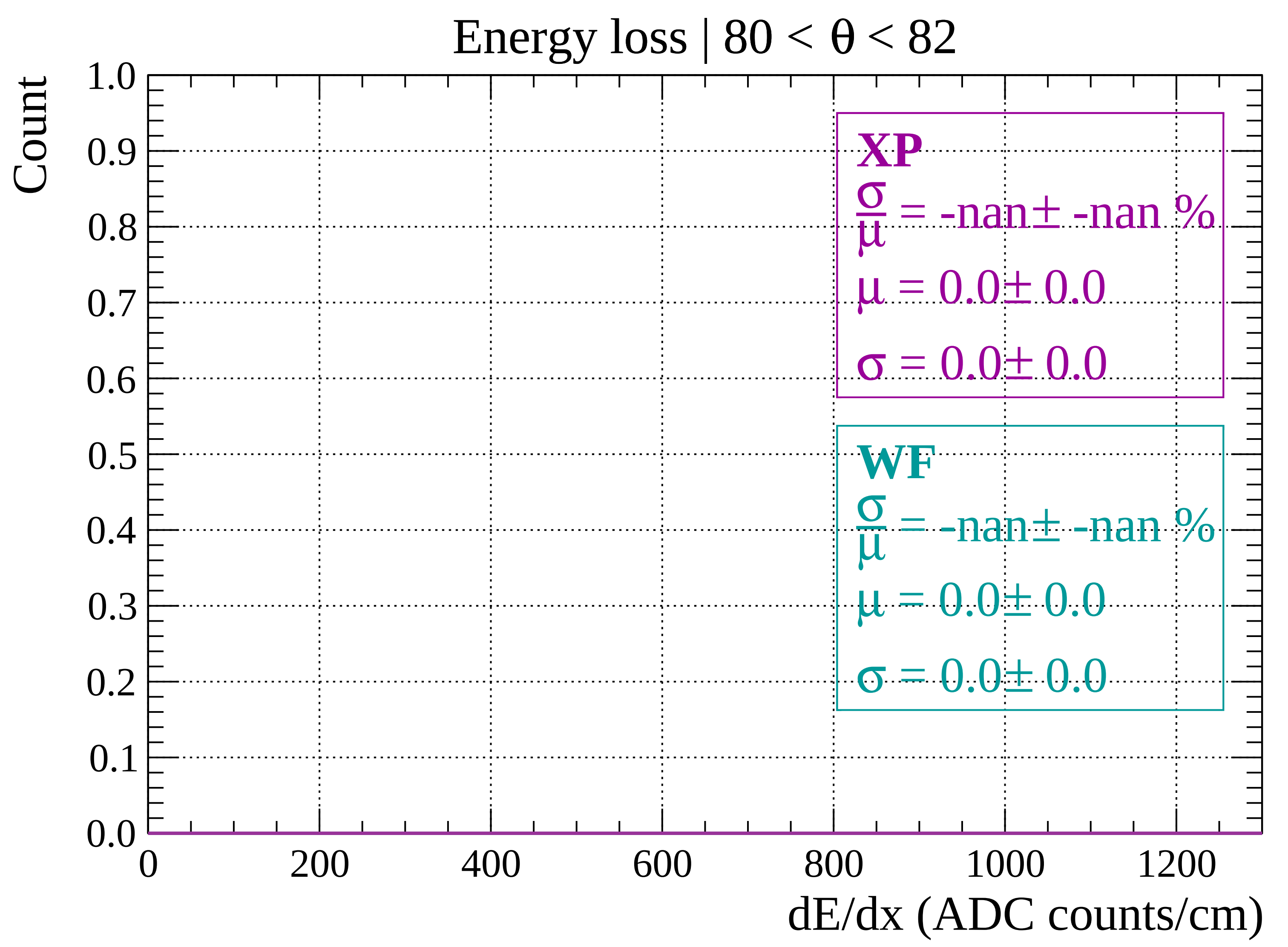


Energy loss | 74 < θ < 76
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

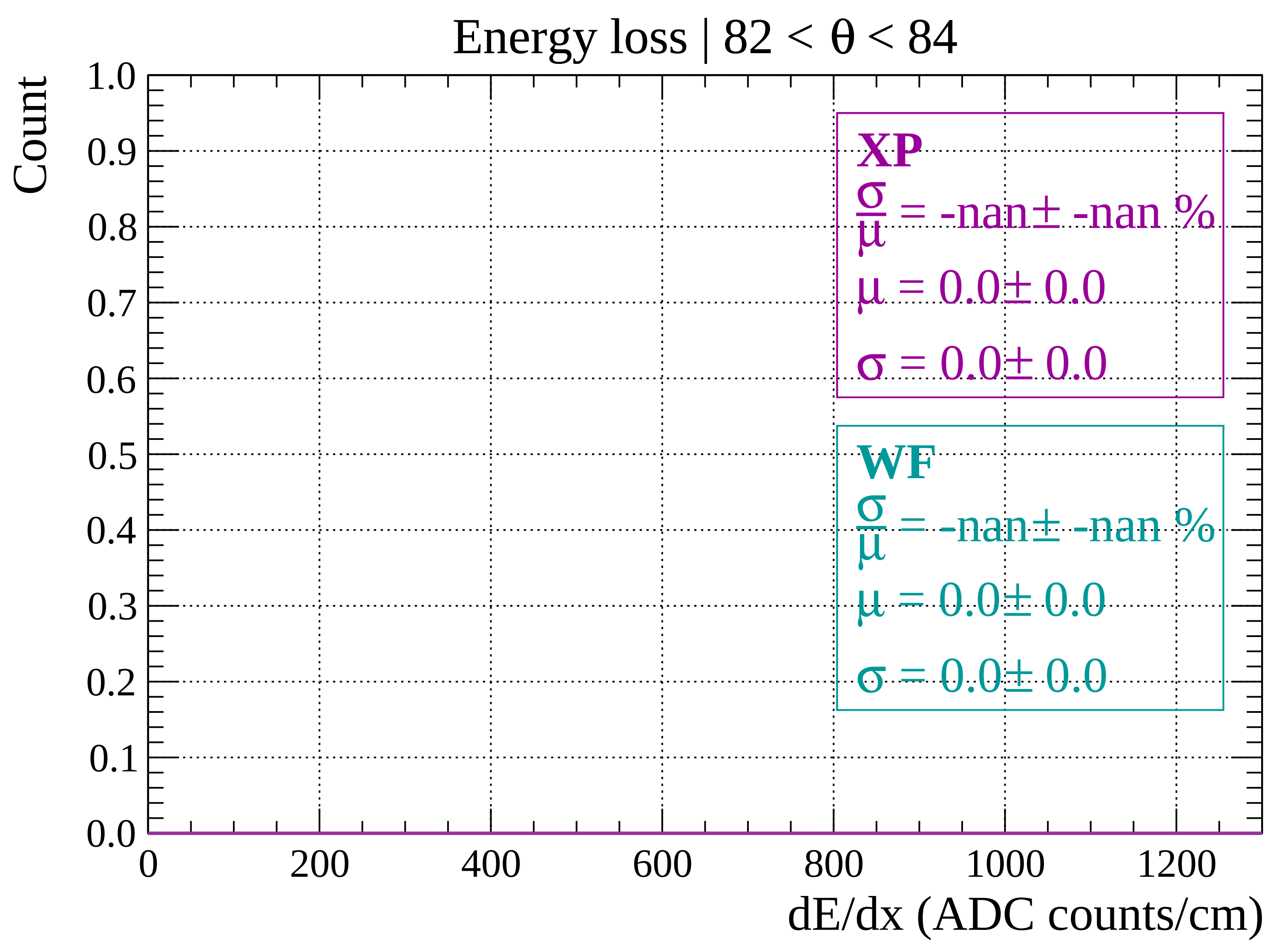

Energy loss | 76 < θ < 78
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

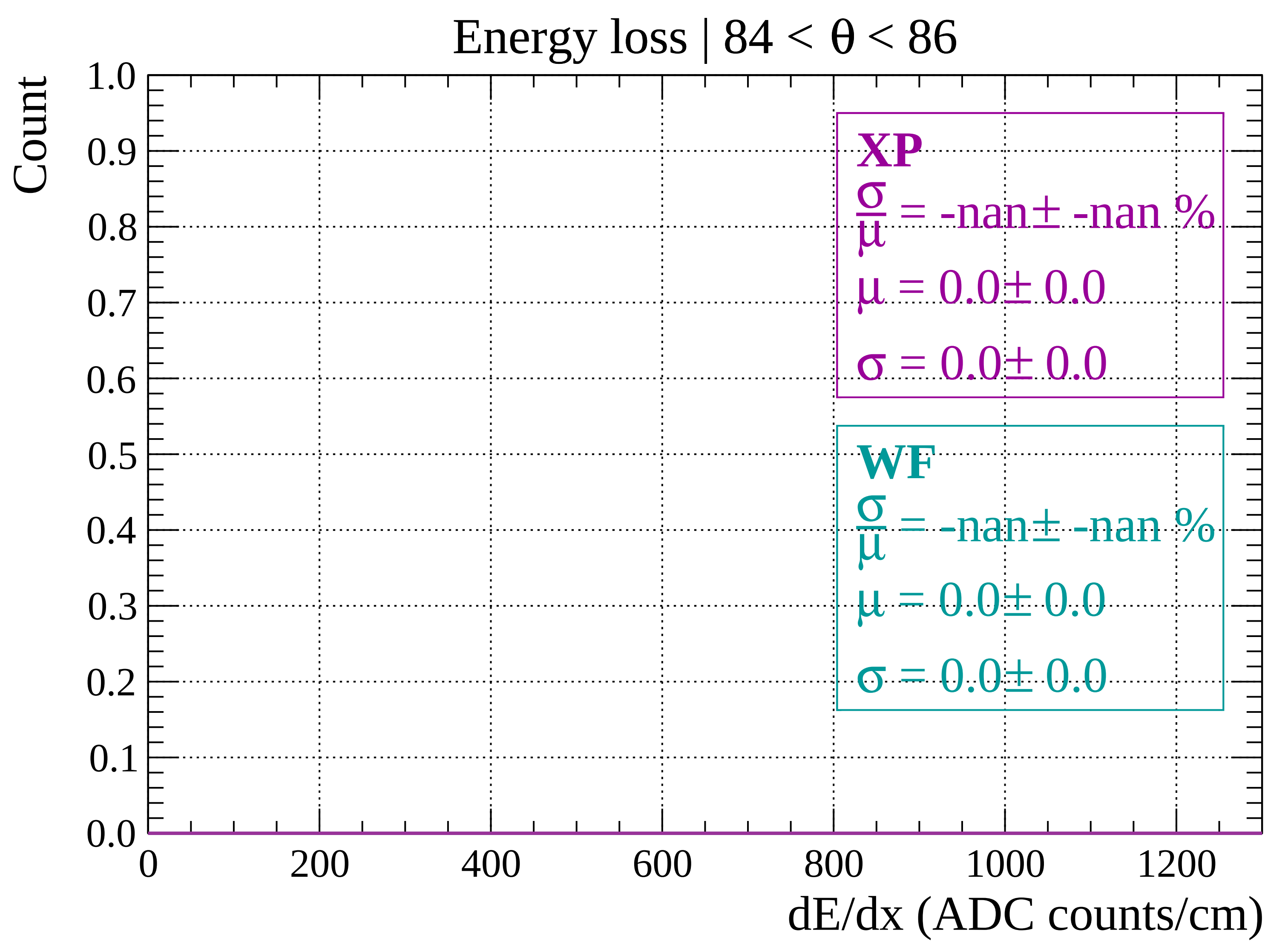


Energy loss | 78 < θ < 80
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

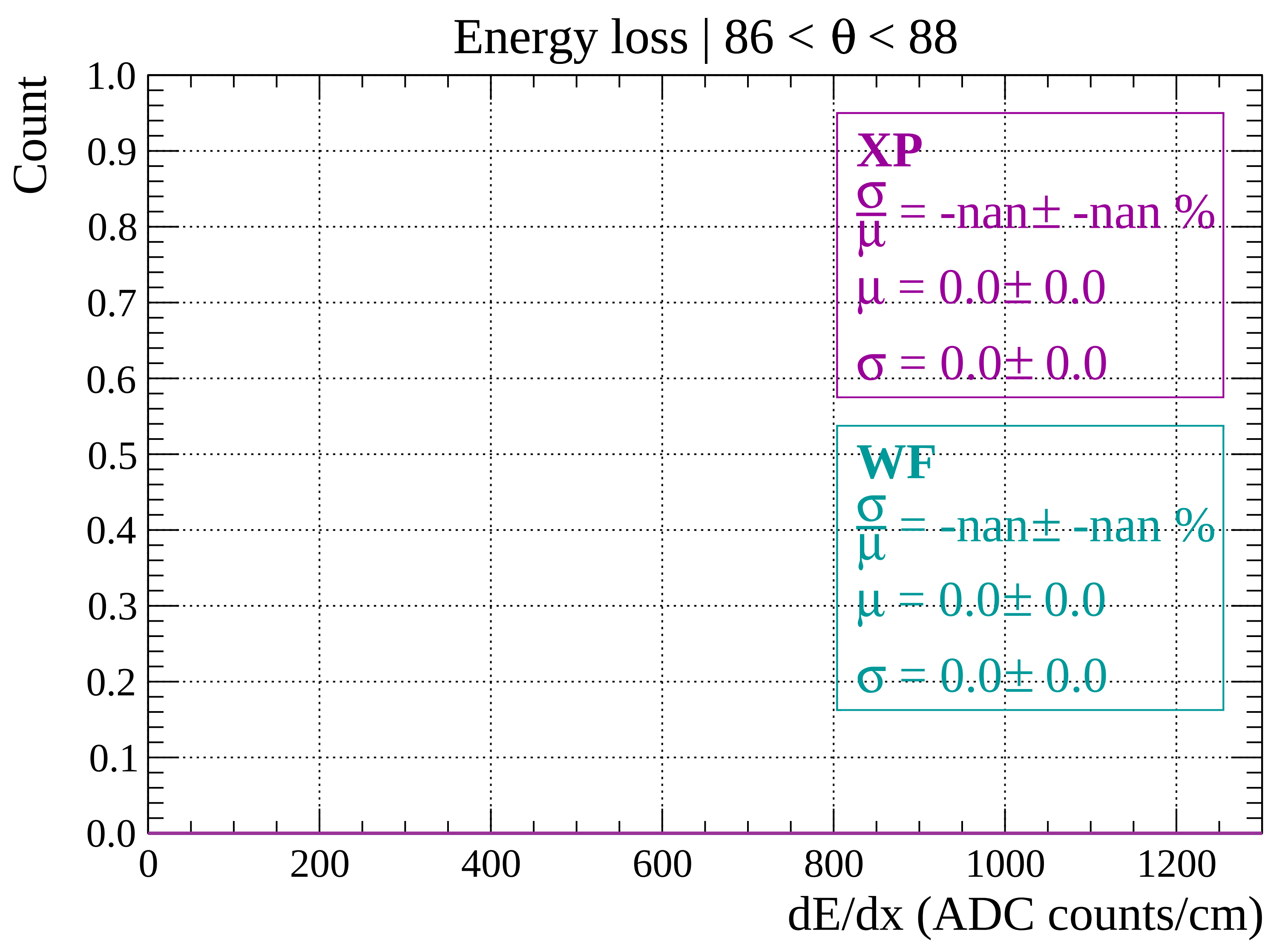


Energy loss | 80 < θ < 82
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

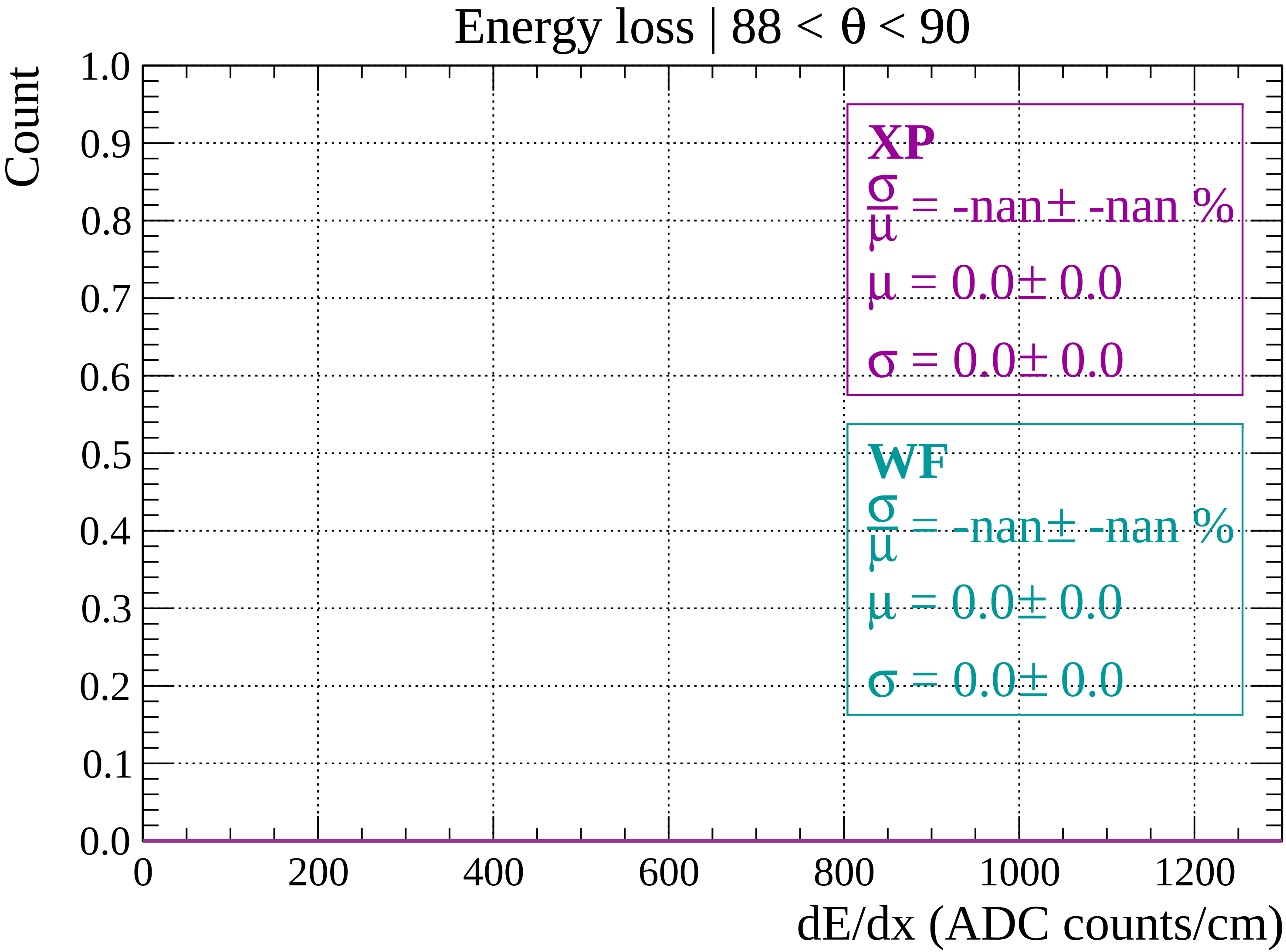

Energy loss | 82 < θ < 84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



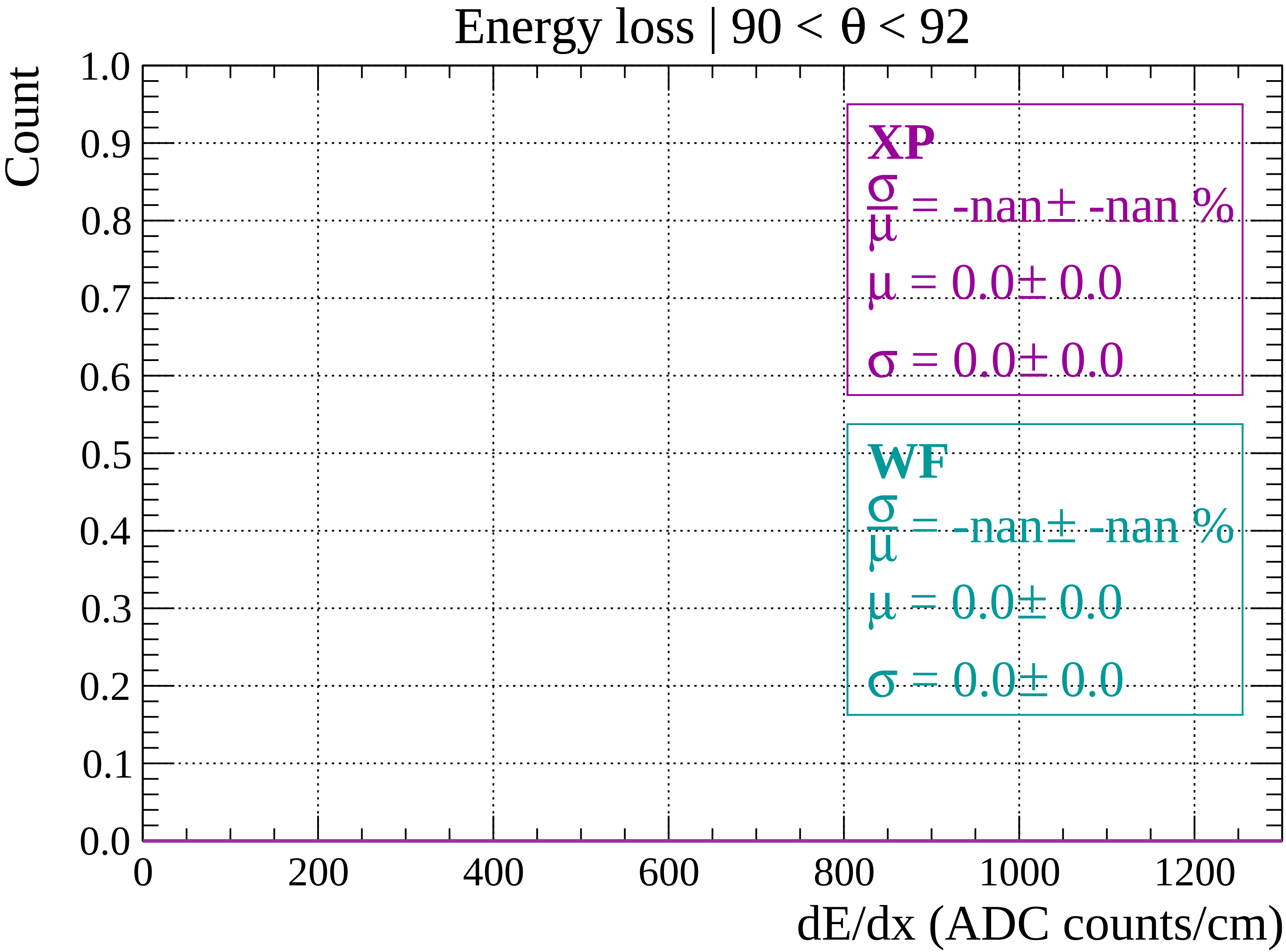
Energy loss | 84 < θ < 86
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



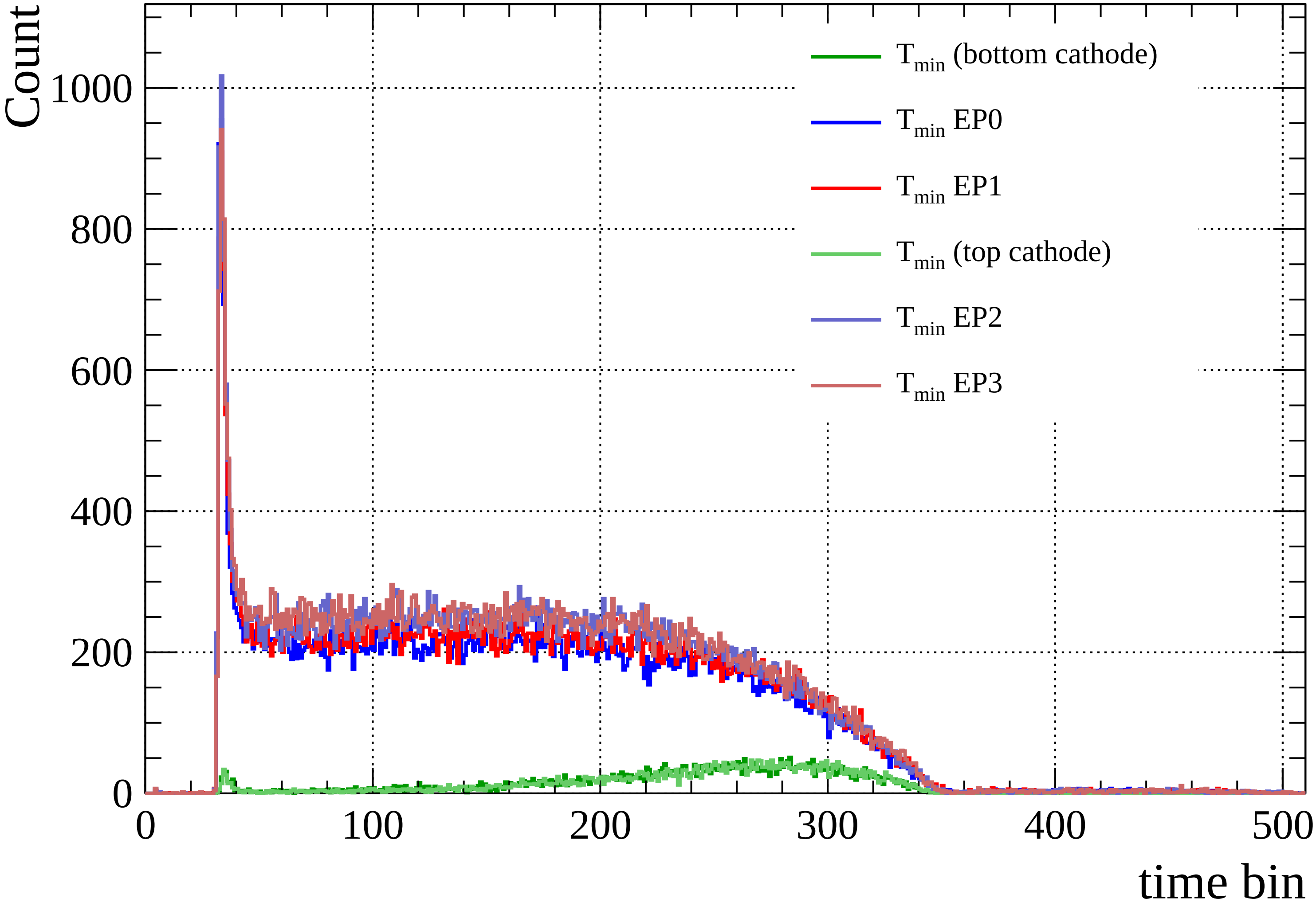
Energy loss | 86 < θ < 88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



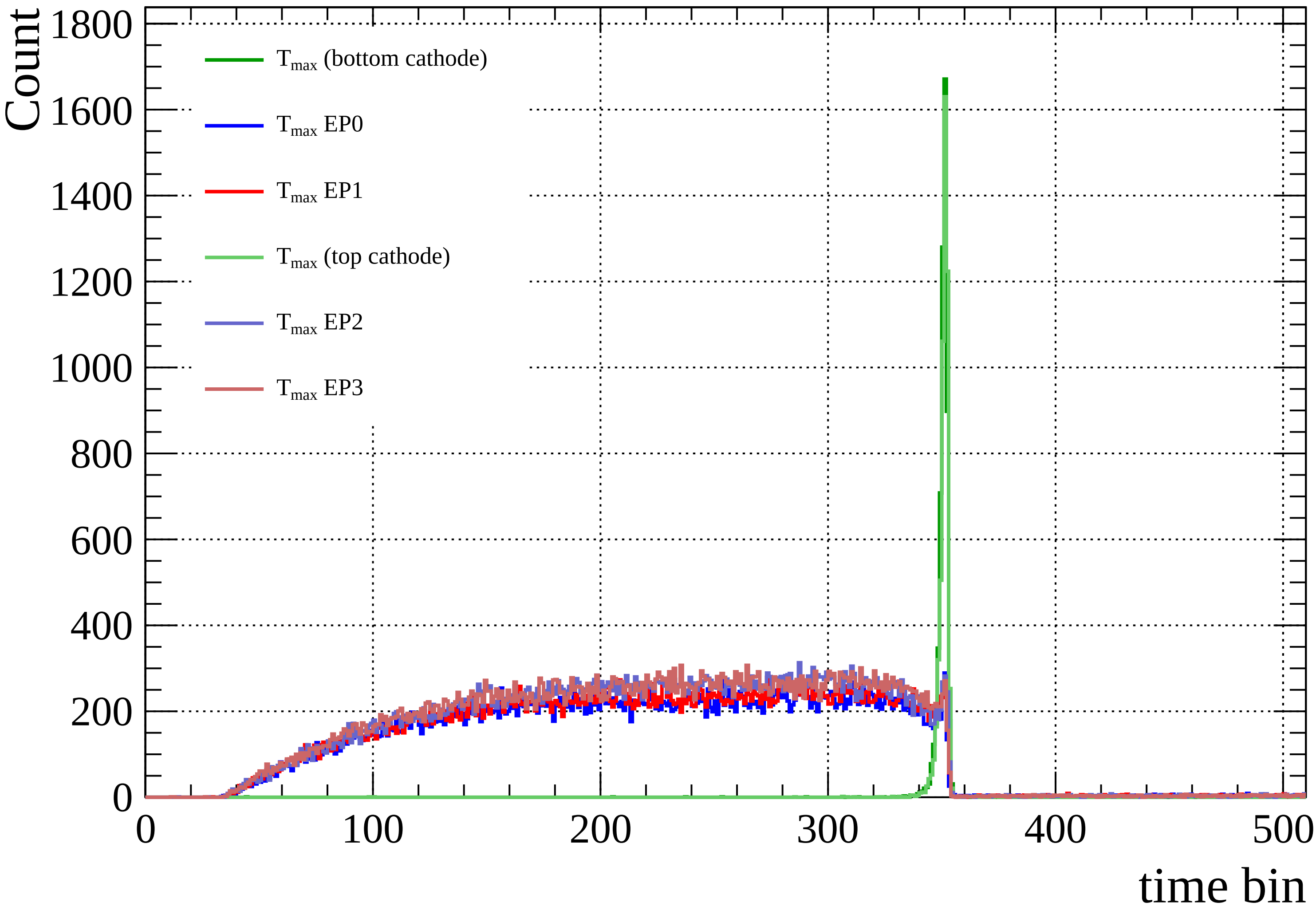
Energy loss | 88 < θ < 90
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Energy loss | 90 < θ < 92
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
1000
800
600
400
200
0
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
200
400
600
800
1000
1200
1400
1600
1800
0
100
200
300
400
500